

FUJITA FREENESS ON COMPLEX PROJECTIVE SEVENFOLDS

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ABSTRACT. We prove that $K_X + 8L$ is globally generated for every smooth complex projective sevenfold X and every ample Cartier divisor L . The proof constructs complete spaces of sections with prescribed point order and controls the entire sequence of log canonical centers arising from their thresholds. Uniform bounds for the full section images, retained adjunction data, and a finite induction on intrinsic sources give the strict coefficient bound needed for point evaluation. The finite inequalities are accompanied by exact rational certificates and independently executable checkers in a documented supplement. The main result of this paper was obtained using the Danus system.

CONTENTS

1. Introduction	1
2. Notation and conventions	4
3. Cutting and point evaluation	5
4. Local centers and persistent adjunction	12
5. Complete descendant costs	23
6. Original section profiles	36
7. Full local images and ambient capacity	49
8. Symbolic images and full tangent components	82
9. Complete original sources	93
10. The exhaustive original-source ledger	116
11. The affine quartic-threefold endpoint	129
12. Cubic-fourfold endpoints	147
13. Triple-fivefold source feedback	159
14. Reduced quadratic tangents at the original source	173
15. Complete local quadratic modules	184
16. The double-fivefold endpoint at order 29/25	198
17. Ambient profile endpoints	212
18. Intrinsic double-fivefold sources	272
19. All-history large-deficit feedback	285
20. Rank-two finite source feedback	297
21. Freeness in dimension seven	308
Appendix A. Details of the use of generative AI	311
References	317

1. INTRODUCTION

Let X be a smooth complex projective variety and let L be an ample Cartier divisor. We write K_X for a canonical divisor and $\mathcal{O}_X(K_X + mL)$ for the corresponding adjoint line bundle. A line

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bundle is globally generated if its global sections generate its fiber at every closed point. Fujita's freeness conjecture predicts that $\mathcal{O}_X(K_X + (n+1)L)$ is globally generated when $\dim X = n$ [Fuj87]. It asks for a sharp uniform coefficient with which an arbitrary ample polarization produces a globally generated adjoint bundle.

The sharp result in dimension two follows from Reider's theorem [Rei88]. Ein and Lazarsfeld prove the three-dimensional statement [EL93, Corollary 2*], and Kawamata proves the four-dimensional statement [Kaw97, Theorem 4.1]. Ye and Zhu establish the five-dimensional statement [YZ20, Main Theorem]. Thus these results give the respective coefficients 3, 4, 5, 6 in dimensions 2, 3, 4, 5.

In arbitrary dimension, Angehrn and Siu obtain the coefficient $1 + n(n+1)/2$ by successively reducing log canonical centers [AS95, Corollary 0.2]. Helmke develops induction criteria that retain the multiplicity and deficit of the center [Hel99]. Heier combines these methods to obtain a bound of order $n^{4/3}$ [Hei02, Theorem 1.4]. Ghidelli and Lacini organize the complete cutting cost to prove generation for integers $m \geq n(\log \log n + 2.34)$, for $n \geq 2$ [GL24, Theorem 1.1]. Their more precise finite bound gives $m \geq 10$ in dimension seven [GL24, Theorem 5.2 and Table 1]. Han proves the linear bound $m \geq \lceil C_0 n \rceil$, with the explicit constant $C_0 = 1.77629988\dots$ [Han26, Theorem 1.1]. His normal Hilbert coefficient estimate provides a further constraint on the multiplicity of an actual minimal log canonical center.

These methods relate the order imposed on a section to the cost of reducing its log canonical center. For the sharp coefficient in the present proof, two requirements must be met together: the complete future cutting cost must fit the remaining coefficient, and the global sections used to achieve that cost must exist on the original polarized variety. A local count alone does not establish the latter requirement. We keep the complete section images and their fixed components throughout the comparison.

We prove the seven-dimensional statement.

Theorem 1.1. *For every smooth complex projective sevenfold X and every ample Cartier divisor L , the line bundle $\mathcal{O}_X(K_X + 8L)$ is globally generated.*

The polarization is arbitrary ample Cartier; its own linear system may have base points. The theorem includes every positive value of the intersection number L^7 . The case $L^7 = 1$ is a reduction within the proof, where the refined source estimates are needed. The coefficient is sharp as a uniform bound: for $X = \mathbb{P}^7$ and $L = \mathcal{O}_{\mathbb{P}^7}(1)$, one has $K_{\mathbb{P}^7} = -8L$, so the coefficient 8 gives the trivial bundle, whereas the coefficient 7 gives $\mathcal{O}_{\mathbb{P}^7}(-1)$ [Hei02, Section 1].

We describe the mechanisms of the proof. Fix a closed point $x \in X$. For an effective rational divisor $B \sim_{\mathbb{Q}} sL$ that is log canonical near x , put

$$b = \text{lct}_x((X, B); \mathfrak{m}_x), \quad M = \text{mult}_x B.$$

Both s and M concern the whole current boundary. If W is its positive-dimensional minimal center, we also retain $d = \dim W$, its multiplicity at x , and its original degree $g = L^d \cdot W$. A successful continuation produces a point center with total coefficient strictly less than 8; Nadel vanishing then gives a section nonzero at x .

First, we compare finite complete section spaces with their asymptotic fixed-order profiles. The multiplier-ideal and valuation foundations are those of Lazarsfeld and Jonsson–Mustață [Laz09, JM12]. For the geometric series used here, we construct a fixed twist and control the error before the later choice of degree, prime or computing valuation. We combine this with bounds for the full ordinary images of primary ideals and for simultaneously spanning tangent components. The local bounds use monomial multiplier ideals and multiplicity–threshold inequalities [How01, dFEM02], but the comparison retains the complete original image, including its conductor and fixed factors. Under hypothetical nongeneration at x in the volume-one case, it yields

$$V(27/25) > 1/32,$$

where $V(u)$ is the asymptotic dimension of $H^0(X, kL \otimes \mathfrak{m}_x^{[ku]})$, normalized by $k^7/7!$. Consequently the original source of order $27/25$ exists in every sufficiently large permitted degree before its threshold is selected. For that threshold the two separate inequalities

$$M \geq \frac{27}{25}s, \quad \frac{27}{25}s + b \leq 7$$

supply different inputs to the later continuations.

Second, we retain the adjunction data as the boundary grows. Fujino–Hashizume adjunction supplies the discriminant and the moduli data on the appropriate higher model [FH23]. Han’s normal Hilbert coefficient estimate and Hilbert-moment inequalities [Han26, Theorem 1.4 and Lemmas 4.2, 4.4] constrain the possible successors. We keep the old discriminant coefficient of a successor prime and the same higher moduli data in the comparison. This gives bounds for complete continuations, including dimension skips. The proof also establishes the required properties of the entire tangent ring and its ordinary powers, including the affine vertex. Every cutting step retains both approximation errors in the published cutting theorem [GL24, Proposition 3.2 and Remark 3.3]; the errors are allocated within a strict total margin.

Third, we bound full symbolic images on the normalizations of all tangent components. The actual component degrees and every conductor enter the count. On a transverse resolution, the intermediate components remain until the intersection comparison is complete. The resulting filtered-quotient estimates give affine continuation bounds for quartic threefolds and cubic fourfolds, which in turn supply the descendant estimates for larger centers. These are statements about the actual singular centers and their incoming boundaries. Their proofs retain the whole source through ordinary restriction to the original center.

Fourth, comparisons of earlier and later ambient sources cover the different initial histories. Complete-series volume theory [LM09, BFJ09] is applied on one moving model, retaining the full fixed divisor, its quadratic and cross terms, and the residual multiplicity. Under hypothetical nongeneration in the volume-one case, take a sufficiently late pure threshold of the complete original source of order $27/25$, at a degree divisible by 25. If its minimal center is a fivefold with $b > 18/5$, one quantity $\mathcal{M}$ controls both the later threshold height and the volume reserve:

$$V(29/25) > (1 - \mathcal{M})_+^7 + \frac{1}{10000}, \quad \mathcal{M} > \frac{341}{1000}.$$

Every sufficiently late pure threshold of the earned $29/25$ -source has computing-profile height at least this same $\mathcal{M}$. A divisorial center would force $V(29/25) \leq (1 - \mathcal{M})_+^7$, contrary to the reserve. The other centers are treated using their own entire threshold boundaries. For double fivefolds of quadratic rank at least three, the intrinsic source endpoint supplies the complementary case $b \leq 18/5$.

Finally, the remaining double fivefolds have quadratic rank two and deficit at most $18/5$. We use their full convergent analytic modules, retaining every factor, branch and opposite character. Analytic splitting, stable matrix-factorization methods and Artin approximation [GP26, Tak08, Art68] enter at distinct steps: the stable data are converted into an actual convergent presentation through a finite exact system. The resulting symbolic-interval estimates, coupled to the old discriminant coefficient and the actual successor deficit, force a stronger source whenever a trial does not yet give an affordable continuation. A finite induction advances the source from $1731/2000$ to $983/1000$ through 235 intervals at one sufficiently large degree divisible by 2000. The terminal normalized leading reserve exceeds $1/2000$. Each trial begins with the same entire incoming boundary, while its threshold prime may change. Only a successful trial enters the additive continuation.

The finite inequalities in these arguments are presented through common certificate definitions and correctness lemmas. The article proves their geometric applicability, exhaustive coverage and error bounds, including the boundary cases of the parameter domains. The accompanying supplement supplies the complete rational data, their formats and independently executable checkers. These programs reconstruct the required domains and algebraic tests from the delivered

records. Their acceptance establishes the finite inequalities under the rules proved in the article. Repeated encodings of the same large certificate are represented by one canonical record stream; the original representations and conversion rules are preserved separately for provenance. Appendix A.4 identifies the computations and their independent checking interface.

The paper is organized as follows. Section 2 fixes the notation. Sections 3–5 develop cutting, retained adjunction and complete continuation costs. Sections 6–8 establish the uniform profiles and full primary and symbolic image bounds. Sections 9 and 10 construct the original source and classify its threshold centers. Sections 11–13 prove the affine descendant estimates. Sections 14–16 develop the double-fivefold geometry, analytic modules and stronger source bound. Sections 17–19 settle the remaining ambient histories and large-deficit ranges. Section 20 proves the finite intrinsic induction, and Section 21 assembles the proof of Theorem 1.1.

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2. NOTATION AND CONVENTIONS

Set-up 2.1. Unless a dimension is explicitly specified, X is a smooth integral complex projective sevenfold, L is ample Cartier and $x \in X$ is a closed point. A boundary is an effective rational divisor. Log canonicity, thresholds and centers through x are local near x ; divisors, section spaces and linear equivalences are global. An assertion stated for ambient dimension n uses the same conventions.

Definition 2.2. For a boundary $B \sim_{\mathbb{Q}} sL$ that is lc near x , set

$$b = \text{def}_x(B) = \text{lct}_x((X, B); \mathfrak{m}_x), \quad M = \text{mult}_x B, \quad \sigma = n - M - b.$$

The deficit is the supremum of the real $t \geq 0$ for which the ideal pair $(X, B; \mathfrak{m}_x^t)$ is lc near x . The coefficient s and multiplicity M include the whole boundary. For a positive-dimensional integral center W through x , put

$$d = \dim W, \quad e = \text{edim } \mathcal{O}_{W,x}, \quad m = \text{mult}_x W, \quad g = L^d \cdot W, \quad r(W) = (m/g)^{1/d}.$$

For $R = \mathcal{O}_{W,x}$ and $\mathfrak{n} = \mathfrak{m}_{W,x}$, the tangent ring is the entire ordinary ring $\text{gr}_{\mathfrak{n}} R$, including its vertex and any nilpotent or embedded structure. Its projective tangent is $\text{Proj}(\text{gr}_{\mathfrak{n}} R)$ with its tautological polarization.

Definition 2.3. For an analytically unramified local ring $(R, \mathfrak{n})$ of dimension $d \geq 1$, the ordinary and normal first Hilbert coefficients are the coefficients e_1 and $\bar{e}_1$ in the eventual expansions

$$\begin{aligned} \text{length}(R/\mathfrak{n}^k) &= m \binom{k+d-1}{d} - e_1 \binom{k+d-2}{d-1} + O(k^{d-2}), \\ \text{length}(R/\overline{\mathfrak{n}^k}) &= m \binom{k+d-1}{d} - \bar{e}_1 \binom{k+d-2}{d-1} + O(k^{d-2}). \end{aligned}$$

A parameter reduction is a reduction generated by a system of parameters. In a Cohen–Macaulay ring these parameters form a regular sequence; the ring itself need not be regular.

Definition 2.4. For every integer $k \geq 1$ and real $t \geq 0$, let

$$U_k(t) = H^0(X, \mathcal{O}_X(kL) \otimes \mathfrak{m}_x^{\lceil kt \rceil}), \quad V(t) = \text{vol}_{\text{Bl}_x X}(\pi^*L - tE_x).$$

The base ideal is the image of the evaluation map after tensoring by $\mathcal{O}_X(-kL)$. It is always the full base ideal. For a nonzero $U_k(t)$ whose full base ideal is proper at x , a pure threshold average is $\Gamma = (\tau/N) \sum_{i=1}^N D_i$, where $\tau = \text{lct}_x(\mathfrak{b}(U_k(t)))$ and the D_i form a sufficiently large sufficiently general tuple of actual members. Its whole coefficient is $s = k\tau$; no earlier boundary is included. For an integral projective center W with $A = L|_W$, write

$$F_p(t) = H^0(W, \mathcal{O}_W(pA) \otimes I_x^{\lceil pt \rceil}).$$

An intrinsic source is complete on the original W , even when W is normal only near x . An ambient inverse image means the full inverse image under $H^0(X, pL) \rightarrow H^0(W, pA)$. An ordinary restriction cutoff is separate from every intrinsic source cutoff. A threshold from B is computed for the whole pair (X, B) and the indicated actual ambient average.

Definition 2.5. An additive continuation from B is a finite chain $B_{i+1} = B_i + t_i H_i$, where $H_i \sim_{\mathbb{Q}} L$ is effective, $t_i > 0$ is rational, the pairs are lc near x , and their minimal centers strictly decrease to $\{x\}$. Its spent coefficient includes $s + \sum_i t_i$. The associated scalar cost is

$$C = \sum_i (b_i - b_{i+1})r(W_i), \quad b_{\text{final}} = 0.$$

A continuation is paid at level $n + 1 - \eta$ if its entire coefficient is less than that number, after all approximation errors. Separate threshold trials from the same B are separate alternatives, not successive terms of one additive chain.

Definition 2.6. A quasi-log structure $[T, \omega]$ is specified by a proper morphism $f : (Y, B_Y) \rightarrow T$ from a globally embedded simple normal crossing pair, with $f^*\omega \sim_{\mathbb{R}} K_Y + B_Y$, such that the natural morphism

$$\mathcal{O}_T \longrightarrow f_* \mathcal{O}_Y(\lceil -B_Y^{\leq 1} \rceil)$$

induces an isomorphism from a coherent ideal $\mathcal{I}_N$ onto $f_* \mathcal{O}_Y(\lceil -B_Y^{\leq 1} \rceil - \lfloor B_Y^{\geq 1} \rfloor)$. The proper closed subscheme N is the non-qlc subscheme. The qlc strata are the images of strata of (Y, B_Y) not contained in N . The qlc centers are those qlc strata which are not irreducible components of T . The structure is qlc when N is empty. The canonical non-quasi-klt subscheme is the canonical union of N with all qlc centers. The notation $B_Y^{\leq 1}$ and $B_Y^{\geq 1}$ selects components by coefficient, retaining their coefficients. A globally embedded simple normal crossing pair is a simple normal crossing divisor in a smooth ambient variety with an ambient divisor having no common component and simple normal crossing total support.

Definition 2.7. For a prime ideal P in a Noetherian domain R , its symbolic power is $P^{(k)} = P^k R_P \cap R$. An ordinary or symbolic graded image below is the image of the entire indicated ideal or module in the entire associated graded ring. A rank-one reflexive module is interpreted through its height-one localizations with their integral lattice. A local ring is G-regular if every finitely generated totally reflexive module is free. A totally reflexive module M is naturally isomorphic to its double dual and satisfies $\text{Ext}^i(M, R) = \text{Ext}^i(M^*, R) = 0$ for every $i > 0$.

3. CUTTING AND POINT EVALUATION

We retain the conventions of Section 2. We record the adjunction and vanishing statements used to lift a section from a minimal center, and prove the point-evaluation criterion. We then identify the two deficit conventions and record the errors in cutting.

For an effective log pair (T, Θ) on a normal variety, choose a log resolution $p : Y \rightarrow T$ and write $K_Y + \Theta_Y = p^*(K_T + \Theta)$. Let $\Theta_Y^{\leq 1}$ denote the reduced sum of the components of coefficient one. The two ideals used below are

$$\mathcal{J}(T, \Theta) = p_* \mathcal{O}_Y(-\lfloor \Theta_Y \rfloor),$$

$$\mathcal{J}_{\text{NLC}}(T, \Theta) = p_* \mathcal{O}_Y(\lceil -\Theta_Y^{\leq 1} \rceil - \lfloor \Theta_Y^{\geq 1} \rfloor) = p_* \mathcal{O}_Y(-\lfloor \Theta_Y \rfloor + \Theta_Y^{\overline{1}}).$$

The second ideal defines the full non-lc subscheme. These are the conventions of [Fuj16, §2.2]. In particular, $\mathcal{J}(T, \Theta) \subseteq \mathcal{J}_{\text{NLC}}(T, \Theta)$, and the non-lc ideal is the unit ideal wherever the pair is lc. A coefficient-one component contributes to the multiplier ideal, whereas the displayed correction removes its contribution to the non-lc ideal.

The algebraic adjunction and vanishing statements are [Fuj16, Theorem 3.8 and Lemma 3.14].

Theorem 3.1 (Canonical quasi-log unions). *Let $[T, \omega]$ be a quasi-log scheme, let $N = T_{-\infty}$ be its non-qlc subscheme with ideal $\mathcal{I}_N$, and let U be the canonical union of N with a union of qlc strata. If $U \neq N$, then U has the induced quasi-log structure with class $\omega|_U$ and non-qlc subscheme N . If $\pi : T \rightarrow S$ is projective and D is Cartier with $D - \omega$ relatively nef and log big, then $R^i \pi_*(\mathcal{I}_U \otimes \mathcal{O}_T(D)) = 0$ for $i > 0$. Here log bigness means bigness on every qlc stratum, relative to its image. For any quasi-log scheme, the reduced closure of its complement of the non-qlc locus inherits its quasi-log structure and non-qlc ideal under the natural inclusion, as in Lemma 3.14.*

For adjunction in the application below, the selected center contains a point outside N , so that $U \neq N$, as required in [Fuj16, Theorem 3.8(i)]. Vanishing in part (ii) also applies when the selected union of strata is empty.

The analytic formulation makes explicit which strata enter the positivity hypothesis; see [Fuj22, Theorems 4.4, 4.5 and 4.8].

Theorem 3.2 (Analytic quasi-log adjunction and vanishing). *Let $[T, \omega]$ be a quasi-log complex analytic space and U the union of its non-qlc subscheme with a union of qlc strata. On every relatively compact open subset of T , the induced quasi-log construction has class $\omega|_U$ and the same non-qlc subscheme. Its defining ideals are independent of that construction and glue to a globally defined coherent ideal. For a projective morphism $\pi : T \rightarrow S$ and a line bundle D , assume $D - \omega$ is relatively nef and its restriction to each qlc stratum not contained in U is big over the image. Then $R^i \pi_*(\mathcal{I}_U \otimes D) = 0$ for $i > 0$.*

The coherent union ideal is canonical even when its construction is made on relatively compact analytic opens. Ampleness of the adjoint difference satisfies all the positivity requirements in this theorem.

The normalization and log-pair constructions are [Fuj21, Theorems 1.9, 1.10 and Lemma 3.10].

Theorem 3.3 (Normalization of a quasi-log scheme). *For an irreducible quasi-log scheme $[T, \omega]$, its normalization $\nu : Z \rightarrow T$ has the induced quasi-log structure and*

$$\nu_* \mathcal{I}_{\text{Nqklt}}(Z, \nu^* \omega) = \mathcal{I}_{\text{Nqklt}}(T, \omega).$$

If T is normal quasi-projective and H is an ample real divisor, there is an effective real divisor Θ with $K_T + \Theta \sim_{\mathbb{R}} \omega + H$ and $\text{Nklt}(T, \Theta) = \text{Nqklt}(T, \omega)$ set-theoretically. If the quasi-log structure and H are rational, Θ can be chosen rational and the equivalence rational. The ample addition remains part of this assertion. For a normal quasi-projective variety T and an effective real divisor Δ with $K_T + \Delta$ real Cartier, there exists a projective birational morphism $g : Y \rightarrow T$ from a normal $\mathbb{Q}$ -factorial variety for which $K_Y + \Delta_Y = g^(K_T + \Delta)$ and (Y, Δ'_Y) is dlt, where Δ'_Y caps the coefficients of Δ_Y at one. Every exceptional prime has coefficient one in Δ'_Y , and $g^{-1} \text{Nklt}(T, \Delta) = \text{Nklt}(Y, \Delta_Y) = \text{Nklt}(Y, \Delta'_Y)$ set-theoretically. The full crepant divisor Δ_Y is retained in pullback identities.*

The normalization equality retains the canonical ideal with its scheme structure. The ordinary representative has the prescribed non-klt support; a comparison of its multiplier ideal with a prescribed ideal requires a separate argument. On the modification, if $\Delta_Y = \sum_j d_j E_j$, then

$$\Delta'_Y = \sum_{d_j < 1} d_j E_j + \sum_{d_j \geq 1} E_j.$$

The full coefficients d_j remain in the crepant pullback identity.

We also use the local statements [Liu19a, Theorem 2.10 and Proposition 3.4].

Theorem 3.4 (Quasi-log inversion and a local cut). *Let $[T, \omega]$ be qlc and $B \geq 0$ real Cartier whose support contains no qlc center. For a union T' of qlc strata, $[T, \omega + B]$ is qlc near T' if and only if its induced structure on T' is qlc. Let T be projective irreducible, ω real Cartier, x in a nonempty qlc open, and Z the positive-dimensional minimal qlc stratum through x , of dimension k . If H is ample with $H^k \cdot \bar{Z} > k^k$, there are $B \geq 0$, $B \sim_{\mathbb{R}} H$, and $0 < c < 1$ such that $[T, \omega + cB]$ remains qlc on a neighborhood of x and its minimal stratum through x has smaller dimension.*

The first assertion concerns the structure induced after adding the specified divisor. The second constructs a divisor under a strict intersection inequality on the closure of the actual minimal stratum.

The following two assertions are [FL20, Theorem 1.4] and [Fuj17, Theorem 1.6], respectively.

Theorem 3.5 (Surface sections with the canonical ideal). *For a normal projective irreducible qlc surface $[T, \omega]$, let D be Cartier and $A = D - \omega$ satisfy $A^2 > 4$ and $A \cdot C \geq 2$ for every curve C . Then $\mathcal{I}_{\text{Nqklt}} \otimes \mathcal{O}_T(D)$ has a section nonzero at any point outside Nqklt . For a complete irreducible normal log surface (T, Δ) with Δ an effective real divisor whose coefficients lie in $[0, 1]$, and $K_T + \Delta$ real Cartier, let D be Cartier and $A = D - (K_T + \Delta)$ nef. If $A^2 > 4$ and $A \cdot C \geq 2$ for every curve through x , then $\mathcal{O}_T(D)$ is generated at x . Global log canonicity is not an additional hypothesis of this second assertion.*

The first assertion supplies a section in the canonical ideal and requires a globally qlc surface. The second gives ordinary evaluation on a log surface and requires the curve inequalities only through the marked point.

In dimension three we use the pure-dimensional case of [Liu19b, Theorems 1.2 and 6.3].

Theorem 3.6 (Quasi-log threefold freeness). *Let $[T, \omega]$ be a projective pure three-dimensional qlc scheme, let D be Cartier, and put $A = D - \omega$. Assume A is ample. If $A^3 \cdot T_i > 27$ on every irreducible component T_i , $A^2 \cdot S \geq 9$ on every integral surface, and $A \cdot C \geq 3$ on every integral curve, then $\mathcal{O}_T(D)$ is globally generated. The numerical hypotheses are required on all the indicated subvarieties.*

All components have dimension three, so the dimension parameter in the source's Fujita condition equals three on every proper positive-dimensional subvariety. This accounts for both the surface and curve constants.

The following disjoint-union version is [Fuj11, Theorem 8.1].

Theorem 3.7 (Vanishing with the non-lc ideal). *Let T be normal, $B \geq 0$ with $K_T + B$ real Cartier, $\pi : T \rightarrow S$ projective, and D Cartier with $D - (K_T + B)$ relatively ample. Let W be a reduced union of lc centers disjoint from $\text{Nlc}(T, B)$. For $\mathcal{I} = \mathcal{I}_W \mathcal{J}_{\text{NLC}}(T, B)$, one has $R^i \pi_*(\mathcal{I} \otimes \mathcal{O}_T(D)) = 0$ for $i > 0$, and the restriction to the canonical union of W and the non-lc subscheme is surjective.*

Disjointness makes the product of the two ideals their intersection. For a center meeting the non-lc subscheme away from the marked point, the next lifting lemma uses the canonical union and computes its intersection ideal directly.

We record the restricted-order form of the deficit inequality from [YZ20, Definition 2.5 and Lemma 2.10].

Theorem 3.8 (Restricted-order deficit drop). *Let (Y, B) be klt, let $G \geq 0$ be a rational Cartier divisor such that $(Y, B + G)$ is lc at x , and let Z be its minimal lc center there. Suppose $D \geq 0$ is rational Cartier, does not contain Z , and $(Y, B + G + D)$ is lc at x with a properly smaller minimal center. Then its deficit is at most the old deficit minus $\text{ord}_x(D|_Z)$, with the deficit convention of the cited definition. Lemma 3.13 identifies that convention with the point-ideal convention used here.*

The order here is the Cartier-power order on the actual center. If f_m is a local equation of $mD|_Z$, its source convention is

$$\mathrm{ord}_x(D|_Z) = \sup_{\substack{m \geq 1 \\ mD|_Z \text{ Cartier at } x}} \frac{1}{m} \max\{j : f_m \in \mathfrak{m}_{Z,x}^j\}.$$

The assumption that D does not contain Z makes this restriction an effective rational Cartier divisor. Both deficits concern the entire pair with boundary $B + G$, before and after the addition.

Lemma 3.9. *Let Z be smooth projective, Δ a boundary lc near x , and W its reduced irreducible minimal lc center through x . Put $\omega = K_Z + \Delta$, let N be its full non-lc subscheme with ideal $\mathcal{I}_N$, and let $U = N \cup W$ be the canonical quasi-log union. Then*

$$\mathcal{I}_U = \mathcal{I}_N \cap \mathcal{I}_W, \quad \mathcal{I}_{\mathrm{Nqlc}(W, \omega|_W)} = (\mathcal{I}_N + \mathcal{I}_W)/\mathcal{I}_W.$$

The quasi-log structure on W is obtained by adjunction first to U , followed by removal of components contained in N . The canonical non-quasi-klt ideal of W is contained in this non-qlc ideal, and its zero scheme avoids x . For every Cartier D with $D - \omega$ ample, restriction gives a surjection

$$H^0(Z, \mathcal{I}_N \otimes \mathcal{O}_Z(D)) \longrightarrow H^0(W, \mathcal{I}_{\mathrm{Nqlc}(W, \omega|_W)} \otimes \mathcal{O}_W(D|_W)).$$

A section on the right nonzero at x therefore lifts to a section of $\mathcal{O}_Z(D)$ nonzero at x .

The lemma identifies the particular ideal on the center through which ambient lifting is possible.

Proof. We compute the canonical union ideal, apply adjunction to this union and then to its surviving reduced space, and use vanishing to lift sections.

Step 1. In this step, we identify the canonical union ideal. A log resolution of the full pair (Z, Δ) , with its crepant boundary, gives the standard quasi-log structure with class ω and non-qlc ideal $\mathcal{I}_N = \mathcal{J}_{\mathrm{NLC}}(Z, \Delta)$. Indeed the negative part of the resolved boundary is exceptional; normality identifies the pushforward allowing these exceptional poles with $\mathcal{O}_Z$. The rounded boundary then gives precisely the ideal in Definition 2.6. The source is globally embedded by placing the smooth resolution in its product with a smooth curve. All coefficients are rational.

Since the pair is lc near x , we have $x \notin \mathrm{Supp} N$. The center W is a reduced irreducible qlc stratum not contained in N . Thus $U \neq N$, and adjunction in Theorem 3.1 gives $[U, \omega|_U]$ with non-qlc subscheme N .

We spell out the ideal in this construction. In the source construction of [Fuj16, Theorem 3.8(i)], modify the globally embedded source $f : (Y, \Theta) \rightarrow Z$ so that the union Y' of strata mapping into U is a union of irreducible components. Write $Y'' = Y - Y'$, $P = \lceil -\Theta^{<1} \rceil$, and $Q = \lfloor \Theta^{>1} \rfloor$. The source restriction sequence is

$$0 \longrightarrow \mathcal{O}_{Y''}(P - Q - Y') \longrightarrow \mathcal{O}_Y(P - Q) \longrightarrow \mathcal{O}_{Y'}(P - Q) \longrightarrow 0.$$

The strict-support argument in that adjunction theorem gives

$$(3.1) \quad 0 \longrightarrow \mathcal{I}_U \longrightarrow \mathcal{I}_N \longrightarrow f_* \mathcal{O}_{Y'}(P - Q) \longrightarrow 0, \quad \mathcal{I}_U = f_* \mathcal{O}_{Y''}(P - Q - Y').$$

The right-hand map is restriction of pulled-back regular functions. These are also the canonical maps in [Fuj22, Theorems 4.4–4.5, equations (4.2)–(4.3)]; their canonicity is the one in Theorem 3.2.

The canonical closed union contains N and the reduced stratum W , so $\mathcal{I}_U \subseteq \mathcal{I}_N \cap \mathcal{I}_W$. Conversely, let h be a local section of $\mathcal{I}_N \cap \mathcal{I}_W$. Every irreducible component of Y' maps into $\mathrm{Supp} N$ or into W . The regular function f^*h therefore vanishes identically on each such component. The space Y' is reduced, so this restriction is zero on Y' . The map to the twisted sheaf in (3.1) is induced by this same restriction; its value on h is zero. Exactness gives $h \in \mathcal{I}_U$, proving $\mathcal{I}_U = \mathcal{I}_N \cap \mathcal{I}_W$.

Step 2. In this step, we identify the induced non-qlc ideal on the center. The non-qlc ideal inside $\mathcal{O}_U$ is $\mathcal{I}_N/\mathcal{I}_U$. We now apply the removal assertion of Theorem 3.1, namely [Fuj16, Lemma 3.14(i)–(ii)], to U . Its surviving reduced space is

$$\overline{U \setminus \text{Supp } N_{\text{red}}} = \overline{W \setminus \text{Supp } N_{\text{red}}} = W,$$

because W is reduced and irreducible and is not contained in $\text{Supp } N$. This gives $[W, \omega|_W]$. The restriction diagram in the proof of that lemma identifies its non-qlc ideal with the image of $\mathcal{I}_N/\mathcal{I}_U$ under $\mathcal{O}_U \rightarrow \mathcal{O}_W$.

To identify the embedding of this ideal, work on an affine open $\text{Spec } R_0 \subseteq Z$, and let $J_0, I_0 \subseteq R_0$ represent $\mathcal{I}_N, \mathcal{I}_W$. The map in question is

$$J_0/(J_0 \cap I_0) \longrightarrow R_0/I_0, \quad h + (J_0 \cap I_0) \longmapsto h + I_0.$$

It is injective and has image $(J_0 + I_0)/I_0$. Hence

$$(3.2) \quad \mathcal{I}_{\text{Nqlc}(W, \omega|_W)} = \frac{\mathcal{I}_N + \mathcal{I}_W}{\mathcal{I}_W} = \mathcal{I}_{N \times_Z W, W}.$$

The last equality follows from $\mathcal{O}_{N \times_Z W} = \mathcal{O}_Z/(\mathcal{I}_N + \mathcal{I}_W)$. Thus the calculation retains the embedded ideal and the entire scheme-theoretic intersection. Rationality of the source boundary is preserved under both adjunction operations.

Let Q_W be the canonical non-quasi-klt subscheme of this structure on W . It contains $N \times_Z W$ and the proper qlc strata of W . Thus $\mathcal{I}_{Q_W, W} \subseteq \mathcal{I}_{\text{Nqlc}(W, \omega|_W)}$. Adjunction and removal preserve exactly the original qlc strata contained in W . None of the proper ones contains x , by minimality of W , and $x \notin \text{Supp } N$. It follows that $x \notin \text{Supp } Q_W$.

Step 3. In this step, we lift sections while preserving their nonzero value at x . Let $i : W \hookrightarrow Z$ be the closed immersion. The quotient calculation gives the exact sequence on Z

$$(3.3) \quad 0 \longrightarrow \mathcal{I}_U \otimes \mathcal{O}_Z(D) \longrightarrow \mathcal{I}_N \otimes \mathcal{O}_Z(D) \longrightarrow i_*(\mathcal{I}_{\text{Nqlc}(W, \omega|_W)} \otimes \mathcal{O}_W(D|_W)) \longrightarrow 0.$$

Tensoring preserves exactness because D is Cartier. The morphism $Z \rightarrow \text{Spec } \mathbb{C}$ is projective, and $D - \omega$ is ample. It is therefore nef and big on every qlc stratum, including all strata outside U . Theorem 3.1 gives $H^1(Z, \mathcal{I}_U \otimes \mathcal{O}_Z(D)) = 0$; these hypotheses also satisfy Theorem 3.2. Taking cohomology of (3.3) proves the claimed surjection. At x , both $\mathcal{I}_N$ and the ideal in (3.2) are unit ideals. Restriction and the inclusion $\mathcal{I}_N \otimes \mathcal{O}_Z(D) \subseteq \mathcal{O}_Z(D)$ therefore preserve the nonzero value at x . $\square$

In ambient dimension n , if the whole boundary is $\Delta \sim_{\mathbb{Q}} sL$ and the target is $D = K_X + (n+1)L$, the adjoint remainder in this proof is

$$D - (K_X + \Delta) \sim_{\mathbb{Q}} (n+1-s)L.$$

Thus lifting uses exactly $s < n+1$, with every old coefficient included in s . For a sevenfold the remainder is $(8-s)L$.

We next record [GL24, Theorem 2.5].

Theorem 3.10 (Nadel vanishing). *Let Z be smooth projective, D an effective rational divisor, and N an integral divisor with $N - D$ big and nef. Then*

$$H^i(Z, \mathcal{O}_Z(K_Z + N) \otimes \mathcal{J}(Z, D)) = 0 \quad (i > 0).$$

Here the vanishing ideal is the multiplier ideal. Its quotient detects the non-klt locus, including the point center needed for evaluation. The intersection and uniqueness statements are [Fuj11, Theorem 9.1(2), (4)], applied on the lc neighborhood. Point tie breaking is [GL24, Lemma 2.8].

Theorem 3.11. *For an effective pair lc near a point, every component through that point of the intersection of two lc centers is an lc center. There is a unique minimal center through a non-klt point. If Z is klt, Δ is effective rational and (Z, Δ) is lc at z with $\{z\}$ an lc center, then for every ample rational Cartier divisor A there is a rational $a > 0$ such that, for every sufficiently*

small rational $\gamma > 0$, some effective $D \sim_{\mathbb{Q}} aA$ makes $(Z, (1 - \gamma)\Delta + \gamma D)$ lc at z with $\{z\}$ its only lc center containing z .

In the point application of [GL24, Lemma 2.8], the finite set is $\{z\}$, so its conditions on other marked points are vacuous. The ample coefficient is fixed before the perturbation parameter is made small.

Proposition 3.12. *Let Z be smooth complex projective, A ample Cartier, and $\Delta \sim_{\mathbb{Q}} cA$, $c \geq 0$, a boundary lc near z with minimal center $\{z\}$. For every integer $a > c$, $\mathcal{O}_Z(K_Z + aA)$ is generated at z .*

The strict difference $a - c$ pays for tie breaking and leaves an ample remainder for Nadel vanishing.

Proof. We first isolate the point center with a small change in coefficient, then lift a section of the multiplier-ideal quotient.

Step 1. In this step, we isolate the point center while keeping the coefficient below a . The ambient variety is smooth, hence klt, and $\{z\}$ is an lc center of the effective rational pair. Apply Theorem 3.11 with ample divisor A . Denote its fixed positive rational coefficient by λ . For sufficiently small rational $\gamma > 0$, it supplies an effective $E \sim_{\mathbb{Q}} \lambda A$ such that

$$\Gamma = (1 - \gamma)\Delta + \gamma E \sim_{\mathbb{Q}} c_{\gamma}A, \quad c_{\gamma} = (1 - \gamma)c + \gamma\lambda,$$

is lc near z with $\{z\}$ its only lc center through z . Choose $\gamma < 1$ sufficiently small that $\gamma|\lambda - c| < a - c$. Then $c_{\gamma} < a$, and Γ is effective and rational.

Step 2. In this step, we lift a nonzero point value from the multiplier-ideal quotient. Put $\mathcal{J}_{\Gamma} = \mathcal{J}(Z, \Gamma)$ and let T_{Γ} be the closed subscheme it defines. The pair is lc but not klt at z , so $(\mathcal{J}_{\Gamma})_z$ is proper and is contained in $\mathfrak{m}_z$. In the lc neighborhood of z , the non-klt locus is the union of lc centers. Its only component through z is therefore $\{z\}$. Consequently T_{Γ} is a disjoint union $T_z \amalg T_1$, where T_z is an Artinian scheme supported at z and $z \notin T_1$.

Let $\mathcal{L} = \mathcal{O}_Z(K_Z + aA)$. The divisor aA is integral and

$$aA - \Gamma \sim_{\mathbb{Q}} (a - c_{\gamma})A$$

is ample, hence big and nef. Theorem 3.10 gives $H^1(Z, \mathcal{J}_{\Gamma} \otimes \mathcal{L}) = 0$. The sequence

$$0 \longrightarrow \mathcal{J}_{\Gamma} \otimes \mathcal{L} \longrightarrow \mathcal{L} \longrightarrow \mathcal{L}|_{T_{\Gamma}} \longrightarrow 0$$

therefore gives a surjection on global sections onto the last term. The invertible sheaf $\mathcal{L}|_{T_z}$ is free of rank one over the Artinian local ring of T_z . Choose a generator, whose residue at z is nonzero, and extend it by zero on T_1 . A global lift has the same nonzero residue at z , as required. $\square$

Lemma 3.13. *The point-ideal threshold of an effective lc pair equals the supremum of $\text{mult}_x D$ over effective rational divisors D for which the same entire pair plus D remains lc near x .*

This identifies the deficit in Definition 2.2 with the ordinary divisor-order deficit on the smooth ambient variety.

Proof. Write the pair as (X, B) and its point-ideal threshold as b . For a prime divisor E over X , let A_E be its log discrepancy with respect to the entire boundary B . The valuative description is

$$(3.4) \quad b = \inf_{\text{ord}_E(\mathfrak{m}_x) > 0} \frac{A_E}{\text{ord}_E(\mathfrak{m}_x)}.$$

The divisors in this infimum have center $\{x\}$. Divisors whose center contains x and has positive dimension have point-ideal order zero and remain subject to the original lc inequalities.

Step 1. In this step, we bound every admissible divisor order by the point-ideal threshold. Let $D \geq 0$ be rational and assume $(X, B + D)$ is lc near x . Set $\alpha = \text{mult}_x D$. Choose a positive integer l such that lD is Cartier at x , with local equation h . Smoothness gives $h \in \mathfrak{m}_x^{l\alpha}$. Thus

$$\text{ord}_E(D) \geq \alpha \text{ord}_E(\mathfrak{m}_x), \quad A_E \geq \text{ord}_E(D).$$

Taking the infimum in (3.4) gives $\alpha \leq b$.

Step 2. In this step, we realize rational orders below the threshold by global effective divisors. Suppose $b > 0$, fix a rational q with $0 < q < b$, and choose an integer $k > q$. Take a simultaneous log resolution $f : Y \rightarrow X$ of B and $\mathfrak{m}_x$, and write

$$K_Y + B_Y = f^*(K_X + B), \quad \mathfrak{m}_x \mathcal{O}_Y = \mathcal{O}_Y(-F).$$

The support of $B_Y + F$, together with the exceptional locus, is simple normal crossing. Choose a very ample divisor P on X . Hyperplanes through the image of x in the embedding defined by P generate $\mathfrak{m}_x \otimes \mathcal{O}_X(P)$. Their degree- k products generate $\mathfrak{m}_x^k \otimes \mathcal{O}_X(kP)$, so the pullback of its complete section space generates $\mathcal{O}_Y(f^*(kP) - kF)$. There is a section of order exactly k : take the k -th power of a hyperplane with nonzero linear term at x .

A general member H of this complete section space therefore has order k at x , and

$$f^*H = kF + \tilde{H},$$

where the moving divisor $\tilde{H}$ is reduced and meets the fixed simple normal crossing strata transversely. Here generality is taken in a base-point-free system on the smooth characteristic-zero variety Y , and also avoids the proper linear subspace of sections with order at least $k + 1$.

Set $D = (q/k)H$. Then D is a global effective rational divisor with $\text{mult}_x D = q$. Its crepant boundary on Y is

$$B_Y + qF + (q/k)\tilde{H}.$$

For each component E of F , $A_E - q \text{ord}_E(\mathfrak{m}_x) > 0$, by $q < b$. The other old components retain their nonnegative log discrepancies over the lc neighborhood, and each new moving component has coefficient $q/k < 1$. The displayed boundary has simple normal crossing support and coefficients at most one over this neighborhood. It is sub-lc there, so $(X, B + D)$ is lc near x .

Every rational $q < b$ is thus realized. Taking their supremum proves the reverse inequality. If $b = 0$, the first part and the admissible divisor $D = 0$ give equality. $\square$

For completeness, the local discrepancy in the cutting source has the same valuative description. For an effective pair (X, B) lc near x in ambient dimension n , write $\pi : \text{Bl}_x X \rightarrow X$ and $\mathfrak{m}_x \mathcal{O}_{\text{Bl}_x X} = \mathcal{O}(-E_x)$. The boundary used in [GL24, Definition 1.4] is

$$\pi^*B - (n - 1 - t)E_x.$$

For a prime divisor E over x , let A_E be its log discrepancy for (X, B) . Its log discrepancy for the displayed boundary is $A_E - t \text{ord}_E(\mathfrak{m}_x)$. The first non-klt place with center contained in E_x occurs at the infimum in (3.4). This also describes the curve case, in which π is the identity and $E_x = \{x\}$.

We use the published cutting statement [GL24, Proposition 3.2 and Remark 3.3].

Theorem 3.14 (One-point cutting). *Let Z be smooth projective, A ample rational Cartier, and B effective rational. Suppose either $B = 0, W = Z$, or (Z, B) is lc but not klt at z and W is its positive-dimensional minimal center. Put $b = \text{lct}_z((Z, B); \mathfrak{m}_z)$ and $r = (\text{mult}_z W / (A^{\dim W} \cdot W))^{1/\dim W}$. For every $\epsilon > 0$ there are an effective rational $H \sim_{\mathbb{Q}} A$ and a rational $t > 0$ such that $B + tH$ is lc at z , its minimal center is properly contained in W , and its deficit b' satisfies*

$$t \leq br + \epsilon, \quad b' \leq b - t/r + \epsilon.$$

Take the two finite point sets in the source to be $\{z\}$. Their cardinality is one, their nonempty successor set is again $\{z\}$, and the avoidance conditions on other marked points are vacuous. The ample rational divisor is the given A , so the source's multiplicity factor is precisely r . Its local discrepancy is the deficit just identified. Strict decrease of the minimal center excludes $t = 0$; for $B = 0$, Remark 3.3 starts with $W = Z$. The addition is to the whole boundary B .

We record explicitly how both errors enter a finite continuation. Fix an ample Cartier divisor A on a smooth projective n -fold Z , a closed point z , and an integer a . Consider a chain of effective rational boundaries lc near z , with $B_0 \sim_{\mathbb{Q}} sA$, $B_{i+1} = B_i + t_i H_i$, and $H_i \sim_{\mathbb{Q}} A$, for

$0 \leq i < h$. Suppose its positive-dimensional minimal centers W_i strictly decrease to the point. Put

$$b_i = \text{lct}_z((Z, B_i); \mathbf{m}_z), \quad r_i = \left(\frac{\text{mult}_z W_i}{A^{\dim W_i} \cdot W_i} \right)^{1/\dim W_i}, \quad b_h = 0.$$

The final equality follows from an lc place with center z in (3.4). If the i -th cut uses $\epsilon_i > 0$, its two bounds give

$$(3.5) \quad \begin{aligned} t_i &\leq b_i r_i + \epsilon_i, \\ t_i &\leq (b_i - b_{i+1}) r_i + r_i \epsilon_i, \\ t_i &\leq \min\{b_i r_i, (b_i - b_{i+1}) r_i\} + (1 + r_i) \epsilon_i. \end{aligned}$$

Consequently, with the scalar cost $C = \sum_{i=0}^{h-1} (b_i - b_{i+1}) r_i$,

$$s + \sum_{i=0}^{h-1} t_i \leq s + C + \sum_{i=0}^{h-1} (1 + r_i) \epsilon_i.$$

There are at most n cuts. Given a strict margin $\delta > 0$, choose, after W_i and r_i are known and before taking that cut,

$$0 < \epsilon_i < \frac{\delta}{2n(1 + r_i)}.$$

The total charged error is then less than $\delta/2$. Whenever the actual resulting chains satisfy $s + C \leq a - \delta$, this gives $s + \sum_i t_i < a$, leaving a positive margin for Proposition 3.12. This calculation charges the coefficient error and the deficit error at every cut.

4. LOCAL CENTERS AND PERSISTENT ADJUNCTION

We retain the boundary and Hilbert notation of Section 2. We first record the adjunction and local algebra inputs. We then express the normal Hilbert bound as an exact intersection identity, retaining the old discriminant coefficients under addition. Finally, we identify the ordinary point blowup and, under the stated strict inequalities, every ordinary point-ideal power.

Theorem 4.1 (Adjunction to a center). *Let T be a normal complex variety, let $\Delta \geq 0$ be a real divisor with $K_T + \Delta$ real Cartier, and let W be a log canonical center. On its normalization $\nu : Z \rightarrow W$ there are discriminant and moduli b -divisors $\mathcal{B}, \mathcal{M}$ such that*

$$\nu^*((K_T + \Delta)|_W) = K_Z + \mathcal{B}_Z + \mathcal{M}_Z.$$

Here a b -divisor specifies compatible traces on birational models; the closure of a Cartier divisor has its pullback as every higher trace. The base trace $\mathcal{B}_Z$ is effective. On a smooth birational model $p : Z' \rightarrow Z$, the b -divisor $\mathcal{M}$ is the closure of a potentially nef divisor $\mathcal{M}_{Z'}$, and $K + \mathcal{B}$ is the closure of $K_{Z'} + \mathcal{B}_{Z'}$. The support of $\mathcal{B}_{Z'}$ is simple normal crossing. The image of its coefficients greater than one is the non-log-canonical locus on W ; the image of its coefficients at least one is that locus together with the intersections with lc centers not containing W . Potential nefness means extension to a nef divisor on a completion; in particular the trace is nef on a projective such model. For a prime P on a higher model of Z , its discriminant coefficient is one minus the infimum, over ambient lc places dominating W , of the generic sub-lc thresholds of the pullback of P on those places with their induced boundaries. Rational input gives rational discriminant and moduli data. No effectivity of every higher discriminant trace or nefness of every lower moduli trace is asserted.

This is [FH23, Theorem 1.2 and Definition 1.3]. The infimum in the coefficient formula ranges over every lc place dominating the center. We use nefness on the model where the moduli b -divisor descends.

Theorem 4.2 (Cartier adjunction). *For an effective pair $(T, W + R)$, with W a prime Cartier divisor not contained in $\text{Supp } R$ and $K_T + W + R$ rational Cartier, adjunction on the normalization of W has zero moduli part and boundary the different. The pair is plt near W if and only if the adjoint pair is klt. If both T and W are smooth near a point $x \in W$, the different is $R|_W$. In this situation $(T, W + R)$ is lc near x if and only if $(W, R|_W)$ is lc near x , by inversion of adjunction.*

This is the divisorial specialization [FH23, Corollary 1.4]. When both varieties are smooth, the restriction includes every component of the remaining boundary.

Theorem 4.3 (Local subadjunction). *Let (T, Δ) be an effective rational lc pair on a normal affine variety with $K_T + \Delta$ rational Cartier. A minimal lc center W admits an effective rational boundary Δ_W such that (W, Δ_W) is klt and $K_W + \Delta_W \sim_{\mathbb{Q}} (K_T + \Delta)|_W$.*

The affine hypothesis is part of [FG12, Theorem 7.2]. Its conclusion supplies a klt boundary locally on the center.

Theorem 4.4. *On a smooth complex variety, the minimal log canonical center of an effective rational pair through a point where the pair is log canonical but not klt is normal and has rational singularities near that point. Its local ring is Cohen–Macaulay.*

We use [dFH09, Corollary 7.12] on the lc neighborhood. The conclusion concerns the local ring at the marked point.

Theorem 4.5. *Let $(R, \mathfrak{n})$ be an analytically unramified Cohen–Macaulay local ring of dimension $d \geq 1$ with infinite residue field. Let h_i be the Hilbert function of a general Artinian minimal reduction of $\mathfrak{n}$, and put $m = e_0(\mathfrak{n})$. Then*

$$h_0 = 1, \quad h_1 = \text{edim } R - d, \quad \sum_i h_i = m, \quad \sum_i i h_i \leq e_1(\mathfrak{n}) \leq \bar{e}_1(\mathfrak{n}).$$

If $h_1 > 0$, then $h_i \leq \binom{h_1 + i - 1}{i}$. If $h_1 = 0$, the Artinian reduction is the residue field.

This is [Han26, Lemmas 4.2 and 4.4]. The reduction generators form a regular sequence in the Cohen–Macaulay ring.

Theorem 4.6 (Associated graded and Rees criteria). *Let $(R, \mathfrak{n})$ be a Cohen–Macaulay local ring of dimension $d \geq 2$ with infinite residue field. If $e_0(\mathfrak{n}) = \text{edim } R - d + 1$, then for a minimal reduction J , one has $\mathfrak{n}^2 = J\mathfrak{n}$ and $\text{gr}_{\mathfrak{n}} R$ is Cohen–Macaulay. More generally, for an $\mathfrak{n}$ -primary ideal I and a minimal reduction J , the Rees algebra $R[It]$ is Cohen–Macaulay if and only if*

$$e_1(I) = \sum_{n=1}^{d-1} \text{length}(I^n / (J \cap I^n)).$$

The two assertions are [Ver08, Theorem 2.7 and Corollary 6.6]. The finite sum in the second assertion controls the Rees algebra itself, including its homogeneous maximal ideal.

Theorem 4.7 (Regular initial forms). *Let I be an ideal in a Noetherian local ring R , and let $x_1, \dots, x_s \in I \setminus I^2$ be an R -regular sequence. Their initial forms are a regular sequence on $\text{gr}_I R$ if and only if $(x_1, \dots, x_s) \cap I^n = (x_1, \dots, x_s)I^{n-1}$ for every $n \geq 1$.*

This is [Ver08, Theorem 5.16]. The intersection equalities are required in every degree.

Theorem 4.8 (Normality of powers). *Let R be an analytically unramified Cohen–Macaulay local ring of dimension at least two, with a canonical module and infinite residue field. An $\mathfrak{n}$ -primary ideal I has every power integrally closed if and only if $R[It]$ satisfies S_2 and $e_1(I) = \bar{e}_1(I)$.*

This is [PUV05, Corollary 2.5]. The canonical-module and S_2 hypotheses will be checked before its application.

Theorem 4.9 (Normal Hilbert inequality). *For a smooth complex projective Z , a boundary Δ lc near x but not klt at x , and its positive-dimensional minimal center W , write $d = \dim W$, $m = \text{mult}_x W$ and $b = \text{lct}_x((Z, \Delta); \mathbf{m}_x)$. Then*

$$2\bar{e}_1(\mathbf{m}_{W,x}) \leq (d - b)m.$$

This is [Han26, Theorem 1.4]. The extension below to a normal center that is no longer minimal retains its divisorial discriminant contribution.

Theorem 4.10 (Multiplicity of lc centers). *If Z is smooth projective of dimension n , (Z, Δ) is lc at x , and W_d is the union of its d -dimensional lc centers through x , then*

$$\text{mult}_x W_d \leq \binom{n - \lceil b_x(Z, \Delta) \rceil}{n - d}.$$

Here b_x is the point deficit in the cited convention.

We use the published form [GL24, Theorem 3.4]. The multiplicity is that of the entire indicated union.

Theorem 4.11 (Negativity). *Let $f : Y \rightarrow T$ be projective birational between normal algebraic varieties, and D real Cartier with $-D$ nef over T . Then $D \geq 0$ if and only if $f_*D \geq 0$. The same assertion holds for a proper bimeromorphic map of normal complex analytic spaces and a Cartier divisor D . Its rational Cartier version is obtained by taking an actual Cartier multiple.*

The algebraic and analytic versions are [Bir12, Lemma 4.15] and [Wan21, Lemma 1.3], respectively. In the analytic version a rational Cartier divisor must first be multiplied to a Cartier divisor.

Theorem 4.12 (Generalized divisorial adjunction). *Let $(T, B + M)$ be generalized lc with its specified higher nef data, and let S be a coefficient-one component of B . Restriction of the full crepant equality on a higher log resolution and pushforward to the normalization S^ν give*

$$(K_T + B + M)|_{S^\nu} = K_{S^\nu} + B_{S^\nu} + M_{S^\nu}.$$

The induced generalized pair is lc, its base boundary is effective, and its moduli data are the restrictions of the prescribed nef data. This construction does not require T to be $\mathbb{Q}$ -factorial.

See [BZ16, Definition 4.7 and Remark 4.8] and [Fil20, Section 2]. The same higher nef data enter the restricted crepant equality.

Theorem 4.13. *Let T be a smooth complex variety, let $\Delta \geq 0$ be a rational divisor, and let $x \in T$ be a closed point such that (T, Δ) is lc near x but not klt at x . Its minimal lc center W through x is normal and has rational singularities near x , and $\mathcal{O}_{W,x}$ is Cohen–Macaulay.*

Proof. Shrink T to an lc neighborhood of x and remove the finitely many lc centers that do not pass through x . The intersection assertion in Theorem 3.11 makes W the minimal center there. Theorem 4.4 gives the normality and rational singularities of W near x , and the Cohen–Macaulay property of $\mathcal{O}_{W,x}$. $\square$

We next isolate the two calculations used by the Hilbert budget. The first applies to a center that is normal at the point, whether or not it is minimal for the current boundary.

Lemma 4.14. *Let Δ be an effective rational divisor on a smooth complex variety T , lc near a closed point x . Let W be a positive-dimensional lc center through x , normal near x , and let $\nu : Z \rightarrow W$ be its normalization. Put $b_\Delta = \text{lct}_x((T, \Delta); \mathbf{m}_x)$. For the discriminant b -divisor $\mathcal{B}$ of Theorem 4.1, every prime divisor F over Z centered at x satisfies*

$$1 - \text{coeff}_F \mathcal{B} \geq b_\Delta \text{ord}_F(\mathbf{m}_{Z,x}).$$

The threshold bounds each exceptional coefficient individually. In particular, the conclusion still applies when $b_\Delta = 0$.

Proof. Fix a model extracting F , and put $r_F = \text{ord}_F(\mathfrak{m}_{Z,x})$. Take any lc place S of (T, Δ) dominating W . Choose a log resolution $\mu : T' \rightarrow T$ containing S , principalizing the point ideal, and resolving the induced map from S to the chosen model of Z . Write

$$K_{T'} + \Delta_{T'} = \mu^*(K_T + \Delta), \quad \mathfrak{m}_x \mathcal{O}_{T'} = \mathcal{O}_{T'}(-D_x).$$

The coefficient of S in $\Delta_{T'}$ is one. Since $\dim W > 0$, $\text{ord}_S(\mathfrak{m}_x) = 0$, so S is not a component of D_x . Normality near x identifies the restricted point ideal with $\mathfrak{m}_{Z,x}$. If f_S denotes the induced map, then over the generic point of F ,

$$D_x|_S \geq r_F f_S^* F.$$

The point-ideal threshold is attained on this principalization: the simple normal crossing criterion is a finite set of closed linear inequalities. Thus $(T', \Delta_{T'} + b_\Delta D_x)$ is sub-lc above a neighborhood of x . Restriction to its coefficient-one component S , followed by decreasing the added effective divisor, gives a sub-lc pair

$$(S, (\Delta_{T'} - S)|_S + b_\Delta r_F f_S^* F)$$

over that generic point. Every threshold entering the infimum in Theorem 4.1 is therefore at least $b_\Delta r_F$. Its coefficient formula proves the assertion. $\square$

Lemma 4.15. *Let Z be a normal complex projective variety of dimension $d \geq 2$, and let $x \in Z$ be a closed point. Let $f : Y \rightarrow Z$ be the normalized blowup of $\mathfrak{m}_{Z,x}$, with*

$$\mathfrak{m}_{Z,x} \mathcal{O}_Y = \mathcal{O}_Y(-E), \quad E = \sum_F r_F F, \quad H = -E, \quad w_F = F \cdot H^{d-1}.$$

Set $m = e_0(\mathfrak{m}_{Z,x})$ and $\bar{e}_1 = \bar{e}_1(\mathfrak{m}_{Z,x})$. Then r_F, w_F are positive integers and

$$-H^d = m = \sum_F r_F w_F, \quad K_Y \cdot H^{d-1} = (d-1)m - 2\bar{e}_1.$$

For a prime divisor $P \subset Z$, its strict transform P^{str} satisfies

$$P^{\text{str}} \cdot H^{d-1} = \begin{cases} -\text{mult}_x P, & x \in P, \\ 0, & x \notin P. \end{cases}$$

If $\mu : V \rightarrow Y$ is a smooth projective model, $H_V = \mu^ H$, and N is nef on V , then $N \cdot H_V^{d-1} \leq 0$.*

These identities use the normalized blowup of the full point ideal. No Cohen–Macaulay hypothesis on Y is needed.

Proof. We compare Euler characteristics with the normal Hilbert polynomial, then compute the strict-transform and nef intersections.

Step 1. In this step, we compute the two leading intersection coefficients from the normal Hilbert polynomial. The ring $\mathcal{O}_{Z,x}$ is excellent and normal, hence analytically unramified. Its normalized Rees algebra is finite and has degree- k piece $\overline{\mathfrak{m}_{Z,x}^k}$. Indeed, the integral dependence equation for a homogeneous element of degree k is exactly the integral dependence equation for its coefficient over $\mathfrak{m}_{Z,x}^k$. For completeness, the needed eventual sheaf comparison can be checked on an affine open of Z . Present this finite graded algebra as a module over a polynomial algebra generated in degree one. Take a finite initial segment of a graded free resolution, longer than the number of projective coordinates. The monomial Cech complex for projective space shows that each of these free terms has no higher cohomology in sufficiently large twists. Dimension shifting in the associated augmented complexes then identifies the module and sheaf sections in large degrees and kills all higher cohomology. A finite affine cover of Z gives a common cutoff. Since the quotient on Z is supported at x ,

$$\chi(Y, \mathcal{O}_Y(kH)) = \chi(Z, \mathcal{O}_Z) - \text{length}(\mathcal{O}_{Z,x} / \overline{\mathfrak{m}_{Z,x}^k}).$$

Resolve Y by $\mu : V \rightarrow Y$, isomorphically over its regular locus. Normality gives $\mu_* \mathcal{O}_V = \mathcal{O}_Y$; its other direct images have support of dimension at most $d-2$. Their Euler characteristics after

twisting by kH are $O(k^{d-2})$. Riemann–Roch on the smooth projective V and projection formula therefore give

$$\chi(Y, \mathcal{O}_Y(kH)) = \frac{H^d}{d!} k^d - \frac{K_Y \cdot H^{d-1}}{2(d-1)!} k^{d-1} + O(k^{d-2}).$$

Here the intersection with the Weil divisor K_Y is the intersection with Cartier factors, equivalently the pushforward of the canonical intersection on V . On the other hand, Definition 2.3 expands the length as

$$\frac{m}{d!} k^d + \frac{(d-1)m - 2\bar{e}_1}{2(d-1)!} k^{d-1} + O(k^{d-2}).$$

Comparison proves the two asserted leading-coefficient identities. Since $H|_F$ is ample, w_F is a positive integer. The normal height-one local ring along F is a discrete valuation ring, so r_F is a positive integer. Finally,

$$\sum_F r_F w_F = E \cdot H^{d-1} = -H^d = m.$$

Step 2. In this step, we compute the intersections of strict transforms with H^{d-1} . Write $k_P = \dim P = d-1$. The restriction of the point ideal to P^{str} is invertible. Thus $P^{\text{str}} \rightarrow P$ factors through the ordinary blowup of the point ideal on P . On that blowup the exceptional Cartier scheme has tautological degree $\text{mult}_x P$, by the leading term of the Hilbert–Samuel polynomial of $\mathcal{O}_{P,x}$. Consequently the top intersection of its tautological divisor $-E_P$ is $-\text{mult}_x P$: one factor is minus the exceptional divisor and the remaining $k_P - 1$ factors give its degree. The factorization is birational, so projection formula gives $(H|_{P^{\text{str}}})^{k_P} = -\text{mult}_x P$. When $x \notin P$, this restriction is trivial and its top intersection is zero.

Step 3. In this step, we prove that the nef contribution has nonpositive intersection with H_V^{d-1} . Write $\mu^* E = \sum_j s_j G_j$ with $s_j > 0$. On the normalization of each G_j , the restrictions of N and H_V are nef. The latter is pulled back from the ample polarization on the point fiber. Hence

$$N \cdot H_V^{d-1} = - \sum_j s_j (N|_{G_j}) \cdot (H_V|_{G_j})^{d-2} \leq 0,$$

as required. $\square$

Theorem 4.16. *For a positive-dimensional minimal center of an effective lc pair, $b > 0$ and $2\bar{e}_1 \leq (d-b)m$. More generally, if W is an lc center normal near x , of dimension $d \geq 2$, and $b > 0$, then every finite set of distinct codimension-one lc centers $T_i \subset W$ through x satisfies*

$$\sum_i \text{mult}_x T_i \leq (d-b)m - 2\bar{e}_1.$$

Here W need not remain minimal for the specified boundary.

The second inequality charges every indicated divisorial center to the same normal Hilbert budget.

Proof. We prove the minimal-center assertion first and then retain the horizontal discriminant to bound codimension-one lc centers.

Step 1. In this step, we prove positivity of the deficit and the minimal-center Hilbert bound. We first check positivity of the deficit at a positive-dimensional minimal center. On a log resolution of the boundary and the point ideal, the deficit is the minimum of the finitely many ratios

$$\frac{A_{X,B}(Q)}{\text{ord}_Q(\mathfrak{m}_x)}, \quad \text{ord}_Q(\mathfrak{m}_x) > 0.$$

Here $A_{X,B}(Q)$ denotes log discrepancy. A zero ratio would give an lc place centered at $\{x\}$, contrary to minimality. Thus $b > 0$. Normality and rational singularities near x follow from Theorem 4.4. Theorem 4.9 now proves the first assertion in every positive dimension.

Step 2. In this step, we derive the discriminant decomposition for the normal center. For the second assertion we retain the horizontal discriminant. Normalize the given center globally as $Z \rightarrow W$ and identify its unique point over x with x . Use the normalized blowup and notation of Lemma 4.15. Write $\mathcal{B}_Z$ for the effective base discriminant, $\beta_F = \text{coeff}_F \mathcal{B}_Y$, and $\mathcal{B}_Z^{\text{str}}$ for its strict transform. On a smooth projective common model V , the compatible moduli trace M_V is nef. The adjunction equality has intersection zero with H_V^{d-1} , since this cycle is supported over x and $d-1 > 0$. Therefore

$$K_Y \cdot H^{d-1} = -M_V \cdot H_V^{d-1} - \mathcal{B}_Z^{\text{str}} \cdot H^{d-1} - \sum_F \beta_F w_F.$$

Substitute the identities of Lemma 4.15 and $m = \sum_F r_F w_F$. This gives the exact decomposition

$$(4.1) \quad \begin{aligned} (d-b)m - 2\bar{e}_1 &= \sum_F (r_F - 1)w_F + \sum_F (1 - \beta_F - br_F)w_F \\ &\quad - M_V \cdot H_V^{d-1} - \mathcal{B}_Z^{\text{str}} \cdot H^{d-1}. \end{aligned}$$

Each term is nonnegative. The exceptional coefficient inequality is Lemma 4.14; the two intersection signs follow from Lemma 4.15 and the effectivity of $\mathcal{B}_Z$. Minimality was not used in this identity or in either of the two lemmas.

Step 3. In this step, we bound the multiplicities of the codimension-one lc centers by the horizontal contribution. Let P_i denote the closure on Z of T_i near x . Its base discriminant coefficient is one. Indeed, the coefficient-at-least-one support assertion of Theorem 4.1 supplies such a divisor above the generic point of P_i . A proper birational morphism to a normal variety is an isomorphism at a codimension-one generic point, so this is the strict transform of P_i . The coefficient-greater-than-one support assertion excludes a coefficient above one, since the ambient pair is lc there. Consequently

$$-\mathcal{B}_Z^{\text{str}} \cdot H^{d-1} = \sum_{P \ni x} (\text{coeff}_P \mathcal{B}_Z) \text{mult}_x P \geq \sum_i \text{mult}_x T_i.$$

Together with the other nonnegative terms in (4.1), this proves the assertion. $\square$

Lemma 4.17. *Let T be a normal complex variety and $\Delta \geq 0$ a rational divisor with $K_T + \Delta$ rational Cartier. Let W be an lc center, $\nu : Z \rightarrow W$ its normalization, and $D \geq 0$ a rational Cartier divisor whose support does not contain W . Put $D_Z = \nu^*(D|_W)$. For each rational $t \geq 0$, let $\mathcal{B}(t)$ be the discriminant b-divisor of $(T, \Delta + tD)$ along W . Then*

$$\mathcal{B}(t) = \mathcal{B}(0) + t \overline{D_Z},$$

where $\overline{D_Z}$ is the Cartier closure. If the representatives of the restricted adjoint classes are chosen by adding tD_Z and its pullbacks, the compatible moduli b-divisors satisfy $\mathcal{M}(t) = \mathcal{M}(0)$.

The equality concerns the actual Cartier divisor and compatible representatives of the moduli data.

Proof. The pairs agree at the generic point of W , so their lc places dominating W are the same. On a higher model of Z , fix a prime P , and let a_P be the coefficient of the pullback of D_Z along P . For every dominating lc place S , its induced boundary changes by the pullback of tD_Z . Its generic threshold along P therefore changes from $\alpha_{P,S}(0)$ to

$$\alpha_{P,S}(t) = \alpha_{P,S}(0) - ta_P.$$

The shift is independent of S . Taking the infimum over all S in Theorem 4.1 gives $\text{coeff}_P \mathcal{B}(t) = \text{coeff}_P \mathcal{B}(0) + ta_P$. This proves the b-divisor equality. The moduli trace is the chosen representative of the restricted adjoint class minus the canonical and discriminant traces. Adding tD_Z to both the representative and the discriminant leaves that trace unchanged on every model. $\square$

Proposition 4.18. *Let W be a minimal center of B of dimension $d \geq 2$. On its global normalization, let Y be the normalized point blowup, $E = \sum_F r_F F$, $H = -E$, and $w_F = F \cdot H^{d-1}$. Use compatible adjunction discriminant and nef moduli data on a common smooth model V . Write the old discriminant primes through x as P_j with coefficients $c_j \in (0, 1)$ and multiplicities k_j , and set*

$$R_0 = \sum_F (r_F - 1)w_F, \quad M_0 = -M_V \cdot H_V^{d-1}, \quad H_0 = \sum_j c_j k_j, \quad K_W = dm - 2\bar{e}_1 - R_0 - M_0 - H_0.$$

These three subtracted quantities are nonnegative and $K_W \geq mb > 0$. For every effective rational D with $(X, B + D)$ lc near x , $W \not\subset \text{Supp } D$. If b_D is the new deficit, D_Z the actual restriction to the normalization and $\beta_F(D)$ the new discriminant coefficients, then

$$K_W = mb_D + \text{mult}_x D_Z + \sum_F (1 - \beta_F(D) - b_D r_F)w_F,$$

with every last summand nonnegative. For distinct codimension-one lc centers $T_i \subset W$ of $B + D$,

$$mb_D + \sum_i (1 - c(T_i)) \text{mult}_x T_i \leq K_W,$$

where $c(T) = 0$ unless T is an old prime. Equality forces the displayed divisor sum to be the entire restriction near x and all exceptional slack to vanish; for a single T , it also forces T to be rational Cartier near x . The compatible moduli data are retained under Cartier addition.

The old horizontal coefficients are part of the fixed parent budget. A later divisorial center pays the difference between its old coefficient and one.

Proof. We retain the same normalized point blowup throughout.

Step 1. In this step, we compute the old discriminant budget K_W . The old base discriminant is effective. Its coefficients through x are less than one: otherwise Theorem 4.1 would give a smaller lc center through x . Thus the positive coefficients are precisely the $c_j \in (0, 1)$ in the statement. Lemma 4.15 gives

$$-\mathcal{B}_Z^{\text{str}} \cdot H^{d-1} = \sum_j c_j k_j = H_0.$$

It also gives $R_0, M_0 \geq 0$, while effectivity gives $H_0 \geq 0$. Rearranging (4.1) for the old boundary yields

$$(4.2) \quad K_W = mb + \sum_F (1 - \beta_F(0) - b r_F)w_F \geq mb > 0.$$

Step 2. In this step, we retain the discriminant and moduli data under the actual addition. For any old lc place S centered at W ,

$$A_{X, B+D}(S) = -\text{ord}_S D.$$

The left side is nonnegative and D is effective, so $\text{ord}_S D = 0$. Thus D does not contain W , and the same place still makes W an lc center. Its normality near x is a property of the old center. By Lemma 4.17, its new discriminant is $\mathcal{B}(0) + \overline{D_Z}$ and its compatible moduli trace is still M_V .

Apply the intersection calculation (4.1) to this same normal center and the new boundary. The proof of that identity used only normality near x , effectivity of the base discriminant, and the point-threshold comparison. These hold even when $b_D = 0$, by Lemma 4.14. The new horizontal contribution is $H_0 + \text{mult}_x D_Z$; all other fixed contributions remain R_0 and M_0 . Rearrangement gives

$$K_W = mb_D + \text{mult}_x D_Z + \sum_F (1 - \beta_F(D) - b_D r_F)w_F.$$

Each final summand is nonnegative by the same threshold comparison. This proves the identity at the actual endpoint, including zero new deficit.

Step 3. In this step, we bound divisorial successors and determine the consequences of equality. For a codimension-one lc center T_i of $B + D$ inside W , Theorem 4.1 gives coefficient one in the new base discriminant, by the same generic-point argument used in the Hilbert budget. The addition identity therefore gives

$$\text{coeff}_{T_i} D_Z = 1 - c(T_i) > 0.$$

Effectivity of the remaining restriction implies

$$\text{mult}_x D_Z \geq \sum_i (1 - c(T_i)) \text{mult}_x T_i.$$

Substitution into the exact identity proves the inequality. If equality holds, every exceptional slack vanishes, since all w_F are positive. Every additional prime of D_Z through x would have positive multiplicity, so none can remain. After removing the finitely many components avoiding x , one has

$$D_Z = \sum_i (1 - c(T_i)) T_i.$$

For a single T , the divisor on the left is rational Cartier and $1 - c(T)$ is a positive rational number. Multiplying this equality by an integer clearing both denominators shows that a positive integral multiple of T is Cartier near x . Thus T is rational Cartier. $\square$

Lemma 4.19. *Let G be a standard graded Cohen–Macaulay algebra over a field, with G_0 that field and $\dim G = d \geq 2$. If $\text{Proj } G$ is reduced, then G is reduced.*

The Cohen–Macaulay assumption controls the affine vertex as well as the projective scheme.

Proof. Every associated prime of G is minimal. No minimal homogeneous prime is the irrelevant maximal ideal: it would be an associated prime of dimension zero, contrary to positive depth there. For a minimal homogeneous prime choose a degree-one element h outside it. Since G is standard graded,

$$G_h = (G_h)_0[h, h^{-1}].$$

The degree-zero ring is the coordinate ring of an open subset of the reduced scheme $\text{Proj } G$, so G_h is reduced. Thus G is generically reduced at every minimal prime. If its nilradical were nonzero, that submodule would have an associated prime in $\text{Ass } G$. Localization at this minimal prime would give a nonzero nilpotent in a reduced zero-dimensional local ring, a contradiction. $\square$

Theorem 4.20. *Let B be an effective rational boundary on the smooth n -fold X , lc near x , and let W be its actual minimal lc center through x , of dimension $d \geq 2$. Put $R = \mathcal{O}_{W,x}$, $\mathfrak{n} = \mathfrak{m}_{W,x}$, $\delta = n - \text{mult}_x B$, and $\sigma = \delta - b$. Write $f : Y = \text{Bl}_{\mathfrak{n}} \text{Spec } R \rightarrow \text{Spec } R$, with exceptional Cartier scheme E . If $0 \leq \sigma < 1$, then Y is normal with rational singularities near its whole exceptional scheme. Its effective strict-pair discriminant satisfies $0 \leq \gamma_F \leq 1 - (1 - \sigma)r_F$, where r_F is the multiplicity of the exceptional scheme along F . If $\sigma < 1/2$, that entire scheme is reduced; if the full tangent ring is also Cohen–Macaulay, that ring is reduced. Under the additional hypothesis $\delta > 1$, for every $q \geq 0$,*

$$f_* \mathcal{O}_Y(-qE) = \mathfrak{n}^q = \overline{\mathfrak{n}}^q, \quad R^i f_* \mathcal{O}_Y(-qE) = 0 \quad (i > 0).$$

Under these same hypotheses $\delta > 1$ and $0 \leq \sigma < 1$, the original Rees ring is normal with rational singularities and the full ordinary tangent ring is Cohen–Macaulay. If R has minimal multiplicity $m = e - d + 1$, then

$$\text{Hilb}(\text{gr}_{\mathfrak{n}} R; z) = \frac{1 + (m - 1)z}{(1 - z)^d}, \quad e_1 = m - 1.$$

If in addition $b > d - 2$, every ordinary power is integrally closed, $\bar{e}_1 = m - 1$, and the Rees ring is normal Cohen–Macaulay. Under this minimal-multiplicity and strict-deficit hypothesis, the full

exceptional cycle $\sum_i \nu_i [C_i]$, with $d_i = \deg C_i$, satisfies

$$\sum_i \nu_i d_i = m, \quad 0 \leq \sum_i (\nu_i - 1) d_i \leq m(d - 2 - b) + 2 < 2.$$

A nonzero sum forces exactly one doubled linear component and $b \leq d - 2 + 1/m$.

The condition on σ controls the effective strict pair on the ordinary blowup. The additional inequality $\delta > 1$ supplies vanishing from degree zero. The minimal-multiplicity conclusions use their separately stated strict bound on b .

Proof. We first identify the ordinary blowup with an actual minimal center and control its exceptional scheme. We then use the defining-ideal vanishing theorem to recover every ordinary power and resolve the original affine Rees ring. Finally, we treat minimal multiplicity and the full exceptional cycle.

Step 1. In this step, we bound discrepancies of the effective strict pair above the point. By Theorem 4.4, R is a normal Cohen–Macaulay local domain with rational singularities. It is excellent, has residue field $\mathbb{C}$, and is analytically unramified. Let $\pi : X_1 = \text{Bl}_x X \rightarrow X$ be the ambient point blowup, with exceptional divisor E_X , and let B_1, W_1 be the strict transforms of B, W . The actual boundary multiplicity gives

$$(4.3) \quad K_{X_1} + B_1 = \pi^*(K_X + B) + (\delta - 1)E_X.$$

Suppose $0 \leq \sigma < 1$, and fix a rational $0 \leq \theta < 1 - \sigma$. For a divisorial valuation v centered above x , the point-ideal definition of b gives

$$(4.4) \quad \begin{aligned} A_{X_1, B_1 + \theta E_X}(v) &= A_{X, B}(v) - (\delta - 1 + \theta)v(\mathfrak{m}_x) \\ &\geq (1 - \sigma - \theta)v(\mathfrak{m}_x) > 0. \end{aligned}$$

If the center of v on X is positive-dimensional or avoids x , then $v(\mathfrak{m}_x) = 0$; its discrepancy is unchanged and is nonnegative over the old lc neighborhood. Thus the effective pair $(X_1, B_1 + \theta E_X)$ is lc near the whole exceptional divisor, and no lc center is contained in E_X . This argument does not require $\delta > 1$.

Step 2. In this step, we identify the ordinary blowup with the minimal center above the point. An old lc place centered at W has point-ideal order zero and remains an lc place centered at W_1 . Fix $y \in W_1 \cap E_X$. Theorem 3.11 gives a minimal lc center C through y , contained in W_1 . If $C \subsetneq W_1$ and $\pi(C) = \{x\}$, it contradicts (4.4). Otherwise its lc place has point-ideal order zero and defines an old lc center $\pi(C)$. Properness makes this image closed; it contains x , and $\dim \pi(C) \leq \dim C < \dim W$. This contradicts the old minimality of W . Hence $C = W_1$. Theorem 4.4 now makes W_1 normal with rational singularities near every point of this fiber.

The quotient $\mathcal{O}_{X, x} \rightarrow R$ gives a surjection of ordinary Rees algebras

$$\mathcal{O}_{X, x}[\mathfrak{m}_x t] \longrightarrow R[\mathfrak{m}_x t].$$

Its Proj is a closed immersion of the ordinary blowup into the ambient point blowup. Since $R[\mathfrak{m}_x t] \subset R[t]$ is a domain, its image is integral and is the scheme-theoretic closure of $W \setminus \{x\}$. It is precisely W_1 . Thus Y in the statement is the ordinary blowup itself, and

$$E_X|_Y = E, \quad E = \text{Proj}(\text{gr}_{\mathfrak{m}_x} R).$$

The ideal $\mathfrak{n}\mathcal{O}_Y$ is invertible and nonzero in this integral scheme, so it defines the full effective Cartier scheme E . Rational singularities give the Cohen–Macaulay property for Y near the fiber; quotient by the Cartier equation gives it for E as well.

Step 3. In this step, we bound exceptional discriminant coefficients and prove the stated reducedness assertions. Apply Theorem 4.1 to the effective strict pair (X_1, B_1) . Since $Y = W_1$ is normal near the fiber, its base discriminant coefficients γ_F there are nonnegative. Addition

of θE_X changes them to $\gamma_F + \theta r_F$, by Lemma 4.17. The pair with this addition is lc, so its discriminant coefficients are at most one. Consequently

$$0 \leq \gamma_F \leq 1 - \theta r_F \quad (0 \leq \theta < 1 - \sigma).$$

Letting rational θ increase to $1 - \sigma$ proves the asserted bound. In particular $(1 - \sigma)r_F \leq 1$. If $\sigma < 1/2$, each positive integer r_F is one. The normal local ring of Y at the generic point of F is a discrete valuation ring; hence the Cartier scheme E is generically reduced. Its Cohen–Macaulay property excludes embedded associated primes, so the entire scheme E is reduced. If the full tangent ring is also Cohen–Macaulay, Lemma 4.19 proves its reducedness, including the vertex.

Step 4. In this step, we recover every ordinary point-ideal power and its higher-direct-image vanishing. Now assume additionally $\delta > 1$, retaining $0 \leq \sigma < 1$. Choose one neighborhood of x on which the strict pair is lc and W_1 is normal with rational singularities over the whole neighborhood. This is possible by the preceding fiberwise conclusions, properness of π , and the corresponding old properties away from x . This neighborhood is fixed independently of q . For an arbitrary integer $q \geq 0$, set $D_q = -qE_X$. Equation (4.3) gives

$$(4.5) \quad D_q - (K_{X_1} + B_1) = -\pi^*(K_X + B) - (q + \delta - 1)E_X.$$

This divisor is π -ample, because $q + \delta - 1 > 0$ and $-E_X$ is π -ample.

Apply Theorem 3.7 to the smooth ambient X_1 , its effective boundary B_1 , the projective morphism π , the Cartier divisor D_q , and the reduced lc center W_1 . On the chosen neighborhood the non-lc ideal is $\mathcal{O}_{X_1}$. Thus the theorem kills every higher direct image of $\mathcal{I}_{W_1} \otimes \mathcal{O}_{X_1}(D_q)$. Its application to the empty union of lc centers also kills every higher direct image of $\mathcal{O}_{X_1}(D_q)$. The exact sequence

$$0 \longrightarrow \mathcal{I}_{W_1} \otimes \mathcal{O}_{X_1}(D_q) \longrightarrow \mathcal{O}_{X_1}(D_q) \longrightarrow \mathcal{O}_{W_1}(D_q|_{W_1}) \longrightarrow 0$$

therefore gives both restriction surjectivity and $R^i f_* \mathcal{O}_Y(-qE) = 0$ for all $i > 0$.

On the smooth ambient point blowup, $\pi_* \mathcal{O}_{X_1}(-qE_X) = \mathfrak{m}_x^q$: exceptional order is ordinary point order in the regular local ring. Since Y is normal and f is proper birational onto the normal $\text{Spec } R$, $f_* \mathcal{O}_Y = R$. The restriction target is consequently an actual ideal in R . If $I_{W,x}$ is the defining ideal of W , the surjective restriction map has image

$$(4.6) \quad \frac{\mathfrak{m}_x^q + I_{W,x}}{I_{W,x}} = \mathfrak{n}^q.$$

This proves $f_* \mathcal{O}_Y(-qE) = \mathfrak{n}^q$ in every degree, including $q = 0$.

Integral closedness follows directly. If $a \in R$ is integral over $\mathfrak{n}^q$, write its monic equation with its i -th coefficient in $\mathfrak{n}^{qi}$. On an open subset of Y where $\mathfrak{n}\mathcal{O}_Y = (h)$, division by the corresponding power of h makes a/h^q integral over $\mathcal{O}_Y$. Normality of Y makes it regular. These local statements give

$$a \in H^0(Y, \mathcal{O}_Y(-qE)) = \mathfrak{n}^q.$$

Thus $\overline{\mathfrak{n}^q} = \mathfrak{n}^q$ for every $q \geq 0$, after the ordinary image has been established.

Step 5. In this step, we prove rationality of the original Rees ring and Cohen–Macaulayness of the full tangent ring. Put $\mathcal{R} = R[\mathfrak{n}t]$ and $\mathcal{L} = \mathcal{O}_Y(-E)$, and form the total space

$$p : \mathcal{T} = \text{Spec}_Y \left(\bigoplus_{q \geq 0} \mathcal{L}^q \right) \longrightarrow Y$$

of the dual line bundle. The preceding identities give

$$H^0(\mathcal{T}, \mathcal{O}_{\mathcal{T}}) = \bigoplus_{q \geq 0} H^0(Y, \mathcal{L}^q) = \mathcal{R}.$$

Since Y is normal, $\mathcal{T}$ is normal. Its rational function field is $\text{Frac}(R)(t)$, which is also the fraction field of $\mathcal{R}$: for $0 \neq a \in \mathfrak{n}$, $t = (at)/a$. An element of this field integral over $\mathcal{R}$ is integral over every local ring of $\mathcal{T}$, hence regular everywhere. The displayed equality of global sections makes it an element of $\mathcal{R}$. This proves normality of the original Rees ring.

The natural map $g : \mathcal{T} \rightarrow \operatorname{Spec} \mathcal{R}$ is proper. To see this, choose generators of $\mathfrak{n}$ and the corresponding embedding of $Y = \operatorname{Proj} \mathcal{R}$ in projective space over R . Its tautological line is a closed subscheme of the product of Y with affine space. The homogeneous equations of $\mathcal{R}$ vanish on this line, so it is a closed subscheme of

$$Y \times_{\operatorname{Spec} R} \operatorname{Spec} \mathcal{R}.$$

Projection to the second factor is projective. Away from the closed point of $\operatorname{Spec} R$, the ideal $\mathfrak{n}$ is the unit ideal; there g is an isomorphism. Thus g is birational.

Let $\mu : Y' \rightarrow Y$ be a projective resolution and $\mathcal{T}'$ the total space of the pulled-back dual line bundle. It is smooth, and $\mathcal{T}' \rightarrow \mathcal{T}$ is the proper base change of μ . Rationality of Y , projection formula, and the established vanishing give, for every $q \geq 0$,

$$H^i(Y', \mu^* \mathcal{L}^q) = 0 \quad (i > 0), \quad H^0(Y', \mu^* \mathcal{L}^q) = \mathfrak{n}^q.$$

A finite affine cover of the separated scheme Y' computes quasi-coherent cohomology by its Čech complex and shows that cohomology commutes with these direct sums. Hence

$$H^i(\mathcal{T}', \mathcal{O}_{\mathcal{T}'}^q) = 0 \quad (i > 0), \quad H^0(\mathcal{T}', \mathcal{O}_{\mathcal{T}'}^q) = \mathcal{R}.$$

The proper birational map $\mathcal{T}' \rightarrow \operatorname{Spec} \mathcal{R}$ is therefore a rational resolution, since its target is affine. It proves that $\mathcal{R}$ has rational singularities and is Cohen–Macaulay of dimension $d + 1$.

Let $\mathcal{R}_+$ be its positive-degree ideal and let $\mathfrak{M} = \mathfrak{n} + \mathcal{R}_+$. At this homogeneous maximal ideal, the two exact sequences

$$0 \longrightarrow \mathcal{R}_+ \longrightarrow \mathcal{R} \longrightarrow R \longrightarrow 0, \quad 0 \longrightarrow \mathcal{R}_+(1) \xrightarrow{t^{-1}} \mathcal{R} \longrightarrow \operatorname{gr}_{\mathfrak{n}} R \longrightarrow 0$$

give the full tangent Cohen–Macaulay property by the depth lemma. More explicitly, the first sequence gives $\operatorname{depth}(\mathcal{R}_+)_{\mathfrak{M}} = d + 1$. In degree $q \geq 0$, the image in the second sequence is exactly $\mathfrak{n}^{q+1}t^q \subset \mathfrak{n}^q t^q$; there is no omitted degree-zero piece. Thus its cokernel is the full ordinary tangent ring of dimension d , and its depth is at least d .

Step 6. In this step, we compute the minimal-multiplicity Hilbert series and prove the additional normal-power assertions. For the remaining assertions we use the minimal-multiplicity assumption, independently of δ and σ . The Cohen–Macaulay local ring R has infinite residue field, so choose a general minimal reduction $J = (a_1, \dots, a_d)$ of $\mathfrak{n}$. Its generators are a system of parameters, hence an R -regular sequence, and their classes in $\mathfrak{n}/\mathfrak{n}^2$ are independent. Theorem 4.6 gives $\mathfrak{n}^2 = J\mathfrak{n}$ and the Cohen–Macaulay property of $G = \operatorname{gr}_{\mathfrak{n}} R$. In every degree one has

$$J \cap \mathfrak{n}^q = J\mathfrak{n}^{q-1} \quad (q \geq 1);$$

for $q \geq 2$ both sides equal $\mathfrak{n}^q$. Theorem 4.7 thus makes the initial forms of the a_i a regular sequence on G . Their quotient is $\operatorname{gr}_{\mathfrak{n}/J}(R/J)$. Its Hilbert function is $(1, m - 1)$, because $(\mathfrak{n}/J)^2 = 0$ and $\dim_{\mathbb{C}} \mathfrak{n}/(J + \mathfrak{n}^2) = e - d = m - 1$. Dividing its series by the d regular linear factors gives

$$\operatorname{Hilb}(G; z) = \frac{1 + (m - 1)z}{(1 - z)^d}.$$

The derivative of this numerator at $z = 1$, or the corresponding cumulative Hilbert polynomial, gives $e_1 = m - 1$.

Moreover,

$$\operatorname{length}(\mathfrak{n}^q / (J \cap \mathfrak{n}^q)) = \begin{cases} m - 1, & q = 1, \\ 0, & q \geq 2. \end{cases}$$

Since $d \geq 2$, the truncated sum in Theorem 4.6 equals e_1 . Thus $R[\mathfrak{n}t]$ is Cohen–Macaulay and satisfies S_2 , without yet using a bound on b .

Assume now $b > d - 2$. Theorem 4.16 and the comparison of ordinary and normal Hilbert coefficients in Theorem 4.5 give

$$m - 1 = e_1 \leq \bar{e}_1 \leq \frac{(d - b)m}{2} < m.$$

The normal Hilbert coefficient is an integer, so $\bar{e}_1 = e_1 = m - 1$. All ring hypotheses of Theorem 4.8 hold: R is analytically unramified and Cohen–Macaulay, $d \geq 2$, its residue field is infinite, and it has a canonical module as a Cohen–Macaulay quotient of the regular local ring $\mathcal{O}_{X,x}$. The ideal $\mathfrak{n}$ is primary for the maximal ideal; its Rees algebra satisfies S_2 , and the two first Hilbert coefficients have just been identified. The theorem proves that every $\mathfrak{n}^q$ is integrally closed. The homogeneous integral-dependence equation identifies the normalization of the Rees ring inside $R[t]$ with $\bigoplus_{q \geq 0} \bar{\mathfrak{n}}^q t^q$. It therefore equals the Rees ring. Since R is normal and the fraction field is $\text{Frac}(R)(t)$, this also proves its normality as a domain.

Step 7. In this step, we bound the excess multiplicity of the full exceptional cycle. The ordinary blowup now equals the normalized blowup. At the generic point of C_i , its normal local ring is a discrete valuation ring. The order of the Cartier equation of E is both $r_i = \text{ord}_{C_i}(\mathfrak{n})$ and the generic scheme multiplicity ν_i . The intersection weight is $w_i = \deg_{\mathcal{O}_{E(1)}} C_i = d_i > 0$. Thus Lemma 4.15 gives

$$\sum_i \nu_i d_i = m, \quad \ell := \sum_i (\nu_i - 1) d_i = m - \sum_i d_i \geq 0.$$

The same entire cycle occurs in the first term of (4.1). Substitution of $\bar{e}_1 = m - 1$ gives

$$0 \leq \ell \leq (d - b)m - 2(m - 1) = m(d - 2 - b) + 2 < 2.$$

The integer ℓ is zero or one. If it is one, exactly one positive integer $(\nu_i - 1)d_i$ equals one. That component has $\nu_i = 2$ and $d_i = 1$; an integral projective variety of degree one is linear. All other generic multiplicities are one. Finally, $1 \leq m(d - 2 - b) + 2$ gives $b \leq d - 2 + 1/m$, as claimed. $\square$

5. COMPLETE DESCENDANT COSTS

We retain the boundary and deficit notation of Section 2. We bound the cost of an entire descending chain before choosing its cuts. The local Hilbert inequalities give a finite set of possible centers; retaining the parent restricts the next center further. We then give the complete rational certificates used in dimension seven.

Set-up 5.1. In the discussion of a root in ambient dimension $n \geq 2$, let X be smooth complex projective, let L be ample Cartier, and let $B \sim_{\mathbb{Q}} sL$, $s \geq 0$, be an effective rational boundary lc near x . Assume its minimal center W through x is positive-dimensional and proper. Its data are $b, d, e, m, g, \bar{e}_1$ as in Definitions 2.2 and 2.3, and $r = r(W) = (m/g)^{1/d}$. Every chain below is an additive continuation in the sense of Definition 2.5. All degree floors refer to intersections with this same L .

Definition 5.2. Fix an integer $n \geq 2$. For $c \geq 1$, set $S_c(j) = \binom{c+j}{j}$, $S_c(-1) = 0$, and

$$T(c, m) = qm - \binom{c+q}{q-1} \quad \text{when } S_c(q-1) < m \leq S_c(q), \quad T(0, 1) = 0.$$

The set $\mathcal{P}_n$ consists of triples $v = (d, e, m)$ with $1 \leq d \leq n - 1$, $d \leq e \leq n$, $c = e - d$, $m = 1$ if $c = 0$, $m \geq c + 1$ otherwise, and $\beta_v = d - 2T(c, m)/m > 0$. In dimension one only $(1, 1, 1)$ is retained. Choose rational degree floors $g_v \in \mathbb{Q}_{\geq 1}$ and let $r_v = (m/g_v)^{1/d}$. A possible successor $w = (f, h, k)$ has $f < d, h \leq e$; set $a_{vw} = (dm - 2T(c, m) - k)/m$ for $f = d - 1$, discarding nonpositive caps, and $a_{vw} = +\infty$ for $f \leq d - 2$. For $0 \leq z \leq \beta_v$, define, in increasing dimension,

$$\mathcal{F}_v(z) = r_v z + \max \left(0, \max_w \max_{0 \leq y \leq \min(z, \beta_w, a_{vw})} (\mathcal{F}_w(y) - r_v y) \right).$$

For an actual root B , its certificate Q_B uses the actual root radius and replaces its adjacent cap by $(dm - 2\bar{e}_1 - k)/m$, leaving descendant envelopes unchanged.

The successor maxima are over states in $\mathcal{P}_n$. The zero alternative represents a direct cut to the point. In particular, $\mathcal{F}_{(1,1,1)}(z) = r_{(1,1,1)}z$. At the root the formula is

$$(5.1) \quad Q_B = rb + \max \left(0, \max_w \max_{0 \leq y \leq \min(b, \beta_w, a_{Bw})} (\mathcal{F}_w(y) - ry) \right),$$

where $a_{Bw} = (dm - 2\bar{e}_1 - k)/m$ in adjacent dimensions and $a_{Bw} = +\infty$ for a dimension skip. Nonpositive adjacent caps are discarded here as well.

Definition 5.3. For a root as in Set-up 5.1 with $d \geq 2$, use the fixed discriminant and moduli data of Proposition 4.18. The *retained candidate list* consists of the following pairs (w, u) :

- (i) Every $w = (f, h, k) \in \mathcal{P}_n$ with $f \leq d - 2$ and $h \leq e$, with $u = +\infty$.
- (ii) Every $w = (d - 1, h, k) \in \mathcal{P}_n$ with $h \leq e$ and $K_W > k$, with $u = (K_W - k)/m$.
- (iii) Each actual old prime P_j whose state $w_j = (d - 1, \text{edim } \mathcal{O}_{P_j, x}, k_j)$ belongs to $\mathcal{P}_n$, has embedding dimension at most e , and has positive cap $u_j = (K_W - (1 - c_j)k_j)/m$, with that cap.

Define its *retained envelope* by

$$Q_{\text{ret}}(B) = rb + \max \left(0, \max_{(w, u)} \max_{0 \leq y \leq \min(b, \beta_w, u)} (\mathcal{F}_w(y) - ry) \right).$$

The old-prime entries name the actual primes of the original discriminant, not merely their numerical states.

Theorem 5.4. *The set $\mathcal{P}_n$ is finite and its envelopes are continuous piecewise affine functions with algebraic breakpoints. If the starting center and every positive-dimensional actual descendant satisfy their assigned degree floors, every entire additive chain from B has scalar cost $C \leq Q_B \leq \mathcal{F}_v(b)$. In particular $s + Q_B < n + 1$ gives an additive point continuation with total coefficient below $n + 1$, including both cutting errors. In the persistent-parent setting, one may instead use the candidate caps $(K_W - k)/m$ for new codimension-one centers, $(K_W - (1 - c_j)k_j)/m$ for each actual old prime, and no additional cap for a dimension skip. The resulting envelope $Q_{\text{ret}}(B)$ bounds the whole cost and satisfies $Q_{\text{ret}}(B) \leq Q_B$.*

Thus a strict bound for the whole tail pays every cutter produced by the construction, even when its next center depends on the chosen error.

Proof. We first obtain the finite state set and its transition caps. We then bound all chains by the recurrence, refine the root using its old discriminant, and choose the errors within a fixed positive margin.

Step 1. In this step, we derive the first-moment bound for every actual state. By Theorem 4.13, each positive-dimensional minimal center has a normal Cohen–Macaulay local ring. This excellent complex local ring is analytically unramified and has infinite residue field. Theorem 4.5 therefore applies to a general Artinian parameter reduction. For $c > 0$, write $a_j = \binom{c+j-1}{j}$, including $a_0 = 1$, for its Hilbert capacities. If h_j is its Hilbert function, then $0 \leq h_j \leq a_j$, $\sum_j h_j = m$, and $\sum_j jh_j \leq \bar{e}_1$. With q as in Definition 5.2,

$$\sum_j jh_j = qm - \sum_{j < q} (q - j)h_j + \sum_{j > q} (j - q)h_j \geq qm - \sum_{j < q} (q - j)a_j = qm - \binom{c+q}{q-1}.$$

For the last equality, sum the cumulative capacities $\sum_{j=0}^i a_j = \binom{c+i}{i}$ for $0 \leq i < q$. Filling degrees below q to capacity and placing the remaining $m - S_c(q - 1)$ in degree q attains this lower bound among bounded sequences. Hence $\bar{e}_1 \geq T(c, m)$. For $c = 0$ the reduction is the residue field and $m = 1$. Theorem 4.16 now gives $0 < b \leq d - 2\bar{e}_1/m \leq \beta_v$. The embedding dimension is at most n ; a normal curve is regular at its closed point and has state $(1, 1, 1)$.

Step 2. In this step, we prove that only finitely many multiplicities have positive cap. Fix $d, c > 0$ and put $\ell = \lfloor d/2 \rfloor + 1$. Every bounded sequence of mass m has first moment at least $\ell(m - S_c(\ell - 1))$, because at most $S_c(\ell - 1)$ of its mass lies below degree ℓ . Apply this to the minimizing sequence above. The inequality $T(c, m) < dm/2$ implies

$$(\ell - d/2)m < \ell S_c(\ell - 1).$$

The coefficient of m on the left is positive. This bounds m for each of the finitely many pairs (d, e) , and proves finiteness.

Step 3. In this step, we show that every actual transition satisfies the recurrence constraints. An lc place of the old boundary centered at a previous minimal center has discrepancy zero. Effectivity of an addition and log canonicity of the new boundary force its order on that addition to vanish. The old place thus remains lc. Its center remains normal near x , so the nonminimal-parent assertion of Theorem 4.16 applies to the new boundary. For an adjacent successor of multiplicity k and deficit $y > 0$, it gives

$$my + k \leq dm - 2\bar{e}_1 \leq dm - 2T(c, m).$$

At the root we retain the first, stronger cap. No adjacent inequality is imposed on a skip of two or more dimensions. The quotient of local rings gives a quotient of cotangent spaces, so embedding dimension cannot increase. Deficits cannot increase under an effective addition, since each discrepancy in their valuative threshold formula decreases. At a point lc center the deficit is zero. Consequently every actual positive successor lies in exactly the domain allowed by the recurrence.

Step 4. In this step, we prove continuity and finite piecewise affinity of the envelopes. Induct on dimension, starting with the curve formula. For a successor w , the function $\mathcal{F}_w(y) - r_v y$ is continuous piecewise affine on its compact allowed interval. Its running maximum up to $\min(z, \beta_w, a_{vw})$ is attained at an endpoint of an affine piece or at the moving endpoint $y = z$. It is therefore continuous piecewise affine with finitely many pieces. The finite maximum over successors has the same properties and vanishes at zero. The caps are rational, and each r_v is algebraic because g_v is rational. Intersections of affine pieces introduce only algebraic breakpoints and coefficients.

Step 5. In this step, we bound the entire scalar chain by the recurrence. Suppose a current center has actual radius ρ , deficit z , and state v . A direct point successor costs $\rho z \leq r_v z$. Otherwise let w, y be the state and deficit of its actual next center. Induction bounds every remaining cut by $\mathcal{F}_w(y)$. The degree floor gives $\rho \leq r_v$, and $y \leq z$, whence

$$C \leq \rho(z - y) + \mathcal{F}_w(y) \leq r_v z + \mathcal{F}_w(y) - r_v y \leq \mathcal{F}_v(z).$$

Keeping the exact radius and normal coefficient at the first step gives $C \leq Q_B$. Its root interval is contained in the state interval, and $r \leq r_v$. Comparing $r(b - y) + \mathcal{F}_w(y)$ with $r_v(b - y) + \mathcal{F}_w(y)$ proves $Q_B \leq \mathcal{F}_v(b)$.

Step 6. In this step, we refine the first-successor bound using the old discriminant. Proposition 4.18 gives $my + k \leq K_W$ for every new adjacent center, and $my + (1 - c_j)k_j \leq K_W$ when that center is the named old prime P_j . Every actual first successor is thus in Definition 5.3; an old prime with an inadmissible state cannot become an actual positive minimal center. Applying the preceding induction after this first step gives $C \leq Q_{\text{ret}}(B)$. For a new prime the difference between the original cap and its retained cap is

$$\frac{R_0 + M_0 + H_0}{m} \geq 0.$$

For the named old prime P_j the difference is

$$\frac{R_0 + M_0 + H_0 - c_j k_j}{m} = \frac{R_0 + M_0 + \sum_{i \neq j} c_i k_i}{m} \geq 0.$$

The radius, descendant envelopes, point alternative, and skip caps are unchanged. This proves $Q_{\text{ret}}(B) \leq Q_B$. All old coefficients and the same compatible moduli data are included in these comparisons.

Step 7. In this step, we allocate both cutting errors within the strict coefficient margin. Write $Q = Q_B$, or $Q = Q_{\text{ret}}(B)$ when that refinement is used, and assume $\Delta = n + 1 - s - Q > 0$. Fix a positive $h < \Delta/2$ before cutting and allot h/n to each of at most n stages. At the actual stage i , after its radius ρ_i is known and before the cutter is chosen, choose

$$0 < \epsilon_i < \frac{h}{n(1 + \rho_i)}.$$

Theorem 3.14 gives an effective rational $H_i \sim_{\mathbb{Q}} L$, a positive rational t_i , and a smaller minimal center for the full boundary $B_i + t_i H_i$, with

$$t_i \leq b_i \rho_i + \epsilon_i, \quad b_{i+1} \leq b_i - t_i / \rho_i + \epsilon_i.$$

The first error has weight one and the second has weight ρ_i after rearrangement. We charge their sum $(1 + \rho_i)\epsilon_i$, even though the rearranged second inequality alone already gives $t_i \leq (b_i - b_{i+1})\rho_i + \rho_i \epsilon_i$. There are at most $d \leq n - 1$ cuts. Thus

$$s + \sum_i t_i \leq s + C + \sum_i \rho_i \epsilon_i < s + Q + h < n + 1.$$

This is one actual finite additive chain. The uniform bound on C holds for its outputs regardless of their dependence on the errors. The point boundary retains the entire initial boundary and every cut. Proposition 3.12 gives generation whenever it is needed. $\square$

Every ample Cartier center degree is a positive integer, so the common choice $g_v = 1$ always applies. A stronger rational floor may be used only when it holds for the original degree of every center assigned that state. It changes a comparison radius, not the polarization or the coefficient already spent.

Definition 5.5. For integers $N, d \geq 1$ and rational $q > 0$, define

$$\text{ROOT}_N(d, q) = \frac{1}{N} \min\{j \in \mathbb{Z}_{\geq 1} : j^d \geq N^d q\}.$$

For the degree floors in Definition 5.2, put $\hat{r}_v = \text{ROOT}_N(d, m/g_v)$, and let $\hat{\mathcal{F}}_v$ be its recurrence with r_v replaced by $\hat{r}_v$. For real $a \geq 0$ and $0 \leq h \leq \beta_v$, define

$$J_v(a, h) = \max_{0 \leq z \leq h} (\hat{\mathcal{F}}_v(z) - az).$$

The rational root is an upper bound checked by integer powers: if $q = A/D$ with positive integers A, D , its value j/N is characterized by $D(j - 1)^d < N^d A \leq D j^d$. The same rule can be used at the root with its actual degree g ; its exact normal-coefficient cap is still retained. Since every deficit drop is nonnegative, replacing any of these radii by upper bounds increases every chain cost.

Lemma 5.6. For the data of Definition 5.5, put $c_{vw} = \min(\beta_w, a_{vw})$, over all permitted successors. Then for every $a \geq 0$ and $0 \leq h \leq \beta_v$,

$$(5.2) \quad J_v(a, h) = (\hat{r}_v - a)_+ h + \max\left(0, \max_w J_w(\max(a, \hat{r}_v), \min(h, c_{vw}))\right).$$

In particular this is an exact finite rational recursion when a, h are rational.

This formula keeps the whole interval of possible deficits in a single quantity. It also covers slopes below one.

Proof. For a fixed successor deficit y , the expression to maximize over $y \leq z \leq h$ is

$$(\hat{r}_v - a)z + \hat{\mathcal{F}}_w(y) - \hat{r}_v y.$$

If $\hat{r}_v \geq a$, its maximum is at $z = h$. If $\hat{r}_v < a$, its maximum is at $z = y$. The point alternative has maximum $(\hat{r}_v - a)_+ h$. Taking all successors gives the identity. Dimensions strictly decrease, and every argument and coefficient on its right is rational when its inputs are rational. $\square$

For example, with degree floor one a curve contributes $J_{(1,1,1)}(a, h) = (1 - a)_+ h$, $0 \leq h \leq 1$. At a baseline $a = 20/21$ this is $h/21$, so a direct curve skip cannot be suppressed merely because it vanishes at baselines greater than one.

Definition 5.7. Take $n = 7$, $g_v = 1$, and $N = 1000$. The following data enumerate $\mathcal{P}_7$. The row indexed by i has state (d_i, e_i, m_i) , moment T_i , cap β_i , rational radius $\hat{r}_i$, and a finite set $\mathcal{L}_i$ of slope–intercept pairs. Subscripts i on $\mathcal{F}$ and $\hat{\mathcal{F}}$ mean this row’s state. Define

$$H_i(z) = \min_{(a,p) \in \mathcal{L}_i} (az + p), \quad 0 \leq z \leq \beta_i.$$

i	d	e	m	T_i	β_i	$\hat{r}_i$	$\mathcal{L}_i$
1	1	1	1	0	1	1	(1, 0)
2	2	2	1	0	2	1	(1, 0)
3	2	3	2	1	1	283/200	(283/200, 0)
4	2	4	3	2	2/3	1733/1000	(1733/1000, 0)
5	2	5	4	3	1/2	2	(2, 0)
6	2	6	5	4	2/5	2237/1000	(2237/1000, 0)
7	2	7	6	5	1/3	49/20	(49/20, 0)
i	d	e	m	T_i	β_i	$\hat{r}_i$	$\mathcal{L}_i$
8	3	3	1	0	3	1	(1, 83/200); (5/4, 33/200); (3/2, 0)
9	3	4	2	1	2	63/50	(63/50, 237/1000); (3/2, 117/1000); (5/3, 17/500); (7/4, 0)
10	3	4	3	3	1	1443/1000	(1443/1000, 0)
11	3	5	3	2	5/3	1443/1000	(5/4, 103/200); (1443/1000, 97/500); (3/2, 167/1000); (2, 0)
12	3	5	4	4	1	397/250	(397/250, 37/1000); (5/3, 17/1000); (7/4, 0)
13	3	5	5	6	3/5	171/100	(171/100, 0)
14	3	5	6	8	1/3	909/500	(909/500, 0)
i	d	e	m	T_i	β_i	$\hat{r}_i$	$\mathcal{L}_i$
15	3	6	4	3	3/2	397/250	(397/250, 103/500); (7/4, 1/8); (2, 3/50); (5/2, 0)
16	3	6	5	5	1	171/100	(171/100, 29/500); (2, 0)
17	3	6	6	7	2/3	909/500	(909/500, 0)
18	3	6	7	9	3/7	1913/1000	(1913/1000, 0)
19	3	6	8	11	1/4	2	(2, 0)
20	3	6	9	13	1/9	2081/1000	(2081/1000, 0)
i	d	e	m	T_i	β_i	$\hat{r}_i$	$\mathcal{L}_i$
21	3	7	5	4	7/5	171/100	(171/100, 211/1000); (2, 19/200); (5/2, 0)
22	3	7	6	6	1	909/500	(909/500, 7/100); (2, 1/25); (5/2, 0)
23	3	7	7	8	5/7	1913/1000	(5/4, 243/500); (7/4, 129/1000); (1913/1000, 13/1000); (2, 0)
24	3	7	8	10	1/2	2	(2, 0)
25	3	7	9	12	1/3	2081/1000	(2081/1000, 0)
26	3	7	10	14	1/5	431/200	(431/200, 0)
27	3	7	11	16	1/11	278/125	(278/125, 0)

i	d	e	m	T_i	β_i	$\widehat{r}_i$	$\mathcal{L}_i$
28	4	4	1	0	4	1	(1, 757/1000); (5/4, 161/500); (5/3, 9/200); (7/4, 0)
29	4	5	2	1	3	119/100	(119/100, 287/500); (5/4, 121/250); (3/2, 1/4); (2, 0)
30	4	5	3	3	2	1317/1000	(1317/1000, 171/500); (3/2, 1/4); (2, 0)
31	4	5	4	6	1	283/200	(283/200, 293/1000); (3/2, 1/4); (2, 0)
i	d	e	m	T_i	β_i	$\widehat{r}_i$	$\mathcal{L}_i$
32	4	6	3	2	8/3	1317/1000	(5/4, 373/500); (1317/1000, 71/125); (3/2, 81/250); (5/3, 229/1000); (7/4, 39/200); (2, 19/200); (5/2, 0)
33	4	6	4	4	2	283/200	(283/200, 379/1000); (3/2, 59/200); (2, 19/200); (5/2, 0)
34	4	6	5	6	8/5	187/125	(5/4, 69/100); (187/125, 297/1000); (3/2, 59/200); (2, 19/200); (5/2, 0)
35	4	6	6	8	4/3	783/500	(5/4, 69/100); (783/500, 269/1000); (7/4, 39/200); (2, 19/200); (5/2, 0)
36	4	6	7	11	6/7	1627/1000	(1627/1000, 61/250); (2, 19/200); (5/2, 0)
37	4	6	8	14	1/2	841/500	(841/500, 111/500); (2, 19/200); (5/2, 0)
38	4	6	9	17	2/9	1733/1000	(1733/1000, 14/125); (2, 53/1000); (5/2, 0)
i	d	e	m	T_i	β_i	$\widehat{r}_i$	$\mathcal{L}_i$
39	4	7	4	3	5/2	283/200	(283/200, 29/50); (5/3, 133/500); (7/4, 117/500); (2, 3/20); (5/2, 0)
40	4	7	5	5	2	187/125	(187/125, 17/40); (3/2, 421/1000); (5/3, 131/500); (2, 3/20); (5/2, 0)
41	4	7	6	7	5/3	783/500	(783/500, 331/1000); (5/3, 131/500); (2, 3/20); (5/2, 0)
42	4	7	7	9	10/7	1627/1000	(5/4, 813/1000); (3/2, 57/125); (1627/1000, 11/40); (2, 3/20); (5/2, 0)
43	4	7	8	11	5/4	841/500	(841/500, 32/125); (2, 3/20); (5/2, 0)
44	4	7	9	13	10/9	1733/1000	(1733/1000, 239/1000); (2, 3/20); (5/2, 0)
45	4	7	10	15	1	1779/1000	(5/3, 42/125); (1779/1000, 28/125); (2, 3/20); (5/2, 0)
i	d	e	m	T_i	β_i	$\widehat{r}_i$	$\mathcal{L}_i$
46	4	7	11	18	8/11	911/500	(7/4, 131/500); (911/500, 21/100); (2, 3/20); (5/2, 0)
47	4	7	12	21	1/2	931/500	(931/500, 49/250); (2, 3/20); (5/2, 0)
48	4	7	13	24	4/13	1899/1000	(1899/1000, 17/100); (2, 139/1000); (5/2, 0)
49	4	7	14	27	1/7	387/200	(7/4, 1/10); (387/200, 37/500); (2, 13/200); (5/2, 0)
i	d	e	m	T_i	β_i	$\widehat{r}_i$	$\mathcal{L}_i$
50	5	5	1	0	5	1	(1, 143/125); (5/4, 103/200); (3/2, 1/4); (2, 0)
51	5	6	2	1	4	1149/1000	(1149/1000, 247/250); (5/4, 92/125); (3/2, 169/500); (5/3, 229/1000); (7/4, 39/200); (2, 19/200); (5/2, 0)
52	5	6	3	3	3	623/500	(623/500, 719/1000); (3/2, 169/500); (5/3, 229/1000); (7/4, 39/200); (2, 19/200); (5/2, 0)
53	5	6	4	6	2	33/25	(33/25, 76/125); (3/2, 169/500); (5/3, 229/1000); (7/4, 39/200); (2, 19/200); (5/2, 0)
54	5	6	5	10	1	69/50	(69/50, 207/500); (3/2, 59/200); (2, 19/200); (5/2, 0)

i	d	e	m	T_i	β_i	$\widehat{r}_i$	$\mathcal{L}_i$
55	5	7	3	2	11/3	623/500	(623/500, 39/40); (5/4, 193/200); (3/2, 101/200); (5/3, 71/250); (7/4, 117/500); (2, 3/20); (5/2, 0)
56	5	7	4	4	3	33/25	(33/25, 77/100); (3/2, 101/200); (5/3, 289/1000); (7/4, 239/1000); (2, 3/20); (5/2, 0)
57	5	7	5	6	13/5	69/50	(69/50, 673/1000); (3/2, 101/200); (5/3, 293/1000); (7/4, 121/500); (2, 3/20); (5/2, 0)
58	5	7	6	8	7/3	1431/1000	(5/4, 128/125); (1431/1000, 301/500); (3/2, 101/200); (5/3, 299/1000); (7/4, 123/500); (2, 3/20); (5/2, 0)
59	5	7	7	11	13/7	369/250	(369/250, 539/1000); (3/2, 101/200); (5/3, 277/1000); (7/4, 237/1000); (2, 3/20); (5/2, 0)
60	5	7	8	14	3/2	379/250	(379/250, 483/1000); (5/3, 34/125); (7/4, 117/500); (2, 3/20); (5/2, 0)
i	d	e	m	T_i	β_i	$\widehat{r}_i$	$\mathcal{L}_i$
61	5	7	9	17	11/9	194/125	(1, 1079/1000); (3/2, 117/250); (5/3, 133/500); (7/4, 117/500); (2, 3/20); (5/2, 0)
62	5	7	10	20	1	317/200	(317/200, 42/125); (5/3, 133/500); (7/4, 117/500); (2, 3/20); (5/2, 0)
63	5	7	11	24	7/11	202/125	(202/125, 139/500); (2, 3/20); (5/2, 0)
64	5	7	12	28	1/3	411/250	(2, 3/20); (5/2, 0)
65	5	7	13	32	1/13	1671/1000	(5/2, 0)
i	d	e	m	T_i	β_i	$\widehat{r}_i$	$\mathcal{L}_i$
66	6	6	1	0	6	1	(1, 198/125); (5/4, 187/250); (3/2, 357/1000); (5/3, 23/100); (7/4, 39/200); (2, 19/200); (5/2, 0)
67	6	7	2	1	5	1123/1000	(1123/1000, 703/500); (5/4, 499/500); (3/2, 101/200); (5/3, 42/125); (7/4, 131/500); (2, 3/20); (5/2, 0)
68	6	7	3	3	4	1201/1000	(1201/1000, 223/200); (5/4, 993/1000); (3/2, 101/200); (5/3, 42/125); (7/4, 131/500); (2, 3/20); (5/2, 0)
69	6	7	4	6	3	63/50	(63/50, 121/125); (3/2, 101/200); (5/3, 42/125); (7/4, 131/500); (2, 3/20); (5/2, 0)
70	6	7	5	10	2	327/250	(327/250, 801/1000); (3/2, 101/200); (5/3, 42/125); (7/4, 131/500); (2, 3/20); (5/2, 0)
71	6	7	6	15	1	1349/1000	(5/3, 42/125); (7/4, 131/500); (2, 3/20); (5/2, 0)

Lemma 5.8. *For every row i of Definition 5.7 and every $0 \leq z \leq \beta_i$,*

$$\mathcal{F}_i(z) \leq \widehat{\mathcal{F}}_i(z) \leq H_i(z).$$

The table includes every possible positive-dimensional minimal-center state on a sevenfold.

The affine minima bound all later choices. They do not assert that each numerical state is geometrically attained. The canonical records form the supplementary family `05-tails/ambient71`, in `supplement/05-tails/data/ambient71.json`; the displayed tables are its deterministic rendering. Each record stores the row index, state, moment, intrinsic cap, radius, and every slope–intercept pair in that order. Lemma 5.8 is the correctness criterion for this family.

Proof. We specify the complete state coverage and the finite comparison that certifies every line of the table.

Step 1. In this step, we enumerate every state with positive cap. For $c > 0$ the moment has the equivalent expression

$$T(c, m) = \sum_{j \geq 1} (m - S_c(j-1))_+.$$

Indeed both sides count the degrees of the first m monomials, listed by increasing degree. Their mean degree $T(c, m)/m$ is nondecreasing with m . Thus after the first multiplicity with $2T(c, m) \geq dm$, no later multiplicity has positive cap. The complete positive ranges are

d	e	m
1	1	1
2	2	1
2	3, 4, 5, 6, 7	2; 3; 4; 5; 6
3	3, 4, 5, 6, 7	1; 2:3; 3:6; 4:9; 5:11
4	4, 5, 6, 7	1; 2:4; 3:9; 4:14
5	5, 6, 7	1; 2:5; 3:13
6	6, 7	1; 2:6

Here a colon denotes every integer in the indicated interval, and semicolon-separated entries correspond to the listed embeddings. Substitution in the moment formula gives the displayed T_i and β_i . At the next integer in each range the cap is nonpositive. These ranges give respectively 1, 6, 20, 22, 16, 6 states, totaling 71. The rational-radius assertions are the integer power comparisons in Definition 5.5. For an explicit bounded enumeration, set $\ell = \lfloor d/2 \rfloor + 1$. The finiteness proof of Theorem 5.4 permits one to test precisely

$$c + 1 \leq m \leq \left\lfloor \frac{2\ell S_c(\ell - 1) - 1}{2\ell - d} \right\rfloor \quad (c > 0),$$

retaining only positive caps, together with the smooth states and the single curve state. Thus the expected indices are reconstructed without using the supplied rows as a stopping rule.

Step 2. In this step, we specify the finite comparisons for every transition and affine interval. For $d_j < d_i$ and $e_j \leq e_i$, put

$$t_{ij} = \begin{cases} \min(\beta_i, \beta_j, \beta_i - m_j/m_i), & d_j = d_i - 1, \\ \min(\beta_i, \beta_j), & d_j \leq d_i - 2. \end{cases}$$

Discard precisely the pairs with $t_{ij} \leq 0$. For any rational a and $t \geq 0$, let $S_j(a, t) = \max_{0 \leq y \leq t} (H_j(y) - ay)$. It is a finite evaluation: test $y = 0, t$ and all numbers

$$y = \frac{p' - p}{u - u'} \in (0, t), \quad (u, p), (u', p') \in \mathcal{L}_j, \quad u \neq u'.$$

These include every breakpoint of the minimum, even when some of the tested intersections are not active. The table satisfies, for every i and every $(a, p) \in \mathcal{L}_i$,

$$(5.3) \quad p \geq (\hat{r}_i - a)_+ \beta_i + \max \left(0, \max_{j: t_{ij} > 0} S_j(\max(a, \hat{r}_i), t_{ij}) \right).$$

All its data, test points, and allowed index ranges have been given; the comparisons are rational comparisons with positive denominators. The minimum of all successor lines is taken before the maximum. If the interval is a point, its single endpoint suffices. Parallel lines create no intersection to divide by, and repeated intersection points are counted once. An empty positive-successor domain contributes only the direct point alternative; in particular, a nonpositive transition cap creates no additional branch.

The portable checker `supplement/05-tails/check.py` requires the complete state list and all 253 affine pairs, with their per-row counts. It rejects missing, repeated, additional, or malformed

records. Rational entries have reduced numerator and positive denominator; all comparisons use integer arithmetic. It reconstructs every moment and the smallest upper radius on the $1/1000$ grid, then performs (5.3) for every pair. There are 1377 allowed state transitions, 6889 line-successor comparisons, and 34839 endpoint or intersection evaluations. Some affine inequalities are equalities; strict payment margins are established separately below.

For instance, row 8 can reach a singular surface only at multiplicity two. That surface's radius bound is $283/200$ and its deficit cap is one. The surpluses over slopes $1, 5/4, 3/2$ are bounded by $83/200, 33/200, 0$, respectively. For row 15, a surface of multiplicity five has cap $\min(2/5, (6-5)/4) = 1/4$. Its surplus over $397/250$ is $649/4000$; multiplicity four has cap $1/2$ and surplus $103/500$, which is larger. Multiplicities one, two, and three have smaller surpluses. These give the radius-line intercept of row 15. Formula (5.3), with the full displayed data, gives the other lines by the same finite test.

Step 3. In this step, we deduce the complete-tail bounds from the line tests. Induct on d_i . For a successor j at deficit $y \leq z$, the total is at most $\hat{r}_i(z-y) + H_j(y)$. If $\hat{r}_i \geq a$, subtracting az bounds this by $(\hat{r}_i - a)\beta_i + S_j(\hat{r}_i, t_{ij})$. If $\hat{r}_i < a$, use $z \geq y$ to bound it instead by $S_j(a, t_{ij})$. The direct point alternative is bounded by the zero term in (5.3). Thus every listed line bounds the entire tail, and so does their minimum. The first inequality follows from increasing the radii in the recurrence. All skips, including a direct point successor, were included in the comparison. $\square$

Proposition 5.9. *At a nongeneration point on a sevenfold, every pure actual threshold of a nonzero complete $U_k(u)$, with uk integral and $u \geq 26/25$, has minimal center of dimension 4, 5, or 6. For dimension four, $b > 3/2$; for dimensions five and six, $b > 7/5$. For $u \geq 29/25$, the nondivisorial states and their deficit caps are*

$$(5, 6, 2; 4), \quad (5, 7, 3; 11/3), \quad (5, 7, 4; 3), \quad (5, 7, 5; 13/5).$$

Their deficits also satisfy $b \geq ((8-p)u-7)/(au-1)$, respectively with

$$(a, p) = (1149/1000, 247/250), (623/500, 39/40), (33/25, 77/100), (69/50, 673/1000).$$

For $u \geq 34/25$, only a divisor remains and the entire source fixed divisor has multiplicity at least $125(u-1)k/73$.

These are necessary conditions on the threshold of each nonzero original source separately; source existence is not assumed at another degree or order.

Proof. We first translate nongeneration into a failed-tail inequality, then apply the complete table to all states. The remaining divisorial cases give the full fixed-divisor bound.

Step 1. In this step, we derive the failed-tail inequality from the actual source coefficient. Fix the given nonzero complete space, its full base ideal I , and $\tau = \text{lct}_x(I)$. Let Γ be the pure threshold average of Definition 2.4, with coefficient $s = k\tau$. Every member has order at least uk , so $\text{mult}_x \Gamma \geq us$. The exceptional divisor of the ordinary point blowup of the smooth sevenfold has discrepancy seven and point-ideal order one. Consequently

$$(5.4) \quad us + b \leq \text{mult}_x \Gamma + b \leq 7, \quad s \leq \frac{7-b}{u}.$$

A point minimal center would have $s \leq 7/u \leq 175/26 < 8$, contradicting Proposition 3.12. At a positive center of row i , Theorem 5.4 and Lemma 5.8 show that $s + H_i(b) < 8$ is a paid point exit for this same Γ . Nongeneration therefore requires

$$(5.5) \quad \frac{7-b}{u} + H_i(b) \geq 8.$$

Step 2. In this step, we exclude all low-dimensional and small-deficit rows by their complete costs. Every slope in the table is at least one. At $u = 26/25$, the left side of (5.5) is increasing in

b. The complete endpoint evaluations are

dimensions	additional cap	maximum	maximizing row
1, 2, 3	none	20773/2600	21
4	3/2	41553/5200	39
5, 6	7/5	38951/4875	60

For each row, substitute the smaller of its own cap and the additional cap, take the minimum of all its affine lines, and then maximize over the indicated rows. The margins below eight are respectively 27/2600, 47/5200, 49/4875, all positive. Since $7 - b > 0$, increasing u preserves all these exclusions.

Step 3. In this step, we identify the four nondivisorial states surviving order 29/25. Omit rows 51, 55, 56, 57 temporarily. At $u = 29/25$ all remaining nondivisorial rows give the following maxima at their caps:

d	$\max_i((7 - \beta_i)/(29/25) + H_i(\beta_i))$	maximizing row
1	179/29	1
2	95919/14500	4
3	43109/5800	21
4	92763/11600	39
5	115473/14500	58

Every entry is below eight; their common margin is at least 37/11600. The four omitted rows are exactly the four states in the statement. Their first listed lines are $ab + p$ with the stated pairs. Since $au - 1 > 0$, (5.5) implies

$$b \geq \frac{(8 - p)u - 7}{au - 1}.$$

At $u = 34/25$, evaluating those four lines at their respective caps gives, in the same order,

$$\frac{33107}{4250}, \quad \frac{135909}{17000}, \quad \frac{13041}{1700}, \quad \frac{127437}{17000}.$$

They are all below eight, with common margin at least 91/17000. Thus only divisors remain for $u \geq 34/25$. In every exclusion the positive gap is fixed before cutting, and the error allocation in Theorem 5.4 applies to it.

Step 4. In this step, we derive the full fixed-divisor multiplicity bound for every surviving divisor. Let P be the actual divisorial minimal center, of multiplicity $m \in \{1, \dots, 6\}$, and put $\nu = \text{ord}_P(I)/k$. General dilution makes each moving coefficient less than one. Since P has coefficient one in Γ , it occurs in the full fixed part and

$$\tau \text{ord}_P(I) = 1, \quad s = 1/\nu, \quad b \leq 7 - u/\nu.$$

Rows 66, ..., 71 give the following single-line bounds:

m	a	p	$D_m = 7a + p - 8$
1	1	198/125	73/125
2	1123/1000	703/500	1267/1000
3	1201/1000	223/200	761/500
4	63/50	121/125	447/250
5	327/250	801/1000	1957/1000
6	5/3	42/125	1501/375

All $D_m > 0$. For $u \geq 1$, nongeneration and $s + C \leq 7a + p - (au - 1)/\nu$ require

$$m\nu \geq \frac{m(au - 1)}{D_m} \geq \frac{125(u - 1)}{73}.$$

For the last inequality, at $u = 1$ the difference is $m(a - 1)/D_m \geq 0$; its slope is $ma/D_m - 125/73 \geq 0$ for every displayed pair. The first row has equal slopes. The entire fixed divisor F contains

$\text{ord}_P(I)P$, so $\text{mult}_x F \geq m \text{ord}_P(I) = mk\nu$, as required. No other fixed component has been removed. $\square$

Lemma 5.10. *For an actual threefold center of embedding dimension at most six, multiplicity eight gives deficit $z \leq 1/4$ and entire descendant scalar cost $C \leq 2z$; multiplicity nine gives $z \leq 1/9$ and $C \leq 9z/4$. All dimension skips and all adjacent caps are included. For embedding dimension seven the corresponding moment bounds are $T(4, 8) = 10, T(4, 9) = 12$, giving caps $1/2, 1/3$, respectively. These latter rows, and the multiplicity-ten and eleven rows, remain in the ambient sevenfold state set.*

The embedding-six rows have smaller caps than their embedding-seven counterparts. They must therefore be checked separately.

Proof. The complete multiplicity ranges above show that a threefold of multiplicity eight or nine and embedding dimension at most six must have embedding dimension six. Thus $c = 3$. The degree capacities begin 1, 3, 6. Filling them gives moments

$$T(3, 8) = 3 + 2 \cdot 4 = 11, \quad T(3, 9) = 3 + 2 \cdot 5 = 13.$$

Theorem 4.16 therefore gives $8z \leq 24 - 22 = 2$ and $9z \leq 27 - 26 = 1$, respectively.

Case 1. The embedding dimension is at most six and the multiplicity is eight. At multiplicity eight an adjacent surface of multiplicity k and positive deficit y satisfies

$$8y + k \leq 24 - 2\bar{e}_1 \leq 2.$$

Hence $k = 1$ and $y \leq 1/8$. Such a surface is smooth: its Cohen–Macaulay Artinian reduction has length one and embedding codimension zero. Its radius, and every later curve radius, is at most one. A skip from the threefold to a curve has the same radius bound, and a point has no remaining cost. The root radius is at most $8^{1/3} = 2$. Every radius in the chain is therefore at most two; telescoping all deficit drops gives $C \leq 2z$.

Case 2. The embedding dimension is at most six and the multiplicity is nine. At multiplicity nine the adjacent inequality is $9y + k \leq 27 - 2\bar{e}_1 \leq 1$, which is impossible for $y > 0$ and $k \geq 1$. Only a curve or point can follow. The root radius is at most $9^{1/3} < 9/4$, because $9 < (9/4)^3$; the curve radius is at most one. Telescoping gives $C \leq 9z/4$.

Case 3. The embedding dimension is seven. At embedding dimension seven the initial capacities are 1, 4, 10. They give $T(4, 8) = 4 + 2 \cdot 3 = 10$ and $T(4, 9) = 4 + 2 \cdot 4 = 12$, hence caps $1/2, 1/3$. For completeness, the adjacent surface restrictions are

m	$T(4, m)$	β	adjacent restriction
8	10	$1/2$	$8y + k \leq 4$
9	12	$1/3$	$9y + k \leq 3$
10	14	$1/5$	$10y + k \leq 2$
11	16	$1/11$	$11y + k \leq 1$

Their positive adjacent multiplicities are respectively $k = 1, 2, 3, k = 1, 2, k = 1$, and none. Every surface also has its own cap $y \leq 2/k$, and every curve skip is retained. These are rows 24–27 of the ambient table, including their rational radius bounds; none is discarded. $\square$

Lemma 5.11. *Let B be an effective rational boundary lc near x on a smooth complex projective n -fold, $n \geq 7$, and let L be ample Cartier. Suppose the minimal center of B through x has dimension five and multiplicity two. For every first positive-dimensional successor U in an additive continuation from B , put $v = (d, e, m) = (\dim U, \text{edim } \mathcal{O}_{U,x}, \text{mult}_x U)$. Let y be its deficit and C the scalar cost of any entire continuation from this successor to the point. Use degree floors one and rational radii $\text{ROOT}_{1000}(d, m)$ in Definition 5.5. Put*

$$C_v = \begin{cases} \min(\beta_v, (8 - m)/2), & d = 4, \\ \beta_v, & d \leq 3. \end{cases}$$

Then its state belongs to the following complete table, its deficit satisfies $0 < y \leq C_v$, and the last column gives the exact value $J_v(50/49, C_v)$.

d	e	m	T_v	C_v	$\hat{r}_v$	$J_v(50/49, C_v)$
1	1	1	0	1	1	0
2	2	1	0	2	1	0
2	3	2	1	1	283/200	3867/9800
2	4	3	2	2/3	1733/1000	11639/24500
2	5	4	3	1/2	2	24/49
2	6	5	4	2/5	2237/1000	59613/122500
3	3	1	0	3	1	3867/9800
3	4	2	1	2	63/50	70137/98000
3	4	3	3	1	1443/1000	20707/49000
3	5	3	2	5/3	1443/1000	8797/9800
3	5	4	4	1	397/250	118353/196000
3	5	5	6	3/5	171/100	10137/24500
3	5	6	8	1/3	909/500	19541/73500
3	6	4	3	3/2	397/250	12953/12250
3	6	5	5	1	171/100	4579/6125
3	6	6	7	2/3	909/500	19541/36750
3	6	7	9	3/7	1913/1000	131211/343000
3	6	8	11	1/4	2	12/49
3	6	9	13	1/9	2081/1000	17323/147000
d	e	m	T_v	C_v	$\hat{r}_v$	$J_v(50/49, C_v)$
4	4	1	0	7/2	1	70137/98000
4	5	2	1	3	119/100	317993/294000
4	5	3	3	2	1317/1000	91599/98000
4	5	4	6	1	283/200	13467/19600
4	6	3	2	5/2	1317/1000	384791/294000
4	6	4	4	2	283/200	57241/49000
4	6	5	6	3/2	187/125	123699/122500
4	6	6	8	1	783/500	49857/61250
4	6	7	11	1/2	1627/1000	10727/19600

In particular, the entire scalar cost after that successor satisfies

$$C - \frac{50}{49}y \leq \frac{384791}{294000} < \frac{1309}{1000} < \frac{131}{100}.$$

This table isolates the inherited double-fivefold cap. Its 28 states are the complete restricted domain for that parent, not a replacement for the 71 ambient states. The coefficient spent to reach the first successor is retained separately. Its canonical records form the supplementary family 05-tails/double-fivefold28, in `supplement/05-tails/data/double-fivefold28.json`, with the displayed table as a deterministic rendering. Each record gives its state and ambient row index, moment, intrinsic cap, first-successor cap, radius, and exact support value. Lemma 5.11 supplies its correctness criterion.

Proof. We first identify the restricted domain, then evaluate the exact support recursion on it.

Step 1. In this step, we determine every first-successor state allowed by the double-fivefold parent. By Theorems 4.13 and 4.5, a fivefold of multiplicity two has embedding dimension six and $\bar{e}_1 \geq 1$. Its positive deficit is at most four. For a fourfold successor, Theorem 4.16 on the persistent parent gives $2y + m \leq 8$. Thus $m \geq 8$ has nonpositive first cap and cannot occur. For dimensions at most three there is no additional first cap. All embeddings are at most six.

The complete first-successor ranges are the curve $(1, 1, 1)$, the smooth surface and surfaces $(2, k + 1, k)$ for $2 \leq k \leq 5$, all threefold rows of embeddings three through six, and fourfolds

$$(4, 4, 1), \quad (4, 5, m), \quad 2 \leq m \leq 4, \quad (4, 6, m), \quad 3 \leq m \leq 7.$$

They give $1 + 5 + 13 + 9 = 28$ states. The fourfold exclusion of $m = 8, 9$ uses the inherited first cap; their positive intrinsic caps alone would not exclude them. All smaller-dimensional states remain available at subsequent steps. The checker reconstructs the 30 intrinsic states with $d \leq 4$ and $e \leq 6$, then applies the first-successor cap to obtain these 28 states. Every later edge uses its own parent's intrinsic cap; the cap inherited from the double fivefold is used only at entry.

Step 2. In this step, we evaluate the support recursion on every allowed deficit interval. For a surface and $a \geq 1$, its support value is $(\hat{r}_v - a)_+ h$, $0 \leq h \leq \beta_v = 2/m$, because later curves have radius one. For a threefold $v = (3, e, m)$, let $T_v = T(e - 3, m)$. Its possible surface children have multiplicities $1 \leq k \leq e - 1$, with

$$h_k = \min\left(\frac{2}{k}, \frac{3m - 2T_v - k}{m}\right) > 0.$$

The smooth surface corresponds to $k = 1$; write $q_k = \text{ROOT}_{1000}(2, k)$. Equation (5.2) becomes, for $a \geq 1$,

$$(5.6) \quad J_v(a, h) = (\hat{r}_v - a)_+ h + \max\left(0, \max_{k: h_k > 0} (q_k - \max(a, \hat{r}_v))_+ \min(h, h_k)\right).$$

For each fourfold row, insert this expression for every threefold of no larger embedding dimension with positive adjacent cap $\min(\beta_w, (4m - 2T_v - m_w)/m)$. Also insert every direct surface with cap $2/k$, $1 \leq k \leq e - 1$. Direct curves and the point contribute zero at these baselines, since all slopes in this evaluation are at least $50/49 > 1$. This lists every term in the recursion and covers the whole continuous interval $0 \leq y \leq C_v$.

The supplementary checker also expands all finite chains as a second evaluation of the same quantity. For a fixed chain $v_0, \dots, v_l$, put $b_0 = C_{v_0}$ and $b_j = \min(b_{j-1}, c_{v_{j-1}v_j})$. Its support is the maximum of

$$(\hat{r}_{v_0} - a)z_0 + \sum_{j=1}^l (\hat{r}_{v_j} - \hat{r}_{v_{j-1}})z_j$$

over $0 \leq z_l \leq \dots \leq z_0$ with $z_j \leq b_j$. At a vertex every maximal block of equal coordinates is anchored at zero or at one of the bounds b_j . Otherwise that block can be perturbed in both directions while preserving all inequalities, which contradicts extremality. It therefore suffices to enumerate the coordinates in $\{0, b_0, \dots, b_l\}$ and retain the feasible tuples. This includes all endpoints, repeated caps, one-dimensional segments, and the single point when $b_0 = 0$. The empty set of positive successors leaves the chain terminating at the point. The 28 starting states have 170 allowed descendant edges and 739 chains in total. For every row, this finite maximum and (5.2) both equal its supplied last entry. The checker also evaluates every zero-length interval and the curve correction at slope $20/21$.

Substituting the printed rational values gives the last column. For its largest entry, $v = (4, 6, 3)$, the successive positive terms are

$$\left(\frac{1317}{1000} - \frac{50}{49}\right) \frac{5}{2} + \left(\frac{397}{250} - \frac{1317}{1000}\right) \frac{4}{3} + \left(2 - \frac{397}{250}\right) \frac{1}{2} = \frac{384791}{294000}.$$

The complete surface ranges and all threefold rows in (5.6) give the other displayed entries and show they are no larger. Thus every comparison involves only the printed rational numbers and maxima over explicitly finite sets. The gap to $131/100$ is $349/294000 > 0$. Finally Theorem 5.4 and the radius comparison give $C - (50/49)y \leq J_v(50/49, C_v)$. $\square$

Corollary 5.12. *On a sevenfold, an effective rational $D \sim_{\mathbb{Q}} tL$, $t > 0$, with $\text{mult}_x D \geq (13/5)t$, gives a point boundary of coefficient less than $8 - 1/100$. Hence $K_X + 8L$ is generated at x .*

The boundary is obtained from the actual threshold of the entire given divisor, followed by additive cuts.

Proof. We bound the coefficient of the threshold boundary, evaluate its complete tail, and reserve the cutting errors.

Step 1. In this step, we bound the coefficient of the actual threshold boundary. Let $\tau = \text{let}_x(D) > 0$, let $\Gamma = \tau D$, and put $s = \tau t$. The threshold is rational, and its pair is lc but not klt at x . The point blowup gives

$$\frac{13}{5}s + b \leq \text{mult}_x \Gamma + b \leq 7, \quad s \leq \frac{7-b}{13/5}.$$

Step 2. In this step, we bound the complete scalar cost over all 71 states. If the minimal center is the point, $s \leq 35/13 < 8 - 1/100$. Otherwise it has one of the 71 states. Evaluating all the affine minima on their full deficit intervals gives

d	$\max_{i:d_i=d} \max_{0 \leq z \leq \beta_i} \left(\frac{7-z}{13/5} + H_i(z) \right)$
1	43/13
2	51/13
3	12879/2600
4	76841/13000
5	11234/1625
6	12949/1625

All slopes of H_i are at least one, so each maximum in z occurs at β_i ; the table supplies every row in the outer maximum. The largest value is the smooth sixfold row at $z = 6$:

$$\frac{83}{13} + \frac{198}{125} = \frac{12949}{1625}, \quad 8 - \frac{12949}{1625} = \frac{51}{1625} > \frac{11}{1000}.$$

The entire scalar tail is consequently at most $12949/1625 - s$.

Step 3. In this step, we allocate the cutting errors and obtain generation at x . At each of at most six cuts choose $0 < \epsilon_i < 1/(6000(1 + \rho_i))$ after its actual radius is known. The two errors charged in the proof of Theorem 5.4 sum to less than $1/1000$. The resulting point coefficient is therefore less than

$$\frac{12949}{1625} + \frac{1}{1000} < 8 - \frac{1}{100}.$$

Proposition 3.12 now gives generation at x . □

6. ORIGINAL SECTION PROFILES

To strengthen the complete sources of Definition 2.4, we compare the finite-degree orders of their full base ideals with convex asymptotic profiles. The comparison is uniform in the valuation after the variety, polarization, point, and compact big interval have been fixed. On a center that is normal only near the point, we keep the normalization conductor in the construction of every original section. We then bound the degree of the full fixed divisor and the multiplicity of the residual ideal by the volume of the moving divisor.

We first collect the external inputs. Restriction is used in its ideal form, with nonzero restricted ideal [Laz09, Theorem 3.2 and Remark 3.7].

Theorem 6.1 (Restriction of multiplier ideals). *Let T be a smooth divisor in a smooth complex variety Z , and let $\mathfrak{a}$ be an ideal with nonzero restriction to T . For every real $c > 0$,*

$$\mathcal{J}(T, (\mathfrak{a}\mathcal{O}_T)^c) \subseteq \mathcal{J}(Z, \mathfrak{a}^c)\mathcal{O}_T.$$

The inclusion is preserved by localization.

The restricted multiplier ideal gives a local comparison; it does not assert that local sections extend to the ambient variety. The following properties of asymptotic ideals concern complete linear systems [Laz09, Theorem 4.7 and Exercise 4.10].

Theorem 6.2 (Asymptotic multiplier ideals). *Let D be a big Cartier divisor on a smooth projective variety. For every positive integer m for which its complete system is nonempty, its base ideal is contained in $\mathcal{J}(\|mD\|)$. For positive integers m, q ,*

$$\mathcal{J}(\|qmD\|) \subseteq \mathcal{J}(\|mD\|)^q.$$

There is a fixed sufficiently positive Cartier divisor A such that $\mathcal{O}(mD + A) \otimes \mathcal{J}(\|mD\|)$ is globally generated for every $m \geq 1$.

For a big integral divisor Q , we use $\mathcal{J}(\|Q\|)$ for the maximal ideal among $\mathcal{J}(\mathfrak{b}(\|mQ\|)^{1/m})$. It is attained for sufficiently divisible m . Below we derive a twist independent of Q , as required when both degree and point order vary. Volume supplies the corresponding global comparison [BFJ09, Theorem A, Proposition 2.12, and Theorems 2.16, 3.1, 3.2].

Theorem 6.3 (Volume and concavity). *On a projective variety of dimension d , volume extends continuously to real divisor classes, and its d -th root is concave on the pseudoeffective cone. On the big cone volume is differentiable, with derivative in direction E equal to $d\langle D^{d-1} \rangle \cdot E$. At a big nef class D this is the usual intersection $dD^{d-1} \cdot E$.*

In particular, volume comparisons extend to real point orders. For a fixed big integral divisor we also use the complete-series limit [LM09, Theorem 2.3 and equation (2.6)].

Theorem 6.4. *If D is a big Cartier divisor on an irreducible projective variety Z of dimension d , then*

$$\lim_{k \rightarrow \infty} \frac{d!h^0(Z, \mathcal{O}_Z(kD))}{k^d} = \text{vol}_Z(D).$$

The next approximation applies at every sufficiently large generating degree p [LM09, Theorem 3.3 and Remark 3.4]. We distinguish this from the divisible-degree statement for a birational graded series [His13, Proposition 17].

Theorem 6.5 (Moving divisors of a full series). *For a big Cartier divisor D on an irreducible projective d -fold, set $V_{k,p} = \text{im}(\text{Sym}^k H^0(pD) \rightarrow H^0(kpD))$. For every $\epsilon > 0$, all sufficiently large p satisfy*

$$\lim_{k \rightarrow \infty} \frac{d! \dim V_{k,p}}{p^d k^d} \geq \text{vol}(D) - \epsilon.$$

For a birational graded series, the top self-intersections of its moving parts divided by the degree to the d -th power converge to its volume along sufficiently divisible degrees. This latter assertion alone gives no assertion at every original degree.

We will use its first assertion to compare the moving divisor of each original complete system, rather than infer an all-degree conclusion from the second assertion.

The valuation inequalities follow from the construction and controlled growth of asymptotic multiplier ideals [JM12, Section 2.3, Proposition 2.13, and Remark 6.3]. We give the derivation below. The computing-valuation assertion is [Xu20, Theorem 1.1 and the following paragraph].

Theorem 6.6 (Valuation approximation). *In the regular excellent characteristic-zero setting, let $\mathfrak{a}_\bullet$ be a graded ideal sequence and $\mathfrak{b}_\bullet$ its asymptotic multiplier sequence. For a valuation v of finite log discrepancy and an integer $k \geq 1$ with $\mathfrak{a}_k \neq 0$,*

$$v(\mathfrak{a}_\bullet) - A(v)/k \leq v(\mathfrak{b}_k)/k \leq v(\mathfrak{a}_\bullet) \leq v(\mathfrak{a}_k)/k.$$

These inequalities use the asymptotic order in the lower bound; they do not bound an arbitrary individual $v(\mathfrak{a}_k)/k$ within $A(v)/k$ of that order. A graded ideal sequence of finite log canonical threshold on a klt pair has a quasi-monomial computing valuation. Given a computing valuation, one may dominate it by such a computing valuation with the same discrepancy; this does not characterize every computing valuation as quasi-monomial.

Derivation of the valuation inequalities. We identify the asymptotic ideal order, prove the multiplier lower bound, and compare the two asymptotic orders.

Step 1. In this step, we identify the asymptotic order with the infimum over eligible degrees. Set $S = \{m \geq 1 : \mathfrak{a}_m \neq 0\}$ and $\alpha = \inf_{m \in S} v(\mathfrak{a}_m)/m$. The graded inclusions make $v(\mathfrak{a}_m)$ subadditive. For a fixed $p \in S$, choose one eligible representative in each residue class modulo p that meets S . Adding multiples of p to those representatives gives $\limsup_{m \rightarrow \infty, m \in S} v(\mathfrak{a}_m)/m \leq v(\mathfrak{a}_p)/p$. Taking the infimum over p proves that this limit is $\alpha = v(\mathfrak{a}_\bullet)$.

Step 2. In this step, we prove the multiplier lower bound for the given valuation. For each integer $k > 0$, choose a sufficiently divisible $m \in S$ for which $\mathfrak{b}_k = \mathcal{J}(\mathfrak{a}_m^{k/m})$. The multiplier-ideal formula on a log resolution gives, for every prime divisor E over the regular base,

$$\text{ord}_E(\mathfrak{b}_k) + A(\text{ord}_E) > \frac{k}{m} \text{ord}_E(\mathfrak{a}_m).$$

To pass to v , choose a common log resolution of $\mathfrak{a}_m \mathfrak{b}_k$, and include its exceptional divisors in the simple-normal-crossing boundary $D = \sum_i D_i$. The monomial retraction $\bar{v}$ with weights $v(D_i)$ has the same value as v on both ideals: their pullbacks are generated locally by monomials in these boundary equations. Moreover

$$A(\bar{v}) = \sum_i v(D_i) A(\text{ord}_{D_i}) \leq A(v),$$

by the definition of discrepancy for general valuations as the supremum over such retractions. Multiplying the divisorial inequalities by these nonnegative weights and adding gives

$$v(\mathfrak{b}_k) + A(v) \geq \frac{k}{m} v(\mathfrak{a}_m) \geq k\alpha.$$

The non-strict inequality also includes a trivial retraction. This proves the required lower bound without replacing the sufficiently divisible index m by k .

Step 3. In this step, we prove equality of the ideal and multiplier asymptotic orders. The construction gives $\mathfrak{a}_m \subseteq \mathfrak{b}_m$, and subadditivity of multiplier ideals gives $\mathfrak{b}_{p+q} \subseteq \mathfrak{b}_p \mathfrak{b}_q$. Thus $v(\mathfrak{b}_m)$ is superadditive and nonnegative, so

$$\gamma := \lim_{m \rightarrow \infty} \frac{v(\mathfrak{b}_m)}{m} = \sup_{m \geq 1} \frac{v(\mathfrak{b}_m)}{m}.$$

Indeed, division by any fixed index and use of the nonnegative remainder gives the lower limit at least the corresponding quotient; the upper limit is at most their supremum. Along $m \in S$, the ideal inclusion gives $\gamma \leq \alpha$. The lower bound just proved and $A(v) < \infty$ give $\gamma \geq \alpha$. Therefore

$$\frac{v(\mathfrak{b}_k)}{k} \leq \gamma = \alpha \leq \frac{v(\mathfrak{a}_k)}{k} \quad (k \in S),$$

which completes the chain. The remaining assertion is the cited computing-valuation theorem of Xu, with its stated domination property. $\square$

The individual-level middle comparison printed in [JM12, Proposition 6.2, equation (6.1)] does not hold for arbitrary graded sequences. For example, on the affine line let $\mathfrak{a}_m = (z^{m+2})$ and $v = \text{ord}_0$. Then $A(v) = 1$, $\alpha = 1$, and $\mathfrak{b}_k = (z^k)$: for sufficiently divisible $m > 2k$, the multiplier exponent is $\lfloor k(m+2)/m \rfloor = k$. Consequently $v(\mathfrak{a}_k)/k - A(v)/k = 1 + 1/k$ exceeds $v(\mathfrak{b}_k)/k = 1$. The derivation above establishes precisely the asymptotic-order comparison used here.

Existence of a suitable computing valuation does not identify all computing valuations. Our finite-source comparison below is proved directly for every divisorial valuation.

For ideal blowups we retain both the full exceptional divisor and the moving bundle [EL99, Proposition 3.1]. The local positivity assertion is [Pan23, Lemma 4.5].

Theorem 6.7 (Exceptional degrees of the full ideal). *Let T be a smooth complex projective variety, A a nef line bundle, and let a family of sections generate $\mathfrak{a} \otimes A$. On its normalized ideal blowup write the full exceptional divisor as $E = \sum r_i E_i$, with distinguished images Z_i . Then $\sum r_i \deg_A Z_i \leq A^{\dim T}$. The construction retains the moving bundle $\nu^* A - E$. For a log resolution of an ideal on a smooth variety, write $\mathfrak{a} \mathcal{O}_Y = \mathcal{O}_Y(-F)$. At the generic point η of a codimension- h component of its zero set,*

$$((-F)|_{E_\eta})^{h-1} \geq 0$$

for each prime E in F , with strict positivity exactly for the Rees valuations centered at η .

An exceptional component with zero tautological degree remains in the resolved ideal. The following vanishing theorem is used for specified coherent sheaves [Sta26, Lemma 30.17.1, Tag 0B5U].

Theorem 6.8 (Serre vanishing). *Let T be proper over a Noetherian affine scheme, A an ample invertible sheaf, and $\mathcal{F}$ a specified coherent sheaf. There is k_0 such that $H^i(T, \mathcal{F} \otimes A^k) = 0$ for every $k \geq k_0$ and $i > 0$.*

Its cutoff depends on the sheaf; uniformity for a varying source will be established separately. We also record the smooth primitive representative supplied by [DN06, Propositions 2.3 and 2.4].

Theorem 6.9 (Primitive Hodge representatives). *Let T be compact Kähler of dimension d , let $p, q \geq 1$ with $p + q \leq d$, and fix Kähler forms $\omega_1, \dots, \omega_{d-p-q+1}$. Put $\Omega = \omega_1 \wedge \dots \wedge \omega_{d-p-q}$. For a smooth closed (p, q) -form f with $[f][\Omega][\omega_{d-p-q+1}] = 0$, there is a smooth $(p-1, q-1)$ -form u satisfying $(f - dd^c u) \wedge \Omega \wedge \omega_{d-p-q+1} = 0$.*

The representative satisfies the primitive equation as a form. For graded centers the same graded sequence must enter both center conditions [Zhu21, Section 2.4 and Lemma 3.2].

Theorem 6.10 (Intersections for a graded source). *For a klt pair with one fixed graded ideal sequence, let Z_1, Z_2 be non-klt centers with nonempty intersection. If the pair with that same graded sequence is klt outside $Z_1 \cup Z_2$, their intersection is a union of non-klt centers of that sequence.*

The intersection assertion retains the condition away from the two centers. Finally, an order increase from a fixed prime uses the entire fixed-prime inequality [Kaw97, Theorem 2.2].

Theorem 6.11 (Order increase from an actual fixed prime). *Let T be normal projective of dimension $d \geq 2$, let x be smooth, and let A be ample Cartier. Let P be a prime divisor of order $r \geq 1$ at x , and $e, k > 0$ rational with $re \leq 1$ and $k^d < A^d$. Suppose there is an effective $\mathbb{Q}$ -divisor $D \sim_{\mathbb{Q}} A$ with $\text{ord}_x D \geq k$, and that every such divisor satisfies $D \geq ekP$. There is a real number λ with*

$$0 \leq \lambda < 1, \quad \lambda \leq \max\{1 - re, (re)^{-1/(d-1)}\}$$

such that every rational $k' > k$ satisfying

$$(\lambda k)^d + \left(\frac{1 - re - \lambda}{1 - \lambda} \right)^{d-1} \left[\left(k' + \frac{\lambda rek}{1 - re - \lambda} \right)^d - \left(\lambda k + \frac{\lambda rek}{1 - re - \lambda} \right)^d \right] < A^d$$

admits an effective $\mathbb{Q}$ -divisor $D' \sim_{\mathbb{Q}} A$ with $\text{ord}_x D' \geq k'$. At $\lambda = 1 - re$, the left-hand side is interpreted by its limiting value.

The prime in this statement belongs to the actual complete source, and its coefficient remains in the numerical comparison.

Lemma 6.12. *Let Y be a smooth complex projective variety of dimension d , and let H_0 be very ample Cartier. Set $B_0 = K_Y + (d+1)H_0$. Then*

$$\mathcal{O}_Y(Q + B_0) \otimes \mathcal{J}(\|Q\|)$$

is globally generated for every big integral Cartier divisor Q on Y .

The same B_0 works while Q varies over the entire lattice of degree and point order.

Proof. We obtain the required cohomology vanishing and then give the regularity argument that turns it into generation.

Step 1. In this step, we obtain the cohomology vanishings required for regularity. Fix a sufficiently divisible m computing $\mathcal{J}(\|Q\|)$. On a log resolution of the full base ideal of $|mQ|$, a sufficiently dilute average of general members has fixed divisor with coefficient $1/m$, and all moving coefficients are less than one. The simple-normal-crossing formula for multiplier ideals therefore gives an effective rational divisor $\Delta \sim_{\mathbb{Q}} Q$ with $\mathcal{J}(Y, \Delta) = \mathcal{J}(\|Q\|)$. For $1 \leq i \leq d$, apply Theorem 3.10 with $N = Q + (d + 1 - i)H_0$. Its difference from Δ is ample. Writing $\mathcal{E} = \mathcal{O}_Y(Q + B_0) \otimes \mathcal{J}(\|Q\|)$, we obtain

$$H^i(Y, \mathcal{E}(-iH_0)) = 0 \quad (i > 0).$$

Step 2. In this step, we deduce global generation from these vanishings. Embed Y by H_0 and regard $\mathcal{E}$ as a coherent sheaf on projective space. We recall why these vanishings imply generation. More generally, suppose a coherent sheaf $\mathcal{G}$ satisfies $H^i(\mathcal{G}(-i)) = 0$ for $i > 0$. Choose a hyperplane avoiding its associated points, so that multiplication by its equation ℓ is injective. The restriction sequence shows that $\mathcal{G}|_{(\ell=0)}$ satisfies the same vanishings. Induction on the support dimension, followed by induction on $j \geq 0$ in that sequence, gives $H^i(\mathcal{G}(j - i)) = 0$ for every $i > 0$. In particular restriction on global sections is surjective in every nonnegative twist. Multiplication by linear forms is surjective on the hyperplane by the dimension induction. Lift those forms and sections to projective space; the difference from any prescribed section of $\mathcal{G}(j + 1)$ is ℓ times a section of $\mathcal{G}(j)$. Hence

$$H^0(\mathcal{G}(j)) \otimes H^0(\mathcal{O}(1)) \longrightarrow H^0(\mathcal{G}(j + 1))$$

is surjective for all $j \geq 0$. The zero-dimensional base case follows by choosing a linear form nonvanishing on the finite support. A finite presentation by twists of $\mathcal{O}$ shows that $\mathcal{G}(j)$ is generated for large j . Iterating the displayed multiplication maps then shows that $\mathcal{G}$ itself is generated. Apply this to $\mathcal{E}$. $\square$

Theorem 6.13. *Let X be smooth projective of dimension n , let $a \geq 1$ be an integer, and suppose $a\pi^*L - \beta E_x$ is big for a rational $\beta > 0$. Fix $0 < b_0 < \beta$. For the complete spaces $V_k(t) = H^0(X, akL \otimes \mathfrak{m}_x^{\lceil kt \rceil})$, write $\mathfrak{a}_k(t)$ for their full base ideals. There are C, M_1, K and a finite set of prime divisors, fixed from (X, L, x, a, b_0, β) , such that every $V_k(t)$ with $k \geq K, 0 \leq t \leq \beta$ is nonzero. For every divisorial valuation w , with homogeneous log discrepancy $A_X(w)$,*

$$\alpha_w(t) = \lim_k \frac{w(\mathfrak{a}_k(t))}{k} = \inf_{V_k(t) \neq 0} \frac{w(\mathfrak{a}_k(t))}{k}$$

is finite, nondecreasing and convex. Simultaneously for $k \geq K, 0 \leq t, u \leq b_0$ and all w ,

$$0 \leq w(\mathfrak{a}_k(t)) - k\alpha_w(t) \leq CA_X(w), \quad |\alpha_w(t) - \alpha_w(u)| \leq M_1 A_X(w) |t - u|.$$

Write $F_k(t)$ for the full divisorial fixed part of $\mathfrak{a}_k(t)$. These divisors have the stated common finite support, and $F_k(t)/k$ converges coefficientwise uniformly to $N(t) = \sum_P \alpha_{\text{ord}_P}(t)P$, with coefficient errors at most C/k . Every actual global section $f \in V_j(t)$, $0 < t \leq b_0$, satisfies

$$w(f) \geq \frac{\text{ord}_x f}{t} \alpha_w(t).$$

For $0 < \epsilon \leq 1$, an effective lc boundary B , and $0 \leq c \leq 1 - \epsilon$, and $0 < t \leq b_0$, put

$$A_{cB}(w) = A_X(w) - cw(B), \quad \Lambda_{cB}(t) = \sup_{x \in cX(w)} \frac{\alpha_w(t)}{A_{cB}(w)}, \quad \tau = \text{lct}_x((X, cB); \mathfrak{a}_k(t)).$$

Then

$$0 \leq \frac{1}{k\tau} - \Lambda_{cB}(t) \leq \frac{C}{\epsilon k}.$$

For any place ρ computing this same threshold and any nonzero actual $f \in V_k(t)$,

$$\frac{\rho(f)}{A_{cB}(\rho)} \geq \frac{\text{ord}_x f}{t} \left(\frac{1}{k\tau} - \frac{C}{\epsilon k} \right).$$

These bounds apply to the actual global images on every subsequent ordinary point-order layer, as upper comparisons with the full valuation ideal. All constants precede the indicated choices.

The error is independent of the divisor computing a later threshold. The assertion also controls transient fixed components, whose asymptotic coefficients may be zero.

Proof. We first compare all integral degree–order pairs with one fixed multiplier-ideal twist. We then pass to real orders, retain all fixed primes, and compare the threshold of the actual source with its own computing places. In the relative assertion we take $0 < \epsilon \leq 1$; the Lipschitz comparison is for $t, u \in [0, b_0]$.

Step 1. In this step, we identify the complete systems and their asymptotic valuative orders. Let $\pi : Y = \text{Bl}_x X \rightarrow X$, let $H = \pi^* L$, and write $E = E_x$. In dimension one, $Y = X$ and $E = x$. For $\ell \geq 0$, the identity $\pi_* \mathcal{O}_Y(-\ell E) = \mathfrak{m}_x^\ell$ identifies the original complete source with the complete system of $kaH - \ell E$. If $\mathfrak{c}(k, \ell)$ is the full base ideal of that system on Y , evaluation gives

$$(6.1) \quad \mathfrak{a}_k(\ell/k) \mathcal{O}_Y = \mathcal{O}_Y(-\ell E) \mathfrak{c}(k, \ell).$$

Put $D_\beta = aH - \beta E$, and clear its denominator by an integer q . A big integral divisor D has a multiple $vD \sim A_0 + T_0$, where A_0 is ample integral and T_0 is effective. For any fixed integral twist R , writing $m = vj + i$ reduces nonvanishing of $|mD + R|$ to that of $|jA_0 + iD + R|$, followed by multiplication by the section of jT_0 . Ampleness gives nonvanishing for all large j , simultaneously over the finitely many residues i . Apply this to

$$kaH - [k\beta]E = m(qD_\beta) + iaH - [i\beta]E, \quad k = qm + i.$$

It proves nonvanishing for every sufficiently large original k at β , hence at every $t \leq \beta$.

For fixed real t , products of actual sections give $V_i(t)V_j(t) \subseteq V_{i+j}(t)$. Thus their minimum valuative orders are subadditive on eligible degrees. To include every such degree in the infimum, fix one eligible p_* and a large eligible r_i in each residue class modulo p_* . Then

$$w(\mathfrak{a}_{rp_*+r_i}(t)) \leq r w(\mathfrak{a}_{p_*}(t)) + w(\mathfrak{a}_{r_i}(t)).$$

After division and passage to the limit this bounds the limsup by $w(\mathfrak{a}_{p_*}(t))/p_*$. Taking the infimum over p_* proves the asserted limit and infimum formula. The same argument for any big integral Q on Y defines

$$\sigma_w(Q) = \lim_m \frac{w(\mathfrak{b}(|mQ|))}{m} = \inf_{|mQ| \neq \emptyset} \frac{w(\mathfrak{b}(|mQ|))}{m}.$$

In particular, for $Q = kaH - \ell E$ big,

$$(6.2) \quad \sigma_w(Q) = k\alpha_w(\ell/k) - \ell w(E).$$

Step 2. In this step, we establish one uniform comparison for all integral degree–order pairs. Fix the twist B_0 of Lemma 6.12 on Y . Choose one integer p , clearing β , and one effective divisor $T \in |pD_\beta - B_0|$. Its existence follows from the finite-twist argument above. Choose $C_T \geq 0$ with $w(T) \leq C_T A_Y(w)$ for every divisorial w . A log resolution of this one divisor supplies such a constant; take $C_T = 0$ if $T = 0$. By Theorem 6.2,

$$\mathfrak{b}(|mQ|) \subseteq \mathcal{J}(\|mQ\|) \subseteq \mathcal{J}(\|Q\|)^m.$$

Consequently $w(\mathcal{J}(\|Q\|)) \leq \sigma_w(Q)$.

Fix integers $k > p$ and $0 \leq \ell \leq k\beta$, and put

$$Q = kaH - \ell E, \quad Q' = Q - pD_\beta = (k-p)aH - (\ell-p\beta)E.$$

Both are big integral divisors. If $\ell \geq p\beta$, put $t' = (\ell - p\beta)/(k - p) \leq \ell/k$. Monotonicity of α_w , which follows from nesting the complete sources, and (6.2) imply

$$\sigma_w(Q') \leq \sigma_w(Q) + p\beta w(E).$$

If $\ell < p\beta$, sections of large free multiples of $(k - p)aL$, multiplied by the section of $(p\beta - \ell)E$, give $\sigma_w(Q') \leq (p\beta - \ell)w(E)$. The same displayed bound follows since $\sigma_w(Q) \geq 0$.

Multiply every generator supplied by Lemma 6.12 for Q' by the fixed section of T . These are actual sections of Q , so

$$\mathcal{O}_Y(-T)\mathcal{J}(\|Q'\|) \subseteq \mathfrak{c}(k, \ell).$$

Taking orders and using (6.1) gives

$$(6.3) \quad 0 \leq w(\mathfrak{a}_k(\ell/k)) - k\alpha_w(\ell/k) \leq w(T) + p\beta w(E).$$

The discrepancy formula on the point blowup is $A_Y(w) = A_X(w) - (n - 1)w(E)$. Moreover $w(E) = w(\mathfrak{m}_x) \leq A_X(w)/n$: the blowup resolves the point ideal and its log canonical threshold is n . Thus the right side of (6.3) is at most $(C_T + p\beta/n)A_X(w)$.

Step 3. In this step, we prove convexity and extend the lattice comparison to real orders. Products of powers of two actual sections, with their degrees chosen in a rational ratio, prove convexity of α_w for rational convex weights. For arbitrary real weights, approximate the resulting argument from above by rational weights and use monotonicity. This proves convexity on $[0, \beta]$. High ample multiples are free, so $\alpha_w(0) = 0$. Choose a nonzero section in one $V_j(\beta)$ and keep its entire divisor. A log resolution of that divisor fixes M_0 such that $\alpha_w(\beta) \leq M_0 A_X(w)$ for all w . For $b_1 = (b_0 + \beta)/2$ and $0 \leq u < t \leq b_1$, convexity gives

$$0 \leq \alpha_w(t) - \alpha_w(u) \leq \frac{M_0}{\beta - b_1} A_X(w)(t - u).$$

For large k , $t_k = \lceil kt \rceil/k$ lies in $[0, b_1]$ whenever $0 \leq t \leq b_0$, and the actual spaces at t and t_k coincide. Apply the lattice estimate at t_k and the Lipschitz estimate between t_k and t . Since $k(t_k - t) < 1$, we obtain

$$0 \leq w(\mathfrak{a}_k(t)) - k\alpha_w(t) \leq \left(C_T + \frac{p\beta}{n} + \frac{M_0}{\beta - b_1} \right) A_X(w).$$

The lower bound uses the infimum at the original real t . This fixes C, M_1, K before any later valuation or order.

Step 4. In this step, we control every fixed prime and every actual global section. Choose an eligible degree d , a nonzero section $f_0 \in V_d(\beta)$, and nonzero sections $f_i \in V_{e_i}(\beta)$ with one large e_i in each residue class modulo d . The divisors of $f_0^r f_i$ contain every fixed prime of $V_{rd+e_i}(\beta)$. Their supports lie in one finite union S , independent of r . Nesting the sources gives the same support for all $t \leq \beta$. For $P \in S$, $A_X(\text{ord}_P) = 1$; hence the uniform estimate gives

$$0 \leq \text{mult}_P(F_k(t)/k) - \alpha_{\text{ord}_P}(t) \leq C/k.$$

For $P \notin S$ the coefficient and the profile are zero. This proves the full fixed-part assertion, including every transient component.

Fix a nonzero actual $f \in V_j(t)$, with $0 < t \leq \beta$, and put $h = \text{ord}_x f \geq jt$. If $h = jt$, the infimum formula proves the section inequality. If $h > jt$, put $k_N = \lfloor Nh/t \rfloor$. For large N , the multiple $a(k_N - Nj)L$ is free. Choose a section g_N nonvanishing at x and at the generic point of $c_X(w)$. The actual global section $f^{k_N} g_N$ belongs to $V_{k_N}(t)$ and has valuation $Nw(f)$. Therefore

$$Nw(f) \geq k_N \alpha_w(t).$$

Division by N and passage to the limit gives $w(f) \geq (h/t)\alpha_w(t)$. This uses global padding sections and makes no assertion about arbitrary local multiples. The same argument applies to an actual section of any higher point order at this fixed comparison order t .

Step 5. In this step, we compare the retained pair's actual threshold with its computing profiles. For $x \in c_X(w)$, log canonicity and effectivity give

$$\epsilon A_X(w) \leq A_{cB}(w) \leq A_X(w).$$

The pair (X, cB) is klt near x , and $\mathfrak{a}_k(t)$ is a nonzero proper ideal there when $t > 0$. A common log resolution gives the positive finite threshold and its valuative expression

$$\frac{1}{k\tau} = \sup_{x \in c_X(w)} \frac{w(\mathfrak{a}_k(t))}{kA_{cB}(w)}.$$

Divide the uniform estimate by $kA_{cB}(w)$ and take this supremum. This proves the bound for $\Lambda_{cB}(t)$. At any place ρ computing this same threshold,

$$\frac{\alpha_\rho(t)}{A_{cB}(\rho)} \geq \frac{1}{k\tau} - \frac{C}{\epsilon k}.$$

The actual-section inequality now gives the claimed comparison for every nonzero $f \in V_k(t)$.

To make the last assertion precise, let $\ell \geq \lceil kt \rceil$ be any integer and set

$$R_\ell = \frac{\ell}{t} \max \left\{ 0, \frac{1}{k\tau} - \frac{C}{\epsilon k} \right\}, \quad \mathcal{I}_{\rho, R_\ell} = \{g \in \mathcal{O}_{X, x} : \rho(g) \geq A_{cB}(\rho) R_\ell\}.$$

Every actual section of degree ak and point order at least ℓ belongs to this full local ideal. Taking its degree ℓ initial form has kernel precisely the sections of point order at least $\ell + 1$. Thus the entire original quotient injects into the degree ℓ image of $\mathcal{I}_{\rho, R_\ell}$ in $\text{gr}_{\mathfrak{m}_x} \mathcal{O}_{X, x}$. This is an upper comparison of an existing global image with a local ideal, at the same chosen place and threshold. $\square$

Lemma 6.14. *Let Y be a smooth integral projective d -fold, and let D be a big integral Cartier divisor. For each nonempty $|pD|$, let $\mu_p : Y_p \rightarrow Y$ resolve its full base ideal and write $\mu_p^*(pD) = M_p + E_p$, with M_p generated by the complete system and E_p its full effective base divisor. Then M_p is big for every sufficiently large integer p , and*

$$\lim_{p \rightarrow \infty} \frac{M_p^d}{p^d} = \text{vol}_Y(D).$$

The limit concerns all large original degrees of this complete series.

Proof. For each r , the algebra generated by $H^0(Y, pD)$ injects in degree r into $H^0(Y_p, rM_p)$. Conversely, multiplying by the section of rE_p embeds the latter space in $H^0(Y, rpD)$. The first assertion of Theorem 6.5, together with Theorem 6.4, therefore gives, for every $\delta > 0$ and all sufficiently large p ,

$$\text{vol}_Y(D) - \delta \leq \frac{\text{vol}_{Y_p}(M_p)}{p^d} \leq \text{vol}_Y(D).$$

Choose $\delta < \text{vol}_Y(D)$. Then M_p is big; it is nef because it is generated. Its volume is M_p^d . One may read this identity from homogeneity and the derivative at a big nef class in Theorem 6.3. Letting δ decrease to zero proves the assertion. $\square$

Lemma 6.15. *Let Y be a smooth projective variety of dimension $d \geq 2$, and let A_0, A_1 be big nef real divisor classes. Put $s_i = A_0^{d-i} A_1^i$ for $0 \leq i \leq d$. Then all s_i are positive and*

$$s_i^2 \geq s_{i-1} s_{i+1} \quad (1 \leq i < d), \quad s_1 \geq s_0^{(d-1)/d} s_d^{1/d}.$$

These comparisons use the two nef divisors on the same model.

Proof. On a smooth projective surface, apply Theorem 6.3 to $((A_0 + tA_1)^2)^{1/2}$. Its second derivative is nonpositive, which gives $(A_0 A_1)^2 \geq A_0^2 A_1^2$. For ample rational classes on a d -fold, restrict to a general smooth complete intersection of $d - i - 1$ sufficiently divisible members of A_0 and $i - 1$ such members of A_1 . The surface inequality gives $s_i^2 \geq s_{i-1} s_{i+1}$, after canceling the degrees of the chosen multiples. Successive ratios are nonincreasing, so $\log s_i$ is concave

and lies above the chord joining $\log s_0$ to $\log s_d$. Finally approximate big nef real classes by ample rational classes. The chord bound gives positive lower limits for every mixed product, and continuity preserves all the inequalities. $\square$

In the intrinsic argument a reached order means a positive rational order with positive complete-source volume. The primes under consideration pass through x ; normality near x makes their generic orders well-defined even when W is nonnormal elsewhere.

Theorem 6.16. *Let W be an integral complex projective variety of dimension $d \geq 2$, normal near x , with every power of $\mathfrak{m}_{W,x}$ integrally closed. Let A be ample Cartier, $v = A^d$, and let P be a prime Weil divisor through x . For rational $t \geq 0$, set*

$$F_p(t) = H^0(W, pA \otimes I_x^{[pt]}), \quad \phi_P(t) = \inf_{F_p(t) \neq 0} \frac{\text{ord}_P F_p(t)}{p}, \quad w(t) = \lim_{p \text{ divisible}} \frac{d! \dim F_p(t)}{p^d}.$$

The volume w has a continuous real extension. For every fixed rational $R_0 > 0$ with $w(R_0) > 0$, there are a cutoff p_0 , an effective Weil divisor Z_0 , and a finite set of primes through x , fixed before p, j, P , such that for integers $p \geq p_0$ and $0 \leq j \leq pR_0$, and every such P , the spaces are nonzero and

$$0 \leq \text{ord}_P F_p(j/p) - p\phi_P(j/p) \leq \text{mult}_P Z_0.$$

Every fixed prime through x belongs to that finite set. The weighted sum of all these errors is at most $A^{d-1} \cdot Z_0$. For finitely many distinct primes, with $g_P = A^{d-1} \cdot P$, take positive rational reached orders $r_{P,i}$ with $w(r_{P,i}) > 0$, heights $h_{P,i} \leq \phi_P(r_{P,i})$, and finite slopes $\lambda_{P,i}$ such that each affine function $h_{P,i} + \lambda_{P,i}(t - r_{P,i})$ supports a lower bound for $\phi_P(t)$. Put

$$T_P(t) = \max(0, \max_i \{h_{P,i} + \lambda_{P,i}(t - r_{P,i})\}).$$

Then for every $t \geq 0$,

$$w(t) \leq v \left(1 - v^{-1} \sum_P g_P T_P(t) \right)_+^d, \quad \sum_P g_P \phi_P(R_0) \leq v - v^{(d-1)/d} w(R_0)^{1/d} < v.$$

The last sum includes every prime through x . Every exact-order original section obeys the simultaneous orders $pT_P(j/p)$. The normalization conductor and all original descent conditions are retained.

The divisor Z_0 bounds the total weighted error of every finite prime history. Its bound is independent of the number of observed primes.

Proof. We compare the original sources with normalization only through their actual conductor, construct one fixed error divisor on a reached interval, and then apply the moving-volume comparison to all retained fixed primes at once.

Step 1. In this step, we compare normalized sections with the original sections while retaining conductor descent. Let $\nu : \bar{W} \rightarrow W$ be the finite normalization and let $\mu : Y \rightarrow \bar{W}$ be a smooth projective resolution that principalizes the ideal of x . Write

$$I_x \mathcal{O}_Y = \mathcal{O}_Y(-F), \quad H = \mu^* \nu^* A.$$

The point has a unique lift, since ν is an isomorphism near x . The divisor F is supported over that point. Pushforward of $\mathcal{O}_Y(-jF)$ computes the integral closure of the j -th point-ideal power: on the normalized ideal blowup, regularity of a section means precisely nonnegative order along every prime, and further resolution does not change this condition. The assumed integral closedness therefore gives

$$\mu_* \mathcal{O}_Y(-jF) = I_{x, \bar{W}}^j \quad (j \geq 0).$$

Away from x both ideals are the unit ideal. Moreover H is nef and big, and $H^d = v$.

Let $\mathfrak{c} \subset \mathcal{O}_{\bar{W}}$ be the actual normalization conductor. It is the unit ideal near x . For sufficiently large a_0 , $\mathfrak{c} \otimes \mathcal{O}_{\bar{W}}(a_0 \nu^* A)$ is generated. Choose a section nonvanishing at x , with effective Cartier

divisor Z . Then Z avoids x and $\mathcal{O}_{\overline{W}}(-Z) \subseteq \mathfrak{c}$. Vanishing along Z will ensure descent of the sections we construct.

The coherent quotient $\mathcal{Q} = \nu_* \mathcal{O}_{\overline{W}} / \mathcal{O}_W$ is supported away from x , in dimension at most $d - 1$. For every p, j , the quotient of the normalized and original point-constrained sheaves is exactly $\mathcal{Q} \otimes \mathcal{O}_W(pA)$. Consequently

$$(6.4) \quad 0 \leq h^0(Y, pH - jF) - \dim F_p(j/p) \leq h^0(W, \mathcal{Q} \otimes \mathcal{O}_W(pA)) = O(p^{d-1}).$$

The constant is independent of j . For rational $t = a/q$, apply the complete-series volume limit to $qH - aF$ when it is big. If its volume is zero, the nonnegative normalized dimensions have limsup zero and hence limit zero. Formula (6.4) thus identifies $w(t) = \text{vol}_Y(H - tF)$ at all rational orders. Theorem 6.3 gives its continuous real extension. This is a dimension comparison; the original finite source still satisfies all its descent conditions.

Step 2. In this step, we construct one error divisor valid on the whole reached interval. Let $\psi_P(t)$ be the infimum defining $\phi_P(t)$, using normalized sections instead. Products and nested point ideals make it nonnegative, nondecreasing, and convex on its rational finite domain, with $\psi_P(0) = 0$. For a big integral $C = eH - jF$, the residue-class argument used in the preceding theorem gives

$$\sigma_P(C) := \lim_m \frac{\text{ord}_{P_Y} \mathfrak{b}(|mC|)}{m} = e\psi_P(j/e),$$

where P_Y is the strict transform of P . Indeed, the e -th power of any eligible normalized degree- p section is a section of pC , and the reverse inclusion is immediate. The divisorial order of F at P_Y is zero, since $d \geq 2$. The asymptotic ideal inclusions give $\text{ord}_{P_Y} \mathcal{J}(\|C\|) \leq \sigma_P(C)$.

Fix R_0 as in the statement. Choose the single twist B_0 of Lemma 6.12 on Y . Bigness of $H - R_0F$ permits a fixed positive integer c_0 , with $b_* = c_0R_0$ integral, and a nonzero section of $c_0H - b_*F - B_0 - \mu^*Z$. Viewed as a section of $c_0H - b_*F - B_0$, it has divisor $T \geq \mu^*Z$. These choices precede p, j, P . For arbitrary $p \geq c_0 + 1$ and $0 \leq j \leq pR_0$, put

$$j' = \max\{j - b_*, 0\}, \quad C = (p - c_0)H - j'F, \quad t' = \frac{j'}{p - c_0}, \quad t = \frac{j}{p}.$$

We have $0 \leq t' \leq t \leq R_0$, so C is big. Multiply the globally generating sections for $\mathcal{O}_Y(C + B_0) \otimes \mathcal{J}(\|C\|)$ by the fixed section of T . As sections of $pH - jF$, their evaluation ideal is

$$(6.5) \quad \mathcal{O}_Y(-T - (j' + b_* - j)F) \mathcal{J}(\|C\|).$$

It is nonzero. The sections vanish along μ^*Z , hence push forward into the actual conductor and descend to W . Their order at x remains at least j , by the point-power identity above. They are thus actual sections of $F_p(j/p)$.

At P_Y , the extra multiple of F in (6.5) has order zero. Global generation and monotonicity yield

$$p\psi_P(t) \leq \text{ord}_P F_p(t) \leq \text{ord}_{P_Y} T + (p - c_0)\psi_P(t') \leq \text{ord}_{P_Y} T + p\psi_P(t).$$

For fixed rational $t \leq R_0$, let p tend to infinity through multiples of its denominator. The infimum definition and inclusion of original sections in normalized sections show $\phi_P(t) = \psi_P(t)$. Set

$$p_0 = c_0 + 1, \quad Z_0 = \nu_* \mu_* T.$$

The maps are isomorphisms at the generic point of any prime through x , so $\text{mult}_P Z_0 = \text{ord}_{P_Y} T$. This proves the all-order, all-original-degree estimate. Proper pushforward keeps an effective Weil divisor; components with smaller-dimensional image have zero pushforward. Summing the prime errors against the positive degrees g_P gives the single bound $A^{d-1} \cdot Z_0$.

Choose one nonzero original section $h_0 \in F_{p_*}(R_0)$ at an eligible degree. The primes through x in its divisor, together with those in Z_0 , form a finite set. Outside this set $\phi_P(R_0) = 0$, and monotonicity and the error estimate force $\text{ord}_P F_p(j/p) = 0$ throughout the reached interval. This proves the common finite support, including primes with zero asymptotic coefficient.

Step 3. In this step, we combine all retained prime observations in the volume comparison. Products of original sections make ϕ_P convex on its rational finite domain. A positive-volume order is interior to that domain, since bigness is open. Thus it admits a finite supporting slope. Ordered secant slopes extend its supporting inequality to every other finite rational order, including an attained endpoint of zero volume. Since $\phi_P(0) = 0$, each supporting line through $(r, \phi_P(r))$ has slope at least $\phi_P(r)/r \geq 0$; lowering its height gives one admissible observation. More generally, the affine functions in the statement lie below ϕ_P by hypothesis. Their values at zero are at most $\phi_P(0) = 0$. Taking their maximum together with zero preserves the lower bound at every attained rational order. Every nonzero original degree- p section of exact order j satisfies

$$\text{ord}_P(f) \geq p\phi_P(j/p) \geq pT_P(j/p)$$

simultaneously for all selected primes. Repeated observations enter by a maximum for that prime.

Fix a rational $t = a/q$ of positive volume and a finite set of distinct primes. Resolve the full base ideal of $|\ell(qH - aF)|$ on Y . With $p = q\ell$, write its moving divisor as

$$M_\ell = pH_\ell - a\ell F_\ell - G_\ell,$$

where H_ℓ, F_ℓ are the pullbacks and G_ℓ is its entire effective base divisor. The coefficient at each strict P is at least $p\phi_P(t)$: equality of normalized and original profiles was proved on any fixed positive-volume interval. All other fixed coefficients stay nonnegative. Projection formula gives

$$(6.6) \quad M_\ell H_\ell^{d-1} \leq p \left(v - \sum_P g_P \phi_P(t) \right).$$

For large ℓ , Lemmas 6.14 and 6.15 give

$$\frac{M_\ell^d}{p^d} \rightarrow w(t), \quad M_\ell H_\ell^{d-1} \geq (M_\ell^d)^{1/d} v^{(d-1)/d}.$$

Combine these with (6.6) and pass to the limit. The bracket is positive when $w(t) > 0$, and replacing $\phi_P(t)$ by the smaller $T_P(t)$ proves the claimed volume upper bound. If the volume is zero the positive-part bound is automatic. Continuity of volume and of the finite maxima T_P extends it to all real $t \geq 0$.

Finally take $t = R_0$ and every prime with nonzero $\phi_P(R_0)$. Their set is finite, as it lies in the divisor of h_0 . Applying the same mixed-intersection comparison to this entire set gives

$$\sum_P g_P \phi_P(R_0) \leq v - v^{(d-1)/d} w(R_0)^{1/d} < v.$$

All omitted profiles are zero. The construction has retained the full conductor, full fixed divisor, and original section spaces at every finite degree. $\square$

Lemma 6.17. *Let R be the regular local ring at a codimension-two generic point of a smooth characteristic-zero variety. Let $J \subset R$ be a nonzero ideal of finite colength, and let $\lambda = \text{lct}_R(J)$. Then*

$$e(J)\lambda^2 \geq 4.$$

The multiplicity here is that of the whole primary ideal.

Proof. We compare colengths of powers with monomial ideals, and then take the Hilbert–Samuel limit. Completion and extension of the residue field preserve colength and multiplicity. They also preserve the threshold: base change of a log resolution preserves the coefficients of the resolved ideal and relative canonical divisor. We may therefore work in $K[[z_1, z_2]]$, with K algebraically closed of characteristic zero. Every ideal used has finite colength, so it is determined modulo a power of (z_1, z_2) and can be represented by a polynomial ideal.

Step 1. In this step, we compare the thresholds and colengths of J^r and its monomial degeneration. Fix an integer $r \geq 1$. A monomial initial degeneration of J^r preserves colength and produces an ideal J'_r primary to (z_1, z_2) . We have

$$\text{lct}(J'_r) \leq \text{lct}(J^r) = \lambda/r.$$

Indeed, the weight family is in the smooth three-dimensional space with coordinates z_1, z_2, s . Localize at the origin of its special fiber. Localization at the generic point of the horizontal point section can only increase the threshold and gives λ/r , since the nonzero fibers are diagonal transforms of J^r . Restriction to the smooth divisor $s = 0$, by Theorem 6.1, can only decrease the threshold. These are comparisons of the full ideals in the flat family.

Step 2. In this step, we bound the colength of a monomial primary ideal using its computing Newton facet. For a monomial primary ideal K_0 , subdivide the normal fan of its Newton polygon into regular cones. The associated toric charts principalize K_0 . A primitive positive ray (a, b) has log discrepancy $a + b$ and ideal order $\min_{z^\alpha \in K_0} (a\alpha_1 + b\alpha_2)$. The simple-normal-crossing criterion on this resolution computes the threshold by one such ray. The coordinate rays have ideal order zero because K_0 is primary. Normalizing the computing ray gives $p_1, p_2 > 0$, $p_1 + p_2 = 1$, such that

$$p_1\alpha_1 + p_2\alpha_2 \geq \eta \quad (z^\alpha \in K_0), \quad \eta = \frac{1}{\text{lct}(K_0)}.$$

Every point in the triangle $\{(x_1, x_2) \in \mathbb{R}_{\geq 0}^2 : p_1x_1 + p_2x_2 < \eta\}$ lies in a unit square based at an exponent absent from K_0 : taking coordinatewise floors can only decrease the weighted sum. The area of this triangle is therefore bounded by the number of absent monomials. Hence

$$\text{length}(R/K_0) \geq \frac{\eta^2}{2p_1p_2} \geq 2\eta^2.$$

Step 3. In this step, we pass from the colength bounds to the Hilbert–Samuel multiplicity. Apply this to J'_r . Colength preservation and the threshold comparison give

$$\text{length}(R/J^r) \geq \frac{2r^2}{\lambda^2}.$$

Divide by $r^2/2$ and pass to the Hilbert–Samuel limit. □

Definition 6.18. For a positive rational point order u , an *allowed degree* is an integer $k \geq 1$ with $ku \in \mathbb{Z}$. At such a degree with $U_k(u) \neq 0$, set

$$I_k = \mathfrak{b}(U_k(u)), \quad \tau = \text{lct}_x(I_k).$$

The divisor P_k is the entire divisorial fixed part of this complete original source.

Proposition 6.19. Suppose $\pi_x^*L - uE_x$ is big on a smooth n -fold, $n \geq 3$. At an allowed degree k , principalize the full original source ideal I_k and its residual ideal $I_k\mathcal{O}_X(P_k)$, where P_k is its entire fixed divisor. On a common smooth resolution write

$$F_k = \pi^*P_k + G_k, \quad M_k = k\pi^*L - F_k, \quad z_k = M_k^n / (k^n L^n).$$

For all sufficiently late allowed degrees, M_k is generated and big, $M_k^n/k^n \rightarrow V(u)$, and

$$\frac{P_k \cdot L^{n-1}}{k} \leq L^n(1 - z_k^{1/n}).$$

If d_k is the sum of the Hilbert–Samuel multiplicities of the full residual ideal at its codimension-two components, weighted by their L -degrees, then

$$\frac{d_k}{k^2} \leq L^n(1 - z_k^{1/(n-1)}).$$

For any distinct codimension-two computing centers W_i of the same I_k, τ , with corresponding places w_i , put $h_k = 1/(k\tau)$ and $\rho_i = w_i(P_k)/(kA_X(w_i))$. Then

$$\sum_i (L^{n-2} \cdot W_i) (h_k - \rho_i)_+^2 \leq d_k/(4k^2).$$

The residual ideal measures the remaining codimension-two contribution. Its removed divisor is still charged in each ρ_i , and the threshold remains that of I_k .

Proof. We first identify the moving-volume limit at every allowed degree. We then keep the two successive intersection losses, and finally charge each distinct computing center to its own residual multiplicity.

Step 1. In this step, we obtain the moving-volume limit for the original complete system. Let q be the denominator of u in lowest terms. Then the allowed degrees are exactly $k = q\ell$, and the source is the complete space of $\ell q(\pi_x^* L - uE_x)$ on the point blowup. The resolved moving bundle M_k is generated by precisely these original sections. Lemma 6.14 gives bigness for every sufficiently large allowed k and

$$\frac{M_k^n}{k^n} \longrightarrow V(u).$$

The imposed point divisor and every further fixed component are included in F_k .

Step 2. In this step, we compute the fixed-divisor and residual intersection losses. Fix such a k , and put

$$A_k = k\pi^* L, \quad s_j = A_k^{n-j} M_k^j \quad (0 \leq j \leq n), \quad C_k = I_k \mathcal{O}_X(P_k).$$

Both A_k and M_k are big and nef. The effective divisor $F_k = A_k - M_k$ gives

$$s_j - s_{j+1} = A_k^{n-j-1} F_k M_k^j \geq 0.$$

The ideal C_k has no codimension-one component. Smoothness of X makes P_k Cartier; its complete removal gives the exact equality $I_k = \mathcal{O}_X(-P_k)C_k$. Since $\pi_* G_k = 0$ as a divisor, projection formula yields

$$(6.7) \quad s_0 - s_1 = k^{n-1}(P_k \cdot L^{n-1}).$$

For the next difference, retain both terms:

$$(6.8) \quad s_1 - s_2 = A_k^{n-2} \pi^* P_k M_k + A_k^{n-2} G_k M_k.$$

The first term is nonnegative.

At a codimension-two component T of C_k , set $R_T = \mathcal{O}_{X, \eta_T}$ and $J_T = C_k R_T$. This is a primary ideal in a regular local surface: a height-one prime over it would be a divisorial component of C_k on X . On the ordinary ideal blowup $\text{Proj } R_T[J_T t]$, its full exceptional Cartier scheme E_T has tautological bundle $\mathcal{O}(-E_T)$. For all sufficiently large m , the definition of Proj and relative Serre vanishing identify

$$\text{length}(J_T^m / J_T^{m+1}) = \chi(E_T, \mathcal{O}_{E_T}(-mE_T)).$$

The left side has leading coefficient $e(J_T)$; the right side has leading coefficient $E_T \cdot (-E_T)$, including all scheme multiplicities. Thus these coefficients are equal. On the common resolution, the restrictions of P_k and L to $\text{Spec } R_T$ are trivial, so M_k restricts to $\mathcal{O}(-G_k)$. Projection formula therefore identifies the coefficient of T in $\pi_*(G_k M_k)$ with $e(J_T)$.

Intersect G_k with one sufficiently general member of the generated system M_k . This gives an effective cycle and treats its finitely many components simultaneously. Any remaining terms have nonnegative L -degree. Consequently (6.8) gives

$$(6.9) \quad s_1 - s_2 \geq k^{n-2} d_k.$$

This argument allows $T \subset P_k$.

By Lemma 6.15, the positive ratios s_j/s_{j-1} are nonincreasing. In particular $s_n/s_0 \leq (s_1/s_0)^n$. Combine this with (6.7) and $s_n/s_0 = z_k$, $s_0 = k^n L^n$, to obtain

$$\frac{P_k \cdot L^{n-1}}{k} \leq L^n(1 - z_k^{1/n}).$$

For the residual estimate put $D_k = k^{n-2}d_k$. Since $s_2 > 0$, (6.9) gives $D_k < s_1$, and the same ratio comparison yields

$$s_n \leq s_1(s_2/s_1)^{n-1} \leq s_1(1 - D_k/s_1)^{n-1} \leq s_0(1 - D_k/s_0)^{n-1}.$$

For the last inequality, the function $x(1 - D_k/x)^{n-1}$ is increasing for $x > D_k$: its derivative is

$$(1 - D_k/x)^{n-2}(1 + (n-2)D_k/x) > 0.$$

Solving the displayed comparison for d_k proves the second degree bound.

Step 3. In this step, we charge each distinct computing center to its residual multiplicity. For a place w_i in the statement, the exact full-ideal factorization gives

$$\frac{w_i(C_k)}{A_X(w_i)} = \frac{w_i(I_k) - w_i(P_k)}{A_X(w_i)} = k(h_k - \rho_i).$$

If C_k is the unit ideal at η_{W_i} , this is zero. Otherwise W_i is one of its codimension-two components and $C_k \mathcal{O}_{X, \eta_{W_i}}$ is primary. Localizing the same place preserves its discrepancy and orders. Its generic threshold λ_i therefore satisfies

$$\lambda_i \leq \frac{A_X(w_i)}{w_i(C_k)}.$$

Lemma 6.17 gives

$$e(C_k \mathcal{O}_{X, \eta_{W_i}}) \geq 4k^2(h_k - \rho_i)^2.$$

The centers W_i are distinct, so these are distinct summands in d_k . Multiply by their original L^{n-2} -degrees, sum, and divide by $4k^2$. The unit-ideal cases contribute zero. This proves the last assertion while keeping every coefficient of P_k in the same-source threshold calculation. $\square$

7. FULL LOCAL IMAGES AND AMBIENT CAPACITY

We retain the source and boundary conventions of Section 2. We first bound the entire ordinary initial image of a primary ideal. The bounds depend on its generic threshold and, in the refined cases, on its full tangent ring. We then apply them to the actual point-order images of one complete global source. Ordinary local images are taken at the closed point x , with residue field $\mathbb{C}$; generic-center localizations are indicated by a prime subscript. Throughout the local estimates, $n \geq 2$ and $1 \leq c \leq n-1$.

Theorem 7.1 (Monomial thresholds). *For a monomial ideal I on a smooth affine space and $c > 0$, the multiplier ideal $\mathcal{J}(I^c)$ is generated by those monomials x^u for which $u + (1, \dots, 1)$ lies in the interior of $c \text{Newt}(I)$. In particular its log canonical threshold is the reciprocal of the first intersection parameter of the positive diagonal with the Newton polyhedron. The threshold formula is unchanged by the characteristic-zero coefficient-field extension and completion used at a generic center.*

This is Howald's formula [How01, Main Theorem]. The coefficient-field and completion passages are the ones used in [dFEM02, proof of Theorem 1.1]; a toric resolution preserves the relevant ideal orders and discrepancies in these passages.

Theorem 7.2 (Thresholds of sums of ideals). *For coherent ideals I, J on a smooth complex variety, $\text{lct}_x(I + J) \leq \text{lct}_x(I) + \text{lct}_x(J)$. In a regular local ring of dimension d , the finite primary approximation consequently satisfies $\text{lct}(I + \mathfrak{m}^N) \leq \text{lct}(I) + d/N$. The generic coefficient-field passage in its use is part of the application, and the chosen approximation retains its strict margin.*

We use the sum inequality in [DK01, Theorem 2.9]. The approximation adds the entire ordinary power of the maximal ideal.

Theorem 7.3 (Primary threshold and multiplicity). *For an ideal I of finite colength in a regular local ring at a codimension- c generic point on a smooth characteristic-zero variety,*

$$e(I) \operatorname{lct}(I)^c \geq c^c.$$

For a finite-colength ideal in a convergent regular c -variable local ring, let $e_i(I)$ be its mixed multiplicities with the maximal ideal, with $e_0 = 1$. Then $\sum_{i=0}^{c-1} e_i(I)/e_{i+1}(I) \leq \operatorname{lct}(I)$. For two finite-colength ideals $I \subseteq J$ in a quasi-unmixed local ring, this sum for I is at most the sum for J ; equality holds exactly when their integral closures agree.

The first assertion is [dFEM02, Theorem 1.2]. For the mixed-multiplicity assertions see [BA17, Remark 7, equation (11), and Corollary 11]. These statements measure the complete primary scheme.

Theorem 7.4 (Homogeneous threshold equality). *Let $V \subseteq \mathbb{A}^d$ be defined by a homogeneous ideal generated in a single degree r . Put $c = \operatorname{lct}(\mathbb{A}^d, V)$ and let e be the codimension of the non-lt locus of $(\mathbb{A}^d, cV)$. Then $c \geq e/r$. If equality holds, that locus is linear and V is pulled back from its quotient, where the induced pair has the corresponding threshold and an isolated non-lt locus.*

The single generating degree is essential in [dFEM03, Theorem 0.2].

Theorem 7.5 (Concavity for coordinate ideals). *Let μ have an unconditional log-concave density on $\mathbb{R}^d$, and let A, B be unconditional measurable sets closed under coordinate contractions. For $0 \leq t \leq 1$,*

$$\mu(tA + (1-t)B)^{1/d} \geq t\mu(A)^{1/d} + (1-t)\mu(B)^{1/d}.$$

Here coordinate contractions include sign changes. This is the class of sets in [LMNZ17, Definitions 1–2 and Theorem 4].

Theorem 7.6 (The exceptional degree of a local ideal). *For a proper ideal I in a Noetherian local ring of dimension $d \geq 1$ $(R, \mathfrak{n})$, its j -multiplicity is the tautological $(d-1)$ -degree, with scheme multiplicities, of the exceptional components of $\operatorname{Bl}_I \operatorname{Spec} R$ supported over $\mathfrak{n}$. Equivalently it is the leading Hilbert coefficient of $H_{\mathfrak{n}}^0(\operatorname{gr}_I R)$ in dimension d . In the equidimensional setting its positivity is equivalent to I having maximal analytic spread.*

We use the degree interpretation in [AM06, Section 3.1] and the torsion definition and positivity criterion in [NU10, Lemmas 2.1 and 3.1]. In particular, a component of zero exceptional degree supplies no positive charge.

Definition 7.7. For a uniform point $Z = (Z_1, \dots, Z_n)$ on the simplex $Z_i \geq 0, \sum Z_i = 1$, define

$$\mathfrak{M}_{n,c}(u) = \max_{\substack{a_i \geq 0 \\ \sum_{i=1}^c a_i = 1}} \Pr\left(\sum_{i=1}^c a_i Z_i \geq u\right).$$

Set

$$B_{n,c}(u) = \begin{cases} 1 - (ncu/(c+1))^c, & 0 \leq u \leq 1/n, \\ 1 - \left(\frac{c(n-1) + nu - 1}{(c+1)(n-1)}\right)^c, & 1/n \leq u \leq 1, \\ 0, & u \geq 1. \end{cases}$$

The sharper envelope $E_{n,c}$ is $B_{n,c}$ on $[0, 1/n]$, the minimum of $B_{n,c}$ and $(n/(n-1))^{n-1}(1-u)^{n-1}$ on $(1/n, 1/2)$, and $(1-u)_+^{n-1}$ on $[1/2, \infty)$. For $c = 1$ use $(1-u)_+^{n-1}$ everywhere.

The unused simplex coordinates record the positive-dimensional support. For a fixed normalized weight vector, its linear form is nonconstant, so each level has measure zero. Dominated convergence on the compact weight simplex shows that the maximum defining $\mathfrak{M}_{n,c}$ exists and is continuous in u . It is nonincreasing, equals one at zero, and vanishes for $u \geq 1$.

Lemma 7.8. *Let $(R, \mathfrak{m})$ be a smooth complex local ring of dimension $n \geq 2$, let P have height c , $1 \leq c < n$, and let I be a proper P -primary ideal. Put $\lambda = \text{lct}_{R_P}(IR_P)$. Then, for every integer $\ell \geq 1$,*

$$\dim_{\mathbb{C}}(\text{in}_{\mathfrak{m}} I)_{\ell} \leq \frac{\ell^{n-1}}{(n-1)!} \mathfrak{m}_{n,c} \left(\frac{1}{\ell\lambda} \right) + C_n(\ell+1)^{n-2},$$

where C_n depends only on n .

The generic localization supplies a constraint on every monomial of a flat initial ideal; its Hilbert function remains the full one.

Proof. We specialize the entire ideals twice, transfer the generic threshold, and count the resulting halfspace with a uniform boundary error.

Step 1. In this step, we transfer the generic threshold bound to the full ordinary tangent specialization. Choose $N \geq 1$ with $P^N \subseteq I \subseteq P$, and put $G = \text{gr}_{\mathfrak{m}} R$, $Q = \text{in}_{\mathfrak{m}} P$, and $J = \text{in}_{\mathfrak{m}} I$. Contract the generic ideals in the extended Rees algebra

$$\mathcal{R} = \sum_{j \in \mathbb{Z}} \mathfrak{m}^j z^{-j} \subseteq R[z, z^{-1}], \quad \mathfrak{m}^j = R \quad (j \leq 0).$$

Their quotients have no z -torsion. The support contraction is prime, and the special ideals satisfy $Q^N \subseteq J \subseteq Q$. Let $\mathfrak{p}$ be any height- c minimal prime of Q . Such a prime exists since $\dim G/Q = n - c$. At its inverse image in $\mathcal{R}$, the quotient by z is the regular local ring $G_{\mathfrak{p}}$ of dimension c . Lifting its parameters and adjoining the nonzerodivisor z proves that the total local ring is regular of dimension $c + 1$.

By Theorem 6.1, restriction to this smooth parameter divisor can only decrease the threshold. Indeed, if the restricted multiplier ideal is the unit ideal, the restriction inclusion and Nakayama's lemma make the ambient multiplier ideal a unit. Localization to the horizontal support prime can only increase the threshold. That localization is R_P after a transcendental coefficient extension. A log resolution base changes with the same orders and discrepancies, so

$$\text{lct}_{G_{\mathfrak{p}}}(JG_{\mathfrak{p}}) \leq \lambda.$$

The power containment ensures that the restricted ideal is nonzero and primary. This argument applies at every height- c component.

Step 2. In this step, we obtain a necessary Newton halfspace for the full monomial specialization. Choose one integral weight realizing homogeneous Gröbner bases for both J and Q . Adding the same integer to all weights makes them positive without changing their homogeneous initials. Write K, H for the full monomial initials. Their Hilbert functions are preserved and $H^N \subseteq K \subseteq H$. Choose a height- c coordinate minimal prime $\mathfrak{p}_0 = (x_1, \dots, x_c)$ of H . In the weight family, any minimal support prime contained in $(z, \mathfrak{p}_0)$ avoids z by flatness. Its height is at least c , by the dimension of the generic support, and at most c , by strict containment in this height- $c + 1$ prime. It therefore comes from a generic component already controlled above. Restriction in the regular total space gives

$$\text{lct}_{G_{\mathfrak{p}_0}}(KG_{\mathfrak{p}_0}) \leq \lambda.$$

The variables outside $\mathfrak{p}_0$ are units. The resulting monomial ideal contains a pure power of every normal variable. Theorem 7.1 supplies a computing compact Newton facet with positive normal weights $a_1, \dots, a_c$, normalized by $\sum a_i = 1$. A zero normal weight would contradict the pure power in that coordinate. Every monomial x^{ν} of the full ideal K consequently satisfies

$$\sum_{i=1}^c a_i \nu_i \geq \frac{1}{\lambda}.$$

The other $n - c$ coordinates are free in this inequality.

Step 3. In this step, we count the halfspace with a dimension-uniform lattice error. Eliminate a free coordinate from $\sum \nu_i = \ell$. The comparison region is an $(n-1)$ -simplex of volume $\ell^{n-1}/(n-1)!$,

cut by one halfspace. Its volume fraction is bounded by the stated capacity. The difference between its lattice count and its volume is bounded by the unit cubes meeting its boundary. For each boundary hyperplane, project along a largest absolute nonzero coefficient. The remaining slopes have absolute values at most one. Over each projected unit cube, only a number depending on n of cubes meets the hyperplane. There are $O_n((\ell + 1)^{n-2})$ projected cubes and a bounded number of faces. This proves the uniform error, also for empty or lower-dimensional regions. Hilbert preservation returns the estimate to J . All embedded and vertex contributions remain in this equality of full Hilbert functions. $\square$

Theorem 7.9. *Let $(R, \mathfrak{m})$ be the local ring of a smooth complex n -fold at a closed point. For every height- c prime P , $1 \leq c \leq n-1$, every proper P -primary ideal I , and $\lambda = \text{lct}_{R_P}(IR_P)$,*

$$\dim_{\mathbb{C}}(\text{in}_{\mathfrak{m}} I)_{\ell} \leq \frac{\ell^{n-1}}{(n-1)!} E_{n,c} \left(\frac{1}{\ell\lambda} \right) + C_n(\ell+1)^{n-2} \quad (\ell \geq 1).$$

The bound also holds with $B_{n,c}$. Constants are independent of the prime, ideal, exponents and computing place. For $n \geq 3$, the exact two-normal capacity is

$$\mathfrak{M}_{n,2}(u) = Q_n(u) = \max\{(1-u)_+^{n-1}, (1-2u)_+^{n-2}(1+2(n-2)u)\},$$

and the same full-primary estimate holds with Q_n . In particular $Q_n(u) = (1-u)_+^{n-1}$ for $u \geq 4/(3n)$.

The first bounds require no tangent hypothesis. The two-normal formula keeps both its balanced and its coordinate branches.

Proof. By Lemma 7.8, it suffices to prove the simplex bounds. We obtain the two general envelopes first, and then optimize the two-normal case.

Step 1. In this step, we derive the codimension envelope from concavity of the simplex distribution. For positive weights a_i with sum one, put $Y = \sum_{i=1}^c a_i Z_i$ and $\Phi(t) = \Pr(Y \leq t)$. The marginal density in the first c coordinates is

$$\frac{(n-1)!}{(n-c-1)!} (1-y_1-\dots-y_c)^{n-c-1}$$

on the positive coordinate simplex. Reflecting it into all orthants and dividing by 2^c gives an unconditional log-concave probability density on $\mathbb{R}^c$: its support is the crosspolytope and the logarithm of its positive density is a constant plus $(n-c-1)\log(1-\sum|y_i|)$. When $n-c=1$, this means the constant density on that convex body.

The sets $A_t = \{\sum a_i |y_i| \leq t\}$ are closed under all coordinate contractions, and $(1-v)A_t + vA_s = A_{(1-v)t+vs}$. Thus Theorem 7.5 makes $\Phi^{1/c}$ concave. The distribution is continuous and strictly increasing on its support, so $\Phi(Y)$ is uniform on $[0, 1]$. Since $\mathbb{E}Y = 1/n$, Jensen's inequality gives

$$\Phi(1/n)^{1/c} \geq \mathbb{E}\Phi(Y)^{1/c} = \int_0^1 v^{1/c} dv = \frac{c}{c+1}.$$

The chords from zero to $1/n$, and from $1/n$ to one, give

$$\Phi(u)^{1/c} \geq \begin{cases} ncu/(c+1), & 0 \leq u \leq 1/n, \\ [c(n-1) + nu - 1]/[(c+1)(n-1)], & 1/n \leq u \leq 1. \end{cases}$$

Taking complements proves $\mathfrak{M}_{n,c} \leq B_{n,c}$. Approximation by positive weight vectors covers zero weights.

Step 2. In this step, we bound the upper simplex tail and its coordinate endpoint range. Convexity of the positive part, followed by the coordinate tail formula, gives for $0 \leq s < u \leq 1$

$$\Pr(Y \geq u) \leq \frac{\mathbb{E}(Y-s)_+}{u-s} \leq \frac{\mathbb{E}(Z_1-s)_+}{u-s} = \frac{(1-s)^n}{n(u-s)}.$$

For $u > 1/n$, choose $s = (nu - 1)/(n - 1)$. This proves the second bound entering $E_{n,c}$. If $1/2 \leq u < 1$, a positive-volume cap has a unique weight $p > u$. The cut simplex has volume fraction

$$\frac{(p - u)^{n-1}}{\prod_{i \neq 1} (p - a_i)} \leq \frac{(p - u)^{n-1}}{p^{n-2}(2p - 1)}.$$

Here zero weights have been appended to length n , and the inequality follows repeatedly from $(p - v)(p - w) \geq p(p - v - w)$. The logarithmic derivative of the last expression has the sign of $[2(n - 1)u - 1]p - (n - 2)u$, which is nonnegative for $p \geq u$. Its maximum is therefore its value $(1 - u)^{n-1}$ at $p = 1$. That value is attained by a coordinate weight. This proves the endpoint part of $E_{n,c}$, including $c = 1$. The envelope is nonincreasing; its possible downward jump at $1/2$ is taken with the stated value.

Step 3. In this step, we compute the exact maximum over two normal weights. Put $N = n - 1 \geq 2$, fix $u \geq 0$, and define $f(t) = t(1 - u/t)_+^N$ for $t > 0$, with $f(0) = 0$. For $p > q = 1 - p$, conditioning on $Z_1 + Z_2$ gives

$$(7.1) \quad \Pr(pZ_1 + qZ_2 \geq u) = \frac{f(p) - f(q)}{p - q}.$$

Indeed the pair sum has density $N(N - 1)t(1 - t)^{N-2}$, its conditional split is uniform, and each summand integrates as

$$N(N - 1) \int_{u/v}^1 (vt - u)(1 - t)^{N-2} dt = v(1 - u/v)^N \quad (v > u).$$

The limiting values at $p = 1/2, 1$ are the two branches of Q_n . We show that an intermediate weight cannot exceed both.

Suppose $0 < u < 1/2$. On $1/2 < p < 1 - u$, both terms are active. The derivative in p of (7.1) has the sign of

$$D(p) = (2p - 1)(f'(p) + f'(q)) - 2(f(p) - f(q)), \quad D'(p) = (2p - 1)(f''(p) - f''(q)), \quad D(1/2) = 0.$$

For $N = 2$ the probability is $1 - u^2/(pq)$, decreasing on this interval. For $N > 2$,

$$f''(t) = N(N - 1)u^2 \frac{(t - u)^{N-2}}{t^{N+1}}.$$

Put $x = p/q$ and $A = (N + 1)/(N - 2) > 1$. The sign of $f''(p) - f''(q)$ is that of $u - C_A(x)$, where

$$C_A(x) = \frac{x(x^{A-1} - 1)}{(1 + x)(x^A - 1)}.$$

This function decreases for $x > 1$. To verify it, set $x = e^{2v}$ and $\kappa = A - 1 > 0$. Then

$$C_A(e^{2v})^{-1} = 2 \cosh^2 v + \sinh(2v) \coth(\kappa v).$$

The derivative of the second term, multiplied by $\sinh^2(\kappa v)$, is $\cosh(2v) \sinh(2\kappa v) - \kappa \sinh(2v) > 0$, since $\sinh(2\kappa v)/\kappa \geq 2v > \tanh(2v)$. Thus D' , and then D , can change sign only from negative to positive. The probability has no interior maximum on the both-active interval.

On $1 - u \leq p \leq 1$, its value is

$$\frac{(p - u)^N}{p^{N-1}(2p - 1)}.$$

The derivative has the sign of $(2Nu - 1)p - (N - 1)u$, so again there is no interior maximum. At the joining endpoint this expression is $(Nu - 1)(1 - 2u)$. If $Nu \leq 1$, the derivative numerator on the both-active interval is nonpositive throughout, so the joining value does not exceed the balanced value. If $Nu > 1$, the one-active expression increases from the joining point to $p = 1$. In both cases only the balanced and coordinate values are needed. The cases $u = 0$, $u \geq 1/2$, and zero weights follow from the already proved endpoint formula and continuity. This proves the exact Q_n .

Step 4. In this step, we locate the range where the coordinate branch gives Q_n . For $0 < u < 1/2$, the ratio of the balanced branch to the coordinate branch is

$$R_n(u) = \frac{(1-2u)^{n-2}(1+2(n-2)u)}{(1-u)^{n-1}}.$$

Its logarithmic derivative has the sign of $1 - 2(n-1)u$. It increases from one, then decreases strictly to zero. To locate its unique positive crossing, put $s = 4/3$. Absolute convergence of the logarithm series gives

$$\log R_n(s/n) = \log(11/3) - 4/3 + \sum_{j \geq 1} \frac{s^j}{jn^j} \left(\frac{(2^{j+1} - 1)(3-j)}{3(j+1)} - \left(\frac{12}{11} \right)^j \right).$$

Each parenthesis is negative: for $j = 1, 2$ it is $1 - 12/11$, respectively $7/9 - 144/121$, and for $j \geq 3$ its first term is nonpositive. The constant is negative as well, because $e^{4/3} > 911/243 > 11/3$. Hence $R_n(4/(3n)) < 1$. This proves the final range in the statement. Applying Lemma 7.8 to all the established simplex bounds proves the full-primary estimates. $\square$

Theorem 7.10 (Reduced schemes of minimal degree). *An integral nondegenerate projective variety of degree equal to its codimension plus one is a linear space, a quadric hypersurface, a rational normal scroll, or a cone over the Veronese surface. A reduced projective scheme is 2-regular if and only if its irreducible components are varieties of minimal degree which admit a linearly joined ordering. In such an ordering, the scheme-theoretic intersection of a component with the preceding union is the intersection of their linear spans. Linear degeneracy is removed by working in the span.*

These are [EGHP04, Theorems 0.1 and 0.4, Proposition 2.4]. The classification applies after reducedness has been established.

Theorem 7.11 (Linear sections and curve regularity). *If $T \subseteq \mathbb{P}^r$ satisfies $N_{2,p}$, $p \geq 1$, and Λ is linear of dimension at most p with $\dim(T \cap \Lambda) \leq 1$, then its section ideal is 2-regular. If the intersection is finite, its length is at most $\dim \Lambda + 1$. For a connected reduced projective curve C over an algebraically closed field,*

$$\text{reg } \mathcal{I}_C \leq \deg C - \dim \langle C \rangle + 2.$$

The section assertion is [EGHP05, Theorem 1.1]; the curve assertion is [Gia06, Main Theorem].

We record the simplex estimates needed for the restricted tangent cases. They are exact integrations over the indicated continuous parameter domains. For a uniform point on the N -coordinate simplex, write $D_N(p, t) = \Pr(pZ_1 + (1-p)Z_2 \geq t)$. We set $D_N(p, t) = 1$ for $t \leq 0$. Formula (7.1) gives its divided-difference expression, including coincident and zero weights by continuity.

Lemma 7.12. *Let Z be uniform on the six-coordinate simplex. For $a_1, a_2, a_3 \geq 0$ with $a_1 + a_2 + a_3 = 1$,*

$$\Pr(a_1Z_1 + a_2Z_2 + a_3Z_3 \geq 1/5) \leq \frac{1053}{3125}.$$

If all three weights are at least $1/5$, the upper bound is $992/3125$.

The first maximum occurs at the balanced two-coordinate weight.

Proof. We remove a weight at most $1/5$, then maximize explicitly when all weights are larger.

Step 1. In this step, we remove a weight at most $1/5$ without decreasing the tail bound. Set $h = 1/5$, fix $p \in [0, 1]$, and let

$$S_z = (1-z)(pZ_1 + (1-p)Z_2) + zZ_3, \quad 0 \leq z \leq h.$$

The density of $AV_1 + BV_2$, for V uniform on the five-coordinate simplex and $A > B > 0$, is

$$f_{A,B}(t) = \frac{4}{A-B} ((1-t/A)_+^3 - (1-t/B)_+^3).$$

It is nonincreasing for $t \geq AB/(A+B+\sqrt{AB})$. This last number is at most $(A+B)/6$, since, with $v = (A+B)/\sqrt{AB} \geq 2$, the comparison is $6 \leq v^2 + v$. For $A = B$ the density has its maximum at $(A+B)/6$; for $B = 0$ it is decreasing.

Differentiating the moving halfspace at a regular parameter gives

$$\frac{d}{dz} \Pr(S_z \geq h) = \frac{f_{S_z}(h)}{1-z} (\mathbb{E}[Z_3 | S_z = h] - h).$$

With $A = (1-z)p$, $B = (1-z)(1-p)$, the conditional density of $Z_3 = t$ on this facet is proportional to

$$(1-t)^3 f_{A,B}\left(\frac{h-zt}{1-t}\right), \quad 0 < t < 1.$$

Its second factor is nonincreasing: its argument increases from $h > (A+B)/6$. The probability density $4(1-t)^3$ has mean h . Multiplication by a nonnegative decreasing function cannot increase that mean, as follows from

$$2 \operatorname{Cov}(t, g(t)) = \iint (t-s)(g(t) - g(s)) d\mu(t) d\mu(s) \leq 0.$$

The derivative is therefore nonpositive. The coarea differentiation holds between the finitely many coefficient and threshold coincidences; continuity extends the conclusion across them and to the boundary weights. A weight $z \leq h$ can thus be removed. The remaining maximum is $Q_6(h)$, whose two branches are 1053/3125 and 1024/3125.

Step 2. In this step, we maximize the tail when all three weights are at least $1/5$. Put $x_i = 5a_i \geq 1$, so $\sum x_i = 5$, and let $s_2 = \sum_{i < j} x_i x_j$, $s_3 = x_1 x_2 x_3$. Successive simplex integration gives, for distinct x_i ,

$$\Pr\left(\sum x_i Z_i \geq 1\right) = \sum_{i=1}^3 \frac{(x_i - 1)^5}{x_i^3 \prod_{j \neq i} (x_i - x_j)} = F(s_2, s_3),$$

where

$$F(s_2, s_3) = 1 - \frac{10}{s_3} + \frac{5s_2 + 5}{s_3^2} - \frac{s_2^2}{s_3^3}.$$

For completeness, the first identity is the simplex tail formula $\sum_i (d_i - t)_+^{N-1} / \prod_{j \neq i} (d_i - d_j)$, with three zero coefficients. Integrating the last coordinate inductively proves it for distinct coefficients; the coefficient identity is the partial-fraction decomposition of $\prod_i (z - d_i)^{-1}$. Continuous limits cover all repeated coefficients. Expanding $(x-1)^5/x^3$ and taking its second divided difference proves the displayed expression for F .

At an interior critical point with three distinct coordinates, the Lagrange equations imply $F_{s_2} = F_{s_3} = 0$. They give $s_3 = 2s_2/5$, then $s_2 = 20/3$. But $x_i = 1 + y_i$, $y_i \geq 0$, $\sum y_i = 2$, gives $s_2 \geq 7$. Thus a maximum is on the boundary or has two equal coordinates. On the boundary $x_3 = 1$, put $q = x_1 x_2 \in [3, 4]$. Then

$$F = 1 - 6/q + 17/q^2 - 16/q^3, \quad F' = 2(3q - 8)(q - 3)/q^4 \geq 0.$$

Its maximum is $5/16$. With $x_1 = x_2 = x$, $x_3 = 5 - 2x$, $1 \leq x \leq 2$, differentiation gives

$$\frac{dF}{dx} = \frac{10x^3(5-3x)(x-1)(x-2)(8x^2-24x+15)}{s_3^4}.$$

The last quadratic and $(x-1)(x-2)$ are both negative on $(1, 2)$, so the maximum occurs at $x = 5/3$. Its value is $992/3125 > 5/16$, proving the second assertion. $\square$

Lemma 7.13. *Let $n \geq 3$ and $u \geq 1/(n+1)$. For Z uniform on the n -coordinate simplex, the supremum over $a, b, d \in [0, 1]$ of*

$$\Pr(aZ_1 + (1-a)Z_2 \geq u, bZ_1 + (1-b)Z_3 \geq u, dZ_2 + (1-d)Z_3 \geq u)$$

is

$$\max \{(1-2u)_+^{n-1}, (1-3u)_+^{n-2}(1+(2n-5)u)\}.$$

For $u \geq 4/(3n+4)$ it equals $(1-2u)_+^{n-1}$.

All three pair constraints enter the estimate simultaneously.

Proof. It suffices first to use positive weights. Write the constraints as

$$Ax_1 + x_2 \geq (A+1)u, \quad Bx_1 + x_3 \geq (B+1)u, \quad Dx_2 + x_3 \geq (D+1)u, \quad A, B, D > 0.$$

Put $N = n-1$, $a_0 = 1-3u$, and

$$F(q) = N \int_0^u (a_0 - qt)_+^{N-1} dt.$$

Step 1. In this step, we express the three-cut probability using the exterior-region integrals. After translating the three coordinates by u , at most one is negative. Integration of the three exterior regions gives the exact probability

$$(a_0)_+^N + F(A+B-1) + F(A^{-1}+D-1) + F(B^{-1}+D^{-1}-1).$$

Set $q_1 = A+B-1$, $q_2 = A^{-1}+D-1$, $q_3 = B^{-1}+D^{-1}-1$, and $x_i = (q_i+2)^{-1}$. Direct common-denominator subtraction gives

$$1 - \sum x_i = \frac{(B-AD)^2}{(1+A+B)(1+A+AD)(B+D+BD)} \geq 0.$$

Step 2. In this step, we prove the ratio inequality needed to merge the exterior contributions. The function $\phi(x) = F(x^{-1}-2)$, with $\phi(0) = 0$, is increasing. If $1/(n+1) \leq u < 1/3$, then $\phi(x)/x$ is nondecreasing on $(0, 1/2]$. To check this, for $q > 0$ use

$$F(q) = \frac{a_0^N - (a_0 - qu)_+^N}{q}.$$

When $0 < q < a_0/u$, put $z = qu/a_0$ and $\rho = u/a_0$. The derivative of $(q+2)F(q)$ has the sign of

$$-2 + (1-z)^{N-1}(2 + 2(N-1)z + (N/\rho)z^2) \leq 0.$$

Indeed $1/\rho \leq N-1$, and $(1-z)^{-(N-1)} \geq 1 + (N-1)z + N(N-1)z^2/2$. For $q \geq a_0/u$, the expression is $a_0^N(1+2/q)$, also decreasing.

Step 3. In this step, we bound the three-cut probability by the exact two-normal maximum. Increase the x_i to numbers of sum one. If all are at most $1/2$, the ratio inequality bounds the sum of their ϕ -values by $2\phi(1/2)$. Otherwise keep the largest, say $p > 1/2$, and merge the other two to get the bound $\phi(p) + \phi(1-p)$. The divided-difference formula gives the identity

$$a_0^N + \phi(p) + \phi(1-p) = (1-u)^N D_n\left(p, \frac{u}{1-u}\right).$$

The exact two-normal maximum in Theorem 7.9 now proves the asserted upper bound. If $u \geq 1/3$, then $F(q) = 0$ for $q \geq 0$. At most one x_i exceeds $1/2$, so the probability is at most $\phi(1) = F(-1) = (1-2u)_+^N$.

Step 4. In this step, we prove sharpness and the stated coordinate-branch range. For sharpness when $u < 1/2$, let A, B tend to infinity. The first two cuts tend to $Z_1 \geq u$; the third remains an arbitrary pair cut in the other two coordinates. Translation and rescaling give exactly $(1-u)^N D_n(p, u/(1-u))$. Dominated convergence proves equality of the suprema and covers zero weights. For $u \geq 1/2$, requiring two coordinates to be at least u has zero simplex measure, so both sides of the asserted equality are zero. The final range for $u < 1/2$ follows from $u/(1-u) \geq 4/(3n)$, equivalent to $u \geq 4/(3n+4)$. $\square$

Lemma 7.14. For Z uniform on the six-coordinate simplex and $p, q \in [0, 1]$,

$$\Pr(pZ_1 + (1-p)Z_2 \geq 1/6, qZ_3 + (1-q)Z_4 \geq 1/6) \leq 13/96.$$

The bound includes independent choices of the two pairs.

Proof. Put $h = 1/6$, $r = 1 - p$, $s = 1 - q$, and use symmetry to assume $p, q \geq 1/2$.

Step 1. In this step, we derive the exact disjoint-pair probability. For $p, q > 1/2$, the probability equals

$$(7.2) \quad \frac{pq(1 - h/p - h/q)_+^5 - ps(1 - h/p - h/s)_+^5 - rq(1 - h/r - h/q)_+^5 + rs(1 - h/r - h/s)_+^5}{(p - r)(q - s)}.$$

To derive it, condition on the pair sums S, T , whose joint density is $120ST(1 - S - T)$. Each pair splits uniformly. Expanding their two split lengths gives four integrals, with

$$120 \int (aS - h)_+(bT - h)_+(1 - S - T) dS dT = ab(1 - h/a - h/b)_+^5.$$

Translation of S, T by $h/a, h/b$ proves this identity.

Step 2. In this step, we bound the probability on the two parameter ranges.

Case 2.1. At least one of p, q is at least $3/4$. If $p \geq 3/4$, the final term in (7.2) is zero. Dropping its negative rq term and putting $L = 1 - h/p$ gives the upper bound $pL^5 D_6(q, h/L)/(2p - 1)$. Here $1/5 \leq h/L \leq 3/14$, and the exact two-normal formula makes this the maximum of

$$B_1(p) = \frac{p}{2p - 1} (5/6 - h/p)^5, \quad B_2(p) = \frac{p}{2p - 1} (2/3 - h/p)^4 (7/3 - h/p).$$

The derivative of $\log B_1$ has the sign of $5p - 4$, so its maximum is at an endpoint. Its endpoint values are $(3/2)(11/18)^5 < 13/96$ and $32/243 < 13/96$. The derivative of $\log B_2$ has positive numerator $4(16p^2 - 13p + 1)$ on $[3/4, 1]$. Thus $B_2(p) \leq B_2(1) = 13/96$. Symmetry handles $q \geq 3/4$.

Case 2.2. Both p and q belong to $[1/2, 3/4]$. It remains to cover $1/2 \leq p, q \leq 3/4$. Put $z = pr$, $w = qs$, both in $[3/16, 1/4]$. Only the small-small argument in (7.2) can be negative. Let D_0 denote the expression with untruncated fifth powers. Expansion gives

$$7776D_0 = 7776 + f(z) + f(w) + \frac{180}{zw} - \frac{10(z + w)}{z^2 w^2}, \quad f(z) = -\frac{2160}{z} + \frac{390}{z^2} - \frac{32}{z^3} + \frac{1}{z^4}.$$

Its z -derivative has the sign of

$$g(z, w) = Az^3 + Bz^2 + 48z - 2, \quad A = 1080 - 90/w + 5/w^2, \quad B = -390 + 10/w.$$

For $v = 1/w \in [4, 16/3]$, $A > 0$, and

$$g_{zz}(3/16, w) = (3480 - 650v + 45v^2)/8 > 0, \quad g_z(3/16, w) = (135v^2 - 1470v + 4008)/256 \geq 1/32.$$

The first numerator decreases to $3880/3 > 0$ and the second to eight on this interval. Hence g increases with z , and a maximum is at a z -endpoint. Symmetry reduces the whole rectangle to its three inequivalent corners. The values of $7776D_0$ there are $992, 77824/81, 74336/81$, all at most 992.

Restoring the positive part adds

$$E(r, s) = \frac{rs}{(1 - 2r)(1 - 2s)} (h/r + h/s - 1)_+^5.$$

On its positive region, its logarithmic r -derivative is nonpositive exactly when $r(1/s + 4) \leq 4$. This holds on $[1/4, 1/2]^2$, as does the symmetric inequality. Its maximum is therefore $E(1/4, 1/4) = 1/972$. Thus the probability is at most $31/243 + 1/972 = 125/972 < 13/96$ on this square. Continuity covers its balanced edges and every zero weight in the preceding cases. $\square$

Lemma 7.15. *Let Z be uniform on the six-coordinate simplex and put $h = 1/6$. For $p \in [0, 1]$, $a, b, c, d \geq 0$, $a + b \leq 1$, $c + d \leq 1$, set $e = 1 - a - b$ and $f = 1 - c - d$. Then*

$$\Pr(pZ_3 + (1 - p)Z_4 \geq h, aZ_1 + bZ_2 + eZ_4 \geq h, cZ_1 + dZ_2 + fZ_3 \geq h) \leq 13/96.$$

The two shared pairs in the last cuts can have unrelated weights.

Proof. We compare the three cuts to a two-cut family, prove monotonicity in one part of its parameter domain, and use convexity on the rest. For $1/2 \leq p \leq 1$, $z \geq 0$, $q \in [0, 1]$, put

$$N(p, z, q) = \Pr(pZ_3 + (1-p)Z_4 \geq h, Z_3 + z(qZ_1 + (1-q)Z_2) \geq h(z+2)).$$

Step 1. In this step, we prove the two density comparisons used in the parameter estimates. The divided-difference formula implies

$$t \mapsto \frac{D_4(q, t)}{(1-t)^3} \text{ is nonincreasing on } [1/6, 1), \quad t \mapsto D_5(q, t) \text{ is convex on } [1/6, \infty).$$

Here are the sign checks. For $q > r = 1 - q > 0$, put $\alpha = r/q$ and $w = t/(1-t)$. The derivative of the first ratio has the sign of $q(1 - w/\alpha)_+^2 - r(1 - \alpha w)_+^2$. If both terms are positive, this is nonpositive for

$$w \geq \frac{\alpha}{1 + \sqrt{\alpha} + \alpha + \alpha^{3/2} + \alpha^2}.$$

The right side is at most $1/5$, by pairing reciprocal terms in its denominator divided by α . The claimed range gives $w \geq 1/5$. If the first term vanishes the sign follows directly. For the second comparison, the density $4((1 - t/q)_+^3 - (1 - t/r)_+^3)/(q - r)$ decreases for $t \geq qr/(1 + \sqrt{qr}) \leq 1/6$. Equal and zero weights follow by limits. We will also use that multiplying a positive measure by a nonnegative increasing function cannot decrease its mean, by the covariance identity in Lemma 7.12.

Step 2. In this step, we prove monotonicity at the coordinate anchor $z = 4, q = 1$. At $z = 4, q = 1$, translation of Z_1 by $(1 - Z_3)/4$ and rescaling give

$$N(p, 4, 1) = (3/4)^4 \Pr(pZ_1 + (3/4)(1-p)Z_2 \geq h).$$

This increases with $p \in [1/2, 1]$. To check the assertion, put $b_0 = (3/4)(1-p)$ and $f_0(t) = t(1 - h/t)_+^5$. The derivative of the last probability is

$$\frac{7}{4(p - b_0)^2} \int_{b_0}^p (t - 3/7) f_0''(t) dt.$$

Write $v_p = 1 - 1/(6p)$, $v_b = (1 - h/b_0)_+$, $\Delta_j = v_p^j - v_b^j$, and $c_* = 55/72$. The integral is $(12/7)(\Delta_5 - c_*\Delta_4)$, by the substitution $v = 1 - h/t$. For $p \geq 12/17$, $\Delta_5/\Delta_4 \geq v_p \geq c_*$. For $1/2 \leq p < 12/17$, the derivative of

$$R(p) = \log \frac{v_b^4(c_* - v_b)}{v_p^4(c_* - v_p)}$$

is

$$20(7p - 3) \left(\frac{1}{p(6p - 1)(12 - 17p)} - \frac{2}{(1 - p)(7 - 9p)(17p - 1)} \right).$$

The bracket has the sign of $W(p) = 357p^3 - 459p^2 + 159p - 7$. With $v = p - 5/8$,

$$W = 121/512 + (231/64)v + (1683/8)v^2 + 357v^3.$$

For $v \geq 0$ this is positive; for $-1/8 \leq v \leq 0$, it is at least $1/5 + 4v + 160v^2 \geq 7/40$. At $p = 1/2$, $\Delta_5/\Delta_4 = 4651/6039 > 55/72$. Thus $R(p) > 0$, and the anchor derivative is nonnegative everywhere.

Step 3. In this step, we bound $N(p, z, q)$ throughout $0 < z \leq 4$. Fix $1/2 < p < 1$, put $r = 1 - p$, $k = h/p$, and $\delta = 1/z \geq 1/4$. On the common facet, parameterized by $C = Z_3 \in [0, k]$, write

$$Z_4 = (h - pC)/r, \quad T(C) = 1 - C - (h - pC)/r, \quad H_\delta(C) = h + \delta(2h - C).$$

Differentiation of the moving lower limit for Z_4 gives

$$(7.3) \quad \partial_p N(p, z, q) = \frac{20}{r^2} \int_0^k (C - h) T(C)^3 D_4(q, H_\delta(C)/T(C)) dC.$$

Terms with $T \leq 0$ have zero mass. The density factor is the four-coordinate simplex slice; differentiation is justified on compact subintervals of $1/2 < p < 1$, where the moving boundary and its derivative are bounded. The switching endpoint has measure zero.

For $q = 1$ the facet density, up to a positive constant, is $L_\delta(C)_+^3$, where

$$L_\delta(C) = a_0 + BC - \delta(2h - C), \quad a_0 = 1 - h/r - h, \quad B = (p - r)/r \geq 0.$$

The anchor shows that its mean at $\delta = 1/4$ is at least h . As δ increases, these densities have nested supports at the upper end of $[0, k]$. For $\delta_2 > \delta_1$,

$$\frac{d}{dC} \frac{L_{\delta_2}(C)}{L_{\delta_1}(C)} = \frac{(\delta_2 - \delta_1)(a_0 + 2hB)}{L_{\delta_1}(C)^2} > 0, \quad 6r(a_0 + 2hB) = 2 - p > 0.$$

Extend the ratio by zero below the smaller support. The mean comparison keeps the mean at least h whenever the density has positive mass. For general q , on this support put $t(C) = H_\delta(C)/T(C)$. It decreases with C and is at least $1/4$, since $H_\delta(C) \geq (1 - C)/4$ and $T(C) \leq 1 - C$. The ratio of the pair density to the coordinate density is $D_4(q, t(C))/(1 - t(C))^3$, an increasing function of C by the first density comparison. Its mean is therefore at least h as well. Formula (7.3) proves $\partial_p N \geq 0$; zero-mass cases give derivative zero. Continuity supplies the endpoint values of p, q .

At $p = 1$, translate Z_3 by h and rescale by $5/6$. The remaining cut is a stochastic three-coordinate cut at $1/5$. Lemma 7.12 gives

$$N(1, z, q) \leq (5/6)^5(1053/3125) = 13/96.$$

Hence the same bound holds for $0 < z \leq 4$. At $z = 0$ it is $\Pr(Z_3 \geq 1/3) = 32/243 < 13/96$.

Step 4. In this step, we bound $N(p, z, q)$ on the unbounded range $z \geq 4$. For fixed $p < 1, q$, integrate the common cut by putting

$$d(C) = \left(\frac{h - pC}{1 - p} \right)_+, \quad T(C) = 1 - C - d(C).$$

For $0 \leq \theta \leq 1/4$, the continuous extension of $N(p, 1/\theta, q)$ is

$$J(\theta) = 5 \int_{T(C) > 0} T(C)^4 D_5 \left(q, \frac{h + \theta(2h - C)}{T(C)} \right) dC.$$

Every threshold in this integral is at least $1/5$. Indeed $T(C) \leq 1 - h/p \leq 5/6$; for $C \leq 2h$ the numerator is at least h , and for $C \geq 2h$ it is at least $(1 - C)/4$, while $T(C) = 1 - C$. The second density comparison therefore makes J convex in θ . At zero it is the disjoint-pair probability of Lemma 7.14, and at $1/4$ it is the already bounded $N(p, 4, q)$. Both are at most $13/96$, so convexity covers all $z \geq 4$. Limits cover $p = 1$.

Step 5. In this step, we compare the three original cuts with the bounded two-cut family. Assume first $e, f > 0$, and exchange coordinates to arrange $p \geq 1/2$. At fixed $Z_1, Z_2, S = Z_3 + Z_4$, put

$$A_0 = (h - aZ_1 - bZ_2)/e, \quad B_0 = (h - cZ_1 - dZ_2)/f, \quad L = (h - (1 - p)S)/(2p - 1).$$

For $p > 1/2$, the allowed interval in Z_3 has length $[S - (A_0)_+ - \max((B_0)_+, L)]_+$, at most $[S - \max((A_0 + B_0)_+, L)]_+$. Set $r_1 = a/e + c/f$, $r_2 = b/e + d/f$, $z = r_1 + r_2$. Then $A_0 + B_0 = h(z + 2) - r_1Z_1 - r_2Z_2$. For $z > 0$, the enlarged interval is exactly that of $N(p, z, r_1/z)$. The simplex density is constant along the interval, so integration gives the bound. For $z = 0$ use the preceding coordinate bound. At $p = 1/2$ the common cut is $S \geq 2h$ and the same comparison uses $(A_0)_+ + (B_0)_+ \geq (A_0 + B_0)_+$. Finally, approximation covers $e = 0$ or $f = 0$; each limiting equality is a proper hyperplane and has measure zero. $\square$

Lemma 7.16. *Let Z be uniform on the six-coordinate simplex. For every nonnegative 4-by-4 matrix A with zero diagonal and row sums one,*

$$\Pr(A(Z_1, Z_2, Z_3, Z_4)^\top \geq (1/6)\mathbf{1}) \leq \frac{138259}{900000}.$$

Every row is retained in the initial comparison.

Proof. First take all off-diagonal entries positive and set $h = 1/6$.

Step 1. In this step, we enlarge the four-row event to a three-cut event with one varying threshold. Some reciprocal pair has $a_{12} + a_{21} \leq 2/3$, after permutation, because the six reciprocal sums total four. Normalize these two rows to write

$$Z_2 \geq A_0 = h(1 + \alpha + \beta) - \alpha Z_3 - \beta Z_4, \quad Z_1 \geq B_0 = h(1 + \gamma + \delta) - \gamma Z_3 - \delta Z_4.$$

Set $z = \alpha + \beta + \gamma + \delta = a_{12}^{-1} + a_{21}^{-1} - 2 \geq 4$, and $Q = ((\alpha + \gamma)Z_3 + (\beta + \delta)Z_4)/z$. On a fiber of fixed $Z_1 + Z_2, Z_3, Z_4$, the first two cuts leave an interval of length $[Z_1 + Z_2 - (A_0)_+ - (B_0)_+]_+$. Replace it by the right-anchored interval of length $[Z_1 + Z_2 - (A_0 + B_0)_+]_+$, which is larger. Reflect each of the other two row intervals to that same endpoint if necessary, by interchanging its two coefficients on Z_1, Z_2 . Each individual length is unchanged, and their new common intersection has the smallest of the three lengths. It therefore contains at least as much one-dimensional measure as the original intersection.

The resulting event has the form

$$Q \geq h + \lambda(2h - Z_1), \quad aZ_1 + bZ_2 + eZ_4 \geq h, \quad cZ_1 + dZ_2 + fZ_3 \geq h,$$

with $0 < \lambda = 1/z \leq 1/4$ and stochastic coefficients in the last two cuts. The fiber density is constant, so this comparison also holds for probabilities.

Step 2. In this step, we bound the probability gained by the varying threshold. At $\lambda = 0$, Lemma 7.15 gives 13/96. A point gained by making $\lambda > 0$ must lie in

$$Z_1 \geq 1/3, \quad (1 - Z_1)/4 \leq Q < h.$$

Write $Q = pZ_3 + (1 - p)Z_4$. Conditioning on Z_1 , and then translating Z_1 by $1/3$ for the upper edge, shows that this strip has probability

$$(2/3)^5 (D_5(p, 1/4) - D_6(p, 1/4)).$$

For $t = 1/4$, $r = 1 - p$, and $\psi(v) = (1 - t/v)_+^4$, the divided-difference formula gives

$$D_5(p, t) - D_6(p, t) = t \frac{\psi(p) - \psi(r)}{p - r}.$$

This is at most 432/3125, since, with $w = t/v$, $t\psi'(v) = 4w^2(1 - w)^3 \leq 432/3125$; the maximum on $[0, 1]$ occurs at $w = 2/5$. Equal or zero weights follow by limits. Consequently the original probability is at most

$$\frac{13}{96} + \left(\frac{2}{3}\right)^5 \frac{432}{3125} = \frac{138259}{900000}.$$

Approximate an arbitrary admissible matrix by positive off-diagonal matrices. The four equality hyperplanes have measure zero, so dominated convergence proves the same bound on the whole closed matrix domain. $\square$

Lemma 7.17. *Let $(R, \mathfrak{m})$ be a smooth complex local ring of dimension $n \geq 3$, and let P be a prime with $d = \dim R/P \geq 2$. Suppose $d < n$ and the full tangent ring of R/P is reduced and Cohen–Macaulay, of embedding dimension e . Put $t = n - e$ and $s = e - d + 1$. For every proper P -primary ideal I , with $\lambda = \text{lct}_{R_P}(IR_P)$, simultaneous homogeneous flat degenerations of the entire support and initial ideals give monomial ideals H, K with their respective full Hilbert functions unchanged, and with*

$$\sqrt{K} = \sqrt{H} = Q_0 = (y_1, \dots, y_t) + (x_i x_j : 1 \leq i < j \leq s).$$

There are $d - 1$ additional free coordinates. For some $N \geq 1$, $Q_0^N \subseteq K \subseteq H \subseteq Q_0$. At every coordinate prime

$$\mathfrak{q}_i = (y_1, \dots, y_t, x_1, \dots, \hat{x}_i, \dots, x_s),$$

the generic threshold of K is at most λ . In particular every monomial of K satisfies a normalized positive weighted inequality of level $1/\lambda$ on the coordinates of $\mathbf{q}_i$, for every i . If $\Psi_{n,t,s}(u)$ is the probability that one of t designated simplex coordinates is at least u , or at least two of s other designated coordinates are at least u , then

$$\Psi_{n,t,s}(u) = \sum_{j=1}^{t+s} (-1)^{j+1} \left(\binom{t+s}{j} - s \binom{t+s-1}{j-1} \right) (1 - ju)_+^{n-1},$$

and, for $\ell \geq 1$,

$$\dim_{\mathbb{C}}(\text{in}_{\mathbf{m}} I)_{\ell} \leq \frac{\ell^{n-1}}{(n-1)!} \Psi_{n,t,s}\left(\frac{1}{\ell\lambda}\right) + C_n(\ell+1)^{n-2}.$$

The reduced tangent hypothesis supplies spanning branches. Collision multiplicities in the full ideals remain part of the comparison.

Proof. We construct a spanning point section by full flat degenerations, collide its points while fixing spanning branches, and retain all their generic inequalities.

Step 1. In this step, we obtain a full reduced spanning point section by flat degeneration. Write $G = \text{gr}_{\mathbf{m}} R$, $Q = \text{in}_{\mathbf{m}} P$, and $J = \text{in}_{\mathbf{m}} I$. The ordinary Rees specialization in Lemma 7.8 gives the generic threshold bound at every component of Q . All these components have codimension $n - d$, since G/Q is Cohen–Macaulay. Choose the y_i to span Q_1 . In the remaining e coordinates, $\text{Proj}(G/Q)$ spans $\mathbb{P}^{e-1}$.

Take $d - 1$ general linear sections. At each stage the linear equation is regular on the Cohen–Macaulay graded ring. The projective section is generically reduced: the incidence of a smooth point of a q -dimensional component and a hyperplane containing its tangent space has dimension at most one less than the dual projective space, and the remaining singular locus drops in dimension. At the curve stage the general hyperplane avoids that remaining finite bad locus. The quotient is Cohen–Macaulay of positive dimension and has no embedded associated primes. Generic reducedness therefore makes the entire quotient reduced, including its vertex. The final quotient is the saturated ring of a finite reduced point set in $\mathbb{P}^{s-1}$. Its degree-one part has dimension s , so it contains s linearly independent points.

Each section is implemented as a degeneration in the original coordinates. If w is its equation and B the current support ideal, then $(B : w) = B$. Give w weight zero and the remaining active variables weight one, taking largest weight initials. For homogeneous $w^a g \in B$ with $w \nmid g$, one has $g \in B$ and initial $w^a(g|_{w=0})$. Thus the new support ideal is the section ideal extended with w free. Apply the same degeneration to the entire current source ideal. The inclusions $B^M \subseteq J \subseteq B$ persist with full initials, and both Hilbert functions are unchanged. The regular-family restriction argument of Lemma 7.8 transfers the generic threshold bound to every new component. Previously freed variables stay free; the construction applies coefficientwise in them.

Step 2. In this step, we retain every collision and fixed branch in the monomial degeneration. Make a linear change taking the selected s points to coordinate vertices in the x_i . Act by $\text{diag}(1, z, \dots, z^{s-1})$. If $[c_1 : \dots : c_s]$ has first nonzero coordinate c_h , the closure of its moving affine cone is defined by the common y_i and

$$x_i - (c_i/c_h)z^{i-h}x_h \quad (c_i \neq 0, i \neq h), \quad x_i \quad (c_i = 0).$$

Its quotient is a polynomial ring in z, x_h and the free variables. Its special prime is $\mathbf{q}_h$, and the selected vertex branches remain fixed throughout. The full support family is the intersection of all these horizontal branch ideals, since contraction from $z \neq 0$ commutes with finite intersections. If there are q points, specialization places its full ideal between the product and the intersection of their special primes. Consequently $Q_0^q \subseteq H_0 \subseteq Q_0$, and $Q_0^{qM} \subseteq K_0 \subseteq H_0$. At $(z, \mathbf{q}_i)$, the total ambient local ring and its parameter divisor are regular. Its fixed horizontal vertex branch has the previously proved threshold bound, so restriction gives it for K_0 at $\mathbf{q}_i$, even when other

points collide there. A final simultaneous monomial degeneration preserves these fixed coordinate primes, the power containments, and their thresholds. It gives the asserted K, H and $N = qM$.

Step 3. In this step, we count the region satisfying all branch inequalities. Theorem 7.1 now gives a positive normalized Newton inequality at every $\mathbf{q}_i$. Every monomial of the full K satisfies them all. If every common y -exponent is below $1/\lambda$, at least two of the active x -exponents must reach $1/\lambda$: zero such exponents violates every row, and just one violates the row omitting it. The resulting union event has the displayed probability by inclusion-exclusion. More explicitly, its complement is the union of the event that none of the $t + s$ coordinates reaches the level and the s disjoint events that exactly one does and is active. For j prescribed coordinates the joint upper tail is $(1 - ju)_+^{n-1}$, which gives the stated coefficients. Count the original region of simultaneous halfspaces before taking this probability bound. There are at most s added faces, so the uniform boundary-cube argument of Lemma 7.8 still applies with C_n . Its constant is independent of q, M , tangent degree and collision lengths. Full Hilbert preservation completes the proof. $\square$

Theorem 7.18. *Let $(S, \mathbf{m})$ be the local ring of a smooth complex n -fold, $n \geq 3$, and let P be prime with $2 \leq d = \dim(S/P) \leq n - 1$. Suppose the entire tangent ring $\text{gr}_{\mathbf{m}}(S/P)$ is reduced and Cohen–Macaulay. Put $e = \text{edim}(S/P)$, $t = n - e$, $s = e - d + 1$, and*

$$\mathcal{S}_{n,t,s}(c) = \sum_{j=1}^{t+s} (-1)^{j+1} \left[\binom{t+s}{j} - s \binom{t+s-1}{j-1} \right] (1 - jc)_+^{n-1}.$$

For every proper P -primary ideal I , put $\lambda = \text{lct}_{S_P}(IS_P) > 0$. There is a constant C_n , independent of the prime, tangent degree, full ideal, primary depth, degeneration weights and computing place, such that for every integer $\ell \geq 1$,

$$\dim_{\mathbb{C}}(\text{in}_{\mathbf{m}} I)_{\ell} \leq \frac{\ell^{n-1}}{(n-1)!} \mathcal{S}_{n,t,s}(1/(\ell\lambda)) + C_n(\ell+1)^{n-2}.$$

When $t = 0, s = 3$, the leading capacity may also be replaced by $2(1 - 2c)_+^{n-1}$, and by $(1 - 2c)_+^{n-1}$ if $c \geq 4/(3n+4)$; one may take the minimum of valid bounds. In particular, for full embedding dimension seven, the height-three capacity is

$$D_4(c) = 6(1 - 2c)_+^6 - 8(1 - 3c)_+^6 + 3(1 - 4c)_+^6,$$

and the height-two capacity may be taken to be

$$K_5(c) = \begin{cases} \min\{3(1 - 2c)_+^6 - 2(1 - 3c)_+^6, 2(1 - 2c)_+^6\}, & 0 \leq c < 4/25, \\ (1 - 2c)_+^6, & c \geq 4/25. \end{cases}$$

Both statements retain the entire ordinary tangent and primary image, with the same uniform-error convention.

The tangent degree places no bound on the collision multiplicities; the preceding lemma retains them in the full monomial ideals.

Proof. Lemma 7.17 proves the general formula, since its probability $\Psi_{n,t,s}$ is precisely $\mathcal{S}_{n,t,s}$. We establish the factor-two refinement at every level and then apply the exact three-pair bound.

Step 1. In this step, we prove the factor-two refinement for three active coordinates. Suppose $t = 0, s = 3$, put $H = 1/\lambda$, and fix the sum S_0 of the three active exponents. The feasible region has a positive normalized inequality on each pair. It is empty for $S_0 < 2H$. For $2H \leq S_0 < 3H$, at least two coordinates are at least H . The three resulting closed triangles are disjoint. Convexity puts the feasible region in just one of them, whose area in the first two coordinates is $(S_0 - 2H)^2/2$.

For $S_0 \geq 3H$, subtract H from each coordinate and put $T = S_0 - 3H$. Write the pair constraints as

$$x'_2 \geq -Ax'_1, \quad x'_3 \geq -Bx'_1, \quad x'_3 \geq -Dx'_2, \quad A, B, D > 0.$$

The nonnegative triangle has area $T^2/2$. At most one translated coordinate is negative. On $x'_1 = -z$, $0 < z \leq H$, the remaining interval has length $[T - (A + B - 1)z]_+$. The other two slopes are $1/A + D - 1$ and $1/B + 1/D - 1$. Each exceeds -1 , and at most one is negative: for example $A + B < 1$ makes both other slopes positive, and permutation proves the other cases. The three exterior areas therefore total at most $3TH + H^2/2$. Hence the whole area is at most

$$T^2/2 + 3TH + H^2/2 \leq (T + H)^2 = (S_0 - 2H)^2,$$

where the difference is $(T - H)^2/2$. This is twice the area of the triangle obtained by requiring two fixed coordinates to be at least H . Integration against the nonnegative slice density of the other $n - 3$ coordinates gives $2(\ell - 2H)_+^{n-1}/(n - 1)!$. Here $n \geq 4$, since the support has dimension at least two.

Step 2. In this step, we obtain the exact three-pair improvement and the sevenfold specializations. If $H/\ell \geq 4/(3n + 4)$, Lemma 7.13 removes the factor two. In both comparisons the original simultaneous halfspace region still has at most three additional faces, so its lattice error is the same uniform $C_n(\ell + 1)^{n-2}$. Substitution of $(n, t, s) = (7, 0, 4)$ and $(7, 0, 3)$ in the general formula gives D_4 and the first branch of K_5 ; the two three-pair improvements give its remaining bounds and endpoint. For the other two sevenfold specializations, direct substitution gives

$$\begin{aligned} \mathcal{S}_{7,1,3}(c) &= (1 - c)_+^6 + 3(1 - 2c)_+^6 - 5(1 - 3c)_+^6 + 2(1 - 4c)_+^6, \\ \mathcal{S}_{7,1,2}(c) &= (1 - c)_+^6 + (1 - 2c)_+^6 - (1 - 3c)_+^6. \end{aligned}$$

These apply to embedding-six fourfold and fivefold centers, respectively, retaining the common linear normal equation. $\square$

Proposition 7.19. *Use the local notation of Theorem 7.9. If $n \geq 6, c = 3$, with no tangent hypothesis, then for $1/(\ell\lambda) \geq 1/5$ the leading capacity can be replaced by $1053/3125$. If $n \geq 7, c = 4$, and the full tangent of R/P is reduced Cohen–Macaulay of embedding dimension n , then for $u = 1/(\ell\lambda) \geq 1/6$ it can be replaced by*

$$B_n^*(u) = \frac{138259}{900000} + \left(1 - \frac{138259}{900000}\right) (1 - u)_+^{n-1}.$$

For the same full-tangent hypothesis with $c = 3, n \geq 6$, the constant $138259/900000$ itself is available at $u \geq 1/6$. If $n \geq 5, c = 3$, the embedding dimension of R/P is $n - 1$, and its full tangent is reduced Cohen–Macaulay, then at $u \geq 4/(3n + 1)$ the leading capacity can be replaced by

$$M_n(u) = (1 - u)_+^{n-1} + (1 - 2u)_+^{n-1} - (1 - 3u)_+^{n-1}.$$

Each bound retains the entire initial image with a uniform $O_n((\ell + 1)^{n-2})$ error.

The common linear normal equation in the final case is included in the capacity M_n .

Proof. We prove the four bounds by conditioning in the simplex obtained from the full degenerations above. In all cases we count the original halfspace region, so its bounded number of faces gives the uniform error independently of the conditional weights.

Case 1. The prime has height three and no tangent hypothesis is imposed. For height three with no tangent hypothesis, Lemma 7.8 gives one stochastic triple. If $n \geq 6$, normalization of the first six simplex coordinates by their sum gives a uniform six-coordinate simplex, and increases each of those coordinates. Lemma 7.12 therefore gives $1053/3125$ at every $u \geq 1/5$.

Case 2. The prime has height three and full embedding dimension n . For height three and full embedding dimension n , Lemma 7.17 gives four active coordinates and four zero-diagonal stochastic rows. Normalization of the first six coordinates and Lemma 7.16 give $138259/900000$ for $n \geq 6$ and $u \geq 1/6$.

Case 3. The prime has height four and full embedding dimension n . For height four and full embedding dimension n , the same lemma gives five active coordinates and five such rows. Put

$y = Z_5$. On $y \geq u$ bound by the entire slice. On $y < u$, a row among the first four with coefficient $a_{i5} = 1$ has no solutions. Otherwise its other coefficients, divided by $1 - a_{i5}$, satisfy

$$\sum_{\substack{j \leq 4 \\ j \neq i}} \frac{a_{ij}}{1 - a_{i5}} Z_j \geq \frac{u - a_{i5}y}{1 - a_{i5}} \geq u.$$

Conditional on y , the other coordinates normalized by $1 - y$ form an $(n - 1)$ -coordinate simplex. The preceding four-row bound applies, since $n - 1 \geq 6$ and $u/(1 - y) \geq 1/6$. Writing $C_4 = 138259/900000$, the total fraction is at most

$$\Pr(Z_5 \geq u) + C_4 \Pr(Z_5 < u) = C_4 + (1 - C_4)(1 - u)_+^{n-1} = B_n^*(u).$$

Case 4. The prime has height three and embedding dimension $n - 1$. Finally, when the embedding dimension is $n - 1$ and the height is three, the full degeneration has one common normal coordinate y and three active coordinates. At the three generic branches every monomial satisfies

$$\alpha_i y + \beta_i x_j + \gamma_i x_k \geq H, \quad \alpha_i + \beta_i + \gamma_i = 1, \quad H = 1/\lambda,$$

where $\{i, j, k\} = \{1, 2, 3\}$ and the weights are positive. For $0 \leq y < H$, division by $1 - \alpha_i$ gives a pair constraint at level at least H on x_j, x_k . The conditional simplex has $n - 1$ coordinates and size $\ell - y$, and

$$\frac{H}{\ell - y} \geq \frac{H}{\ell} \geq \frac{4}{3n + 1} = \frac{4}{3(n - 1) + 4}.$$

Lemma 7.13 bounds its volume by $(\ell - y - 2H)_+^{n-2}/(n - 2)!$. For $y \geq H$ use the whole slice. Integrating these two pieces gives

$$\frac{1}{(n - 1)!} ((\ell - H)_+^{n-1} + (\ell - 2H)_+^{n-1} - (\ell - 3H)_+^{n-1}).$$

This is the asserted M_n bound. Its nonnegativity and monotonicity in u also follow from its interpretation as $\Pr(Z_1 \geq u \text{ or } (Z_2 \geq u, Z_3 \geq u))$. All four comparisons concern the full ideals supplied by the degenerations, so no additional error depends on their exponents. $\square$

Lemma 7.20. *For a uniform point Z on the seven-coordinate simplex,*

$$\max_{a_i \geq 0, \sum_{i=1}^3 a_i = 1} \Pr \left(\sum_{i=1}^3 a_i Z_i \geq 1/6 \right) = \frac{256}{729}.$$

If all three coefficients are at least $1/6$, the maximum is $11/32$. At every level $c \geq 0$ the three-coordinate tail is at most $\min\{1, 256/729 + 18(1/6 - c)_+\}$.

The unrestricted maximum is attained at $(1/2, 1/2, 0)$.

Proof. We remove a small coefficient, optimize the remaining compact three-coefficient region, and control changes of level uniformly.

Step 1. In this step, we remove a coefficient at most $1/6$ and obtain the two-coordinate bound. Set $h = 1/6$. The density of $AV_1 + BV_2$ on a six-coordinate simplex, for $A > B > 0$, is

$$f_{A,B}(t) = \frac{5}{A - B} ((1 - t/A)_+^4 - (1 - t/B)_+^4).$$

This follows by differentiating (7.1) and scaling the coefficients. It decreases for $t \geq B$. Below B , its derivative is nonpositive precisely when $(1 - t/A)^3/A \geq (1 - t/B)^3/B$. With $r = (A/B)^{1/3}$, the equality level is

$$t_0 = \frac{Br^3}{r^3 + r^2 + r + 1} \leq \frac{A + B}{8}.$$

After division by B , the last comparison follows from

$$(r^3 + 1)(r^3 + r^2 + r + 1) - 8r^3 = (r^6 + 1 - 2r^3) + (r^5 + r - 2r^3) + (r^4 + r^2 - 2r^3) \geq 0.$$

For $A = B$ the density is $20t(1 - t/A)^3/A^2$, with maximum at $A/4$; for $B = 0$ it decreases. Thus the same monotonicity range holds in these limiting cases.

Fix $p \in [0, 1]$ and put $S_z = (1 - z)(pZ_1 + (1 - p)Z_2) + zZ_3$, $0 \leq z \leq h$. The moving-facet calculation in Lemma 7.12 gives

$$\frac{d}{dz} \Pr(S_z \geq h) = \frac{f_{S_z}(h)}{1 - z} (\mathbb{E}[Z_3 \mid S_z = h] - h).$$

With $A = (1 - z)p$, $B = (1 - z)(1 - p)$, the conditional density of $Z_3 = t$ on the facet is proportional to

$$(1 - t)^4 f_{A,B} \left(\frac{h - zt}{1 - t} \right).$$

The second factor decreases: its argument increases from $h > (A + B)/8$. The density $5(1 - t)^4$ has mean h , and the covariance calculation in that lemma shows that a nonnegative decreasing weight cannot increase this mean. The derivative is nonpositive. The simplex divided-difference formula has only finitely many coefficient and level coincidences; continuity extends the conclusion across them, and approximation includes zero or repeated weights. Thus any coefficient at most h can be removed and the others renormalized. The resulting two-coordinate maximum is

$$Q_7(h) = \max \left\{ \frac{15625}{46656}, \frac{16384}{46656} \right\} = \frac{256}{729}.$$

Step 2. In this step, we maximize the tail when all three coefficients are at least $1/6$. Put $x_i = 6a_i \geq 1$, so $\sum x_i = 6$, and write $s_2 = \sum_{i < j} x_i x_j$, $s_3 = x_1 x_2 x_3$. The simplex formula justified in Lemma 7.12 expresses the probability as the second divided difference of $(x - 1)^6/x^4$. Expanding it gives the continuous expression

$$F(s_2, s_3) = 1 - \frac{20}{s_3} + \frac{15s_2 + 37}{s_3^2} - \frac{6s_2^2 + 12s_2}{s_3^3} + \frac{s_2^3}{s_3^4}.$$

For example the divided differences of x^{-j} , $j = 1, 2, 3, 4$, are respectively

$$\frac{1}{s_3}, \quad \frac{s_2}{s_3^2}, \quad \frac{s_2^2 - 6s_3}{s_3^3}, \quad \frac{s_2^3 - 12s_2 s_3 + s_3^2}{s_3^4},$$

as follows by a common denominator. These identities and $(x - 1)^6/x^4 = x^2 - 6x + 15 - 20/x + 15/x^2 - 6/x^3 + 1/x^4$ prove the formula including coincident coordinates.

Writing $x_i = 1 + y_i$ gives $s_3 \geq 4$. Moreover

$$F_{s_2} = \frac{3[5s_3^2 - 4(s_2 + 1)s_3 + s_2^2]}{s_3^4} > 0,$$

since the bracket divided by s_3^2 is $(s_2/s_3 - 2)^2 + 1 - 4/s_3$. Equality $s_3 = 4$ forces (x_1, x_2, x_3) to be a permutation of $(1, 1, 4)$, where the bracket is again positive. At an interior constrained critical point the difference of two multiplier equations is $(x_j - x_i)(F_{s_2} + x_k F_{s_3}) = 0$. Three distinct coordinates would force both partial derivatives to vanish. A maximum therefore lies on the boundary or has repeated coordinates.

On $x_3 = 1$, put $v = x_1 x_2 \in [4, 25/4]$. Then

$$F = 1 - 10/v + 55/v^2 - 135/v^3 + 125/v^4, \quad \frac{dF}{dv} = \frac{5(v - 4)(2v^2 - 14v + 25)}{v^5} \geq 0.$$

The quadratic has negative discriminant and positive leading coefficient. This boundary maximum is $1053/3125$. For $(x_1, x_2, x_3) = (x, x, 6 - 2x)$, $1 \leq x \leq 5/2$, one has $s_2 = 3x(4 - x)$, $s_3 = 2x^2(3 - x)$ and

$$\frac{dF}{dx} = \frac{12x^4(2 - x)(x - 1)(5 - 2x)R(3 - x)}{s_3^5}, \quad R(z) = 40z^4 - 200z^3 + 315z^2 - 162z + 27.$$

To check the factorization, use $s'_2 = 6(2 - x)$, $s'_3 = 6x(2 - x)$ and $s_3^5(F_{s_2} + xF_{s_3}) = 2x^4(x - 1)(5 - 2x)R(3 - x)$. The second derivative of R is $30(16z^2 - 40z + 21)$, with roots $3/4, 7/4$. Its first

derivative at $1/2, 3/4, 7/4, 2$ is respectively $23, 81/2, -79/2, -22$. It follows that R has just one interior extremum on $[1/2, 2]$, a maximum. Its endpoint values are $9/4$ and 3 , so it is positive throughout. The displayed probability increases to $x = 2$ and then decreases. Its maximum is $F(12, 8) = 11/32$. Since $1053/3125 < 11/32 < 256/729$, both asserted maxima follow.

Step 3. In this step, we bound the change in probability uniformly as the level varies. Choose a largest coefficient $a_i \geq 1/3$ and an unused coordinate Z_7 . Conditional on the other coordinates, Z_i is uniform on $[0, S]$, where $S = Z_i + Z_7$. An interval of length ϵ for the weighted form has conditional probability at most $\epsilon/(a_i S)$. The density of S is $30S(1 - S)^4$, whence $\mathbb{E}(1/S) = 6$. Every interval therefore has probability at most 18ϵ . Comparison with the tail at $1/6$ proves the final assertion for lower levels; monotonicity covers higher levels. $\square$

Proposition 7.21. *Let $(R, \mathfrak{m})$ be the local ring at a closed point of a smooth complex n -fold, $n \geq 7$. Let P be prime, let I be a proper P -primary ideal, and put $\lambda = \text{lct}_{R_P}(IR_P)$. Then the following hold.*

(1) *If $\text{ht } P = 3$, then for every integer $\ell \geq 1$, with $c = 1/(\ell\lambda)$,*

$$\dim_{\mathbb{C}}(\text{in}_{\mathfrak{m}} I)_{\ell} \leq \frac{\ell^{n-1}}{(n-1)!} \min\{1, 256/729 + 18(1/6 - c)_+\} + C_n(\ell + 1)^{n-2}.$$

(2) *If $\text{ht } P = 4$, the full tangent ring $\text{gr}_{\mathfrak{m}}(R/P)$ is reduced Cohen–Macaulay, and $\text{edim}(R/P) \in \{n-1, n\}$, then for every integer $\ell \geq 1$ with $c = 1/(\ell\lambda) \geq 1/6$,*

$$\dim_{\mathbb{C}}(\text{in}_{\mathfrak{m}} I)_{\ell} \leq \frac{\ell^{n-1}}{(n-1)!} \left[\frac{138259}{900000} + \left(1 - \frac{138259}{900000}\right) (1 - c)_+^{n-1} \right] + C_n(\ell + 1)^{n-2}.$$

The constant C_n depends only on n , uniformly in the prime, full ideal, primary exponents, tangent degree, collision schemes, weights and computing place.

The first estimate has no tangent hypothesis. The second includes the common linear normal equation when the embedding dimension is $n - 1$.

Proof. Case 1. The prime has height three. For height three, Lemma 7.8 places the full monomial ideal in a stochastic three-coordinate cap. For $n \geq 7$, dividing the first seven simplex coordinates by their sum gives a uniform seven-coordinate point and increases each active coordinate. Lemma 7.20 therefore bounds the volume. The original halfspace count retains its dimension-only error.

Case 2. The prime has height four. For height four and embedding dimension n , use Proposition 7.19. In embedding dimension $n - 1$, Lemma 7.17 has $t = 1, s = 4$. With common normal coordinate y , every monomial satisfies four inequalities

$$\alpha_i y + \sum_{j \neq i} \beta_{ij} x_j \geq 1/\lambda, \quad \alpha_i + \sum_{j \neq i} \beta_{ij} = 1.$$

On the normalized slice $y \geq c$, use the whole simplex. On $y < c$, a row with $\alpha_i = 1$ has no solutions; otherwise division by $1 - \alpha_i$ leaves a stochastic three-coordinate inequality at level at least c . Conditional normalization of the other $n - 1 \geq 6$ coordinates raises this level to at least $c/(1 - y) \geq 1/6$. All four zero-diagonal rows are present. By Lemma 7.16, their conditional fraction is at most $C_4 = 138259/900000$. Their total fraction is therefore at most $C_4 + (1 - C_4) \Pr(y \geq c) = C_4 + (1 - C_4)(1 - c)_+^{n-1}$. Count the original four-halfspace region, before conditioning, to obtain the uniform lattice error even at limiting weights. $\square$

Proposition 7.22. *Let $(R, \mathfrak{m})$ be the local ring at a closed point of a smooth complex sevenfold, let P have height $1 \leq r \leq 5$, and let I be a proper P -primary ideal. Put $\lambda = \text{lct}_{R_P}(IR_P)$, and, for $1/6 \leq c \leq 1/2$,*

$$Q_5(c) = \max \left\{ (1 - c)^6, 1 - \left(\frac{29 - 28c}{36 - 42c} \right)^5 \right\}, \quad T_5(c) = \max \left\{ (1 - c)^6, \left(\frac{7}{6} \right)^6 \left(\frac{1 - 2c}{1 - c} \right)^5 \right\}.$$

For every integer $\ell \geq 1$, put $c = 1/(\ell\lambda)$. If $1/6 \leq c \leq 1/2$, then

$$\dim_{\mathbb{C}}(\text{in}_{\mathbf{m}} I)_{\ell} \leq \frac{\ell^6}{6!} \min\{E_{7,5}(c), Q_5(c), T_5(c)\} + C_7(\ell + 1)^5.$$

If $c \geq 1/3$, then the same full image satisfies

$$\dim_{\mathbb{C}}(\text{in}_{\mathbf{m}} I)_{\ell} \leq \frac{\ell^6}{6!} (1 - c)_+^6 + C_7(\ell + 1)^5.$$

The constant depends only on the ambient dimension and is independent of P, I, ℓ, λ and the computing place.

Both refinements depend on the largest normal weight, so adjoining zero weights includes every height at most five.

Proof. Use Lemma 7.8 and append zeros to its r weights to obtain five stochastic weights, then two further zeros. Write $Y = \sum w_i Z_i$ and $p = \max w_i$. The form has no atoms, since some weight is positive and some is zero. We prove the two new bounds on $1/6 \leq c < 1/2$; Theorem 7.9 supplies $c = 1/2$ and all larger levels.

Step 1. In this step, we prove the coordinate bound when the largest weight exceeds $1 - c$. If $p > 1 - c$, every other weight is less than c . The one-vertex simplex calculation gives

$$\Pr(Y \geq c) = \frac{(p - c)^6}{\prod_{i \neq i_0} (p - w_i)} \leq G(p) := \frac{(p - c)^6}{p^5(2p - 1)}.$$

The inequality merges the other weights using $(p - a)(p - b) \geq p(p - a - b)$. The logarithmic derivative of G has the sign of $(12c - 1)p - 5c$. Its value at $p = 1 - c$ is $(6c - 1)(1 - 2c) \geq 0$, and its slope is positive. Thus $G(p) \leq G(1) = (1 - c)^6$.

Step 2. In this step, we derive the Q_5 bound when every weight is at most $1 - c$. Suppose $p \leq 1 - c$, and set $f(t) = \Pr(Y \leq t)^{1/5}$. The concavity and centroid calculation in the proof of Theorem 7.9 give $f(1/7) \geq 5/6$. Zero weights are included by approximation; the atomless limiting distribution gives convergence of every distribution value and hence preserves concavity. Since $f(1 - c) = 1$, its chord from $1/7$ to $1 - c$ gives

$$f(c) \geq \frac{5}{6} + \frac{c - 1/7}{6(1 - c - 1/7)} = \frac{29 - 28c}{36 - 42c}.$$

This proves the second branch of Q_5 in the capped case.

Step 3. In this step, we derive the T_5 bound and its coordinate endpoint range. Put $p_0 = 1 - c$, $q_0 = c$. The vertices of $\{w : 0 \leq w_i \leq p_0, \sum w_i = 1\}$ are exactly the permutations of $(p_0, q_0, 0, \dots, 0)$. Indeed two coordinates strictly between their bounds can be perturbed in opposite directions; a vertex thus has at most one such coordinate. It has exactly one coordinate equal to $p_0 > 1/2$, and the remaining mass is q_0 . The same perturbations, iterated until a bound is reached, express every capped vector as a convex combination of these vertices.

For $0 \leq v < c$, the expectation $\mathbb{E}[(\sum w_i Z_i - v)_+]$ is convex in the weights and symmetric. Thus, with $Y_0 = p_0 Z_1 + q_0 Z_2$,

$$\mathbb{E}[(Y - v)_+] \leq \mathbb{E}[(Y_0 - v)_+] = \frac{1}{7(p_0 - q_0)} \left(\frac{(p_0 - v)^7}{p_0^5} - \frac{(q_0 - v)^7}{q_0^5} \right).$$

The equality integrates (7.1) from v to infinity. Discarding its nonnegative second term and using $(Y - v)_+ \geq (c - v)\mathbf{1}_{Y \geq c}$, choose $v = (8c - 1)/6 \in [0, c)$. Then $c - v = (1 - 2c)/6$, $p_0 - v = 7(1 - 2c)/6$, and

$$\Pr(Y \geq c) \leq \left(\frac{7}{6} \right)^6 \left(\frac{1 - 2c}{1 - c} \right)^5.$$

This proves T_5 , with the large-maximum case already covered.

The ratio of this last expression to $(1 - c)^6$ is

$$\mathcal{R}(c) = \left(\frac{7}{6}\right)^6 \frac{(1 - 2c)^5}{(1 - c)^{11}}.$$

At $c = 1/3$, the inequality $\mathcal{R}(c) < 1$ reduces to $7^6 = 117649 < 131072 = 2^{17}$. Its logarithmic derivative has numerator $1 - 12c$ over $(1 - c)(1 - 2c) > 0$, so this comparison persists to $1/2$. The coordinate weight attains $(1 - c)^6$. This proves the exact endpoint from $1/3$ onward. The rational chord endpoint has derivative $210/(36 - 42c)^2 > 0$, and $(1 - 2c)/(1 - c)$ decreases. Thus both Q_5 and T_5 are nonincreasing on their stated interval. Finally apply each valid probability bound to the same full monomial image and take their minimum; the uniform error remains unchanged. $\square$

Theorem 7.23. *Let q, r be positive integers with $r < q \sup\{\text{ord}_x f/a : 0 \neq f \in H^0(X, aL), a \geq 1\}$. For $U_j = H^0(X, qjL \otimes \mathfrak{m}_x^{rj})$, fix the uniform constants before an actual sufficiently late pure threshold τ with a positive-dimensional center of codimension c . For every $0 < h < 1/(j\tau) - C/j$, there is a finite $a > h/r$ such that, for every rational $R_0 > 0$,*

$$\text{vol}(q\pi^*L - R_0E_x) \geq q^n L^n - n \int_0^{R_0} t^{n-1} \mathfrak{m}_{n,c} \left(\frac{(h + a(t - r))_+}{t} \right) dt.$$

The integrand is zero at $t = 0$. The explicit applicable capacities in this section may replace $\mathfrak{m}_{n,c}$. The same selected place and profile control every ordinary point-order layer, on both sides of the observed order. Bounds from different actual observations may be combined by their pointwise minimum on the same original images.

The line in the integrand is chosen once. In particular the layers below the observation at r belong to the same estimate as the later layers.

Proof. We first choose a supporting line for one actual valuation, then sum the full ordinary point filtration.

Step 1. In this step, we choose a supporting line for one actual computing profile. The strict inequality on r supplies an actual section with normalized order greater than r/q . On the point blowup, a convex combination of its effective divisor and the big class $q\pi^*L$ shows that $q\pi^*L - rE_x$ is big. Choose rationals $r < b_0 < \beta$ with $q\pi^*L - \beta E_x$ big and apply Theorem 6.13 on $[0, b_0]$. Its cutoff and error constant are fixed before the actual degree j .

There is a place v centered on the observed center with $\tau v(\mathfrak{b}(U_j)) = A_X(v)$. To see this for the actual average, principalize its full base ideal and choose sufficiently many general members transverse to the resolved fixed boundary. Their individual moving coefficients are less than one. The coefficient-one strata are then those of the ideal pair, with its entire fixed part. Write $A_v = A_X(v)$, and define for rational $t \geq 0$

$$\phi(t) = \inf \left\{ \frac{v(f)}{kA_v} : 0 \neq f \in H^0(X, qkL), \text{ord}_x f \geq kt \right\},$$

using $+\infty$ for an empty set. Products of actual sections make this function convex on rational parameters. It is nonnegative, $\phi(0) = 0$, and it is finite on an open interval containing r . Uniform approximation gives

$$\phi(r) \geq \frac{1}{j\tau} - \frac{C}{j} > h.$$

Choose a finite supporting slope a at r . Its inequality at zero implies $a \geq \phi(r)/r > h/r$. Convex secant inequalities extend that support line to every rational argument at which an actual section exists. Thus every actual section of degree qk and point order at least ℓ satisfies

$$v(f) \geq kA_v(h + a(\ell/k - r))_+.$$

This uses only products and powers of complete global sections.

Step 2. In this step, we sum the full ordinary layers using the selected profile. Put $T(t) = h + a(t - r)$, and write P for the center prime. When $T(\ell/k) > 0$, the full valuation ideal

$$I_{k,\ell} = \{f \in \mathcal{O}_{X,x} : v(f) \geq kA_v T(\ell/k)\}$$

is proper P -primary: elements outside P have value zero, and a sufficiently high power of its finitely generated ideal lies in $I_{k,\ell}$. Its generic threshold is at most $1/(kT(\ell/k))$. The germ map gives an injection

$$\frac{H^0(X, qkL \otimes \mathfrak{m}_x^\ell)}{H^0(X, qkL \otimes \mathfrak{m}_x^{\ell+1})} \hookrightarrow (\text{in}_{\mathfrak{m}} I_{k,\ell})_\ell.$$

Lemma 7.8 bounds this actual image with capacity argument $T(\ell/k)_+ / (\ell/k)$. If $T(\ell/k) \leq 0$, use the entire polynomial layer, corresponding to $\mathfrak{m}_{n,c}(0) = 1$. At $\ell = 0$ its dimension is at most one. These are all the ordinary layers, with their full scheme structure.

For rational $R_0 > 0$, sum over $0 \leq \ell < \lceil kR_0 \rceil$. The total error is $O_{n,R_0}(k^{n-1})$, and the leading sum tends to the stated integral. Rounding the upper endpoint changes only that same error. The ample Hilbert polynomial has normalized limit $q^n L^n$. Subtracting the upper loss bounds the liminf of the actual final spaces from below. Along divisible degrees these spaces are exactly the complete sections of $k(q\pi^*L - R_0E_x)$; their volume is at least that liminf, whether or not this divisor was known to be big. This proves the assertion for every R_0 , with the same already chosen v, a .

The other capacities apply to these same valuation ideals wherever their stated tangent and threshold hypotheses hold. Each explicit envelope is bounded and piecewise continuous with finitely many transition values. Where $T(t) > 0$, its argument is $T(t)/t = a + (h - ar)/t$, which is strictly increasing since $a > h/r$; on the remaining interval the argument is zero. The composed leading functions are therefore Riemann integrable, including at their transition values. For finitely many actual observations on the same complete section images, every corresponding bound holds on each layer. Taking their minimum before summation is therefore valid; the constants are the maximum of the finitely many fixed errors. No sum of the trial boundaries enters this argument. $\square$

Corollary 7.24. *Let X be smooth projective of dimension n , let L be ample Cartier, and fix integers $p, k \geq 1$ and rational $0 < u < v$ with ku, kv integral. Suppose the complete space $U_u = H^0(X, pkL \otimes \mathfrak{m}_x^{ku})$ is nonzero. Let $B_0 \sim_{\mathbb{Q}} s_0 L$ be effective and lc near x , and let $0 \leq \theta < 1$ be rational. Take the actual threshold τ of its full source ideal against $(X, \theta B_0)$, and an actual sufficiently general threshold average with minimal center W of positive dimension and deficit b . Put $\eta = (1 - \theta)/(k\tau)$ and $U_v = H^0(X, pkL \otimes \mathfrak{m}_x^{kv})$. Then the following bounds hold, with a uniform $O((k+1)^{n-1})$ error on each fixed bounded interval of orders:*

(i) *If $\text{codim } W = 3$, $n \geq 6$, and $\eta/v \geq 1/5$, the leading upper bound for $\dim U_u - \dim U_v$ is*

$$\frac{k^n}{n!} \frac{1053}{3125} (v^n - u^n).$$

(ii) *If $\text{codim } W = 4$, $n \geq 7$, its full tangent is reduced Cohen–Macaulay of embedding dimension $n - 1$ or n , and $\eta/v \geq 1/6$, the leading upper bound is*

$$\frac{k^n}{n!} \left[\frac{138259}{900000} (v^n - u^n) + \left(1 - \frac{138259}{900000} \right) ((v - \eta)_+^n - (u - \eta)_+^n) \right].$$

(iii) *If $\text{codim } W = 3$, $n \geq 5$, its full tangent is reduced Cohen–Macaulay of embedding dimension $n - 1$, and $\eta/v \geq 4/(3n + 1)$, the leading upper bound is*

$$\frac{k^n}{n!} \sum_{j=1}^3 \epsilon_j ((v - j\eta)_+^n - (u - j\eta)_+^n), \quad (\epsilon_1, \epsilon_2, \epsilon_3) = (1, 1, -1).$$

(iv) If $n \geq 3$, $d = \dim W \geq 2$, and the full tangent is reduced Cohen–Macaulay of embedding dimension e , put $t = n - e$, $s = e - d + 1$, and

$$a_j = (-1)^{j+1} \left[\binom{t+s}{j} - s \binom{t+s-1}{j-1} \right].$$

The leading upper bound is

$$\frac{k^n}{n!} \sum_{j=1}^{t+s} a_j ((v - j\eta)_+^n - (u - j\eta)_+^n).$$

The signed sum is nonnegative. When $t = 0, s = 3$, it may also be replaced by $2[(v - 2\eta)_+^n - (u - 2\eta)_+^n]$, and the factor two may be replaced by one if $\eta/v \geq 4/(3n + 4)$.

(v) If $n \geq 7$, $\text{codim } W = 3$, and $\eta/v \geq 497/3000$, the leading upper bound is

$$\frac{k^n}{n!} \frac{3}{8} (v^n - u^n).$$

The whole threshold boundary has coefficient $\theta s_0 + pk\tau$, and

$$\eta \geq \frac{(1 - \theta)u}{n - b - \theta \text{mult}_x B_0} > 0.$$

No nonvanishing hypothesis is imposed on U_v .

These are finite estimates at the original ambient degree, with the entire incoming boundary retained.

Proof. We compare the actual shell layers with a full valuation ideal and then integrate each applicable capacity.

Step 1. In this step, we place every actual shell layer in the full primary comparison image. Let w be an actual computing place centered on W , and let I be the full ideal of U_u . Then

$$(1 - \theta)A_X(w) \leq A_X(w) - \theta w(B_0) = \tau w(I).$$

The full valuation ideal $J = \{f : w(f) \geq w(I)\}$ contains I , is primary for the center prime, and has generic threshold $\lambda \leq \tau/(1 - \theta)$. Every actual layer $ku \leq \ell < kv$ maps into $(\text{in}_{\mathfrak{m}} J)_\ell$, where $1/(\ell\lambda) \geq k\eta/\ell \geq \eta/v$.

Step 2. In this step, we integrate the five capacity bounds and obtain the lower bound for η . Proposition 7.19 applies in (i) and (iii), and Proposition 7.21 applies in (ii). The constant bound integrates to (i), and the two increasing terms of the B_n^* bound give (ii). For (iii), the signed leading expression is the volume of $\{y_1 \geq k\eta\} \cup \{y_2, y_3 \geq k\eta\}$ in an expanding simplex. It is nonnegative and increasing with ℓ , so its left-endpoint sum is bounded by its integral, giving exactly (iii). In (iv), Theorem 7.18 gives the union event with one common coordinate or two active coordinates at absolute level $k\eta$. Eliminating a free coordinate identifies its leading count with the volume in an expanding simplex. It is nonnegative and increasing in ℓ , and termwise integration gives the stated signed sum. The two three-pair improvements integrate in the same way. In (v), Proposition 7.21 gives the uniform fraction $256/729 + 9/500 < 3/8$ on every layer, proving the bound. All errors sum to $O((k + 1)^{n-1})$. Finally the point blowup gives $b + \theta \text{mult}_x B_0 + ku\tau \leq n$; its last positive term justifies division and proves the lower bound for η . $\square$

Proposition 7.25. *At a nongeneration point on a sevenfold, the nonzero original $U_k(u)$, for rational $u, u + h$ with integral $ku, k(u + h)$, satisfies*

$$\dim U_k(u) - \dim U_k(u + h) \leq a((u + h)^7 - u^7) \frac{k^7}{7!} + C(k + 1)^6$$

when $0 < h \leq 1/100$, with $a = 11/25$ on $[51/50, 26/25]$, and $a = 17/50$ on $[26/25, 29/25]$. The next source is not assumed nonzero. The constant is uniform in the actual source, center and place on each fixed finite mesh.

Both endpoints of each shell lie in its indicated closed interval. The estimate can therefore be iterated on a fixed mesh after its common error allowance and original degree have been chosen.

Proof. We use the complete scalar census to obtain tangent hypotheses for every surviving low-dimensional center. We then apply an appropriate full-primary bound to each ordinary layer of the actual source.

Step 1. In this step, we derive the scalar restrictions for the actual source. Fix a nonzero $U_k(u)$, its full ideal I , its own threshold τ , and its sufficiently general pure average Γ . Write $s = k\tau$, $M = \text{mult}_x \Gamma$, and let b be its deficit. The complete source and the point blowup give

$$(7.4) \quad M \geq us, \quad M + b \leq 7, \quad s \leq (7 - b)/u.$$

The first inequality concerns the entire average. If its minimal center is the point, then $s \leq 7/u < 8$ and Proposition 3.12 contradicts nongeneration.

Use the complete rational supersolutions for $\mathcal{F}_v$ in the scalar table supporting Proposition 5.9, with degree floors one. Write $H_v(b)$ for the minimum of its certified affine lines. Every such line $ab + p$ bounds the whole descendant cost by Theorem 5.4. Failure of payment therefore requires

$$(7.5) \quad 8 \leq s + ab + p, \quad 0 \leq \sigma = 7 - M - b \leq 7 - b + u(ab + p - 8).$$

These implications include both cutting errors. Indeed, a strict gap in $s + H_v(b) < 8$ is fixed before the cuts, and at each actual radius r_i the error in the two cutting inequalities is chosen smaller than its allotted gap divided by $1 + r_i$. Theorem 5.4 then spends only that allotted amount. The census includes all adjacent caps and dimension skips, including the embedding-specific multiplicity-eight and nine rows of Lemma 5.10.

Step 2. In this step, we prove the shell rate on $[51/50, 26/25]$. Put $u_* = 51/50$. Every affine slope in the complete census is at least one. Hence $(7 - b)/u_* + H_v(b)$ increases on the entire deficit interval of a state, and its maximum occurs at the cap. Substitution into that complete table gives the following upper maxima after retaining the states in the next display:

dimension	maximum for omitted states	state attaining it
1	117/17	(1, 1, 1)
2	376/51	(2, 5, 4)
3	101747/12750	(3, 6, 4)
4	815567/102000	(4, 7, 8)

All are below eight, with smallest gap 433/102000. This calculation uses the minimum of every row's available lines before evaluating its cap. For example, the last two values are $275/51 + 647/250$ and $575/102 + 4717/2000$. Increasing u only decreases the source term. Thus these states are all paid, including their complete tails.

The surviving threefold and fourfold states, their caps, and chosen complete-tail lines are exactly

(d, e, m)	β_v	a	p
(3, 7, 5)	7/5	171/100	211/1000
(4, 5, 2)	3	119/100	287/500
(4, 6, 3)	8/3	1317/1000	71/125
(4, 6, 4)	2	283/200	379/1000
(4, 7, 4)	5/2	283/200	29/50
(4, 7, 5)	2	187/125	17/40
(4, 7, 6)	5/3	783/500	331/1000
(4, 7, 7)	10/7	1627/1000	11/40

Dimensions five and six remain included. For each displayed row, (7.4)–(7.5) give

$$b \geq \beta(u) = \frac{(8-p)u-7}{au-1}.$$

Its denominator is positive and its derivative has the sign of $7a+p-8 > 0$. The upper bound for σ increases with b , and its value at β_v decreases with u , since $a\beta_v + p < 8$. The exact endpoint values are

(d, e, m)	$\beta(u_*)$	upper bound for σ
(3, 7, 5)	47239/37210	971/10000
(4, 5, 2)	14363/5345	209/3125
(4, 6, 3)	29032/17167	628/1875
(4, 6, 4)	38671/22165	5659/50000
(4, 7, 4)	5684/4433	10797/20000
(4, 7, 5)	36325/26296	16267/50000
(4, 7, 6)	41119/29866	25973/150000
(4, 7, 7)	43975/32977	4389/70000

In particular the embedding-seven rows have $b > 5/4$, the embedding-six fourfolds have $b > 5/3$, and the embedding-five fourfold has $b > 8/3$.

All these rows have $b > 1$ and $\sigma < 1$, so $\delta = 7 - M = b + \sigma > 1$. Theorem 4.20 gives the full ordinary tangent ring Cohen–Macaulay and every ordinary power integrally closed. Every displayed slack except that of (4, 7, 4) is below $1/2$, so the same theorem gives full reducedness. In that remaining row, if $b \leq 9/4$, substituting $9/4$ instead of the cap into (7.5) gives $\sigma \leq 17161/40000 < 1/2$. If $b > 9/4$, its minimal multiplicity and the full-cycle assertion of the same theorem exclude a nonzero doubled-component contribution. Thus it is generically reduced on every component; the full Cohen–Macaulay ring has no embedded associated primes, so it is reduced. This handles its affine vertex as well as its projective tangent.

Choose an actual computing place w centered on the current minimal center. The full valuation ideal $J_w = \{f : w(f) \geq w(I)\}$ contains I and has generic threshold $\lambda_w \leq \tau$. At every actual shell layer $ku \leq \ell < k(u+h)$, its capacity argument satisfies

$$(7.6) \quad \frac{1}{\ell\lambda_w} \geq \frac{u}{(u+h)(7-b)}.$$

For $u \geq 51/50$, $h \leq 1/100$, one has $u/(u+h) \geq 102/103$. For the threefold and the embedding-seven fourfolds the argument is greater than $408/2369 > 1/6$. The threefold uses B_7^* , and the fourfolds use $138259/900000$. To check the common rate, put $C_4 = 138259/900000 < 31/200$. Since $(5/6)^6 < 67/200$,

$$B_7^*(1/6) < 31/200 + (169/200)(67/200) = 17523/40000 < 11/25.$$

For the embedding-six fourfolds, the argument is greater than $153/824 > 2/11$, and

$$M_7(2/11) = \frac{9^6 + 7^6 - 5^6}{11^6} = \frac{633465}{1771561} < 2/5 < 11/25.$$

The embedding-five fourfold has argument greater than $306/1339 > 1/5$, so its rate is at most $1053/3125 < 11/25$. All uses of Proposition 7.19 thus have their full tangent and interval hypotheses established.

Every fivefold or divisor has complete tail at most $(5/2)b$ in the same census. Failure of payment gives

$$b \geq \frac{8u-7}{(5/2)u-1} \geq \frac{116}{155}.$$

Its argument in (7.6) is at least $15810/99807 > 3/20$. The height-two rate is Q_7 , with

$$Q_7(3/20) = \max\{24137569/64000000, 16807/40000\} < 11/25.$$

The height-one rate is the first branch and obeys the same bound. This covers every center on $[51/50, 26/25]$.

Step 3. In this step, we prove the shell rate on $[26/25, 29/25]$. Proposition 5.9 gives dimensions four, five, or six, with $b > 3/2$ in dimension four and $b > 7/5$ in dimensions five and six. Among fourfold rows with cap greater than $3/2$, the additional types $(e, m) = (4, 1), (5, 2), (5, 3), (6, 5)$ have respective maximum source-plus-tail bounds

$$75/26 + 4757/1000, \quad 12984/1625, \quad 125/26 + 372/125, \quad 135/26 + 13453/5000.$$

Each is strictly below eight; their smallest gap is $16/1625$. The five remaining types and their exact slack caps are

(e, m)	upper bound for σ
$(6, 3)$	$481/1875$
$(6, 4)$	$217/12500$
$(7, 4)$	$2311/5000$
$(7, 5)$	$2921/12500$
$(7, 6)$	$2699/37500$

They follow from the same lines above at $u = 26/25$, and all are below $1/2$. Since $\delta = b + \sigma > 1$, Theorem 4.20 gives the full reduced Cohen–Macaulay tangent in every case. In the two embedding-six rows, the lower bounds $\beta(26/25)$ are respectively $9116/4621$ and $11573/5895$, both greater than $19/10$.

Now $u/(u+h) \geq 104/105$. An embedding-seven fourfold has argument greater than $208/1155 > 9/50$. The full four-branch bound of Lemma 7.17 gives

$$\Psi_{7,0,4}(9/50) = 6(32/50)^6 - 8(23/50)^6 + 3(14/50)^6 = \frac{5280752440}{15625000000} < 17/50.$$

An embedding-six fourfold has argument greater than $1040/5355 > 19/100 > 2/11$, so its common-linear-normal rate is

$$M_7(19/100) = \frac{332908409016}{10^{12}} < 17/50.$$

Every fivefold has argument greater than $26/147 > 7/40$, and both branches are retained in

$$Q_7(7/40) = \max\{1291467969/4096000000, 4084223/12800000\} < 17/50.$$

A divisor uses the smaller coordinate rate. These conclusions cover every actual center on $[26/25, 29/25]$.

Step 4. In this step, we sum all ordinary layers of the finite shell with a uniform error. For the appropriate rate a , each actual layer has dimension at most $a\ell^6/6! + C(\ell+1)^5$. The constants depend only on the fixed order range and the finite list of applicable local bounds. Summing every layer $ku \leq \ell < k(u+h)$, and bounding the increasing leading power sum by its integral, gives

$$\dim U_k(u) - \dim U_k(u+h) \leq a((u+h)^7 - u^7) \frac{k^7}{7!} + C(k+1)^6.$$

The later space may be zero; the injection and telescoping use only the earlier nonzero space. On a fixed finite mesh, take the common error bound before choosing the original degree, divisible by all mesh denominators. A starting dimension exceeding the total leading loss and this error then propagates nonvanishing at that same degree, using each successive source's own threshold. Every fixed component and every intermediate ordinary layer remains in this calculation. $\square$

Proposition 7.26. *Suppose $K_X + 8L$ is not generated at x . Let $u \geq 29/25$ be rational, let $k \geq 1$ be an integer with ku integral and $U_k(u) \neq 0$, and take a pure actual threshold average Γ of this full source. If its minimal lc center W through x is nondivisorial, then $\dim W = 5$. With*

$R = \mathcal{O}_{W,x}$, $\mathbf{n} = \mathbf{m}_{W,x}$ and $G = \text{gr}_{\mathbf{n}} R$, the necessary types are exactly the following:

$\text{edim } R$	$\text{mult}_x W$	$(1-z)^5 \text{Hilb}(G; z)$	$e_1(\mathbf{n}) = \bar{e}_1(\mathbf{n})$	upper bound for σ
6	2	$1+z$	1	617/3125
7	3	$1+2z$	2	36299/75000
7	4	$1+2z+z^2$	4	517/2500
7	5	$1+2z+2z^2$	6	1569/25000

In every row $0 \leq \sigma < 1/2$, the entire ring G is reduced Cohen–Macaulay, every positive power of $\mathbf{n}$ is integrally closed, and $R[\mathbf{n}T]$ is a normal Cohen–Macaulay domain. The ordinary point blowup is normal and rational near its entire exceptional Cartier scheme $E = \text{Proj } G$, which is reduced Cohen–Macaulay with its actual polarization $\mathcal{O}_E(1)$. For the effective strict transform of the same Γ on $\text{Bl}_x X$, every component F of E has discriminant coefficient $0 \leq \gamma_F \leq \sigma$. Every conclusion refers to this same finite source, threshold, boundary and original degree.

The ordinary powers and affine vertex are part of the conclusion.

Proof. We derive the slack bounds from this source’s full price, then determine the entire Hilbert series and Rees algebra.

Step 1. In this step, we derive the point-slack bounds for the actual threshold boundary. Write $s = k\tau$, $M = \text{mult}_x \Gamma$, $b = \text{def}_x \Gamma$, and $\sigma = 7 - M - b$. Proposition 5.9 leaves precisely the four fivefold states in the statement. The complete tail lines from its proof have the following data:

(e, m)	β_v	a	p	$a\beta_v + p$
(6, 2)	4	1149/1000	247/250	698/125
(7, 3)	11/3	623/500	39/40	16631/3000
(7, 4)	3	33/25	77/100	473/100
(7, 5)	13/5	69/50	673/1000	4261/1000

By (7.4)–(7.5),

$$0 \leq \sigma \leq 7 - b + u(ab + p - 8).$$

Here $au > 1$, so the right side increases with $b \leq \beta_v$. Its value at β_v decreases with u , since the last column is less than eight. Substituting $u = 29/25$ gives in order

$$\frac{617}{3125}, \quad \frac{36299}{75000}, \quad \frac{517}{2500}, \quad \frac{1569}{25000},$$

all strictly less than $1/2$. The failed-tail inequalities also give the increasing lower bounds $b \geq ((8-p)u - 7)/(au - 1)$. Their values at $29/25$ are

$$\frac{28348}{8321}, \quad \frac{28725}{11134}, \quad \frac{3467}{1328} > \frac{5}{2}, \quad \frac{37483}{15020} > \frac{11}{5}.$$

In particular every row has $b > 1$. The strict point-payment implication used here is Theorem 5.4, with both cutting errors and every old coefficient included.

Step 2. In this step, we determine the full ordinary tangent rings, Hilbert moments, and Rees algebras. Theorem 4.13 makes R a normal rational Cohen–Macaulay local domain. Now $\delta = 7 - M = b + \sigma > 1$, so Theorem 4.20 applies with $\sigma < 1/2$. It gives the full Cohen–Macaulay tangent ring, normal ordinary powers and the normal original Rees ring. It also gives reducedness of the entire exceptional scheme and of G . This use of the $\delta > 1$ clause concerns the current Γ and needs no larger deficit assumption.

For $(e, m) = (6, 2), (7, 3)$, the minimal-multiplicity assertion of the same theorem gives the numerators $1+z$, $1+2z$ and first coefficients one and two. For the other rows, Theorem 4.16 gives $2\bar{e}_1(\mathbf{n}) \leq (5-b)m$. The displayed strict lower bounds on b , and integrality of the coefficient, imply respectively $\bar{e}_1 \leq 4$ and $\bar{e}_1 \leq 6$.

Choose a general minimal reduction J of $\mathfrak{n}$ with five independent linear initial forms. Its parameters form a regular sequence on R , and its Artinian quotient has length m . Its Hilbert function satisfies $h_0 = 1, h_1 = 2$. Theorem 4.5 therefore gives

$$2m - 4 = 2 + 2(m - 3) \leq \sum_{j \geq 1} jh_j \leq e_1(\mathfrak{n}) \leq \bar{e}_1(\mathfrak{n}) \leq 2m - 4.$$

Equality forces all remaining length into degree two. Thus the Artinian Hilbert functions are $(1, 2, 1)$ and $(1, 2, 2)$. Because G is Cohen–Macaulay, the five initial parameters are regular. Their quotient is the associated graded ring of R/J : for each regular initial form the identity $\text{ord}(fg) = \text{ord}(g) + 1$ gives $(f) \cap \mathfrak{n}^j = f\mathfrak{n}^{j-1}$, and one iterates through the parameters. Dividing the Artinian series by $(1 - z)^5$ proves the two remaining full Hilbert series.

In every row $h_j = 0$ for $j \geq 3$, and $e_1 = \sum jh_j$. The quotient $\mathfrak{n}^j/(J \cap \mathfrak{n}^j)$ identifies with the j -th power of the maximal ideal of R/J , so

$$e_1 = \sum_{j=1}^4 \text{length}(\mathfrak{n}^j/(J \cap \mathfrak{n}^j)).$$

Theorem 4.6 now proves Cohen–Macaulayness of the whole Rees algebra. Its normality and the equality $\mathfrak{n}^j = \overline{\mathfrak{n}^j}$ were established above; in particular the ordinary and normal Hilbert coefficients agree at every row, giving exactly 1, 2, 4, 6.

Step 3. In this step, we identify the entire exceptional scheme and its discriminant bounds. The ordinary blowup is normal and rational near all of E by Theorem 4.20; away from E it is the normal punctured center. Its Cartier exceptional scheme is $\text{Proj } G$, with its original tautological polarization. It is Cohen–Macaulay and reduced. More explicitly, reducedness on $\text{Proj } G$ detects every minimal homogeneous prime of the positive-dimensional Cohen–Macaulay ring G . There are no embedded primes, so no nilpotent submodule can remain at the vertex or at component intersections. Finally every exceptional multiplicity is $r_F = 1$; the strict-pair inequality of that theorem becomes $0 \leq \gamma_F \leq 1 - (1 - \sigma) = \sigma$. All statements concern the same boundary used in the first display. $\square$

Lemma 7.27. *Suppose $K_X + 8L$ is not generated at x , and let $u \geq 99/100$ be rational. For a nonzero complete source $U_k(u)$, $k \geq 1$, ku integral, take its own pure actual threshold average with minimal center W and deficit b . Then $\dim W \in \{3, 4, 5, 6\}$. In dimension three,*

$$(\text{edim } W, \text{mult}_x W) \in \{(6, 4), (7, 5)\}, \quad b > 51/50,$$

and the full tangent is reduced Cohen–Macaulay with respective Hilbert numerators $1 + 3z$, $1 + 4z$ over $(1 - z)^3$. In dimension four every entire descendant scalar cost satisfies $\mathcal{C} \leq (5/3)b + 42/125$, and

$$b \geq \frac{(958/125)u - 7}{(5/3)u - 1} \geq \frac{7342}{8125} > \frac{9}{10}.$$

In dimensions five and six,

$$b \geq \frac{8u - 7}{(5/2)u - 1} \geq \frac{184}{295} > \frac{3}{5}.$$

For $u \geq 1$ these last two deficit bounds improve to $249/250$ and $2/3$, respectively.

The threefold tangent hypotheses follow from its own price; the fourfold bound is a bound on the whole descendant cost.

Proof. We apply the complete scalar table, check its needed threefold row, and then check every fourfold state.

Step 1. In this step, we classify the surviving low-dimensional states and their full tangents. Use (7.4)–(7.5), now at $u_* = 99/100$. A point center costs at most $700/99 < 8$. Row numbers below refer to the complete scalar table supporting Proposition 5.9. Apart from rows 1, 2, 8, every line in dimensions at most three has slope greater than $100/99$. Therefore $(7 - b)/u_* + H_v(b)$ increases

to the row's cap. Rows 1, 2 have $H_v(b) = b$ and maximum $700/99$. For row 8, the breakpoints of the three-line minimum are $33/50, 1$; its source-plus-tail maximum is $200/33 + 283/200 < 8$ at $b = 1$.

For row 22, of type $(3, 7, 6)$ and cap one, we use the complete recurrence directly with radius upper bound $909/500$. Its cube exceeds six. An adjacent surface of multiplicity j has deficit

$$y \leq \min\{2/j, 1 - j/6\}.$$

Multiplicity six has no positive adjacent cap. For $j \leq 3$ the surface radius is at most $1733/1000 < 909/500$. A surface's whole tail is at most its radius times y , since its further positive-dimensional center is a smooth curve of radius at most one. At $j = 4$, the possible surplus is at most $(2 - 909/500)/3 = 91/1500 < 7/100$; at $j = 5$, it is at most $(2237/1000 - 909/500)/6 = 419/6000 < 7/100$. A jump to a curve or point gives no surplus. Thus the entire row has $C \leq 909b/500 + 7/100$, with maximum total cost $200/33 + 236/125 = 32788/4125 < 8$.

Using this strengthened line, substitution in all remaining rows gives the following maxima after retaining rows 15, 21:

$\dim W$	maximum of source plus entire tail
1	$700/99$
2	$749/99$
3	$474341/59400$

For the last maximum, row 11 has cap $5/3$ and value $H_v(5/3) = 1559/600$, giving $1600/297 + 1559/600 = 474341/59400 < 8$. All other unretained rows give smaller values by the same cap substitution in the displayed table. This includes all the embedding-specific multiplicity-eight and nine cases of Lemma 5.10 and all dimension skips. Theorem 5.4 absorbs both cutting errors in these strict gaps. Increasing u only lowers the source term, so none of the excluded states returns.

For rows 15, 21, use respectively

$$(\beta_v, a, p) = (3/2, 397/250, 103/500), \quad (7/5, 171/100, 211/1000).$$

Their failed-tail deficit bounds increase with u . At u_* they give $b \geq 35803/28606$ and $b \geq 71111/69290$, both greater than $51/50$. Maximizing the slack inequality at the cap gives respectively $\sigma \leq 3553/25000$ and $\sigma \leq 5179/20000$, both less than $1/2$. Since $b > 1$, the $\delta > 1$ clause of Theorem 4.20 gives the full reduced Cohen–Macaulay rings. Their multiplicities are minimal, so the same theorem gives the asserted numerators.

Step 2. In this step, we bound every fourfold tail and derive the remaining deficit bounds. Put $a_0 = 5/3$, $p_0 = 42/125$. For the following rows the indicated existing line has $a \leq a_0, p \leq p_0$:

rows	(a, p)
29, 30, 31	$(3/2, 1/4)$
32	$(5/3, 229/1000)$
33, 34	$(3/2, 59/200)$
35	$(783/500, 269/1000)$
36	$(1627/1000, 61/250)$
39	$(5/3, 133/500)$
40, 41	$(5/3, 131/500)$
42	$(1627/1000, 11/40)$
45	$(5/3, 42/125)$

Every other fourfold row is covered by this full list:

row	β_v	a	p	$(a - a_0)_+ \beta_v + p$
28	4	7/4	0	1/3
37	1/2	841/500	111/500	689/3000
38	2/9	1733/1000	14/125	1711/13500
43	5/4	841/500	32/125	1651/6000
44	10/9	1733/1000	239/1000	8443/27000
46	8/11	7/4	131/500	5323/16500
47	1/2	931/500	49/250	881/3000
48	4/13	1899/1000	17/100	4709/19500
49	1/7	7/4	1/10	47/420

Every last entry is less than p_0 . Since $b \leq \beta_v$, these inequalities prove $\mathcal{C} \leq a_0 b + p_0$ for all rows 28–49, including multiplicity ten and all possible dimension jumps. Combining it with failed payment gives the displayed increasing deficit bound. At $u \geq 1$, the inequality $8 \leq 7 + (2/3)b + 42/125$ gives $b \geq 249/250$.

Every fivefold or divisorial row 50–71 has complete tail at most $(5/2)b$: row 50 has the stronger line $2b$, and all the others contain $(5/2)b$. Their failed payment gives $b \geq (8u - 7)/((5/2)u - 1)$, which increases from $184/295$ at $99/100$ and equals $2/3$ at one. This exhausts the complete state table. $\square$

Proposition 7.28. *Let X be a smooth complex projective sevenfold, let L be ample Cartier, and suppose $K_X + 8L$ is not generated at the closed point x . For integers $q \geq 1$ and $\ell \geq 0$, let $A_{q,\ell}$ be the full image of $U_q(\ell/q)$ in $\mathfrak{m}_x^\ell / \mathfrak{m}_x^{\ell+1}$, after a local trivialization of qL . Define*

$$B(t) = \begin{cases} 1, & 0 \leq t < 9/10, \\ 61/100, & 9/10 \leq t < 99/100, \\ 11/25, & 99/100 \leq t < 26/25, \\ 17/50, & 26/25 \leq t < 29/25, \\ \left(\frac{(125 - 52t)_+}{73t} \right)^6, & 29/25 \leq t \leq 13/5. \end{cases}$$

There are constants $J \geq 1$ and $C > 0$, fixed before the degree, layer, source, center and computing place, such that for every integer $q \geq J$ and every integer $1 \leq \ell \leq \lfloor 13q/5 \rfloor$,

$$\dim_{\mathbb{C}} A_{q,\ell} \leq \frac{\ell^6}{6!} B(\ell/q) + C(\ell + 1)^5.$$

Also $\dim_{\mathbb{C}} A_{q,0} \leq 1$. The constants depend only on the original data and the fixed upper point-order bound. Every image is the complete original image; a zero current source has zero image.

Each bound is for one actual ordinary layer. It can therefore be combined with other bounds on that image before any summation.

Proof. We compare each nonzero layer with its own full valuation ideal, apply the early classifications on the first four intervals, and use the same source's fixed factor or fivefold tangent on the last.

Step 1. In this step, we compare each actual layer with its full primary image and uniform local error. Fix an integer $q \geq 1$ and an integer $1 \leq \ell \leq \lfloor 13q/5 \rfloor$, and put $u = \ell/q$. If $U_q(u) = 0$, its full image is zero. Otherwise take the full source ideal I , its own pure threshold τ , and a sufficiently large general actual threshold average Γ . The average has the same coefficient-one strata as the ideal pair: on a simultaneous resolution its entire fixed divisor is retained, and enough general moving members have coefficients below one and meet the resolution strata transversely. For $u \geq 9/10$, a point center would cost $q\tau \leq 7/u \leq 70/9 < 8$, contrary to Proposition 3.12.

Choose a computing place w centered on its minimal center W , and write P for the corresponding prime. The ideal

$$J_w = \{f \in \mathcal{O}_{X,x} : w(f) \geq w(I)\}$$

is proper and P -primary. A factor outside P has value zero; sufficiently high powers of its finite set of generators have value at least $w(I) > 0$. It contains the entire I , and its generic threshold λ_w obeys $\lambda_w \leq A_X(w)/w(J_w) \leq \tau$. Every actual source germ maps into its full initial image, giving

$$(7.7) \quad A_{q,\ell} \hookrightarrow (\text{in}_{\mathfrak{m}} J_w)_{\ell}, \quad \frac{1}{\ell \lambda_w} \geq \frac{1}{\ell \tau} \geq \frac{1}{7-b}.$$

The last inequality is the point-blowup bound $\ell \tau + b \leq 7$ for this same source. This fixes the exact rational current order u , with no next order required. Every local comparison used below has a dimension-only error $C_7(\ell + 1)^5$, independent of q, ℓ, I, P, w .

Step 2. In this step, we bound the original layers at orders below $26/25$. For $u < 9/10$, the full image is contained in the polynomial layer of dimension $\binom{\ell+6}{6}$, which is $\ell^6/6! + O((\ell + 1)^5)$. For $9/10 \leq u < 99/100$, append zero weights to reach six normal weights in Lemma 7.8. Since $b > 0$, the argument in (7.7) exceeds $1/7$, and Theorem 7.9 gives the upper fraction

$$B_{7,6}(1/7) = 1 - (6/7)^6 < 61/100.$$

This includes every positive-dimensional center.

For $99/100 \leq u < 26/25$, apply Lemma 7.27 at this own source. A threefold has $b > 51/50$, so its argument exceeds $50/299 > 1/6$. Its full tangent is reduced Cohen–Macaulay of embedding dimension six or seven. Proposition 7.21 therefore gives its height-four bound, with

$$\frac{138259}{900000} + \left(1 - \frac{138259}{900000}\right) (5/6)^6 < \frac{17523}{40000} < \frac{11}{25}.$$

For a fourfold, $b > 9/10$ makes the argument exceed $10/61 > 49/300 = 1/6 - 1/300$. The height-three bound of that proposition gives fraction at most $256/729 + 3/50 < 21/50 < 11/25$, with no tangent hypothesis. For a fivefold or divisor, $b > 3/5$ makes the argument exceed $5/32 > 3/20$. The two-normal rate and its coordinate branch satisfy

$$Q_7(3/20) = \max\{24137569/64000000, 16807/40000\} < 11/25.$$

Thus every full layer on this interval has the required rate.

Step 3. In this step, we bound the original layers between $26/25$ and $29/25$. The own-source part of the proof of Proposition 7.25 gives dimensions four, five, or six. It proves the full reduced Cohen–Macaulay tangent for every fourfold by its five remaining states and their exact slack bounds, independently of the shell width. It gives $b > 19/10$ in embedding dimension six, $b > 3/2$ in embedding dimension seven, and $b > 7/5$ in dimensions five and six. Apply these conclusions to the current $u = \ell/q$.

The argument in (7.7) is greater than $10/51 > 19/100$ for an embedding-six fourfold and than $2/11 > 9/50$ for an embedding-seven fourfold. The respective fractions are at most

$$M_7(19/100) = \frac{332908409016}{10^{12}} < 17/50, \quad D_4(9/50) = \frac{5280752440}{15625000000} < 17/50.$$

The first is in the required interval $c \geq 2/11$. For a fivefold or divisor the argument exceeds $5/28 > 7/40$, and

$$Q_7(7/40) = \max\{1291467969/4096000000, 4084223/12800000\} < 17/50.$$

These are direct bounds on the current full image. No shell error has been differentiated or multiplied by a growing number of steps.

Step 4. In this step, we prove the final baseline branch for each source of order at least $29/25$. Proposition 5.9 leaves a divisor or the four fivefold states. Put

$$\varphi(u) = \frac{125(u-1)}{73}, \quad u - \varphi(u) = \frac{125 - 52u}{73}.$$

For a divisorial center of multiplicity m , put $\nu = \text{ord}_P(I)/q$. Its coefficient in the threshold average is one, so $s = q\tau = 1/\nu$. The complete scalar table has the following full-tail lines in every possible multiplicity:

m	a	p
1	1	198/125
2	1123/1000	703/500
3	1201/1000	223/200
4	63/50	121/125
5	327/250	801/1000
6	5/3	42/125

Failed payment and $b \leq 7 - u/\nu$ give

$$8 \leq 7a + p - (au - 1)/\nu, \quad m\nu \geq \frac{m(au - 1)}{7a + p - 8} \geq \varphi(u) \quad (u \geq 1).$$

Every denominator is positive. The last difference is affine in u ; at one its value is $m(a-1)/(7a+p-8) \geq 0$, and its slope is nonnegative because $ma/(7a+p-8) \geq 125/73$ for each of the six displayed pairs. For the first pair that slope difference is zero. Consequently the entire fixed divisor has point multiplicity at least $q\varphi(u)$. Its nonzero initial equation divides every actual source germ. The degree $\ell = qu$ image is therefore at most

$$\frac{q^6}{6!}(u - \varphi(u))_+^6 + O((\ell + 1)^5),$$

and is zero when the fixed factor's degree exceeds ℓ . All other fixed factors remain in this upper comparison.

For the embedding-six double fivefold, the scalar bound is

$$b \geq \beta_D(u) = \frac{(1753/250)u - 7}{(1149/1000)u - 1}, \quad b \leq 4.$$

It follows that $u \leq 375/302$. On this interval,

$$c_D = \frac{1}{7 - \beta_D(u)} = \frac{1149u - 1000}{1031u} \geq \frac{4}{21}, \quad c_D \geq \frac{\varphi(u)}{u}.$$

The last comparison is precisely $55875 \geq 44998u$, with equality at $375/302$. The exact two-normal endpoint in Theorem 7.9 gives fraction at most $(1 - c_D)^6 \leq (1 - \varphi(u)/u)_+^6$.

For the embedding-seven fivefolds of multiplicity three, four and five, Proposition 7.26 gives the full reduced Cohen–Macaulay tangents for the present Γ . Theorem 7.18 therefore gives fraction at most $2(1 - 2c)_+^6$, where $c = 1/(7 - \beta(u))$ and $\beta(u)$ is this row's scalar lower bound. In the three rows respectively,

$$1 - 2c = \frac{2000 - 795u}{1697u}, \quad \frac{200 - 63u}{201u}, \quad \frac{2000 - 427u}{2333u}.$$

All these rows are excluded at $u \geq 34/25$. On $[29/25, 34/25]$ the three displayed quantities are positive and satisfy

$$\frac{9}{8}(1 - 2c) \leq \frac{125 - 52u}{73u} = 1 - \frac{\varphi(u)}{u}.$$

After clearing positive denominators, the three numerator comparisons are

$$383000 - 183637u > 0, \quad 69600 - 42225u > 0, \quad 1019000 - 689989u > 0.$$

They decrease and remain positive at $34/25$, proving the inequalities on the entire interval. Since $(9/8)^6 > 2$, the resulting fractions are at most $(1 - \varphi(u)/u)^6$. After $34/25$ the divisorial argument

covers every source. This proves the last branch through $13/5$, including the vanishing positive part.

Step 5. In this step, we fix the common error bound for all original degrees and images. Only finitely many dimension-uniform local comparisons were used. Take their maximum error constant together with the polynomial layer constant before choosing q, ℓ or any threshold. The fixed-factor comparison is uniform as its degree varies; for nonnegative residual degree it lies between zero and ℓ . The argument applies at every positive original degree, so one may take $J = 1$. At $\ell = 0$ the target is the residue field and has dimension one.

In particular these bounds hold separately on the full images $A_{q,\ell}$. If other actual profiles bound the same images with leading terms F_i and errors $C_i(\ell+1)^5$, taking the pointwise minimum of these leading terms and the maximum error constant remains valid. Summing over $\ell \leq 13q/5$ has total error $O(q^6)$. The thresholds establishing the baseline may vary with the layer; each selected profile still uses its own single place as in Theorem 7.23. Every source keeps its original degree and full image throughout. $\square$

7.1. Exact arithmetic for the capacity estimates. The proofs above reduce several comparisons to rational arithmetic. We collect that arithmetic in the supplement **07-capacities**, without replacing the simplex optimizations or the full-image arguments. The package instructions are in `supplement/07-capacities/README.md`, and its independent checker is `supplement/07-capacities/check.py`. Its canonical data are `supplement/07-capacities/data/certificate.json`; the accompanying `supplement/07-capacities/data/bindings.json` locates each input and displayed value in the formulas above. The small tables remain in the article. Extracting their named displays reproduces their data exactly, including the row parameters and the denominators as printed. Each family identifier is **07-capacities/** followed by the family name below.

Here is the finite scope of the certificate. Coefficients of a polynomial are listed in increasing order of degree. All arithmetic is in $\mathbb{Q}$, and every signed term is retained. The family **spanning** contains all four sevenfold coefficient lists with $(t, s) = (1, 2), (1, 3), (0, 4), (0, 3)$, including their zero coefficients. The family **pairs** contains both branches of $Q_6(1/5)$, $Q_7(1/6)$, $Q_7(3/20)$, and $Q_7(7/40)$. For the unique positive crossing of the two branches of Q_7 , the family **crossings** uses the exact bracket

$$c_- = \frac{16020581569}{90194313216}, \quad c_+ = \frac{8010290789}{45097156608}.$$

The balanced branch is larger at c_- , and the coordinate branch is larger at c_+ . The uniqueness argument is the one in Theorem 7.9; these two sign checks only locate its crossing.

The family **constants** consists of the thirteen values used above: the boundary and interior values for the six- and seven-coordinate three-weight problems, C_4 , the strip maximum $432/3125$, the disjoint-pair correction and square bound, the three-cut anchor, the rounded B_7^* bound, $M_7(2/11)$, $M_7(19/100)$, and $D_4(9/50)$. The three records in **polynomial** retain R , its four derivative values and its two endpoint values from Lemma 7.20. The family **envelopes** checks all six joins of $B_{7,r}$ at $1/7$, for $1 \leq r \leq 6$, and retains the competing expressions in $E_{7,5}, Q_5, T_5$ at $c = 1/6, 1/3, 1/2$. It also retains all three candidate expressions for K_5 at $c = 0, 4/25, 1/2$; the single-pair expression is enabled only on its stated range. These endpoint records do not assert an optimization over unsampled weights or levels.

The family **source-tables** contains the eight rows at $51/50$, the five fourfold rows at $26/25$, and the four fivefold rows at $29/25$. Its parameters are precisely the displayed (d, e, m, β_v, a, p) ; the last four rows also retain their full cap costs, Hilbert numerators and first moments. Each row checks

$$\beta(u) = \frac{(8-p)u-7}{au-1}, \quad \sigma_* = 7 - \beta_v + u(a\beta_v + p - 8).$$

The family **fourfold-majorant** includes every row $28, \dots, 49$, with the two types of comparisons in Lemma 7.27. The family **baseline** keeps all six divisorial rows and all three embedding-seven

fivefold branches in the last proof. Thus a missing competing branch or state is a missing certificate record. The complete scalar census establishing that these rows apply is still Proposition 5.9.

Finally, **integrals** contains five coefficient vectors, one for each $j = 0, \dots, 4$, for the primitive $(t - j)^7$. They check the sevenfold specialization of the signed integrals in Corollary 7.24. The normalization $\eta = 1$ is removed by scaling; $j = 0$ covers the constant part of B_7^* . The formula for arbitrary dimension, and the monotonicity that compares ordinary-layer sums with these integrals, were proved in that corollary.

Lemma 7.29. *For the finite families just specified, equality of every coefficient vector and every rational value with its defining formula, together with the stated rational sign comparisons, certifies all their tabulated arithmetic. For the integral family it certifies, for $0 \leq u \leq v$, $\eta > 0$, and $0 \leq j \leq 4$,*

$$7 \int_u^v (t - j\eta)_+^6 dt = (v - j\eta)_+^7 - (u - j\eta)_+^7.$$

The comparison includes intervals crossing $j\eta$, intervals on either side, and point intervals.

Proof. We describe the checks and their implication, separating finite algebra from the geometric premises of the estimates.

Step 1. In this step, we verify the finite polynomial identities and capacity comparisons. The spanning vectors are computed from $(-1)^{j+1}[(\binom{t+s}{j} - s(\binom{t+s-1}{j-1})]$ for every indicated index. Each pair value is evaluated from both positive-part branches of Q_n . Subtracting those branches at the two bracket endpoints gives strict rational signs; no decimal approximation to the root is used. The constant records substitute the displayed arguments into the formulas already derived. For the quartic identity, substitute $s_2 = 3x(4 - x)$, $s_3 = 2x^2(3 - x)$ into F , differentiate its rational expression and clear s_3^5 . The resulting polynomial must equal $12x^4(2 - x)(x - 1)(5 - 2x)R(3 - x)$ coefficient by coefficient. Here $s_3 > 0$ on $[1, 5/2]$. Together with the two zeros of R'' and all four recorded values of R' , the endpoint values give the positivity argument used in that proof.

At a join or a positive-part zero, each candidate expression is evaluated with its stated closed endpoint convention. In particular the value of E at $1/2$ is its coordinate branch. Both maxima Q_5, T_5 , their coordinate competitor and the minimum with $E_{7,5}$ are retained. Their denominators are positive on $[1/6, 1/2]$. The comparisons $7^6 < 2^{17}$ and the numerator $1 - 12c$ of the logarithmic derivative give the coordinate endpoint range already proved.

Step 2. In this step, we verify the source-row signs and the baseline comparisons on their full intervals. For every source row check $au - 1 > 0$, $7a + p - 8 > 0$, and $a\beta_v + p < 8$ before taking a quotient. These signs justify the endpoint substitutions. Check its entire stated slack bound; the exceptional row $(4, 7, 4)$ at $51/50$ retains the split at $b = 9/4$, where its slack is $17161/40000 < 1/2$. The geometric argument for $b > 9/4$ remains necessary. For the fourfold majorant, either compare a, p separately or check $(a - 5/3)_+ \beta_v + p \leq 42/125$. This covers the entire closed interval $[0, \beta_v]$, including its endpoints and the equality row. It never discards a zero endpoint or replaces a complete scalar tail by one of its descendants.

For a divisorial baseline row the difference from the proposed affine lower bound is affine in u , with nonnegative value at one and nonnegative slope. This proves the assertion on the whole unbounded interval. For each fivefold branch, clearing its positive denominator gives exactly the displayed decreasing affine numerator; its value at $34/25$ is positive. Check all three, and retain the factor $(9/8)^6 > 2$. The double-fivefold comparison and the geometric exclusion beyond its own endpoint remain as proved above.

Step 3. In this step, we verify the positive-part integral identities and account for their error conventions. For the integral records, expand $7(t - j)^6$ and integrate term by term. The primitive has derivative $7(t - j)^6$ and value zero at j , so it is $(t - j)^7$. For $t \leq j$, use the zero primitive instead. Splitting an interval at the finitely many contained knots gives a complete partition, with matching values at the knots. An empty active interval and a point interval both have

integral zero; they require no division by their length. Scaling by η proves the stated formula. Applying all signed coefficients only after these individual integrations gives the full signed sum.

Every operation is an integer or rational operation. There is no floating-point or quadrature error, and each upper rounding is checked in its original direction. For example the retained gaps include $11/25 - 17523/40000 = 77/40000$, $17/50 - D_4(9/50) > 0$, and $3/8 - 256/729 - 9/500 > 0$; the source-row slack gaps are recorded separately. The two cutting errors still consume only the strict gaps assigned in the geometric proofs. The uniform lattice and degree-rounding errors are still $O((\ell + 1)^5)$ per layer and $O(q^6)$ after summation. No finite arithmetic check changes their quantifiers, nor proves the zero-weight and coincident-weight limits, the continuum maxima, or the applicability to the full ordinary primary image. Those arguments were retained above. $\square$

The portable checker requires exactly these named families and their complete index sets. It rejects malformed rationals, extra or missing fields, duplicate records, missing branches and failed inequalities. Its report records exact margins and the number of records checked in each family. The extraction program merely transcribes the named article displays and produces the primitive coefficient witnesses; the independent checking program does not import it. Thus acceptance has exactly the arithmetic scope of Lemma 7.29.

8. SYMBOLIC IMAGES AND FULL TANGENT COMPONENTS

For intrinsic cuts on a center, we need bounds for the entire symbolic image of its original sections. We retain the ordinary tangent ring and the original section spaces of section 2. We first obtain these bounds on the normalizations of all tangent components. We then express their rational component degrees on a resolution of every conductor. Ordinary and symbolic exponents in this section are nonnegative integers, with $P^{(0)} = R$, and a prime on a center means a prime divisor.

We use the following forms of the projective and local results. The node statement is [Sta26, Lemma 53.20.11, Tag 0CBX]. It preserves the specified base parameter; nonvanishing of the deformation term must be checked in an application.

Theorem 8.1 (A split node in a flat family). *Let $f : Y \rightarrow S$ be locally of finite type over a locally Noetherian scheme, flat at y , and suppose that y is a split node of its fiber over s . Then there is $h \in \widehat{\mathcal{O}}_{S,s}$ with*

$$\widehat{\mathcal{O}}_{Y,y} \simeq \widehat{\mathcal{O}}_{S,s}[[u, v]]/(uv - h).$$

The proper-morphism statement is [Sta26, Theorem 37.53.4, Tag 03H0].

Theorem 8.2 (Stein factorization). *Let $f : T \rightarrow S$ be a proper morphism with S locally Noetherian. There is a factorization $T \xrightarrow{g} S' \rightarrow S$ such that $S' = \operatorname{Spec}_S(f_*\mathcal{O}_T)$, the second morphism is finite, g is proper with geometrically connected fibers, and $g_*\mathcal{O}_T = \mathcal{O}_{S'}$.*

The next assertion is the very ample specialization of [Sta26, Lemma 37.32.3, Tag 0G4F].

Theorem 8.3 (Bertini irreducibility). *Let T be an integral projective complex variety of dimension at least two, with a fixed closed immersion into projective space. A general hyperplane section of T is geometrically irreducible.*

For minimal singularities we use [Kol85, Corollary 3.4.8 and the second assertion of Theorem 3.4.9] and [Bon16, Theorems 2.2, 3.2 and Remark 2.3].

Theorem 8.4 (Minimal singularities and surface resolutions). *A minimal singularity means a reduced Cohen–Macaulay singularity of multiplicity $\operatorname{edim} - \dim + 1$ with reduced tangent cone. A normal such singularity in characteristic zero is rational. Its point blowup has minimal singularities and rational exceptional divisors. In dimension two, minimal normal singularities are exactly rational surface singularities with reduced fundamental cycle. Their minimal resolution factors through the point blowup, and a resolution component E_i is contracted by this factor*

exactly when $E_i \cdot Z = 0$, where Z is the fundamental cycle. Its weighted graph satisfies weight at least valence.

The three surface assertions below are [Lip69, Proposition 17.1], [YZ20, Appendix B, Theorem B.1, by Jun Lu], and [Che24, Lemma 2.25].

Theorem 8.5 (Rational surface invariants). *A normal two-dimensional local ring admitting a desingularization and having a rational singularity has finite divisor class group. For a rational normal surface singularity of multiplicity m , its minimal log discrepancy is at most $2/m$. On a resolution on which its maximal ideal is invertible, that ideal is $\mathcal{O}(-Z)$, with Z the corresponding fundamental cycle. These are assertions about the specified ordinary surface; an induced boundary retains its actual different.*

We also use the module and ring criteria in [Sta26, Tags 0AVB, 0AY2 and 031S].

Theorem 8.6 (Reflexivity and normality). *A finite module over a Noetherian normal domain is reflexive if and only if it is torsion-free and S_2 , equivalently torsion-free and equal to the intersection of its height-one localizations in its vector space over the fraction field. A coherent torsion-free sheaf on an integral locally Noetherian scheme injects into its double dual. A Noetherian ring is normal if and only if it satisfies R_1 and S_2 .*

The following comparison results retain their smooth-model and jet-surjectivity hypotheses; see [Leh19, Lemma 4.1] and [AI20, Lemma 2.24].

Theorem 8.7 (Section counts on a smooth model). *Let T be a smooth projective d -fold, D Cartier with Iitaka dimension r , and A the pullback of a very ample divisor under a birational morphism. If a positive integer s satisfies $D \cdot A^{d-1} < sA^d$, then $h^0(T, D) < s^r A^d + 1$. For an ample line bundle L on a proper variety and a fixed point x with Seshadri constant ϵ_x , there are $c_1, c_2 \geq 0$ such that the full map $H^0(qL) \rightarrow \mathcal{O}/\mathfrak{m}_x^{p+1}$ is onto whenever $p, q \geq 1$, $p \geq c_1$ and $q\epsilon_x - p \geq c_2$.*

The estimate needed for symbolic images concerns arbitrary Weil divisors on normalized components. We give its finite form.

Lemma 8.8. *Let T be a normal integral complex projective variety of dimension $q \geq 1$, and let $f : T \rightarrow T_0 \subseteq \mathbb{P}^N$ be finite and birational onto an integral projective variety. Put $J = f^*\mathcal{O}_{T_0}(1)$ and $a = J^q$. For an integral Weil divisor D , put $\eta = J^{q-1} \cdot D$. If $\eta < 0$, then $H^0(T, \mathcal{O}_T(D)) = 0$. If $\eta = ah + v \geq 0$, with integers $h \geq 0$ and $0 \leq v < a$, then*

$$h^0(T, \mathcal{O}_T(D)) \leq a \binom{h+q-1}{q} + (v+1) \binom{h+q-1}{q-1}.$$

Binomial coefficients with a nonnegative upper argument smaller than the lower argument are zero. The right-hand side is

$$\frac{\eta^q}{a^{q-1}q!} + O_{q,a}((h+1)^{q-1}).$$

Proof. We restrict the full divisor sheaf along a general hyperplane and normalize the section at each stage. A nonzero section of $\mathcal{O}_T(D)$ gives an effective divisor linearly equivalent to D . Since J is ample, that divisor has nonnegative degree. This proves the assertion for $\eta < 0$. If $q = 1$, the normal curve is smooth. Subtracting $\eta + 1$ closed points from a divisor of degree $\eta \geq 0$ leaves negative degree and loses at most one section at each subtraction. Thus $h^0(T, \mathcal{O}_T(D)) \leq \eta + 1$, as required.

Step 1. In this step, we preserve the divisor degree under restriction and normalization. Suppose $q \geq 2$ and write $\mathcal{G} = \mathcal{O}_T(D)$. A general hyperplane section of T_0 is irreducible by Theorem 8.3. Choose it to contain no component of the complement of the open set on which f is an isomorphism. Its inverse image T' has one component: any component has dimension $q - 1$,

whereas the inverse image of the excluded locus in a general hyperplane has dimension at most $q - 2$, because f is finite.

A general member of this generating system is smooth on the smooth locus of T . To check this without an immersion hypothesis, the universal incidence over that locus is the projective bundle of the kernel of the surjective evaluation map, so it is smooth. Its generic fiber over the parameter space is regular by localization, and is smooth over the characteristic-zero function field. The nonsmooth locus has constructible image missing the generic point; after shrinking the parameter space it is empty. The chosen member also contains no component of the singular locus of T . Thus it is generically reduced. A Cartier quotient of an S_2 scheme is S_1 , so there are no embedded associated points and T' is integral. The same avoidance and smoothness show that T' is regular in codimension one.

Let $\nu : T'' \rightarrow T'$ be the finite normalization. It is an isomorphism in codimension one. The composite map from T'' to the hyperplane section of T_0 is finite and birational, and $J'' = \nu^*(J|_{T'})$ has degree $(J'')^{q-1} = a$. Choose the hyperplane also to contain no component of the nonfree locus of $\mathcal{G}$, which has codimension at least two. This choice works for every twist $\mathcal{G}(-\ell J)$. Multiplication by its equation gives

$$0 \longrightarrow \mathcal{G}(-J) \longrightarrow \mathcal{G} \longrightarrow \mathcal{G}|_{T'} \longrightarrow 0.$$

At every nongeneric point of T' , the S_2 condition on $\mathcal{G}$ gives depth at least two before the cut and at least one after it. Therefore $\mathcal{G}|_{T'}$ is torsion-free of rank one. Pull it to T'' , remove torsion, and take its double dual. By [Theorem 8.6](#) this gives a divisor sheaf $\mathcal{O}_{T''}(D'')$, and the natural map

$$\mathcal{G}|_{T'} \longrightarrow \nu_* \mathcal{O}_{T''}(D'')$$

is injective: its kernel is generically zero and hence torsion. In codimension one this is ordinary line-bundle restriction. Consequently

$$(J'')^{q-2} \cdot D'' = \eta.$$

One can compute this equality on a further general complete intersection curve, avoiding the two exceptional sets of codimension at least two in T' . For $q = 2$, those sets are empty. A twist by $-\ell J$ changes the degree by exactly $-\ell a$.

Step 2. In this step, we sum the restriction bounds to obtain the section estimate. Iterate the restriction inequality until the degree becomes negative. If the bound in dimension $q - 1$ is denoted by B_{q-1} , this gives

$$h^0(T, \mathcal{G}) \leq \sum_{\ell=0}^h B_{q-1}(\eta - \ell a).$$

The curve bound starts the induction. Substitution of $\eta = ah + v$ and the identities

$$\sum_{\ell=0}^h \binom{\ell + q - 2}{q - 1} = \binom{h + q - 1}{q}, \quad \sum_{\ell=0}^h \binom{\ell + q - 2}{q - 2} = \binom{h + q - 1}{q - 1}$$

give the stated expression. They follow by telescoping Pascal's identity. Expanding the two finite products gives the stated error, uniformly for $0 \leq v < a$. No bound for the support of D enters. $\square$

Theorem 8.9. *Let $(R, \mathfrak{n})$ be a normal Cohen–Macaulay complex algebraic local domain of dimension $d \geq 2$, with residue field $\mathbb{C}$, multiplicity m , and full reduced Cohen–Macaulay tangent ring. Suppose its actual ordinary point blowup is normal with rational singularities near the entire exceptional Cartier scheme. Let N_i be the normalizations of its integral exceptional components, H_i their pulled-back tautological divisors, and $m_i = H_i^{d-1}$, so $\sum_i m_i = m$. For every height-one prime P , put $k = \text{mult}(R/P)$. The full strict transform determines effective rational divisors*

$C_{P,i}$ on N_i , with $k_i = H_i^{d-2} \cdot C_{P,i} \geq 0$ and $\sum_i k_i = k$. For every $r, j \geq 0$, the full ordinary initial image of $P^{(r)}$ injects into

$$\bigoplus_i H^0(N_i, \mathcal{O}_{N_i}(jH_i - \lceil rC_{P,i} \rceil)).$$

Uniformly in the germ, prime, exponents, Cartier indices and supports,

$$\dim \operatorname{im}(P^{(r)} \cap \mathfrak{n}^j \longrightarrow \mathfrak{n}^j / \mathfrak{n}^{j+1}) \leq \sum_i \frac{m_i}{(d-1)!} (j - rk_i/m_i)_+^{d-1} + C_{d,m}(j+1)^{d-2},$$

and for $0 \leq h \leq l$,

$$\operatorname{length} \frac{P^{(r)} \cap \mathfrak{n}^h}{P^{(r)} \cap \mathfrak{n}^l} \leq \frac{1}{d!} \sum_i m_i ((l - rk_i/m_i)_+^d - (h - rk_i/m_i)_+^d) + C'_{d,m}(l+1)^{d-1}.$$

All conductor branches and residue-field degrees are included. The leading sum is at most $(m-1)(l^d - h^d) + (l - rk)_+^d - (h - rk)_+^d$. For a minimal-multiplicity actual center with $b > d-2$ and reduced full tangent, the stated blowup hypotheses hold; its components are normal arithmetically Cohen–Macaulay varieties of minimal degree in their spans. If the tangent is integral, the sum has one term with shift rk/m .

Proof of Theorem 8.9. We construct the rational restriction cycles on the actual blowup, bound the whole initial image, and then sum every ordinary layer. The minimal-multiplicity specialization is treated at the end.

Step 1. In this step, we identify the full exceptional scheme and the strict prime transform. Put $G = \operatorname{gr}_{\mathfrak{n}} R$, $Y = \operatorname{Bl}_{\mathfrak{n}} \operatorname{Spec} R$, $E = \operatorname{Proj} G$, and $q = d-1$. The Cohen–Macaulay ring G has depth $d \geq 2$ at its irrelevant ideal. Hence

$$G_j \longrightarrow H^0(E, \mathcal{O}_E(j))$$

is injective for every j : an element in its kernel is killed by a power of the irrelevant ideal, and such torsion is zero. Reducedness and unmixedness give an injection

$$\mathcal{O}_E \longrightarrow \bigoplus_i (\nu_i)_* \mathcal{O}_{N_i}, \quad \nu_i : N_i \longrightarrow E_i.$$

All E_i have dimension q ; their multiplicities in E are one. Their degrees are the positive integers m_i , whose sum is the leading degree m of G . Each H_i is ample and generated, and is the pullback under the finite birational map $N_i \rightarrow E_i$ of the given hyperplane class.

Let D_P be the strict transform of $\operatorname{Spec}(R/P)$ on Y . The surjection of ordinary Rees algebras

$$R[\mathfrak{n}T] \longrightarrow (R/P)[(\mathfrak{n}(R/P))T]$$

has degree- j kernel $(P \cap \mathfrak{n}^j)T^j$. Its target is a domain, so its Proj is exactly D_P , with its full scheme structure. Its intersection with E is $\operatorname{Proj}(\operatorname{gr}_{\mathfrak{n}(R/P)}(R/P))$. The degree of the fundamental cycle of this scheme is $k = \operatorname{mult}(R/P)$, including its generic scheme multiplicities. In dimension two this is the length of the entire zero-dimensional exceptional scheme.

Step 2. In this step, we construct the rational restriction divisor at every conductor branch. At a codimension-two point η of Y on E , the ring $A_\eta = \mathcal{O}_{Y,\eta}$ is a normal excellent surface local ring with a rational singularity. A resolution and its vanishing higher direct images localize to this ring. Thus the finiteness assertion of Theorem 8.5, namely [Lip69, Proposition 17.1], applies. It does not require its residue field to be algebraically closed or the ring to be henselian. Consequently, near η , some positive multiple $a_\eta D_P$ is principal. Choose an equation f_η with precisely that divisor. It is regular because its divisor is effective and A_η is normal.

For each codimension-one point z of N_i above η , put

$$c_{i,z} = \frac{1}{a_\eta} \operatorname{ord}_z(\nu_i^*(f_\eta|_{E_i})).$$

Set this coefficient to zero where D_P is absent. These coefficients are nonnegative. Two choices, after passage to a common multiple, differ by a unit on the normal surface local ring, so they give the same coefficient. They define a finite effective rational divisor $C_{P,i} = \sum_z c_{i,z}[z]$.

We check its entire degree. At a generic point η of $D_P \cap E$, let e_η be the Cartier equation of E , and put $f = f_\eta$, $a = a_\eta$. In the two-dimensional Cohen–Macaulay ring A_η , these form a parameter sequence. The two computations of its intersection length give

$$(8.1) \quad \text{length } A_\eta/(e_\eta, f) = a \text{ length } A_\eta/(I_{D_P}, e_\eta),$$

$$(8.2) \quad \text{length } A_\eta/(e_\eta, f) = \sum_{i, z \mapsto \eta} [\kappa(z) : \kappa(\eta)] \text{ord}_z(\nu_i^*(f|_{E_i})).$$

For the first equality, a prime filtration of $A_\eta/(f)$ has generic multiplicity a on D_P . The parameter Euler characteristic is additive, and a finite-length factor contributes zero, so only that generic multiplicity remains. The branch ring A_η/I_{D_P} is a one-dimensional domain and hence Cohen–Macaulay; its parameter multiplicity is the length displayed. For the second equality, normalize the reduced curve $A_\eta/(e_\eta)$. The normalization quotient has finite length. Multiplication by f on this quotient has kernel and cokernel of equal length, so it contributes zero to the same Euler characteristic. Every normalized branch is a discrete valuation ring, and its length over A_η is its order times the displayed residue-field degree. This proves both equalities without deleting the conductor quotient.

Divide by a , take the full cycles, and intersect with the remaining hyperplanes. Projection of the branch cycles includes the same residue-field degrees. We obtain

$$k_i = H_i^{d-2} \cdot C_{P,i} \geq 0, \quad \sum_i k_i = k.$$

The construction allows several branches on one normalization, arbitrarily large denominators, and zero degree on a component.

Step 3. In this step, we bound the entire symbolic initial image by divisor section spaces. Take $\xi \in P^{(r)} \cap \mathfrak{n}^j$. On a local trivialization of $\mathcal{O}_Y(-jE)$, the divided pullback ξ/e_η^j is regular and has order at least r along D_P . The equation of E is a unit at the generic point of D_P . Normality therefore gives

$$\frac{(\xi/e_\eta^j)^{a_\eta}}{f_\eta^r} \in A_\eta :$$

its order at D_P is nonnegative, and the denominator has no other divisorial zeros. On every normalized branch, either the initial section is zero, or its integral order is at least $\lceil rc_{i,z} \rceil$. The codimension-one description of reflexive sheaves in [Theorem 8.6](#) gives the asserted inclusion in the direct sum of divisor section spaces. The two injections at the beginning of the proof make this an injection of the whole original initial image. In particular no symbolic quotient is assumed Cohen–Macaulay and no saturation of that image is taken.

Apply [Lemma 8.8](#) to $jH_i - \lceil rC_{P,i} \rceil$. Its integral degree η_i satisfies

$$\eta_i \leq m_i j - rk_i.$$

If $\eta_i < 0$, it has no sections. Otherwise its integer quotient by m_i is between zero and j . The lemma gives

$$h^0(N_i, \mathcal{O}_{N_i}(jH_i - \lceil rC_{P,i} \rceil)) \leq \frac{m_i}{(d-1)!} (j - rk_i/m_i)_+^{d-1} + O_{d,m_i}((j+1)^{d-2}).$$

Rounding upward only increases the degree subtracted; no estimate of the number of prime coefficients is needed. Since there are at most m components and $1 \leq m_i \leq m$, their errors sum to the asserted constant $C_{d,m}$.

Step 4. In this step, we sum the ordinary layers and bound every allocation of restriction degrees. The successive quotients of the filtration by $P^{(r)} \cap \mathfrak{n}^j$, for $h \leq j \leq l$, are exactly the

initial images just bounded. For any $c \geq 0$,

$$\sum_{j=h}^{l-1} (j-c)_+^{d-1} \leq \int_h^l (t-c)_+^{d-1} dt = \frac{(l-c)_+^d - (h-c)_+^d}{d}.$$

The errors sum to $O_{d,m}((l+1)^{d-1})$. This proves the quotient-length estimate, also when $h = 0$, $h = l$, or $r = 0$.

For the last bound put $\Phi(c) = (l-c)_+^d - (h-c)_+^d$. For $d \geq 2$ this is nonnegative, decreasing, and convex on $[0, \infty)$: its derivative is $-d((l-c)_+^{d-1} - (h-c)_+^{d-1})$ and is increasing, including across the two junctions. Convexity on the simplex $k_i \geq 0$, $\sum k_i = k$, bounds the leading sum by the largest of

$$(m - m_i)\Phi(0) + m_i\Phi(rk/m_i).$$

The function $\Phi(0) - \Phi(c)$ is concave and vanishes at zero. Since $m_i \geq 1$,

$$m_i(\Phi(0) - \Phi(rk/m_i)) \geq \Phi(0) - \Phi(rk).$$

This proves the stated bound $(m-1)\Phi(0) + \Phi(rk)$ for the sum before its factor $1/d!$. When the tangent is integral, there is just one normalized component and $k_1 = k$.

Step 5. In this step, we prove the blowup and component assertions for minimal multiplicity. For the actual center in the last assertion, [Theorem 4.20](#) gives the normal Cohen–Macaulay ordinary Rees ring and the full Hilbert series

$$\text{Hilb}(G; z) = \frac{1 + (m-1)z}{(1-z)^d}.$$

The reduced tangent and minimal multiplicity meet the definition in [Theorem 8.4](#). Its blowup has minimal singularities, and the already established normality makes them rational. This holds near the entire exceptional scheme. The displayed Cohen–Macaulay Hilbert series gives regularity at most one for G , hence 2-regularity of the projective ideal. [Theorem 7.10](#) gives minimal-degree components and their linearly joined ordering.

For completeness the normal and arithmetically Cohen–Macaulay properties of the classified integral components can be checked on their coordinate rings. Linear spaces and integral quadrics have polynomial and hypersurface coordinate rings; an integral quadric has singular locus of codimension at least two, so it is normal. For a rational normal scroll, remove its cone variables and write the remaining smooth scroll, of dimension q_0 , as $\mathbb{P}_{\mathbb{P}^1}(\bigoplus_{a=1}^{q_0} \mathcal{O}(a_a))$, with $a_a > 0$, polarized by its tautological bundle J . Products of binary monomials show that the section ring is generated in degree one. It is the full section ring on a normal variety, hence is integrally closed. For $\ell \geq 0$, projection to $\mathbb{P}^1$ computes its cohomology from $\text{Sym}^\ell(\bigoplus_a \mathcal{O}(a_a))$, a sum of nonnegative-degree line bundles. For $-q_0 < \ell < 0$ all fiber cohomology vanishes. For $\ell \leq -q_0$, the only possibly nonzero direct image is in degree $q_0 - 1$ and is a sum of negative line bundles, so it has no global sections. These calculations give $H^i(\mathcal{O}(\ell J)) = 0$ for $0 < i < q_0$ and every ℓ ; the section ring is Cohen–Macaulay. The curve case is the Veronese section ring of $\mathbb{P}^1$ and has the same conclusion. For the Veronese surface the ring is the second Veronese subring of $\mathbb{C}[x_0, x_1, x_2]$; it is integrally closed, and $H^1(\mathbb{P}^2, \mathcal{O}(2\ell)) = 0$ for every ℓ proves the Cohen–Macaulay assertion. Forming a cone adjoins polynomial variables, preserving both properties. Thus all the components have the properties claimed, without an assumption that the original germ is a cone. $\square$

Proposition 8.10. *In [Theorem 8.9](#), suppose $R = \mathcal{O}_{W,x}$ for an original integral projective W , and let A be ample Cartier with $g = A^d$. Fix rational $0 \leq \lambda < \theta$ with $g - m\lambda^d > 0$. If the full original $F_p(\lambda)$ lies in $P^{(r)}$, then*

$$\dim F_p(\theta) \geq \frac{p^d}{d!} \left[g - m\lambda^d - \sum_i m_i \left(\left(\theta - \frac{rk_i}{m_i p} \right)_+^d - \left(\lambda - \frac{rk_i}{m_i p} \right)_+^d \right) \right] - K(p+1)^{d-1}.$$

Here K depends only on the fixed original data and the interval, and precedes the choices of P, r, p . For an original subspace $S_p \subset F_p(\lambda)$, the same interval bounds $\dim S_p - \dim(S_p \cap F_p(\theta))$; no complete-space lower mass is assigned to a proper image. No ambient lifting cutoff is included in this intrinsic assertion.

Proof of Proposition 8.10. All sections in this proof stay on the original W .

Step 1. In this step, we obtain the complete-source lower dimension bound. By its Hilbert polynomial and Theorem 6.8, applied to the fixed coherent sheaf $\mathcal{O}_W$ and ample A ,

$$h^0(W, \mathcal{O}_W(pA)) = \frac{g}{d!} p^d + O_{W,A}((p+1)^{d-1}).$$

The ordinary local Hilbert polynomial similarly gives, for $h_p = \lceil \lambda p \rceil$,

$$\text{length}(R/\mathfrak{n}^{h_p}) = \frac{m\lambda^d}{d!} p^d + O_{R,\lambda}((p+1)^{d-1}).$$

The kernel of point evaluation is the complete $F_p(\lambda)$. Its dimension is therefore at least the difference of these two quantities; surjectivity of point evaluation is unnecessary.

Step 2. In this step, we bound the complete-source loss by the full symbolic interval. Put $l_p = \lceil \theta p \rceil$ and trivialize $\mathcal{O}_W(pA)$ at x . Under the stated symbolic containment, the actual jet map gives an injection

$$F_p(\lambda)/F_p(\theta) \longrightarrow \frac{P^{(r)} \cap \mathfrak{n}^{h_p}}{P^{(r)} \cap \mathfrak{n}^{l_p}}.$$

Its kernel before taking the quotient is exactly the original space with point order at least l_p . Subtract the interval bound of Theorem 8.9 from the complete-space lower bound. Replacing h_p, l_p in its leading powers by $\lambda p, \theta p$ costs a uniform $O((p+1)^{d-1})$: for every shift $c \geq 0$ and $0 \leq t \leq \theta p + 1$,

$$0 \leq \frac{d}{dt} (t - c)_+^d \leq d(\theta p + 1)^{d-1}.$$

The local interval constant depends only on d, m . The Hilbert constants involve only the fixed original data. Their sum, with the ceiling error, is the required K , chosen before P, r, p . Increasing it accommodates the finitely many degrees preceding the Hilbert-polynomial cutoffs as well.

Step 3. In this step, we retain the actual dimension of a subspace and specify the intrinsic nonvanishing cutoff. For a subspace S_p , use the same map with kernel $S_p \cap F_p(\theta)$. In particular its exact finite loss is at most

$$\frac{1}{d!} \sum_i m_i \left((l_p - rk_i/m_i)_+^d - (h_p - rk_i/m_i)_+^d \right) + C'_{d,m} (l_p + 1)^{d-1}.$$

This uses its actual dimension and retains every condition defining S_p . The polynomial lower bound was used only for the complete original space. If the bracket in the proposition exceeds a fixed $\epsilon > 0$, choosing p so that $\epsilon p^d/d! > K(p+1)^{d-1}$ proves nonvanishing at that same degree. This cutoff is intrinsic; no ambient restriction map has been used. $\square$

We now express the full rational restriction degrees on the minimal transverse resolutions of all conductors.

Proposition 8.11. *Let W be an actual minimal center of dimension $d \geq 3$, minimal multiplicity and full reduced tangent, with $b > d - 2$. Take the minimal resolutions of the generic transverse surface singularities of every conductor on its ordinary point blowup. The resulting graph, with original components attached, is a finite tree, and the pullback of the actual exceptional Cartier equation is reduced on every vertex. Give original vertices their component degrees m_i and every internal resolved vertex mass zero. With valence v_i , define the matrix and vector*

$$M_{ii} = m_i + v_i, \quad M_{ij} = -1 \text{ on an edge, } \quad M_{ij} = 0 \text{ otherwise,} \quad \kappa_i = (d-2)m_i + 2 - v_i.$$

The matrix is positive definite with strictly positive inverse. For every prime P through x , its full nonnegative intersection vector p , original discriminant coefficient α and original exceptional log discrepancies u obey

$$\sum_i p_i = \text{mult}_x P, \quad u_i \geq b, \quad \alpha p + Mu \leq \kappa.$$

Partition the vertices into original A and internal B . For $B \neq \emptyset$, put

$$C = -M_{AB}M_{BB}^{-1}, \quad S = M_{AA} - M_{AB}M_{BB}^{-1}M_{BA}, \quad \hat{\kappa} = \kappa_A + C\kappa_B.$$

Then the full rational restriction degrees on original components are $\delta = p_A + Cp_B \geq 0$, with $\sum \delta_i = \text{mult}_x P$, and $\alpha\delta + Su_A \leq \hat{\kappa}$. The matrix S has row sums m_i and nonpositive off-diagonal entries. For $B = \emptyset$ take $S = M$ and $\hat{\kappa} = \kappa$. Let H be an actual rational Cartier addition not containing W , whose restriction $H|_W$ is effective rational Cartier and has ordinary order at least γ . If it selects P at its own threshold t , with deficit $z > 0$, set $\rho_i = u_i - t \text{ord}_i(H|_W)$. Then

$$\rho_i \geq z, \quad p + M\rho \leq \kappa, \quad \delta + S\rho_A \leq \hat{\kappa}.$$

For every nonempty proper $I \subset A$ and bounds $U_j \geq u_j$,

$$\sum_{i \in I} \delta_i \leq \sum_{i \in I} \hat{\kappa}_i - z \sum_{i \in I} m_i + \sum_{\substack{i \in I \\ j \notin I}} (-S_{ij})(U_j - \gamma t - z).$$

All old boundary and compatible moduli terms remain in these inequalities.

Proof. We first determine the transverse resolutions with their actual Cartier equation. We then intersect the retained adjunction equality with complete curves on every vertex, and eliminate the internal vertices in both the discrepancy vector and the prime vector. Write Δ for the original ambient boundary in this proof.

Step 1. In this step, we determine the conductor incidence tree and its transverse Cartier quotients. By [Theorem 8.9](#) the ordinary blowup Y is normal and rational near its entire reduced exceptional scheme E . Its components E_i are normal and arithmetically Cohen–Macaulay of minimal degree. Let their number be s_0 . They have a linearly joined ordering by [Theorem 7.10](#). The vector dimensions of their spans are $d + m_i - 1$, whereas the whole tangent spans a vector space of dimension $d + m - 1$. Thus the sum of the vector dimensions of the successive joining spaces is

$$\sum_i (d + m_i - 1) - (d + m - 1) = (s_0 - 1)(d - 1).$$

Each joining has projective dimension at most $d - 2$, since it is a proper closed subset of the new integral component. Hence every joining is scheme-theoretically a linear $\mathbb{P}^{d-2}$. Inductively the pairwise intersections are linear as well: a new joining is contained in an old component, and its intersections with the other old components are intersections with their already linear pairwise joins.

Make a bipartite graph whose vertices are components and distinct codimension-one conductors. A new joining contained in only one old component adds one conductor vertex and one component vertex. A joining contained in several old components equals their existing codimension-one intersection, so it attaches the new component to the existing conductor vertex. This constructs a tree. If $s_0 = 1$, it is the single original vertex. Smaller intersections are retained in the scheme but introduce no codimension-one edge. At a conductor on s components, all these components are smooth generically and their transverse tangent directions are independent. The full transverse Cartier quotient is therefore

$$(8.3) \quad K[[x_1, \dots, x_s]] / (x_i x_j : i \neq j),$$

where K is the conductor function field. In particular the branches, and not just their support, are specified.

Step 2. In this step, we resolve each transverse surface while retaining the actual exceptional equation. Let A_0 be the completed normal surface local ring of Y at the conductor, let $\mathfrak{a}$ be its maximal ideal, and let e_0 define the actual exceptional Cartier scheme. Excellence preserves normality on completion. Its quotient by e_0 is (8.3).

For $s = 2$, apply Theorem 8.1 to the morphism defined by e_0 before completion. More explicitly, the linear conductor has rational function field K ; lifts of its affine coordinates identify the generic transverse local ring with an essentially finite-type K -algebra of dimension two and residue field K . The map from the discrete valuation ring $K[e_0]_{(e_0)}$ is flat because it is torsion-free, and its fiber is the split node above. Thus

$$A_0 \simeq K[[u, v, e_0]]/(uv - h(e_0)).$$

The series h cannot be zero, since A_0 is a normal local domain. Absorbing a unit in v gives $uv = e_0^{a_0}$, for an integer $a_0 \geq 1$, with e_0 unchanged. Its minimal resolution is a chain of $a_0 - 1$ curves. One can check the equation and the multiplicities on its smooth charts

$$u = \xi^{\ell+1}\zeta^\ell, \quad v = \xi^{a_0-\ell-1}\zeta^{a_0-\ell}, \quad e_0 = \xi\zeta, \quad 0 \leq \ell < a_0.$$

Successive charts have transition $\xi_{\ell+1} = 1/\zeta_\ell$, $\zeta_{\ell+1} = \xi_\ell\zeta_\ell^2$. They arise by the successive blowups of the ideals (u, e_0) in the remaining singular charts. The transition gives self-intersection -2 on each internal curve. The pullback of e_0 has coefficient one on every curve and on both strict branches. When $a_0 = 1$, there is no internal curve.

For $s \geq 3$, we verify the minimal-surface hypotheses for this particular e_0 . If $e_0 \in \mathfrak{a}^a$, then

$$\text{length } A_0/(e_0, \mathfrak{a}^N) \geq \text{length } A_0/\mathfrak{a}^N - \text{length } A_0/\mathfrak{a}^{N-a}.$$

Comparison of the linear leading terms yields $s \geq a \text{ mult } A_0$. If $e_0 \in \mathfrak{a}^2$, then $\text{edim } A_0 = s$, and the Cohen–Macaulay multiplicity bound gives $s \geq 2(s-1) > s$, a contradiction. This multiplicity bound also follows from a parameter reduction: its Artinian quotient has a constant class and at least $\text{edim } A_0 - 2$ independent linear classes. We obtain $e_0 \notin \mathfrak{a}^2$, $\text{edim } A_0 = s+1$, and $\text{mult } A_0 = s$.

Choose ℓ_0 nonzero on each of the s axes and put $Q = (e_0, \ell_0)$. Then $\text{length}(A_0/Q) = s$, already accounted for by its constant and $s-1$ linear classes. Consequently $\mathfrak{a}^2 \subset Q$; independence of the two linear initial forms gives $Q \cap \mathfrak{a}^2 = Q\mathfrak{a}$, hence $\mathfrak{a}^2 = Q\mathfrak{a}$. Induction gives

$$(8.4) \quad (e_0) \cap \mathfrak{a}^{N+1} = e_0\mathfrak{a}^N \quad (N \geq 0).$$

The case $N = 0$ is immediate. For $N \geq 1$, write $e_0c = e_0c_1 + \ell_0c_2$ with $c_1, c_2 \in \mathfrak{a}^N$. The parameters form a regular sequence, so $c_2 = e_0c_3$ and $c - c_1 = \ell_0c_3$. The preceding induction case gives $c_3 \in \mathfrak{a}^{N-1}$, proving the claim. Thus e_0^* is regular on $\text{gr}_{\mathfrak{a}} A_0$, and its quotient is the reduced Cohen–Macaulay axis ring. A homogeneous nilpotent in $\text{gr}_{\mathfrak{a}} A_0$ would be divisible by e_0^* , and regularity would make the quotient nilpotent too. Iteration in the positively graded ring makes it zero. Hence the full tangent is reduced and Cohen–Macaulay.

Theorem 8.4 now applies to this normal minimal surface. On its minimal resolution the fundamental cycle is reduced, and by Theorem 8.5 it is the maximal-ideal cycle. The resolution factors the point blowup. The equation $e_0^* = 0$ cuts its projective tangent in s distinct reduced points by (8.4). At each point the parameter quotient has length one, so the tangent curve is regular there. The point blowup is regular there too: a lift of a regular parameter of its Cartier quotient, together with its exceptional equation, generates the surface maximal ideal. The blowup is Cohen–Macaulay by Theorem 8.4; it is regular at the generic points of its reduced exceptional curve and off that curve, so Theorem 8.6 also gives its normality. A minimal resolution changes none of the regular marked points. In these coordinates e_0 is the exceptional parameter times a function with a simple zero on the exceptional curve. The strict branches therefore meet it transversely and avoid the remaining singularities. On the factoring minimal resolution this

proves, for the actual function,

$$(8.5) \quad \operatorname{div}(e_0) = \sum_a F_a + \sum_{i=1}^s C_i.$$

Here every F_a is an exceptional curve and every C_i is a strict branch; each branch meets one exceptional curve once. Intersecting this principal divisor with F_a gives

$$-F_a^2 = \text{internal valence of } F_a + \text{number of attached strict branches.}$$

Thus its self-intersection is minus its valence in the attached tree. The internal weights are at least two, since the resolution is minimal.

Choose the algebraic minimal resolution of each generic transverse surface before completing. Completion is flat and preserves the intersection numbers and the orders of the Cartier equation on its exceptional curves; it also preserves minimality. Thus the completed resolution is the minimal resolution computed above. The computations can then be made at geometric generic points and spread over a common open subset of each conductor. The marked strict branches are the original components and are defined over K . Every leaf of the attached tree is one of these branches, because each internal full valence is at least two. An automorphism of a finite tree fixing all leaves fixes every vertex and edge. Thus splitting the generic resolution introduces no permutation of unmarked vertices. The intersection points are defined over K as well, so each genus-zero internal curve has a rational point and is $\mathbb{P}_K^1$. The intersection identities descend. Equivalently, one includes all conjugate branches and their residue degrees before descent. A common proper model realizes the finitely many generic resolutions; exceptional loci over the remaining smaller subsets will be avoided by the test curves. Replacing each conductor vertex by its attached tree preserves the incidence tree. Equations above show that the pullback of the original point ideal has value one at every vertex, including the vertices with mass zero.

Step 3. In this step, we derive the full adjunction matrix and prove positivity of its inverse. Fix this model before choosing P . On an original component choose a general complete curve cut by $d - 2$ hyperplanes; on an internal conductor divisor choose a general $\mathbb{P}^1$ fiber. These curves cover their respective divisors. The first curves avoid the singular loci of the normal components and the nongeneric parts of their conductors, and meet every conductor generically. The component's Cohen–Macaulay minimal-degree Hilbert series is $(1 + (m_i - 1)z)/(1 - z)^d$. A general hyperplane curve has Hilbert polynomial $m_i j + 1$, hence is smooth of genus zero. Adjunction along this flag gives

$$H_i \cdot F_i = m_i, \quad -K_{E_i} \cdot F_i = (d - 2)m_i + 2.$$

Every conductor has degree one on this curve. On an internal fiber the canonical degree on the transverse surface is $v_i - 2$, since its self-intersection is $-v_i$.

Use the discriminant $\mathcal{B}$ and the same compatible moduli $\mathcal{M}$ from [Theorem 4.1](#) and [Proposition 4.18](#). Write the exceptional coefficients as $1 - u_i$. To check $u_i \geq b$ also on the internal vertices, principalize the ambient point ideal on a common resolution carrying an lc place dominating W . The divisor of that ideal has no such place as a component. Its restriction there dominates the pullback of any base prime times the base prime's point-ideal value. The pair with point-ideal exponent b is lc; adjunction to the coefficient-one place is sub-lc. In the threshold definition of $\mathcal{B}$ in [Theorem 4.1](#), this makes every threshold at least b times that value. Each value here is one. Taking the infimum over all the dominating places proves $u_i \geq b$.

Let p_i be the intersection of the entire strict prime with the indicated general curve. It is nonnegative, since the prime contains no exceptional component and the curves are general. Retain its old coefficient $\alpha \geq 0$. Every remaining strict boundary term has nonnegative intersection with these curves. The same is true of the moduli trace, with the following precise comparison. The present model is regular near the chosen curves. On a common projective model $\pi : V \rightarrow Y'$ carrying the original potentially nef trace, set

$$N = \pi^* \mathcal{M}_{Y'} - \mathcal{M}_V.$$

This is an exceptional rational Cartier divisor in the neighborhood under consideration, and $-N$ is relatively nef. By [Theorem 4.11](#), $N \geq 0$. A general curve covering a divisor is not contained in its exceptional image. Its strict transform therefore has nonnegative intersection with N and with $\mathcal{M}_V$, which is nef on complete curves by potential nefness. This proves the required nonnegativity of $\mathcal{M}_{Y'}$ on these curves, without declaring its lower trace nef everywhere.

The pullback of E restricts to $-H_i$ on an original component and has degree zero on an internal fiber. Ordinary adjunction on the resolved transverse crossings contributes each conductor with coefficient one. Intersect the whole equality $K_{Y'} + \mathcal{B}_{Y'} + \mathcal{M}_{Y'} \equiv_W 0$ with each test curve. Retaining the strict prime and all exceptional coefficients gives

$$(8.6) \quad \alpha p_i + (m_i + v_i)u_i - \sum_{j \text{ adjacent to } i} u_j \leq (d-2)m_i + 2 - v_i.$$

For an internal vertex $m_i = 0$; its right side is exactly $2 - v_i$, including the negative contributions at branching vertices. This proves $\alpha p + Mu \leq \kappa$.

The quadratic form is

$$\mathbf{y}^\top M \mathbf{y} = \sum_{i \in A} m_i y_i^2 + \sum_{ij \text{ an edge}} (y_i - y_j)^2.$$

It is positive for nonzero $\mathbf{y}$, since the tree is connected and has an original vertex of positive mass. If $M \mathbf{y} \geq 0$ and $\mathbf{y}$ had a negative minimum, a positive-mass row at that minimum would be negative. At a zero-mass minimum the inequality forces all neighboring values to equal it. Propagation reaches a positive-mass vertex, a contradiction. Thus $M^{-1} \geq 0$. A zero entry of one of its nonnegative columns would likewise propagate to every vertex and contradict its nonzero right-hand side. The inverse is strictly positive. Each principal internal block is also positive definite with nonnegative inverse.

Step 4. In this step, we eliminate internal vertices without losing any prime restriction degree. If $B \neq \emptyset$, the old strict prime on the ordinary blowup is rational Cartier at every generic conductor, as proved in [Theorem 8.9](#). Its numerical pullback to the transverse resolution is its strict transform plus $\sum_{j \in B} q_j F_j$. Intersecting with each internal fiber gives

$$q = M_{BB}^{-1} p_B.$$

Restriction of the same local principal multiple to an original component consequently has degree

$$\delta = p_A - M_{AB} M_{BB}^{-1} p_B = p_A + C p_B.$$

This identifies δ with the rational restriction degrees of the entire prime in [Theorem 8.9](#). In particular $\delta \geq 0$ and $\sum_i \delta_i = \text{mult}_x P$. The zero masses give

$$M_{BA} \mathbf{1}_A + M_{BB} \mathbf{1}_B = 0, \quad \mathbf{1}_A^\top C = \mathbf{1}_B^\top.$$

It follows that $\sum_i p_i = \sum_{i \in A} \delta_i$; no internal prime intersection is lost or counted twice.

The internal rows of (8.6) imply

$$u_B \leq M_{BB}^{-1} (\kappa_B - M_{BA} u_A - \alpha p_B).$$

Since $M_{AB} \leq 0$, substitution into the original rows weakens their left sides and gives

$$\alpha \delta + S u_A \leq \hat{\kappa}.$$

The Schur complement is positive definite. Its off-diagonal entries are nonpositive since $M_{BB}^{-1} \geq 0$, and the block row-sum identity gives $S \mathbf{1}_A = (m_i)_{i \in A}$. When B is empty, the same conclusions hold with $\delta = p$, $S = M$, and $\hat{\kappa} = \kappa$.

Step 5. In this step, we derive the actual-threshold inequalities for every original-vertex subset. For the actual addition tH , compatible Cartier adjunction in [Proposition 4.18](#) keeps the moduli and adds the actual restriction $tH|_W$ to the discriminant. The new exceptional discrepancies are therefore $\rho_i = u_i - t \text{ord}_i(H|_W)$. Since the point ideal has value one at every vertex, $\text{ord}_i(H|_W) \geq \gamma$. The point-ideal comparison just given applies to the new pair with

exponent z , so $\rho_i \geq z$; it does not require W to remain minimal. The coefficient of P in the new base discriminant is exactly one. Indeed, its lc-center property forces a coefficient at least one over its generic point by [Theorem 4.1](#), a birational map to normal W is an isomorphism there, and lc bounds that coefficient by one. Thus $\alpha + t \operatorname{coeff}_P(H|_W) = 1$, retaining the original coefficient.

Repeat (8.6) with this entire coefficient-one prime and the new exceptional discrepancies. The same residual and moduli comparisons give

$$p + M\rho \leq \kappa, \quad \delta + S\rho_A \leq \hat{\kappa}.$$

Write $w_{ij} = -S_{ij} \geq 0$ for $i \neq j$. The row sums express S as the diagonal matrix of the masses plus the weighted graph Laplacian. Summing the last inequality over $I \subset A$, all edges internal to I cancel:

$$\sum_{i \in I} \delta_i + \sum_{i \in I} m_i \rho_i + \sum_{\substack{i \in I \\ j \notin I}} w_{ij} (\rho_i - \rho_j) \leq \sum_{i \in I} \hat{\kappa}_i.$$

Use $\rho_i \geq z$ and $\rho_j \leq U_j - \gamma t$. This is the claimed subset inequality. The proof applies to disconnected subsets and their complements, and retains every vertex of mass zero in the elimination which defines S and δ . $\square$

9. COMPLETE ORIGINAL SOURCES

We retain the original section spaces of [Definition 2.4](#). The two numerical comparisons below concern the images of these spaces in the ordinary point filtration. The multiplicity of a threshold average and its affine price enter as separate inequalities.

The finite certificates used here are supplied in `supplement/09-original-sources/`. Each identifier below names one exact rational data file. The proofs specify the inequalities to check, the full parameter covers, and why those checks give the claimed bounds. The accompanying independent checker expands these formulas and checks every record. It also reconstructs the covers, rejects missing or repeated records, and compares the small tables printed here with their canonical data.

Proposition 9.1. *For $u > 0$ and a nonzero complete $U_k(u)$, with ku integral, a sufficiently large sufficiently general pure average at its actual full-ideal threshold has precisely the threshold ideal's lc centers near x . Its coefficient $s = k\tau$, multiplicity M and deficit b obey*

$$M \geq us, \quad M + b \leq 7, \quad us + b \leq 7.$$

Every divisor of the source, including remote fixed components, remains in the average.

Proof. We realize the full-ideal threshold by a dilute average and then compute its whole coefficient and multiplicity.

Step 1. In this step, we construct an average with precisely the full threshold ideal's lc centers. Put $I = \mathfrak{b}(U_k(u))$ and $\tau = \operatorname{lct}_x(I)$. Principalize I and the point ideal on a common smooth model $f : Y \rightarrow X$. Write

$$I\mathcal{O}_Y = \mathcal{O}_Y(-G), \quad K_Y = f^*K_X + \sum_E (A_X(E) - 1)E.$$

Since $I \neq 0$, a log resolution gives a positive rational local threshold. Since $I \subseteq \mathfrak{m}_x^{ku}$ at x , the exceptional divisor of the point blowup gives $\tau \leq 7/(ku)$, so this threshold is finite.

The pullbacks of the actual members of the source have the form

$$f^*D_i = G + H_i,$$

where the H_i belong to the generated moving system on Y . Choose an integer $N > \tau$, and then a general N -tuple whose moving divisors and the fixed resolution divisors have simple normal

crossings near $f^{-1}(x)$. The crepant boundary of $\Gamma = (\tau/N) \sum_i D_i$ is

$$\tau G - \sum_E (A_X(E) - 1)E + \frac{\tau}{N} \sum_i H_i.$$

All its coefficients are at most one near $f^{-1}(x)$. Every moving coefficient is strictly below one. Consequently its coefficient-one strata are exactly those of the resolved ideal pair (X, I^τ) . Their images give the same lc centers.

Step 2. In this step, we bound the coefficient, multiplicity, and deficit of the entire average. Each actual D_i has point multiplicity at least ku and is linearly equivalent to kL . Thus

$$\Gamma \sim_{\mathbb{Q}} k\tau L, \quad \text{mult}_x \Gamma \geq \frac{\tau}{N} Nku = us.$$

For each $0 \leq t < b$, the point blowup applied to $(X, \Gamma; \mathfrak{m}_x^t)$ gives $M + t \leq 7$. Letting t increase to b gives $M + b \leq 7$, and then $us + b \leq 7$. The divisor G is the entire fixed divisor on the resolution; no component of any D_i has been removed. $\square$

Lemma 9.2. *Let X be a smooth integral complex projective sevenfold, L ample Cartier, and $x \in X$ a closed point. Put $v = L^7$. Fix a positive rational number T , an integer $D \geq 1$ with DT integral, a bounded piecewise continuous function $\mathcal{A} : [0, T] \rightarrow [0, 1]$, and constants $C \geq 0$ and $k_0 \geq 1$. For integers $\ell, k \geq 0$, with $k \geq 1$, put*

$$S_{\ell, k} = H^0(X, kL \otimes \mathfrak{m}_x^\ell).$$

Assume, for every D -divisible $k \geq k_0$ and $1 \leq \ell < Tk$, that

$$\dim(S_{\ell, k}/S_{\ell+1, k}) \leq \frac{\ell^6}{6!} \mathcal{A}(\ell/k) + C(\ell + 1)^5.$$

Then

$$\liminf_{\substack{k \rightarrow \infty \\ D|k}} \frac{7! \dim U_k(T)}{k^7} \geq v - 7 \int_0^T t^6 \mathcal{A}(t) dt.$$

If the right side is positive, every sufficiently large D -divisible original degree has $U_k(T) \neq 0$. If additionally $T = 13/5$, the positive lower bound gives generation of $K_X + 8L$ at x .

This is a statement about complete global kernels. It assigns no lower mass to an intrinsic image or a proper subspace.

Proof. The degree-zero quotient has dimension at most one. Telescoping the actual filtration gives

$$\dim U_k(T) \geq h^0(X, kL) - 1 - \frac{1}{6!} \sum_{\ell=1}^{Tk-1} \ell^6 \mathcal{A}(\ell/k) - C(Tk + 1)^6.$$

Here Tk is an integer. In particular there is no rounding of the final kernel and no assumption that a later kernel is nonzero. The complete-volume limit in [Theorem 6.4](#) ([\[LM09, Theorem 2.3 and equation \(2.6\)\]](#)), applied to the ample divisor L , and the Riemann sum give the displayed lower bound. Positivity of a liminf gives one cutoff valid for all the indicated original degrees. When $T = 13/5$, a nonzero section at one such degree has its entire effective divisor $D_k \in |kL|$ of multiplicity at least $13k/5$. Apply [Corollary 5.12](#) to D_k . $\square$

Lemma 9.3. *Let X be a smooth complex projective sevenfold, L ample Cartier, and $\Gamma \sim_{\mathbb{Q}} sL$ an effective rational boundary that is lc near x . Suppose its minimal lc center through x is a divisor smooth at x , and put $b = \text{def}_x(\Gamma)$. The whole scalar cost of every additive point continuation is at most $5b/4 + 187/250$. Consequently*

$$s + \frac{5}{4}b + \frac{187}{250} < 8 \implies \mathcal{O}_X(K_X + 8L) \text{ is generated at } x.$$

Proof. We specialize [Theorem 5.4](#) using a rational certificate for its finite recurrence. All degree floors are one: the restriction of an ample Cartier divisor has positive integral degree on every actual center.

Step 1. In this step, we specify the complete rational state and line certificate. Index the states of $\mathcal{P}_7$ as in `09-original-sources/scalar-tail`, stored in `data/scalar-tail.json` within this section's supplement. For row i , write (d_i, e_i, m_i) for its state, $T_i = T(e_i - d_i, m_i)$, and $\beta_i = d_i - 2T_i/m_i$. The column R_i is a rational upper bound for m_i^{1/d_i} ; the check is $R_i^{d_i} \geq m_i$. Define $H_i(z)$, for $0 \leq z \leq \beta_i$, as the minimum of all the affine lines $az + p$ in that row. For a possible successor j , with $d_j < d_i$ and $e_j \leq e_i$, put

$$t_{ij} = \begin{cases} \min\{\beta_i, \beta_j, \beta_i - m_j/m_i\}, & d_j = d_i - 1, \\ \min\{\beta_i, \beta_j\}, & d_j \leq d_i - 2. \end{cases}$$

Discard a successor only when $t_{ij} \leq 0$. For $t \geq 0$ set

$$G_j(v, t) = \max_{0 \leq y \leq t} \{H_j(y) - vy\}.$$

This maximum is checked at $0, t$ and at all intersections $y = (p' - p)/(a - a')$ in $(0, t)$ of distinct-slope lines from row j . Those points include every breakpoint of the minimum, even if some intersections are redundant.

The complete data satisfy, for each line (a, p) of each row i , the rational inequality

$$p \geq (R_i - a)_+ \beta_i + \max \left\{ 0, \max_{j: t_{ij} > 0} G_j(\max\{a, R_i\}, t_{ij}) \right\}.$$

Together with the displayed intersection rule and the radius power checks, this is the finite verification rule for all 71 rows. The columns retain every line, multiplicity and embedding dimension.

The data contain all 71 distinct states and all 253 affine lines. Completeness is checked by enumerating $(1, 1, 1)$ and, for $2 \leq d \leq 6$, $d \leq e \leq 7$, all positive integers $m \geq e - d + 1$ with $2T(e - d, m) < dm$; when $e = d$ only $m = 1$ is allowed. The moment formula in [Theorem 5.4](#) makes this a finite enumeration. The row identifiers run from one to 71, with no omissions; each line of each row is tested against every permitted successor.

Step 2. In this step, we deduce the complete recurrence bound from the finite tests. Here is why these tests certify the entire recurrence. If the current deficit is z and the next one is y , then $0 \leq y \leq \min\{z, t_{ij}\}$. Induction bounds the remaining scalar cost by $H_j(y)$. The total cost is at most $R_i(z - y) + H_j(y)$. After subtracting az , this is bounded by $(R_i - a)\beta_i + G_j(R_i, t_{ij})$ when $R_i \geq a$, and by $G_j(a, t_{ij})$ when $R_i < a$, since then $z \geq y$. The zero term handles a direct point successor. The test therefore gives every listed line, and their minimum, by induction on dimension. Both adjacent successors and all dimension skips occur in this induction.

For example, the row $(3, 6, 4)$ has $\beta = 3/2$ and $R = 397/250$. An adjacent surface of multiplicity five has deficit at most $1/4$, so its surplus over this radius is at most $(2237/1000 - 397/250)/4 = 649/4000$. For multiplicity four the surplus is $(2 - 397/250)/2 = 103/500$, the larger value recorded in that row. The same finite rule includes the larger threefold multiplicities. In embedding dimension at most six, multiplicity eight gives $\beta \leq 1/4$ and an adjacent positive-deficit surface has multiplicity less than two. All subsequent radii are at most two, giving whole cost at most $2z$. Multiplicity nine gives $\beta \leq 1/9$ and no adjacent positive-deficit surface; a skip to a curve has radius at most one. Its whole cost is at most $9z/4$, since $9^{1/3} < 9/4$. The separate embedding-seven rows, including multiplicities eight through eleven, remain in the table.

Step 3. In this step, we specialize the bound to the smooth divisorial root and reserve both cutting errors. A smooth divisorial root is row 66, with $(d, e, m) = (6, 6, 1)$. Its line $(5/4, 187/250)$ gives the claimed cost bound. If the displayed total coefficient is below eight, fix a smaller positive margin. [Theorem 5.4](#) pays both cutting errors within that margin while retaining the entire

incoming boundary and every subsequent cut. The resulting point boundary gives generation by [Proposition 3.12](#). $\square$

For the first integral comparison, set $r = 29/25$ and define

$$\mathcal{B}_{\geq 2}(t) = \begin{cases} 1, & 0 \leq t < 9/10, \\ 61/100, & 9/10 \leq t < 26/25, \\ 14/25, & 26/25 \leq t < r, \\ ((125 - 52t)_+ / (73t))^6, & t \geq r. \end{cases}$$

For $c \geq 0$, define three numerical capacities and their heights:

ν	h_ν	$K_\nu(c)$
div	3/10	$(1 - c)_+^6$
quad	8/25	$\max\{(1 - c)_+^6, (1 - 2c)_+^5(1 + 10c)\}$
span	257/1000	$\min\{1, 2(1 - 2c)_+^6\}$.

Lemma 9.4. *For each of the three pairs (h_ν, K_ν) just defined, and every real $a > h_\nu/r$, one has*

$$7 \int_0^{13/5} t^6 \min \left\{ \mathcal{B}_{\geq 2}(t), K_\nu \left(\frac{(h_\nu + a(t - r))_+}{t} \right) \right\} dt < \frac{199}{100}.$$

The integrand at zero is defined to be zero.

The bound is uniform over every supporting slope, including the unbounded ranges.

Proof. We integrate two affine kernels, reduce the bounded slope ranges to polynomial signs, and give the complete coefficients certifying those signs.

Step 1. In this step, we integrate common upper bounds for the incoming and outgoing kernels. Put

$$\kappa = \frac{37}{40}, \quad \beta = \frac{52}{73}, \quad q_0 = \frac{1617}{1825}, \quad C_0 = \frac{9}{50}, \quad D_0 = \frac{181}{100}.$$

An upper bound for $t^6 \mathcal{B}_{\geq 2}(t)$ is t^6 before $9/10$, $(\kappa t)^6$ between $9/10$ and r , and $(q_0 - \beta(t - r))_+^6$ after r . Indeed $\kappa^6 > 61/100 > 14/25$, and the last kernel is the one in the definition of $\mathcal{B}_{\geq 2}$. The contribution of replacing the first interval by $(\kappa t)^6$ is at most $(1 - \kappa^6)(9/10)^7 < C_0$.

For $p, z > 0$, put

$$\mathcal{R}_p(z) = \frac{\kappa^6(p + rz)^7}{z(z + \kappa)^6}.$$

In both applications p is $p_D = 43/50$ or $p_H = 1267/1500$; these satisfy

$$(9/10)\kappa < p < \kappa r, \quad p < q_0.$$

The crossing of $p - z(t - r)$ with κt is $t_1 = (p + rz)/(z + \kappa) \in (9/10, r)$. Integrating the two kernels on either side gives

$$(*) \quad 7 \int_0^r \min\{t^6 \mathcal{B}_{\geq 2}(t), (p - z(t - r))_+^6\} dt < C_0 + \mathcal{R}_p(z) - \frac{p^7}{z}.$$

On the outgoing interval, if $0 < z < \beta$, an upper bound is

$$(**) \quad 7 \int_r^\infty \min\{(q_0 - \beta(t - r))_+^6, (p - z(t - r))_+^6\} dt \leq \frac{p^7}{z} - \frac{(\beta p - z q_0)^7}{\beta z(\beta - z)^6}.$$

When $\beta p - z q_0 > 0$, this follows by integration at the positive crossing of the two lines. When that numerator is nonpositive, the right side is at least p^7/z , which already bounds the integral. At every slope the outgoing integral is also bounded by q_0^7/β ; for $z > 0$ it is bounded by p^7/z .

Step 2. In this step, we reduce the divisorial-kernel bound to endpoint comparisons and polynomial positivity. Write $z = a - 1$. The untruncated kernel is $p_D - z(t - r)$. Taking its positive part only enlarges the original comparison when the underlying profile is negative. For

$a \leq 1$, the incoming kernel is at most its value at $a = 1$. The total integral is therefore strictly below

$$C_0 + 7rp_D^6 - \frac{6p_D^7}{\kappa} + \frac{q_0^7}{\beta} < \frac{199}{100}.$$

For $z \geq 7/10$, (*) and the bound p_D^7/z give $C_0 + \mathcal{R}_{p_D}(z)$. The sign of the derivative of $\mathcal{R}_{p_D}$ is the sign of

$$(6r\kappa - 7p_D)z - p_D\kappa = \frac{209}{500}z - \frac{1591}{2000}.$$

Thus its supremum on this half-line is attained at its left endpoint or its limit $\kappa^6 r^7$. Both give a total strictly below $199/100$. For $0 < z \leq 7/10 < \beta$, (*) and (**) reduce the desired inequality to $P_D(z) > 0$, where

$$P_D(z) = z^{-1}[D_0\beta z(\beta - z)^6(z + \kappa)^6 - \beta\kappa^6(p_D + rz)^7(\beta - z)^6 + (\beta p_D - zq_0)^7(z + \kappa)^6].$$

The numerator vanishes at zero, so this is a polynomial of degree twelve.

Step 3. In this step, we reduce the two-normal-kernel bound to explicit polynomial inequalities. For $\theta_0 = 7/6$, weighted arithmetic-geometric mean gives

$$((1 - 2c)^5(1 + 10c))^{1/6} \leq A_0 - B_0c \quad (0 \leq c \leq 1/2),$$

where

$$A_0 = \frac{5\theta_0 + \theta_0^{-5}}{6} \leq \frac{21}{20}, \quad B_0 = \frac{5}{3}(\theta_0 - \theta_0^{-5}) \geq \frac{7}{6}.$$

Thus $K_{\text{quad}}(c)$ is at most $\max\{(1 - c)_+^6, (21/20 - (7/6)c)_+^6\}$, and is also at most one. Put

$$p_1 = \frac{21}{25}, \quad p_H = \frac{1267}{1500}, \quad w = \frac{7}{6}z + \frac{7}{60}, \quad z = a - 1.$$

For $z \geq 0$, the two affine kernels are $p_1 - z(t - r)$ and $p_H - w(t - r)$. Their maximum on the incoming interval is the second, since $p_H - p_1 = 7/1500 > 0$ and $w \geq z$. On the outgoing interval both are at most $p_H - z(t - r)$. For $a \leq 1$, the incoming maximum is bounded by its value at $a = 1$, giving the total bound

$$C_0 + \mathcal{R}_{p_H}(7/60) - \frac{p_H^7}{7/60} + \frac{q_0^7}{\beta} < \frac{199}{100}.$$

For $0 < z \leq 17/25 < \beta$, applying (*) with w and (**) with z reduces the claim to positivity of

$$P_H(z) = z^{-1}[D_0\beta zw(\beta - z)^6(w + \kappa)^6 - \beta z\kappa^6(p_H + rw)^7(\beta - z)^6 + \beta p_H^7(z - w)(\beta - z)^6(w + \kappa)^6 + w(\beta p_H - zq_0)^7(w + \kappa)^6].$$

Again the numerator vanishes at zero; this polynomial has degree thirteen. For $z \geq 17/25$, use the outgoing bound p_H^7/z . The required positive polynomial is

$$P_L(z) = D_0zw(w + \kappa)^6 - z\kappa^6(p_H + rw)^7 + p_H^7(z - w)(w + \kappa)^6.$$

Step 4. In this step, we bound the spanning kernel by the divisorial comparison. Since $2 < (9/8)^6$, its kernel is at most the sixth power of the positive part of $(9/8)[t - 2(h_{\text{span}} + a(t - r))]$. Its anchor is

$$p' = \frac{9}{8}(r - 2h_{\text{span}}) = \frac{2907}{4000} < p_D, \quad a' = \frac{9}{4}a - \frac{1}{8}.$$

The assumption on a gives $a' > 1 - p'/r > 1 - p_D/r = h_{\text{div}}/r$. Increasing the anchor from p' to p_D increases the kernel everywhere. The divisorial comparison already proved therefore bounds this integral as well.

Step 5. In this step, we verify polynomial positivity on all bounded intervals and the unbounded half-line. The data `09-original-sources/volume-two` specify a positive scale A_* , degree N , and coefficient tuple for each bounded interval $[l, u]$. They satisfy the identity

$$A_*(1+y)^N P\left(\frac{l+uy}{1+y}\right) = \sum_{i=0}^N c_i y^i.$$

For the unbounded interval the identity is $A_* P_L(17/25 + y) = \sum_{i=0}^8 c_i y^i$. All coefficients are positive. The substitutions cover the closed bounded intervals, using the leading coefficient at the right endpoint, and the entire unbounded half-line. The tuples are exact binomial expansions of the displayed polynomials.

The file `data/volume-two.json` contains the six bounded tuples and the unbounded tuple, comprising 90 integer coefficients, their exact positive scales, and the eight rational differences used here. The degrees are twelve for P_D , thirteen for P_H , and eight for P_L . The bounded intervals are $[0, 1/2]$, $[1/2, 3/5]$, $[3/5, 7/10]$ for P_D , and $[0, 1/2]$, $[1/2, 3/5]$, $[3/5, 17/25]$ for P_H . Checking the complete coefficient identities, including the constant and leading terms, proves positivity also at every shared endpoint.

The three intervals for P_D cover $[0, 7/10]$; the three for P_H cover $[0, 17/25]$. The four positive endpoint differences in the data certify, in order, the two bounds for $a \leq 1$ and the two endpoints for the divisorial half-line. The remaining differences certify the common envelope and the weighted-mean bounds. This proves the strict inequality in every case. $\square$

Theorem 9.5. *If X is a smooth complex projective sevenfold, L is ample Cartier and $L^7 \geq 2$, then $\mathcal{O}_X(K_X + 8L)$ is globally generated.*

Proof. Assume nongeneration at a closed point x . We first reach the original observation source, then select one of its actual profiles and apply the integral comparison.

Step 1. In this step, we obtain the complete observation source before selecting its threshold. The full bound of [Proposition 7.28](#) is at most $\mathcal{B}_{\geq 2}$ at every order up to $13/5$. Summing its complete layers gives

$$V(29/25) \geq L^7 - \frac{143360580567427}{78125000000000} > 0 \quad (L^7 \geq 2).$$

The subtracted rational number is $7 \int_0^{29/25} t^6 \mathcal{B}_{\geq 2}(t) dt < 46/25$. In particular $25\pi^*L - 29E_x$ is big before any center is selected. Choose a rational $\beta > 29/25$ still in the big interval and fix the constants of [Theorem 6.13](#) with $a = 1$ and $b_0 = 29/25$. Include the baseline cutoff at this stage. Choose one original degree $k_* = 25j$ beyond these cutoffs with $C/k_* < 10^{-5}$.

Let τ_* be the actual threshold of the full base ideal I_* of $U_{k_*}(29/25)$. Choose its pure threshold average and put $s = k_*\tau_*$, with whole multiplicity M and deficit b . [Proposition 9.1](#) gives $rs + b \leq 7$, where $r = 29/25$. By [Proposition 5.9](#) and [Proposition 7.26](#), its minimal center is a divisor, a double fivefold of embedding dimension six, or a fivefold of embedding dimension seven and multiplicity three, four or five. The entire tangents in the last two cases are reduced and Cohen–Macaulay.

Step 2. In this step, we obtain a profile height for each actual center alternative. Fix a computing place w centered on this actual minimal center and let $A = A_X(w)$. Its profile over all original degrees is

$$\phi_w(t) = \inf_{\substack{q \geq 1, \\ 0 \neq f \in H^0(X, qL) \\ \text{ord}_x f \geq qt}} \frac{w(f)}{qA}.$$

The uniform comparison chosen before w gives

$$\phi_w(r) \geq \frac{1}{s} - \frac{C}{k_*} \geq \frac{r}{7-b} - \frac{C}{k_*}.$$

For the double fivefold the census gives $b \geq 28348/8321$, hence

$$\frac{r}{7-b} \geq \frac{241309}{747475} > \frac{8}{25} + 10^{-5}.$$

For the three embedding-seven types its lower deficits are $28725/11134$, $3467/1328$, and $37483/15020$. Their minimum is the last one, and therefore

$$\frac{r}{7-b} \geq \frac{87116}{338285} > \frac{257}{1000} + 10^{-5}.$$

For a divisorial center P , choose $w = \text{ord}_P$ and put $m_P = \text{mult}_x P$. Use the profile $\phi = m_P \phi_w$. The coefficient of P in the actual threshold average is one, so $m_P \leq M \leq 7$. If $m_P \geq 2$, then

$$\phi(r) \geq \frac{2r}{7} - \frac{7C}{k_*} > \frac{3}{10}.$$

If $m_P = 1$, the divisor is smooth at x . Nongeneration and [Lemma 9.3](#) give $8 \leq s + 5b/4 + 187/250$. Together with $rs + b \leq 7$ this yields

$$b \geq \frac{17654}{5625}, \quad \frac{r}{7-b} \geq \frac{6525}{21721} > \frac{3}{10} + 10^{-5}.$$

Thus the same preselected cutoff supplies the height h_ν in each of the three classes.

Step 3. In this step, we bound every original layer using the fixed computing profile. Use $\phi = \phi_w$ in the nondivisorial cases. Products and powers of original sections, or [Theorem 7.23](#), give a finite supporting slope $a > h_\nu/r$ such that $\phi(t) \geq T(t) = (h_\nu + a(t-r))_+$ throughout its domain. The place and slope now remain fixed. For a nondivisorial center, each original degree- k germ of point order at least ℓ belongs to

$$J_{k,\ell} = \{g : w(g) \geq kAT(\ell/k)\}.$$

When $T(\ell/k) > 0$, this ideal is primary for the selected center prime: w is positive on that prime and zero on its complement. A power of the prime lies in $J_{k,\ell}$. Its generic threshold is at most $1/(kT(\ell/k))$. The actual layer is consequently contained in the full degree- ℓ initial ideal.

For an embedding-seven fivefold the comparison is [Theorem 7.18](#) with $n = 7, t = 0, s = 3$. Its full kernel K_5 is bounded above by $\min\{1, 2(1-2c)_+^6\}$, the function used in [Lemma 9.4](#). The exact two-normal kernel in [Theorem 7.9](#) gives the double-fivefold comparison. For the divisorial profile, every germ at degree k and order ℓ contains that equation to exponent at least $\lceil kT(\ell/k)/m_P \rceil$. Its full point initial form therefore has a factor of degree at least $kT(\ell/k)$. The resulting layer bound is

$$\frac{(\ell - kT(\ell/k))_+^6}{6!} + O((\ell + 1)^5).$$

The actual germ retains every other fixed component. When $T = 0$, the full polynomial image gives the same bound. All local errors are uniform in the primary depth, degree, layer and computing place.

Take the minimum with [Proposition 7.28](#) on these same complete images. Then [Lemma 9.2](#) applies with

$$\mathcal{A}(t) = \min \left\{ \mathcal{B}_{\geq 2}(t), K_\nu \left(\frac{(h_\nu + a(t-r))_+}{t} \right) \right\}, \quad T = 13/5, \quad D = 25.$$

By [Lemma 9.4](#), the residual normalized dimension is greater than $1/100$. This gives an actual section of order $13k/5$ for all sufficiently large 25-divisible k . [Corollary 5.12](#) contradicts nongeneration at x . Since the point was arbitrary, the adjoint bundle is globally generated. $\square$

For the second integral, put $r_0 = 9/10$, $r_1 = 26/25$ and $Z = 27/25$. Define

$$\mathcal{B}_1(t) = \begin{cases} 1, & 0 \leq t < r_0, \\ 61/100, & r_0 \leq t < 99/100, \\ 11/25, & 99/100 \leq t < r_1, \\ 17/50, & r_1 \leq t \leq Z. \end{cases}$$

For $c \geq 0$, set $E_j(c) = (1 - jc)_+^6$ and $N_q(c) = E_{7,q}(c)$, for $q = 3, 5$, in the notation of [Definition 7.7](#). Put

$$\begin{aligned} S(c) &= \mathcal{S}_{7,1,3}(c) = E_1(c) + 3E_2(c) - 5E_3(c) + 2E_4(c), \\ R(c) &= E_1(c) + E_2(c) - E_3(c), \\ H(c) &= \min\{1, 256/729 + 18(1/6 - c)_+\}, \\ D(c) &= \mathcal{S}_{7,0,4}(c) = 6E_2(c) - 8E_3(c) + 3E_4(c). \end{aligned}$$

Define

$$\begin{aligned} K_Q(c) &= \max\{E_1(c), (1 - 2c)_+^5(1 + 10c)\}, \\ K_M(c) &= \begin{cases} \min\{S(c), N_3(c), H(c)\}, & c < 2/11, \\ \min\{S(c), N_3(c), H(c), R(c)\}, & c \geq 2/11, \end{cases} \\ K_D(c) &= \begin{cases} D(c), & c < 1/6, \\ \min\{D(c), 138259/900000\}, & c \geq 1/6. \end{cases} \end{aligned}$$

The functions S and D are the full capacities in [Theorem 7.18](#) for fourfolds of embedding dimension six and seven, respectively. Their application requires the entire ordinary tangent to be reduced and Cohen–Macaulay. The additional R comparison for $c \geq 2/11$, and the height-three constant for $c \geq 1/6$, are provided by [Proposition 7.19](#). In particular K_D retains D on its whole domain and adds the constant only where that separate bound applies. The function H is supplied by [Proposition 7.21](#) for every height-three primary ideal. Thus all the local capacities are available once the actual center types are established.

Lemma 9.6. *Put $h_0 = 131/1000$. Choose*

$$(\nu, h_1) \in \{(Q, 37/200), (M, 41/200), (D, 189/1000)\}.$$

For real $a_0 > h_0/r_0$ and $a_1 > h_1/r_1$, put $T_i(t) = (h_i + a_i(t - r_i))_+$, $i = 0, 1$. Then

$$7 \int_0^Z t^6 \min\{\mathcal{B}_1(t), N_5(T_0(t)/t), K_\nu(T_1(t)/t)\} dt < \frac{31}{32}.$$

The integrand at zero is defined to be zero.

The complete finite comparison includes the incoming intervals before both observations.

Proof. We give a cover of the full slope domain and the exact polynomial integrals on every member of that cover.

Step 1. In this step, we cover the full two-slope domain by bounded rectangles and unbounded comparisons. The functions S, R, D are nonincreasing by inclusion–exclusion on the seven-coordinate unit simplex: S is the probability that one designated coordinate is at least c or at least two of three other coordinates are at least c ; R is the probability that one designated coordinate is at least c or two other designated coordinates are both at least c ; D is the probability that at least two of four designated coordinates are at least c . For a set of j coordinates, the probability that they all exceed c is $E_j(c)$, by translation of the simplex. These identities give exactly the displayed polynomials. The other capacities are nonincreasing directly from their formulas; adding a term to a minimum at a branch endpoint can only decrease it.

On a rectangle $[l_0, u_0] \times [l_1, u_1]$, use slopes (u_0, u_1) before r_0 , (l_0, u_1) between r_0 and r_1 , and (l_0, l_1) after r_1 . Each replacement lowers its profile height on the indicated interval. The

capacities are nonincreasing, so the substituted integrand bounds the original integrand from above throughout the rectangle.

The first slope grid is $(131/900, 1, 2, 4, 8, 16)$; the second is $(h_1/r_1, 1, 2, 4, 8, 16)$. Each family starts with their twenty-five product rectangles. A tag $i.j$ identifies the two initial intervals. At a recorded split, coordinate zero or one is bisected at the recorded rational midpoint; suffixes zero and one identify the lower and upper children. Thus the leaf rectangles in the data cover the full bounded domain, with shared endpoints retained.

If $a_0 \geq 16$, discard the second profile, use the baseline before r_0 , and use slope 16 for the first profile after r_0 . If $a_1 \geq 16$, discard the first profile, use the baseline before r_1 , and use slope 16 for the second after r_1 . Increasing a slope beyond its anchor increases its height, so these bounds cover both unbounded ranges and their intersection.

Step 2. In this step, we specify the polynomial bounds and their full validity intervals. For one of the endpoint profiles with height h , anchor v , and slope a , put $d = av - h$, $Y(t) = at - d$, and

$$P_j(t) = 7((1 - ja)t + jd)_+^6.$$

The prefixes 0 : and 1 : in the data specify the corresponding endpoint profile. The names mean

Bq	$7qt^6$
E1	P_1
Nqlo	$7t^6 - 7(7q/(q+1))^q t^{6-q} Y(t)^q$
Nqhi	$7t^6 - 7[6(q+1)]^{-q} t^{6-q} ((6q-1+7a)t - 7d)^q$
C	$(7/6)^6 P_1$
Qbal	$7((1-2a)t + 2d)^5((1+10a)t - 10d)$
Mspan	$P_1 + 3P_2 - 5P_3 + 2P_4$
Mref	$P_1 + P_2 - P_3$
H3	$7(256/729)t^6$
H3lip	$7(256/729 + 3 - 18a)t^6 + 126dt^5$
Dlo	$6P_2 - 8P_3 + 3P_4$
Dhi	$7(138259/900000)t^6.$

Here $q = 3, 5$ in the generic envelopes; in Bq it is the displayed baseline rate. The table entries split every positive part at its rational zero. Thus a named expression is a polynomial on its entire open integration cell, with endpoint values understood by continuity. Only the applicable branches with $Y(t) > 0$ are used for profile capacities.

For the crossing in K_Q , put

$$c_- = \frac{16020581569}{90194313216}, \quad c_+ = \frac{8010290789}{45097156608}.$$

The logarithmic derivative of $(1 - c)^6 / ((1 - 2c)^5(1 + 10c))$ has the sign of $12c - 1$. The ratio starts at one, decreases, and then increases to infinity. The rational comparisons at c_- and c_+ give, respectively,

$$(1 - c_-)^6 < (1 - 2c_-)^5(1 + 10c_-), \quad (1 - c_+)^6 > (1 - 2c_+)^5(1 + 10c_+).$$

They bracket its unique positive crossing. The data use Qbal only below c_- and E1 only above c_+ . Inside the bracket another profile or the baseline is used.

Step 3. In this step, we sum the exact cell bounds and prove the strict integral margin. A row (l, r, name, N) asserts

$$\int_l^r P(t) dt < \frac{N}{10^9}, \quad \int_l^r \sum_i c_i t^i dt = \sum_i \frac{c_i}{i+1} (r^{i+1} - l^{i+1}),$$

where P is the polynomial from the dictionary. Every named polynomial is an upper bound for the minimum on its full cell; it need not attain that minimum. The certificate `09-original-sources/two-profile` gives every split, leaf rectangle, integration cell and numerator, including both unbounded slope comparisons.

Its file `data/two-profile.json` contains exactly the families Q, M, D , with respectively 31, 13, 0 splits, 477, 321, 201 bounded-domain integration cells, and 9, 11, 12 unbounded-domain cells. These are 1,031 cells in total. Each leaf tag must occur once in the tree and once in the cell data; a split replaces an active rectangle by its two exact halves. Each ordered cell list must start at zero, end at Z , and have consecutive endpoints equal. Both unbounded slope ranges are required for each family. For every cell the stated integer is $N = \lfloor 10^9 \int_l^r P(t) dt \rfloor + 1$, so the inequality is strict even when the scaled integral is an integer. The degree-six integrals can equivalently be computed by exact Lagrange interpolation at the seven rational nodes $j/6$, $0 \leq j \leq 6$, after rescaling the cell to $[0, 1]$; the integration weights satisfy all seven moment identities $\sum_j w_j (j/6)^m = 1/(m+1)$, $0 \leq m \leq 6$.

There are 56, 38, 25 leaf rectangles for Q, M, D , respectively. The largest sums of upper numerators are

$$968716327, \quad 968546109, \quad 937247717.$$

The first unbounded range has numerator 502032677. The second has numerators

$$946259716, \quad 944079086, \quad 932247877.$$

Every integer is below 968750000, the numerator of $31/32$. The smallest stated margin is $33673/10^9$. The contiguous integration cells cover $[0, Z]$, and the rectangle trees and the two unbounded ranges cover all admissible slopes. This proves the claim. $\square$

For a prime divisor P on X , we use its profile on the original ambient section ring:

$$\phi_P(t) = \inf_{\substack{q \geq 1, \\ 0 \neq f \in H^0(X, qL) \\ \text{ord}_x f \geq qt}} \frac{\text{ord}_P(f)}{q}.$$

Lemma 9.7. *Let X be a smooth integral complex projective sevenfold, L ample Cartier with $L^7 = 1$, and $x \in X$ a closed nongeneration point of $K_X + 8L$. Assume $V(27/25) > 1/32$. Then there is a cutoff J , fixed before the following degree, average, center and computing place, such that:*

- (i) *For every 25-divisible $k \geq J$ and every real $0 \leq t \leq 27/25$, the complete $U_k(t)$ is nonzero and its full fixed divisor obeys*

$$F_k(t) \cdot L^6/k < 2/5.$$

- (ii) *If a pure actual threshold average of $U_k(27/25)$ at any such degree has a divisorial minimal center P smooth at x , its data satisfy*

$$L^6 \cdot P = 1, \quad b \geq 1486/625, \quad s \leq 107/25, \quad \phi_P(27/25) > 7/30.$$

The volume hypothesis supplies mass for the original source; the smooth-divisor conclusion uses the complete continuation bound in [Lemma 9.3](#).

Proof. We first fix all degree cutoffs, then bound the whole fixed divisor, and finally use the actual coefficient of a smooth divisorial center.

Step 1. In this step, we fix one cutoff controlling the complete sources and their entire fixed divisors. Put $Z = 27/25$. The Cartier divisor $25\pi^*L - 27E_x$ on the point blowup is big. Its complete series therefore gives $U_k(Z) \neq 0$ at every sufficiently large 25-divisible k . Equivalently, this also follows from [Theorem 6.13](#), after choosing a rational $\beta > Z$ still in the big interval and taking $a = 1$, $b_0 = Z$. Fix its constants C, K at this stage. Choose $J \geq K$ such that $C/k < 10^{-6}$ for all $k \geq J$. These choices precede any new threshold, tuple, prime or place.

Apply [Proposition 6.19](#) to the complete source at Z . On its full principalization let M_k be the generated moving divisor and set $z_k = M_k^7/k^7$. Since $z_k \rightarrow V(Z) > 1/32 > (3/5)^7$, enlarge J so that $z_k > (3/5)^7$ for every allowed $k \geq J$. The comparison is exact: $1/32 > (3/5)^7$ is

78125 > 69984 after multiplication by the positive common denominator. Writing $F_k(Z)$ for the full fixed divisor on X , the same proposition gives

$$\frac{F_k(Z) \cdot L^6}{k} \leq 1 - z_k^{1/7} < \frac{2}{5}.$$

For every real $0 \leq t \leq Z$, $\lceil kt \rceil \leq kZ$. Hence $U_k(Z) \subseteq U_k(t)$, the latter space is nonzero, and $F_k(t) \leq F_k(Z)$ coefficientwise. This proves the bound simultaneously throughout the interval, including all remote fixed primes.

Step 2. In this step, we derive the degree and profile height of the actual smooth prime. Now choose an allowed $k \geq J$, its full ideal I , its own threshold τ , and a sufficiently general pure average. Suppose its minimal center is a divisor P smooth at x . At the generic point of P , log canonicity with this divisorial center gives

$$\tau \operatorname{ord}_P(I) = 1, \quad \frac{\operatorname{ord}_P(I)}{k} = \frac{1}{s}.$$

[Proposition 9.1](#) gives $Zs + b \leq 7$. Nongeneration and [Lemma 9.3](#) imply

$$8 \leq s + \frac{5}{4}b + \frac{187}{250} \leq \frac{7-b}{Z} + \frac{5}{4}b + \frac{187}{250}.$$

Indeed a strict inequality in the other direction leaves a positive margin for both cutting errors and gives generation. Solving this affine inequality gives

$$b \geq \frac{1486}{625}, \quad s \leq \frac{107}{25}, \quad \frac{1}{s} \geq \frac{25}{107}.$$

For the positive integer $g = L^6 \cdot P$, the whole fixed-divisor estimate now implies

$$\frac{25g}{107} \leq \frac{g}{s} \leq \frac{F_k(Z) \cdot L^6}{k} < \frac{2}{5}.$$

If $g \geq 2$, this would give $50/107 < 2/5$, contradicting $250 > 214$. Therefore $g = 1$.

For the original profile at rational Z , taking the twenty-fifth power of any actual section shows that this infimum is unchanged on the 25-divisible progression. The uniform comparison already fixed before P gives

$$\phi_P(Z) \geq \frac{\operatorname{ord}_P(I)}{k} - \frac{C}{k} \geq \frac{25}{107} - \frac{C}{k} > \frac{7}{30},$$

because $25/107 - 7/30 = 1/3210 > 10^{-6}$. Thus every cutoff used in the conclusion was chosen before the source degree and its actual smooth prime. $\square$

Lemma 9.8. *Let X be a smooth integral complex projective sevenfold, let L be ample Cartier with $L^7 = 1$, and suppose $K_X + 8L$ is not generated at x . There is a cutoff J_0 , depending only on the original data, such that $U_{20i}(9/10) \neq 0$ for every $i \geq J_0$. Every sufficiently large sufficiently general pure average at its actual full-ideal threshold has minimal-center dimension in $\{2, 3, 4, 5\}$ and deficit $b_0 > 4/25$. The cutoff precedes the degree, average, center and computing place.*

Proof. We obtain the initial source and fixed-degree bound, derive its deficit bound, and exclude divisorial centers.

Step 1. In this step, we obtain the nonzero initial source and its fixed-degree bound. Write $r = 9/10$. The ordinary jet count and [Lemma 9.2](#) give $V(r) \geq 1 - r^7 > r^7$; the last inequality is $2 \cdot 9^7 < 10^7$. Thus every sufficiently late original degree $20i$ has a nonzero source. By [Proposition 6.19](#), its entire fixed divisor has degree less than $20i/10$, after one fixed enlargement of the cutoff.

Step 2. In this step, we bound the deficit and exclude point and curve centers. Fix an actual threshold average, of whole coefficient s and deficit b . By [Proposition 9.1](#), $s \leq (10/9)(7 - b)$. Every radius $m^{1/d}$ of the 71 states in [Lemma 9.3](#) is at most $\sqrt{6}$. Indeed the multiplicities in dimensions one through six are bounded respectively by 1, 6, 11, 14, 19, 6, and $11^2 < 6^3$, $14 < 6^2$,

and $19^2 < 6^5$. Since every original center degree is a positive integer, induction in [Theorem 5.4](#) bounds the entire remaining scalar cost by $\sqrt{6}b$: each drop from b to y costs at most $\sqrt{6}(b - y)$, including every skip. If $b \leq 4/25$, the whole cost is therefore at most

$$\frac{70}{9} + \left(\sqrt{6} - \frac{10}{9} \right) \frac{4}{25} < 8.$$

The strict margin is $(5/2 - \sqrt{6})4/25$, which pays both cutting errors in [Theorem 5.4](#). Hence $b > 4/25$. A curve has remaining scalar cost at most b , so its whole cost is at most $70/9 - b/9 < 8$. A point has $s \leq 70/9 < 8$ and is paid by [Proposition 3.12](#).

Step 3. In this step, we exclude a divisorial center using the preselected full fixed-degree bound. If the minimal center were a divisor P , its coefficient in the pure average would be one. For the full base ideal I , this says $\text{ord}_P I = 1/\tau = 20i/s$. Its original degree $L^6 \cdot P$ is a positive integer, so the entire fixed divisor has normalized degree at least $1/s \geq 9/70 > 1/10$, contrary to the preselected bound. These exclusions leave exactly the stated dimensions. No generic primarity of I was used. $\square$

Lemma 9.9. *Let X be a smooth integral complex projective sevenfold, let L be ample Cartier, and suppose $K_X + 8L$ is not generated at x . Let $k \geq 1$ be divisible by 25, with $U_k(26/25) \neq 0$. If a sufficiently large sufficiently general pure average at its actual full-ideal threshold has a four-dimensional minimal center W , then*

$$(\text{edim } \mathcal{O}_{W,x}, \text{mult}_x W) \in \{(6, 3), (6, 4), (7, 4), (7, 5), (7, 6)\}.$$

The entire ordinary tangent ring of W at x is reduced and Cohen–Macaulay. Its deficit satisfies $b > 39/20$ in embedding dimension six and $b > 3/2$ in embedding dimension seven.

Proof. We first exclude four scalar states, then bound the slack for the five remaining states. Put $u = 26/25$, and let s , M , and b be the whole source coefficient, point multiplicity, and deficit. The census [Proposition 5.9](#) gives $b > 3/2$.

Step 1. In this step, we exclude the four scalar states whose complete prices are below eight. The state list in [Lemma 9.3](#) then leaves only rows 28, 29, 30, 32, 33, 34, 39, 40, 41. For a scalar tail line $ab + p$ and deficit upper bound β , the whole cost is at most $(7 - b)/u + ab + p$. In the following four rows this increases with b , and its value at β is $8 - \eta$:

row	(e, m)	a	p	β	η
28	(4, 1)	1	757/1000	4	4659/13000
29	(5, 2)	119/100	287/500	3	16/1625
30	(5, 3)	1317/1000	171/500	2	703/3250
34	(6, 5)	187/125	297/1000	8/5	7611/65000

The positive gaps pay both cutting errors, excluding these rows.

Step 2. In this step, we bound the slack and prove full tangent reducedness for the five remaining states. For a remaining row set $\delta = 7 - M$ and $\sigma = \delta - b \geq 0$. Since $M \geq us$, $s \leq (7 - b - \sigma)/u$. Nongeneration and the complete tail line imply $8 \leq s + ab + p$. Thus

$$\sigma \leq 7 - 8u + (ua - 1)b + up \leq 7 - 8u + (ua - 1)\beta + up.$$

The last inequality uses $ua > 1$. Substitution gives every row of the following table:

row	(e, m)	a	p	β	upper bound for σ
32	(6, 3)	5/4	373/500	8/3	1599/6250
33	(6, 4)	283/200	379/1000	2	217/12500
39	(7, 4)	283/200	29/50	5/2	2311/5000
40	(7, 5)	187/125	17/40	2	2921/12500
41	(7, 6)	783/500	331/1000	5/3	2699/37500

All five bounds are below $1/2$, and $\delta = b + \sigma > 1$. The $\delta > 1$ clause of [Theorem 4.20](#) therefore makes the full ordinary tangent ring Cohen–Macaulay. Its $\sigma < 1/2$ clause makes the entire exceptional scheme reduced and then the full tangent ring reduced. This applies to the whole ring, including its vertex.

Step 3. In this step, we prove the stronger deficit bound in embedding dimension six. Finally, if $b \leq 39/20$ in row 32, use its line $(1317/1000)b + 71/125$; in row 33 use $(283/200)b + 379/1000$. The source price plus these whole tails is at most $8 - 2101/260000$ and $8 - 311/52000$, respectively. Both strict margins pay the two cutting errors. Hence $b > 39/20$ in the two embedding-six cases, as required. $\square$

Lemma 9.10. *Let X be a smooth integral complex projective sevenfold, let L be ample Cartier with $L^7 = 1$, and suppose $K_X + 8L$ is not generated at x . Put $U = 26/25$. Then $V(U) > 3/40$. There is a cutoff J_1 , depending only on the original data and fixed source comparisons, such that every $U_{25j}(U)$, $j \geq J_1$, is nonzero. If its pure actual threshold average has divisorial minimal center P , then P is smooth at x , its full original profile satisfies $\phi_P(U) > 1/5$, and $V(U) < (4/5)^7$. The cutoff precedes every such observation.*

Proof. We establish the source first, exclude excessive fixed-prime height, and then apply the scalar deficit bounds to the actual divisor.

Step 1. In this step, we obtain the source volume and fix its comparison cutoff. The full baseline and [Lemma 9.2](#) give

$$V(U) \geq 1 - \frac{39}{100}(9/10)^7 - \frac{17}{100}(99/100)^7 - \frac{11}{25}U^7 > \frac{3}{40}.$$

Consequently the source is big before any degree or prime is chosen. Fix its comparison constants from [Theorem 6.13](#), with normalized error $C/(25j) < 1/10000$, and include the nonzero-source cutoff.

Step 2. In this step, we exclude excessive multiplicity-weighted fixed-prime height. For any prime P through x , put $m = \text{mult}_x P$. We claim that $m\phi_P(U) < 37/100$. Otherwise the finite convex function $m\phi_P$, zero at zero, has a finite supporting slope $a \geq h/U$ at U , where $h = 37/100$. Its value on every order carrying an actual section is at least $T(t) = (h + a(t - U))_+$. Indeed products prove convexity, powers permit any fixed divisible progression, and endpoint secants retain the line at a finite endpoint. The same prime is fixed in all these steps.

Every actual section of original degree k and point order ℓ is locally divisible by a power of a full equation of P with point order at least $kT(\ell/k)$. Multiplication by its nonzero full initial form is injective in the ambient polynomial ring. Thus its degree- ℓ image has dimension at most

$$\frac{k^6}{6!} (\ell/k - T(\ell/k))_+^6 + C'(\ell + 1)^5.$$

All other fixed factors remain in the original image. Take the minimum with the full baseline, weakened here to

$$B_*(t) = \begin{cases} 1, & t < 9/10, \\ 61/100, & 9/10 \leq t < U, \\ 14/25, & U \leq t < 29/25, \\ ((125 - 52t)_+ / (73t))^6, & t \geq 29/25. \end{cases}$$

Through $Q = 13/5$, the total leading loss is at most

$$I(a) = 7 \int_0^Q \min\{t^6 B_*(t), (t - T(t))_+^6\} dt.$$

Here is an exact bound for the entire slope domain. Write $p = U - h = 67/100$, $r = 9/10$, $\kappa = 461/500$, and $\gamma = p/U = 67/104$.

For $h/U \leq a \leq 11/10$, the incoming loss is at most $11(387/550)^7 - 10p^7 < 17/50$. For outgoing orders use $(\gamma t)^6$ up to $t_* = 125/(52 + 73\gamma) = 13000/10299$, and then the late baseline kernel. The point t_* lies between $29/25$ and $125/52$. Integration gives the outgoing bound

$$\frac{125}{52}(\gamma t_*)^6 - Up^6 < \frac{7}{10} - \frac{9}{100}.$$

The three strict comparisons just used have integer cross-products, respectively,

$$\begin{aligned} 1476089891693672584850 &< 1505826850000000000000, \\ 431339178891181945800781250 &< 434381911923555432668905164, \\ 1125000000000000 &< 117595896819700. \end{aligned}$$

Thus $I(a) < 19/20$.

For $11/10 \leq a \leq 11/5$, put $z = a - 1$. The whole profile kernel integrated to infinity is

$$U_1(z) = \frac{(p + Uz)^7}{z(1+z)^6}.$$

Its derivative has the sign of $(6U - 7p)z - p$, so its maximum on $[1/10, 6/5]$ is at an endpoint. The endpoint bounds $U_1(1/10) < 19/20$ and $U_1(6/5) < 3/4$ are certified by

$$\begin{aligned} 26001999810956733660 &< 26296608593750000000, \\ 2983941066843017652476 &< 3188809800000000000000. \end{aligned}$$

For $a \geq 11/5$, let $z = a - 1 \geq 6/5$. The crossing $(p + Uz)/(z + \kappa)$ lies in (r, U) , and $61/100 < \kappa^6$. Integrating the trivial kernel, then $(\kappa t)^6$, then the profile kernel gives

$$I(a) \leq C_0 + \frac{\kappa^6(p + Uz)^7}{z(z + \kappa)^6}, \quad C_0 = (1 - \kappa^6)r^7.$$

The rational function increases on this range: its derivative has the sign of $(6U\kappa - 7p)z - p\kappa$, positive already at $z = 6/5$. It stays below its limit $\kappa^6 U^7$. The exact remaining bound is

$$C_0 + \kappa^6 U^7 < 199/200,$$

whose cross-product is

$$2423979354162758872184776736600 < 24291992187500000000000000000000.$$

This proves $I(a) < 199/200$ for every supporting slope. The uniform errors sum to $O(k^6)$ at fixed Q . By [Lemma 9.2](#), the actual source at $13/5$ has positive asymptotic mass, contradicting [Corollary 5.12](#). The height claim follows.

Step 3. In this step, we prove smoothness and the profile bounds for the observed divisor. Let $s = 25j\tau$ be the whole coefficient of the actual average. Its divisorial minimal center has $\text{ord}_P \mathbf{b}(U_{25j}(U)) = 1/\tau$. The census gives $b > 7/5$, and the state list gives $1 \leq m \leq 6$. If $m \geq 2$, the preselected comparison and $Us + b \leq 7$ imply

$$m\phi_P(U) \geq \frac{m}{s} - \frac{mC}{25j} > \frac{13}{35} - \frac{6}{10000} > \frac{37}{100},$$

contrary to the height claim. Hence $m = 1$, so a full local equation of P has a nonzero linear term.

For this smooth divisor, nongeneration and [Lemma 9.3](#) give

$$8 \leq \frac{7-b}{U} + \frac{5}{4}b + \frac{187}{250}, \quad b \geq \frac{3388}{1875}, \quad \frac{1}{s} \geq \frac{150}{749}.$$

It follows that $\phi_P(U) \geq 150/749 - 1/10000 > 1/5$, since $1498749 < 1500000$. Put $g = L^6 \cdot P \geq 1$. Every complete source at U has normalized full fixed degree at least $g\phi_P(U)$. Taking the moving-volume limit in [Proposition 6.19](#) gives

$$V(U) \leq (1 - g\phi_P(U))^7 < (4/5)^7.$$

The positive volume ensures $g\phi_P(U) < 1$. The argument counts the original positive integral degree; it needs no degree-one conclusion for P . $\square$

Lemma 9.11. *Let R be the local ring at a closed point of a smooth complex n -fold X , $n \geq 3$, with maximal ideal $\mathfrak{m}$. Let p be a regular parameter and P_W a height- c prime containing p , where $2 \leq c \leq n-1$, and write $W = V(P_W)$. Let w be a positive multiple of a divisorial valuation centered at P_W , put $A = A_X(w)$, and set $\alpha = w(p)/A$. Then $0 < \alpha < 1$. For $q > 0$ set $J_q = \{f \in R : w(f) \geq qA\}$. For $u, v \geq 0$ and a uniform n -coordinate simplex vector $(Z_0, \dots, Z_{n-1})$, define*

$$C_{n,c}(u, v; \alpha) = \sup_{\substack{\beta \geq \alpha, b_i \geq 0 \\ \beta + \sum_{i=1}^{c-1} b_i = 1}} \Pr \left\{ Z_0 \geq u, \beta Z_0 + \sum_{i=1}^{c-1} b_i Z_i \geq v \right\}.$$

For every integer $N \geq 0$ and $\ell \geq 1$,

$$\dim_{\mathbb{C}}(\text{in}_{\mathfrak{m}}((p^N) \cap J_q))_{\ell} \leq \frac{\ell^{n-1}}{(n-1)!} C_{n,c}(N/\ell, q/\ell; \alpha) + C_n(\ell+1)^{n-2}.$$

The constant depends only on n , before every other choice. Independently, let P_W be any prime of height $2 \leq c \leq n-1$, let w be a positive multiple of a divisorial valuation centered there, and set $A = A_X(w)$, $J_q = \{f : w(f) \geq qA\}$ for $q > 0$. Then the ordinary bound, with no regular parameter hypothesis, is

$$\dim_{\mathbb{C}}(\text{in}_{\mathfrak{m}} J_q)_{\ell} \leq \frac{\ell^{n-1}}{(n-1)!} \mathbb{M}_{n,c}(q/\ell) + C_n(\ell+1)^{n-2}.$$

Proof. We preserve the smooth parameter through two degenerations, use relative thresholds to obtain a common separating normal, and count the full intersection image.

Step 1. In this step, we preserve the relative threshold bounds through both full-ideal degenerations. An element outside P_W has value zero. Positive values on a finite generating set of P_W give $P_W^a \subseteq J_q \subseteq P_W$ for some integer a . Thus J_q is P_W -primary. At the generic point of W , complete p to regular parameters $p, z_2, \dots, z_c$. The coordinate simple-normal-crossing pair gives $A \geq w(p) + \sum_{i=2}^c w(z_i) > w(p) > 0$. For $0 \leq \eta < 1$, the same valuation gives

$$(9.1) \quad \text{lct}_{\eta W}((X, \eta(p=0)); J_q) \leq \frac{1 - \eta\alpha}{q}.$$

Consider the extended point Rees algebra $\mathcal{D} = \sum_{j \in \mathbb{Z}} \mathfrak{m}^j t^{-j} \subset R[t, t^{-1}]$, with $\mathfrak{m}^j = R$ for $j \leq 0$. Contract the entire ideals J_q, P_W from the generic fiber. Their quotients have no t -torsion; the special ideals are $J = \text{in}_{\mathfrak{m}} J_q$ and $Q = \text{in}_{\mathfrak{m}} P_W$, with $Q^a \subseteq J \subseteq Q$. The horizontal support is prime. At every height- c minimal prime T of Q , the total local ring at (t, T) is regular: its quotient by the nonzerodivisor t is the dimension- c regular local ring $(\text{gr}_{\mathfrak{m}} R)_T$, and its parameters lift together with t . The horizontal generic localization is R_{P_W} after a transcendental coefficient-field extension.

The retained element is p/t . Its special fiber is the nonzero linear form $p_0 = \text{in}_{\mathfrak{m}} p$. Since $p_0 \in T$, it is a regular parameter of the special local ring, and $p/t, t$ are part of a regular parameter system of the total ring. The smooth restriction theorem [Theorem 6.1](#) ([\[Laz09, Theorem 3.2 and the ideal version after Remark 3.7\]](#)) applies with both this retained divisor and the special fiber. Its mixed nonnegative rational form follows by clearing a common denominator in a product of integer ideal powers. A unit multiplier ideal on the special fiber forces a unit ambient multiplier ideal by the restriction inclusion and Nakayama's lemma. Localizing at the horizontal generic point would then contradict [\(9.1\)](#). Thus that inequality holds at every height- c component of the full point-initial support, with p_0 retained. The coefficient-field passage preserves these multiplier ideals and discrepancies: extend a log resolution and its divisor coefficients over that field.

Now take a common homogeneous monomial term order for J, Q in coordinates including p_0 . One integer weight realizes both finite Gröbner bases; adding a common constant makes all

weights positive without changing the homogeneous initials. The full special ideals K, H preserve every graded Hilbert function and satisfy $H^a \subseteq K \subseteq H$. At a height- c coordinate minimal prime T' of H , localize the weight family at its special-fiber parameter and T' . A horizontal minimal support prime has height at least c by the generic support dimension and at most c by strict containment in this height- $c + 1$ prime. It is therefore one of the components already controlled above. Both the special and total ambient local rings are regular. The distinguished coordinate is still p_0 ; under generic diagonal identification its extra parameter factor is a unit. The same mixed restriction argument proves (9.1) for $K_{T'}$. It first gives rational $\eta < 1$. On a fixed log resolution the threshold is the minimum of finitely many affine functions of η , so continuity gives every real $\eta < 1$ and the limiting inequality at one.

Step 2. In this step, we obtain one separating normal for the full relative threshold family. The ideal $K_{T'}$ is primary in its c normal coordinates, over the coefficient field of the remaining coordinates. By Theorem 7.1 ([How01, Main Theorem]), its Newton polyhedron has interior disjoint from the segment

$$[q(1, \dots, 1), (0, q/(1 - \alpha), \dots, q/(1 - \alpha))].$$

In fact the relative toric discrepancy subtracts η times the first normal weight. The corresponding test point is $q(1 - \eta, 1, \dots, 1)/(1 - \eta\alpha)$; these points traverse the segment up to its last endpoint, which cannot be interior either.

Separate this compact segment from the open convex Newton interior. Upward closure gives a nonzero normal $v_i \geq 0$ and a minimum h_v on the polyhedron satisfying

$$h_v \geq q \sum_i v_i, \quad h_v \geq \frac{q}{1 - \alpha} \sum_{i>0} v_i.$$

All v_i are positive: otherwise a pure power of the zero-weight coordinate in the primary ideal would have value zero, contrary to $h_v \geq q \sum_i v_i > 0$. Set $d_i = qv_i/h_v$, retain $b_i = d_i$ for $i > 0$, and increase the first coefficient to $\beta = 1 - \sum_{i>0} d_i$. Then $\beta \geq \alpha$, the coefficients sum to one, and every monomial of the whole K , before localization, satisfies

$$\beta u_0 + \sum_{i>0} b_i u_i \geq q.$$

Localization was used only to obtain a necessary inequality for all those original monomials.

Step 3. In this step, we bound the full intersection image with a uniform lattice remainder. The full point-initial image of $(p^N) \cap J_q$ is contained in $(p_0^N) \cap J$. Applying the same monomial order preserves its degree- ℓ dimension and puts every leading monomial in $(p_0^N) \cap K$. Its exponent therefore obeys $u_0 \geq N$, the preceding normal inequality, and $\sum u_i = \ell$. This is an inclusion, so it does not require initials to commute with intersection.

Eliminate an unused, zero-weight coordinate, which exists because $c \leq n - 1$. The resulting simplex region has at most $n + 2$ faces. Project each boundary face along its largest absolute normal coefficient. All its other slopes then have absolute value at most one, so above each unit $(n - 2)$ -cube it meets only a dimension-dependent number of unit cubes. The boundary count is $O_n((\ell + 1)^{n-2})$, uniformly in every threshold and normal. Scaling gives the asserted coefficient and remainder.

For later integration the exact capacity is continuous in its two offsets, for fixed $\alpha > 0$. A first-coordinate strip has probability at most $(n - 1)$ times its width. For the other strip condition on all but that coordinate and an unused coordinate. Their remaining sum S has $\mathbb{E}(1/S) = n - 1$; the conditional first coordinate is uniform on an interval of length S . Since $\beta \geq \alpha$, the strip probability is at most $(n - 1)/\alpha$ times its width. Suprema retain these bounds. The auxiliary β can vary between levels and need not equal α . No actual division or valuation debit has been made in this proof. For the final ordinary assertion, run the same two full-ideal degenerations with the single test $\eta = 0$ and no retained coordinate. The Newton interior then avoids $q(1, \dots, 1)$. A separating nonnegative normal v , divided by $\sum v_i$, puts every monomial

in a region $\sum a_i u_i \geq q$, with $a_i \geq 0$ and $\sum a_i = 1$. The same lattice estimate, with the extra coordinate floor omitted, proves the bound by the exact $\mathbb{M}_{n,c}$. The exact capacity obeys the envelopes $B_{n,c}, E_{n,c}$ of [Theorem 7.9](#). One can recover this consequence from that theorem by taking positive rational weights a_i of sum one and the full monomial ideal $I_N = (x^u : \sum a_i u_i \geq N)$. Its Newton polyhedron lies between the half-spaces with thresholds N and $N+1$: rounding each coordinate down loses at most $\sum a_i = 1$. Hence $1/(N+1) \leq \text{lt}(I_N) \leq 1/N$. Let N, ℓ tend to infinity proportionally in the full-primary estimate; the lattice limit is the simplex probability for these weights. Continuity of that probability gives arbitrary weights, zero weights, and boundary thresholds by one-sided limits. Taking the supremum proves the assertion. $\square$

Lemma 9.12. *Let X be a smooth integral complex projective sevenfold, let L be ample Cartier with $L^7 = 1$, and suppose $K_X + 8L$ is not generated at x . There is a cutoff J , depending only on the original data and fixed complete-source comparisons, with the following property. For all $i, j \geq J$, the complete original sources $U_{20i}(9/10)$ and $U_{25j}(26/25)$ are nonzero. Let sufficiently large sufficiently general pure averages at their respective actual full-ideal thresholds have minimal centers W and P , and suppose P is a divisor. Every place w computing the former full-ideal threshold and centered on W then satisfies*

$$W \subset P, \quad \frac{w(P)}{A_X(w)} > \frac{3}{5}.$$

No primarity or tangent hypothesis is imposed on the original base ideal or on W . The cutoff precedes both original degrees, averages, centers and computing places.

Proof. Put $r = 9/10$, $U = 26/25$, $h = 263/2000$, and $h_P = 1/5$. Fix the comparison constants for both big original sources before any observation. Include the cutoffs of [Lemma 9.8](#) and [Lemma 9.10](#), and make the initial normalized error at most $1/40000$. For the actual coefficient $s_0 = 20i\tau_0$, we have $s_0 < 38/5$. Thus the full normalized profile

$$G(t) = \inf_{\substack{k \geq 1, 0 \neq f \in H^0(X, kL) \\ \text{ord}_x f \geq kt}} \frac{w(f)}{kA_X(w)}$$

satisfies $G(r) > h$, since $5/38 - h = 3/38000 > 1/40000$. Put $F(t) = \phi_P(t)$. It satisfies $F(U) > h_P$ and $V(U) < (4/5)^7$. Fix supporting slopes a at r and d at U ; then

$$G(t) \geq H_a(t) = (h + a(t - r))_+, \quad F(t) \geq F_d(t) = (h_P + d(t - U))_+, \quad a \geq 263/1800.$$

These lines hold at every order carrying an original section, on both sides of the observation orders.

Step 1. In this step, we bound partial local division using the actual valuation debit. Suppose $\theta = w(P)/A_X(w) \leq 3/5$. The initial-center lemma gives $2 \leq \text{codim } W \leq 5$. For any piecewise affine lower profiles $H \leq G$, $f \leq F$ and any $0 \leq q(t) \leq f(t)$, every actual point-order ℓ section at original degree k is divisible by p^N , where p is the smooth equation of P and $N = \lceil kq(\ell/k) \rceil$. Division sends its full point layer injectively into degree $\ell - N$, and the residual value is at least

$$kH(\ell/k) - N\theta \geq k(H(\ell/k) - (3/5)q(\ell/k)) - 1.$$

This uses the actual θ , including $\theta = 0$; it does not require containment in P in advance. Positive residual thresholds define primary comparison ideals for W , to which the ordinary assertion of [Lemma 9.11](#) applies. Add zero weights to reach five normal coordinates. This proves the comparison for every initial dimension two through five, without the stronger smaller-center classification.

Write $C(x, y) = x^6 \mathbb{M}_{7,5}(y/x)$ for $x > 0, y \geq 0$, with $C(x, y) = 0$ for $x \leq 0$. This exact function is nondecreasing in x : eliminate an unused simplex coordinate and expand the remaining simplex at fixed absolute threshold. Also $\mathbb{M}_{7,5}$ is 30-Lipschitz. Choose a largest normal weight, at least $1/5$, and an unused coordinate. Conditional on all others their sum S has $\mathbb{E}(1/S) = 6$, so a

threshold strip of width ϵ has probability at most 30ϵ . Therefore rounding N changes the bound by at most $30k^5x^5$. The full leading image is consequently bounded by

$$\frac{k^6}{6!}C(t - q(t), (H(t) - (3/5)q(t))_+) + O((\ell + 1)^5), \quad t = \ell/k.$$

Negative residual degree gives zero image. The degree-zero case is included in the remainder. The error is uniform before any selection among polynomial envelopes.

Step 2. In this step, we exclude every remaining slope pair by the finite rational comparison. The upper envelopes of Definition 7.7 give the following applicable polynomials for $C(x, (y)_+)$:

range	upper bound
$y \leq 0, x > 0$	x^6
$x \leq 0$ or $y \geq x$	0
$0 \leq 7y \leq x$	$x^6 - (35/6)^5xy^5$
$x \leq 7y, 0 \leq y \leq x$	$x^6 - x((29x + 7y)/36)^5$
$x \leq 7y, 0 \leq y \leq x$	$(7/6)^6(x - y)^6$
$x \leq 2y \leq 2x$	$(x - y)^6$.

At individual boundary points either adjacent valid branch can be used. In particular no continuity is asserted for the selected majorant at $y/x = 1/2$. On the same images use the baseline rate $S(t) = 1, 61/100, 11/25, 17/50, 1$, with successive breaks $r, 99/100, U, 29/25$; the final value one is the ordinary jet bound, valid at all higher orders.

If $a \geq 3$, set $D_a = ar - h$ and $t_0 = D_a/(a - 1/7) < r$. The bound $\mathbb{M}_{7,5}(z) \leq \min\{1, (7/6)^6(1 - z)_+^6\}$ and direct seventh-power integration give total loss at most

$$t_0^7 \left(1 + \frac{6}{7(a-1)}\right) < r^7 \frac{10}{7} < \frac{5}{7}.$$

In particular $V(U) > 2/7 > (4/5)^7$, a contradiction. If $d \geq 5$, the same prime's outgoing loss is at most $(U - h_P)^7/(d - 1) \leq (21/25)^7/4 < 3/40$, contrary to $V(U) > 3/40$ and zero volume beyond the finite maximum original point order. It remains to consider $263/1800 \leq a \leq 3, 0 < d \leq 5$.

Put $d_* = 241/100$. On a rectangle $[a_0, a_1] \times [d_0, d_1]$, choose the lower profile $H = (h + a_1(t - r))_+$ before r , and $H = (h + a_0(t - r))_+$ afterwards; choose $f = (h_P + d_1(t - U))_+$ before U , and $f = (h_P + d_0(t - U))_+$ afterwards. Use all three legal portions

$$q_0 = 0, \quad q_1 = f, \quad q_2 = \min\{f, \max\{(7H - t)/(16/5), 0\}\}.$$

For each portion set $x_j = t - q_j, y_j = H - (3/5)q_j$. Here is the exact rule for every entry of the following complete tables. Split $[0, T]$ at the four baseline breaks, the zeros of the affine pieces of H, f , and where $(7H - t)/(16/5)$ equals zero or f . On each resulting interval all q_j, x_j, y_j are affine. Insert every interior root of $x_j, y_j, y_j - x_j, 7y_j - x_j, 2y_j - x_j$, for all three j . This fixes the domain of each applicable polynomial above. Only isolated roots are inserted; an identically zero expression uses its indicated boundary branch. Bisect each resulting interval. On each half $[l, v]$ integrate every applicable polynomial and $t^6 S(t)$, take the smallest integral, sum over all halves, and multiply by seven. For a polynomial $\sum c_m t^m$, the integral is $\sum c_m (v^{m+1} - l^{m+1})/(m + 1)$. This specifies a finite rational expression for each row, using only affine roots and binomial expansion. Each candidate bounds the entire half on which it is integrated.

For incoming orders take $d_0 = d_1 = d_*$, $T = U$. The last column bounds that rational expression from above in units of 10^{-6} :

a_0	a_1	N
263/1800	623/1920	787771
623/1920	42517/115200	781712
42517/115200	23827/57600	784110
23827/57600	7241/14400	790166
7241/14400	3789/6400	785541
3789/6400	19619/28800	777696
19619/28800	2063/2400	776559
2063/2400	5663/3600	787788
5663/3600	3	754890

For outgoing orders take $T = (d_0 U - h_P)/(d_0 - 1)$. At $t \geq T$ the prime lower bound is at least t , so all later leading layers vanish. The entire rectangle table is

a_0	a_1	d_0	d_1	N
263/1800	623/1920	241/100	823/320	998684
263/1800	623/1920	823/320	2187/800	978712
263/1800	7241/14400	2187/800	1223/400	999409
263/1800	7241/14400	1223/400	741/200	978277
263/1800	5663/3600	741/200	5	994678
623/1920	42517/115200	241/100	7971/3200	992893
623/1920	23827/57600	7971/3200	823/320	992851
623/1920	23827/57600	823/320	2187/800	986367
42517/115200	23827/57600	241/100	7971/3200	995259
23827/57600	17597/38400	241/100	7971/3200	994889
23827/57600	7241/14400	7971/3200	823/320	992398
23827/57600	7241/14400	823/320	2187/800	986921
17597/38400	7241/14400	241/100	7971/3200	993485
7241/14400	3789/6400	241/100	823/320	998556
7241/14400	19619/28800	823/320	2187/800	989973
7241/14400	2063/2400	2187/800	1223/400	992431
7241/14400	2063/2400	1223/400	741/200	978213
3789/6400	19619/28800	241/100	823/320	988370
19619/28800	2063/2400	241/100	2187/800	987879
2063/2400	5663/3600	241/100	741/200	992675
5663/3600	3	241/100	5	833099

The certificate `09-original-sources/incidence` contains these nine incoming and twenty-one outgoing records. Both tables are derived from its file `data/incidence.json`. For each record the checker rebuilds the full affine arrangement just described, integrates all applicable competitors on each half, and verifies $N = \lceil 10^6 I \rceil$ for the resulting rational total I . The fixed-slope incoming segments $d_0 = d_1 = d_*$ are retained. Shared endpoints use the same closed bounds; they do not create additional positive-length integration cells. The first table covers the full initial-slope interval; the second covers its product with $[d_*, 5]$. Coverage can be checked by slicing at the displayed a -endpoints: the successive d -intervals join from d_* to five. All entries are upward rational enclosures of the explicitly specified integrals, not values at sampled slopes.

If $d \leq d_*$, the actual incoming prime profile is at least F_{d_*} , so the first table's portions remain legal. It bounds the loss by $790166/10^6 < 61741/78125 = 1 - (4/5)^7$, contrary to the upper volume bound. If $d_* \leq d \leq 5$, the applicable rectangle bounds the loss through every subsequent order by $999409/10^6 < 1$. Sum actual layers to a fixed rational order beyond both T and the finite maximum section order. The uniform error is $O(k^6)$; the terminal source is zero, whereas

the complete original leading dimension is $k^7/7!$. This contradicts the strict loss bound. All slopes have now been treated. Therefore $\theta > 3/5$. Positive $w(p)$ means $p \in P_W$, giving $W \subset P$. The division only bounded local images; coefficient images would carry $(kL - NP)|_P$, and no positivity or lifting assertion for these twisted spaces was used. $\square$

Lemma 9.13. *Let X be a smooth integral complex projective sevenfold, let L be ample Cartier with $L^7 = 1$, and suppose $K_X + 8L$ is not generated at x . There is a cutoff J_2 , depending only on the original data and fixed complete-source comparisons, such that for every $j \geq J_2$, the complete original $U_{25j}(26/25)$ is nonzero and every sufficiently large sufficiently general pure average at its actual full-ideal threshold has minimal center of dimension four or five. The cutoff precedes the degree, average, center and computing place.*

Proof. The census leaves dimensions four, five, and six. We exclude dimension six by keeping its smooth divisor coordinate in the comparison for an independent initial source. Fix all cutoffs in the preceding lemmas before either source. If a divisorial observation exists, let P be that actual prime. It is smooth, and $F(U) = \phi_P(U) > 1/5$, where $U = 26/25$; also $V(U) < (4/5)^7$. Choose independently any sufficiently late actual initial source of degree $20i$ at $r = 9/10$, with a computing place w centered on W . By Lemma 9.12, $W \subset P$, $2 \leq c = \text{codim } W \leq 5$, and $\alpha = w(P)/A_X(w) > 3/5$. The full initial profile G in that lemma has $G(r) > h = 263/2000$. Fix its supporting slope $a \geq 263/1800$ and the supporting slope $d > 0$ of F at U . Write

$$H_a(t) = (h + a(t - r))_+, \quad F_d(t) = (1/5 + d(t - U))_+.$$

Step 1. In this step, we obtain the joint retained-parameter bound on every original layer. For any lower profiles $0 \leq f \leq F$, $0 \leq H \leq G$, every original degree- k point-layer germ at order ℓ belongs to

$$(p^{\lceil kf(\ell/k) \rceil}) \cap J_{kH(\ell/k)}.$$

Apply Lemma 9.11, pad the other normal weights with zeros to reach five, and relax α to $3/5$. For a uniform seven-coordinate simplex vector Z , put

$$\begin{aligned} \mathcal{K}(t, f, H) = t^6 \sup_{\substack{3/5 \leq \beta \leq 1 \\ \lambda_i \geq 0, \sum_{i=1}^4 \lambda_i = 1}} \Pr\{tZ_0 \geq f, \\ t[\beta Z_0 + (1 - \beta) \sum_{i=1}^4 \lambda_i Z_i] \geq H\}. \end{aligned}$$

The full layer is bounded by $k^6 \mathcal{K}(t, f, H)/6! + C(\ell + 1)^5$. For $H = 0$ use ordinary divisibility directly. The ceiling strengthens the floor, and a real valuation threshold has no rounding loss. With $S = 1, 61/100, 11/25$ on the successive intervals with endpoints $0, r, 99/100, U$, actual telescoping gives

$$(9.2) \quad 1 - V(U) \leq 7 \int_0^U \min\{t^6 S(t), \mathcal{K}(t, f(t), H(t))\} dt.$$

All images here are full original images. At fixed upper order the uniform errors sum to $O(k^6)$, and zero current sources give zero images. The observations and slopes remain fixed while this later counting degree tends to infinity.

Step 2. In this step, we bound the joint capacity on all sixteen coordinate-weight intervals. For $j = 0, \dots, 15$, put $b_j = 3/5 + j/40$, $B_j = b_{j+1}$, and $C_j = 1 - b_j$. Then $B_j > C_j > 0$. For $b_j \leq \beta \leq b_{j+1}$ and nonnegative coordinates, enlarge the weighted sum to $B_j e + C_j \sum \lambda_i z_i$. These enlarged coefficients sum to $41/40$; they are not renormalized. Conditional on $e = tZ_0$, the other six coordinates have sum $x = t - e$. Put $y = (H - B_j e)/C_j$. The exact remaining four-normal

capacity is bounded by $Q(x, y) = x^5 E_{6,4}(y/x)$, with the following branches:

range	$Q(x, y)$
$y \leq 0, x > 0$	x^5
$x \leq 0$ or $y \geq x$	0
$0 \leq 6y \leq x$	$x^5 - (24/5)^4 xy^4$
$x \leq 6y, 2y \leq x$	$\min\{x^5 - x((19x + 6y)/25)^4, (6/5)^5(x - y)^5\}$
$x \leq 2y \leq 2x$	$(x - y)^5$.

Integrating the conditional simplex measure gives

$$\mathcal{K}(t, f, H) \leq \max_{0 \leq j < 16} K_j(t, f, H),$$

where

$$K_j = (t - \max\{f, H/B_j\})_+^6 + 6 \int_f^{\min(t, H/B_j)} Q(t - e, (H - B_j e)/C_j) de.$$

An empty integration interval contributes zero. The whole enlarged region is empty if $f \geq t$ or $H \geq B_j t$; for $H = 0$ use $(t - f)_+^6$. This integration occurs after the full ambient image inclusion. Its $B_j e$ is an auxiliary coordinate contribution, not a debit from w .

Step 3. In this step, we verify the complete bounded-slope integral certificate. For $a \in [a_0, a_1] \subset [263/1800, 3]$, use the broken lower profile $H = (h + a_1(t - r))_+$ before r and $H = (h + a_0(t - r))_+$ afterwards. Test separately $f = F_3$ and $f = 0$. Split $[0, U]$ at $r, 99/100$ and at zeros of these affine profiles. For every coordinate interval use the four affine edges

$$e_z(t) = \frac{H(t) - C_j z t}{B_j - C_j z}, \quad z = 1, 1/2, 1/6, 0.$$

Insert roots of $H - B_j t$, $f - t$, and $f - e_z$. All denominators are positive. Only isolated roots of the affine differences are inserted; zero differences are boundary equalities. When $H < B_j t$, the four edges are ordered as displayed and below t , since their derivative with respect to z has sign $H - B_j t < 0$. The intervals thus fix every clipping and branch choice. Refine them into equal rational pieces of length at most $1/40$.

On $[e_1, e_{1/2}]$ integrate the last, axis branch of Q , and on $[e_{1/6}, e_0]$ its first positive-threshold branch; clip each lower endpoint at f . Bisect the clipped middle slice $[e_{1/2}, e_{1/6}]$. On each half use one of its two displayed polynomial bounds, retaining the choice specified in the supplementary **09-original-sources/retained-divisor** records. Either polynomial bounds its whole slice for every t in the integration cell. Including the terminal $y \leq 0$ term gives a degree-at-most-six polynomial $P_j(t)$ bounding K_j on that whole cell. When $H = 0$, these incoming profiles have $f = 0$, so the ordinary polynomial applies directly.

For a cell $[l, v]$, write

$$P_j(l + (v - l)z) = \sum_{i=0}^6 c_{ji} \binom{6}{i} z^i (1 - z)^{6-i}.$$

If $P_j(t) = \sum_m p_{jm} t^m$, its exact coefficients are

$$c_{ji} = \sum_{k=0}^i \frac{\binom{i}{k}}{\binom{6}{k}} (v - l)^k \sum_{m \geq k} p_{jm} \binom{m}{k} l^{m-k}.$$

The nonnegative Bernstein basis gives

$$7 \int_l^v \max_j P_j(t) dt \leq (v - l) \sum_{i=0}^6 \max_j c_{ji}.$$

Every recorded B -cell supplies seven integer enclosures $c_{ji} \leq M_i/10^6$ for all sixteen j , and an integer D satisfying

$$(v-l) \sum_i M_i/10^6 \leq D/10^8.$$

An S -cell instead supplies its rational rate s and $s(v^7-l^7) \leq D/10^8$. These rational inequalities are checked by the displayed coefficient formula, integration of powers, and multiplication of positive denominators. The sixteen-character middle-choice string uses $0 = BB, 1 = BC, 2 = CB, 3 = CC$, where B is the centroid polynomial and C the convex-order polynomial, in the order of the two middle halves. A dash means an empty middle slice; X means the enlarged region is empty.

The complete certificate is **09-original-sources/retained-divisor**, stored in **data/retained-divisor.json**. It contains each closed beta interval, every slope interval, every cell's endpoints and type, all branch strings and seven coefficient enclosures for the B -cells, and all cell and row integers. The checking rule rebuilds the affine-root partition for all sixteen beta intervals before comparing the complete ordered cell list. It then tests every branch string, expands each polynomial, checks all its Bernstein coefficients, integrates the shell cells, and sums every cell integer. Thus no cell or beta interval can be omitted.

These are all 1,099 cells. The ten F_3 rows contain 97, 96, 97, 90, 84, 79, 80, 76, 75, 70 cells; the four zero-floor rows contain 58, 62, 65, 70. Each cell partition covers $[0, U]$, and the successive slope intervals in each case cover $[263/1800, 3]$. There are 622 B -cells and 477 S -cells. The former require $622 \cdot 16 \cdot 7 = 69664$ coefficient inequalities. Their exact sums of the last column, also rendered from the canonical data, are

row	$f = F_3$	$f = 0$
1	78813928	83874182
2	78316263	84492074
3	78478991	77012365
4	78479182	65467130
5	78294000	
6	78461598	
7	77466558	
8	76864869	
9	75668322	
10	65463863	

Consequently the incoming loss in (9.2) is less than 789/1000 for $f = F_3$, and less than 169/200 for $f = 0$. The respective strict integer margins before these enlarged bounds are $78900000 - 78813928 > 0$ and $84500000 - 84492074 > 0$.

Step 4. In this step, we exclude all old and prime supporting slopes. For $a \geq 3$, the ordinary-capacity integration in Lemma 9.12 already gives $V(U) > 2/7 > (4/5)^7$. Thus suppose $a \leq 3$. If $d \leq 3$, then $F_d \geq F_3$ on $[0, U]$. The first certificate gives

$$V(U) > 211/1000 > (4/5)^7,$$

with strict gap 803/625000, a contradiction. If $d \geq 3$, use the second certificate for incoming orders, and retain the same prime and slope on every outgoing layer. For any fixed rational terminal order Q ,

$$V(U) - V(Q) \leq 7 \int_U^Q (t - F_d(t))_+^6 dt \leq \frac{(U - 1/5)^7}{d - 1} \leq \frac{(21/25)^7}{2}.$$

Choose Q beyond the finite maximum original section order, so the actual terminal space and its volume are zero. This would force

$$1 < \frac{169}{200} + \frac{(21/25)^7}{2} < 1.$$

The last strict gap is $364005211/48828125000$. This treats every $d \geq 3$, including unbounded slopes.

All comparisons used the original $20i$ and $25j$ observations and their whole coefficients. Relative local threshold tests add no global boundary. The undivided intersection retains every fixed, embedded and remote condition. In particular no coefficient image with normal twist $(kL - NP)|_P$ was declared positive or lifted. The strict scalar inputs already paid both cutting errors; no additional uncharged cut occurs here. The hypothetical divisorial observation is impossible, completing the census. $\square$

Theorem 9.14. *Suppose $L^7 = 1$ and $K_X + 8L$ is not generated at x . Then $V(t) > 1/32$ for every $0 \leq t \leq 27/25$. There is a cutoff J , depending only on the original data and fixed source comparisons, such that every $k \geq J$ divisible by 25 has $U_k(t) \neq 0$ throughout that interval, and its full fixed divisor $F_k(t)$ satisfies*

$$F_k(t) \cdot L^6/k < 2/5.$$

If the actual pure threshold at $27/25$ has a divisorial minimal center P smooth at x , then

$$L^6 \cdot P = 1, \quad b \geq 1486/625, \quad s \leq 107/25, \quad \phi_P(27/25) > 7/30.$$

The same preselected cutoff, enlarged using only the fixed data, works before the degree, average, center and computing place.

Proof. We apply the two-profile integral to independently chosen original sources, then use the full fixed-degree estimate. Put $r_0 = 9/10$, $r_1 = 26/25$, and $Z = 27/25$.

Step 1. In this step, we identify the two source reserves and their complete center alternatives. The initial source and its center are supplied by Lemma 9.8: its minimal center has dimension in $\{2, 3, 4, 5\}$ and deficit $b_0 > 4/25$. The source at $26/25$ has volume greater than $3/40$ by Lemma 9.10. Its exhaustive center alternatives, by Lemma 9.13, Lemma 9.9 and the fivefold clause of Proposition 5.9, are

ν	$\dim W_1$	$(\text{edim } W_1, \text{mult}_x W_1)$	b_1
Q	5	unrestricted	$> 7/5$
M	4	$(6, 3), (6, 4)$	$> 39/20$
D	4	$(7, 4), (7, 5), (7, 6)$	$> 3/2$.

The entire fourfold tangents are reduced and Cohen–Macaulay, including the common linear normal direction in case M. In case Q we use the full height-two capacity; no tangent hypothesis is needed.

Step 2. In this step, we choose the two actual profiles after fixing all comparison cutoffs. Fix the complete-source comparison constants for both big sources, and all preceding source cutoffs, before either observation. Enlarge one cutoff using only these fixed constants so that each comparison error is less than 10^{-6} in original L -units at every eligible degree beyond it. Then choose an original degree $20i$ for the first source and, independently, an original degree $25j$ for the second, both beyond that cutoff. Choose their own actual thresholds, pure averages and computing places w_0, w_1 , and put $A_i = A_X(w_i)$. All boundaries and their fixed components stay in the inequalities for s_i and b_i . No containment between the two minimal centers is needed.

The separate affine prices give

$$\frac{1}{s_0} > \frac{5}{38} > \frac{131}{1000} + 10^{-6}, \quad \frac{1}{s_1} > \begin{cases} 13/70 > 37/200 + 10^{-6}, & \nu = \text{Q}, \\ 104/505 > 41/200 + 10^{-6}, & \nu = \text{M}, \\ 52/275 > 189/1000 + 10^{-6}, & \nu = \text{D}. \end{cases}$$

For these now fixed places, let $\phi_i(t)$ be the infimum of $w_i(f)/(qA_i)$ over all original sections of degree q and point order at least qt . Taking twentieth or twenty-fifth powers shows that the infimum is unchanged on the corresponding progression. Choose $h_0 = 131/1000$, and respectively $h_1 = 37/200, 41/200, 189/1000$ in cases Q, M, D. By Theorem 6.13, supporting lines

give $\phi_i(t) \geq T_i(t) = (h_i + a_i(t - r_i))_+$, with the same finite $a_i > h_i/r_i$ on both sides of the observation order. These slopes are fixed before summing any subsequent degree or point layer.

Step 3. In this step, we sum the common full-image bounds and obtain the 27/25 reserve. For an arbitrary original degree k and layer ℓ , every actual germ belongs, for each i , to the full ideal

$$J_{i,k,\ell} = \{f : w_i(f) \geq kA_iT_i(\ell/k)\}.$$

When the right side is positive, this is primary for the prime P_i of the actual center: the valuation is positive on P_i , zero off P_i , and a power of P_i lies in the ideal. Its generic threshold λ_i satisfies

$$\lambda_i \leq \frac{A_i}{w_i(J_{i,k,\ell})} \leq \frac{1}{kT_i(\ell/k)}.$$

For zero right side the comparison uses the whole local ring. Thus the degree- ℓ image of the complete source is contained in both full initial ideals. Applying [Theorem 7.9](#), [Theorem 7.18](#) and the indicated refinements, including [Proposition 7.21](#), bounds this same image by each capacity, with uniform $O((\ell + 1)^5)$ error. At the first center, padding the normal weights with zeros gives the N_5 bound for every allowed codimension. This argument never requires the original base ideal itself to be primary.

Take the minimum with [Proposition 7.28](#) on these same images; its restriction to $[0, Z]$ is $\mathcal{B}_1$. The degree, source, center and place cutoffs were fixed in advance, and a zero current source has zero layer image. Summing the complete filtration, with its uniform $O(k^6)$ total remainder, gives

$$V(Z) \geq 1 - 7 \int_0^Z t^6 \min\{\mathcal{B}_1(t), N_5(T_0(t)/t), K_\nu(T_1(t)/t)\} dt.$$

The polynomial comparison in [Lemma 9.6](#) now gives $V(27/25) > 1/32$. Monotonicity gives the same bound for every smaller real order. [Lemma 9.7](#) then gives the remaining conclusions. $\square$

10. THE EXHAUSTIVE ORIGINAL-SOURCE LEDGER

Theorem 10.1. *Assume $L^7 = 1$ and nongeneration at x . After one cutoff chosen before a 25-divisible degree k and its pure actual 27/25 threshold, the minimal center is one of the following:*

d	e	m
6	6	1
4	6	3
4	7	4, 5
5	6	2, 3
5	7	3.

The divisor is smooth at x and has original degree one. All fourfold rows have full reduced Cohen–Macaulay tangent. For the fivefold rows the precise data are

(e, m)	b	σ	Hilbert numerator
(6, 2)	[10564/4375, 4]	[0, 1734/3125]	$1 + z$
(6, 3)	[21587/8642, 3]	[0, 4339/25000]	$1 + z + z^2$
(7, 3)	[7973/3100, 10/3]	[1/2, 14581/15000]	$1 + 2z.$

Every ordinary maximal-ideal power is integrally closed and the original Rees ring is normal with rational singularities. Each full tangent is Cohen–Macaulay; the (6, 3) tangent is reduced, the (7, 3) tangent is nonreduced, and reducedness is not yet asserted for (6, 2). For (7, 3) the full exceptional cycle is $2[\Pi] + [\Pi']$ for distinct linear components, or $3[\Pi]$; the latter requires $b \leq 3$. These cycles do not identify the full ideal or the analytic germ. Every earlier source history is included.

We classify the actual thresholds of the complete source constructed in [Theorem 9.14](#). We first give the complete scalar reduction and the resulting ordinary tangent information. The geometric exclusion then uses two profiles on the original section ring to rule out singular divisors and full reduced embedding-seven fivefold tangents. The remaining cubic cycle completes the ledger.

10.1. The scalar bounds and the actual boundary. For a state $v = (d, e, m) \in \mathcal{P}_7$, take the degree floor to be one in [Definition 5.2](#). This is valid because $L^d \cdot W$ is a positive integer. Write $\mu_v = T(e - d, m)$ and $B_v = d - 2\mu_v/m$. In the following certificate, $\hat{r}_v$ is a rational upper bound for $m^{1/d}$, and H_v is the minimum of all the listed affine functions. The column headed r records $\hat{r}_v$. The numerical cap B_v is not a boundary divisor. The canonical data are `10-ledger/scalar`, in `supplement/10-ledger/data/scalar.json`; the following short table is its complete rendered view. It contains seventy-one states and all 253 affine witnesses. The exact checking rule and its implication are proved in [Lemma 10.2](#).

10.2. The complete scalar certificate. The columns are the row number, dimension, embedding dimension, multiplicity, minimum Hilbert moment, deficit cap, radius bound, and all affine upper bounds. A continued line belongs to the preceding row.

id	d	E	m	mu	B	r	upper lines (slope,intercept)
1	1	1	1	0	1	1	(1,0)
2	2	2	1	0	2	1	(1,0)
3	2	3	2	1	1	1	(283/200,0)
4	2	4	3	2	2/3	1733/1000	(1733/1000,0)
5	2	5	4	3	1/2	2	(2,0)
6	2	6	5	4	2/5	2237/1000	(2237/1000,0)
7	2	7	6	5	1/3	49/20	(49/20,0)
8	3	3	1	0	3	1	(1,83/200); (5/4,33/200); (3/2,0)
9	3	4	2	1	2	63/50	(63/50,237/1000); (3/2,117/1000); (5/3,17/500); (7/4,0)
10	3	4	3	3	1	1443/1000	(1443/1000,0)
11	3	5	3	2	5/3	1443/1000	(5/4,103/200); (1443/1000,97/500); (3/2,167/1000); (2,0)
12	3	5	4	4	1	397/250	(397/250,37/1000); (5/3,17/1000); (7/4,0)
13	3	5	5	6	3/5	171/100	(171/100,0)
14	3	5	6	8	1/3	909/500	(909/500,0)
15	3	6	4	3	3/2	397/250	(397/250,103/500); (7/4,1/8); (2,3/50); (5/2,0)
16	3	6	5	5	1	171/100	(171/100,29/500); (2,0)
17	3	6	6	7	2/3	909/500	(909/500,0)
18	3	6	7	9	3/7	1913/1000	(1913/1000,0)
19	3	6	8	11	1/4	2	(2,0)
20	3	6	9	13	1/9	2081/1000	(2081/1000,0)
21	3	7	5	4	7/5	171/100	(171/100,211/1000); (2,19/200); (5/2,0)
22	3	7	6	6	1	909/500	(909/500,7/100); (2,1/25); (5/2,0)
23	3	7	7	8	5/7	1913/1000	(5/4,243/500); (7/4,129/1000); (1913/1000,13/1000); (2,0)
24	3	7	8	10	1/2	2	(2,0)
25	3	7	9	12	1/3	2081/1000	(2081/1000,0)
26	3	7	10	14	1/5	431/200	(431/200,0)
27	3	7	11	16	1/11	278/125	(278/125,0)
28	4	4	1	0	4	1	(1,757/1000); (5/4,161/500); (5/3,9/200); (7/4,0)
29	4	5	2	1	3	119/100	(119/100,287/500); (5/4,121/250); (3/2,1/4); (2,0)
30	4	5	3	3	2	1317/1000	(1317/1000,171/500); (3/2,1/4); (2,0)
31	4	5	4	6	1	283/200	(283/200,293/1000); (3/2,1/4); (2,0)
32	4	6	3	2	8/3	1317/1000	(5/4,373/500); (1317/1000,71/125); (3/2,81/250); (5/3,229/1000); (7/4,39/200); (2,19/200); (5/2,0)
33	4	6	4	4	2	283/200	(283/200,379/1000); (3/2,59/200); (2,19/200); (5/2,0)
34	4	6	5	6	8/5	187/125	(5/4,69/100); (187/125,297/1000); (3/2,59/200); (2,19/200); (5/2,0)
35	4	6	6	8	4/3	783/500	(5/4,69/100); (783/500,269/1000); (7/4,39/200); (2,19/200); (5/2,0)
36	4	6	7	11	6/7	1627/1000	(1627/1000,61/250); (2,19/200); (5/2,0)
37	4	6	8	14	1/2	841/500	(841/500,111/500); (2,19/200); (5/2,0)
38	4	6	9	17	2/9	1733/1000	(1733/1000,14/125); (2,53/1000); (5/2,0)
39	4	7	4	3	5/2	283/200	(283/200,29/50); (5/3,133/500); (7/4,117/500); (2,3/20); (5/2,0)
40	4	7	5	5	2	187/125	(187/125,17/40); (3/2,421/1000); (5/3,131/500); (2,3/20); (5/2,0)
41	4	7	6	7	5/3	783/500	(783/500,331/1000); (5/3,131/500); (2,3/20); (5/2,0)
42	4	7	7	9	10/7	1627/1000	(5/4,813/1000); (3/2,57/125); (1627/1000,11/40); (2,3/20);

(5/2,0) |
43	4	7	8	11	5/4	841/500	(841/500,32/125); (2,3/20); (5/2,0)
44	4	7	9	13	10/9	1733/1000	(1733/1000,239/1000); (2,3/20); (5/2,0)
45	4	7	10	15	1	1779/1000	(5/3,42/125); (1779/1000,28/125); (2,3/20); (5/2,0)
46	4	7	11	18	8/11	911/500	(7/4,131/500); (911/500,21/100); (2,3/20); (5/2,0)
47	4	7	12	21	1/2	931/500	(931/500,49/250); (2,3/20); (5/2,0)
48	4	7	13	24	4/13	1899/1000	(1899/1000,17/100); (2,139/1000); (5/2,0)
49	4	7	14	27	1/7	387/200	(7/4,1/10); (387/200,37/500); (2,13/200); (5/2,0)
50	5	5	1	0	5	1	(1,143/125); (5/4,103/200); (3/2,1/4); (2,0)
51	5	6	2	1	4	1149/1000	(1149/1000,247/250); (5/4,92/125); (3/2,169/500);
(5/3,229/1000); (7/4,39/200); (2,19/200); (5/2,0)							
52	5	6	3	3	3	623/500	(623/500,719/1000); (3/2,169/500); (5/3,229/1000); (7/4,39/200);
(2,19/200); (5/2,0)							
53	5	6	4	6	2	33/25	(33/25,76/125); (3/2,169/500); (5/3,229/1000); (7/4,39/200);
(2,19/200); (5/2,0)							
54	5	6	5	10	1	69/50	(69/50,207/500); (3/2,59/200); (2,19/200); (5/2,0)
55	5	7	3	2	11/3	623/500	(623/500,39/40); (5/4,193/200); (3/2,101/200); (5/3,71/250);
(7/4,117/500); (2,3/20); (5/2,0)							
56	5	7	4	4	3	33/25	(33/25,77/100); (3/2,101/200); (5/3,289/1000); (7/4,239/1000);
(2,3/20); (5/2,0)							
57	5	7	5	6	13/5	69/50	(69/50,673/1000); (3/2,101/200); (5/3,293/1000); (7/4,121/500);
(2,3/20); (5/2,0)							
58	5	7	6	8	7/3	1431/1000	(5/4,128/125); (1431/1000,301/500); (3/2,101/200);
(5/3,299/1000); (7/4,123/500); (2,3/20); (5/2,0)							
59	5	7	7	11	13/7	369/250	(369/250,539/1000); (3/2,101/200); (5/3,277/1000);
(7/4,237/1000); (2,3/20); (5/2,0)							
60	5	7	8	14	3/2	379/250	(379/250,483/1000); (5/3,34/125); (7/4,117/500); (2,3/20);
(5/2,0)							
61	5	7	9	17	11/9	194/125	(1,1079/1000); (3/2,117/250); (5/3,133/500); (7/4,117/500);
(2,3/20); (5/2,0)							
62	5	7	10	20	1	317/200	(317/200,42/125); (5/3,133/500); (7/4,117/500); (2,3/20);
(5/2,0)							
63	5	7	11	24	7/11	202/125	(202/125,139/500); (2,3/20); (5/2,0)
64	5	7	12	28	1/3	411/250	(2,3/20); (5/2,0)
65	5	7	13	32	1/13	1671/1000	(5/2,0)
66	6	6	1	0	6	1	(1,198/125); (5/4,187/250); (3/2,357/1000); (5/3,23/100); (7/4,39/200);
(2,19/200); (5/2,0)							
67	6	6	2	1	5	1123/1000	(1123/1000,703/500); (5/4,499/500); (3/2,101/200); (5/3,42/125);
(7/4,131/500); (2,3/20); (5/2,0)							
68	6	6	3	3	4	1201/1000	(1201/1000,223/200); (5/4,993/1000); (3/2,101/200);
(5/3,42/125); (7/4,131/500); (2,3/20); (5/2,0)							
69	6	6	4	6	3	63/50	(63/50,121/125); (3/2,101/200); (5/3,42/125); (7/4,131/500);
(2,3/20); (5/2,0)							
70	6	6	5	10	2	327/250	(327/250,801/1000); (3/2,101/200); (5/3,42/125); (7/4,131/500);
(2,3/20); (5/2,0)							
71	6	6	6	15	1	1349/1000	(5/3,42/125); (7/4,131/500); (2,3/20); (5/2,0)

Lemma 10.2. *For each of the seventy-one rows in [subsection 10.2](#), the complete descendant envelope with degree floors one satisfies*

$$\mathcal{F}_v(z) \leq H_v(z) \quad (0 \leq z \leq B_v).$$

These bounds include every dimension skip and every embedding-compatible adjacent successor.

Proof. We verify the finite certificate and then apply induction on dimension.

Step 1. In this step, we verify the complete state enumeration. The state set in [Definition 5.2](#) gives precisely the displayed rows. To check its multiplicity cutoffs, fill the Hilbert function in successive degrees up to the capacities $\binom{e-d+j-1}{j}$; the resulting minimum moment is $T(e-d, m)$. Retain exactly those rows for which $2T(e-d, m) < dm$. The numbers of rows in dimensions one through six are 1, 6, 20, 22, 16, 6, respectively. This enumeration stops, for each positive codimension, at the first multiplicity failing the strict moment inequality. Indeed the minimum moment divided by the multiplicity is the mean degree of the filled Hilbert entries. Successive entries have nondecreasing degrees, so this mean cannot decrease; it tends to infinity. In codimension zero only multiplicity one occurs, and the curve state is $(1, 1, 1)$.

Step 2. In this step, we verify every affine pair on its full successor domains. Here is an exact verification rule for every affine pair in the table. For a possible successor $w = (f, h, k)$ of $v = (d, e, m)$, set

$$T_{vw} = \begin{cases} \min(B_v, B_w, B_v - k/m), & f = d - 1, h \leq e, \\ \min(B_v, B_w), & f \leq d - 2, h \leq e. \end{cases}$$

Discard nonpositive caps: they allow no positive-deficit successor. The direct point successor is accounted for separately below. For a rational number a , compute

$$S_w(a, T) = \max_{0 \leq y \leq T} \{H_w(y) - ay\}$$

at $y = 0, T$ and at all pairwise intersections $y = (p' - p)/(a' - a'')$ in $(0, T)$, where $(a', p), (a'', p')$ are distinct-slope lines of row w . These include every breakpoint of the minimum. Each displayed line (a, p) of row v satisfies the rational inequality

$$(10.1) \quad p \geq (\hat{r}_v - a)_+ B_v + \max \left(0, \max_w S_w(\max(a, \hat{r}_v), T_{vw}) \right).$$

Parallel lines contribute no intersection, and coincident intersection points are counted once. At a zero cap the interval is a single point; it introduces no positive-dimensional successor. Thus no empty interval, endpoint, dimension skip, or competing affine function is omitted. This rule involves only the entries in the certificate. The radius check is $\hat{r}_v^d \geq m$; the table chooses the smallest multiple of $1/1000$ with this property. Thus all checks reduce to comparisons of integers after positive denominators are cleared.

For example, in row $(3, 6, 4)$, a multiplicity-five surface has deficit at most $1/4$, and its excess over slope $397/250$ is

$$\left(\frac{2237}{1000} - \frac{397}{250} \right) \frac{1}{4} = \frac{649}{4000}.$$

A multiplicity-four surface has deficit at most $1/2$ and excess $(2 - 397/250)/2 = 103/500$, which is the largest surplus for this line. The other lines and all other rows use exactly (10.1); the minimum of a successor's lines is taken before the maximum over its deficit.

Step 3. In this step, we deduce the complete descendant bound by induction on dimension. Suppose the current deficit is z , and the next positive-dimensional center has row w and deficit y . The inductive upper bound for the complete cost is $\hat{r}_v(z - y) + H_w(y)$, with $0 \leq y \leq \min(z, T_{vw})$. If $\hat{r}_v \geq a$, subtracting az bounds this expression by $(\hat{r}_v - a)B_v + S_w(\hat{r}_v, T_{vw})$. If $\hat{r}_v < a$, use $z \geq y$ to obtain $S_w(a, T_{vw})$. A direct point successor costs at most $\hat{r}_v z$. Equation (10.1) therefore proves every line, and hence their minimum. The same induction bounds the envelope in Definition 5.2, so it also bounds its actual-root certificate in Theorem 5.4.

In particular, the threefold rows of multiplicities eight and nine have not been omitted. For embedding dimension at most six, Lemma 5.10 gives the caps $1/4, 1/9$ and the complete costs $2z, 9z/4$, including all skips. The corresponding embedding-seven rows instead have moments $10, 12$ and caps $1/2, 1/3$. The embedding-seven multiplicities ten and eleven also remain in the certificate. No embedding-six cap is imposed on them. $\square$

Lemma 10.3. *Let $\Gamma \sim_{\mathbb{Q}} sL$ be the pure actual threshold of a nonzero original complete source $U_k(u)$, where ku is integral, on a sevenfold at a nongeneration point. Put $M = \text{mult}_x \Gamma$, $b = \text{def}_x(\Gamma)$, and $\sigma = 7 - M - b$. If its state has an affine complete-tail bound $az + p$, then*

$$(10.2) \quad M \geq us, \quad M + b \leq 7, \quad 8 \leq s + ab + p, \quad \sigma \leq 7 - b + u(ab + p - 8).$$

If $au > 1$, then

$$(10.3) \quad b \geq \beta(a, p; u) := \frac{(8 - p)u - 7}{au - 1}, \quad \frac{1}{s} \geq \frac{u}{7 - b}.$$

Proof. The first two inequalities are the separate multiplicity and point discrepancy assertions of Proposition 9.1 for the entire boundary Γ . If $s + ab + p < 8$, Theorem 5.4 and Lemma 10.2 give an actual additive point continuation of coefficient less than eight, contradicting Proposition 3.12.

More explicitly, for a strict margin $\Delta = 8 - s - ab - p > 0$, there are at most six cuts. After the actual radius ρ_i of a cut is known, choose its approximation parameter less than $\Delta/(12(1 + \rho_i))$. Its two errors together cost $(1 + \rho_i)\varepsilon_i$; their sum is less than $\Delta/2$. This charges both errors and every intervening boundary coefficient.

Now $s \geq 8 - ab - p$ and $us + b \leq 7$ give (10.3). Substituting the former inequality into $\sigma \leq 7 - us - b$ proves the last part of (10.2). The source, its full ideal, and its threshold have remained the same throughout. $\square$

From now on, fix $u = 27/25$ and a sufficiently late actual source as in Theorem 9.14. Write $R = \mathcal{O}_{W,x}$, $\mathfrak{n} = \mathfrak{m}_{W,x}$, $G = \text{gr}_{\mathfrak{n}} R$, and $\delta = 7 - M = b + \sigma$. The ring R is normal and Cohen–Macaulay by Theorem 4.13. The moment and deficit estimates below are Theorem 4.5 ([Han26, Lemmas 4.2 and 4.4]) and Theorem 4.16. The scalar census, Proposition 5.9, gives $d \in \{4, 5, 6\}$, with $b > 3/2$ for $d = 4$ and $b > 7/5$ for $d = 5, 6$.

Lemma 10.4. *Assume $L^7 = 1$ and nongeneration at x . Let Γ be a pure actual threshold of a nonzero original $U_k(27/25)$, with $25 \mid k$, whose minimal center W has dimension four. Put $e = \text{edim } \mathcal{O}_{W,x}$ and $m = \text{mult}_x W$. Then the only states are $(e, m) = (6, 3), (7, 4), (7, 5)$. Their full ordinary tangents are reduced Cohen–Macaulay. Every ordinary maximal-ideal power is integrally closed, and the original Rees ring is normal with rational singularities. Their Hilbert numerators are, respectively, $1 + 2z, 1 + 3z, 1 + 3z + z^2$.*

Proof. We exhaust the scalar states and then apply the ordinary-power theorem.

Step 1. In this step, we enumerate the surviving fourfold states and bound their slack. The possible multiplicities before using $b > 3/2$ are

e	4	5	6	7
m	1	2, 3, 4	3, ..., 9	4, ..., 14.

The cap $B_v \leq 3/2$ excludes $(5, 4)$, all $(6, m)$ with $m \geq 6$, and all $(7, m)$ with $m \geq 7$. For the remaining nine states use the following lines:

e	m	B_v	a	p
4	1	4	1	757/1000
5	2	3	119/100	287/500
5	3	2	1317/1000	171/500
6	3	8/3	1317/1000	71/125
6	4	2	283/200	379/1000
6	5	8/5	187/125	297/1000
7	4	5/2	283/200	29/50
7	5	2	187/125	17/40
7	6	5/3	783/500	331/1000

For the first three states and $(6, 5)$, respectively, the upper bounds for $s + ab + p$, already using the weaker source order $26/25$, are

$$\frac{75}{26} + \frac{4757}{1000} < 8, \quad \frac{50}{13} + \frac{518}{125} = \frac{12984}{1625} < 8, \quad \frac{125}{26} + \frac{372}{125} < 8, \quad \frac{135}{26} + \frac{13453}{5000} < 8.$$

Indeed $(7 - b)/(26/25) + ab + p$ increases with b in these rows, so its maximum occurs at B_v . These inequalities contradict (10.2).

At the actual order $27/25$, the remaining substitutions are

e	m	$\beta(a, p; 27/25)$	$(27/25)/(7 - \beta)$
6	3	25664/10559	10559/44675
6	4	30767/13205 > 2	excluded
7	4	5068/2641	2641/12425
7	5	29525/15392	15392/72425
7	6	32063/17282 > 5/3	excluded.

For the three survivors, (10.2) yields

e	m	affine upper bound for σ	upper bound at B_v
6	3	$(10559b - 25664)/25000$	187/1875
7	4	$(2641b - 5068)/5000$	3069/10000
7	5	$(15392b - 29525)/25000$	1259/25000.

Step 2. In this step, we obtain the full tangent, ordinary-power, and Hilbert-series conclusions. All three bounds are less than $1/2$, while $\delta \geq b > 3/2$. Thus Theorem 4.20 applies to the actual pair and gives all the stated ordinary-power, Rees, and full tangent conclusions, including the affine vertex. The first two rows have minimal multiplicity. In the last row a general Artinian linear reduction of the Cohen–Macaulay graded ring has length five, with Hilbert function beginning 1, 3. Its one remaining dimension lies in degree two: a graded algebra generated in degree one cannot have a zero graded piece followed by a nonzero one. This gives $1 + 3z + z^2$. $\square$

Lemma 10.5. *Assume $L^7 = 1$ and nongeneration at x . Let Γ be a pure actual threshold of a nonzero original $U_k(27/25)$, with $25 \mid k$, whose minimal center W has dimension five. Put $e = \text{edim } \mathcal{O}_{W,x}$ and $m = \text{mult}_x W$. Before any geometric exclusion, it belongs to one of the following seven states:*

$$(e, m) = (6, 2), (6, 3), (7, 3), (7, 4), (7, 5), (7, 6), (7, 7).$$

The last four states have full reduced Cohen–Macaulay tangent, as does $(7, 3)$ when $b \leq 2$. Every such fivefold satisfies $1/s \geq 124/601$. The two embedding-six states have exactly the deficit intervals, slack bounds, ordinary-power assertions, and Hilbert numerators stated in Theorem 10.1.

Proof. We enumerate the surviving fivefold states and then determine their slack and full-ring properties.

Step 1. In this step, we obtain the seven surviving states and their inverse-coefficient bounds. The initial multiplicity ranges are $(e, m) = (5, 1), (6, m)$ for $2 \leq m \leq 5$, and $(7, m)$ for $3 \leq m \leq 13$. The cap one excludes $(6, 5)$, and the caps at most $11/9$ exclude all $(7, m)$ with $m \geq 9$. For every remaining state the exact complete-tail line and its consequence are as follows:

e	m	B_v	a	p	$\beta(a, p; 27/25)$	$(27/25)/(7 - \beta)$
5	1	5	1	143/125	$632/125 > 5$	—
6	2	4	$5/4$	92/125	10564/4375	175/743
6	3	3	623/500	719/1000	21587/8642	8642/36025
6	4	2	33/25	76/125	$439/190 > 2$	—
7	3	11/3	$3/2$	101/200	5473/3100	124/601
7	4	3	33/25	77/100	2021/1064	1064/5025
7	5	13/5	69/50	673/1000	22829/12260	2452/11665
7	6	7/3	1431/1000	301/500	24746/13637	13637/65475
7	7	13/7	369/250	539/1000	26447/14852	14852/71775
7	8	3/2	379/250	483/1000	$27959/15932 > 3/2$	—

The three rows whose lower bounds exceed their caps are excluded. The last column for every surviving row is at least $124/601$; this is a comparison of the displayed positive rational numbers.

Step 2. In this step, we determine the slack and full-ring conclusions in each remaining family.

Case 2.1. The embedding dimension is seven and the multiplicity is not three. The slack bounds for the noncubic embedding-seven rows are

m	affine upper bound for σ	upper bound at B_v
4	$(266/625)b - 2021/2500$	1171/2500
5	$(613/1250)b - 22829/25000$	9047/25000
6	$(13637/25000)b - 12373/12500$	21221/75000
7	$(14852b - 26447)/25000$	3257/25000 (strict).

For the last entry use $b \leq 13/7 < 2$. Each bound is less than $1/2$, and every row has $\delta \geq b > 1$. The ordinary-power theorem therefore gives full reduced Cohen–Macaulay tangent and normal rational original Rees rings. For multiplicity seven the moment bound is eleven, whereas $2\bar{e}_1 \leq 7(5-b) < 24$ because $b \geq 26447/14852 > 11/7$. Integrality gives $e_1 = \bar{e}_1 = 11$. An Artinian linear reduction has length seven and embedding dimension two. Its minimum moment is attained only by $(h_0, h_1, h_2, h_3) = (1, 2, 3, 1)$, so the full numerator is $1 + 2z + 3z^2 + z^3$.

Case 2.2. The embedding dimension is seven and the multiplicity is three. For the embedding-seven cubic, the same calculation gives

$$(10.4) \quad \sigma \leq \frac{31}{50}b - \frac{5473}{5000}.$$

If $b \leq 2$, this is at most $727/5000 < 1/2$, and $\delta \geq 5473/3100 > 1$. Thus its full tangent is also reduced Cohen–Macaulay, with minimal-multiplicity numerator $1 + 2z$.

Case 2.3. The embedding dimension is six. Finally, for $(6, 2)$ and $(6, 3)$, respectively,

$$\sigma \leq \frac{7}{20}b - \frac{2641}{3125} \leq \frac{1734}{3125} < 1, \quad \sigma \leq \frac{4321}{12500}b - \frac{21587}{25000} \leq \frac{4339}{25000} < \frac{1}{2}.$$

Their lower bounds for b exceed one, so [Theorem 4.20](#) supplies every ordinary power, the original Rees ring, and full Cohen–Macaulayness. It also gives reducedness for the triple, but does not give reducedness for the double from these inequalities. A general linear Artinian reduction of the full graded ring has embedding dimension one and length two or three; its Hilbert function is respectively $(1, 1)$ or $(1, 1, 1)$. Hence the numerators are $1 + z$ and $1 + z + z^2$, and the first Hilbert coefficients are one and three. Equality of all ordinary powers with their integral closures identifies the normal first coefficients as well. This argument makes no factoriality assumption on the center. $\square$

10.3. Two profiles on the full original images. Put $r = 9/10$, $u = 27/25$, and $Q = 13/5$. For an original degree q , let $A_{q,\ell}$ denote the actual image of $U_q(\ell/q)$ in $\mathfrak{m}_x^\ell/\mathfrak{m}_x^{\ell+1}$, after a local trivialization of qL . Thus

$$\dim A_{q,\ell} = \dim U_q(\ell/q) - \dim U_q((\ell+1)/q).$$

Write $E_j(c) = (1 - jc)_+^6$, and define

$$(10.5) \quad K_T(c) = \begin{cases} 3E_2(c) - 2E_3(c), & 0 \leq c < 4/25, \\ E_2(c), & c \geq 4/25. \end{cases}$$

This kernel is a specialization of [Theorem 7.18](#) with $(n, d, e) = (7, 5, 7)$: the spanning function is $\mathcal{S}_{7,0,3} = 3E_2 - 2E_3$. The improved range starts at $4/(3n+4) = 4/25$. In particular, if P has height two in the regular local ring $\mathcal{O}_{X,x}$, and the full tangent of $\mathcal{O}_{X,x}/P$ is reduced Cohen–Macaulay with embedding dimension seven, then every proper P -primary ideal I , with $\lambda = \text{lct}_{(\mathcal{O}_{X,x})_P}(I_P)$, satisfies

$$(10.6) \quad \dim_{\mathbb{C}}(\text{in}_{\mathfrak{m}_x} I)_\ell \leq \frac{\ell^6}{6!} K_T\left(\frac{1}{\ell\lambda}\right) + C(\ell+1)^5 \quad (\ell \geq 1),$$

with a constant independent of P, I, λ . The theorem also permits taking the minimum with $2E_2$ below $4/25$; the displayed envelope is sufficient for the certificate.

For the complete original images, use the baseline of [Proposition 7.28](#):

$$(10.7) \quad B(t) = \begin{cases} 1, & 0 \leq t < 9/10, \\ 61/100, & 9/10 \leq t < 99/100, \\ 11/25, & 99/100 \leq t < 26/25, \\ 17/50, & 26/25 \leq t < 29/25, \\ \left(\frac{(125-52t)_+}{73t}\right)^6, & 29/25 \leq t \leq 13/5. \end{cases}$$

That proposition gives

$$(10.8) \quad \dim A_{q,\ell} \leq \frac{\ell^6}{6!} B(\ell/q) + C(\ell+1)^5 \quad (1 \leq \ell \leq \lfloor qQ \rfloor),$$

uniformly in every sufficiently large original degree and in the actual source, center, and computing place. A zero source has zero image, and $\dim A_{q,0} \leq 1$. Its constants and cutoff precede every degree and layer. This estimate applies to each full image before any summation.

Definition 10.6. Let D be a closed rational rectangle of positive side lengths, and let f_λ be a Riemann-integrable function on $[0, Q]$ for each $\lambda \in D$. A *rectangular integration certificate* consists of rational grids covering both sides of D , a finite list of bisections of current grid rectangles, and one record for every terminal rectangle. Each bisection replaces its rectangle by both closed halves in one coordinate at its exact midpoint.

The record of a terminal rectangle D' consists of a partition $0 = t_0 < \cdots < t_m = Q$, rational polynomials p_j of degree at most six, and positive integers n_j , satisfying, for $1 \leq j \leq m$,

$$f_\lambda(t) \leq p_j(t) \quad (\lambda \in D', \ t_{j-1} < t < t_j), \quad n_j = \left\lfloor 10^9 \int_{t_{j-1}}^{t_j} p_j(t) dt \right\rfloor + 1.$$

Its total numerator is $s_{D'} = \sum_j n_j$.

Lemma 10.7. *For a rectangular integration certificate in Definition 10.6,*

$$\int_0^Q f_\lambda(t) dt < \frac{\max_{D'} s_{D'}}{10^9} \quad (\lambda \in D).$$

Proof. Each bisection preserves the union of the closed rectangles, so induction over the split list proves that the terminal rectangles cover D . Their interiors are disjoint. A parameter on a shared edge or vertex belongs to at least one terminal rectangle, and its estimate is valid there as well. No separate sampling of these segments or points is needed. For that rectangle, integrate the pointwise polynomial bounds over all its order intervals and add their strict rounded bounds. The finitely many order endpoints do not change a Riemann integral. The rounding remains strict for an integer multiple of 10^{-9} , including a zero integral. $\square$

Lemma 10.8. *Assume $L^7 = 1$ and nongeneration at x . After one cutoff chosen before the original degree k divisible by twenty-five, its pure actual $27/25$ threshold has neither a singular divisorial minimal center nor a fivefold minimal center of embedding dimension seven whose full tangent is reduced Cohen–Macaulay. The assertion is independent of every earlier source history.*

Proof. We obtain two actual supporting profiles, dispose of their unbounded slopes, and use the complete rational certificate for the remaining bounded ranges.

Step 1. In this step, we fix all uniform cutoffs before selecting the actual observations. By Theorem 9.14, $V(u) > 1/32$. Fix rational numbers $u < \gamma < \beta$ such that $\pi^*L - \beta E_x$ is big; these exist because the big cone is open. Apply Theorem 6.13 with original degree multiplier one and compact point-order interval $[0, \gamma]$. Fix its constant C_{prof} and degree cutoff. Take a common maximum J with the cutoffs of Theorem 9.14 and Proposition 7.28, enlarged so that $C_{\text{prof}}/j < 10^{-6}$ for every $j \geq J$. All these choices depend only on the original data and the fixed big interval. Now choose an old degree $k_0 \geq J$ divisible by twenty and an arbitrary new degree $k \geq J$ divisible by twenty-five. Both complete sources are nonzero. Only after these choices take their pure actual threshold averages $\Gamma_0 \sim_{\mathbb{Q}} s_0 L$ and $\Gamma \sim_{\mathbb{Q}} s L$, with $s_0 = k_0 \tau_0$ and $s = k \tau$.

Step 2. In this step, we obtain a uniform positive height for the earlier profile. Write b_0 and W_0 for the deficit and minimal center of the old boundary Γ_0 . A point center would already contradict point evaluation, since $s_0 \leq 7/r < 8$. If $b_0 \leq 4/25$, all radii in the complete state certificate are at most $49/20 < 5/2$. The deficit drops telescope, so the complete envelope is at most $(49/20)b_0$. On the other hand,

$$s_0 + \frac{5}{2}b_0 \leq \frac{70}{9} + \frac{25}{18}b_0 \leq 8.$$

Since $b_0 > 0$ by the Hilbert budget, the complete envelope is strictly less than $(5/2)b_0$. Thus complete cutting, including both errors within the resulting strict margin, contradicts nongeneration. We have $b_0 > 4/25$. A curve center is also impossible, since its radius is at most one, its deficit is at most one, and $s_0 + b_0 \leq 70/9 - b_0/9 < 8$. Consequently W_0 has codimension at most five, and

$$\frac{1}{s_0} > \frac{5}{38} > \frac{131}{1000} + 10^{-6}.$$

This uses no prescribed dimension, degree, or tangent of an earlier center.

Step 3. In this step, we obtain the new profile's height from its own full threshold price. For a singular divisor, $b > 7/5$ and $us + b \leq 7$ imply

$$\frac{1}{s} > \frac{27}{140} > \frac{24}{125} + 10^{-6}.$$

For an embedding-seven fivefold, [Lemma 10.5](#) instead gives

$$\frac{1}{s} \geq \frac{124}{601} > \frac{103}{500} + 10^{-6}.$$

No price at a later source order enters these inequalities.

For these actual boundaries, fix a computing place w_0 centered on W_0 , and a place w_1 centered on the new center. For a divisor P , take $w_1 = \text{ord}_P$, with $A_X(w_1) = 1$. Let $A_i = A_X(w_i)$.

The normalized profiles on the original complete section ring are

$$\phi_i(t) = \inf \left\{ \frac{w_i(f)}{qA_i} : q \geq 1, 0 \neq f \in H^0(X, qL), \text{ord}_x f \geq qt \right\}.$$

An empty infimum is $+\infty$. These profiles are finite on the fixed big interval containing both anchors. Put $v_0 = r, v_1 = u, k_1 = k, \tau_1 = \tau$, and $s_1 = s$, and let I_i be the full ideal of $U_{k_i}(v_i)$. Since w_i computes that same finite threshold, $w_i(I_i)/A_i = 1/\tau_i$. The complete-source comparison in [Theorem 6.13](#) therefore gives

$$(10.9) \quad 0 \leq \frac{1}{s_i} - \phi_i(v_i) = \frac{w_i(I_i)}{k_i A_i} - \phi_i(v_i) \leq \frac{C_{\text{prof}}}{k_i} < 10^{-6} \quad (i = 0, 1).$$

Here the asymptotic order in that theorem is the infimum over the full original section spaces defining ϕ_i . Their restrictions to degrees divisible by twenty or twenty-five give the same infima: taking a suitable power of any original section leaves both normalized orders unchanged. Set

$$(10.10) \quad h_0 = \frac{131}{1000}, \quad h = \begin{cases} 24/125, & \text{singular divisor,} \\ 103/500, & \text{full reduced fivefold tangent.} \end{cases}$$

The preceding strict prices and (10.9) give $\phi_0(r) > h_0$ and $\phi_1(u) > h$. The supporting-profile assertion of [Theorem 7.23](#), normalized to original degree, gives finite slopes $a_0 > h_0/r$ and $a_1 > h/u$. Put

$$(10.11) \quad H_0(t) = (h_0 + a_0(t - r))_+, \quad H_1(t) = (h + a_1(t - u))_+.$$

Each H_i is a lower bound for the same ϕ_i on every original degree and every layer, including the layers below its anchor.

Step 4. In this step, we impose both profile constraints on each full original image. Every actual germ in $U_q(\ell/q)$ lies in both valuation comparison ideals

$$J_{i,q,\ell} = \{f \in \mathcal{O}_{X,x} : w_i(f) \geq qA_i H_i(\ell/q)\}.$$

For a positive threshold this is primary for the center prime: if a factor is outside that prime its valuation is zero. The generic threshold of the ideal is at most $1/(qH_i(\ell/q))$, using w_i itself. At threshold zero the ideal is the whole local ring. Thus the primary estimates bound the entire actual initial image, without asserting that the original base ideal is primary.

The codimension-five envelope from [Theorem 7.9](#) is

$$(10.12) \quad N(c) = \begin{cases} 1 - (35c/6)^5, & 0 \leq c \leq 1/7, \\ \min\{1 - ((29 + 7c)/36)^5, (7/6)^6 E_1(c)\}, & 1/7 < c < 1/2, \\ E_1(c), & c \geq 1/2. \end{cases}$$

It also bounds smaller codimensions: append zero weights in the simplex-tail formulation. For the new divisor, a local equation in the regular ambient ring has ordinary order $m \geq 2$. Every germ in its valuation ideal is divisible by that equation to exponent $\lceil qH_1(\ell/q) \rceil$. Multiplication by its entire initial equation is injective in the polynomial tangent ring of the smooth ambient variety, and therefore gives capacity $K_P(c) = E_2(c)$, with uniform error $O((\ell + 1)^5)$. Only the ambient regular ring is used for this divisibility argument. For the new fivefold apply [\(10.6\)](#), the full-image bound of [Theorem 7.18](#), giving K_T .

The functions N, K_P, K_T are nonincreasing. For each full original layer take the minimum of the two primary bounds and [\(10.8\)](#), with a common maximum of their uniform error constants. These are three estimates of the same $A_{q,\ell}$. Summing all layers, including those below both anchors, and applying [Theorem 7.23](#) with ample asymptotics gives

$$(10.13) \quad V(Q) \geq 1 - I_K(a_0, a_1),$$

$$I_K(a_0, a_1) = 7 \int_0^Q t^6 \min\{B(t), N(H_0(t)/t), K(H_1(t)/t)\} dt.$$

The degree-zero layer has no leading contribution. The uniform remainders sum to $O(q^6)$, so they disappear after normalization by $q^7/7!$. Summation only after the second anchor also gives

$$(10.14) \quad V(Q) \geq V(u) - 7 \int_u^Q t^6 K(H_1(t)/t) dt.$$

These are bounds on the same original complete images. No relation between the two centers, and no nonzero source at $29/25$, is assumed. Positive $V(Q)$ supplies an actual effective rational divisor of point order at least $(13/5)$ times its coefficient, contradicting [Corollary 5.12](#).

Step 5. In this step, we exclude both unbounded supporting-slope ranges. If $a_0 \geq 16$, use rate one before r and the old slope sixteen afterwards. The old height increases with its slope after the anchor. The following cell integrals have the strict upper numerators shown, with denominator 10^9 :

0	9/10	B1	478296901
9/10	14269/15750	0 : N5hi	12938722
14269/15750	14269/15500	0 : C	10505221
14269/15500	14269/15000	0 : E1	291832
14269/15000	13/5	0 : E1	1.

The last integrand is zero. The polynomial dictionary below defines every name. These five rows are also the data `10-ledger/old-tail`. Their sum is $502032677/10^9 < 1$, so [\(10.13\)](#) gives volume greater than $497967323/10^9$.

For $t \geq u$, the new normalized height is at least $h/u > 4/25$, so both new kernels are E_2 . If $a_1 > 1/2$, direct integration gives

$$7 \int_u^\infty (t - 2H_1(t))_+^6 dt = \frac{(u - 2h)^7}{2a_1 - 1}.$$

Here $u - 2h > 0$; the affine positive part vanishes at $2(a_1u - h)/(2a_1 - 1) > u$ and is zero afterwards. The denominator is positive and increases with a_1 . For the divisor with $a_1 \geq 5$, this is at most

$$\frac{4191719943087}{476837158203125} < \frac{1}{100};$$

for the fivefold with $a_1 \geq 4$, it is at most

$$\frac{3622557586593623}{427246093750000000} < \frac{1}{100}.$$

The denominators $2a_1 - 1$ are at least nine and seven, respectively. These are the two separate data records `10-ledger/P2-new-tail` and `10-ledger/T-new-tail`; their checks include the branch, the positive denominator, the antiderivative and both displayed strict comparisons. The lower bound $V(u) > 1/32$ from [Theorem 9.14](#) makes (10.14) greater than $17/800$ in both ranges.

Step 6. In this step, we verify a strict integral margin on the complete bounded-slope domain. The remaining closed rectangles of parameters are

$$[131/900, 16] \times [8/45, 5] \quad \text{and} \quad [131/900, 16] \times [103/540, 4].$$

On a parameter rectangle $[l_0, u_0] \times [l_1, u_1]$, use the following endpoint slopes in (10.11):

$$(10.15) \quad (u_0, u_1) \quad (0 < t < r), \quad (l_0, u_1) \quad (r < t < u), \quad (l_0, l_1) \quad (u < t < Q).$$

Below an anchor, the upper slope minimizes the height; above it, the lower slope does. Non-increasing capacities then give a pointwise upper integrand on the entire parameter rectangle, including its boundary.

The full data `10-ledger/P2` and `10-ledger/T`, specified in [subsection 10.5](#), start with old-slope grid $131/900, 1, 2, 4, 8, 16$ and new-slope grids $8/45, 1, 2, 4, 5$ and $103/540, 1, 2, 3, 4$. Indices in the tags start at one. A split replaces its tagged rectangle by the lower and upper halves in the indicated coordinate, at the recorded midpoint, appending respectively zero and one to the tag. The complete lists of splits and leaves certify coverage of every initial grid rectangle, with disjoint interiors. There are forty-four divisor leaves and eighty-two fivefold leaves. Their largest total upper numerators, with denominator 10^9 , are respectively

$$999872274 < 9999000000, \quad 999778839 < 9999000000.$$

Thus every bounded pair has $I_K < 9999/10000$, leaving a margin greater than $1/10000$. More precisely, the upper bounds lie below $9999/10000$ by $27726/10^9$ and $121161/10^9$; the resulting volume margins are greater than $127726/10^9$ and $221161/10^9$.

Here is the exact polynomial and integration rule for those full tables. Set $v_0 = r, v_1 = u$, and for the endpoint slopes selected by (10.15), put

$$d_i = a_i v_i - h_i, \quad Y_i(t) = a_i t - d_i, \quad P_{i,j}(t) = 7((1 - ja_i)t + jd_i)_+^6, \quad (h_1 = h).$$

The table names mean

Bq	$7qt^6 \quad (q = 1, 61/100, 11/25, 17/50)$
$Bhigh$	$7((125 - 52t)/73)^6 \quad (29/25 \leq t \leq 125/52)$
$Bzero$	$0 \quad (t \geq 125/52)$
$0 : N5lo$	$7t^6 - 7(35/6)^5 t Y_0(t)^5$
$0 : N5hi$	$7t^6 - (7/36^5)t((29 + 7a_0)t - 7d_0)^5$
$0 : C$	$(7/6)^6 P_{0,1}(t)$
$0 : E1$	$P_{0,1}(t)$
$1 : E2$	$P_{1,2}(t)$
$1 : Tlow$	$3P_{1,2}(t) - 2P_{1,3}(t).$

The q in the name Bq is its listed rational rate, not a section degree. A constant baseline rate must dominate every baseline branch met by its interval. Beyond $29/25$, the high baseline is decreasing, with initial value $(1617/2117)^6 < 17/50$, so the constant rate $17/50$ remains a valid upper bound there. A capacity name is used only on its branch of (10.5) or (10.12), with $Y_i > 0$. The cell endpoints include each necessary anchor, branch change, and positive-part zero of the selected polynomial. Indeed $d_i \geq 0$ makes $a_i - d_i/t$ nondecreasing, so its endpoint values certify its whole branch interval. Every positive-part base is affine; its endpoint signs certify the entire cell, including an identically zero piece. Crossing an anchor is allowed only for an expression

whose selected slope does not change there. Values at order endpoints do not affect the integral. Consecutive cells are merged only when the polynomial stays the same.

A row $l \mid r' \mid \text{name} \mid N'$ asserts that the named polynomial has integral strictly less than $N'/10^9$ on $[l, r']$. After expansion of its linear powers, the check is exactly

$$\int_l^{r'} \sum_j c_j t^j dt = \sum_j \frac{c_j}{j+1} ((r')^{j+1} - l^{j+1}) < \frac{N'}{10^9}.$$

The recorded numerator is exactly the floor of 10^9 times this integral, plus one. Each cell therefore introduces a positive rounding error at most 10^{-9} ; all such errors are included in its rectangle's total. An equivalent exact integration rule uses the nodes $j/6$, $0 \leq j \leq 6$, on the unit interval with weights

$$(41/840, 9/35, 9/280, 34/105, 9/280, 9/35, 41/840).$$

Their moments of degrees zero through six are $1/(j+1)$; affine rescaling thus integrates every polynomial in the dictionary exactly. This is polynomial integration after full branch validation.

Each named expression is one of the valid individual bounds for the minimum; it need not minimize them. The data contain all 340 divisor cells and 717 fivefold cells. For each terminal rectangle they partition $[0, Q]$, and their numerators sum to its recorded total. The exact tests therefore give a rectangular integration certificate for the integrand in (10.13). Lemma 10.7 proves the claimed bound for every bounded pair, including every rectangle edge and vertex. The domains here have positive side lengths; no empty terminal rectangle or interval is admitted. Together with the unbounded-slope bounds, they force $V(Q) > 0$ for either supposed center type. This is the required contradiction. $\square$

10.4. The remaining cubic cycle and the assembly.

Lemma 10.9. *Assume $L^7 = 1$ and nongeneration at x . Take $25 \mid k$ beyond the cutoff of Lemma 10.8. If a pure actual threshold of $U_k(27/25)$ has a minimal center that is a fivefold of embedding dimension seven, it has multiplicity three and satisfies*

$$\frac{7973}{3100} \leq b \leq \frac{10}{3}, \quad \frac{1}{2} \leq \sigma \leq \frac{14581}{15000} < 1.$$

Its full tangent is Cohen–Macaulay and nonreduced, with Hilbert numerator $1 + 2z$. Every ordinary maximal-ideal power is integrally closed, and its original Rees ring is normal with rational singularities. Its ordinary exceptional scheme has full fundamental cycle $2[\Pi] + [\Pi']$ for distinct linear four-planes or $3[\Pi]$; the latter requires $b \leq 3$.

Proof. We isolate the cubic state, bound its deficit and slack, and then determine its full exceptional cycle.

Step 1. In this step, we isolate the cubic state and its nonreduced Cohen–Macaulay tangent. By Lemma 10.5, multiplicities four, five, six, and seven have full reduced Cohen–Macaulay tangent. They are all excluded by Lemma 10.8, applied to this very same actual source. Only multiplicity three remains. Since R is Cohen–Macaulay of minimal multiplicity, Theorem 4.6 ([Ver08, Theorem 2.7]) makes its full associated graded ring Cohen–Macaulay; Theorem 4.20 gives its numerator $1 + 2z$ and $e_1 = 2$. This argument precedes any restriction on σ . The full tangent cannot be reduced, since otherwise the same profile exclusion would apply.

Step 2. In this step, we bound the deficit and slack and obtain the normal-power conclusions. If $b < 7973/3100$, (10.4) gives $\sigma < 1/2$, and the ordinary-power theorem makes the full Cohen–Macaulay tangent reduced, a contradiction. Thus $b \geq 7973/3100$. If $b > 10/3$, the minimal-multiplicity, strict-deficit clause of that theorem applies with $d = 5, m = 3$. Its exceptional cycle excess is a nonnegative integer at most

$$3(3 - b) + 2 = 11 - 3b < 1.$$

It is therefore zero. All minimal components of the full graded ring occur on its projectivization and are generically reduced. The ring is Cohen–Macaulay, so it has no embedded associated primes and is reduced. This is again a contradiction. Hence $b \leq 10/3$.

Substitution into (10.4) gives $\sigma \leq 14581/15000 < 1$. The value $\sigma < 1/2$ is impossible by the preceding reducedness argument. Also $\delta \geq b > 1$. The ordinary-power theorem now gives $\mathfrak{n}^q = \bar{\mathfrak{n}}^q$ for every q , and normal rationality of the original Rees ring. In particular $\bar{e}_1 = e_1 = 2$, and its normalized blowup is the actual ordinary blowup near the entire exceptional scheme.

Step 3. In this step, we determine the possible full exceptional cycles. Write the full exceptional cycle as

$$[E] = \sum_i r_i [C_i], \quad d_i = \deg_{\mathcal{O}_E(1)} C_i.$$

Both r_i and d_i are positive integers, and $\sum_i r_i d_i = 3$. Set $\ell_E = \sum_i (r_i - 1) d_i = 3 - \sum_i d_i$. Because the full graded ring is Cohen–Macaulay and nonreduced, some minimal component is generically nonreduced. It has positive degree on $\text{Proj } G$, so $1 \leq \ell_E \leq 2$. The exact decomposition in Proposition 4.18, on this same ordinary blowup, gives

$$(10.16) \quad \ell_E \leq (5 - b)3 - 2\bar{e}_1 = 11 - 3b.$$

Indeed its exceptional weights are exactly the tautological degrees d_i . The old-prime and moduli terms remain nonnegative in the decomposition; dropping them only weakens this upper bound.

If $\ell_E = 1$, exactly one term $(r_i - 1)d_i$ equals one. It has $r_i = 2, d_i = 1$, and the remaining total degree is one, giving a distinct component with $r_i = d_i = 1$. Both reduced degree-one components are linear four-planes, so the full cycle is $2[\text{II}] + [\text{II}']$. If $\ell_E = 2$, the sum of the reduced component degrees is one. There is a single linear component with multiplicity three, giving $3[\text{II}]$. Equation (10.16) then forces $b \leq 3$. The calculation retains the full scheme and all its nilpotent structure; it identifies only its fundamental cycle. $\square$

Proof of Theorem 10.1. We combine the center exclusions and then collect the full-ring conclusions.

Step 1. In this step, we fix the common cutoff and identify the surviving divisorial state. Choose the common cutoff of Lemma 10.8 and Theorem 9.14, before the degree, average, center, or place. The scalar census restricts the dimension to four, five, or six. The full six-dimensional state list has $(e, m) = (6, 1)$ or $(7, m)$ with $2 \leq m \leq 6$; the cap one already excludes the last multiplicity. The profile exclusion eliminates every remaining singular divisor. Multiplicity one makes the remaining divisor smooth at x , and the original degree-one conclusion is precisely Theorem 9.14.

Step 2. In this step, we collect the full tangent and Rees conclusions for every surviving state. The fourfold states and their full tangent and ordinary-power properties are Lemma 10.4. The embedding-six fivefolds and all their numerical bounds are Lemma 10.5. The other fivefold possibility, including the entire exceptional cycle, is Lemma 10.9. A smooth divisor has a regular local ring, whose ordinary powers are integrally closed and whose full tangent is a polynomial ring. For its Rees assertion one can also apply Theorem 4.20: here $m = 1, e = d = 6, b \geq 1486/625$, and the scalar line $(1, 198/125)$ gives

$$0 \leq \sigma \leq \frac{2}{25}b + \frac{221}{3125} \leq \frac{1721}{3125} < 1, \quad \delta \geq b > 1,$$

using the cap $b \leq 6$. Thus its original Rees ring is normal with rational singularities as well.

All inequalities used a source’s own full coefficient and actual threshold. The old observation required only a complete big source at $9/10$, a deficit bound obtained from the full scalar table, and codimension at most five. Both fixed places constrained all layers of every original degree. Thus every earlier source history is included, with no selected later center or discarded fixed, residual, embedded, or remote component. $\square$

The supplement interface for the complete profile data follows.

10.5. The complete profile certificate. The canonical profile data are in `supplement/10-ledger/data/profiles.json`. The families `10-ledger/P2` and `10-ledger/T` have the following sizes and largest upper numerators:

family	initial rectangles	splits	leaves and cells	largest numerator
<i>P2</i>	20	24	44 and 340	999872274
<i>T</i>	20	62	82 and 717	999778839

Every leaf has its tag, all four rational endpoints, its total numerator, and its complete ordered list of integration cells. Each cell records its two endpoints, a name from the polynomial dictionary in Lemma 10.8, and its strict upper numerator with denominator 10^9 . The split lists record the parent tag, coordinate and exact midpoint. Thus the file contains the full witnesses for every order interval and parameter rectangle, including the splits needed to certify coverage.

The same file contains the five cells of `10-ledger/old-tail`, and the separate records `10-ledger/P2-new-tail` and `10-ledger/T-new-tail`. Each new-tail record specifies its slope threshold, denominator, exact loss fraction, comparison target and available volume floor. The profile heights and anchors are explicit; in particular the fivefold height is $103/500$. The family `10-ledger/source-margins` records the three height comparisons that absorb the error in (10.9).

The portable checker is `supplement/10-ledger/check.py`. From that supplement directory its entry point is `python3 check.py -data data -report report.json`. It uses rational arithmetic, checks all records of both data files, and accepts only after checking the complete state set, every scalar recurrence, both split trees, every branch and cell integral, all totals, and all three unbounded slope ranges. It rejects extra or missing records, duplicate identifiers, gaps, overlaps, illegal branches and failed inequalities. The scalar implication is Lemma 10.2; the rectangular implication is Lemma 10.7, applied with the branch and source arguments of Lemma 10.8. The supplement's manifest specifies the formats and provenance; its reproduction records include exact agreement with the former tables and rejection checks for damaged certificates.

11. THE AFFINE QUARTIC-THREEFOLD ENDPOINT

The higher-dimensional endpoints will use a continuation from a quartic threefold. We retain the point-ideal deficit and complete-source conventions of section 2. After treating nonreduced and integral tangents, we derive geometric bounds on the full symbolic images of every remaining tangent configuration. One exact certificate protocol turns these bounds into a strict estimate for the complete filtration on the original threefold. The numerical records are supplied as four supplementary datasets; the domain, coverage, and correctness arguments are given here.

Theorem 11.1. *Let $n \geq 7$, let X be smooth complex projective of dimension n , and let L be ample Cartier. Suppose $D \sim_{\mathbb{Q}} cL$, $c \geq 0$, is effective rational and lc near x , with actual minimal center of dimension three, embedding dimension six and multiplicity four. If*

$$b > 1, \quad c + \frac{5000}{3997}b \leq n + \frac{49267}{107919},$$

the entire D has an additive point continuation of coefficient less than $n + 1 - 1/200000$. Every positive original center degree and every full quartic tangent scheme is allowed.

The continuation in this statement contains every component of D with its original coefficient. Its different source trials all start from D .

Definition 11.2. A *quartic datum* consists of the objects and hypotheses of Theorem 11.1, together with the notation W for the actual minimal center, $A = L|_W$, and $g = A^3$. We set

$$a_* = \frac{5000}{3997}, \quad c_* = \frac{804700}{107919}, \quad T_* = \frac{799999}{100000}, \quad K_* = T_* - c_*, \quad G_* = \frac{18}{25}.$$

Thus $c \leq n - 7 + c_* - a_*b$. For integers $p \geq 1$ and $j \geq 0$, we write

$$F_{p,j} = H^0(W, \mathcal{O}_W(pA) \otimes I_x^j).$$

These are spaces on the original projective W .

11.1. The complete lower-dimensional continuation.

Lemma 11.3. *Let (X, L, x, D, W, b) be a quartic datum. Then $1 < b \leq 3/2$. Every surface which is the actual first smaller minimal center, with positive deficit z and multiplicity k , has $1 \leq k \leq 5$ and*

$$z \leq d_k := \min \left\{ \frac{2}{k}, \frac{6-k}{4} \right\}.$$

Its complete scalar continuation costs at most $q_k z$, where

k	1	2	3	4	5
q_k	1	283/200	1733/1000	2	2237/1000
d_k	5/4	1	2/3	1/2	1/4.

A curve continuation costs at most its deficit. If an effective rational Cartier $H \sim_{\mathbb{Q}} L$, not containing W , has $\text{ord}_x(H|_W) \geq G_$, its own threshold from D admits an additive point continuation of total coefficient less than $n + 1 - 1/200000$.*

The last assertion will provide a terminal zero space when a proof is conducted under failure of the endpoint.

Proof. We first bound every smaller center and then charge the two cutting errors.

Step 1. In this step, we bound every smaller center and its complete scalar continuation. Minimal multiplicity gives $e_1 = 3$, and [Theorem 4.20](#) gives $\bar{e}_1 = 3$, since $b > 1$. [Theorem 4.16](#) gives $4b \leq 6$. A surface inside W has embedding dimension at most six. If its embedding codimension is c_P , its Artinian first moment is at least $c_P + 2(k - 1 - c_P) = 2k - c_P - 2$, by [Theorem 4.5](#) [[Han26](#), Lemmas 4.2 and 4.4]. Here the actual surface is normal and Cohen–Macaulay by [Theorem 4.13](#); its local ring is analytically unramified and has infinite residue field. The strict surface Hilbert inequality gives

$$2(2k - c_P - 2) \leq 2\bar{e}_1(\mathfrak{m}_{P,x}) \leq (2 - z)k < 2k.$$

Together with $k \geq c_P + 1$, this implies $k = c_P + 1 \leq 5$. For $c_P = 0$, the surface is regular and $k = 1$. Its intrinsic cap is consequently $2/k$.

For the additional parent cap, write the old horizontal term in [Proposition 4.18](#) as H_0 , and let α be the old coefficient of the actual surface prime. That proposition gives

$$4z + (1 - \alpha)k \leq 6 - R_0 - M_0 - H_0.$$

Since $H_0 - \alpha k \geq 0$, we obtain $4z + k \leq 6$. Thus the prime is fully charged even if it was already present.

Every original ample Cartier degree is a positive integer. Hence the initial radius is at most $4^{1/3} < 397/250$, a surface radius is at most $\sqrt{k} \leq q_k$, and a curve radius is at most one. The last assertion uses normality of an actual curve center. Deficit drops along a surface–curve–point chain telescope to the initial surface deficit. Its whole scalar cost is therefore at most $q_k z$. The same estimate includes a direct surface–point cut. The displayed list includes every possible surface, and direct curve and point exits are retained. There is no threefold strictly below the starting threefold; the threefold rows of [Lemma 5.10](#) do not occur in this tail.

Step 2. In this step, we bound the whole price of the actual threshold and its continuation. At the threshold t of H , let z denote the new deficit. [Theorem 3.8](#) [[YZ20](#), Definition 2.5 and Lemma 2.10] applies to the smooth klt background, the entire old boundary D , and the actual addition tH . It gives

$$(11.1) \quad z \leq b - \gamma t, \quad \gamma = \text{ord}_x(H|_W).$$

At $\gamma = G_*$, the scalar whole price is at most

$$n - 7 + c_* - a_* b + \frac{b - z}{G_*} + q_k z.$$

The coefficient $1/G_* - a_*$ is positive. Its maximum over $1 \leq b \leq 3/2$ and $0 \leq z \leq \min(b, d_k)$ is obtained at $b = 3/2$; the sign of $q_k - 1/G_*$ then determines the endpoint of the z -interval. The five resulting values, after subtracting $n - 7$, are

$$\frac{3308125}{431676}, \quad \frac{165969827}{21583800}, \quad \frac{70982389}{8993250}, \quad \frac{1720013}{215838}, \quad \frac{3399652303}{431676000}.$$

The fourth is the largest, and

$$\frac{1720013}{215838} = 8 - \frac{6691}{215838} < T_*.$$

Point and curve exits satisfy the first bound. The price decreases as γ increases, so the assertion holds for $\gamma \geq G_*$.

Step 3. In this step, we allocate both cutting errors within the required final margin. For later use fix a total weighted error allowance of 10^{-6} . At each actual cutting stage, with known radius ρ , choose the error in [Theorem 3.14](#) [[GL24](#), Proposition 3.2 and Remark 3.3] less than $1/(3 \cdot 10^6(1 + \rho))$. Both inequalities

$$t_i \leq b_i \rho_i + \varepsilon_i, \quad b_{i+1} \leq b_i - t_i / \rho_i + \varepsilon_i$$

are retained. In particular $t_i \leq (b_i - b_{i+1})\rho_i + \rho_i \varepsilon_i$, and the sum of the coefficient and weighted deficit allowances is below 10^{-6} over at most three dimension drops. A scalar whole price at most $n - 7 + T_*$ therefore gives actual price below

$$n - 7 + T_* + 10^{-6} = n + 1 - 9 \cdot 10^{-6} < n + 1 - 1/200000.$$

The complete-tail construction also supplies arbitrarily small remaining cost at a zero-deficit limit. All these choices follow the actual radii and precede the corresponding cuts. $\square$

Lemma 11.4. *Let (X, L, x, D, W, b) be a quartic datum whose full projective tangent is nonreduced. Then the entire D has an additive point continuation of coefficient less than $n + 1 - 1/100$.*

The nonreduced exceptional component improves the parent budget. We keep its actual point-ideal multiplicity in that budget.

Proof. We use the nonreduced cycle to improve the parent budget and then bound the complete continuation.

Step 1. In this step, we identify the doubled component and bound the deficit. By [Theorem 4.20](#), the ordinary blowup is normal near its entire exceptional Cartier scheme $E = \sum_i \nu_i E_i$, and

$$\sum_i \nu_i \deg E_i = 4, \quad \ell := \sum_i (\nu_i - 1) \deg E_i \leq 6 - 4b < 2.$$

The full tangent ring is Cohen–Macaulay. A generically reduced Cohen–Macaulay ring has no nilpotent supported only on a proper closed subset: its associated primes are minimal, and localization at all of them detects every nonzero submodule of the ring. The assumed nonreduced projective tangent thus forces $\ell > 0$. Consequently $\ell = 1$, one plane is generically doubled, and $b \leq 5/4$. Its actual point-ideal value is two.

Step 2. In this step, we derive the improved surface-successor caps from the persistent budget. The term R_0 in the persistent budget equals $\ell = 1$, because the ordinary and normalized blowups agree with their tautological polarizations. Thus

$$K_W = 5 - M_0 - H_0.$$

For an actual surface successor with old coefficient α , we retain $H_0 \geq \alpha k$ and obtain

$$4z + (1 - \alpha)k \leq 5 - M_0 - H_0, \quad 4z + k \leq 5.$$

In particular $k \leq 4$ if $z > 0$, with $z \leq 1/2$ for $k = 3$, and $z \leq 1/4$ for $k = 4$.

Step 3. In this step, we bound the complete scalar cost and reserve both cutting errors. Put $r_* = 397/250$. A direct point or curve exit has scalar cost at most $r_* b$. A surface exit has scalar

cost at most $r_*b + (q_k - r_*)_+z$. For $k \leq 2$ the excess vanishes; for $k = 3$ it is at most $145/2000$, and for $k = 4$ at most $103/1000$. Every complete chain therefore costs at most $r_*b + 103/1000$. Since $r_* > a_*$,

$$c + r_*b + \frac{103}{1000} \leq n - 7 + c_* + (r_* - a_*)\frac{5}{4} + \frac{103}{1000} = n + 1 - \frac{258391}{13489875}.$$

The last margin exceeds $1/100 + 1/1000$; the integer difference $1000 \cdot 258391 - 11 \cdot 13489875 = 110002375$ is positive. At each of the at most three cuts choose $\varepsilon_i < 1/(3000(1 + \rho_i))$. The two errors have total weighted allowance below $1/1000$, proving the stated bound. $\square$

11.2. Original source trials and the integral tangent.

Construction 11.5. For a quartic datum, [Theorem 6.8](#) [[Sta26](#), Tag 0B5U, Lemma 30.17.1] applied to the fixed sheaf I_W supplies an ordinary restriction cutoff p_{lift} such that

$$H^0(X, pL) \longrightarrow H^0(W, pA)$$

is onto for every $p \geq p_{\text{lift}}$. Given a nonzero $F_{p,j}$, we take its full ambient inverse image and a sufficiently dilute general average $H \sim_{\mathbb{Q}} L$. This restriction does not contain W , and hence it contains no old lc center through x . Positive restricted point order makes the threshold from D positive and finite. Its minimal center is properly contained in W , and [\(11.1\)](#) holds with $\gamma = j/p$. This is ordinary lifting of the entire intrinsic space; the condition on j is imposed on the original W .

Lemma 11.6. *Let (X, L, x, D, W, b) be a quartic datum whose full projective tangent is integral. Then the entire D has an additive point continuation of coefficient less than $n + 1 - 1/200000$.*

Five original symbolic intervals suffice in this case. Each interval uses the fixed prime selected by that interval's own threshold.

Proof. Suppose the asserted continuation fails. At one sufficiently large original degree p , divisible by 100, use the successive levels

$$\gamma_0 = \frac{31}{50}, \quad \gamma_1 = \frac{63}{100}, \quad \gamma_2 = \frac{16}{25}, \quad \gamma_3 = \frac{33}{50}, \quad \gamma_4 = \frac{17}{25}, \quad \gamma_5 = \frac{18}{25}.$$

All these orders are integral after multiplication by p .

Step 1. In this step, we obtain the initial source mass and the full symbolic interval estimate. The ordinary jet map and the full Hilbert polynomial give

$$\dim F_{p,p\gamma_0} \geq (g - 4\gamma_0^3) \frac{p^3}{6} - O((p+1)^2) \geq \frac{1459}{31250} \frac{p^3}{6} - O((p+1)^2).$$

Write $\mathfrak{m}_W = \mathfrak{m}_{W,x}$ in the local ring $\mathcal{O}_{W,x}$. [Theorem 8.9](#) applies with one component, so every prime of multiplicity k and symbolic order r has full interval bound

$$(11.2) \quad \text{length} \frac{P^{(r)} \cap \mathfrak{m}_W^h}{P^{(r)} \cap \mathfrak{m}_W^l} \leq \frac{4}{6} \left((l - rk/4)_+^3 - (h - rk/4)_+^3 \right) + C_1(l+1)^2.$$

The constant precedes every prime and order.

Step 2. In this step, we force a full fixed surface prime at every unpaid tested source. At any nonzero tested source, a point or curve exit costs at most

$$c + b/\gamma \leq n - 7 + c_* + (50/31 - a_*)\frac{3}{2} = n + 1 - \frac{1787}{3345489} < n - 7 + T_*.$$

Failure therefore selects a surface prime P . Let α be its old discriminant coefficient and η its coefficient in the restricted average. By [Theorem 4.1](#) [[FH23](#), Theorem 1.2 and Definition 1.3], the new discriminant has coefficient at least one at the generic point of the lc prime P , and local log canonicity bounds that coefficient by one. The compatible Cartier-addition law of [Proposition 4.18](#) therefore gives

$$(11.3) \quad \alpha k \leq 6 - 4b, \quad \alpha + t\eta = 1.$$

The old horizontal contribution contains αk ; it is never removed before the threshold is taken.

At the first five levels set, respectively,

$$u_0 = \frac{3109}{2500}, \quad u_1 = \frac{1533}{1000}, \quad u_2 = \frac{1139}{625}, \quad u_3 = \frac{1379}{625}, \quad u_4 = \frac{12411}{5000}.$$

If $k\eta \leq u_i$, (11.1) and (11.3) imply

$$(11.4) \quad (4\gamma_i - u_i)b + u_i z \leq \gamma_i(6 - k).$$

Indeed, when $k - 6 + 4b > 0$, they give $t \geq (k - 6 + 4b)/u_i$. When that numerator is nonpositive, the same inequality follows from $z \leq b$.

The following table gives the maximum of $c_* - a_*b + (b - z)/\gamma_i + q_k z$ on $1 \leq b \leq 3/2$, $0 \leq z \leq \min(b, d_k)$, and (11.4). Each listed pair (b, z) attains the maximum. Unlisted k give an empty polygon. The bounds follow by intersecting the boundary lines of these displayed halfplanes and evaluating the affine objective at every feasible intersection.

i	k	maximum	b	z
0	1, 2	26762125/3345489	3/2	0
0	3	5368314237929/671040342000	3/2	27/6218
0	4	2623634842/335520171	1	9/3109
1	1, 2	859150/107919	3/2	0
1	3	126049121453/15756174000	3/2	39/146
1	4	39998900/5072193	60/47	0
2	1, 2	27364325/3453408	3/2	0
2	3	1966710878659/245839482000	3/2	1017/2278
2	4	979121723/122919741	3/2	217/2278
3	1, 2	9322175/1187109	3/2	0
3	3	169742651291/21260043000	3/2	831/1379
3	4	170079929/21260043	3/2	837/2758
3	5	83509225987/10630021500	3/2	6/1379
4	1, 2	28568725/3669246	3/2	0
4	3	7302656803/917311500	3/2	2/3
4	4	170079929/21260043	3/2	10033/24822
4	5	335305594573/42520086000	3/2	3233/24822.

Every maximum is less than T_* . By Lemma 11.3, this would pay the whole continuation. Hence $k\eta > u_i > 1$. Before selecting P , choose the general tuple so every nonfixed prime coefficient is less than $1/10$. On a principalization, the general moving members have no common prime and have normalized coefficients at most $1/(pN)$. As $k \leq 5$, the inequality $k\eta > 1$ forces P to be fixed in the entire source. Thus $\eta = r/p$, where r is its exact minimum symbolic order.

Step 3. In this step, we propagate the complete sources at one degree and obtain the paid final threshold. The kernel of the germ map from the current source to (11.2) is exactly the next original source. The normalized leading loss is therefore bounded by

$$4 \left((\gamma_{i+1} - u_i/4)_+^3 - (\gamma_i - u_i/4)_+^3 \right).$$

This expression decreases with u_i . All five losses and the remaining masses are

i	loss	remaining mass
0	29600143/2500000000	87119857/2500000000
1	3042547/400000000	272415753/10000000000
2	709213/78125000	181636489/10000000000
3	263473/78125000	29582389/2000000000
4	7734763/2500000000	116972893/10000000000.

These are positive. Choose the common intrinsic cutoff from these fixed margins and the five uniform quadratic errors, before any threshold, and then choose one 100-divisible p above it and p_{lift} . The full source F_{p,pG_*} is nonzero. Its own threshold is paid by Lemma 11.3, contradicting failure. Only the final successful trial is included in the resulting additive continuation. $\square$

11.3. Full component and conductor bounds. We next retain the restrictions of every actual prime on every component. For a reduced tangent, [Theorem 8.9](#) applies to the normal rational neighborhood of the entire exceptional scheme. For a prime of multiplicity k , its rational restriction degrees are denoted by δ_i , so $\delta_i \geq 0$ and $\sum_i \delta_i = k$. They include the contribution of a prime meeting an internal conductor component.

Lemma 11.7. *Let (X, L, x, D, W, b) be a quartic datum with reduced full tangent. Let P be an actual surface minimal center of an effective lc addition to D , with positive deficit. If E_i is an integral tangent component of degree m_i , and δ_i is the full rational restriction degree of the strict transform of P on E_i , then $\delta_i \leq m_i + 1$.*

This bound includes a surface with nonreduced tangent, and conductors shared by more than two components.

Proof. We compare the rational restriction with the scheme intersection and then take a finite linear section.

Step 1. In this step, we bound the rational restriction cycle by the full scheme intersection. The surface argument in [Lemma 11.3](#) gives minimal multiplicity for P . Its full tangent curve C_P is Cohen–Macaulay with Hilbert series

$$\frac{1 + (k - 1)z}{(1 - z)^2}.$$

Its homogeneous ideal in its linear span is 2-regular. Indeed the graded ring is Cohen–Macaulay, with numerator of degree one, so its regularity is one and that of the ideal is two. The entire ordinary point blowup P_Y is normal near C_P , by [Theorem 4.20](#) with dimension two and positive deficit. The Rees quotient embeds it as the full strict transform in $Y = \text{Bl}_x W$, and $C_P = P_Y \cap E$ scheme-theoretically.

At the generic point of a curve S in $C_P \cap E_i$, let $R_S = \mathcal{O}_{Y, \eta_S}$. This is a normal rational surface local ring, hence has finite divisor class group by [Theorem 8.5](#) [[Lip69](#), Proposition 17.1]. The quotients by the primes of P_Y and E_i are discrete valuation rings, since these two divisors are normal. Choose positive integers N_P, N_i and regular functions f, g_i with

$$\text{div}(f) = N_P P_Y, \quad \text{div}(g_i) = N_i E_i$$

on $\text{Spec } R_S$. The two prime divisors are distinct. Associativity of intersection multiplicities, applied in both orders to the parameter ideal (f, g_i) in the Cohen–Macaulay ring R_S , gives

$$N_P \text{ord}_{P_Y}(g_i) = N_i \text{ord}_{E_i}(f).$$

The coefficient of the actual rational restriction along S is therefore

$$d_S = \frac{\text{ord}_{P_Y}(g_i)}{N_i} = \frac{\text{ord}_{E_i}(f)}{N_P}.$$

For every $h \in I_{E_i}$, the divisor of h^{N_i}/g_i is effective. Normality puts this quotient in R_S . Restricting to the discrete valuation ring of P_Y gives $\text{ord}_{P_Y}(h) \geq d_S$. The actual equation e of E belongs to I_{E_i} . Thus the full intersection scheme $Z_i = C_P \cap E_i$ has generic length

$$\text{length} \frac{R_S/I_{P_Y}}{(I_{E_i}, e)(R_S/I_{P_Y})} = \text{ord}_{P_Y}(I_{E_i}) \geq d_S.$$

Consequently the rational restriction cycle is coefficientwise at most $[Z_i]_1$. No reduced support has replaced Z_i .

Step 2. In this step, we bound the restriction degree by a finite linear section. The span of E_i is $\mathbb{P}^{m_i+1}$. Choose a general hyperplane $\Lambda_i \simeq \mathbb{P}^{m_i}$ of this span that contains no component of C_P and avoids the finite associated points of Z_i . Then

$$\delta_i \leq \deg[Z_i]_1 = \text{length}(Z_i \cap \Lambda_i) \leq \text{length}(C_P \cap \Lambda_i) \leq m_i + 1.$$

For the final inequality apply the finite-scheme clause of [Theorem 7.11](#) [[EGHP05](#), Theorem 1.1]: the full 2-linear ideal resolution supplies $N_{2,p}$ through every required stage. If C_P spans a smaller linear space, replace Λ_i by its intersection with that span; its dimension can only decrease. The finite intersection retains all nilpotent lengths, as required by the cited theorem. $\square$

Lemma 11.8. *Let (X, L, x, D, W, b) be a quartic datum with reduced full tangent and at least two components. Its components are normal surfaces of minimal degree in their spans. The following list is exhaustive: two quadrics; a cubic surface of minimal degree and a plane; a chain of two planes and one quadric, in either order; two planes and one quadric sharing a line; or four planes.*

Let u_i denote the original component discrepancies. In the tables below, U_i bounds u_i , and $B(b)$ bounds αk for every later-selected prime of original discriminant coefficient α and multiplicity k . Coordinates in a chain are in chain order, starting with coordinate zero. The chain data are

family	(m_i)	b range	(U_i)	$B(b)$
PPC ₀	(1, 1, 2)	(1, 4/3]	(2, 18/13, 3/2)	$6 - 4b$
PPC ₁	(1, 1, 2)	(4/3, 26/19]	(22/13, 18/13, 3/2)	$9 - 13b/2$
PPC ₂	(1, 1, 2)	(26/19, 18/13]	(22/13, 18/13, 19/13)	$9 - 13b/2$
PQP ₀	(1, 2, 1)	(1, 6/5]	(2, 4/3, 2)	$6 - 4b$
PQP ₁	(1, 2, 1)	(6/5, 22/17]	(2, 4/3, 2)	$8 - 6b$
PQP ₂	(1, 2, 1)	(22/17, 4/3]	(5/3, 4/3, 5/3)	$8 - 6b$

For the remaining families above $b = 8/7$, the data are

family	(m_i)	b range	(U_i)	$B(b)$
COMMON	(1, 1, 2)	(8/7, 6/5]	(8/5, 8/5, 7/5)	$6 - 5b$
STAR	(1, 1, 1, 1)	(8/7, 6/5]	(6/5, 8/5, 8/5, 8/5)	$6 - 5b$
PATH0	(1, 1, 1, 1)	(8/7, 6/5]	(2, 4/3, 4/3, 2)	$6 - 4b$
PATH1	(1, 1, 1, 1)	(6/5, 9/7]	(2, 4/3, 4/3, 2)	$8 - 6b$
PATH2	(1, 1, 1, 1)	(9/7, 4/3]	(5/3, 4/3, 4/3, 5/3)	$8 - 6b$

In COMMON the Schur weights between each pair of original components are $1/3$, and $\hat{\kappa} = (5/3, 5/3, 8/3)$. In STAR the first component is central, all three edge weights are one, and $\hat{\kappa} = (0, 2, 2, 2)$. In a binary chain or path, the actual edge weights are reciprocals of positive integers, at most one, and $\hat{\kappa}_i = m_i + 2 - v_i$, where v_i is the original graph valence. Four planes sharing a line are impossible. The configuration with one triple conductor and one binary conductor has $b \leq 9/8$.

Every strict lower endpoint in these tables records where an integral crossing index has been forced to equal one. At equality we use the preceding row.

Proof. We classify the joins, eliminate conductor arms with their full mass, and retain the old prime in the discrepancy inequalities. The Hilbert series is $(1 + 3z)/(1 - z)^3$. [Theorem 7.10](#) [[EGHP04](#), Theorem 0.4 and Proposition 2.4] supplies a linearly joined ordering of the components. Their positive degrees sum to four. A component of degree m_i spans $\mathbb{P}^{m_i+1}$. If there are r components, the vector-space dimensions of the spans give total joining dimension $2(r - 1)$, with each join contributing at most two. Hence each join is a full line. This gives the listed partitions and

$$\sum_{\text{conductor lines}} (s - 1) = r - 1,$$

where s is the number of components containing that line. Additional isolated intersections are retained in the full schemes.

Use the entire tree of [Proposition 8.11](#), with zero mass on internal vertices and positive mass m_i on original vertices. Denote its matrix by $\mathcal{M}_T$ and its right side by κ . Its two inequalities are

$$(11.5) \quad \alpha p + \mathcal{M}_T u \leq \kappa, \quad p + \mathcal{M}_T \rho \leq \kappa, \quad u \geq b, \quad \rho \geq z.$$

All prime mass on internal vertices is transferred by the stated Schur complement; its total remains k . An internal vertex of a minimal conductor resolution has weight and valence at least two. With two attached branches the internal graph is a chain of weight-two vertices. Eliminating a chain gives weight $1/q$, where q is its edge length, a positive integer. With three branches there

is one trivalent internal vertex; with four there is one four-valent vertex or two trivalent vertices joined by a chain. This follows by summing $v - 2$ over the internal tree. Thus arbitrary arm lengths are included in the following eliminations.

Case 1. The tangent has two components. For masses m_1, m_2 and $w = 1/q$, elimination of the binary conductor gives

$$(m_i + w)u_i - wu_j \leq m_i + 1 \quad (i \neq j).$$

The inverse matrix has nonnegative entries and determinant $m_1m_2 + w(m_1 + m_2)$. Therefore

$$(11.6) \quad u_i \leq 1 + \frac{m_j + 2w}{m_1m_2 + w(m_1 + m_2)}, \quad m_1u_1 + m_2u_2 \leq 6.$$

For two quadrics this gives $u_i \leq 3/2$. For masses $(3, 1)$ the first bound for the cubic increases with w and is at most $10/7$; the plane bound is at most two. The summed inequality also gives $u_{\Pi} \leq 6 - 3b$. Thus

$$b \leq 10/7, \quad u_C \leq 10/7, \quad u_{\Pi} \leq \min(2, 6 - 3b).$$

These arguments apply to smooth quadrics and quadric cones, and to smooth cubic scrolls and cubic cones.

Case 2. A plane is the middle component of a three-component chain. Let the edge weights be a_1, c_1 , and put $A_1 = a_1/(1 + a_1) \leq 1/2$ and $Q_1 = c_1/(2 + c_1) \leq 1/3$. The middle row after eliminating the leaves gives

$$u_1 \leq \frac{1 + 2A_1 + 3Q_1}{1 + A_1 + 2Q_1} \leq 18/13.$$

The derivative numerators are $1 + Q_1$ and $1 - A_1$, both positive. The leaves give $u_0 \leq 2$ and $u_2 \leq 3/2$. If the plane-plane index is at least two, then $A_1 \leq 1/3$, giving $b \leq 4/3$. Above this value $a_1 = 1$, and $u_0 \leq 22/13$. If the other index is at least two, then $Q_1 \leq 1/5$, giving $b \leq 26/19$. Above that value $c_1 = 1$, and $u_2 \leq 19/13$.

For $a_1 = 1$, the middle inverse row of the original-component matrix is

$$\frac{(2 + c_1, 2(2 + c_1), 2c_1)}{6 + 7c_1},$$

whose smallest entry is $2c_1/(6 + 7c_1)$. Retaining $\alpha\delta$ in the old inequality gives

$$\alpha k \leq \frac{8 + 10c_1 - (6 + 7c_1)b}{2c_1} = \frac{10 - 7b}{2} + \frac{8 - 6b}{2c_1} \leq \frac{18 - 13b}{2} \quad (b > 4/3).$$

The sign $8 - 6b < 0$ justifies the last comparison.

Case 3. A quadric is the middle component of a three-component chain. Put $A_i = a_i/(1 + a_i) \leq 1/2$, $S = A_1 + A_2$, and $r_0 = \min(A_1, A_2)$. Elimination gives

$$u_1 \leq \frac{2 + 2S}{2 + S} \leq 4/3,$$

and its middle inverse row is $(A_1, 1, A_2)/(2 + S)$. Since $S \leq r_0 + 1/2$, retaining the original prime gives

$$\alpha k \leq \frac{2 + 2S - (2 + S)b}{r_0} \leq \frac{3 - 5b/2}{r_0} + 2 - b \leq 8 - 6b \quad (b > 6/5).$$

If either index is at least two, $S \leq 5/6$, and the first fraction is at most $22/17$. Above this value both indices are one, and the leaves have bound $5/3$. This proves all six simple-chain rows.

Case 4. Three components share a line. Suppress only the weight-two arms, keeping their transferred prime mass. Let their weights be $t_i = 1/q_i$, and put $A_i = t_i/(m_i + t_i)$, $S = A_1 + A_2$, $Q = A_3$, with $(m_1, m_2, m_3) = (1, 1, 2)$. The central row gives

$$u_c \leq \frac{-1 + 2S + 3Q}{S + 2Q}.$$

If a plane index is at least two, $S \leq 5/6$, $Q \leq 1/3$, and this is at most $10/9$. If the quadric index is at least two, $S \leq 1$, $Q \leq 1/5$, and it is at most $8/7$. For $b > 8/7$, all three indices

therefore equal one. The central bound is $6/5$, and the leaves give $(8/5, 8/5, 7/5)$. The full inverse central row, including the mass-zero vertex, is $(3/10, 3/10, 1/5, 3/5)$. Its product with κ is $6/5$. Retaining the full nonnegative vector p , of sum k , yields $\alpha k \leq 6 - 5b$. The Schur weights and right side are respectively $1/3$ and $(5/3, 5/3, 8/3)$.

Case 5. Four planes share a line. For a four-valent internal vertex, put $A_i = t_i/(1 + t_i) \leq 1/2$ and $S = \sum A_i \leq 2$. Leaf elimination gives $Su_c \leq -2 + 2S$, so $u_c \leq 1$. If there are two trivalent vertices, let $S, T \leq 1$ be their two leaf sums and $w > 0$ the connecting weight. Their rows are

$$(S + w)u_c - wu_d \leq -1 + 2S, \quad (T + w)u_d - wu_c \leq -1 + 2T.$$

At the larger of u_c, u_d , its own row bounds that value by $2 - 1/S$ or $2 - 1/T$, hence by one. Both possibilities contradict $u \geq b > 1$.

Case 6. The configuration has one triple conductor line and one binary conductor line. Number the three planes on the triple line by 1, 2, 3, with plane 4 attached to plane 3. Let $t_1, t_2, t_3 \leq 1$ be the arm weights and $w \leq 1$ the binary weight. Set

$$A_0 = \frac{w}{1 + w} \leq \frac{1}{2}, \quad S = \frac{t_1}{1 + t_1} + \frac{t_2}{1 + t_2} \leq 1, \quad B_0 = \frac{t_3}{1 + A_0 + t_3} \leq \frac{1}{2 + A_0}.$$

The original plane right sides are $(2, 2, 1, 2)$, and the central right side is -1 . Elimination gives

$$u_c \leq \frac{-1 + 2S + (1 + 2A_0)B_0}{S + (1 + A_0)B_0} \leq \frac{3 + 3A_0}{3 + 2A_0} \leq \frac{9}{8}.$$

The derivative numerators for the first fraction in S, B_0 are $1 + B_0$ and $1 + A_0 - S$, respectively, so the two upper bounds have the stated direction. This entire configuration lies below $8/7$.

Case 7. The binary conductor graph is a star. The central vertex is an original plane. Its right side is zero, and each leaf has right side two. Put $A_i = t_i/(1 + t_i)$, $S = \sum A_i \leq 3/2$. Then $u_c \leq 2S/(1 + S) \leq 6/5$. If any index is at least two, $S \leq 4/3$ and $u_c \leq 8/7$. Above $8/7$ all three weights are one. The central inverse row has entry $2/5$ at the center and $1/5$ at each leaf, giving the stated bounds and $\alpha k \leq 6 - 5b$.

Case 8. The binary conductor graph is a path. Order the four vertices 0, 1, 2, 3. Let the three weights be t_0, w, t_3 , and set $A_0 = t_0/(1 + t_0)$, $A_3 = t_3/(1 + t_3)$, both at most $1/2$. Eliminating the endpoints gives

$$(1 + A_0 + w)u_1 - wu_2 \leq 1 + 2A_0, \quad (1 + A_3 + w)u_2 - wu_1 \leq 1 + 2A_3.$$

The larger inner value is at most $4/3$, and the outer values are at most two. If $A_0 \leq 1/3$, use $u_2 \leq 4/3$ and $w \leq 1$ in the first row to get $u_1 \leq 9/7$. The other endpoint is symmetric. Above $9/7$ both endpoint indices are one, and their bounds improve to $5/3$.

Retain the old prime in these same eliminations and add the two inner rows. The middle edge cancels, giving

$$\alpha(A_0\delta_0 + \delta_1 + \delta_2 + A_3\delta_3) + (1 + A_0)u_1 + (1 + A_3)u_2 \leq 2 + 2(A_0 + A_3).$$

With $r_0 = \min(A_0, A_3)$ and $S = A_0 + A_3 \leq r_0 + 1/2$, the same signed comparison as for the middle quadric gives $\alpha k \leq 8 - 6b$ when $b > 6/5$. The middle index remains arbitrary. Together with the general bound $\alpha k \leq 6 - 4b$, this proves the three path rows. $\square$

11.4. The full source under failure. The remaining arguments have one source construction and one final summation. We record both before specifying the finite bounds.

Definition 11.9. For a reduced, nonintegral quartic datum, choose a closed or left-open rational deficit bin with endpoints B_-, B_+ , within one of the ranges in [Lemma 11.8](#), or within the low range $(1, 8/7]$. In the low range we may use the LOW family for any reduced configuration and take $B(b) = 6 - 4b$. For two quadrics the range is $[1, 3/2]$, and for a cubic plus a plane it is $[1, 10/7]$. Let $B(b)$ be the corresponding old-prime bound, using $6 - 4b$ in these two-component cases. Set

$$\beta_k = 1 - \frac{B(B_-)}{k}, \quad D_* = K_* + a_*B_-.$$

The initial source level γ_0 is the smallest multiple of $1/10000$ strictly greater than $B_+/(K_* + a_*B_+)$. The certificate tables give its value in each bin. All these values satisfy

$$(11.7) \quad \frac{1}{2} < \gamma_0 < \frac{31}{50}, \quad c_* + (1/\gamma_0 - a_*)B_+ < T_*.$$

For $k \in \{3, 4, 5\}$ and $\gamma \in [\gamma_0, G_*]$, the *failure domain* in (h, v) is

$$(11.8) \quad h \geq 0, \quad v \geq 0, \quad v \leq d_k h, \quad \gamma + v \leq B_+ h, \quad D_* h - q_k v \leq 1.$$

For a component allocation $x_i = h\delta_i$, define

$$(11.9) \quad f(\gamma, h, v, x) = \sum_i \frac{m_i}{2} (\gamma - \beta_k x_i / m_i)_+^2, \quad x_i \geq 0, \quad \sum_i x_i = kh.$$

Here δ_i denotes the full rational restriction degree.

The bins will be fixed numerical intervals, independent of p , the selected prime, and the threshold. Their endpoints and every subsequent source breakpoint occur in the complete supplementary tables.

Lemma 11.10. *Let (X, L, x, D, W, b) be a quartic datum with reduced, nonintegral full tangent. Suppose it has no additive point continuation of coefficient less than $n + 1 - 1/200000$. Fix a bin containing b as in Definition 11.9. There is a constant C_2 , depending only on the fixed original data, such that, for every p above the ordinary restriction cutoff p_{lift} of Construction 11.5 and every nonzero $F_{p,j}$ with $\gamma = j/p \in [\gamma_0, G_*]$, its sufficiently dilute full lifted system has an own threshold $t > 0$ whose minimal center is a fixed surface prime P , with multiplicity $k \in \{3, 4, 5\}$, positive deficit z , and exact full symbolic order r . For $h = 1/t$, $v = z/t$, one has*

$$(h, v) \text{ satisfies (11.8),} \quad \frac{r}{p} \geq \beta_k h.$$

The full source shell obeys

$$(11.10) \quad \dim(F_{p,j}/F_{p,j+1}) \leq p^2 f(\gamma, h, v, x) + C_2(j + 1),$$

Here $x_i = h\delta_i$ uses the full restriction degrees from Theorem 8.9. The same constant works for every prime, symbolic order, and conductor index. Moreover $F_{p,[G_*p]} = 0$.

Every shell uses its own prime and its own threshold. No coincidence between the primes at different shells is asserted.

Proof. We exclude paid successors, identify the full fixed prime, and then bound its source shell.

Step 1. In this step, we exclude point, curve, and low-multiplicity surface exits. At γ_0 , a point, curve, or surface of multiplicity at most two has a remaining radius bound $q \leq 1/\gamma_0$. Its scalar whole price is at most

$$n - 7 + c_* - a_* b + (b - z)/\gamma_0 + qz \leq n - 7 + c_* + (1/\gamma_0 - a_*)B_+ < n - 7 + T_*.$$

For $\gamma \geq \gamma_0$, the same bound follows by replacing $(b - z)/\gamma$ by $(b - z)/\gamma_0$. A zero-deficit outcome has arbitrarily small remaining cost and satisfies the same strict comparison. Lemma 11.3 pays all these exits. Failure leaves precisely the positive-deficit surfaces $k = 3, 4, 5$.

Step 2. In this step, we identify the selected surface as a full fixed prime with its exact symbolic order. For the actual selected prime, let α and η be the old and added restricted coefficients. Coefficient-one detection for the new lc prime in the same discriminant gives $\alpha + t\eta = 1$. With the old-prime bound from the chosen family,

$$\eta = (1 - \alpha)h \geq \left(1 - \frac{B(B_-)}{k}\right)h = \beta_k h.$$

Here $\beta_k \geq 1/3$, while $h \geq \gamma/B_+ > 1/3$. Thus $\eta > 1/9$. Before the prime is selected, choose the tuple so each nonfixed coefficient is less than $1/20$. This is possible on a principalization of the full system, where sufficiently many general moving members have coefficients at most $1/(pN)$.

Hence P is fixed in the entire $F_{p,j}$, and its coefficient is exactly r/p . This uses the full minimum order, rather than an order of a selected section.

Step 3. In this step, we derive the failure domain, the full shell bound, and the terminal-source vanishing. The first four inequalities of (11.8) follow from the surface cap and (11.1). If $c + t + q_k z \leq n - 7 + T_*$, the reserved two cutting errors give the excluded point continuation. Its failure, the old upper price, and $b \geq B_-$ therefore imply the weaker inequality $t + q_k z \geq K_* + a_* B_-$. Multiplication by h gives the fifth inequality.

[Theorem 8.9](#) bounds the entire symbolic initial image by

$$\sum_i \frac{m_i}{2} (j - r\delta_i/m_i)_+^2 + C_2(j+1).$$

The germ map from $F_{p,j}$ has kernel exactly $F_{p,j+1}$. Substituting $r/p \geq \beta_k h$ increases this upper bound, since each $\delta_i \geq 0$. This proves (11.10). Finally, a nonzero $F_{p,[G_*p]}$ would have a threshold paid by [Lemma 11.3](#). Thus it is zero. $\square$

Lemma 11.11. *Let (X, L, x, D, W, b) be a quartic datum with reduced, nonintegral full tangent. Let H be effective rational Cartier, not containing W , of restricted order at least $\gamma > 0$. Suppose its actual threshold $t > 0$ from D has a surface minimal center P , with multiplicity k and positive deficit z . Write δ_i for its full rational restriction degrees, $h = 1/t$, $v = z/t$, and $x_i = h\delta_i$. Then the allocations satisfy the following bounds. For two quadrics,*

$$\delta_1 + \delta_2 = k, \quad 0 \leq \delta_i \leq \frac{9}{2} - \gamma t - 3z.$$

For a cubic component and a plane, in that order,

$$\delta_C + \delta_\Pi = k, \quad \delta_C, \delta_\Pi \geq 0, \quad \delta_C \leq 4 + \min(2, 6 - 3b) - \gamma t - 4z, \quad \delta_\Pi \leq \frac{24}{7} - \gamma t - 2z.$$

For a chain, star, common-line configuration, or four-plane path, use the relevant $m_i, U_i, \hat{\kappa}_i$ from [Lemma 11.8](#). Put $r_i = \sum_j w_{ij}$. In a binary tree the weights in the following upper bounds may all be replaced by one. Define

$$(11.11) \quad \mathcal{U}_i(h, v, \gamma) = \left(\hat{\kappa}_i + \sum_j w_{ij} U_j \right) h - (m_i + r_i)v - r_i \gamma,$$

$$(11.12) \quad \mathcal{L}_i(h, v, \gamma) = \left(k - \sum_j \hat{\kappa}_j + \hat{\kappa}_i - r_i U_i \right) h + (4 - m_i + r_i)v + r_i \gamma.$$

Then $0 \leq x_i \leq \mathcal{U}_i$, and $x_i \geq \mathcal{L}_i$. For the simple three-component chains we retain all lower bounds. For the four-plane paths we retain the two inner lower bounds; for COMMON and STAR only the upper bounds are required.

These are necessary inequalities on the actual full restriction degrees. Using fewer of them enlarges the domain to be bounded.

Proof. For two components the post-threshold matrix from (11.5) is

$$\delta_i + (m_i + w)\rho_i - w\rho_j \leq m_i + 1, \quad z \leq \rho_i \leq u_i - \gamma t.$$

With any earned old bound $U_j \geq u_j$, this gives

$$\delta_i \leq m_i + 1 - m_i z + w(U_j - \gamma t - z) \leq m_i + 1 + U_j - \gamma t - (m_i + 1)z.$$

The last bracket before replacement is nonnegative because $U_j - \gamma t \geq \rho_j \geq z$; hence $w \leq 1$ has the correct direction. Substitute $U_j = 3/2$ for the two quadrics, and $U_C = 10/7$, $U_\Pi = \min(2, 6 - 3b)$ for the other case. This proves both two-component assertions.

For a general original subset I , the full Schur inequality in [Proposition 8.11](#) gives

$$\sum_{i \in I} \delta_i \leq \sum_{i \in I} \hat{\kappa}_i - z \sum_{i \in I} m_i + \sum_{\substack{i \in I \\ j \notin I}} w_{ij} (U_j - \gamma t - z).$$

Again every bracket is nonnegative, so the replacement of a binary weight by one is allowed. The singleton $I = \{i\}$, multiplied by h , gives [\(11.11\)](#). The complementary subset, with $\sum_i x_i = kh$, gives [\(11.12\)](#). The complement of an inner path vertex contains both sides of that vertex; both boundary edges and their old budgets occur in this calculation. $\square$

11.5. A common certificate protocol. The geometric estimates now reduce the problem to bounding a convex capacity on explicit rational domains. We specify these domains, give finite evidence covering every feasible point and every competing function, and prove the implication from that evidence to the required integral bound. The four complete datasets are identified in [subsection 11.6](#). All arithmetic in them is exact.

Definition 11.12. A *source certificate* for a reduced, nonintegral quartic datum fixes a family and an interval of incoming deficits, called a *bin*, contained in one of the ranges already established. It consists of an initial source level γ_0 as in [\(11.7\)](#), a finite partition of $[\gamma_0, G_*]$ into *pieces*, nonnegative rational endpoint bounds V_l, V_u on each piece $[l, u]$, and a rational mass ceiling R . Let $\mathcal{M}$ be the affine interpolant of these endpoint bounds on each open piece. The certificate also supplies the finite domain and upper-bound records specified below. Indices and rational numbers in a record have precisely those stated meanings.

11.5.1. Two-component domains. For two quadrics put $M_* = 9/2 - (\gamma + 3v)/h$. Convexity in δ_1 bounds the capacity by the endpoints of $[\max(0, k - M_*), \min(k, M_*)]$. Region zero has $M_* \geq k$ and capacity

$$f_0(\gamma, h, v) = \gamma^2 + (\gamma - k\beta_k h/2)_+^2.$$

Region one has $k/2 \leq M_* \leq k$ and capacity

$$f_1(\gamma, h, v) = \left(\gamma - \frac{\beta_k}{2} \left(\frac{9}{2}h - \gamma - 3v \right) \right)_+^2 + \left(\gamma - \frac{\beta_k}{2} \left(\left(k - \frac{9}{2} \right)h + \gamma + 3v \right) \right)_+^2.$$

In both regions impose [\(11.8\)](#) and $M_* \geq k/2$. A row (A_0, B_0, R_0, R_1) means

$$(11.13) \quad A_0 h + B_0 v \leq R_0 + R_1 \gamma.$$

The seven rows, numbered from zero, are

$$(11.14) \quad \begin{array}{c|cccc} 0 & -1 & 0 & 0 & 0 \\ 1 & 0 & -1 & 0 & 0 \\ 2 & -d_k & 1 & 0 & 0 \\ 3 & -B_+ & 1 & 0 & -1 \\ 4 & D_* & -q_k & 1 & 0 \\ 5 & k/2 - 9/2 & 3 & 0 & -1 \\ 6 \text{ (region 0)} & k - 9/2 & 3 & 0 & -1 \\ 6 \text{ (region 1)} & 9/2 - k & -3 & 0 & 1. \end{array}$$

For the cubic plus plane put

$$A_C = 4 + \min(2, 6 - 3B_-), \quad A_{\Pi} = 24/7, \quad M_C = A_C - (\gamma + 4v)/h, \quad M_{\Pi} = A_{\Pi} - (\gamma + 2v)/h.$$

Keep all four endpoints of the allowed interval for δ_C . Writing $s_C = h\delta_C$ and $s_\Pi = h\delta_\Pi$, their descriptions are

region	extra condition	(s_C, s_Π)
0	$M_C \geq k$	$(kh, 0)$
1	$M_\Pi \geq k$	$(0, kh)$
2	$M_C \leq k$	$(A_C h - \gamma - 4v, (k - A_C)h + \gamma + 4v)$
3	$M_\Pi \leq k$	$((k - A_\Pi)h + \gamma + 2v, A_\Pi h - \gamma - 2v)$

Every region also imposes $M_C, M_\Pi \geq 0$ and $M_C + M_\Pi \geq k$. The capacity in each is

$$f(\gamma, h, v) = \frac{3}{2}(\gamma - \beta_k s_C/3)_+^2 + \frac{1}{2}(\gamma - \beta_k s_\Pi)_+^2.$$

Rows zero through four are those of (11.14). Rows five through eight are

$$(11.15) \quad \begin{array}{c|ccc} 5 & -A_C & 4 & 0 & -1 \\ 6 & -A_\Pi & 2 & 0 & -1 \\ 7 & k - A_C - A_\Pi & 6 & 0 & -2 \\ 8 \text{ (region 0)} & k - A_C & 4 & 0 & -1 \\ 8 \text{ (region 1)} & k - A_\Pi & 2 & 0 & -1 \\ 8 \text{ (region 2)} & A_C - k & -4 & 0 & 1 \\ 8 \text{ (region 3)} & A_\Pi - k & -2 & 0 & 1. \end{array}$$

This includes the endpoints at which one component has zero allocation and both endpoints when the component degrees differ.

For any two rows (A_0, B_0, R_0, R_1) and (D_0, E_0, S_0, S_1) , let $\Delta = A_0 E_0 - B_0 D_0$. When $\Delta \neq 0$, their intersection is

$$(11.16) \quad \begin{aligned} h &= h_0 + h_1 \gamma, & h_0 &= (R_0 E_0 - B_0 S_0)/\Delta, & h_1 &= (R_1 E_0 - B_0 S_1)/\Delta, \\ v &= v_0 + v_1 \gamma, & v_0 &= (A_0 S_0 - R_0 D_0)/\Delta, & v_1 &= (A_0 S_1 - R_1 D_0)/\Delta. \end{aligned}$$

Its feasible interval is $[\gamma_0, G_*]$ intersected with every row inequality

$$(11.17) \quad (A_i h_1 + B_i v_1 - R_{1,i})\gamma \leq R_{0,i} - A_i h_0 - B_i v_0.$$

A positive coefficient supplies an upper endpoint, a negative one a lower endpoint, and a zero coefficient with negative right side excludes the pair. These rules include every row pair, every degeneracy, and every possible multiplicity $k = 3, 4, 5$. The domains are bounded because

$$h \geq \gamma/B_+ > 0, \quad (D_* - q_k d_k)h \leq 1, \quad D_* - q_k d_k > 0.$$

For the sign, $K_* > 0$, $B_- \geq 1$, and $a_* > 5/4 > \max_{k=3,4,5} q_k d_k$. Thus a nonempty domain has a vertex among those row pairs.

Partition at all endpoints of (11.17) and at $\gamma_0 + j/200$ inside the source interval. On each open piece the active row-pair list is fixed. Every substituted candidate is a sum of positive multiples of squared positive affine functions, hence is convex in γ . The maximum of the full list is convex as well. The complete tables give rational endpoint upper bounds V_l, V_u , rounded upward to multiples of 10^{-8} , for every active candidate on the closure of the piece. Their labels identify $(k, \text{region}, \text{row pair})$, and (11.16)–(11.17) specify every competitor, including those not attaining a maximum. The bound on the integral over $[l, u]$ is

$$(11.18) \quad \frac{u-l}{2}(V_l + V_u).$$

A vertex supported only at a breakpoint affects finitely many ordinary shells; those shells are retained with their full ordinary bound. An empty interior-active list has zero capacity on its piece.

The two-quadric table has 25 bins and 693 pieces, and its largest normalized mass ceiling is 49900103/50000000. The cubic-plus-plane table has 14 bins and 436 pieces, and its largest ceiling

is 19979391/20000000. Both are strictly less than 999/1000. Each normalized mass is computed from the full table by

$$(11.19) \quad \mathfrak{M} = 4\gamma_0^3 + 3 \sum_{[l,u]} (u-l)(V_l + V_u).$$

The factor 3 here incorporates the factor 6 in the threefold Hilbert normalization and the 1/2 in (11.18).

11.5.2. Conductor allocation domains. For a simple three-component chain we use every bound $\max(0, \mathcal{L}_i) \leq x_i \leq \mathcal{U}_i$. At fixed (h, v, γ) , the allocation set is the intersection of a box with $\sum_i x_i = kh$. All but at most one coordinate of each vertex is an endpoint of its allowed interval. To see this, if two coordinates are strictly between their endpoints, increasing one and decreasing the other gives a nontrivial segment through the point within the set. Conversely, the endpoint enumeration contains every vertex, including degenerate boxes.

For a three-component chain, choose the free coordinate $f \in \{0, 1, 2\}$. At each of the other two coordinates, in increasing coordinate order, choose $0, \mathcal{L}_i, \mathcal{U}_i$, with labels 0, 1, 2, respectively. Set $x_f = kh - \sum_{i \neq f} x_i$. For each $k = 3, 4, 5$ this gives 27 descriptions. The full description index is

$$\ell = 27(k-3) + 9f + 3e_0 + e_1 \quad (0 \leq \ell \leq 80).$$

All constraints are reimposed on the substituted coordinates; choosing a negative lower endpoint does not make it feasible.

For the remaining reduced configurations with $1 < b \leq 8/7$, use Lemma 11.7. Split a summand of integer mass m_i in (11.9) into m_i identical numerical summands. There are four summands of weight one, with allocation degrees in $[0, 2]$ and total degree k , because $\delta_i/m_i \leq 1 + 1/m_i \leq 2$. This enlarges the domain of the same full symbolic bound. Its convex symmetric objective is maximized, up to permutation, at

$$k = 3: (2, 1, 0, 0), \quad k = 4: (2, 2, 0, 0), \quad k = 5: (2, 2, 1, 0).$$

Indeed a vertex of this box slice has all but at most one coordinate equal to zero or two. These are the three descriptions of the LOW family, with x_i equal to h times the displayed entries and $B(b) = 6 - 4b$. This is a numerical relaxation; the tangent components and their full images are retained.

Above 8/7, use COMMON, STAR, or one of the three PATH tables. For COMMON and STAR, the endpoint labels are 0, 2, for zero and $\mathcal{U}_i$. For a path use these labels at the outer coordinates, and 0, 1, 2 at each inner coordinate, where label one means $\mathcal{L}_i$. For $k = 3, 4, 5$ in that order, increase the free coordinate and then the ordered endpoint tuple lexicographically, starting the description index at zero. The numbers of descriptions are

family	LOW	COMMON	STAR	PATH
descriptions	3	36	96	180.

These enumerate every vertex of the corresponding enlarged allocation domain. Convexity of (11.9) reduces its maximum to these descriptions.

After substitution write

$$x_i = A_i h + B_i v + D_i \gamma, \quad \mathcal{U}_i = V_i h + W_i v + Z_i \gamma, \quad \mathcal{L}_i = E_i h + J_i v + N_i \gamma.$$

The first five row quadruples are always

$$(11.20) \quad (-1, 0, 0, 0), \quad (0, -1, 0, 0), \quad (-d_k, 1, 0, 0), \quad (-B_+, 1, 0, -1), \quad (D_*, -q_k, 1, 0).$$

For a simple three-component chain append, for $i = 0, 1, 2$, the three rows in this exact order:

$$(11.21) \quad (-A_i, -B_i, 0, D_i), \quad (E_i - A_i, J_i - B_i, 0, D_i - N_i), \quad (A_i - V_i, B_i - W_i, 0, Z_i - D_i).$$

There are fourteen rows, indexed zero through thirteen. For a remaining-family description append instead, in increasing coordinate order,

$$(11.22) \quad (-A_i, -B_i, 0, D_i), \quad (A_i - V_i, B_i - W_i, 0, Z_i - D_i),$$

and immediately afterwards append $(E_i - A_i, J_i - B_i, 0, D_i - N_i)$ if i is an inner path coordinate. The LOW descriptions have only the first five rows. This specifies the row indices in every data entry.

11.5.3. Checking finite domains and upper bounds. We describe the common verification rule for the 46-bin simple-chain table and the 40-bin remaining-family table. It uses empty-domain certificates, complete polygon boundaries, all clipped vertices, and upper bounds for every competing function.

Step 1. In this step, we bound and classify the full master domains. Substitute $h = \gamma H$ and $v = \gamma V$. At γ_0 , a row becomes

$$(11.23) \quad A_i H + B_i V \leq r_i^0 := R_{0,i}/\gamma_0 + R_{1,i}.$$

All rows except row four are independent of the source level. Row four subsequently tightens to

$$(11.24) \quad D_* H - q_k V \leq 1/\gamma.$$

The full domains are bounded, since $H > 0$ and $(D_* - q_k d_k)H \leq 1/\gamma_0$, with positive coefficient. Later domains are nested inside the initial one.

The lists **EMPTY**, **POLYGONS**, and **SEGMENTS**, together with **POINTS** when present, partition the entire description set of each bin. For a rational number r , write $\text{num}(r)$ for its reduced numerator with positive denominator.

An **EMPTY** entry gives row indices i_j , positive rational weights w_j , and the negative integer

$$\text{num} \left(\sum_j w_j r_{i_j}^0 \right), \quad \sum_j w_j (A_{i_j}, B_{i_j}) = (0, 0).$$

Summing these halfplanes is impossible. This excludes the domain at γ_0 , and nesting excludes it at every later source order.

A **POLYGONS** entry gives a cyclic edge-row list $i_0, \dots, i_{s-1}$. Let p_j be the intersection of rows i_{j-1}, i_j in (11.23), and $t_j = (-B_{i_j}, A_{i_j})$. The positive integers supplied are exactly the numerators of

$$\lambda_j = \frac{(p_{j+1} - p_j) \cdot t_j}{t_j \cdot t_j}.$$

The nonzero normals have positive consecutive cross products and make one full turn in cyclic increasing angle order. Their quadrant signs and cross products check this without trigonometry. The sides $p_{j+1} - p_j = \lambda_j t_j$ consequently form a closed strictly convex counterclockwise polygon. This is the intersection of the inward halfplanes of its selected rows. Since every selected row is an original row, the polygon contains the entire feasible set. Any unselected row can only reduce that set; no assertion that it is redundant is needed.

A **SEGMENTS** entry gives positive weights for which both the weighted normal and the weighted right side vanish. Every feasible point must therefore have equality on every supported row. The specified nonzero line row l is among them. Its two printed endpoint row pairs give p, q . When $p \neq q$, use the direction $t = q - p$. The first endpoint normal has negative scalar product with t , and the second has positive scalar product. Writing a point on the line as $p + st$, the two inequalities give $s \geq 0$ and $s \leq 1$, respectively. Thus all solutions lie in $[p, q]$. The simple-chain records use this nonzero direction. The remaining-family format also permits $p = q$: in that case use $t = (-B_l, A_l)$, require the same strict signs, and obtain $s \geq 0$ and $s \leq 0$. The only possible point is then p . A separate **POINTS** entry uses a zero-weighted-sum certificate with two independent supported normals; their printed intersection is the only possible point. All these conditions are exact rational identities in the row data and (11.16).

Step 2. In this step, we enumerate every vertex created or retained by the moving constraint. For a master vertex p , define $J(p) = D_* H_p - q_k V_p$. All printed vertices have $J(p) > 0$. Its

candidate of type V is $(h, v) = \gamma p$, on

$$[\gamma_0, \min(G_*, 1/J(p))].$$

For a master edge $[p, q]$ on row $i \neq 4$, with $J(p) \neq J(q)$, its candidate of type E is the intersection of rows four and i from (11.16), on

$$[\max(\gamma_0, \min(1/J(p), 1/J(q))), \min(G_*, \max(1/J(p), 1/J(q)))].$$

The CANDIDATES list contains every interval with positive length. These are all the possibilities: clipping a polygon by one halfplane preserves some old vertices and creates new vertices only where its line meets an old edge. The same rule applies to a segment and its endpoints, or to a master point. Intervals of negative length contribute nothing. Zero-length intervals concern finitely many source shells and are retained with the ordinary $O(p^2)$ bound. Hence no empty, segment, point, or changed-vertex domain is discarded without its corresponding argument.

Step 3. In this step, we identify the capacity function attached to every candidate. Substitution in (11.9) gives

$$(11.25) \quad F(\gamma) = \sum_i \frac{m_i}{2} (\xi_i + \zeta_i \gamma)_+^2.$$

Sort the triples (m_i, ξ_i, ζ_i) lexicographically. The FUNCTIONS list supplies a representative description, type, and row pair for each distinct triple list. The candidate list keeps every interval even when identical functions are grouped. For a V candidate the triples are

$$\left(m_i, 0, 1 - \frac{\beta_k}{m_i} (A_i H_p + B_i V_p + D_i) \right).$$

For an E candidate they are

$$\left(m_i, -\frac{\beta_k}{m_i} (A_i h_0 + B_i v_0), 1 - \frac{\beta_k}{m_i} (A_i h_1 + B_i v_1 + D_i) \right).$$

In the simple-chain and COMMON tables the sorted weights are $(1, 1, 2)$; in every other remaining family they are $(1, 1, 1, 1)$. This fixes which square has which weight in each rounding witness.

Step 4. In this step, we verify upper covers for every candidate function. The GAMMA dictionary lists all rational endpoints. Each PIECES entry gives its left and right endpoint indices, followed by two integers which represent $10^8 V_l, 10^8 V_u$. Let $\mathcal{M}(\gamma)$ be the affine interpolant of the actual rational values V_l, V_u on the open piece. Its value at the finitely many breakpoints is immaterial to the leading sum.

For each function the intervals in UPPER COVERS cover the union of all its candidate intervals, up to these endpoints. There are two types. A C entry supplies a constant B_0 with $F(l), F(u) \leq B_0$, and $B_0 \leq \mathcal{M}$ on every piece or subpiece it meets. The latter condition is checked at both endpoints of every overlap. An L entry lies in one piece and supplies its exact chord values $\mathcal{M}(l), \mathcal{M}(u)$, each bounding F at the corresponding endpoint. Convexity of (11.25) gives $F \leq \mathcal{M}$ throughout every cover.

Every endpoint inequality has an integer witness $(e, v_0, u_0, \dots, u_{N-1}, S)$. With $D_0 = 10^e$, its meaning at the given rational endpoint γ and rational upper value B_0 is

$$(11.26) \quad \begin{aligned} v_0 &= \lfloor D_0 B_0 \rfloor, & u_i &= \lceil D_0 (\xi_i + \zeta_i \gamma)_+ \rceil, \\ S &= 2D_0 v_0 - \sum_i m_i u_i^2 \geq 0. \end{aligned}$$

The floor and ceiling assertions mean the corresponding two integer inequalities, so they are checkable without a floating-point approximation. They imply

$$F(\gamma) \leq \frac{\sum_i m_i u_i^2}{2D_0^2} = \frac{v_0}{D_0} - \frac{S}{2D_0^2} \leq B_0.$$

The simple-chain data contain all 14,504 such endpoint witnesses. For example their first left witness is $(2, 42, 57, 57, 25, 652)$, giving

$$2 \cdot 100 \cdot 42 - 57^2 - 57^2 - 2 \cdot 25^2 = 652 > 0.$$

The corresponding right witness is $(2, 42, 59, 59, 25, 188)$. Both bound the function by $42/100 < 21322219/50000000$. The remaining-family data include all 15,062 endpoint witnesses, with the same rule and the appropriate sorted weights. The rules apply to every competitor, including nonmaximizing candidates and zero terms.

Step 5. In this step, we bound the whole masses and verify coverage of every required bin. Formula (11.19) uses the rational endpoint values; the integers in **PIECES** are divided by 10^8 before that sum is taken. Each bin supplies a rational ceiling R and the exact positive integer $\text{num}(R - \mathfrak{M})$. This certifies the entire sum, including every zero-capacity piece. The simple-chain table has 46 bins and 1,618 pieces. Its largest ceiling and strict comparison are

$$R \leq \frac{24966091}{25000000}, \quad \frac{999}{1000} - \frac{24966091}{25000000} = \frac{8909}{25000000} > 0.$$

The bins cover $(1, 18/13]$ for PPC and $(1, 4/3]$ for PQP, and no bin crosses a strict index or old-budget boundary of Lemma 11.8.

The remaining-family table has 40 bins and 2,481 pieces. Its largest ceiling and strict comparison are

$$R \leq \frac{19975087}{20000000}, \quad \frac{999}{1000} - \frac{19975087}{20000000} = \frac{4913}{20000000} > 0.$$

Its LOW bins cover $(1, 8/7]$; COMMON and STAR each cover $(8/7, 6/5]$; and its three path groups cover $(8/7, 6/5]$, $(6/5, 9/7]$, and $(9/7, 4/3]$. Successive endpoints in each group coincide. Thus the finite certificate covers every allowed deficit, every allocation description, and every source-order interval required by the geometry.

Lemma 11.13. *Let a source certificate in the sense of Definition 11.12 satisfy all the indexing, domain, candidate, upper-bound, rounding, and mass checks specified above for its family. For $\gamma \in [\gamma_0, G_*]$, let $\Psi(\gamma)$ be the supremum of that family's capacity over all its feasible (h, v) and allocation coordinates, with value zero when the feasible set is empty. Except at finitely many source levels, $\Psi(\gamma) \leq \mathcal{M}(\gamma)$, and*

$$4\gamma_0^3 + 6 \int_{\gamma_0}^{G_*} \Psi(\gamma) d\gamma \leq \mathfrak{M} \leq R,$$

where $\mathfrak{M}$ is defined by (11.19). All coefficients, subdivisions, and exceptional levels in this assertion depend only on the fixed bin and its certificate.

Proof. Convexity first reduces an allocation box slice to its listed endpoint descriptions. For two components the same reduction gives the two or four displayed regions. Boundedness then reduces the resulting capacity to vertices of the corresponding (h, v) domain. The exhaustive row-pair intervals cover these vertices in the two-component case. In the conductor cases, the Farkas records exclude precisely the descriptions declared empty, and the checked polygons, segments, and points contain every other feasible domain. Clipping these sets gives only their surviving vertices and the listed moving-edge intersections. Thus every feasible capacity is bounded by one of the candidate functions whose complete intervals are retained in the certificate.

Each candidate is convex in γ . On a two-component piece, its two endpoint bounds give the asserted interpolant bound. For a conductor candidate, the checked rounding identity supplies both endpoint inequalities on each of its covers. A constant cover lies below the interpolant on every overlap; a linear cover is its chord on a single piece. The complete cover union therefore gives the same bound everywhere except at the finitely many partition or isolated candidate endpoints. All functions are bounded on their compact domains, so these finitely many points do not affect the integral. Integrating each affine piece gives (11.18), and the checked aggregate

rounding or positive mass-slack numerator gives $\mathfrak{M} \leq R$. Every object used here is fixed by the finite certificate before any geometric trial is selected. $\square$

11.6. Supplementary certificates. The four complete datasets have identifiers `11-quartic-threefold/` followed by the family key in the table. Their exact records satisfy the preceding protocol, with the following maximum mass ceilings and strict margins:

family key	bins	pieces	max R	$999/1000 - \max R$
two-quadrics	25	693	49900103/500000000	49897/500000000
cubic-plane	14	436	19979391/200000000	609/200000000
simple-chains	46	1618	24966091/250000000	8909/250000000
remaining	40	2481	19975087/200000000	4913/200000000

The full data and their checker are in `supplement/11-quartic-threefold/`. Each family key names its `.txt` file under `data/`. The first two files give the complete bin and piece tables, including active counts and maximizing labels. The last two give every domain description, function, candidate interval, cover, and rounding witness. The simple-chain data contain 92 segment records and three separate point records. The remaining data contain ten segment-format records with coincident endpoints; their point interpretation is included above.

The remaining-family file is one ordinary, unencoded stream of 17,822 records, including all 2,481 pieces and all 15,062 rounding witnesses. Its former dictionary encoding expands to this same stream. The archive retains that encoding and the earlier presentation; the decoding rule and record equality are documented in the supplement. Repeated witness occurrences belonging to different records remain in the canonical data.

The portable `check.py` reads only these four data files and performs exact rational checks of the predicates used in Lemma 11.13. It reconstructs the rows from the displayed family formulas, checks complete bin and description coverage, and checks every competing function and endpoint witness. The `README.md` specifies the file grammar and reproduction command, and `manifest.json` maps the family keys to the files and protocol. Missing or repeated identifiers, invalid indices, incomplete domains or covers, and failed inequalities cause rejection. This finite validation supplies the numerical premise of the lemma; the full-image geometry and uniform source estimates remain the preceding mathematical arguments.

11.7. Summation on the original threefold.

Proof of Theorem 11.1. We combine the tangent cases and then compare the whole source filtration with its original Hilbert polynomial.

Step 1. In this step, we reduce to the full reduced nonintegral tangent configurations. The nonreduced case is paid by Lemma 11.4, and the integral case by Lemma 11.6. Suppose that the full tangent is reduced and nonintegral and that the asserted continuation fails.

Lemma 11.8 gives the exhaustive remaining configurations. For two components, the two-quadric or cubic-plus-plane table applies. The 46-bin table applies to both simple three-component chains. If three components share a line, or if there are four planes, the 40-bin table applies, including the low range and the mixed triple and binary conductor. The four-plane configuration on one line has already been excluded by its original discrepancy inequality.

Step 2. In this step, we telescope the complete source filtration using the applicable certificate. Choose the applicable deficit bin and its full source-order partition. For every sufficiently large original p , Lemma 11.10 gives $F_{p,[G_*p]} = 0$, and bounds every nonzero shell above γ_0 using its own fixed prime. Lemma 11.13, applied to the corresponding supplementary certificate, bounds that shell's leading term by $p^2 \mathcal{M}(j/p)$, apart from the finitely many exceptional source levels retained below. Below γ_0 , the entire ordinary tangent shell has dimension $2j^2 + 3j + 1$.

Telescope all quotients from $F_{p,0} = H^0(W, pA)$ to $F_{p,[G_*p]} = 0$. The ordinary initial part contributes $4\gamma_0^3 p^3/6 + O(p^2)$. On each fixed later piece, the affine upper interpolant contributes its trapezoid integral times p^3 . The uniform symbolic errors $O(j+1)$ sum to $O(p^2)$. There

are finitely many breakpoint shells, each with its full ordinary $O(p^2)$ bound. Integer ceilings and the Riemann-sum errors also contribute $O(p^2)$: each fixed piece has a finite list of affine-square functions of bounded slope. All these constants precede the adaptive primes, orders, and conductor indices. We obtain

$$h^0(W, pA) \leq \mathfrak{M} \frac{p^3}{6} + O(p^2) \leq \frac{999}{1000} \frac{p^3}{6} + O(p^2).$$

Step 3. In this step, we contradict the original Hilbert polynomial and obtain the additive continuation. On the original projective W , the ample Cartier divisor A has positive integral degree g . Its Hilbert polynomial and Serre vanishing give

$$h^0(W, pA) = g \frac{p^3}{6} + O(p^2), \quad g \geq 1.$$

The strict gap $1/1000$ contradicts the preceding upper bound for all sufficiently large p . Choose the intrinsic cutoff from this fixed gap, the original Hilbert error, and the finite certificate errors before any threshold is selected. Then choose one original degree beyond this cutoff and the separate ordinary restriction cutoff. All trials used above are at that degree. No lower mass has been assigned to a proper image or to a tangent component, and the original fixed and remote conditions remain in every space and in every kernel.

Failure is therefore impossible. Some actual trial from the entire D , followed by its complete smaller-center continuation, has scalar coefficient at most $n - 7 + T_*$. The two-error reserve of [Lemma 11.3](#) makes its actual coefficient less than $n + 1 - 1/200000$, as required. $\square$

The positive residual coefficient also gives generation of $\mathcal{O}_X(K_X + (n+1)L)$ at x , by [Proposition 3.12](#). This uses the successful additive continuation and its remaining ample difference.

12. CUBIC-FOURFOLD ENDPOINTS

We retain the boundary, deficit and original-source conventions of [Section 2](#). We first prove an endpoint from an affine bound on the whole coefficient and deficit. We then prove an endpoint from the actual multiplicity of the whole boundary. The latter uses the geometry of a reduced reducible tangent and two successive improvements of a complete intrinsic source.

Definition 12.1. Let $v = (f, h, k)$ run through the states in the following table. Write $T_v = T(h - f, k)$, in the notation of [Definition 5.2](#), and $\beta_v = f - 2T_v/k$. The entries in each list correspond in order.

f	h	k	T_v	q_v
1	1	1	0	1
2	$k+1$	1, 2, 3, 4, 5	0, 1, 2, 3, 4	1, $\frac{283}{200}$, $\frac{1733}{1000}$, 2, $\frac{2237}{1000}$
3	3	1	0	1
3	4	2, 3	1, 3	$\frac{63}{50}$, $\frac{1443}{1000}$
3	5	3, 4, 5, 6	2, 4, 6, 8	$\frac{1443}{1000}$, $\frac{397}{250}$, $\frac{171}{100}$, $\frac{909}{500}$
3	6	4, 5, 6, 7, 8, 9	3, 5, 7, 9, 11, 13	$\frac{397}{250}$, $\frac{171}{100}$, $\frac{909}{500}$, $\frac{1913}{1000}$, 2, $\frac{2081}{1000}$

For a lower-dimensional state $w = (f', h', k')$ with $h' \leq h$, set

$$c_{vw} = \begin{cases} \min\{\beta_w, (fk - 2T_v - k')/k\}, & f' = f - 1, \\ \beta_w, & f' \leq f - 2. \end{cases}$$

Retain precisely the transitions with $c_{vw} > 0$. Let $\hat{\mathcal{F}}_v$ be the envelope of [Definition 5.2](#) with radius q_v at each state. For $\kappa \geq 1$ and $0 \leq a \leq \beta_v$, define its *complete surplus* by

$$J_v(\kappa, a) = \max_{0 \leq z \leq a} (\hat{\mathcal{F}}_v(z) - \kappa z).$$

For a first successor of a cubic fourfold, set

$$C_v = \begin{cases} \min\{\beta_v, (8-k)/3\}, & f = 3, \\ \beta_v, & f \leq 2. \end{cases}$$

Only $C_v > 0$ gives a positive-dimensional first successor.

These envelopes include every dimension skip. Each $q_v^f \geq k$, so the radii bound the actual radii for every positive integral original degree. Replacing the radii by these upper bounds only increases the cost of each nonnegative deficit drop in the recurrence. The definition of T gives exactly the displayed positive states in dimensions at most three and embedding dimension at most six. Expanding the first deficit drop gives the finite rule

$$(12.1) \quad J_v(\kappa, a) = (q_v - \kappa)_+ a + \max \left(0, \max_{v \rightarrow w} J_w(\max\{\kappa, q_v\}, \min\{a, c_{vw}\}) \right).$$

Indeed, after fixing a child deficit z , the maximizing outer deficit is a if $q_v \geq \kappa$, and is z otherwise. The zero option is the direct point. Thus the rule computes maxima on the entire closed deficit intervals.

For multiplicities eight and nine at embedding at most six, the moments are respectively 11 and 13. Their intrinsic caps are $1/4$ and $1/9$, and Lemma 5.10 bounds their entire costs by $2z$ and $9z/4$. At the cubic fourfold their additional first caps are respectively 0 and $-1/3$, so neither is a first positive-deficit successor. The embedding-seven caps $1/2, 1/3$ in that lemma belong to a different ambient state set: a quotient of $\mathcal{O}_{W,x}$ has embedding dimension at most six here.

For a cubic fourfold the normal first Hilbert coefficient is at least two, by Theorems 4.13, 4.20 and 4.16. Consequently its deficit is at most $8/3$. Proposition 4.18 gives the first caps above even for an old prime. More explicitly, if its multiplicity is k , old coefficient is α , and new deficit is z , then

$$3z + (1 - \alpha)k \leq K_W \leq 8 - \alpha k, \quad z \leq (8 - k)/3.$$

The cancellation uses only this prime's own old contribution; all other old contributions remain nonnegative deductions.

Lemma 12.2. *Use the states, radii and transition caps of Definition 12.1. Fix a state v and rational numbers $\kappa \geq 1$ and $0 \leq a \leq \beta_v$. For every path $\mathbf{v} = (v_0, \dots, v_\ell)$ starting at v , with strictly decreasing dimensions and nonincreasing embedding dimensions, put*

$$\mathcal{D}_{\mathbf{v}}(a) = \{(y_0, \dots, y_\ell) : 0 \leq y_\ell \leq \dots \leq y_0 \leq a, \quad y_i \leq c_{v_{i-1}v_i} \ (1 \leq i \leq \ell)\}, \quad y_{\ell+1} = 0.$$

Paths may end at any state. Then $J_v(\kappa, a)$ is the maximum, over the vertices of all nonempty $\mathcal{D}_{\mathbf{v}}(a)$, of

$$\sum_{i=0}^{\ell} q_{v_i}(y_i - y_{i+1}) - \kappa y_0.$$

Every vertex is obtained by selecting $\ell + 1$ linearly independent active inequalities, solving them as equalities, and retaining the solutions satisfying all the inequalities.

Proof. Expanding a complete tail along a path gives precisely the displayed linear function and deficit constraints. Ending the path means taking the next center to be a point. Thus all dimension skips and the direct point occur among these paths. A negative transition cap makes its domain empty. A zero cap forces that child and all later deficits to zero, giving the same value as a shorter path; it creates no additional positive-deficit transition.

Each domain is a bounded closed rational polyhedron in at most three variables. At a vertex its active normals span the full coordinate space: otherwise a nonzero vector annihilating them permits sufficiently small feasible perturbations in both directions. Conversely, a feasible solution of a full-rank active system is a vertex. This proves the stated enumeration rule even when the domain lies in a proper affine subspace. Dependent active systems need not determine a point; every vertex they contain is still recovered from a full-rank subset of its active constraints. A nonempty compact polyhedron attains a linear maximum at a vertex. The argument includes

segments, singletons and $a = 0$, retains their endpoints, and retains ties between different paths. Taking the finite maximum therefore gives the complete surplus. $\square$

We use this lemma to check the three tables below independently of the recurrence (12.1). Their exact data are in the following supplement families.

Family identifier	Contents
12-cubic-fourfold/states	State census
12-cubic-fourfold/affine	Affine corrections
12-cubic-fourfold/integral-prices	Integral-tangent prices
12-cubic-fourfold/reduced-prices	Reduced-tangent prices
12-cubic-fourfold/claims	Masses and margins

The state family includes all nineteen states, the first excluded multiplicity for each of the seven positive-codimension pairs (f, h) , and the separate embedding-seven comparisons. To check completeness, fill each successive Artinian degree up to its binomial capacity and stop at the first nonpositive intrinsic cap. The minimum average degree is nondecreasing with the length, so every subsequent multiplicity is excluded. The displayed radii are the upward multiples of $1/1000$, checked by $(q_v - 1/1000)^f < k \leq q_v^f$.

For each required source order and deficit cap, enumerate every compatible path and apply Lemma 12.2. The two price families contain seventeen rows each, with three exact rational costs and their integers N characterized by

$$N - 1 < 10^4 \text{ cost} \leq N.$$

This is ordinary upward rounding, including integral costs. The affine family records the sixteen nonquartic corrections and the quartic split at deficit one. The claims family records the source masses, forced-order bounds and strict margins used below. The checking command reads these records, requires the complete expected state and parameter sets, recomputes every maximum and identity with rational arithmetic, and rejects missing, duplicate, unrecognized or invalid records. The printed tables are rendered from the same records, and their rendered text is compared after newline normalization. Thus acceptance verifies the finite inequalities by the lemma; the following proofs supply their geometric applicability, the reductions over all b, σ, n , and the uniform source and cutting errors. No geometric conclusion is inferred from arithmetic alone.

Theorem 12.3. *For $n \geq 7$, an effective rational $D \sim_{\mathbb{Q}} cL$, $c \geq 0$, lc near x , with actual minimal center of state $(4, 6, 3)$ and deficit y , has an additive point continuation of coefficient less than $n + 1 - 1/400000$ whenever*

$$c + \frac{1317}{1000}y \leq n + \frac{51}{100}.$$

The bound concerns the entire incoming divisor and permits every original center degree and tangent scheme.

Proof. We pay the complete scalar tails, then use the quartic-threefold endpoint for the one remaining successor. Put $q = 1317/1000$; the first cutting radius is at most q , since $q^4 \geq 3$.

Step 1. In this step, we pay every nonquartic successor and every quartic successor of deficit at most one. For every nonquartic first successor, the exact correction $J_v(q, C_v)$ from (12.1) is

v	$J_v(q, C_v)$	v	$J_v(q, C_v)$
(1, 1, 1)	0	(2, 2, 1)	0
(2, 3, 2)	49/500	(2, 4, 3)	104/375
(2, 5, 4)	683/2000	(2, 6, 5)	46/125
(3, 3, 1)	49/500	(3, 4, 2)	26/125
(3, 4, 3)	63/500	(3, 5, 3)	121/300
(3, 5, 4)	1229/4000	(3, 5, 5)	1179/5000
(3, 5, 6)	167/1000	(3, 6, 5)	451/1000
(3, 6, 6)	167/500	(3, 6, 7)	149/750

For a surface the correction is $(q_v - q)_+ C_v$. For a threefold the rule adds the largest permitted surface correction, with its inherited cap; direct curves contribute zero. This describes every comparison in the table. The largest entry is 451/1000, which is less than 977/2000. For the quartic threefold and $z \leq 1$, the correction is $J_{(3,6,4)}(q, 1) = 477/1000$. Indeed its full tail satisfies

$$(12.3) \quad \hat{\mathcal{F}}_{(3,6,4)}(z) \leq \frac{397}{250}z + \frac{103}{500}, \quad 0 \leq z \leq 4/3.$$

The constant is the quartic-surface surplus $(2 - 397/250)/2 = 103/500$; the quintic-surface cap is $1/4$, and every other surface and direct curve gives a smaller surplus.

Apply Theorem 3.14, that is, [GL24, Proposition 3.2 and Remark 3.3], to D . If the new deficit is z , its coefficient is at most $c + q(y - z)$ plus the first cutting error. For a nonquartic successor, Theorem 5.4 and the table bound the whole ideal cost by

$$c + qy + 977/2000 \leq n + 1997/2000.$$

For a quartic successor with $z \leq 1$, the bound is $c + qy + 477/1000 \leq n + 987/1000$. The direct point costs at most $c + qy$.

Fix a total weighted error allowance $1/2000$. At each of the at most four actual cuts, after its radius r_i is known, choose its published error smaller than $1/(8000(1 + r_i))$. This charges both the coefficient error and the deficit error in the finite-chain estimate. The resulting nonquartic coefficient is less than $n + 1 - 1/1000$, and the low-deficit quartic coefficient is less than $n + 1 - 1/80$. Both suffice for the asserted margin.

Step 2. In this step, we pay the quartic successor of deficit greater than one at its whole coefficient. It remains that the actual first center has state $(3, 6, 4)$ and $1 < z \leq 4/3$. Put $a = 5000/3997$ and $\epsilon_0 = 49267/107919$. For the entire new boundary, of coefficient c' , we have $c' + az \leq c + q(y - z) + az + \text{error} \leq n + 51/100 + (a - q)z + \text{error} < n + 51/100 + a - q + \text{error}$. Here $a < q$, and the exact entry margin is

$$\epsilon_0 - 51/100 - a + q = \frac{1357633}{107919000} > \frac{1}{2000}.$$

The same first-cut error allowance therefore gives $c' + az < n + \epsilon_0$. Theorem 11.1 applies to this actual whole boundary and gives coefficient less than $n + 1 - 1/200000$. Its continuation includes the first cut and all of D , and proves the required weaker margin. $\square$

We next record how complete intrinsic sources are used at one original degree. Let $B \sim_{\mathbb{Q}} sL$ be effective rational and lc near x , with actual minimal center W of state $(4, 6, 3)$. Put $b = \text{def}_x(B)$, $A = L|_W$, and $g = A^4 \geq 1$. For each fixed rational order γ , the source is $F_p(\gamma) = H^0(W, pA \otimes I_x^{[\gamma p]})$. Its full ordinary point-jet kernel gives

$$(12.4) \quad \dim F_p(\gamma) \geq \frac{p^4}{24}(g - 3\gamma^4) - K_0(p + 1)^3.$$

This requires neither global normality of W nor surjectivity onto the point jets. Separately, the exact restriction sequence and Theorem 6.8, [Sta26, Lemma 30.17.1, Tag 0B5U], give $H^0(X, pL) \rightarrow H^0(W, pA)$ for all sufficiently large original integers p .

Whenever $F_p(\gamma) \neq 0$, take its full ambient inverse image. A sufficiently dilute general tuple of actual members, divided by p times the tuple length, gives $H \sim_{\mathbb{Q}} L$ not containing W . At its own positive threshold t from the whole B , let z be the new deficit. Apply Theorem 3.8, [YZ20, Definition 2.5 and Lemma 2.10], with smooth klt background $(X, 0)$, old boundary B , and addition tH . Lemma 3.13 identifies the deficit conventions. It gives

$$(12.5) \quad z \leq b - \gamma t.$$

Every source used below has its own threshold from this same B . On a principalization of each full system, general moving members are generically reduced and have no common prime. Thus, before choosing the threshold or its center, dilution makes every nonfixed prime coefficient less than any prescribed positive constant. A fixed prime retains its exact full minimum order divided by p .

All source counts below concern every sufficiently large original integer p . At an interval $[\lambda, \theta]$ we use precisely $h = \lceil \lambda p \rceil$ and $l = \lceil \theta p \rceil$. Replacing either endpoint by its unrounded value changes a shifted fourth power by at most $4(l+1)^3$, uniformly in every nonnegative shift. The finitely many rounding terms therefore belong to the fixed $O((p+1)^3)$ errors. Degrees divisible by 100 or 500 give the exact integral exponents in the respective meshes, but no proper subsystem is substituted for the original complete source.

Lemma 12.4. *Let $n \geq 7$, let X be smooth complex projective of dimension n , and let L be ample Cartier. Let $B \sim_{\mathbb{Q}} sL$, $s \geq 0$, be effective rational and lc near x , with actual minimal center W of state $(4, 6, 3)$. Assume*

$$\frac{27}{25}s + b \leq n.$$

If $b \leq 2$, or if the full projective tangent of W is integral, then the entire B has an additive point continuation of coefficient less than $n + 1 - 1/2000$.

Here the affine price suffices because integrality supplies one symbolic shift with the full prime multiplicity.

Proof. We first price every possible changed center. If the first source does not pay, its actual fixed prime supplies an interval estimate that earns a source whose every outcome pays.

Step 1. In this step, we compute the complete successor prices at both source orders. Put $\lambda = 3/4$ and $\theta = 41/50$. The first source is nonzero for all sufficiently large original p , since $1 - 3\lambda^4 = 13/256 > 0$ in (12.4). For $\gamma \in \{\lambda, \theta\}$ and a possible successor v , (12.5) bounds the whole scalar coefficient by

$$(12.6) \quad \frac{25}{27}(n - b) + \frac{b}{\gamma} + J_v(1/\gamma, \min\{b, C_v\}).$$

Since $1/\gamma > 25/27$ and J_v is nondecreasing in its second argument, replacing b by an upper bound b_* gives a bound over the whole domain $0 < b \leq b_*$.

The next table gives the ceilings of 10^4 times these exact prices at $n = 7$, for the three indicated pairs (γ, b_*) . Each entry is specified by (12.1) and (12.6); in particular every root interval is

intersected with its intrinsic cap through C_v .

v	$(3/4, 2)$	$(3/4, 8/3)$	$(41/50, 8/3)$
(1, 1, 1)	72963	75680	72644
(2, 2, 1)	72963	75680	72644
(2, 3, 2)	73780	76496	74599
(2, 4, 3)	75628	78344	76068
(2, 5, 4)	76297	79013	76547
(2, 6, 5)	76578	79294	76714
(3, 3, 1)	73780	76496	74599
(3, 4, 2)	74962	77678	75819
(3, 4, 3)	74060	76776	74879
(3, 5, 3)	76725	79441	78302
(3, 5, 4)	75873	78589	76692
(3, 5, 5)	75223	77940	75587
(3, 5, 6)	74579	77295	74639
(3, 6, 4)	78419	81135	79617
(3, 6, 5)	77310	80026	78129
(3, 6, 6)	76195	78911	76634
(3, 6, 7)	74896	77612	74956

The direct point is bounded by the curve row. A surface contributes $(q_v - 1/\gamma)_+ \min\{b_*, C_v\}$; a threefold adds the largest of the permitted surface surpluses with the prescribed child caps. Thus the table contains every finite comparison, including all skips. The exact column maxima are

$$(12.8) \quad \frac{21173}{2700}, \quad \frac{65719}{8100}, \quad \frac{2644079}{332100}.$$

Increasing n adds $25(n - 7)/27$ to each price, whereas the available coefficient increases by $n - 7$.

Reserve total weighted cutting error less than $1/10000$. At each of at most four cuts choose the two published errors with common parameter less than $1/(40000(1 + r_i))$, after the actual radius r_i is known. The first and last maxima are less than $7999/1000$. They therefore give coefficient less than $n + 1 - 1/2000$ after all errors. The first column proves the assertion for $b \leq 2$. At $b > 2$, only the two middle rows $(3, 6, 4)$ and $(3, 6, 5)$ can remain unpaid. Their complete tails satisfy

$$(12.9) \quad \begin{aligned} \hat{\mathcal{F}}_{(3,6,4)}(z) &\leq \frac{397}{250}z + \frac{103}{500}, \quad 0 \leq z \leq 4/3, \\ \hat{\mathcal{F}}_{(3,6,5)}(z) &\leq \frac{171}{100}z + \frac{29}{500}, \quad 0 \leq z \leq 1. \end{aligned}$$

The first bound was checked in (12.3). In the second, the quartic-surface cap is $1/5$, giving $(2 - 171/100)/5 = 29/500$; the quintic-surface cap is zero and the remaining descendants give smaller surpluses.

Step 2. In this step, we force a full fixed prime from failure of the first threshold. Assume that the endpoint fails, so $b > 2$, and assume the full projective tangent integral. The full tangent ring is Cohen–Macaulay with Hilbert series $(1 + 2\zeta)/(1 - \zeta)^4$. It has no irrelevant torsion, so integrality of its projectivization makes the entire tangent ring integral. The strict-deficit clause of Theorem 4.20 and the last clause of Theorem 8.9 supply all the ordinary blowup hypotheses for the latter theorem.

The own threshold of $F_p(\lambda)$ must select an actual prime threefold P of multiplicity $k = 4$ or 5 and embedding dimension six. Let α be its coefficient in the original direct discriminant, zero if absent, and let η be its coefficient in $H|_W$. The retained budget and the Cartier-addition identity give

$$(12.10) \quad 3b + k\alpha \leq 8, \quad \alpha + t\eta = 1.$$

Indeed all other old coefficients and moduli remain in Proposition 4.18. The coefficient detection in Theorem 4.1, [FH23, Theorem 1.2 and Definition 1.3], makes the actual divisorial lc subcenter's discriminant coefficient one.

Set $(q_4, p_4) = (397/250, 103/500)$ and $(q_5, p_5) = (171/100, 29/500)$. Failure implies $s + t + q_k z + p_k \geq n + 1 - 1/1000$, since a smaller price pays with the fixed error reserve. Define

$$A_k = q_k - 25/27, \quad D_k = 8 - 1/1000 - 175/27 - p_k.$$

Using (12.5), the original price, and $n \geq 7$, we obtain

$$(12.11) \quad 0 < (q_k \lambda - 1)t \leq A_k b - D_k.$$

The omitted dimension term is $-2(n-7)/27 \leq 0$. The constants are

k	A_k	D_k
4	4469/6750	35411/27000
5	2117/2700	39407/27000

and $D_k > ((8-k)/3)A_k > 0$. Hence $b > (8-k)/3$, and (12.10) gives

$$\eta \geq \frac{3}{k} \frac{(b - (8-k)/3)(q_k \lambda - 1)}{A_k b - D_k}.$$

The derivative of $(b - (8-k)/3)/(A_k b - D_k)$ has the sign $((8-k)/3)A_k - D_k < 0$. Its minimum on the allowed interval is therefore at $b = 8/3$. Consequently

$$\eta \geq \begin{cases} 15471/36775, & k = 4, \\ 45765/102278, & k = 5, \end{cases} \quad \eta > 21/50.$$

Choose the earlier dilution so that every nonfixed prime has coefficient less than $21/50$. The selected prime is then a full fixed prime, with exact integer order

$$r = \min_{0 \neq f \in F_p(\lambda)} \text{ord}_P(f), \quad \eta = r/p > 21/50.$$

The prime and its order may depend on p .

Step 3. In this step, we use the complete symbolic interval to obtain the paid final source. At this same degree, the full point-jet map induces an injection

$$F_p(\lambda)/F_p(\theta) \hookrightarrow \frac{P^{(r)} \cap \mathfrak{n}^{\lceil \lambda p \rceil}}{P^{(r)} \cap \mathfrak{n}^{\lceil \theta p \rceil}}, \quad \mathfrak{n} = \mathfrak{m}_{W,x}.$$

Theorem 8.9, or equivalently Proposition 8.10, gives the one-component shift $rk/(3p) > 14/25$. The function $(\theta - u)_+^4 - (\lambda - u)_+^4$ is nonincreasing in $u \geq 0$, as follows by integrating $4(t - u)_+^3$ from λ to θ . The normalized loss is therefore at most

$$3[(41/50 - 14/25)^4 - (3/4 - 14/25)^4] = \frac{195993}{20000000}.$$

Subtract this from the actual complete-source lower mass:

$$(12.12) \quad \dim F_p(41/50) \geq \frac{p^4}{24} \frac{51227}{1250000} - K_1(p+1)^3, \quad \frac{13}{256} - \frac{195993}{20000000} = \frac{51227}{1250000} > \frac{1}{25}.$$

The interval and rounding errors are uniform in the selected prime and order. Fix a common intrinsic cutoff absorbing these errors and the initial error, together with the separate ordinary lifting cutoff, before choosing p or any threshold. This proves nonvanishing for every sufficiently large original p .

The own threshold of the full inverse image of $F_p(41/50)$, taken again from B , has every outcome paid by the last column of (12.7). Its maximum and the fixed error reserve give the asserted continuation, a contradiction. The successful chain retains all of B ; the unsuccessful trial is a separate alternative and is not added to its cost. $\square$

Theorem 12.5. *For $n \geq 7$, let $B \sim_{\mathbb{Q}} sL$, $s \geq 0$, be effective rational and lc near x , with actual minimal center of state $(4, 6, 3)$. If the multiplicity of the whole boundary satisfies*

$$\text{mult}_x B \geq (27/25)s,$$

then the entire B has an additive point continuation of coefficient less than $n + 1 - 1/2000$. No tangent integrality or branch-balance assumption is imposed.

The actual multiplicity supplies an additional nonnegative slack in the coefficient bound. That slack controls the whole reducible tangent before any distribution of a prime among its branches is used.

Proof. We first constrain the full tangent by the actual point slack. We then earn two complete intrinsic sources at one degree. The second source selects a center whose reduced tangent supplies the component degrees needed for the final interval estimate. Write W for the given center, $b = \text{def}_x(B)$, $M = \text{mult}_x B$, and $\sigma = n - M - b$. Set

$$\lambda = 3/4, \quad \theta_0 = 381/500, \quad \theta_1 = 39/50.$$

Step 1. In this step, we prove that failure forces $b > 19/8$ and point slack below $1/8$. The ambient point blowup gives $M + b \leq n$, hence

$$(12.13) \quad \sigma \geq 0, \quad s \leq \frac{25}{27}(n - b - \sigma).$$

Lemma 12.4 pays $b \leq 2$ and every integral full tangent. Assume that the asserted continuation does not exist. Thus $b > 2$ and the full tangent is not integral. The minimal-multiplicity and Hilbert-budget statements give

$$\text{Hilb}(\text{gr}_{\mathbf{n}} \mathcal{O}_{W,x}; \zeta) = \frac{1 + 2\zeta}{(1 - \zeta)^4}, \quad \bar{e}_1 = 2, \quad b \leq 8/3, \quad \mathbf{n} = \mathbf{m}_{W,x}.$$

The original degree and lifting cutoffs are as above. For any earned source of order γ , every possible first successor v has entire scalar price at most

$$(12.14) \quad \frac{25}{27}(n - b - \sigma) + \frac{b}{\gamma} + J_v(1/\gamma, \min\{b, C_v\}).$$

We retain total weighted cutting error less than $1/10000$, with the per-cut choice made in the preceding proof.

At $\gamma = \lambda$, all the surplus terms in this formula are constant for $b \geq 2$. Every nonsmooth root has cap at most two; for the smooth roots the radius is one, and the positive child surpluses have already reached their full caps. The table (12.7) consequently gives, at $n = 7$,

$$\begin{aligned} b \leq 19/8 : \quad & \frac{21173}{2700} + \frac{11}{27} \frac{3}{8} = \frac{43171}{5400} < \frac{7999}{1000}, \\ \sigma \geq 1/8 : \quad & \frac{65719}{8100} - \frac{25}{216} = \frac{129563}{16200} < \frac{7999}{1000}. \end{aligned}$$

The source at λ exists by (12.4), independently of reducedness. Both ranges therefore pay, also for $n > 7$ by the favorable $25/27$ slope. Failure forces

$$(12.15) \quad b > 19/8, \quad 0 \leq \sigma < 1/8.$$

Step 2. In this step, we classify the full reducible tangent and exclude a common plane of three components. Let $\pi : X_1 = \text{Bl}_x X \rightarrow X$ have exceptional divisor E_X , let B_{str} be the effective strict transform, and let Y be the strict transform of W . For every valuation ν mapping to x and every rational $0 \leq a < 1 - \sigma$, the full discrepancy identity gives

$$A_{X_1, B_{\text{str}} + aE_X}(\nu) = A_{X, B}(\nu) - (b + \sigma - 1 + a)\nu(\mathbf{m}_x) \geq (1 - \sigma - a)\nu(\mathbf{m}_x) > 0.$$

All other discrepancies are unchanged near the fiber. These pairs are lc, and Y is minimal at every point above x : a smaller lc center contained in E_X contradicts the strict inequality,

while any other smaller center maps properly to an old lc center through x , contradicting the minimality of W . Center intersection, as in Theorem 3.11, [GL24, Lemma 2.7], also excludes a crossing center through such a point.

The ordinary Rees quotient identifies Y with the actual point blowup of W near the entire fiber. By Theorem 4.20 it is normal with rational singularities there, and the entire exceptional scheme $E = \text{Proj}(\text{gr}_n \mathcal{O}_{W,x})$ and tangent ring are reduced and Cohen–Macaulay. The full tangent has degree three and spans $\mathbb{P}^5$. Its homogeneous ideal is 2-regular. Theorem 7.10, [EGHP04, Theorem 0.4 and Proposition 2.4], shows that its components are linearly joined varieties of minimal degree. Since the integral case is paid, the alternatives are an integral quadric threefold and a linear $\mathbb{P}^3$, meeting in a reduced plane, or three linear $\mathbb{P}^3$'s. In the latter case the plane-incidence graph is connected by the linearly joined description and Cohen–Macaulayness.

We exclude a common plane of three components before using a nodal calculation. If such a plane exists, its generic transverse curve is the union of three coordinate axes: the three transverse directions are independent because E spans $\mathbb{P}^5$. Choose $7/8 < a < 1 - \sigma$. Take two general very ample sections of the smooth X_1 , transverse to that plane and to the strata on fixed log resolutions of (B_{str}, E_X) . At a resulting general point o , their ambient intersection V is smooth and their intersection T with Y is an actual minimal surface center of $(V, B_{\text{str}}|_V)$. Restricting the simple normal crossing crepant boundaries verifies lc after adding $aE_X|_V$, with unchanged coefficients and no smaller coefficient-one stratum. Theorem 4.13 makes T normal and Cohen–Macaulay.

Let $S_o = \mathcal{O}_{T,o}$, and let ξ be the Cartier equation of $E_X|_T$. Its completed quotient is

$$\widehat{S_o/(\xi)} = \mathbb{C}[[z_1, z_2, z_3]]/(z_1 z_2, z_1 z_3, z_2 z_3).$$

A general linear parameter has colength three. For example, quotienting by $z_1 + z_2 + z_3$ gives $\mathbb{C}[[z_1, z_2]]/(z_1^2, z_1 z_2, z_2^2)$, with two-dimensional socle. Thus $S_o/(\xi)$ is not Gorenstein. Its embedding dimension is three, so that of S_o is three or four. The former would make the Cohen–Macaulay surface a hypersurface, and its nonzerodivisor quotient Gorenstein. Hence $\text{edim } S_o = 4$. If ℓ is the above parameter, the multiplicity inequalities give

$$3 = \text{edim } S_o - 2 + 1 \leq \text{mult } S_o \leq \text{length } S_o/(\xi, \ell) = 3.$$

This is a minimal-multiplicity surface. Its actual point deficit b_o is at least a , since $\nu(\xi) \geq \nu(\mathfrak{m}_o)$ for every valuation centered at o . Theorems 4.20 and 4.16 give $\bar{e}_1(\mathfrak{m}_{T,o}) = 2$ and $4 \leq 3(2 - b_o)$, so $b_o \leq 2/3$, a contradiction.

The three-plane graph is therefore a path. Indeed a third plane edge with two different intersection planes inside one component would put that component in the $\mathbb{P}^4$ spanned by the other two, contradicting the total span $\mathbb{P}^5$. Coincident intersection planes are the common plane just excluded. Intersections of smaller dimension remain in the schemes.

Step 3. In this step, we derive the whole-boundary component inequalities and the geometric deficit cap. At a generic conductor plane there are exactly two smooth branches. Their transverse curve is a split node. To apply the flat-family statement, take two general transverse sections at a general closed point of the plane. The preceding surface construction gives a normal minimal surface center there. Its exceptional equation ξ defines a morphism to the affine line, locally of finite type, with the split node as its fiber at that point. The local ring is torsion-free over the discrete valuation ring of the affine line at zero, hence flat. Its residue field is $\mathbb{C}$. Theorem 8.1, [Sta26, Lemma 53.20.11, Tag 0CBX], gives the completed transverse local model

$$(12.16) \quad u_1 u_2 = \xi^N, \quad N \geq 1, \quad E = (\xi = 0).$$

Indeed its deformation parameter is a series in ξ ; normality excludes the zero series, and its leading unit is absorbed in one coordinate. We use this completed model only to compute generic intersection lengths and residues. Completion preserves those lengths, and residue-field degrees are retained on descent.

All needed rational Cartier restrictions exist in the original local rings. They are two-dimensional normal local rings with rational singularities, so Theorem 8.5, [Lip69, Proposition 17.1], gives finite class groups. For any fixed finite collection of Weil divisors their failure to be rational Cartier is therefore confined to subsets of dimension at most one in E . This earns the restrictions; no analytic identification is inferred from the completed model.

Write E_i for the components, $m_i = \deg E_i$, and t_i for the number of conductor planes on E_i . Intersect every component with the same two general hyperplanes of $\mathbb{P}^5$. Their curves Γ_i avoid the preceding bad sets and are smooth rational complete curves: the components are normal arithmetically Cohen–Macaulay of minimal degree by Theorem 8.9, and their general curve sections have Hilbert polynomial $m_i j + 1$. The model (12.16) gives

$$(12.17) \quad w_{ij} = E_j \cdot \Gamma_i = E_i \cdot \Gamma_j = 1/N_{ij} \leq 1 \quad \text{on an edge,} \quad E_i \cdot \Gamma_i = -m_i - \sum_{j \neq i} w_{ij}.$$

For other pairs put $w_{ij} = 0$. Resolving such a node retains the chain of $N_{ij} - 1$ internal vertices of mass zero. Eliminating its equations by the Schur complement in Proposition 8.11 gives exactly the edge weight $1/N_{ij}$; these internal vertices have not been discarded. The residue formula on each branch has conductor coefficient one, independently of N_{ij} . Since $K_{E_i} \cdot \Gamma_i = -2 - 2m_i$, adjunction gives

$$(12.18) \quad K_Y \cdot \Gamma_i = -m_i - 2 + t_i.$$

Apply Theorem 4.1, [FH23, Theorem 1.2 and Definition 1.3], to the effective strict pair. Retaining its full discriminant and compatible moduli, we have, near these complete curves,

$$(12.19) \quad K_Y + \sum_i \gamma_i E_i + D_0 + \mathcal{M}_Y \sim_{\mathbb{Q}} f^*((K_X + B)|_W) + (b + \sigma - 1)E, \quad 0 \leq \gamma_i \leq \sigma.$$

Here $f : Y \rightarrow W$ is the point blowup and $D_0 \geq 0$ has no exceptional component. The bounds on γ_i are the strict-pair bounds in Theorem 4.20. No old horizontal coefficient or moduli contribution is removed.

The moduli term has nonnegative degree on these curves for a specific reason. On a higher smooth model $h : V' \rightarrow Y$ its prescribed trace $\mathcal{M}_{V'}$ is potentially nef. Where $\mathcal{M}_Y$ is rational Cartier, let $N' = h^*\mathcal{M}_Y - \mathcal{M}_{V'}$. Its pushforward is zero and $-N'$ is relatively nef, so Theorem 4.11, [Bir12, Lemma 4.15], gives $N' \geq 0$. Generality keeps the strict transforms of Γ_i out of N' . Hence $\mathcal{M}_Y \cdot \Gamma_i \geq 0$; likewise $D_0 \cdot \Gamma_i \geq 0$. This uses the higher potentially nef data and the earned local Cartierness, not nefness of the lower trace.

Put $\beta = b - 2$ and $a_i = \sigma - \gamma_i \in [0, \sigma]$. Intersecting (12.19) with Γ_i , and using (12.17)–(12.18), gives the full component inequalities

$$(12.20) \quad \beta m_i + a_i \left(m_i + \sum_j w_{ij} \right) - \sum_j a_j w_{ij} \leq 2 - t_i.$$

For the middle component of a three-plane path this forces $\beta \leq \sigma(w_{\text{left}} + w_{\text{right}}) \leq 2\sigma$, whereas (12.15) gives $\beta > 3/8 > 2\sigma$. Thus this case is excluded. The only remaining tangent is $E = Q \cup \Pi$, with Q an integral quadric threefold and $\Pi = \mathbb{P}^3$, meeting along a reduced plane. Its quadric component satisfies $2\beta \leq 1 - a_Q(2 + w) + a_{\Pi}w \leq 1 + \sigma$, so

$$(12.21) \quad b \leq 5/2 + \sigma/2.$$

Step 4. In this step, we compute every changed-center price after imposing the geometric cap. The right side of (12.14) is increasing in b . At the bound (12.21), all its surplus terms are constant in b . The remaining coefficient of σ is

$$\frac{1}{2} \left(\frac{1}{\gamma} - \frac{25}{27} \right) - \frac{25}{27} < 0 \quad \text{for } \gamma \in \{\lambda, \theta_0, \theta_1\}.$$

Thus the worst price is at $n = 7, \sigma = 0, b = 5/2$. The ceilings of 10^4 times the exact complete prices

$$\frac{9}{2} \frac{25}{27} + \frac{5}{2\gamma} + J_v(1/\gamma, \min\{5/2, C_v\})$$

are the following, over the entire state set of Definition 12.1 with positive first cap:

(12.22)

v	λ	θ_0	θ_1
(1, 1, 1)	75000	74476	73718
(2, 2, 1)	75000	74476	73718
(2, 3, 2)	75817	75502	75048
(2, 4, 3)	77665	77280	76725
(2, 5, 4)	78334	77914	77308
(2, 6, 5)	78615	78174	77538
(3, 3, 1)	75817	75502	75048
(3, 4, 2)	76999	76579	75973
(3, 4, 3)	76097	75782	75328
(3, 5, 3)	78762	78587	78334
(3, 5, 4)	77910	77595	77140
(3, 5, 5)	77260	76862	76286
(3, 5, 6)	76616	76161	75505
(3, 6, 4)	80456	80211	79858
(3, 6, 5)	79347	79032	78578
(3, 6, 6)	78232	77847	77291
(3, 6, 7)	76933	76478	75822

The direct point is bounded by the curve row. The same finite surface and threefold comparisons in (12.1) specify every entry, and give the exact maxima

(12.23)

$$\frac{7241}{900}, \quad \frac{916807}{114300}, \quad \frac{93433}{11700}.$$

Only (3, 6, 4) is unpaid at λ or θ_0 . For that row at θ_0 , restricting its new deficit to $z \leq 5/4$ and using the complete tail (12.3) gives

(12.24)

$$\frac{9}{2} \frac{25}{27} + \frac{5}{2\theta_0} + \left(\frac{397}{250} - \frac{1}{\theta_0} \right) \frac{5}{4} + \frac{103}{500} = \frac{1015757}{127000} < \frac{7999}{1000}.$$

The last maximum in (12.23) is also less than $7999/1000$. Thus every θ_1 outcome pays, whereas a failed θ_0 trial must select state (3, 6, 4) with deficit greater than $5/4$. These comparisons hold over all b, σ, n in their allowed ranges, by the monotonicities just proved.

Step 5. In this step, we obtain the complete source at θ_0 from the first symbolic interval. Fix the finite set of intrinsic Hilbert and interval constants for the orders $\lambda, \theta_0, \theta_1$, together with their rounding constants. They are uniform in all later primes and symbolic orders by Theorem 8.9. The positive rational masses computed below fix one intrinsic cutoff; independently fix the ordinary ambient lifting cutoff. Choose any original p above both. Every space in what follows is the complete original F_p at this same degree.

The source at λ has leading mass at least $13/256$. Take its own sufficiently dilute lifted threshold from B . The table forces an actual prime P of state (3, 6, 4). Let α and η be its old discriminant and current restricted coefficients. Equations (12.10) and (12.5) give

$$3b + 4\alpha \leq 8, \quad \alpha + t\eta = 1, \quad t \leq b/\lambda.$$

Consequently

$$4\eta \geq \lambda \frac{3b - 4}{b} = \frac{9}{4} - \frac{3}{b} > \frac{75}{76},$$

using $b > 19/8$. Choose the tuple before its threshold so that every nonfixed coefficient is less than $1/5$. Since $\eta > 75/304 > 1/5$, this is an actual full fixed prime, with $\eta = r/p$, where r is the exact minimum order of $F_p(\lambda)$ along P .

The original finite jet map injects $F_p(\lambda)/F_p(\theta_0)$ into the entire symbolic interval of $P^{(r)}$ between the corresponding ceiling orders. Use the worst-concentration clause of Theorem 8.9, with $d = 4, m = 3, k = 4$. Since $4r/p > 75/76 > \theta_0$, its normalized leading loss is at most $2(\theta_0^4 - \lambda^4)$. Thus

$$(12.25) \quad \begin{aligned} \dim F_p(\theta_0) &\geq c_0 \frac{p^4}{24} - K_2(p+1)^3, \\ c_0 &= 1 - \lambda^4 - 2\theta_0^4 = \frac{581177533}{6250000000} > \frac{9}{1000}. \end{aligned}$$

This estimate permits any rational distribution of the first prime's degree, including zero degree on one component. It retains all original shells, symbolic conditions and conductor restrictions.

Step 6. In this step, we obtain the component restriction degrees from the actual successor's full reduced tangent. Take the own threshold of the full ambient inverse image of this earned $F_p(\theta_0)$, again from B . Write H' for its average and t' for its threshold. By (12.24), its unpaid center P' has state $(3, 6, 4)$ and deficit $z > 5/4$. It need not equal P . The minimal-multiplicity clause of Theorem 4.20, applied to this actual center, makes its full tangent reduced and Cohen–Macaulay: the nonzero exceptional excess would require $z \leq 3 - 2 + 1/4 = 5/4$. Its Hilbert series is $(1 + 3\zeta)/(1 - \zeta)^3$.

Put $R = \mathcal{O}_{W,x}$ and

$$Z = \text{Proj}(\text{gr}_{\mathfrak{n}(R/P')}(R/P')).$$

The graded quotient embeds this entire scheme in $E = Q \cup \Pi$. It is a reduced pure surface of degree four and spans $\mathbb{P}^5$, because both local embedding dimensions are six. The degreewise ordinary Rees quotient shows that Z is the whole exceptional scheme of the strict transform of P' .

Let k_Q, k_Π be the full rational restriction degrees of that strict transform on Q, Π , as in Theorem 8.9. We need the cycle coefficients, including those on the conductor. At a generic surface of Z , let $\mathcal{A}$ be its two-dimensional normal local ring on Y , with exceptional equation ξ . Choose an actual local equation f_0 of a positive multiple a_0 of the strict prime, using the finite local class group. Multiplicity associativity in the Cohen–Macaulay ring gives

$$a_0 \text{length}(\mathcal{A}/(I_{\tilde{P}'}, \xi)) = \text{length}(\mathcal{A}/(f_0, \xi)) = \sum_i \text{ord}(f_0|_{E_i}).$$

Dividing by a_0 says that the sum of the actual branch restriction coefficients is the full coefficient of Z . That coefficient is one, since Z is reduced. Each individual coefficient is therefore between zero and one, also at the conductor. In particular $k_Q + k_\Pi = 4$.

The scheme Z is not contained in Q , whose span is $\mathbb{P}^4$. It therefore has a surface component in Π outside Q ; its Π -restriction coefficient is one. This proves $k_\Pi \geq 1$. For the upper bound, the Cohen–Macaulay Hilbert series of Z gives a 2-regular homogeneous ideal, generated by quadrics. One can see the generation directly by reducing by three regular linear parameters: the quotient has Hilbert function $(1, 3)$, and lifting its quadratic relations along the regular sequence generates the homogeneous ideal. Choose a general line in Π meeting Z in a finite scheme. Some defining quadric is nonzero on this line, so that scheme has length at most two. The line meets every surface component of Z contained in Π in its degree, with disjoint general intersections. All contributing restriction coefficients are at most one. Hence

$$(12.26) \quad 1 \leq k_\Pi \leq 2, \quad k_Q = 4 - k_\Pi \geq 2, \quad \frac{k_\Pi}{m_\Pi} \geq 1, \quad \frac{k_Q}{m_Q} \geq 1.$$

This argument uses the actual successor's full reduced scheme, including its conductor cycle; it asserts no balance for an arbitrary prime on E .

Let α', η' be the old discriminant and restricted H' -coefficients of P' . The same full identities give

$$3b + 4\alpha' \leq 8, \quad \alpha' + t'\eta' = 1, \quad t' < \frac{b - 5/4}{\theta_0}.$$

Therefore

$$\eta' > \theta_0 \frac{3b - 4}{4(b - 5/4)} > \theta_0 \frac{25}{36} = \frac{127}{240} > \frac{1}{2}.$$

Here $(3b - 4)/(b - 5/4) = 3 - (1/4)/(b - 5/4)$ is increasing, and its value at $19/8$ is $25/9$. The preselected dilution again makes P' a full fixed prime. If r' is its exact minimum order in $F_p(\theta_0)$, then $r'/p = \eta'$. By (12.26), both actual component shifts satisfy $r'k_i/(m_i p) > 1/2$.

Step 7. In this step, we obtain the complete source at θ_1 and pay its own threshold. Set $h = \lceil \theta_0 p \rceil$ and $l = \lceil \theta_1 p \rceil$. After a local trivialization of pA , taking germs gives an injection

$$\frac{F_p(\theta_0)}{F_p(\theta_1)} \hookrightarrow \frac{P'(r') \cap \mathfrak{n}^h}{P'(r') \cap \mathfrak{n}^l}.$$

Indeed, all source germs lie in the numerator, and the kernel before quotienting is precisely the complete source imposing ordinary order at least l , namely $F_p(\theta_1)$. Thus $\dim F_p(\theta_0) - \dim F_p(\theta_1)$ is at most the length of this entire symbolic interval. Apply Theorem 8.9 to that length. Since every component shift is greater than $p/2$ and $\sum_i m_i = 3$, monotonicity in the shift bounds it by

$$\frac{p^4}{24} 3[(\theta_1 - 1/2)^4 - (\theta_0 - 1/2)^4] + K(p+1)^3 = \frac{p^4}{24} \frac{268980237}{62500000000} + K(p+1)^3.$$

Here K includes the endpoint-rounding errors and is uniform in the actual prime and symbolic order. Subtracting this bound from the already earned dimension in (12.25) gives

$$(12.27) \quad \dim F_p(\theta_1) \geq c_1 \frac{p^4}{24} - K_3(p+1)^3, \\ c_1 = c_0 - \frac{268980237}{62500000000} = \frac{19512331}{3906250000} > \frac{1}{250}.$$

Both interval errors, all endpoint roundings, and the initial Hilbert error are uniform in P, P', r, r' and their rational restriction indices. Choose the common intrinsic cutoff so their sum is smaller than $p^4/(24 \cdot 500)$ for every p above it. The displayed positive masses then establish each next source before its own threshold is taken. The separate ordinary lifting cutoff was also fixed before choosing p . Thus the argument applies to every sufficiently large original degree; no normalization of the original projective W has supplied extra sections.

Take the own threshold of the full ambient inverse image of $F_p(\theta_1)$, from the entire original B . Every changed-center case is paid by (12.22). Its exact maximum is $93433/11700 < 7999/1000$, with favorable dimension slope $25/27$. After both published cutting errors, whose total weighted contribution is less than $1/10000$, the coefficient is less than $n + 1 - 1/2000$. This contradicts failure. The unsuccessful threshold trials were separate alternatives; the successful additive chain keeps all original coefficients, discriminant and compatible moduli data. Every original fixed, residual, embedded, symbolic and remote condition remained in the complete source spaces throughout. $\square$

In either endpoint the final boundary has a point minimal center and positive remaining adjoint coefficient. Proposition 3.12 therefore gives generation of $\mathcal{O}_X(K_X + (n+1)L)$ at x .

13. TRIPLE-FIVEFOLD SOURCE FEEDBACK

We retain the original sources and the whole-boundary deficit convention of Section 2. We treat the cubic hypersurface tangent in full. For an integral tangent, successive symbolic intervals either reach a source with a complete continuation of cost below eight or produce a quartic threefold paid by the affine endpoint. For a reducible tangent, two intrinsic sources determine one fixed prime, whose convex order profile gives the contradiction.

Theorem 13.1. *Let $L^7 = 1$. There is a cutoff, fixed before the source degree, general tuple and threshold center, such that every nonzero complete $U_k(27/25)$ with $25 \mid k$ above that cutoff has this property: if a pure average at its actual full-ideal threshold has minimal center of state $(5, 6, 3)$, then $K_X + 8L$ is generated at x . All original degrees, earlier histories, analytic presentations and full tangent schemes are allowed.*

The conclusion concerns the original ambient source. The intrinsic sources used in its proof remain complete on the particular center selected by that source.

13.1. The parent and its complete scalar bounds.

Set-up 13.2. Assume $L^7 = 1$ and that $K_X + 8L$ is not generated at x . Fix the cutoff in Theorem 10.1 before choosing k , and let $B \sim_{\mathbb{Q}} sL$ be a sufficiently dilute pure average at the full-ideal threshold of a nonzero $U_k(27/25)$, with $25 \mid k$ above that cutoff. Suppose its minimal center is W of state $(5, 6, 3)$. Write

$$A = L|_W, \quad R = \mathcal{O}_{W,x}, \quad \mathbf{n} = \mathbf{m}_{W,x}, \quad G = \text{gr}_{\mathbf{n}} R, \quad \lambda = \frac{3997}{5000}, \quad \theta = \frac{803}{1000}.$$

We use $F_p(\gamma)$ for the complete original intrinsic space in Definition 2.4. Each new threshold trial starts from the same entire B .

For the scalar calculations, retain every state $v = (d, e, \mu) \in \mathcal{P}_7$ with $d \leq 4$ and $e \leq 6$. The following table specifies a rational upper radius q_v and a function

$$H_v(y) = \min_{(a,c) \text{ in row } v} (ay + c), \quad 0 \leq y \leq \beta_v, \quad \beta_v = d - \frac{2T(e - d, \mu)}{\mu}.$$

The moment column is $T(e - d, \mu)$. The three panels form one table; all degrees in this table have lower bound one. They are the readable view of the canonical dataset 13-triple-fivefold/tails; all thirty rows and all seventy-seven affine pairs are retained.

d	e	μ	T	β	q	Affine pairs defining H_v
1	1	1	0	1	1	(1, 0)
2	2	1	0	2	1	(1, 0)
2	3	2	1	1	$\frac{283}{200}$	$(\frac{283}{200}, 0)$
2	4	3	2	$\frac{2}{3}$	$\frac{1733}{1000}$	$(\frac{1733}{1000}, 0)$
2	5	4	3	$\frac{1}{2}$	2	(2, 0)
2	6	5	4	$\frac{2}{5}$	$\frac{2237}{1000}$	$(\frac{2237}{1000}, 0)$
3	3	1	0	3	1	$(1, \frac{83}{200}); (\frac{5}{4}, \frac{33}{200}); (\frac{3}{2}, 0)$
3	4	2	1	2	$\frac{63}{50}$	$(\frac{63}{50}, \frac{237}{1000}); (\frac{3}{2}, \frac{117}{1000}); (\frac{5}{3}, \frac{17}{500}); (\frac{7}{4}, 0)$
3	4	3	3	1	$\frac{1443}{1000}$	$(\frac{1443}{1000}, 0)$
3	5	3	2	$\frac{5}{3}$	$\frac{1443}{1000}$	$(\frac{5}{4}, \frac{103}{200}); (\frac{1443}{1000}, \frac{97}{500}); (\frac{3}{2}, \frac{167}{1000}); (2, 0)$
d	e	μ	T	β	q	Affine pairs defining H_v
3	5	4	4	1	$\frac{397}{250}$	$(\frac{397}{250}, \frac{37}{1000}); (\frac{5}{3}, \frac{17}{1000}); (\frac{7}{4}, 0)$
3	5	5	6	$\frac{3}{5}$	$\frac{171}{100}$	$(\frac{171}{100}, 0)$
3	5	6	8	$\frac{1}{3}$	$\frac{909}{500}$	$(\frac{909}{500}, 0)$
3	6	4	3	$\frac{3}{2}$	$\frac{397}{250}$	$(\frac{397}{250}, \frac{103}{500}); (\frac{7}{4}, \frac{1}{8}); (2, \frac{3}{50}); (\frac{5}{2}, 0)$
3	6	5	5	1	$\frac{171}{100}$	$(\frac{171}{100}, \frac{29}{500}); (2, 0)$
3	6	6	7	$\frac{2}{3}$	$\frac{909}{500}$	$(\frac{909}{500}, 0)$
3	6	7	9	$\frac{3}{7}$	$\frac{1913}{1000}$	$(\frac{1913}{1000}, 0)$
3	6	8	11	$\frac{1}{4}$	2	(2, 0)
3	6	9	13	$\frac{1}{9}$	$\frac{2081}{1000}$	$(\frac{2081}{1000}, 0)$
4	4	1	0	4	1	$(1, \frac{757}{1000}); (\frac{5}{4}, \frac{161}{500}); (\frac{5}{3}, \frac{9}{200}); (\frac{7}{4}, 0)$

d	e	μ	T	β	q	Affine pairs defining H_v
4	5	2	1	3	$\frac{119}{100}$	$(\frac{119}{100}, \frac{287}{500}); (\frac{5}{4}, \frac{121}{250}); (\frac{3}{2}, \frac{1}{4}); (2, 0)$
4	5	3	3	2	$\frac{1317}{1000}$	$(\frac{1317}{1000}, \frac{171}{500}); (\frac{3}{2}, \frac{1}{4}); (2, 0)$
4	5	4	6	1	$\frac{283}{200}$	$(\frac{283}{200}, \frac{293}{1000}); (\frac{3}{2}, \frac{1}{4}); (2, 0)$
4	6	3	2	$\frac{8}{3}$	$\frac{1317}{1000}$	$(\frac{5}{4}, \frac{373}{500}); (\frac{1317}{1000}, \frac{71}{125}); (\frac{3}{2}, \frac{81}{250}); (\frac{5}{3}, \frac{229}{1000}); (\frac{7}{4}, \frac{39}{200}); (2, \frac{19}{200}); (\frac{5}{2}, 0)$
4	6	4	4	2	$\frac{283}{200}$	$(\frac{283}{200}, \frac{379}{1000}); (\frac{3}{2}, \frac{59}{200}); (2, \frac{19}{200}); (\frac{5}{2}, 0)$
4	6	5	6	$\frac{8}{5}$	$\frac{187}{125}$	$(\frac{5}{4}, \frac{69}{100}); (\frac{187}{125}, \frac{297}{1000}); (\frac{3}{2}, \frac{59}{200}); (2, \frac{19}{200}); (\frac{5}{2}, 0)$
4	6	6	8	$\frac{4}{3}$	$\frac{783}{500}$	$(\frac{5}{4}, \frac{69}{100}); (\frac{783}{500}, \frac{269}{1000}); (\frac{7}{4}, \frac{39}{200}); (2, \frac{19}{200}); (\frac{5}{2}, 0)$
4	6	7	11	$\frac{6}{7}$	$\frac{1627}{1000}$	$(\frac{1627}{1000}, \frac{61}{250}); (2, \frac{19}{200}); (\frac{5}{2}, 0)$
4	6	8	14	$\frac{1}{2}$	$\frac{841}{500}$	$(\frac{841}{500}, \frac{111}{500}); (2, \frac{19}{200}); (\frac{5}{2}, 0)$
4	6	9	17	$\frac{2}{9}$	$\frac{1733}{1000}$	$(\frac{1733}{1000}, \frac{14}{125}); (2, \frac{53}{1000}); (\frac{5}{2}, 0)$

Lemma 13.3. *For every state in this table, $\mathcal{F}_v(y) \leq H_v(y)$ on $[0, \beta_v]$, with degree floors one. For the actual parent of Setup 13.2, its original degree is $g = A^5 = 1$, and*

$$(13.1) \quad \frac{21587}{8642} \leq b \leq 3, \quad M \geq \frac{27}{25}s, \quad M + b \leq 7, \quad \frac{27}{25}s + b \leq 7.$$

Its full tangent is a reduced Cohen–Macaulay cubic hypersurface with Hilbert series $(1 + \zeta + \zeta^2)/(1 - \zeta)^5$, and its ordinary and normal first Hilbert coefficients equal three. Its ordinary point blowup is normal with rational singularities near its entire exceptional scheme, and every ordinary point-ideal power is integrally closed.

The degree-one assertion is earned from the complete descendant costs. It is not a consequence of $L^7 = 1$ alone.

Proof. We first verify the rational scalar bounds and then apply them to the original degree.

Step 1. In this step, we verify the complete rational scalar bounds. For a smaller state $w = (f, h, \nu)$, with $h \leq e$, put

$$c_{vw} = \begin{cases} \min(\beta_v, \beta_w, \beta_v - \nu/\mu), & f = d - 1, \\ \min(\beta_v, \beta_w), & f \leq d - 2. \end{cases}$$

Omit only nonpositive caps. Each printed radius satisfies $q_v^d \geq \mu$. Every printed pair (a, c) satisfies the finite rational inequality

$$(13.2) \quad c \geq (q_v - a)_+ \beta_v + \max \left(0, \max_w \max_{0 \leq y \leq c_{vw}} \{H_w(y) - \max(a, q_v)y\} \right).$$

Here and below $(u)_+ = \max(u, 0)$. This check uses zero, the cap, and every intersection $(c' - c)/(a - a')$ of two distinct-slope lines in the descendant row that lies in the interval. Between these points the expression is affine. Thus the printed lines, this complete rule, and the table specify the check on the entire domain, including every dimension skip. Induction in dimension in Definition 5.2 proves the assertion: subtract ay_0 from $q_v(y_0 - y) + H_w(y)$, and use $y_0 \leq \beta_v$ when $q_v \geq a$, or $y_0 \geq y$ otherwise.

In particular, the multiplicity-eight threefold has moment eleven, cap $1/4$, and adjacent cap $(2 - \nu)/8$. Only a smooth surface can have positive adjacent cap; all direct curve skips have radius one. Its full tail is at most $2y$. The multiplicity-nine threefold has moment thirteen, cap $1/9$, and adjacent cap $(1 - \nu)/9$, so no surface is eligible. Its direct curve skips give the full bound $(2081/1000)y$, in particular $9y/4$, as required by Lemma 5.10. Embedding-seven rows have different caps and cannot occur as descendants here, since the embedding dimension of a quotient of R is at most six.

Step 2. In this step, we collect the parent deficit, multiplicity, tangent, and Rees data. The ledger gives the deficit interval and the tangent and Rees conclusions. The Hilbert numerator gives $e_1 = 3$; equality of ordinary and normal powers gives $\bar{e}_1 = 3$. The pure ambient average has multiplicity at least $27s/25$. The ordinary point blowup of the smooth sevenfold bounds its

point-ideal threshold by $7 - M$. This proves the three separate inequalities in (13.1). The local blowup assertion also follows from Theorem 4.20, since the ledger gives $0 \leq \sigma \leq 4339/25000 < 1$ and $7 - M = b + \sigma > 1$.

Step 3. In this step, we exclude original degree at least two by the complete scalar price. Suppose now that $g \geq 2$. The parent radius is at most $217/200$, since

$$\left(\frac{217}{200}\right)^5 - \frac{3}{2} = \frac{1170140857}{320000000000} > 0.$$

Its actual normal moment gives the adjacent fourfold cap $(9 - \mu)/3$. Applying the endpoint and line-intersection rule above to all thirty rows gives

$$\max_v \max_{0 \leq y \leq c_v} \left(H_v(y) - \frac{217}{200}y \right) \leq \frac{129}{125}, \quad c_v = \begin{cases} \min(3, \beta_v, (9 - \mu)/3), & d = 4, \\ \min(3, \beta_v), & d \leq 3. \end{cases}$$

The largest value is the cubic fourfold at $y = 2$, where $H_{(4,6,3)}(2) = 1601/500$. The fourfold of multiplicity nine has cap zero. Theorem 5.4 therefore bounds the whole coefficient by

$$\frac{100}{27} + 3\frac{217}{200} + \frac{129}{125} = \frac{215749}{27000} = 8 - \frac{251}{27000}.$$

Its strict margin pays both cutting errors and gives generation by Proposition 3.12, a contradiction. As g is a positive integer, $g = 1$. $\square$

We specify how numerical margins below pay approximate cuts. At an actual cut with radius r_i , Theorem 3.14, namely [GL24, Proposition 3.2 and Remark 3.3], retains both

$$t_i \leq b_i r_i + \varepsilon_i, \quad b_{i+1} \leq b_i - t_i / r_i + \varepsilon_i.$$

Once that radius is known, choose the error small enough that the weighted allowance $(1 + r_i)\varepsilon_i$ fits its assigned share. All fixed scalar price exclusions below have gap greater than $1/10000$. Reserve a total less than $1/100000$, divided over the at most five further dimension drops. The strict-budget construction in Theorem 5.4 absorbs these allowances, including changes of the deficit caps. A threshold trial itself is taken exactly; it is not one of these approximate cuts.

13.2. Actual intrinsic sources and their successor primes. For a state $v = (d, e, \mu)$ in the scalar table set

$$C_v = \begin{cases} \min(\beta_v, (9 - \mu)/3), & d = 4, \\ \beta_v, & d \leq 3, \end{cases} \quad \Phi_{\gamma,v}(b, z) = \frac{25(7 - b)}{27} + \frac{b - z}{\gamma} + H_v(z).$$

These functions will be used on $0 \leq z \leq b \leq 3$ with $z \leq C_v$. A point has $z = 0$ and no remaining scalar cost. For a fourfold and a proposed bound $u > 0$, the constrained domain has the additional inequality

$$(13.3) \quad (3\gamma - u)b + uz \leq \gamma(9 - \mu).$$

The finite calculation uses the following common rule. It applies both to the first source and to each later integral-tangent transition.

Lemma 13.4. *Let v be a row of the scalar table and let $\gamma > 0$ and $C \leq \beta_v$ be rational. Let $\mathcal{D}$ be the closed domain*

$$0 \leq z \leq b \leq 3, \quad z \leq C,$$

with or without the additional inequality (13.3), for a specified rational $u > 0$. Let $\mathcal{V}$ consist of all feasible intersections of two boundary lines with independent normals. Adjoin every feasible intersection of a boundary line with a horizontal equality between two distinct-slope affine functions in H_v , and call the resulting set $\mathcal{Q}$. If $\mathcal{D} \neq \emptyset$, then

$$\max_{\mathcal{D}} \Phi_{\gamma,v} = \max_{\mathcal{Q}} \Phi_{\gamma,v}.$$

Moreover, $\mathcal{D} = \emptyset$ if and only if $\mathcal{V} = \emptyset$. These assertions include domains that are segments or points.

The record therefore carries the entire finite set, rather than only one proposed maximizing point.

Proof. Every domain is bounded. At a vertex its active boundary normals span the plane: otherwise a small displacement in both directions orthogonal to those normals would preserve all constraints and contradict extremality. Thus intersections with nonzero determinant find every vertex. A nonempty bounded polyhedron has a vertex, including when its affine dimension is zero or one. This also justifies the emptiness test.

Subdivide by every horizontal equality of distinct-slope tail lines, whether or not those two lines are minimal there. On each closed piece their ordering is fixed, and $\Phi_{\gamma,v}$ is affine. Every vertex of a subdivided piece is an original vertex or the intersection of a boundary with one of these horizontal lines. Two distinct horizontal lines cannot create another vertex. An equality that coincides with an entire edge only retains its endpoints. Equal-slope competing functions have constant ordering; coincident functions cause no additional cut. Hence all extrema lie in $\mathcal{Q}$.

For a segment this argument is the subdivision of that segment at every admissible crossing, with both endpoints included. For a point it is evaluation at that point. Singular line pairs are skipped only in computing intersections; all their original inequalities remain in the feasibility test. Finally, a cap $C < 0$ gives an empty domain, while $C = 0$ gives an empty positive-deficit domain whose closed segment or point is still recorded. No positive-deficit case is removed on the basis of zero area. $\square$

The dataset `13-triple-fivefold/first-source`, described in Section 13.6, supplies all first-source vertices, all subdivided vertices and their exact values. Lemma 13.4 explains why checking those records proves an upper bound on the whole domain.

Lemma 13.5. *Let B, W be as in Setup 13.2. There are separate intrinsic and ordinary ambient restriction cutoffs such that every 5000-divisible p above them has $F_p(\lambda) \neq 0$. Let $\gamma \geq \lambda$ be rational, with $p\gamma$ integral and $F_p(\gamma) \neq 0$. Take a sufficiently dilute general average $H \sim_{\mathbb{Q}} L$ from its full ambient inverse image, and take its own threshold t from the entire B . Then the actual minimal center P of $B + tH$ is a fourfold of embedding dimension six and multiplicity $\mu \in \{3, 4, 5, 6\}$. Its deficit z obeys*

$$0 < z \leq b - \gamma t, \quad z \leq C_{(4,6,\mu)}.$$

If α is the coefficient of P in the old direct discriminant and η its coefficient in $H|_W$, then

$$(13.4) \quad 0 \leq \alpha < 1, \quad \alpha\mu \leq 9 - 3b, \quad \alpha + t\eta = 1.$$

At $\gamma = \lambda$, the prime is fixed in the full original space, and its minimum order r satisfies $\eta = r/p$ and $\mu r/p > 27/25$. At that same degree,

$$(13.5) \quad \dim F_p(\theta) \geq \frac{18145162291861493}{3125000000000000000} \frac{p^5}{120} - O(p^4).$$

The error and the intrinsic cutoff precede all the indicated primes and orders. These assertions apply to integral and reducible full tangents alike.

The later source in (13.5) is the kernel of a map on the first complete source. In particular its positive mass has not been assigned to a proper restriction image.

Proof. We first construct the original source, then exclude the other actual successors, and finally estimate its full symbolic loss.

Step 1. In this step, we obtain the complete intrinsic source and its ordinary ambient lifting. The ample Hilbert polynomial on the original W and its ordinary point Hilbert polynomial give

$$(13.6) \quad \dim F_p(\lambda) \geq \left(1 - 3\lambda^5\right) \frac{p^5}{120} - O(p^4), \quad 1 - 3\lambda^5 = \frac{64502732955140729}{3125000000000000000} > 0.$$

This is the kernel estimate for point evaluation and needs no surjectivity to jets. Fix its intrinsic cutoff. Independently, Theorem 6.8, applied to the fixed coherent ideal $I_W \subset \mathcal{O}_X$ and the ample L , gives ordinary restriction surjectivity for every sufficiently large p from

$$0 \longrightarrow I_W(pL) \longrightarrow \mathcal{O}_X(pL) \longrightarrow \mathcal{O}_W(pA) \longrightarrow 0.$$

Use the full inverse image of $F_p(\gamma)$ under this map. General members have nonzero restriction to W ; its moving part has no fixed divisor after principalizing the full ideal. The threshold is positive because its restriction is nonzero at the generic point of the minimal center, and finite because every member vanishes at x . At the threshold a new lc center meets W ; center intersection places the minimal one properly inside W .

Apply Theorem 3.8, that is [YZ20, Definition 2.5 and Lemma 2.10], with klt background $(X, 0)$, old boundary B , and actual rational Cartier addition tH . Its restriction has order at least γt and does not contain W . Lemma 3.13 identifies the deficit conventions. Thus $z \leq b - \gamma t$. This restricted order is intrinsic; the construction has asserted only ordinary ambient restriction surjectivity.

Step 2. In this step, we classify every actual successor and retain its old discriminant coefficient. The persistent budget in Proposition 4.18 gives the fourfold cap $3z + \mu \leq 9$. Indeed, if the same prime already has coefficient α , its old contribution $\alpha\mu$ and its newly spent contribution $(1 - \alpha)\mu$ add to μ . All other discriminant, ramification and compatible moduli deductions remain nonnegative. A dimension skip keeps its own intrinsic cap. Every whole scalar coefficient is bounded by $\Phi_{\gamma,v} \leq \Phi_{\lambda,v}$. The unrestricted table in Section 13.6 bounds every row other than

$$(3, 6, 4), \quad (4, 6, 3), \quad (4, 6, 4), \quad (4, 6, 5), \quad (4, 6, 6)$$

strictly below $8 - 1/1000$. The point price is at most $100/27 + 3/\lambda < 8$. For $(3, 6, 4)$ and $z \leq 1$, the complete list of endpoints and tail intersections gives

$$\Phi_{\lambda,(3,6,4)}(b, z) \leq \frac{431653343}{53959500} = 8 - \frac{22657}{53959500} < 8 - \frac{1}{3000}.$$

For a quartic threefold with $z > 1$, put $c = s + t$. Then

$$(13.7) \quad c + \frac{z}{\lambda} \leq s + \frac{b}{\lambda} \leq \frac{100}{27} + \frac{3}{\lambda} = 7 + \frac{49267}{107919}.$$

Theorem 11.1 applies to this actual whole $B + tH$, at every positive original degree and for every full quartic tangent. Each excluded row therefore gives generation, with the error allowances specified above. The surviving fourfold has positive deficit. Its old coefficient is less than one because W was minimal. The old budget gives $\alpha\mu \leq 9 - 3b$. Under Cartier addition the discriminant coefficient increases by $t\eta$, with the same moduli data. At the generic point of the new prime it equals one. This proves (13.4); it also follows directly by subtracting $t\eta$ from each generic threshold in Theorem 4.1.

Step 3. In this step, we force a full fixed prime and obtain the next complete source at the original degree. At $\gamma = \lambda$, suppose $\mu\eta \leq u = 27/25$. Combining the deficit drop with (13.4) gives (13.3). If $\mu - 9 + 3b > 0$, use $t \geq (\mu - 9 + 3b)/u$; if the numerator is nonpositive, $z \leq b$ gives the same inequality. The constrained polygon certificate bounds every fourfold price by

$$\frac{430631}{54000} = 8 - \frac{1369}{54000}.$$

This contradicts nongeneration, so $\mu\eta > 27/25$. Before the prime is chosen, dilute the general tuple so every nonfixed prime in $H|_W$ has coefficient below $1/10$: each such coefficient on the moving part is at most $1/(pN)$ for a tuple of size N . Since $\mu \leq 6$, the selected prime is fixed and its coefficient is the exact minimum r/p . All original germs in $F_p(\lambda)$ belong to $P^{(r)}$.

Apply Proposition 8.10 using the normal rational blowup and the full reduced tangent supplied by Lemma 13.3. The complete distribution-free loss between λ and θ is at most $2(\theta^5 - \lambda^5)$

in normalized leading degree, since $\mu r/p > 27/25 > \theta$. The map has kernel literally $F_p(\theta)$. Subtraction from (13.6) gives

$$1 - \lambda^5 - 2\theta^5 = \frac{18145162291861493}{312500000000000000} > \frac{1}{200}.$$

All local errors are uniform over the prime and symbolic order, and the fixed Hilbert error concerns the original W . Their sum is $O(p^4)$. Enlarge the intrinsic cutoff using this positive coefficient, separately from the ordinary restriction cutoff. Every sufficiently large allowed original degree now has the claimed source. Global normalization or deletion of a remote condition has not entered this kernel calculation. $\square$

13.3. The integral tangent.

Lemma 13.6. *The full projective tangent of the parent in Setup 13.2 cannot be integral.*

The argument uses seventeen transitions at one original intrinsic degree. The fixed primes may change at every transition.

Proof. We keep the complete interval estimate on the normalization of the integral tangent, including its conductor, and force a source whose own threshold is paid.

Step 1. In this step, we fix the source levels and the two independent degree cutoffs. List the following pairs of successive levels and proposed multiplicity-weighted fixed orders:

i	γ_i	γ_{i+1}	u_i
1	3997/5000	4/5	2829/2500
2	4/5	161/200	11367/10000
3	161/200	81/100	11793/10000
4	81/100	163/200	611/500
5	163/200	41/50	6323/5000
6	41/50	33/40	817/625
7	33/40	83/100	6749/5000
8	83/100	167/200	557/400
9	167/200	21/25	14351/10000
10	21/25	169/200	14777/10000
11	169/200	17/20	3801/2500
12	17/20	171/200	16083/10000
13	171/200	43/50	1697/1000
14	43/50	87/100	1116/625
15	87/100	22/25	1963/1000
16	22/25	9/10	21403/10000
17	9/10	23/25	24949/10000

Fix the seventeen interval-error constants before any threshold. Together with the initial Hilbert error, they determine an intrinsic cutoff. The ordinary restriction cutoff remains separate. Take any 5000-divisible p above their maximum. All eighteen point orders are integral at this same p .

Step 2. In this step, we propagate the complete sources using the forced full symbolic orders. Suppose that $F_p(\gamma_i) \neq 0$. At its own threshold from B , Lemma 13.5 leaves a fourfold prime P_i . Write its multiplicity as μ_i , its old discriminant coefficient as α_i , and its restricted cutter coefficient as η_i . If $\mu_i \eta_i \leq u_i$, the same calculation as (13.3) gives

$$(3\gamma_i - u_i)b + u_i z_i \leq \gamma_i(9 - \mu_i).$$

The complete fourfold comparisons in the dataset 13-triple-fivefold/integral in Section 13.7 give $\Phi_{\gamma_i, v} < 79999/10000$ on each of these domains. The certificate includes all ten positive-cap fourfold rows, every interval, and the empty multiplicity-nine row. To check an entry, apply Lemma 13.4 to all its constraints and all the tail lines, and compare the largest vertex value with

the stored upper bound and attaining point. The point, curve, surface and threefold alternatives are already controlled by the unrestricted first-source certificate, since $\Phi_{\gamma_i, v} \leq \Phi_{\lambda, v}$. A quartic threefold with deficit greater than one uses exactly (13.7) and Theorem 11.1.

Thus $\mu_i \eta_i > u_i > 1$. Preselected dilution below $1/10$ for every nonfixed prime forces P_i to be fixed. With its exact minimum order r_i , we have $\eta_i = r_i/p$. The full symbolic interval in Theorem 8.9 has one component of degree three, hence shift $r_i \mu_i/3$, even if the integral tangent is not normal. Its normalization does not replace the original global source. The map on $F_p(\gamma_i)$ has kernel exactly $F_p(\gamma_{i+1})$, so its normalized leading loss is at most

$$(13.8) \quad \mathcal{L}_i = 3 \left[\left(\gamma_{i+1} - \frac{u_i}{3} \right)_+^5 - \left(\gamma_i - \frac{u_i}{3} \right)_+^5 \right].$$

The positive-part difference decreases with the shift; thus the strict actual order may be replaced by u_i for this upper bound. Subtract the seventeen losses from the initial mass in (13.6). All successive masses are recorded in 13-triple-fivefold/integral, identified in Section 13.7, and the last is

$$1 - 3\lambda^5 - \sum_{i=1}^{17} \mathcal{L}_i = \frac{51447479205198179}{4500000000000000000} > 0.$$

Every loss is nonnegative, so this positive final value also bounds all preceding masses from below. Choose the already specified intrinsic cutoff so the total normalized $O(p^4)$ error is less than half this value. Induction makes every complete source nonzero at the chosen original p .

Step 3. In this step, we pay every outcome of the terminal source's own threshold. At $\gamma_{18} = 23/25$, apply the same endpoint and line- intersection rule to $\Phi_{23/25, v}$ on every unrestricted domain. Its largest value is

$$\frac{2481721}{310500} < \frac{7997}{1000},$$

attained in row $(4, 6, 3)$ at $(b, z) = (3, 2)$. Equivalently, for every row all endpoint and line-intersection values of $100/27 + 75/23 + H_v(z) - (25/23)z$ are at most this fraction. This formulation, with all the tail lines and caps above, checks the whole final domain. Its actual own threshold and complete tail therefore give a point continuation with the reserved cutting errors. Proposition 3.12 contradicts nongeneration.

Each source trial in this argument begins at B . The interval losses are losses of section dimensions; they are not a sum of the costs of the alternative threshold trials. In particular, any actual quartic jump encountered during the construction is paid with its own entire coefficient $s + t_i$ by (13.7). $\square$

13.4. Full component restrictions for a cubic fourfold.

Lemma 13.7. *Let B, W be as in Setup 13.2, and assume its full tangent is reducible. Let P be any actual fourfold selected at the own threshold of a source as in Lemma 13.5, and write $\mu = \text{mult}_x P$. Its normal first Hilbert coefficient E_P satisfies*

$$E_P = 2 \quad (\mu = 3), \quad T(2, \mu) \leq E_P \leq T(2, \mu) + 1 \quad (\mu = 4, 5, 6).$$

If $\mu = 3$, its full tangent ring is Cohen–Macaulay with Hilbert series $(1 + 2\zeta)/(1 - \zeta)^4$, all ordinary point powers are integrally closed, and its original Rees ring is normal Cohen–Macaulay. Let E_i be the components of the parent's full projective tangent, with degrees m_i , and let $C_{P,i}$ be the full rational restrictions of Theorem 8.9. Then

$$(13.9) \quad (m_i) = (1, 1, 1) \text{ or } (1, 2), \quad 0 \leq \delta_i := \mathcal{O}_{E_i}(1)^3 \cdot C_{P,i} \leq m_i + 1, \quad \sum_i \delta_i = 3.$$

The component caps concern the full rational restriction cycle of the actual prime. The cubic hypersurface parent itself has Hilbert numerator $1 + \zeta + \zeta^2$; no minimal-multiplicity formula is applied to that parent.

Proof. We first earn the normal moment of the successor, and then compare its complete tangent intersection with the original restriction cycle.

Step 1. In this step, we determine the successor's normal moments and the cubic Rees conclusions. Theorem 4.13 makes the local ring of P normal, rational and Cohen–Macaulay. The moment bounds in Theorem 4.5 [Han26, Lemmas 4.2 and 4.4], together with Theorem 4.16, give $E_P \geq T(2, \mu)$ and $z \leq 4 - 2E_P/\mu$. Test the following lower values for E_P :

μ	$E_{\min}$	q	Y	S
3	3	1317/1000	2	29/75
4	6	283/200	1	411/1250
5	8	187/125	4/5	741/2500
6	10	783/500	2/3	671/2500

Here $Y = \min((9 - \mu)/3, 4 - 2E_{\min}/\mu)$. For every smaller state $w = (f, h, \nu)$, with $h \leq 6$, use

$$A_w = \begin{cases} \min(Y, \beta_w, (4\mu - 2E_{\min} - \nu)/\mu), & f = 3, \\ \min(Y, \beta_w), & f \leq 2. \end{cases}$$

Every endpoint and pairwise line intersection of $H_w(y) - qy$ on $[0, A_w]$, including every nonpositive-cap exclusion, is recorded in `13-triple-fivefold/first-source`, identified in Section 13.6. Their maximum together with zero is the displayed S . Theorem 5.4 therefore gives the complete root bound $qz + S$. Both $1/\lambda - 25/27$ and $q - 1/\lambda$ are positive. The four resulting whole prices are bounded by

$$\frac{100}{27} + \frac{3 - Y}{\lambda} + qY + S \in \left\{ \frac{430343663}{53959500}, \frac{4289445761}{539595000}, \frac{2144611627}{269797500}, \frac{305833177}{38542500} \right\}.$$

All are less than $8 - 1/100$, so these four integer lower moments are impossible. This gives exactly the stated windows; in particular no equality is inferred for $\mu = 4, 5, 6$.

For $\mu = 3$, put $Q = \mathcal{O}_{P,x}$ and $\mathfrak{q} = \mathfrak{m}_{P,x}$. It has minimal multiplicity $3 = 6 - 4 + 1$. Theorem 4.20 gives its full tangent Hilbert series and $e_1 = 2$. For a minimal reduction J of $\mathfrak{q}$, Theorem 4.6 [Ver08, Theorem 2.7 and Corollary 6.6] gives

$$\mathfrak{q}^2 = J\mathfrak{q}, \quad \text{length}(\mathfrak{q}/J) = 2, \quad \mathfrak{q}^a/(J \cap \mathfrak{q}^a) = 0 \quad (a \geq 2).$$

Its Rees algebra is therefore Cohen–Macaulay, hence S_2 . This applies in dimension four with infinite residue field $\mathbb{C}$. The excellent normal local ring Q is analytically unramified and has a canonical module. The equality $E_P = e_1 = 2$ and Theorem 4.8 [PUV05, Corollary 2.5] make every ordinary power normal. Consequently $Q[\mathfrak{q}T]$ is normal as well as Cohen–Macaulay. The parameters in J are a regular sequence in this Cohen–Macaulay ring; no regular-local-ring assertion is used.

Step 2. In this step, we bound the full rational restriction degrees by the complete tangent intersection. The full parent tangent is a squarefree cubic in six variables. Indeed a cubic relation in its homogeneous defining ideal gives a quotient with the same Hilbert series, so there are no further relations. Reducibility gives either three distinct linear factors or a linear factor and an irreducible quadratic factor. Thus each E_i is a linear four-space or an integral quadric fourfold. An integral quadric has rank at least three; its singular locus has codimension at least two. The hypersurface is Cohen–Macaulay, so it is normal. These are statements about individual components.

Let Y be the actual ordinary blowup of $\text{Spec } R$, with full exceptional scheme E , and let P_Y be the full strict transform of P . The surjection of ordinary Rees rings

$$R[\mathfrak{n}T] \longrightarrow Q[\mathfrak{q}T]$$

identifies P_Y with the normal ordinary blowup of P . Its intersection $C = P_Y \cap E$ is scheme-theoretically its full projective tangent. The Cohen–Macaulay Hilbert series $(1 + 2\zeta)/(1 - \zeta)^4$

makes its coordinate ring have regularity one, and its saturated ideal in $\mathbb{P}^5$ have regularity two. This retains any nilpotent successor structure.

At the generic point of any three-dimensional component of $C \cap E_i$, the local ring S of Y is a normal rational surface ring. By Theorem 8.5 [Lip69, Proposition 17.1], its divisor class group is finite. Let I_P, I_i be the distinct primes defining P_Y, E_i there. Their quotients are DVRs, by the normality just established. Choose regular local equations f, h with $\text{div}(f) = N_P P_Y$ and $\text{div}(h) = N_i E_i$, for positive integers N_P, N_i . Parameter-intersection associativity gives

$$\text{length } S/(f, h) = N_P \text{ord}_{S/I_P}(h \bmod I_P) = N_i \text{ord}_{S/I_i}(f \bmod I_i).$$

The coefficient of the original rational restriction at this point is therefore

$$d_S = \frac{\text{ord}_{S/I_i}(f \bmod I_i)}{N_P} = \frac{\text{ord}_{S/I_P}(h \bmod I_P)}{N_i}.$$

For every $l \in I_i$, the divisor of l^{N_i}/h is effective. Normality makes it regular. On the DVR of P_Y this gives $\text{ord}_{S/I_P}(l \bmod I_P) \geq d_S$. Thus $\text{length } S/(I_P + I_i) \geq d_S$. The equation of the entire Cartier E belongs to I_i , so this length is precisely the generic length of $C \cap E_i$. We obtain the full cycle inequality

$$(13.10) \quad C_{P,i} \leq [C \cap E_i]_3.$$

All conductor branches, indices and residue multiplicities in the restriction remain in this calculation. No crossing model has been assumed for the parent.

Let Λ_i span E_i . Cut it by three general hyperplanes to a linear space F_i of dimension m_i . Choose them also filter-regular on the full $C \cap E_i$. Its lower-dimensional associated strata miss this final cut, and the length of the resulting zero-dimensional scheme is the degree of its full three-dimensional cycle. This scheme is contained in $C \cap F_i$. The ideal of C is 2-regular and has a linear resolution, so it satisfies the $N_{2,p}$ hypotheses for $p \geq m_i$. The intersection $C \cap F_i$ is finite. Theorem 7.11, in its finite-scheme form [EGHP05, Theorem 1.1], gives

$$\text{length}(C \cap F_i) \leq \dim F_i + 1 = m_i + 1.$$

Taking degrees in (13.10) proves the upper caps. Nonnegativity and the full equality $\sum \delta_i = 3$ are those of Theorem 8.9. This proves (13.9). $\square$

13.5. One original fixed prime and its profile.

Proof of Theorem 13.1. Assume nongeneration and choose the original ambient cutoff as in Setup 13.2. The preceding lemmas apply to every allowed k , every sufficiently general pure threshold average and its actual center. Lemma 13.6 excludes the integral tangent. We prove that the remaining full reducible tangent is impossible by identifying the two source primes, bounding their common profile, and checking its full capacity.

Step 1. In this step, we fix the original profile comparison before selecting either prime. Fix the test interval $[0, 1]$ before any supporting slope or test point is chosen. For every original prime $P \subset W$ through x , put

$$d_P = A^4 \cdot P, \quad \phi_P(v) = \inf_{F_q(v) \neq 0} \frac{\text{ord}_P F_q(v)}{q}, \quad w(v) = \lim_{q \text{ divisible}} \frac{120 \dim F_q(v)}{q^5}.$$

Theorem 6.16 applies to the original W : local normality and all closed ordinary powers were earned in Lemma 13.3. Its volume is continuous. Lemma 13.5 gives

$$(13.11) \quad w(\theta) \geq \frac{18145162291861493}{3125000000000000000} > \frac{1}{200}.$$

Apply the fixed-profile theorem on the reached interval $[0, \theta]$. It supplies one fixed effective error divisor Z_0 , a finite support of primes through x , and a cutoff such that

$$(13.12) \quad 0 \leq \text{ord}_P F_p(v) - p\phi_P(v) \leq \text{mult}_P Z_0 \quad (pv \in \mathbb{Z}, 0 \leq v \leq \theta)$$

for every sufficiently large original p . Since $d_P \geq 1$, each error is at most $A^4 \cdot Z_0$. Fix a 5000-divisible p above this cutoff, the intrinsic cutoff and the independent ordinary lifting cutoff, with

$$\frac{A^4 \cdot Z_0}{p} < \frac{1}{10000}.$$

Then both $F_p(\lambda)$ and $F_p(\theta)$ are nonzero. Fix the general-tuple dilutions to put every nonfixed prime coefficient below $1/10$ before selecting either prime.

Step 2. In this step, we obtain a uniform fixed height for each actual prime. At either level $\gamma \in \{\lambda, \theta\}$, use its own threshold t from B , its actual prime P , and its deficit z . Lemma 13.5 gives $\mu = \text{mult}_x P \in \{3, 4, 5, 6\}$, $z \leq b - \gamma t \leq b - \lambda t$, and (13.4). The following single lines are among the complete tail bounds already proved:

μ	a_μ	c_μ	η_μ	h_μ
3	3/2	81/250	53757/142480	377/1000
4	283/200	379/1000	3541077/8848000	2/5
5	187/125	297/1000	3305853/8246875	2/5
6	783/500	269/1000	17000577/45272500	3/8

Here $H_{(4,6,\mu)}(z) \leq a_\mu z + c_\mu$. If $s + t + H_{(4,6,\mu)}(z) < 8$, its actual positive gap pays the cutting errors by Theorem 5.4 and gives generation. Consequently nongeneration implies

$$8 \leq s + t + H_{(4,6,\mu)}(z) \leq \frac{175}{27} + c_\mu + (a_\mu - 25/27)b - (a_\mu \lambda - 1)t.$$

Set $A_\mu = a_\mu - 25/27$, $C_\mu = 175/27 + c_\mu - 8$, and $D_\mu = a_\mu \lambda - 1$. Both A_μ, D_μ are positive. The actual numerator $A_\mu b + C_\mu$ is positive, and the old-prime inequality gives

$$(13.13) \quad \eta \geq \frac{D_\mu(3b + \mu - 9)}{\mu(A_\mu b + C_\mu)}.$$

The derivative of its varying ratio has the sign of $3C_\mu - A_\mu(\mu - 9)$. The exact identities are

μ	D_μ	$3C_\mu - A_\mu(\mu - 9)$	$\eta_\mu - h_\mu$
3	1991/10000	-313/2250	1051/3562000
4	131151/1000000	-6569/6750	1877/8848000
5	122439/625000	-299/216	7103/8246875
6	629651/2500000	-1097/600	46779/90545000

Thus (13.13) is minimized at $b = 3$, where it equals $D_\mu/(3A_\mu + C_\mu) = \eta_\mu$. Each displayed difference is greater than $1/10000$. In particular $\eta > 3/8$, so the preselected dilution makes P a fixed prime of the full source, with $\eta = \text{ord}_P F_p(\gamma)/p$. Equation (13.12) now gives

$$(13.14) \quad \phi_P(\gamma) \geq h_\mu \geq 3/8.$$

This argument uses the actual old coefficient even when the prime was already in the discriminant.

Step 3. In this step, we prove that the two selected primes coincide and have original degree one. The full fixed-divisor bound of Theorem 6.16 and (13.11) give

$$(13.15) \quad \sum_{P \ni x} d_P \phi_P(\theta) \leq 1 - w(\theta)^{1/5} < \frac{3}{4},$$

because $1/200 > 1/1024 = (1/4)^5$. This sum includes every prime through x , and its support is finite. The prime of the λ -trial has height at least $3/8$ at θ , by monotonicity; the prime of the θ -trial has the same lower height. Distinct primes would contribute at least $3/4$, contradicting (13.15). They are therefore the same original prime, which we now denote by P . Its degree d_P is a positive integer; $d_P \geq 2$ would give the same contradiction. Hence $d_P = 1$.

Step 4. In this step, we bound the full allowed range of supporting slopes. Write $\mu = \text{mult}_x P$ and $h = h_\mu$. Products of original sections make ϕ_P convex on its rational finite domain, nested

point ideals make it nondecreasing, and ampleness gives $\phi_P(0) = 0$. The positive volume at $\theta > \lambda$ makes λ interior to that finite interval. Thus it has a finite supporting slope a at λ . Ordered secant slopes extend the supporting inequality to every attained original order. In particular

$$(13.16) \quad T_P(v) := (h + a(v - \lambda))_+ \leq \phi_P(v), \quad a \geq \frac{\phi_P(\lambda)}{\lambda} \geq \frac{h}{\lambda}.$$

At θ , (13.15) therefore gives $h + a(\theta - \lambda) < 3/4$. Since $\theta - \lambda = 9/2500$, the closed interval $[h/\lambda, 104]$ contains every possible slope for $\mu = 3, 4, 5$.

For $\mu = 6$, use the own threshold t_1 and deficit z_1 of the θ -trial at this same prime. Its adjacent cap is $z_1 \leq 1$. The inequality $b \geq z_1 + \theta t_1$ yields

$$\begin{aligned} 8 &\leq s + t_1 + H_{(4,6,6)}(z_1) \\ &\leq \frac{175}{27} + \frac{269}{1000} + (1 - 25\theta/27)t_1 + (783/500 - 25/27)z_1 \\ &\leq \frac{50}{9} + \frac{367}{200} + \frac{277}{1080}t_1. \end{aligned}$$

It follows that $t_1 \geq 3291/1385$. The finite source order, without an asymptotic error subtraction, then gives

$$\phi_P(\theta) \leq \frac{\text{ord}_P F_p(\theta)}{p} = \frac{1 - \alpha}{t_1} \leq \frac{1385}{3291} < \frac{421}{1000}.$$

Consequently $a < 115/9$. The closed slope domain in this case is $[h/\lambda, 115/9]$. No unbounded slope tail remains outside these proved bounds.

Step 5. In this step, we derive the lower volume estimate on the original fivefold. On the fixed interval $[0, 1]$, let m_i be the degrees of the parent's full tangent components and $\delta_i \geq 0$ the rational restriction degrees of the same P . They include the full conductor and residue contributions, and $\sum \delta_i = \mu$. Every original section of exact point order j in any degree p' has prime order at least $\lceil p'T_P(j/p') \rceil$. Apply Theorem 8.9 to the entire original layer. Its dimension is at most

$$\frac{(p')^4}{24} \sum_i m_i \left(\frac{j}{p'} - \frac{\delta_i}{m_i} T_P(j/p') \right)_+^4 + \kappa(j+1)^3.$$

Rounding the symbolic order upward strengthens the required vanishing. Summing all layers below $\lceil p'v \rceil$ and using the original Hilbert polynomial gives, for $0 \leq v \leq 1$,

$$(13.17) \quad w(v) \geq 1 - 5 \int_0^v \sum_i m_i \left(\xi - \frac{\delta_i}{m_i} T_P(\xi) \right)_+^4 d\xi.$$

Indeed $0 \leq \delta_i/m_i \leq 6$ and $0 \leq a \leq 104$, so these functions have a common Lipschitz bound on $[0, 1]$. Their positive parts are at most one. The Riemann-sum and endpoint rounding errors, after normalization, are $O(1/p')$, uniformly in the prime, slope, test point and rational restriction degrees. The summed local errors and the original Hilbert error are $O((p')^4)$. This proves the finite version before passing to the divisible-degree limit. Bigness above θ is not required. All original fixed and remote descent conditions remain in the spaces being telescoped.

The global upper comparison of Theorem 6.16, with $d_P = 1$, is $w(v) \leq (1 - T_P(v))_+^5$. Combining it with the separately proved five-dimensional lower bound gives the necessary inequality

$$(13.18) \quad 1 \leq 5 \int_0^v \sum_i m_i \left(\xi - \frac{\delta_i}{m_i} T_P(\xi) \right)_+^4 d\xi + (1 - T_P(v))_+^5 \quad (0 \leq v \leq 1).$$

Step 6. In this step, we bound every allocation of the prime's component restriction degrees. For $\mu = 4, 5, 6$, convexity in (δ_i) gives

$$\sum_i m_i \left(\xi - \frac{\delta_i}{m_i} T_P(\xi) \right)_+^4 \leq 2\xi^4 + (\xi - \mu T_P(\xi))_+^4.$$

For three unit components this is the vertex maximum on their simplex. For degrees $(1, 2)$, the other vertex gives $\xi^4 + 2(\xi - \mu T_P(\xi)/2)_+^4$, bounded by the displayed expression by midpoint convexity. This is an upper bound on every actual allocation, not an assumption on its support.

For $\mu = 3$, Lemma 13.7 gives $\delta_i \leq m_i + 1$. The vertices for three unit components are permutations of $(0, 1, 2)$; for degrees $(1, 2)$ the endpoints are $(0, 3)$ and $(2, 1)$. The full integral is convex in the allocation, so it is bounded by the largest of the following three patterns. A pair (ω, χ) means the term $\omega(\xi - \chi T_P(\xi))_+^4$:

$$(13.19) \quad \begin{aligned} &((1, 0), (1, 1), (1, 2)), \quad ((1, 0), (2, 3/2)), \quad ((1, 2), (2, 1/2)) \quad (\mu = 3), \\ &((2, 0), (1, \mu)) \quad (\mu = 4, 5, 6). \end{aligned}$$

The test point in each supplementary bin record is the same for every pattern.

Step 7. In this step, we verify the complete rational slope certificate and contradict the volume comparison. Use the dataset `13-triple-fivefold/slopes` in Section 13.8. The families of multiplicities 3, 4, 5, 6 contain, respectively, 4, 11, 11, 13 closed bins. Consecutive endpoints agree, and their first and last endpoints are exactly the domains proved above. For a bin $[\ell, u]$ containing a , put

$$T_0(\xi) = \begin{cases} (h + u(\xi - \lambda))_+, & 0 \leq \xi \leq \lambda, \\ h + \ell(\xi - \lambda), & \lambda \leq \xi \leq 1. \end{cases}$$

Both branches equal h at λ , and $T_0 \leq T_P$ throughout. Replacing T_P by T_0 increases both the local integral and the global term in (13.18).

Here is the exact evaluation rule for every record. Define $o = \lambda - h/u$, so $0 \leq o \leq \lambda$, and set

$$\mathcal{K}(a_0, c_0; x_0, y_0) = \begin{cases} \frac{(a_0 y_0 + c_0)_+^5 - (a_0 x_0 + c_0)_+^5}{a_0}, & a_0 \neq 0, \\ 5(y_0 - x_0)(c_0)_+^4, & a_0 = 0. \end{cases}$$

This equals $5 \int_{x_0}^{y_0} (a_0 \xi + c_0)_+^4 d\xi$, including zero crossings and negative slopes. At the rational test point $\vartheta \in [\lambda, 1]$ recorded for the bin, put

$$(13.20) \quad \mathcal{I}_\chi = o^5 + \mathcal{K}(1 - \chi u, \chi(u\lambda - h); o, \lambda) + \mathcal{K}(1 - \chi \ell, \chi(\ell\lambda - h); \lambda, \vartheta).$$

For $\chi = 0$, this is ϑ^5 . The local value for a pattern is $\sum_{(\omega, \chi)} \omega \mathcal{I}_\chi$, and the global value is $(1 - h - \ell(\vartheta - \lambda))_+^5$. Each bin record gives all local values in the order of (13.19), the global value, and their combined maximum. The exact combined upper values in each family are at most, respectively,

$$(13.21) \quad \frac{49884763}{50000000}, \quad \frac{24992633}{25000000}, \quad \frac{99899271}{100000000}, \quad \frac{19998699}{20000000}.$$

These comparisons involve only the rational positive-part rule (13.20). The records give every numerator and denominator, as well as every input bin and test point; no slope sampling or omitted tail is involved. Each ceiling is strictly less than $19999/20000$. Thus its written test point has the entire right side of (13.18) less than one, for every real slope in the bin and every actual allocation. This contradicts (13.18).

The integral and reducible full cubic tangents exhaust the reduced cubic hypersurface supplied by the ledger. Both have now been excluded under nongeneration. The ambient cutoff was fixed before the original degree and threshold center. Every subsequent intrinsic cutoff was fixed on the resulting original W before its adaptive degrees and primes, and its ambient restriction cutoff was fixed separately. This proves the theorem for every original degree and source history in its statement. $\square$

13.6. Source and normal-moment certificates. The complete finite data accompanying this section are in `supplement/13-triple-fivefold/`. The four stable dataset identifiers are `13-triple-fivefold/tails`, `13-triple-fivefold/first-source`, `13-triple-fivefold/integral`, and `13-triple-fivefold/slopes`. Their canonical files are, respectively, `data/tails.json`, `data/first-source.json`, `data/integral.json`, and `data/slopes.json`. The small scalar table above is a derived view of the first file. Each rational number is an integer string or a signed numerator and positive denominator separated by a slash. No floating-point number is used.

The scalar dataset gives the state, moment, cap, radius and every affine pair. Its thirty row indices are 1–6, 8–20, and 28–38. Completeness can be checked independently from $2T(e-d, \mu) < d\mu$, as follows. For $c = e - d > 0$, choose $\ell > d/2$. The capacity of the degrees below ℓ is $\binom{c+\ell-1}{\ell-1}$, so

$$T(c, \mu) \geq \ell \left(\mu - \binom{c+\ell-1}{\ell-1} \right).$$

Thus every admissible multiplicity obeys

$$\mu < \frac{2\ell}{2\ell-d} \binom{c+\ell-1}{\ell-1}.$$

The choice $\ell = \lfloor d/2 \rfloor + 1$ gives a finite enumeration, after which the exact moment test selects the rows. For $c = 0$ the only multiplicity is one; the curve is the smooth row. Each of the seventy-seven affine pairs is checked by (13.2), with all descendants and all eligible skips. This also checks the degree-two exclusion and its margin 251/27000.

The first-source dataset contains twenty-nine unrestricted polygon records and ten constrained fourfold records, all at $\gamma = \lambda$, with $u = 27/25$ in the latter records. Each record lists its row and cap, every vertex of $\mathcal{D}$, all 466 subdivided witnesses in total, their exact values, and their maximum. Separate summary records retain each supplied attaining point. The two multiplicity-nine records have empty positive-deficit domains: their closed domains are a segment and a point, respectively. Both are checked explicitly. Lemma 13.4 proves the completeness of this test on every domain.

There are also four normal-moment records, one for each $\mu = 3, 4, 5, 6$, with nineteen descendant rows each. They contain every inherited cap, every interval endpoint and tail-line intersection, its exact surplus, and the resulting whole cost. Nonpositive caps remain explicit empty positive-deficit cases. The one-dimensional version of the same affine-subdivision argument checks all these values, including ties. The record for the quartic low-deficit interval retains all its eight endpoints and intersections. Consequently these checks establish precisely the scalar alternatives, the 1369/54000 fixedness margin, the 22657/53959500 quartic low-deficit margin, and all four normal-moment cost gaps greater than 1/100. The two exact source masses are checked by the identities in (13.6) and (13.5). Their use for the original complete spaces and their uniform errors was proved above.

13.7. Integral-tangent transition certificates. The integral dataset gives the nineteen first-level comparisons, the seventeen intervals displayed in the proof, all 170 constrained fourfold comparisons, and every exact loss and remaining mass. The interval indices must be exactly $1, \dots, 17$; each has exactly rows $28, \dots, 37$. The additional zero-cap row 38 is checked on its closed domain at every interval. Every left endpoint equals the preceding right endpoint, the first is λ , and the last is $23/25$. Each endpoint times 5000 is integral. For each interval the checker requires $1 < u_i < 3\gamma_i$, reconstructs every feasible polygon candidate by Lemma 13.4, and compares the entire maximum with both the stored bound and its attaining point. This recomputes 2886 candidates for the 170 comparisons. Empty, segment and point cases use the same feasibility test.

Each whole cost is strictly below 79999/10000; the smallest actual gap below eight among these comparisons is $5471/53001000 > 1/10000$. The seventeen exact losses are evaluated by

(13.8), and every prefix reserve is required to be positive. Their final reserve is the fraction displayed in Lemma 13.6. All thirty unrestricted terminal rows are also evaluated, giving the maximum 2481721/310500. Thus the data certify the whole chain of section-dimension estimates and the final strict scalar comparison. The proof has separately established that the trials start from the same B , that each uses its own threshold, and that the interval errors can be absorbed at a single original degree.

13.8. Slope certificates and independent checking. The slope dataset gives the four height identities and all thirty-nine closed bins, in families of 4, 11, 11, 13. Each bin supplies its two endpoints, its common test point, every local pattern integral, the global term, and their combined maximum. There are 47 local integrals. The first and last endpoints are required to equal the proved full slope domain, and successive endpoints must agree. Hence no open endpoint, interior gap or unbounded slope tail remains. The box-slice vertex enumeration in the proof verifies the three cubic patterns; the other three multiplicities use the full concentration bound.

The independent evaluation splits each component at the profile onset, at λ , and at every zero of the resulting affine function. On a positive interval it integrates the fourth-degree polynomial by expansion:

$$5 \int_{x_0}^{y_0} (a_0 \xi + c_0)^4 d\xi = 5 \sum_{j=0}^4 \binom{4}{j} a_0^j c_0^{4-j} \frac{y_0^{j+1} - x_0^{j+1}}{j+1}.$$

On a nonpositive interval the contribution is zero. A zero affine slope requires no division, and a zero-length interval contributes zero. In particular a zero shift contributes the full ϑ^5 . This evaluation is independent of the positive-part primitive (13.20). It checks each stored local and global value and their combined maximum against (13.21). The gaps of the four ceilings below 19999/20000 are

$$\frac{112737}{50000000}, \quad \frac{6117}{25000000}, \quad \frac{95729}{100000000}, \quad \frac{301}{20000000}.$$

These are exact rational comparisons. No sampling error or floating-point tolerance enters them. The finite-source rounding, the uniform lower-order errors, and both cutting errors remain the separate estimates in the proof.

The supplement specifies the data schema, source mapping and conversion in its `README.md` and `manifest.json`. Its independent entry point is

```
python3 check.py -data data -report report.json.
```

It rejects malformed rationals, unrecognized fields, duplicate or missing records, incomplete index sets, missing polygon witnesses, uncovered slope intervals and failed inequalities. The printed scalar view is compared with its canonical rows. The checker reads only the delivered data and uses integer and rational arithmetic; candidate generation is not part of acceptance. Together with the coverage arguments just given, acceptance proves exactly the finite inequalities used above. Their geometric applicability is the content of the preceding lemmas.

14. REDUCED QUADRATIC TANGENTS AT THE ORIGINAL SOURCE

We retain the boundary data of Definition 2.2 and the complete original spaces of Definition 2.4. The multiplicity slack σ improves the price of the same boundary. We first use that improvement to exclude a nonreduced quadratic tangent and to determine the exceptional cycle of the remaining cubic fivefold of embedding dimension seven. We then combine three actual threshold profiles on the same original images. Their complete rational comparison earns the original source of order 29/25.

Lemma 14.1. *Use Definition 5.2 with $n = 7$ and all degree floors equal to one. For every state in $\mathcal{P}_7$, its complete scalar envelope is bounded by the minimum of the affine lines in its row of*

the rational dataset 14-double-geometry/tail-table. In particular,

$$\mathcal{F}_{(5,6,2)}(b) \leq \frac{5}{4}b + \frac{92}{125}, \quad \mathcal{F}_{(5,7,3)}(b) \leq \frac{623}{500}b + \frac{39}{40}$$

for every b in the cap interval of the indicated state. Every smaller dimension and every allowed embedding dimension is included.

These bounds concern the whole continuation. They impose no order on the source that produced B .

Proof. We evaluate the finite recurrence of Definition 5.2 by a rational upper table. The full supplementary table records a state (d_i, e_i, m_i) , its moment $\mu_i = T(e_i - d_i, m_i)$, its cap $\beta_i = d_i - 2\mu_i/m_i$, a radius upper bound ρ_i , and a list of affine pairs. In that table the columns E, B, r mean, respectively, e_i, β_i, ρ_i ; they do not denote a boundary or an exceptional divisor. Let $H_i(z)$ be the minimum of its affine functions. The radius satisfies $\rho_i^{d_i} \geq m_i$.

Step 1. In this step, we enumerate all states and verify their rational radius bounds. The state enumeration consists of the curve state $(1, 1, 1)$ and, for $2 \leq d \leq 6$ and $d \leq e \leq 7$, every state allowed by Definition 5.2. For $c = e - d = 0$, retain just $m = 1$. For $c > 0$, start at $m = c + 1$, fill the lowest monomial degrees in c variables to recover $T(c, m)$, and increase m while $2T(c, m) < dm$. The average filled degree $T(c, m)/m$ is nondecreasing and tends to infinity, so the first excluded multiplicity ends that list. This gives all 71 states. Each recorded radius is the smallest multiple of $1/1000$ whose d -th power is at least m .

Step 2. In this step, we verify the complete successor tests and deduce both stated tail bounds. For $d_j < d_i$ and $e_j \leq e_i$, put

$$c_{ij} = \begin{cases} \min(\beta_i, \beta_j, \beta_i - m_j/m_i), & d_j = d_i - 1, \\ \min(\beta_i, \beta_j), & d_j \leq d_i - 2. \end{cases}$$

Discard exactly the nonpositive caps. For every affine pair (a, p) in row i , the rational table satisfies

$$(14.1) \quad p \geq (\rho_i - a)_+ \beta_i + \max \left(0, \max_{j: c_{ij} > 0} \max_{0 \leq z \leq c_{ij}} \{H_j(z) - \max(a, \rho_i)z\} \right).$$

All the maxima in this test are evaluated at $0, c_{ij}$ and the intersections $(p' - p)/(a - a')$ of distinct-slope lines in row j that lie in the interval. Thus the table and (14.1) specify finite rational inequalities, including all breakpoints. Parallel affine lines have no isolated intersection; their minimum is already evaluated at the same endpoints. Ties at an endpoint are retained. A negative successor cap gives an empty domain, and a zero cap permits only zero deficit, whose continuation cost is zero. Thus discarding the nonpositive caps loses no positive-deficit continuation. Every remaining domain is a closed segment, with both endpoints included.

To see that this test is sufficient, let the first successor have deficit $z \leq b$ and state j . Induction in dimension bounds its entire remaining cost by $H_j(z)$. The cost from the current state is at most $\rho_i(b - z) + H_j(z)$. After subtracting ab , use $b \leq \beta_i$ if $\rho_i \geq a$, and $b \geq z$ if $\rho_i < a$. The result is bounded by the right side of (14.1). A direct point successor is its zero alternative. The test therefore gives $\mathcal{F}_i \leq H_i$.

Rows 51 and 55 contain the two displayed lines. The table includes the embedding-six threefold multiplicities eight and nine, with moments 11, 13 and caps $1/4, 1/9$. It also includes their embedding-seven counterparts, with moments 10, 12 and caps $1/2, 1/3$, and the multiplicity-ten and eleven rows. In particular no adjacent cap is assigned to a dimension skip. The explicit eight and nine estimates in Lemma 5.10 remain available throughout. $\square$

Proposition 14.2. *At a nongeneration point with $L^7 = 1$, consider any sufficiently late pure actual threshold average of the complete original $U_k(27/25)$, where $25 \mid k$. If its minimal center*

has state $(5, 6, 2)$, its full tangent ring is reduced. The nonreduced case has an additive point continuation of coefficient below $79/10$. If the original center has state $(5, 7, 3)$, then

$$27175/8642 \leq b \leq 10/3, \quad 1/2 \leq \sigma \leq 8479/15000 < 2/3, \quad 1/s \geq 4321/13425,$$

and its entire exceptional cycle is $2[\Pi] + [\Pi']$ for distinct linear components. Such an observation earns

$$V(t) > 1/1000 \quad (0 \leq t \leq 29/25), \quad \dim U_q(t) \geq q^7/(2000 \cdot 7!)$$

for all sufficiently late original degrees and every t in that interval. The cutoff precedes any subsequent $29/25$ observation.

The first assertion excludes the square of a linear form without imposing conditions on the higher terms of the germ. The cycle in the second assertion records all exceptional multiplicities; it does not identify the full exceptional scheme.

Proof of the tangent, coefficient, and exceptional-cycle assertions. We first keep the full slack in the price, then pay the nonreduced double, and finally use the cubic tail to bound every exceptional multiplicity.

Step 1. In this step, we retain the full point slack in the original threshold price. Choose the cutoff in [Theorem 10.1](#) before the observing degree k , the general tuple and the computing place. The hypothesis $25 \mid k$ makes this degree eligible for the ledger and makes $27k/25$ integral. Let $\Gamma \sim_{\mathbb{Q}} sL$ be the specified pure threshold average and W its minimal center. By [Proposition 9.1](#), $M \geq (27/25)s$. Since $M + b + \sigma = 7$, we have

$$(14.2) \quad \frac{27}{25}s + b + \sigma \leq 7.$$

The multiplicity M includes every fixed component of Γ . The ledger gives the full Cohen–Macaulay tangent ring $G = \operatorname{gr}_{\mathfrak{m}_{W,x}} \mathcal{O}_{W,x}$ in both states under consideration. If this ring is nonreduced, [Theorem 4.20](#) excludes $\sigma < 1/2$. Thus

$$(14.3) \quad \sigma \geq \frac{1}{2}, \quad s \leq \frac{25}{27} \left(\frac{13}{2} - b \right).$$

Step 2. In this step, we pay the nonreduced double-fivefold case by an additive point continuation. Suppose the state is $(5, 6, 2)$ and G is nonreduced. The local ring is normal and Cohen–Macaulay by [Theorem 4.13](#). Its minimal completed presentation is a hypersurface in six variables, of order two. The initial ideal of its defining principal ideal is generated by its quadratic initial form: multiplication of initial forms in the regular graded polynomial ring preserves order. A nonreduced quadratic over $\mathbb{C}$ is the square of a linear form. This identifies only the initial quadratic.

We have $b \leq 7/2$. Indeed, if $b > 7/2$, the minimal-multiplicity clause of [Theorem 4.20](#) applies with $d = 5$, $m = 2$ and $b > d - 2$. Its other hypotheses hold because $7 - M = b + \sigma > 1$ and the ledger gives $0 \leq \sigma \leq 1734/3125 < 1$. It bounds the nonreduced cycle defect by

$$0 \leq \sum_i (\nu_i - 1) d_i \leq 2(3 - b) + 2 < 1.$$

This integer is zero. Every component is generically reduced, and the full Cohen–Macaulay tangent ring has no embedded associated primes, including at its vertex. Its nilradical must therefore vanish, a contradiction.

By [Lemma 14.1](#) and (14.3),

$$\begin{aligned} s + \mathcal{F}_{(5,6,2)}(b) &\leq \frac{25}{27} \left(\frac{13}{2} - b \right) + \frac{5}{4}b + \frac{92}{125} \\ &\leq \frac{25}{9} + \frac{35}{8} + \frac{92}{125} = \frac{70999}{9000} < \frac{79}{10}. \end{aligned}$$

The second inequality uses $5/4 - 25/27 > 0$ and $b \leq 7/2$. The gap to $79/10$ is $101/9000$. Apply [Theorem 3.14](#) and [Theorem 5.4](#) with a total weighted error smaller than $1/1000$. More explicitly, at each of at most five cuts, after its actual radius ρ_i is known, choose $0 < \epsilon_i < 1/(5000(1 + \rho_i))$. Retain both inequalities

$$t_i \leq b_i \rho_i + \epsilon_i, \quad b_{i+1} \leq b_i - t_i / \rho_i + \epsilon_i.$$

Their total weighted error is less than $1/1000$, and hence the actual additive point boundary has coefficient below $79/10$. Every term of Γ remains in it. By [Proposition 3.12](#), this implies generation of $K_X + 8L$ at x . The contradiction proves the full reducedness assertion.

Step 3. In this step, we bound the cubic height and exclude a triple exceptional component. Now let the state be $(5, 7, 3)$. The ledger gives $b \leq 10/3$, a full nonreduced Cohen–Macaulay tangent ring, and the two cycles $2[\Pi] + [\Pi']$ and $3[\Pi]$. Put $a_* = 623/500$ and $p_* = 39/40$. Nongeneration and [Lemma 14.1](#) imply

$$(14.4) \quad 8 \leq s + a_* b + p_*.$$

A strict reverse inequality would leave a positive margin for both cutting errors and give a point continuation by [Theorem 5.4](#).

Combining (14.2), (14.3), and (14.4) gives

$$\left(\frac{27}{25}a_* - 1\right)b \geq \frac{27}{25}(8 - p_*) - \frac{13}{2}, \quad \frac{27}{25}a_* - 1 = \frac{4321}{12500}.$$

Consequently

$$b \geq \frac{27175}{8642}, \quad \frac{1}{s} \geq \frac{27/25}{13/2 - b} \geq \frac{4321}{13425}.$$

The denominator is positive since $b \leq 10/3$. The same inequalities give the upper slack bound

$$\sigma \leq \left(\frac{27}{25}a_* - 1\right)b + 7 - 8\frac{27}{25} + \frac{27}{25}p_* \leq \frac{8479}{15000} < \frac{2}{3}.$$

Here every quantity belongs to the same Γ .

The ordinary powers and original Rees ring already have the properties in [Theorem 10.1](#). Thus the exceptional Cartier scheme is the actual $\text{Proj } G$. The strict-pair inequality in [Theorem 4.20](#) gives $(1 - \sigma)r_F \leq 1$ for every exceptional component, with its full scheme multiplicity r_F . Multiplicity three would imply $\sigma \geq 2/3$. It is impossible, so the full cycle is $2[\Pi] + [\Pi']$, with distinct linear $\mathbb{P}^4$'s. No nilpotent structure, embedded structure, gluing or conormal data is replaced by this cycle. $\square$

It remains to prove the source assertions. We establish the linked heights at three original orders, apply every capacity to their common global images, and then sum the complete rational certificate. The earlier source mass also controls every unbounded supporting slope.

14.1. The three-profile comparison. We specify the constants, parameter domains and exact checking rules for this comparison. The complete rational records are identified in [subsection 14.2](#). Write

$$r_0 = \frac{9}{10}, \quad r_1 = \frac{26}{25}, \quad r_2 = \frac{27}{25}, \quad T = \frac{29}{25}, \quad h_0 = \frac{131}{1000}, \quad h_* = \frac{1609}{5000}, \quad \epsilon = \frac{1}{100000}.$$

For $c \geq 0$, put $E_j(c) = (1 - jc)_+^6$ and

$$N(c) = E_{7,5}(c), \quad Q(c) = \max\{E_1(c), (1 - 2c)_+^5(1 + 10c)\},$$

using the envelope from [Definition 7.7](#). Define

$$K_M(c) = \min\{E_1(c) + 3E_2(c) - 5E_3(c) + 2E_4(c), E_{7,3}(c), \\ 1, 256/729 + 18(1/6 - c)_+\}.$$

For $c \geq 2/11$, also include $E_1(c) + E_2(c) - E_3(c)$ in this minimum. Put

$$K_D(c) = 6E_2(c) - 8E_3(c) + 3E_4(c),$$

and for $c \geq 1/6$ replace this by its minimum with 138259/900000. Set

$$B_*(t) = \begin{cases} 1, & 0 \leq t < 9/10, \\ 61/100, & 9/10 \leq t < 99/100, \\ 11/25, & 99/100 \leq t < 26/25, \\ 17/50, & 26/25 \leq t \leq 29/25. \end{cases}$$

For $t < T$, this is the original-layer baseline of [Proposition 7.28](#); at T we use its left limit.

The functions with coefficients $(1, 3, -5, 2)$ and $(6, -8, 3)$ are nonnegative and nonincreasing. On the uniform seven-coordinate simplex they are, respectively, the probabilities that one designated coordinate is at least c or two of three other coordinates are at least c , and that at least two of four designated coordinates are at least c . Inclusion–exclusion gives the displayed expressions. Likewise $E_1 + E_2 - E_3$ is the probability of the union of one coordinate event and the intersection of two other coordinate events.

Definition 14.3. Fix K equal to Q , K_M , or K_D . Its corresponding height $h_{\min}$ is $37/200$, $41/200$, or $189/1000$. Let finite slopes a_0, a_1, a_2 and linked heights satisfy

$$\begin{aligned} a_0 &\geq h_0/r_0, & h_1 &= \max\{h_{\min}, h_0 + (r_1 - r_0)a_0 - \epsilon\}, & a_1 &\geq h_1/r_1, \\ h_2 &= \max\{h_*, h_0 + (r_2 - r_0)a_0 - \epsilon, h_1 + (r_2 - r_1)a_1 - \epsilon\}, & a_2 &\geq h_2/r_2. \end{aligned}$$

For $t > 0$, put

$$c_i(t) = \frac{(h_i + a_i(t - r_i))_+}{t}, \quad f(t) = \min\{B_*(t), N(c_0(t)), K(c_1(t)), Q(c_2(t))\}.$$

At $c_i(t) = 0$ the corresponding bound is one. The *complete rational data* for these three choices form the dataset **14-double-geometry/three-profile**. Its three families' grids, split trees, box heights, polynomial intervals, integer upper bounds, and unbounded tails are all part of the data.

Lemma 14.4. *Let X be a smooth complex projective sevenfold, L ample Cartier with $L^7 = 1$, and x a nongeneration point. Fix a sufficiently late pure actual observation at $r_2 = 27/25$ in degree k_2 with $25 \mid k_2$, whose minimal center has state $(5, 7, 3)$. Then sufficiently late pure actual observations at $r_0 = 9/10$ in degrees k_0 with $50 \mid k_0$, and at $r_1 = 26/25$ in degrees k_1 with $25 \mid k_1$, together with the fixed r_2 observation, have the linked heights and slopes of [Definition 14.3](#). Their cutoffs precede the three degrees and computing places. The first center has codimension at most five. The middle center belongs to one of the following families:*

family	dimension and embedding	deficit	$h_{\min}$
Q	$d = 5$ or 6	$b_1 > 7/5$	$37/200$
M	$d = 4, e = 6$	$b_1 > 39/20$	$41/200$
D	$d = 4, e = 7$	$b_1 > 3/2$	$189/1000$.

Both fourfold families have a full reduced Cohen–Macaulay tangent ring. No relation between their centers, or equality of their degrees, is required.

The first family includes a middle divisorial observation. By [Theorem 7.9](#), the universal two-normal capacity also bounds a one-normal ideal, so including this extra alternative needs no exclusion of a divisor.

Proof. Choose the auxiliary degrees with $50 \mid k_0$ and $25 \mid k_1$. All three degrees are divisible by 25, so [Theorem 9.14](#) supplies the three nonzero complete sources after its common cutoff. Moreover, every $k_i r_i$ is an integer, as required by [Proposition 9.1](#). The positive volume at r_2 , together with [Theorem 6.3](#), permits fixed rational numbers $r_2 < \beta_0 < \beta$ for which $\pi^*L - \beta E_x$ is big. Apply [Theorem 6.13](#) on $[0, \beta_0]$. The choices depend only on the original data and precede every observation. For the place w_i computing the observation at r_i , write $A_i = A_X(w_i)$ and

$\phi_i(t) = \alpha_{w_i}(t)/A_i$. These profiles are finite and convex on a neighborhood of all three anchors. They vanish at zero because high ample multiples are generated. Write s_i, b_i for the coefficient and deficit of the observation at r_i . The full source multiplicity gives $r_i s_i + b_i \leq 7$.

Step 1. In this step, we obtain a uniform positive height for the first observation. At r_0 , a point center has coefficient at most $70/9 < 8$. A curve center has entire tail at most b_0 , so its whole price is at most $(7 - b_0)/r_0 + b_0 \leq 70/9$. Both contradict [Proposition 3.12](#). Thus its codimension is at most five. Every row of the complete table in [Lemma 14.1](#) is bounded by $(5/2)b_0$. Nongeneration implies $8 \leq s_0 + (5/2)b_0$. Combining the two inequalities gives $b_0 \geq 4/25$, and hence

$$s_0 \leq \frac{38}{5}, \quad \frac{1}{s_0} \geq \frac{5}{38} > h_0.$$

Step 2. In this step, we determine the full tangent and deficit bounds for every middle fourfold. The dimension and deficit gates follow from [Proposition 5.9](#). For a fourfold, the states with cap exceeding $3/2$ are exactly

$$(4, 4, 1), (4, 5, 2), (4, 5, 3), (4, 6, 3), (4, 6, 4), (4, 6, 5), (4, 7, 4), (4, 7, 5), (4, 7, 6).$$

The table bounds the whole prices for states $(4, 4, 1), (4, 5, 2), (4, 5, 3), (4, 6, 5)$ by, respectively,

$$\frac{99341}{13000}, \quad \frac{12984}{1625}, \quad \frac{25297}{3250}, \quad \frac{512389}{65000},$$

all less than eight. Each strict margin pays both cutting errors. For the five remaining states, combine $8 \leq s_1 + ab_1 + p$ with $r_1 s_1 + b_1 + \sigma_1 \leq 7$, using the indicated full-tail lines:

(e, m)	a	p	upper bound for σ_1
$(6, 3)$	$5/4$	$373/500$	$1599/6250$
$(6, 4)$	$283/200$	$379/1000$	$217/12500$
$(7, 4)$	$283/200$	$29/50$	$2311/5000$
$(7, 5)$	$187/125$	$17/40$	$2921/12500$
$(7, 6)$	$783/500$	$331/1000$	$2699/37500$

Every last entry is less than $1/2$. Also $7 - \text{mult}_x \Gamma_1 = b_1 + \sigma_1 > 3/2 > 1$. Theorem [4.20](#) therefore supplies all ordinary normal powers, the original Rees conclusions, and the full reduced Cohen–Macaulay tangent ring, including its vertex. For $(e, m) = (6, 3), (6, 4)$, the lines $(1317/1000)b_1 + 71/125$ and $(283/200)b_1 + 379/1000$ would leave respective price margins $2101/260000$ and $311/52000$ if $b_1 \leq 39/20$. This proves the stronger middle deficit gate.

Step 3. In this step, we obtain the linked supporting lines from the actual finite thresholds. The middle inverse coefficients consequently satisfy, in families Q, M, D , respectively,

$$\frac{1}{s_1} > \frac{13}{70}, \quad \frac{1}{s_1} > \frac{104}{505}, \quad \frac{1}{s_1} > \frac{52}{275}.$$

These bounds strictly exceed their chosen $h_{\min}$. The already proved coefficient bound at r_2 gives $1/s_2 \geq 4321/13425 > h_*$. Choose the common finite-degree cutoff so the uniform errors in [Theorem 6.13](#) are smaller than ϵ and than these fixed positive height gaps. Choose a finite supporting slope a_i at each anchor. At another observation their heights compare by

$$\phi_j(r_i) \leq \frac{w_j(I_i)}{k_i A_X(w_j)} \leq \frac{1}{k_i \tau_i} \leq \phi_i(r_i) + \frac{C_i}{k_i},$$

where I_i is that observation's full base ideal. Applying this inequality at each later anchor to the earlier supporting lines gives the two linked maxima. Convexity and $\phi_i(0) = 0$ give $a_i \geq \phi_i(r_i)/r_i > h_i/r_i$. This proves the claimed comparison for the actual three profiles. $\square$

Lemma 14.5. *In the setting of [Lemma 14.4](#), choose the three actual profiles and supporting lines as in its proof, and let $f(t)$ be the function in [Definition 14.3](#). Let $A_{q,\ell}$ be the full original*

image defined in [Proposition 7.28](#). There are constants J, C , fixed before the observation degrees, centers and computing places, such that for every integer $q \geq J$ and $1 \leq \ell < qT$,

$$(14.5) \quad \dim_{\mathbb{C}} A_{q,\ell} \leq \frac{\ell^6}{6!} f(\ell/q) + C(\ell + 1)^5.$$

The same estimate holds with any one of the three profile capacities alone. Also $\dim A_{q,0} \leq 1$. A zero current source has zero image; no nonzero next source is required.

Proof. We extend the supporting lines to every actual section, form the full valuation ideals, and apply the local capacities to the same image.

Step 1. In this step, we extend each supporting-line inequality to every original section. Fix a nonzero $g \in H^0(X, qL)$, put $u = \text{ord}_x g/q$, and fix one of the three places. We claim

$$(14.6) \quad \frac{w_i(g)}{qA_i} \geq \phi_i(r_i) + a_i(u - r_i) \geq h_i + a_i(u - r_i).$$

This also holds when u is outside the fixed big interval. For a small rational $0 < \theta < 1$, the point $r_i + \theta(u - r_i)$ is inside that interval. Take sections at r_i whose normalized w_i -orders tend to $\phi_i(r_i)$. Powers of those sections and of g can be chosen with degree proportions $1 - \theta$ and θ . Their products belong to the complete original space at order $r_i + \theta(u - r_i)$. The infimum defining the profile therefore gives

$$\phi_i(r_i + \theta(u - r_i)) \leq (1 - \theta)\phi_i(r_i) + \theta \frac{w_i(g)}{qA_i}.$$

The supporting-line inequality on the left proves (14.6). This argument uses actual products only to compare valuations; every image to be counted remains the image of the complete original space.

Step 2. In this step, we place the original source germs in each full valuation ideal. Put $R = \mathcal{O}_{X,x}$, $P_i = I_{W_i,x}$, and

$$L_i(t) = (h_i + a_i(t - r_i))_+.$$

When $L_i(\ell/q) > 0$, define the entire ideal

$$J_{i,q,\ell} = \{g \in R : w_i(g) \geq qA_i L_i(\ell/q)\}.$$

It is proper and P_i -primary. Indeed w_i is zero on $R \setminus P_i$, is positive on every element of P_i , and the positive minimum of its values on finitely many generators of P_i puts a power of P_i in $J_{i,q,\ell}$. Multiplication by an element outside P_i preserves membership. With the generic threshold $\lambda_{i,q,\ell} = \text{lct}_{R_{P_i}}(J_{i,q,\ell} R_{P_i})$, the valuative formula gives

$$\lambda_{i,q,\ell} \leq \frac{A_i}{w_i(J_{i,q,\ell})} \leq \frac{1}{qL_i(\ell/q)}, \quad \frac{1}{\ell\lambda_{i,q,\ell}} \geq c_i(\ell/q).$$

Localization at the actual center preserves the discrepancy of w_i . Since $a_i \geq 0$, (14.6) puts every germ from $U_q(\ell/q)$ in this ideal. The degree- ℓ germ map has kernel exactly $U_q((\ell + 1)/q)$. Consequently the very same $A_{q,\ell}$ is contained in $(\text{in}_{\mathfrak{m}_x} J_{i,q,\ell})_\ell$ for each i . No assertion that a source ideal itself is primary is used.

Step 3. In this step, we obtain the stated capacities for all center alternatives on the common images. For the first place the codimension is at most five. The local comparison of [Theorem 7.23](#) and [Theorem 7.9](#) gives the bound N : append zero weights to the simplex comparison when the codimension is smaller. For the final place the codimension is two, so the exact capacity is Q , with no tangent hypothesis. The same bound covers the middle family Q ; in codimension one its capacity E_1 is at most Q .

For a middle fourfold of embedding dimension six, [Theorem 7.18](#) applies with $(n, t, s) = (7, 1, 3)$. Its formula gives

$$\mathcal{S}_{7,1,3}(c) = E_1(c) + 3E_2(c) - 5E_3(c) + 2E_4(c).$$

The generic height-three envelope, the first clause of [Proposition 7.21](#), and the $c \geq 2/11$ clause of [Proposition 7.19](#) supply every other term in K_M . For a middle fourfold of embedding dimension seven, use $(n, t, s) = (7, 0, 4)$, obtaining

$$\mathcal{S}_{7,0,4}(c) = 6E_2(c) - 8E_3(c) + 3E_4(c).$$

The height-three clause of [Proposition 7.19](#) supplies the additional constant for $c \geq 1/6$. This proves the bound K_D . The full reduced Cohen–Macaulay tangent hypotheses in these applications were established in [Lemma 14.4](#).

Each applicable capacity is nonincreasing. Its local threshold argument is at least $c_i(\ell/q)$, so the full-primary estimate gives the stated profile bound with error $C_7(\ell+1)^5$. If a supporting line is nonpositive, use all degree- ℓ polynomials; their dimension is $\ell^6/6! + O((\ell+1)^5)$, corresponding to capacity one. Finally [Proposition 7.28](#) bounds this same $A_{q,\ell}$ by $B_*(\ell/q)$. Taking the largest of the finitely many uniform error constants permits the pointwise minimum of their leading terms. This is [\(14.5\)](#), including all changes of center and every original fixed or embedded condition. $\square$

Lemma 14.6. *For the data of [Definition 14.3](#), suppose $a_0 \leq 16$, $a_1 \leq 8$, and $a_2 \leq 8$. Then*

$$7 \int_0^T t^6 f(t) dt < \frac{999}{1000}.$$

If $a_0 \geq 16$, the same bound holds using only the baseline below r_0 and the first profile above r_0 . If $a_1 \geq 8$, then

$$\frac{3}{40} - 7 \int_{r_1}^T t^6 K(c_1(t)) dt > \frac{1}{1000}.$$

If $a_2 \geq 8$, then

$$\frac{1}{32} - 7 \int_{r_2}^T t^6 Q(c_2(t)) dt > \frac{1}{1000}.$$

The finite boxes and these three unbounded cases cover all finite supporting slopes. No center is chosen at T in this comparison.

Proof. We describe the polynomial rule, its coverage of each box, and the three unbounded ranges. This also fixes the meaning of every recorded integer and gives the correctness criterion for the supplementary checker.

Step 1. In this step, we specify the polynomial bound and validity range for every recorded interval. For a profile with height h , anchor r , and slope a , write $d_0 = ar - h$. Define

$$P_j(t) = 7((1 - ja)t + jd_0)_+^6.$$

On an interval avoiding its zero this is an ordinary polynomial or zero. The label $E1$ means P_1 , and C means $(7/6)^6 P_1$. A prefix 0, 1, 2 specifies the profile. A label Bq means $7qt^6$.

For $q = 3, 5$, the labels $Nqlo$ and $Nqhi$ mean, respectively,

$$7t^6 - 7 \left(\frac{7q}{q+1} \right)^q t^{6-q} (at - d_0)^q \quad (0 < c \leq 1/7),$$

and

$$7t^6 - \frac{7}{(6(q+1))^q} t^{6-q} ((6q-1+7a)t - 7d_0)^q \quad (1/7 < c < 1/2).$$

On the latter range C is another available upper bound. For $c \geq 1/2$, use $E1$.

For Q , set

$$q_- = \frac{16020581569}{90194313216}, \quad q_+ = \frac{8010290789}{45097156608}.$$

The label $Qbal$ is

$$7((1-2a)t + 2d_0)^5((1+10a)t - 10d_0) \quad (0 < c \leq q_-).$$

Here the balanced expression is the larger expression in Q . For $c \geq q_+$, $E1$ is the larger expression. These comparisons amount to the signs of $(1-2c)^5(1+10c) - (1-c)^6$, with its positive parts

retained. For $0 < c < 1/2$, setting $y = (1 - 2c)/(1 - c)$ turns the ratio of the balanced expression to $E_1(c)$ into $y^5(12 - 11y)$. Its derivative is $6y^4(10 - 11y)$, so there is exactly one crossing in this interval. Substitution of the two displayed rational endpoints puts that ratio above one at q_- and below one at q_+ . For $c \geq 1/2$ the balanced expression vanishes. Thus the indicated polynomial is valid on each whole range. No Q option is used in the interval (q_-, q_+) .

The remaining labels are

label	polynomial	validity
$Mspan$	$P_1 + 3P_2 - 5P_3 + 2P_4$	$c > 0$
$Mref$	$P_1 + P_2 - P_3$	$c \geq 2/11$
$H3$	$7(256/729)t^6$	$c \geq 1/6$
$H3lip$	$7(256/729 + 3 - 18a)t^6 + 126d_0t^5$	$0 < c < 1/6$
Dlo	$6P_2 - 8P_3 + 3P_4$	$c > 0$
Dhi	$7(138259/900000)t^6$	$c \geq 1/6$.

Positive-part zeros of every P_j are subdivision points. Signed expressions are kept intact; their negative summands are not estimated independently.

Step 2. In this step, we verify coverage of the bounded domain and sum every exact interval bound. The first and third grids are

$$(131/900, 1/2, 1, 2, 4, 8, 16), \quad (1609/5400, 1/2, 1, 2, 4, 8).$$

The middle grid is $(h_{\min}/r_1, 1/2, 1, 2, 4, 8)$. Consecutive grid entries determine closed root boxes. A tag $i.j.k$ denotes the root with these consecutive intervals, numbered starting at one. Each recorded split gives its coordinate, numbered 0, 1, 2, and its rational midpoint; appending 0 or 1 means the lower or upper child. The checker starts with all 150 root boxes in each family, replaces only an existing leaf by its two children at each listed split, and compares the resulting leaf addresses and coordinates with the entire box list. Each split must lie strictly between the two endpoints of its coordinate. Both closed children cover the parent. The lists have 172, 188, and 150 leaf boxes in families Q, M, D , respectively. The last family needs no split. Thus they cover even the rectangular relaxation of the linked slope constraints, including every face. Even a box whose intersection with the feasible slope constraints is empty is checked on the whole box; no feasibility test discards it. Intersections that are segments or points, as well as equality in a linked-height maximum, are therefore covered by the same endpoint substitutions.

For a box $[l_0, u_0] \times [l_1, u_1] \times [l_2, u_2]$, evaluate h_1 at l_0 and h_2 at (l_0, l_1) . Their defining maxima are increasing in these arguments. For each profile use u_i before its anchor and l_i afterwards. This lowers every supporting line throughout the box. Since the capacities decrease with c , it gives an upper bound for the loss.

Subdivide time at the baseline endpoints, the anchors, the multiples of $1/100$, and all rational solutions of

$$c(t) \in \{0, 1/7, 4/25, 1/6, 2/11, q_-, q_+, 1/4, 1/3, 1/2, 1\}.$$

These include every validity boundary and every positive-part root above. If the slope equals a listed value, no division by their zero difference is made: the corresponding normalized line is either constantly that value or never reaches it. A vanishing supporting line uses the bound one. The recorded interval may merge adjacent pieces having the same polynomial. For checking such a merged interval, subdivide again at every anchor, baseline jump and relevant rational root. On each open subinterval, $c(t) = \max(0, a - d_0/t)$ has a fixed branch and monotonicity, so one exact interior value determines its whole validity range. Reconstructing the polynomial on every such subinterval must give the same coefficients. At a positive-part zero and at an anchor the limiting formulas agree. At a baseline jump use the defined endpoint value; changing a value at finitely many endpoints changes neither integral. At $t = 0$, use the bounded capacity and the zero limit

of its weighted density $7t^6f(t)$. For a polynomial $p(t) = \sum_j p_j t^j$ on $[l, r]$, its recorded integer is

$$(14.7) \quad N_{l,r} = 1 + \left\lfloor 10^{10} \sum_j p_j \frac{r^{j+1} - l^{j+1}}{j+1} \right\rfloor.$$

Hence $N_{l,r}/10^{10}$ is a strict upper integral bound. Taking one available polynomial on each interval bounds the pointwise minimum even if that polynomial is selected by comparing its integral with the others. Each box's full interval list must start at zero, end at T , and have consecutive equal endpoints, with strictly positive interval lengths. The checker consumes every record, reconstructs each polynomial by the dictionary above, checks its strict numerator and sums all its intervals. It rejects missing, repeated, extra or malformed records, invalid branches, and any gap or overlap. The counts and maximum sum numerators are

family	boxes	integration intervals	maximum numerator
Q	172	1445	9986634132
M	188	1827	9989635195
D	150	1442	9796511857.

Every numerator is less than 9990000000. In particular the smallest reserve below the desired upper bound is $364805/10^{10}$. The definitions, the split trees, and (14.7) give an exact rational check of every integration interval. The 510 parameter boxes contain 4714 bounded integration records. The five first-slope tail records in each family bring the total to 4729. All arithmetic is rational; there is no floating-point error. The only enlargement is the explicit strict rounding in (14.7), already included in every sum. No search for an optimal polynomial is a premise: any admissible choice whose checked sum meets the strict bound proves the required inequality.

Step 3. In this step, we bound all three unbounded supporting-slope ranges. For $a_0 \geq 16$, discard the other profiles, use the baseline below r_0 , and use the first profile of slope 16 above it. The five recorded intervals have total numerator 5020326735, less than 9990000000.

For $a_1 \geq 8$, start at the already reached order r_1 . Lower the height to $h_{\min}$ and the slope to eight. In family Q , $h_{\min}/r_1 > q_+$, and the entire remaining loss is at most $(r_1 - h_{\min})^7/7$. In family M , $h_{\min}/r_1 > 2/11$, and it is at most

$$\frac{(r_1 - h_{\min})^7}{7} + \frac{(r_1 - 2h_{\min})^7}{15} - \frac{(r_1 - 3h_{\min})^7}{23}.$$

This integrates the full nonnegative capacity $E_1 + E_2 - E_3$. In family D , use $K_D \leq 6E_2$ and obtain $6(r_1 - 2h_{\min})^7/15$. The respective differences from $3/40$ are

$$\frac{2444639182386909}{8960000000000000}, \quad \frac{165134863173366609}{515200000000000000}, \quad \frac{1029536443789265789}{1953125000000000000},$$

each greater than $1/1000$. For $a_2 \geq 8$, use $h_*/r_2 > q_+$. Its post- r_2 loss is at most $(r_2 - h_*)^7/7$, and the difference from $1/32$ is

$$\frac{5836632000426184167212369}{54687500000000000000000000} > \frac{1}{1000}.$$

The integration rule is

$$\int_r^\infty 7(t - j[h + a(t - r)])_+^6 dt = \frac{(r - jh)^7}{ja - 1} \quad (ja > 1, r - jh > 0).$$

Every argument here satisfies these inequalities. Increasing a slope after its anchor decreases the whole applicable capacity. This covers all unbounded ranges and also the portions where a capacity vanishes. $\square$

Lemma 14.7. *Let X be a smooth complex projective sevenfold, L ample Cartier with $L^7 = 1$, and x a nongeneration point. If a sufficiently late pure actual original 27/25 observation in a degree k divisible by 25 has minimal-center state $(5, 7, 3)$, then*

$$V(29/25) > \frac{1}{1000}, \quad \dim U_q(29/25) \geq \frac{q^7}{2000 \cdot 7!}$$

for every sufficiently large original degree q . The degree cutoff depends only on the original data and the fixed comparisons, and precedes any subsequent threshold or computing place at 29/25.

Proof. Use the three profiles of Lemma 14.4. We first obtain the earlier source reserves, then telescope the common images of Lemma 14.5 and apply the complete certificate to every slope range.

Step 1. In this step, we obtain the earlier source reserves in every original degree. Sum Proposition 7.28 only as far as r_1 . Ample Riemann–Roch and the uniform layer errors give

$$(14.8) \quad V(r_1) \geq A := 1 - \frac{39}{100} \left(\frac{9}{10}\right)^7 - \frac{17}{100} \left(\frac{99}{100}\right)^7 - \frac{11}{25} \left(\frac{26}{25}\right)^7 > \frac{3}{40},$$

and $\dim U_q(r_1) \geq Aq^7/7! - O(q^6)$ in every sufficiently large original degree. The last strict rational inequality follows by writing $A = A_0 + (17/100)[(26/25)^7 - (99/100)^7]$, where

$$A_0 = 1 - \frac{39}{100} \left(\frac{9}{10}\right)^7 - \frac{61}{100} \left(\frac{26}{25}\right)^7 > \frac{1}{100}.$$

The two powers are, respectively, greater than 13159/10000 and less than 9321/10000. Their difference exceeds 1919/5000 $> 13/34$, so $A > 1/100 + (17/100)(13/34) = 3/40$.

At r_2 , Theorem 9.14 gives $V(r_2) > 1/32$. To obtain its finite floor in all original degrees, let m be a denominator of a rational big anchor r and write $q = mp + j$, $0 \leq j < m$. On the original point blowup,

$$q\pi^*L - [qr]E_x = p(m\pi^*L - mrE_x) + j\pi^*L - [jr]E_x.$$

Put $D = m\pi^*L - mrE_x$ and denote the finitely many twists by F_j . Bigness supplies a fixed p_0 for which both $p_0D + F_j$ and $p_0D - F_j$ are effective for all j . Multiplication by their nonzero sections gives

$$h^0((p - p_0)D) \leq h^0(pD + F_j) \leq h^0((p + p_0)D).$$

The complete-volume limit in Theorem 6.4 therefore gives the same leading count in each residue class. In particular, for every sufficiently large q , $\dim U_q(r_2) \geq q^7/(32 \cdot 7!)$. This cutoff depends only on the original data and the fixed anchor.

Step 2. In this step, we telescope the common images with a uniform finite-degree error. For any $0 \leq u < v \leq T$, the exact kernel of each point-order map gives

$$\dim U_q(u) - \dim U_q(v) = \sum_{\ell=\lceil qu \rceil}^{\lceil qv \rceil-1} \dim A_{q,\ell}.$$

The initial space $U_q(0) = H^0(X, qL)$ has dimension $q^7/7! + O(q^6)$. Since $a_i r_i \geq h_i$, each $c_i(t)$ is nondecreasing. Every capacity, and hence their minimum f , is bounded by one and nonincreasing. Multiplication by $7t^6$ gives a function with total variation bounded solely in terms of T , uniformly in all three slopes and linked heights. Its Riemann-sum error is $O(1/q)$. The full-primary errors sum to $O(q^6)$, and the endpoint ceilings contribute another $O(q^6)$. Thus Lemma 14.5 gives

$$(14.9) \quad \dim U_q(T) \geq \frac{q^7}{7!} \left(1 - 7 \int_0^T t^6 f(t) dt\right) - C'q^6.$$

The constant is independent of the observations. The same argument starting at r_1 or r_2 bounds the remaining loss using just that anchor's capacity. None of these equalities assumes that the next complete space is nonzero.

Step 3. In this step, we combine every slope range to obtain the uniform later-source reserve. When $a_0 \leq 16, a_1 \leq 8, a_2 \leq 8$, Lemma 14.6 bounds the full integral in (14.9) by a number strictly below 999/1000. Its 510 closed boxes include every linked-height alternative. When $a_0 \geq 16$, the first unbounded case gives the same bound with the first profile alone after r_0 . When $a_1 \geq 8$, start with (14.8) and apply the appropriate Q, M, D remaining-loss bound. When $a_2 \geq 8$, start with the original 1/32 floor at r_2 and apply the final unbounded case. These alternatives exhaust all finite supporting slopes and all changes of center between the three observations.

The finitely many strict certificate margins, including the unbounded cases, have a positive minimum. Consequently a fixed $\delta > 0$ and a fixed C'' work in every case in the estimate

$$\dim U_q(T) \geq \frac{q^7}{7!} \left(\frac{1}{1000} + \delta \right) - C'' q^6.$$

Taking the normalized volume limsup gives $V(T) > 1/1000$. Increasing the cutoff using only these constants gives the claimed finite lower bound $q^7/(2000 \cdot 7!)$ in every original degree. No threshold at T has been selected. The three observations are alternative trials; their coefficients are not added, and the cubic boundary retains all its original committed cost. $\square$

The preceding proof of the tangent and cycle assertions pays the nonreduced double and establishes the cubic bounds. Lemma 14.7 applies to the same 25-divisible observing degree and supplies the cubic source. Since $U_q(T) \subseteq U_q(t)$ and $V(t) \geq V(T)$ for every $0 \leq t \leq T$, these results prove every assertion of Proposition 14.2, with one cutoff before any later 29/25 observation. The source conclusion holds for every sufficiently large original degree q and every real $0 \leq t \leq T$.

14.2. Supplementary rational certificates. The dataset 14-double-geometry/tail-table is the complete 71-state table in `supplement/14-double-geometry/data/tail-table.txt`. Its columns and the test for every affine line are defined in Lemma 14.1. The dataset 14-double-geometry/three-profile is in `supplement/14-double-geometry/data/three-profile.txt`. For each of Q, M, D , its ordered stream contains the root grids, all binary splits, each leaf's six endpoints, two linked heights and total numerator, every labelled integration interval, and all three unbounded alternatives. Thus the data retain every witness used in Lemma 14.6.

The local `README.md` describes the data format and explains how to reproduce the checks. The file `manifest.json` maps both datasets to the statements they certify. The portable `check.py` reads these data and verifies the finite conditions of the two lemmas using integer and rational arithmetic. State enumeration, the closed split trees, interval partitions, polynomial admissibility and strict rounding are checked independently of the procedure that selected the witnesses. The preceding lemmas explain why those finite checks imply the loss bounds; the source-transfer proof supplies their geometric application and the uniform estimates in every late original degree.

15. COMPLETE LOCAL QUADRATIC MODULES

We retain the point-ideal and symbolic-power conventions of Definition 2.7. We first estimate symbolic images on an integral quadratic tangent. We then treat the two branches of a rank-two hypersurface, keeping the complete convergent divisorial module. The coefficient estimates at the end use the direct adjunction of the whole ambient boundary.

Theorem 15.1 (Analytic splitting and approximation). *For a convergent complex power series f of order at least two and Hessian rank r , an analytic change of coordinates writes*

$$f = x_1^2 + \cdots + x_r^2 + g(x_{r+1}, \dots, x_d), \quad g \in (x_{r+1}, \dots, x_d)^3.$$

No isolatedness assumption is required. A formal solution without constant term to a finite system of convergent analytic equations over $\mathbb{C}$ has, for every integer c , an exact convergent solution agreeing modulo the c -th power of the variable ideal.

The splitting assertion is [GP26, Theorem 2.4]; the exact approximation assertion is [Art68, Theorem (1.2)]. In particular, splitting retains a convergent residual series, with all its higher terms.

A matrix factorization of an element f is a pair of square matrices (ϕ, ψ) satisfying $\phi\psi = \psi\phi = fI$. Its trivial factors are $(1, f)$ and $(f, 1)$. Quotienting by the first removes its zero cokernel; quotienting by both gives the stable category, in which maps through finite free modules are zero. A factorization over a local ring is *reduced* when both matrices have entries in the maximal ideal.

Theorem 15.2 (Matrix factorizations and periodicity). *Let S be a G -regular local ring and f a nonunit nonzerodivisor. Matrix factorizations of f , modulo the specified trivial factorizations, correspond by cokernel to totally reflexive modules over $S/(f)$; stable categories further quotient by free summands. For $B = S[[u, v]]/(f + uv)$, the block construction*

$$(\phi, \psi) \mapsto \left(\begin{pmatrix} \phi & u \\ v & -\psi \end{pmatrix}, \begin{pmatrix} \psi & u \\ v & -\phi \end{pmatrix} \right)$$

is fully faithful on stable categories. It is an equivalence when $S/(f)$ is henselian and $1/2, \sqrt{-1} \in S$. These statements are in the formal category; convergence of a particular module isomorphism is a separate assertion.

We use the cokernel and stable-category assertions of [Tak08, Theorems 2.10 and 3.12]. We will descend the particular isomorphisms needed here by encoding inverse module maps.

Theorem 15.3 (Factoriality and analytic extension). *A Noetherian local complete intersection is factorial if each localization of dimension at most three is factorial. Convergent complex power-series rings are factorial. If A is an analytic subset of a complex space T with $\dim A \leq p$, and $Z \subseteq T \setminus A$ is analytic with $\dim_z Z > p$ at every point, then its closure in T is analytic.*

The factoriality criterion is [GR05, Exposé XI, Corollary 3.14], also stated in [AH20, Theorem 1.4]. The analytic assertions are [Dem12, Chapter II, Theorems 2.10 and 8.7].

Theorem 15.4 (Determinantal presentations and powers). *If an ideal I in a Noetherian ring has projective dimension exactly one and a free resolution $0 \rightarrow R^t \xrightarrow{A} R^{t+1} \rightarrow I \rightarrow 0$, then $I = aI_t(A)$ for a nonzerodivisor a , with $\text{grade } I_t(A) = 2$; conversely these data give the resolution. If I itself has grade two, the scalar factor is a unit. For an m by n matrix A , $m \leq n$, suppose*

$$\text{grade } I_m(A) \geq n - m + 1, \quad \text{grade } I_j(A) \geq (m + 1 - j)(n - m) + 2 \quad (1 \leq j < m).$$

Then $\text{Ass}(R/I_m(A)^k) = \text{Ass}(R/I_m(A))$ for every $k \geq 1$.

The presentation and power criteria are the Hilbert–Burch theorem [ES04, Theorem 2.2] and [BCV15, Lemma 3.3], respectively.

Theorem 15.5 (Symbolic containment). *Let R be regular and essentially of finite type over a characteristic-zero field, and suppose R/I has log-terminal singularities. If $h \geq 2$ is the largest height of a minimal prime of I , then for every $k \geq 1$,*

$$I^{((h-1)k+1)} \subseteq I^{k+1}.$$

This is [GH19, Theorem 5.7], with the height restriction stated in Theorem 1.3 and the underlying Theorem 4.1 of that paper. Its ambient ring is regular.

Theorem 15.6 (Small models and one-dimensional contractions). *On a variety of klt type, a Weil divisor admits a small projective birational model where its strict transform is $\mathbb{Q}$ -Cartier and relatively ample; its relevant section algebra is finitely generated. For a proper contraction from a smooth variety to a normal base, with connected fibers and relatively ample anticanonical divisor, a one-dimensional fiber component forces that fiber to be pure one-dimensional. In the birational case the fiber is a reduced $\mathbb{P}^1$, its anticanonical degree is one, and its image is a smooth point of the base.*

The two inputs are [Zhu24, Lemma 2.5] and [AW96, Proposition 4.1]. The local models used below will be constructed explicitly.

Theorem 15.7 (Volume under a birational map). *For a birational morphism $f : Y \rightarrow T$ of normal varieties and the same valuation v centered at a closed point y over x , $\text{vol}_x(v) \leq \text{vol}_y(v)$. If both varieties are klt, $K_Y \leq f^*K_T$, and v is divisorial, then $A_T(v) \leq A_Y(v)$, as in the proof of the cited lemma. The discrepancy and volume refer to the same valuation in both comparisons.*

The volume comparison is [LX19, Lemma 2.9(1)]. The discrepancy comparison for the same divisorial valuation is recorded in the proof of Lemma 2.9(2) of that paper.

Theorem 15.8 (Negative analytic neighborhoods). *Fix a compact complex submanifold C , its restricted ambient tangent bundle and its first-order neighborhood. If its normal bundle is weakly negative or 2-positive, the compatible holomorphic neighborhood classes inject into a subset of the finite-dimensional space*

$$\bigoplus_{\ell \geq 2} H^1(C, T_C M \otimes \text{Sym}^\ell N_C^*).$$

Identification with a particular split neighborhood additionally requires its specified first-order identification and vanishing of the relevant cohomology groups.

These are the neighborhood conventions of [GS24, Definition 2.1, Theorem 1.1 and Corollary 5.5].

Theorem 15.9. *Let R be a normal complex analytic hypersurface local domain of dimension $d \geq 3$, multiplicity two and quadratic initial rank at least three. For any height-one prime P , set $k = \text{mult}(R/P)$. For all integers $r, j \geq 0$,*

$$P^{(r)} \subset \mathfrak{m}^{\lceil rk/2 \rceil}, \quad \dim \text{im}(P^{(r)} \cap \mathfrak{m}^j \rightarrow \mathfrak{m}^j / \mathfrak{m}^{j+1}) \leq \frac{2}{(d-1)!} (j - rk/2)_+^{d-1} + C_d(j+1)^{d-2}.$$

For $0 \leq h \leq l$ the entire interval has length at most

$$\frac{2}{d!} ((l - rk/2)_+^d - (h - rk/2)_+^d) + C'_d(l+1)^{d-1}.$$

The constants are independent of the prime, symbolic exponent and higher analytic terms. No factoriality, normality of the whole blowup or parity of k is assumed.

The degree of the full tangent intersection, rather than the number of its components, supplies the shift $rk/2$.

Proof. We restrict the symbolic initial image to the exceptional quadric of the ordinary blowup, and then count sections in its divisor classes.

Step 1. In this step, we restrict the full symbolic initial image to the exceptional quadric. Write $R = \mathbb{C}\{z_0, \dots, z_d\}/(f)$, and let q be the quadratic initial form of f . Orders add in the regular ambient ring, so the entire initial ideal of (f) is (q) . Thus

$$\text{gr}_{\mathfrak{m}} R = \mathbb{C}[z_0, \dots, z_d]/(q).$$

This graded ring is a domain. Its projective quadric E is normal: it is Cohen–Macaulay, and its singular locus has codimension $\text{rank}(q) - 1 \geq 2$. Set $Y = \text{Bl}_{\mathfrak{m}} \text{Spec } R$ and $H = \mathcal{O}_E(1)$. The exceptional scheme is this entire quadric, and it is an effective Cartier divisor on the integral scheme Y .

At a codimension-one point ξ of E , the ring $\mathcal{O}_{E,\xi}$ is regular of dimension one. Its Cartier equation in $\mathcal{O}_{Y,\xi}$, together with a lift of a uniformizer, generates the maximal ideal by Nakayama's lemma. Hence $\mathcal{O}_{Y,\xi}$ is regular of dimension two. This assertion concerns only these local rings of the blowup.

The graded surjection

$$R[\mathfrak{m}\tau] \longrightarrow (R/P)[(\mathfrak{m}/P)\tau]$$

has degree- j kernel $P \cap \mathfrak{m}^j$. Its projective image is the integral strict transform P_Y . Its intersection with E is $\text{Proj gr}_{\mathfrak{m}/P}(R/P)$, with its scheme structure. In the regular two-dimensional rings just found, P_Y is Cartier. Its full restriction defines an effective Weil divisor C_P on E , and the Hilbert–Samuel degree gives

$$H^{d-2} \cdot C_P = k.$$

In particular, the intersection lengths are retained.

For $g_0 \in P^{(r)} \cap \mathfrak{m}^j$, divide its pullback by the j -th power of a local equation of E . The quotient is regular because $g_0 \in \mathfrak{m}^j$. In each of the same two-dimensional regular local rings, it is still divisible to order at least r by the equation of P_Y . Restriction to E and normality of E therefore give

$$(15.1) \quad \text{im}(P^{(r)} \cap \mathfrak{m}^j \longrightarrow \mathfrak{m}^j / \mathfrak{m}^{j+1}) \subseteq H^0(E, \mathcal{O}_E(jH - rC_P)).$$

The initial image injects into sections because the full graded hypersurface ring has no irrelevant torsion.

Step 2. In this step, we count sections in the three possible divisor-class configurations.

Case 2.1. The quadratic rank is at least five. In rank at least five, put $q = uv - q'(z)$, with q' an irreducible quadratic form. The chart $u \neq 0$ is affine $(d-1)$ -space, and its complement is one reduced prime hyperplane. Divisor localization shows that the class group is generated by H . There is no relation: a rational function with divisor supported on the complement is a unit on that affine chart, hence a constant. Consequently C_P has class $(k/2)H$. In this case k is even. For the integer $T = j - rk/2 \geq 0$,

$$h^0(E, \mathcal{O}_E(T)) = \binom{T+d}{d} - \binom{T+d-2}{d} = \frac{2T^{d-1}}{(d-1)!} + O_d((T+1)^{d-2}).$$

The space is zero for $T < 0$.

Case 2.2. The quadratic rank is four. In rank four write $q = uv - s_1s_2$, with $d-3$ additional coordinates. The complement of the same affine chart has primes $D_1 = (u = s_1 = 0)$ and $D_2 = (u = s_2 = 0)$. Divisor localization and the constant-unit argument give $\text{Cl}(E) = \mathbb{Z}D_1 \oplus \mathbb{Z}D_2$ and $H = D_1 + D_2$. For $t \geq 1$ put $v_t(\ell) = \binom{\ell+t-1}{t-1}$; put $v_0(0) = 1$ and $v_0(\ell) = 0$ for $\ell > 0$. The affine monomials in $s_1/u, s_2/u, z_i/u$, counted by their poles along both primes, give

$$h^0(E, \mathcal{O}_E(A_1D_1 + A_2D_2)) = \sum_{\ell \geq 0} v_{d-3}(\ell) (A_1 - \ell + 1)_+ (A_2 - \ell + 1)_+.$$

The monomials of maximal pole have independent residues, so their coefficients cannot cancel. An effective class has both coefficients nonnegative. Write the class of C_P as $a_1D_1 + a_2D_2$, where $a_1 + a_2 = k$, and substitute $A_i = j - ra_i$. Each nonzero product is at most $(T - \ell + 1)^2$, where $T = j - rk/2$. There is no nonzero summand when $T < 0$. For $d \geq 4$, the leading integral is

$$\int_0^T \frac{\ell^{d-4}(T-\ell)^2}{(d-4)!} d\ell = \frac{2T^{d-1}}{(d-1)!}.$$

For $d = 3$, the sole term is at most $(T+1)^2$.

Case 2.3. The quadratic rank is three. In rank three write $q = uv - w^2$. The complementary prime $D_0 = (u = w = 0)$ generates $\text{Cl}(E)$, and $H = 2D_0$. The coordinate w/u has pole one and each of the other $d-2$ affine coordinates has pole two. Hence, for an integer $A_0 \geq 0$,

$$h^0(E, \mathcal{O}_E(A_0D_0)) = \sum_{\ell=0}^{\lfloor A_0/2 \rfloor} \binom{\ell+d-3}{d-3} (A_0 - 2\ell + 1).$$

The space is zero for $A_0 < 0$. The class of C_P is kD_0 ; thus $A_0 = 2j - rk = 2T$. The leading integral is

$$\int_0^T \frac{2\ell^{d-3}(T-\ell)}{(d-3)!} d\ell = \frac{2T^{d-1}}{(d-1)!}.$$

In all three counts, a nonzero term has $0 \leq T \leq j$. Replacing polynomial sums by these integrals and accounting for their endpoints costs $O_d((j+1)^{d-2})$, uniformly in all divisor coefficients. Equation (15.1) proves the layer estimate. The common zero range gives the asserted containment, by applying it at the ordinary order of any nonzero element of $P^{(r)}$.

Step 3. In this step, we sum the actual layer estimates over the full symbolic interval. Finally, sum the actual layer dimensions over $h \leq j < l$. The sum of the errors is $O_d((l+1)^{d-1})$. The shifted nonnegative power has the same sum-to-integral error, uniformly in the shift. Its integral is the stated difference of positive parts. $\square$

Theorem 15.10. *Let $d \geq 2$, $0 \neq \psi \in \mathbb{C}\{z_1, \dots, z_{d-1}\}$ have order $D \geq 3$, and set $R = \mathbb{C}\{u, v, z\}/(uv - \psi)$. Let ν_i have weights $(i, D-i, 1, \dots, 1)$. For $f \neq 0$, set*

$$\alpha = \max(\nu_1 f, \nu_{D-1} f), \quad \beta = \min(\nu_1 f, \nu_{D-1} f) = \text{ord}_{\mathfrak{m}} f, \quad \zeta = ((D-1)\alpha - \beta)/(D-2).$$

For $N \geq 0, 0 \leq h \leq l$, write $\Phi_{h,l}(z) = (l-z)_+^d - (h-z)_+^d$. Then

$$\text{length} \frac{(f^N) \cap \mathfrak{m}^h}{(f^N) \cap \mathfrak{m}^l} = \frac{D \Phi_{h,l}(N\beta) + (D-2)\Phi_{h,l}(N\zeta)}{(D-1)d!} + O_{d,D}((l+1)^{d-1}).$$

Its leading term is at most $2\Phi_{h,l}(N \text{ord}_{\mathfrak{m}} f)/d!$. The error is uniform in f, N, h, l and the higher terms. The initial form of f need not be a nonzerodivisor.

The endpoint valuations describe the filtration on the entire principal ideal even when its initial equation vanishes on a tangent component.

Proof. We describe the ordinary filtration by both endpoint valuations and then count its two complete normal-form branches.

Step 1. In this step, we express the exact ordinary filtration using both endpoint valuations. Convergent division by $uv - \psi$ gives the unique normal form

$$H = \sum_{q \geq 0} u^q H_q(z) + \sum_{q \geq 1} v^q K_q(z).$$

The analogous formal decomposition gives the same finite quotients. After substituting $v = \psi/u$, it embeds the completion in $\mathbb{C}((u))[[z]]$: each fixed total z -degree has only finitely many negative u -powers. This proves that the ring is a domain. At weight $(i, D-i, 1, \dots, 1)$ the initial equation is $uv - \psi_D$, an irreducible polynomial. Thus the weight filtration is a valuation on the actual ring.

The normal-form monomials have endpoint weights

$$u^q z^I : (q + |I|, (D-1)q + |I|), \quad v^q z^I : ((D-1)q + |I|, q + |I|).$$

They are independent in each weighted initial ring as well as in $\text{gr}_{\mathfrak{m}} R = \mathbb{C}[u, v, z]/(uv)$. It follows that

$$(15.2) \quad \text{ord}_{\mathfrak{m}} H = \min\{\nu_1(H), \nu_{D-1}(H)\}.$$

Injective multiplication by f^N pulls its ordinary filtration back to

$$\text{ord}_{\mathfrak{m}}(f^N H) = \min\{N\nu_1(f) + \nu_1(H), N\nu_{D-1}(f) + \nu_{D-1}(H)\}.$$

Step 2. In this step, we count both full branches and derive the principal interval bound. Interchange the branches if necessary so that the larger endpoint of f is paired with the smaller u -weight. At a cutoff $0 \leq T \leq l$, put $X_T = T - N\alpha$ and $Y_T = T - N\beta$. The first branch is the union

$$q + |I| < X_T \quad \text{or} \quad (D-1)q + |I| < Y_T, \quad q \geq 0.$$

The second is the simplex $q + |I| < Y_T$, with $q \geq 1$. Replacing this last range by $q \geq 0$ changes the count by $O_d((l+1)^{d-1})$.

In the first branch the two boundaries meet at

$$q_0 = \frac{N(\alpha - \beta)}{D - 2}, \quad Z_T = \frac{(D - 1)X_T - Y_T}{D - 2} = T - N\zeta.$$

Integrating the larger of $(X_T - q)_+^{d-1}$ and $(Y_T - (D - 1)q)_+^{d-1}$, divided by $(d - 1)!$, gives

$$\frac{(Y_T)_+^d + (D - 2)(Z_T)_+^d}{(D - 1)d!}.$$

This remains valid if the crossing height is nonpositive. The second branch contributes $(Y_T)_+^d/d!$. Thus the whole quotient $(f^N)/((f^N) \cap \mathfrak{m}^T)$ has leading volume

$$(15.3) \quad \frac{D(T - N\beta)_+^d + (D - 2)(T - N\zeta)_+^d}{(D - 1)d!}.$$

Every nonempty region has intercepts at most l and belongs to a fixed finite set of facet directions depending only on d, D . Unit cubes meeting its boundary lie in bounded-width facet slabs, whose total number is $O_{d,D}((l + 1)^{d-1})$. Interior cubes give the exact volume count. This proves the uniform error for the union and its intersection, without a bound on f or N .

Subtract the complete counts at $T = l$ and $T = h$. Their difference is the stated ideal interval. Finally,

$$\Phi_{h,l}(z) = \int_h^l d(t - z)_+^{d-1} dt$$

is nonnegative and decreasing in z . Since $\zeta \geq \beta = \text{ord}_{\mathfrak{m}} f$ and the two coefficients in the formula sum to two, replacing both shifts by $N\beta$ proves the last bound. $\square$

Lemma 15.11. *Let $d \geq 4$, let*

$$R = \mathbb{C}\{u, v, z_1, \dots, z_{d-1}\}/(uv - \psi), \quad \psi = \prod_{\lambda=1}^{s_0} a_{\lambda}^{M_{\lambda}}, \quad D = \text{ord } \psi \geq 3,$$

where the a_{λ} are pairwise nonassociate irreducible convergent nonunits and $M_{\lambda} \geq 1$. Let P be a non-Cartier height-one prime such that R/P is normal Cohen–Macaulay and has embedding dimension $d + 1$. Then there are integers $0 \leq q_{\lambda} \leq M_{\lambda}$, nonunits

$$a_+ = \prod_{\lambda} a_{\lambda}^{q_{\lambda}}, \quad a_- = \psi/a_+,$$

and a meromorphic function γ such that, as convergent ideals,

$$P = \gamma(u, a_+), \quad \gamma = f_+(u, z) + (a_-/u)f_-(v, z), \quad f_+, f_- \in \mathfrak{m}.$$

Set $\delta = \text{ord } a_+$, $h_1 = \nu_1(\gamma)$, $h_2 = \nu_{D-1}(\gamma)$, and $\rho_+ = h_1$, $\rho_- = h_2 + \delta - 1$, using the endpoint valuations of [Theorem 15.10](#). Then

$$\rho_+, \rho_- \geq 1, \quad \text{ord } f_+ \geq \rho_+, \quad \text{ord } f_- \geq \rho_-, \quad \text{mult}(R/P) = h_1 + h_2 + \delta.$$

For every integer $N \geq 1$, the complete normal forms of the divisorial module $\gamma^{-N}P^{(N)}$ are

$$\sum_{q \geq 0} u^q A_{N,q}(z) F_q(z) + \sum_{q \geq 1} v^q a_+(z)^N G_q(z), \quad A_{N,q} = \prod_{\lambda} a_{\lambda}^{(Nq_{\lambda} - qM_{\lambda})_+}.$$

These completed modules have the same finite point quotients as the convergent symbolic ideals.

This presentation records both branches and every irreducible factor of the actual residual series.

Proof. We first obtain an isomorphism in the complete category, then descend that isomorphism exactly, and finally compute its divisorial powers.

Step 1. In this step, we identify the completed rank-one module by matrix periodicity and cancellation. The ring R is a normal hypersurface domain. The normal-form embedding used in [Theorem 15.10](#) proves integrality, also after completion. Its singular locus is contained in

$u = v = \psi = 0$, of codimension two; the hypersurface is Cohen–Macaulay, so the height-one regularity criterion proves normality. The exact sequence

$$0 \longrightarrow P \longrightarrow R \longrightarrow R/P \longrightarrow 0$$

makes P a rank-one maximal Cohen–Macaulay module. Complete over $\mathbb{C}[[z_1, \dots, z_{d-1}]]$. This base is regular, hence G-regular; its quotient by the nonzero nonunit ψ is complete, hence henselian, and contains $1/2$ and $\sqrt{-1}$. Over these hypersurfaces the totally reflexive modules are the maximal Cohen–Macaulay modules. Thus [Theorem 15.2](#) applies in its stable category.

We spell out the cancellation needed here. A unit entry in a factorization splits off one of $(1, \psi)$ or $(\psi, 1)$ by invertible row and column operations. The remaining matrices are reduced. If maps between two reduced factorizations are inverse modulo maps through trivial factors, their compositions are identities modulo the maximal ideal. Indeed the two components of a map through either trivial factor contain an entry of one of the reduced matrices. Reducing modulo the maximal ideal makes those components zero. The two maps therefore have square invertible component matrices by Nakayama’s lemma; they give an actual isomorphism of the reduced factorizations. Consequently stable module isomorphism identifies the reduced parts after the free cokernels have been split off.

A reduced factorization of size t over the base maps to a reduced block factorization of size $2t$. After inverting u , elementary operations change its presenting matrix to $\text{diag}(uI_t, 0)$, so its cokernel has rank t . The nonfree rank-one module $\hat{P}$ has no free summand. The preceding cancellation therefore forces $t = 1$. Its source is a scalar factorization $\psi = a_+a_-$ into nonunits. The matrix

$$\begin{pmatrix} -a_- & u \\ v & -a_+ \end{pmatrix}$$

has cokernel (a_+, u) : the row (a_+, u) kills its columns, and reduction modulo u , where the annihilator of a_+ is (a_-) , proves exactness at the middle module.

Step 2. In this step, we descend the completed module isomorphism to an exact convergent isomorphism. A convergent irreducible a_λ remains irreducible in the complete base. Otherwise a formal solution of $UV - a_\lambda = 0$ with U, V nonunits, approximated modulo the maximal ideal by [Theorem 15.1](#), gives an exact convergent factorization into nonunits. Unique factorization in the complete base therefore lets us choose the displayed convergent product a_+ .

For completeness, choose analytic presentation matrices C_P, C_I for P and $I = (u, a_+)$. Matrices F, G inducing inverse maps are encoded by the finite equations

$$FC_P = C_I U, \quad GC_I = C_P V, \quad GF - \text{id} = C_P A, \quad FG - \text{id} = C_I B.$$

All matrix sizes are those of the chosen finite presentations. Lift the entries to the regular convergent ambient ring and add an unknown multiple of $uv - \psi$ to each scalar equation. The completed isomorphism supplies a formal solution. Subtract its constant entries from the unknowns before applying [Theorem 15.1](#); all new unknown series then have zero constant term. Exact approximation yields convergent matrices satisfying all four equations. They induce mutually inverse analytic module maps. An isomorphism between rank-one torsion-free modules is multiplication in the fraction field, so $P = \gamma I$.

The full inverse ideal is

$$I^{-1} = R + (a_-/u)R.$$

To see this, write $u\gamma = H \in R$. The condition $a_+H \in uR$, reduced modulo u , says $H \in (u, a_-)$; the reverse inclusion gives the same conditions. Thus

$$P = (uf_+ + a_-f_-, a_+f_+ + vf_-).$$

A nonzero constant in either coefficient produces an uncanceled linear u - or v -term in this ideal. The maximal embedding dimension forces both coefficients to vanish. Their full u, v normal forms, together with $uv = a_+a_-$, absorb the unwanted branches into the other coefficient. This gives precisely $f_+(u, z)$ and $f_-(v, z)$, still vanishing at the origin.

Step 3. In this step, we compute all divisorial powers, endpoint orders, and the prime multiplicity. The primes over u are $Q_\lambda = (u, a_\lambda)$; at their generic points u has order M_λ . The ideal I has divisor $\sum_\lambda q_\lambda Q_\lambda$. At such a point substitute $u = \psi/v$ into the complete normal form. The negative powers of v , the constant term, and its positive powers have independent coefficients over the base. Thus divisorial order at least Nq_λ requires the coefficient of u^q to be divisible by $a_\lambda^{(Nq_\lambda - qM_\lambda)_+}$, and every coefficient of v^q , $q \geq 1$, to be divisible by $a_\lambda^{Nq_\lambda}$. Intersect these conditions in the factorial base. Normality identifies their intersection with the entire divisorial module in the statement. Multiplication by γ^N gives $P^{(N)}$. Finite Hom commutes with flat completion; applying it twice shows that reflexive powers commute with completion. All finite point quotients therefore agree.

The separated branches in $u\gamma = uf_+ + a_-f_-$ give

$$h_1 = \min\{\text{ord } f_+, D - \delta - 1 + \nu_1(f_-)\}, \quad h_2 = \min\{\nu_{D-1}(f_+), 1 - \delta + \text{ord } f_-\}.$$

In particular $h_1 \geq 1$, $h_2 \geq 2 - \delta$, with the claimed lower bounds on the two coefficient orders. For $N = 1$, $A_{1,0} = a_+$ and $A_{1,q} = 1$ for $q \geq 1$. The eventual point thresholds on the two branches of P are $h_1 + q$ and $h_2 + \delta + q$. Finitely many exceptional thresholds change the cumulative count by $O_P(l^{d-2})$: each is a bounded shift of the Hilbert function in $d - 1$ base variables. Hence

$$\text{length} \frac{P}{P \cap \mathfrak{m}^l} = \binom{l - h_1 + d - 1}{d} + \binom{l - h_2 - \delta + d - 2}{d} + O_P(l^{d-2}).$$

Subtract this from

$$\text{length}(R/\mathfrak{m}^l) = \binom{l + d - 1}{d} + \binom{l + d - 2}{d}.$$

The coefficient of $l^{d-1}/(d-1)!$ is $h_1 + h_2 + \delta$, proving the multiplicity assertion. $\square$

Lemma 15.12. *Let $d \geq 4$, let $a \in \mathbb{C}\{z_1, \dots, z_{d-1}\}$ be an irreducible nonunit, and let $m_0 \geq 2$. Then*

$$S = \mathbb{C}\{r, t, z\}/(rt - a)$$

is factorial. The action of μ_{m_0} with weights $(1, -1, 0, \dots, 0)$ has invariant ring

$$R_0 = \mathbb{C}\{u, v, z\}/(uv - a^{m_0}), \quad u = r^{m_0}, \quad v = t^{m_0},$$

and is unramified in codimension one. If P_0 is a height-one prime of R_0 , its full reduced pullback has a semi-invariant generator f_0 , of some character k_0 modulo m_0 . Writing S_j for the entire character- j submodule, pullback identifies

$$P_0^{(N)} = f_0^N S_{-Nk_0} \quad (N \geq 1).$$

Every P_0 is rational Cartier, and it is Cartier if and only if $k_0 = 0$.

The pullback here includes every prime above P_0 , even when that pullback is reducible.

Proof. We prove factoriality of the cover, identify its invariants, and then determine the full symbolic character modules.

Step 1. In this step, we prove factoriality of the convergent cover. If a has order one, S is a convergent regular power-series ring and Theorem 15.3 applies. Otherwise Artin approximation, as in the preceding lemma, shows that a is irreducible in $\mathbb{C}[[z]]$. Put $T = \mathbb{C}[[r, t, z]]/(rt - a)$. The elements r and $rt - a$ are relatively prime in the ambient regular factorial ring, so r is a nonzerodivisor in T . Since $T/(r)$ is a domain and the r -adic filtration is separated with graded ring $(T/(r))[R_1]$, T is a domain. The ideal (r, t) has height two. At a height-one prime one of r, t is a unit, and the corresponding partial derivative of $rt - a$ is a unit. Thus T is regular in codimension one; its hypersurface structure gives Serre's condition S_2 , so it is normal.

Apply Theorem 15.2 over the complete regular base $\mathbb{C}[[z]]$. Its hypotheses are the same as in the preceding lemma. Every maximal Cohen–Macaulay module is, up to free summands, a block cokernel. On inverting r , its presentation becomes $\text{diag}(rI, 0)$, so it becomes stably free.

For any rank-one reflexive T -module, take its $2d$ -th syzygy. The depth lemma makes that syzygy maximal Cohen–Macaulay. The determinant relations of the $2d$ free-cover sequences identify its divisor class with that of the original module. Thus every class restricts trivially to $T[1/r]$. Divisor localization makes it a sum of classes of height-one primes over r . There is just (r) , with coefficient one in the divisor of r , because $T/(r)$ is a domain. This class is principal, so T is factorial.

Completion sends every finite rank-one reflexive S -module to a reflexive module over T , by the double-Hom base-change identity. That completion is free. Faithful flatness and Nakayama’s lemma give one generator over S ; torsion-freeness makes it free. This proves factoriality of S .

Step 2. In this step, we identify the invariant ring and a semi-invariant generator of the full pullback. The complete r - and t -branch normal forms show that invariants are exactly R_0 . Convergent division gives the same assertion analytically. A nonidentity group element fixes precisely $r = t = a = 0$, of codimension two. Hence no height-one ramification occurs. Factoriality makes the full reduced pullback of P_0 principal. The group acts on its one-dimensional generator space modulo the maximal ideal. Projection of a generator to that character remains a generator by Nakayama’s lemma; denote it by f_0 .

Step 3. In this step, we identify the full symbolic modules and the Cartier character criterion. An invariant function has order at least N along P_0 if and only if its pullback is divisible by f_0^N on every component. Normality makes this divisorial divisibility equivalent to membership in $f_0^N S$. Invariance forces the remaining factor to have character $-Nk_0$. This proves the equality of the full symbolic modules. The invariant $f_0^{m_0}$ has divisor $m_0 P_0$ downstairs. Character zero gives a Cartier equation; conversely a Cartier equation generates the pullback with trivial character on its generator space. $\square$

Theorem 15.13. *Let W be an actual d -dimensional minimal center of the entire boundary B , $d \geq 4$, with analytic ring $R = \mathbb{C}\{u, v, z_1, \dots, z_{d-1}\}/(uv - \psi)$, $D = \text{ord } \psi \geq 3$. Let P be a height-one prime with R/P normal Cohen–Macaulay and embedding dimension $d + 1$. Write $b = \text{def}_x(B)$ and c_P for the coefficient of P in the same original direct discriminant. For its full symbolic powers there are numbers*

$$\alpha + \beta = \text{mult}(R/P), \quad \alpha \geq \beta \geq 1 + 1/D, \quad \zeta = ((D-1)\alpha - \beta)/(D-2) \geq \beta$$

such that for all integers $N \geq 1$ and $0 \leq h \leq l$, uniformly in P, N, h, l at fixed d, D ,

$$\text{length} \frac{P^{(N)} \cap \mathfrak{m}^h}{P^{(N)} \cap \mathfrak{m}^l} \leq \frac{D \Phi_{h,l}(N\beta) + (D-2)\Phi_{h,l}(N\zeta)}{(D-1)d!} + C_{d,D}(l+1)^{d-1}.$$

For singular irreducible powers, and for smooth powers with $d = 5$, this is equality up to the stated error. It includes the full opposite-character modules and every pullback component. Every later-deficit assertion below assumes an effective lc addition to the entire B selects P as its actual codimension-one minimal center, with positive deficit y .

If ψ has at least two distinct irreducible factors, then for non-Cartier P ,

$$2b + (\text{mult}(R/P) + 1)c_P \leq 2d - 2, \quad \text{mult}(R/P) + 2y \leq 2d - 3.$$

For Cartier P the corresponding bounds are

$$b + \lceil \text{mult}(R/P)/2 \rceil c_P \leq d - 1, \quad y \leq d - 1 - \lceil \text{mult}(R/P)/2 \rceil.$$

If $\psi = a^M$ with a irreducible of order $r_a \geq 2$, $M \geq 2$, the cover $rt' = a$, with $u = r^M, v = (t')^M$, is factorial and unramified in codimension one. For a full pullback generator f of character $k \bmod M$, let ν_i have weights $(i/M, r_a - i/M, 1, \dots, 1)$, $1 \leq i \leq D - 1$, and let α, β be the maximum and minimum of $\nu_1(f), \nu_{D-1}(f)$. Then

$$b + c_P \nu_i(f) \leq d - 1, \quad b + \alpha c_P \leq d - 1, \quad y \leq d - 1 - \alpha.$$

For non-Cartier P , $1 \leq k < M$, and also

$$b + \left(1 + \frac{r_a k(M-k)}{M}\right) c_P \leq d-1, \quad y \leq d-2 - \frac{r_a k(M-k)}{M} \leq d-3.$$

For smooth powers restrict to $d = 5$ and write $\psi = z^M$, $M = D \geq 3$. The cover $\mathbb{C}\{a, c, t_1, t_2, t_3\}$, with $u = a^M, v = c^M, z = ac$, is regular and every P is rational Cartier. Use weights $\nu_i = (i/M, (M-i)/M, 1, 1, 1)$ and the same endpoint maximum/minimum convention for α, β . Then

$$b + c_P \nu_i(f) \leq 4, \quad b + \alpha c_P \leq 4, \quad y \leq 4 - \alpha.$$

For non-Cartier character $k \neq 0$ one additionally has

$$b + \left(1 + \frac{k(M-k)}{M}\right) c_P \leq 4, \quad b + (2 - 1/M) c_P \leq 4, \quad y \leq 3 - \frac{k(M-k)}{M} \leq 2 + 1/M.$$

An embedding-six cubic prime in this smooth-power case is non-Cartier. If ψ is irreducible of first power, the original ring is factorial.

The interval estimate concerns the original symbolic ideal. The covers and small modifications below are used to describe that ideal and to measure the same old discriminant coefficient.

Proof. We first bound the full symbolic intervals. We then identify the exact cover modules in the power cases, and finally compare the old and new discriminants on the required analytic models. All assertions involving y use the effective lc addition and positive later deficit specified in the statement.

Step 1. In this step, we bound the full symbolic interval for an arbitrary residual factorization. Factor the actual convergent ψ , absorbing its unit into one factor. If ψ is irreducible, the first assertion of [Lemma 15.12](#), with $a = \psi$, makes R factorial. If $P = (g)$ is Cartier, the maximal embedding dimension implies $g \in \mathfrak{m}^2$. The endpoint orders of g are integers at least two. The branch multiplicity calculation in [Lemma 15.11](#), with the divisorial module R in place of (u, a_+) , gives

$$\text{mult}(R/(g)) = \nu_1(g) + \nu_{D-1}(g).$$

Thus their maximum and minimum are admissible α, β , and [Theorem 15.10](#) supplies the required full interval, with equality up to its uniform error. This argument also excludes a Cartier prime of multiplicity three and maximal embedding dimension.

For non-Cartier P , use the notation of [Lemma 15.11](#), and put

$$e_1 = h_1 + \frac{\delta}{D}, \quad e_2 = h_2 + \frac{(D-1)\delta}{D}, \quad \alpha = \max(e_1, e_2), \quad \beta = \min(e_1, e_2).$$

That lemma gives

$$e_1 + e_2 = \text{mult}(R/P), \quad e_1 = \rho_+ + \delta/D, \quad e_2 = \rho_- + 1 - \delta/D.$$

Since $1 \leq \delta \leq D-1$, both e_i are at least $1 + 1/D$.

Write $d_\lambda = \text{ord } a_\lambda$ and

$$c_N(q) = \sum_{\lambda} d_\lambda (Nq_\lambda - qM_\lambda)_+.$$

The two endpoint valuations extend to the fraction field. By [\(15.2\)](#), the exact point thresholds of the coefficient series in the full module are

$$\begin{aligned} c_N(q) + \min\{Nh_1 + q, Nh_2 + (D-1)q\}, & \quad q \geq 0, \\ N\delta + \min\{Nh_1 + (D-1)q, Nh_2 + q\}, & \quad q \geq 1. \end{aligned}$$

There is no cancellation between these coefficient spaces: in each weighted initial ring their distinct normal-form powers are independent, and multiplication by γ^N is injective.

For $s \geq 1$ put $J(s) = \binom{s+d-2}{d-1}$, and put $J(s) = 0$ for $s \leq 0$. A coefficient space with integer point shift t contributes

$$J(l-t) - J(h-t).$$

This is decreasing in t : its difference at $t, t+1$ is the difference of the corresponding nondecreasing binomial functions of degree $d-2$. Since

$$c_N(q) \geq (N\delta - Dq)_+,$$

replacing $c_N(q)$ by the right-hand side bounds each entire coefficient interval from above, before any summation.

The resulting reference module is the full divisorial power of (u, z_1^δ) on $uv = z_1^D$, with the external shifts Nh_1, Nh_2 . On its regular cover

$$u = p_1^D, \quad v = p_2^D, \quad z_1 = p_1 p_2,$$

it is $p_1^{N\delta}$ times the entire character- $-N\delta$ module. This is a comparison of explicit coefficient spaces; it does not replace the original germ. The character condition on the exponents is one coset of the index- D lattice $\{\ell_{p_1} - \ell_{p_2} \equiv 0 \pmod{D}\}$. The total endpoint shifts are Ne_1, Ne_2 .

At a cutoff T , put

$$X_T = T - N\alpha, \quad Y_T = T - N\beta, \quad Z_T = \frac{(D-1)X_T - Y_T}{D-2} = T - N\zeta.$$

The quotient is counted in the union of the two weighted d -simplices with these intercepts. After division by the character index, the larger simplex has volume $DY_{T,+}^d / ((D-1)d!)$. At a fixed sum w of the other $d-2$ exponents, the part of the other triangle outside it has area

$$\frac{D(D-2)}{2(D-1)} (Z_T - w)_+^2.$$

Integration against $w^{d-3} dw / (d-3)!$, followed by division by D , gives $(D-2)Z_{T,+}^d / ((D-1)d!)$. Thus the reference volume is (15.3).

All total shifts are positive. Every nonempty simplex for $T \leq l$ lies in a box of side at most $D(l+1)$. Fundamental boxes for the character lattice have fixed size; the boxes meeting the fixed facet directions number $O_{d,D}((l+1)^{d-1})$, independently of the chosen coset and all shifts. Subtract the two complete reference counts at l and h . The termwise interval comparison already proved gives the asserted upper bound with uniform error. This works also for a single smooth factor, in any dimension in the hypotheses of the theorem.

Step 2. In this step, we identify the power covers and compute their exact full character intervals. For $\psi = a^M$ with a singular and irreducible, Lemma 15.12 gives the factorial cover $rt' = a$, its codimension-one unramifiedness, and the exact equality

$$(15.4) \quad P^{(N)} = f^N S_{-Nk}.$$

Here f generates the full reduced pullback. The same lemma proves factoriality of the original ring when ψ itself is irreducible, by taking $a = \psi$ in its factoriality assertion. For a smooth power with $d=5$, the cover is exactly the five-variable regular ring $\mathbb{C}\{a, c, t_1, t_2, t_3\}$ in the statement. Its invariants are R , its nontrivial fixed loci have codimension two, and the proof of the lemma gives (15.4) in this case as well.

These covers give the same endpoint numbers chosen above. For a non-Cartier prime in a single-power case the presentation has $a_+ = a^q$, $1 \leq q < M$, and $\delta = q r_a$. The full divisorial pullback of (u, a_+) is (r^q) . Hence f differs from γr^q by a unit, and

$$\nu_1(f) = h_1 + \frac{q}{M} = e_1, \quad \nu_{D-1}(f) = h_2 + \frac{(D-1)q}{M} = e_2.$$

For the smooth cover use $r_a = 1$ and its coordinate a in place of r . Cartier generators agree up to a unit as well. Thus neither α, β nor their associated shift changes when we pass to the cover description.

For the singular cover the weighted initial equation is $rt' - a_{r_a}(z)$, a domain. Thus all the displayed rational weights give valuations on the actual cover. On the original ring their weights are $(i, D-i, 1, \dots, 1)$; the weighted initial equation $uv - a_{r_a}^M$ is again a domain. In the smooth

case the specified weights are monomial valuations on the regular cover and restrict to the same original weights with $D = M$. In both cases weighted hypersurface adjunction gives

$$(15.5) \quad A_W(\nu_i) = (D + d - 1) - D = d - 1, \quad \nu_i(\mathfrak{m}) = 1.$$

The base coordinates of weight one fix the primitive normalization.

The invariant f^M has divisor MP downstairs. Applying the principal multiplicity calculation above and the associativity formula for multiplicity gives

$$M \operatorname{mult}(R/P) = \nu_1(f^M) + \nu_{D-1}(f^M) = M(\alpha + \beta).$$

This uses every component of the pullback. If $k \neq 0$, the full normal form on the singular cover is

$$(15.6) \quad f = r^k F(u, z) + (t')^{M-k} G(v, z).$$

If F is a unit, the invariant $r^{M-k} f \in P$ has a nonzero linear u -term. If G is a unit, an invariant multiple has a nonzero linear v -term. Both contradict maximal embedding dimension. Hence F, G vanish. Every nonconstant invariant variable has ν_i -weight at least one, so $\beta \geq 1 + 1/M \geq 1 + 1/D$. For character zero f is a Cartier equation in $\mathfrak{m}^2$, giving $\beta \geq 2$. The identical argument in the smooth cover replaces r, t' by a, c and uses the three free parameters.

The exact interval count in the singular cover uses all normal forms $r^q z^I$, $q \geq 0$, and $(t')^q z^I$, $q \geq 1$. In a fixed character space each q lies in one residue class modulo M . Write $q/M = j + \epsilon/M$, $0 \leq \epsilon < M$. The endpoint weights on one branch are $(j + |I|, (D-1)j + |I|)$, up to shifts bounded by D , and are reversed on the other branch. For every element H of the opposite-character module,

$$\operatorname{ord}_{\mathfrak{m}}(f^N H) = \min\{N\nu_1(f) + \nu_1(H), N\nu_{D-1}(f) + \nu_{D-1}(H)\}.$$

The proof of (15.3) now counts this entire module. Residue shifts and the possible zero-exponent overlap affect only $O_{d,D}((l+1)^{d-1})$ boundary boxes. It gives equality up to that uniform error. For the smooth five-dimensional cover, one can either use the same two-branch count or the full index- M lattice in the exponents of a, c, t_1, t_2, t_3 . At a fixed sum w of the three free exponents, integrate the same triangle areas against $w^2 dw/2$ and divide by M . The result is precisely the asserted equality with denominator $5! = 120$. No character subspace is omitted.

Step 3. In this step, we compare the original point ideal with the retained direct adjunction. We make explicit the comparison used for every weight. Let $\Delta_W, \mathcal{M}$ be the direct discriminant and nef moduli data for B . For $0 < t < b$, the ambient ideal pair $(X, B; \mathfrak{m}_x^t)$ is lc near x . Principalize this ideal on the models entering the threshold definition of Theorem 4.1. At a prime valuation v of the base, adding this ideal subtracts $t v(\mathfrak{m})$ from every generic threshold on an ambient lc place dominating W . The infimum over all those places therefore gives

$$A_{\text{gen}}(v) \geq t v(\mathfrak{m}).$$

Let t increase to b . On a fixed model where the boundary is simple normal crossing this lc inequality also holds after analytification and on higher analytic models. For the weights in (15.5) it gives $A_{\text{gen}}(\nu_i) \geq b$. After the stipulated addition, Proposition 4.18 retains W and the compatible moduli data. Repeating the same argument gives $A_{\text{gen,new}}(\nu_i) \geq y$. The ideal is always the pullback of the original point ideal.

Step 4. In this step, we construct the small analytic model for distinct residual factors. Suppose now that $s_0 \geq 2$, and first take P non-Cartier. In the notation of Lemma 15.11, order the complete blocks $a_\lambda^{M_\lambda}$ so that q_λ/M_λ is nonincreasing. Successively blowing up the first variable and the next complete block gives charts

$$U_j V_j = a_j^{M_j}, \quad u = U_j \prod_{\lambda < j} a_\lambda^{M_\lambda}, \quad v = V_j \prod_{\lambda > j} a_\lambda^{M_\lambda}.$$

For the first step the incidence equations are

$$uT_0 = a_1^{M_1} S_0, \quad (\psi/a_1^{M_1})T_0 = vS_0.$$

Both affine charts are normal domains containing the dense graph, so there is no extra component. Nontrivial fibers lie over the common zero set of $u, v, a_1, \psi/a_1^{M_1}$. The two base blocks are coprime, so that common base set has codimension two. Its inverse image has dimension $d - 2$; the modification is small. On the overlap the next first variable is a unit, so the next blowup ideal glues to the unit ideal on the chart already constructed. Induction gives a projective small analytic map $\pi : Y \rightarrow W$ whose central fiber is a chain of projective lines.

Each chart origin is rational factorial by [Lemma 15.12](#). For $M_j = 1$, the chart itself is the factorial ring $U_j V_j = a_j$; for $M_j \geq 2$, its cyclic cover $rt' = a_j$ is factorial. At all other points of the central fiber one of U_j, V_j is nonzero, so the chart is smooth. The charts are Gorenstein and the map is small; hence $K_Y = \pi^* K_W$.

Let $\mathcal{D}_+$ be the divisor of (u, a_+) . Its strict transform on chart j is

$$\sum_{\lambda < j} q_\lambda \operatorname{div}(a_\lambda) + \frac{q_j}{M_j} \operatorname{div}(U_j).$$

At a weight ν_i , its value is the concave piecewise-linear function

$$(15.7) \quad F_+(i) = \min_{0 \leq t \leq 1} \left\{ it + \sum_{\lambda} d_\lambda (q_\lambda - M_\lambda t)_+ \right\}, \quad F_+(0) = 0, \quad F_+(D) = \delta.$$

Indeed, in the chart for which $\sum_{\lambda < j} M_\lambda d_\lambda \leq i \leq \sum_{\lambda \leq j} M_\lambda d_\lambda$, its value is

$$\sum_{\lambda < j} q_\lambda d_\lambda + \frac{q_j}{M_j} \left(i - \sum_{\lambda < j} M_\lambda d_\lambda \right).$$

The sorted slopes give the displayed minimum and $F_+(i) \geq \delta i/D$.

Step 5. In this step, we retain the nef moduli correction in the weight inequalities. Fix the actual finite divisors $\Delta_Y, \mathcal{M}_Y, P_Y$. They are rational Cartier at every point of the compact central fiber just described. For each divisor, an appropriate reflexive power of its coherent divisorial sheaf is invertible on a neighborhood of each such point. A finite covering and a common multiple make these fixed divisors rational Cartier on one neighborhood of the whole fiber. Properness permits shrinking the base so that its full inverse image lies there. This argument does not require every divisor on that neighborhood to be rational Cartier, and it does not change any original local ideal.

Take a common analytic resolution $\rho : V \rightarrow Y$ carrying the original nef moduli trace $\mathcal{M}_Y$. The divisor

$$E_{\mathcal{M}} = \rho^* \mathcal{M}_Y - \mathcal{M}_V$$

has zero pushforward and relatively nef negative. Relative nefness comes from the same projective nef moduli model: a compact contracted curve has point or compact algebraic-curve image there, on which the nef degree is nonnegative. Apply the analytic assertion of [Theorem 4.11](#) to a Cartier multiple to get $E_{\mathcal{M}} \geq 0$. Since the old trace is effective and contains $c_P P_Y$, on V the difference between the generalized boundary and the boundary of $(Y, c_P P_Y)$ is

$$\rho^*(\Delta_Y - c_P P_Y) + E_{\mathcal{M}} \geq 0.$$

Combining this with the point-ideal comparison proves

$$(15.8) \quad b + c_P(\nu_i(\gamma) + F_+(i)) \leq d - 1.$$

The same inequality after addition has y in place of b and coefficient one on P_Y , by coefficient-one detection in [Theorem 4.1](#). All other old components and the compatible moduli correction remain in this comparison.

Step 6. In this step, we prove the stronger coefficient and later-deficit bounds for distinct factors. Interchanging u, v and replacing a_+ by a_- interchanges ρ_+, ρ_- . Assume $\rho_+ \leq \rho_-$, and

put $j_0 = \rho_- - \rho_+$. These are integers, and $\text{mult}(R/P) = \rho_+ + \rho_- + 1$. Define

$$\mathcal{G}(t) = \sum_{\lambda} d_{\lambda} \max\{q_{\lambda}(1-t), (M_{\lambda} - q_{\lambda})t\}.$$

One has $\mathcal{G}(t) \geq \delta(1-t)$ and $\mathcal{G}(t) \geq 1$. For the second inequality, two partially chosen factors each contribute at least $\max(t, 1-t)$. With one partial factor, any other complete factor supplies the complementary t or $1-t$. With no partial factor, the two nonunits a_+, a_- supply one complete factor of each type. Multiplication by the positive integers d_{λ} preserves these bounds.

If $\delta \geq j_0 + 1$, use the integer $i = D - \delta + j_0$, which lies between 1 and $D - 1$. The coefficient orders from the presentation give

$$\nu_i(\gamma) \geq \min\{\rho_+, D - \delta - i + \rho_-\} = \rho_+.$$

By (15.7),

$$F_+(i) = \min_{0 \leq t \leq 1} \{j_0 t + \mathcal{G}(t)\} \geq \min_{0 \leq t \leq 1} \{j_0 t + \max(1, \delta(1-t))\} = j_0 + 1 - \frac{j_0}{\delta} \geq j_0 + \frac{1}{j_0 + 1}.$$

If instead $\delta \leq j_0$, use $i = D - 1$. Here $\nu_i(\gamma) = \rho_- + 1 - \delta$ and $F_+(i) \geq \delta - \delta/D$. Since $D \geq \delta + 1$, the sum is at least $\rho_- + 1/(j_0 + 1)$. In either case some actual weight satisfies

$$\nu_i(\gamma) + F_+(i) \geq \rho_- + \frac{1}{j_0 + 1} \geq \frac{\text{mult}(R/P) + 1}{2}.$$

The last difference is $j_0(j_0 - 1)/(2(j_0 + 1)) \geq 0$. Equation (15.8), before and after addition, now gives the two non-Cartier distinct-factor inequalities.

For Cartier $P = (g)$, use the same small model and moduli comparison. Its strict divisor has value $\nu_i(g)$, so the endpoint maximum gives

$$b + \max\{\nu_1(g), \nu_{D-1}(g)\} c_P \leq d - 1.$$

The maximum of the two integers whose sum is $\text{mult}(R/P)$ is at least $\lceil \text{mult}(R/P)/2 \rceil$. The comparison after addition proves the stated Cartier bound on y .

Step 7. In this step, we derive the endpoint and character coefficient bounds for both power families. In either power case, the cover has already made every selected prime rational Cartier. The same finite-divisor argument on W itself, followed by the same nef moduli correction, gives

$$b + c_P \nu_i(f) \leq d - 1, \quad y + \nu_i(f) \leq d - 1.$$

Here the second inequality uses the new coefficient one, and $\nu_i(f)$ is the value of P , as seen from its Cartier multiple $\text{div}(f^M) = MP$. Taking the maximum of the two endpoint values gives the asserted α -bounds.

For a non-Cartier singular power, apply (15.6) at $i = r_a(M - k)$. Both prefactors have weight $r_a k(M - k)/M$, while F, G have weight at least one. Consequently

$$\nu_i(f) \geq 1 + \frac{r_a k(M - k)}{M}.$$

Substitution gives both its old coefficient inequality and $y \leq d - 2 - r_a k(M - k)/M$. Since $r_a \geq 2$ and $k(M - k) \geq M - 1$, this last upper bound is at most $d - 3$.

For a smooth power we retain $d = 5$ and use $i = M - k$. The corresponding bound is $1 + k(M - k)/M$. Thus

$$b + \left(1 + \frac{k(M - k)}{M}\right) c_P \leq 4, \quad y \leq 3 - \frac{k(M - k)}{M}.$$

Finally,

$$k(M - k) - (M - 1) = (k - 1)(M - k - 1) \geq 0.$$

This proves the weaker old coefficient $2 - 1/M$ and the upper bound $2 + 1/M$ for y . The principal multiplicity calculation at the start excludes the Cartier cubic. Together, the general interval

comparison, the exact power counts, and these retained coefficient comparisons prove all the assertions. $\square$

16. THE DOUBLE-FIVEFOLD ENDPOINT AT ORDER 29/25

We retain the notation for the entire boundary and its original section spaces. We prove an endpoint whose hypothesis is the affine price $(29/25)s + b \leq n$. The proof uses intrinsic sections on the actual fivefold and keeps this price when their thresholds change the center.

The finite descendant calculations use the following elementary certificate criterion. It also specifies how the supplementary data are checked, including cases where a deficit interval collapses.

Lemma 16.1. *Let a finite directed acyclic graph have rational vertex weights $q_i \geq 1$ and positive rational edge caps h_{ij} . Fix a starting vertex i , a rational $a \geq 1$, and a rational C . For a directed flag $(i_0 = i, i_1, \dots, i_r)$, including the flag of length zero, put $C_0 = C$ and $C_j = \min(C, h_{i_0 i_1}, \dots, h_{i_{j-1} i_j})$. On the closed polytope*

$$0 \leq y_r \leq \dots \leq y_0, \quad y_j \leq C_j,$$

consider

$$(q_{i_0} - a)y_0 + \sum_{j=1}^r (q_{i_j} - q_{i_{j-1}})y_j.$$

For $C \geq 0$, the maximum over all flags and their polytopes is attained at a feasible tuple whose coordinates belong to $\{0, C_0, \dots, C_r\}$. Denote this maximum by $H_i(a, C)$. Then

$$H_i(a, C) = (q_i - a)_+ C + \max(\{0\} \cup \{H_j(\max(a, q_i), \min(C, h_{ij})) : i \rightarrow j\}).$$

For $C = 0$ every flag polytope is the single zero tuple. For $C < 0$ it is empty and supplies no admissible comparison.

Proof. For $C \geq 0$ each polytope is nonempty and compact. At a vertex, partition a feasible tuple into maximal consecutive blocks of equal coordinates. A positive block whose value is different from every active upper cap can be moved slightly in both directions, preserving its strict separations from neighboring blocks and all its upper bounds. Such a tuple is not a vertex. Every positive block at a vertex is therefore anchored at one of the C_j ; a zero block is anchored at zero. This proves the claimed exhaustive finite set. Equal caps and ties require no deletion or perturbation. For a flag of length zero the polytope is the closed segment $[0, C]$, with both endpoints retained.

To obtain the recurrence, fix the first child deficit z . If $q_i \geq a$, the outer deficit is maximized at C , and the remaining functional has baseline q_i . If $q_i < a$, the outer deficit is maximized at its smallest allowed value z , and the remaining baseline is a . The option with no child contributes $(q_i - a)_+ C$. Acyclicity makes this a finite recursion. The zero and empty cases follow directly from the displayed inequalities. $\square$

Theorem 16.2. *For $n \geq 7$, let $B \sim_{\mathbb{Q}} sL$, $s \geq 0$, be effective rational and lc near x , with a five-dimensional minimal center of multiplicity two and deficit b . If $(29/25)s + b \leq n$, then $K_X + (n+1)L$ is generated at x . No tangent, analytic factorization, quadratic-rank, factoriality or original-degree hypothesis is imposed.*

The assertion concerns the given boundary. In particular, it does not require that this boundary arise from an ambient point-order source.

Proof. We first establish the complete costs of the possible successor centers and the source comparison that retains old discriminant coefficients. We then treat a square quadratic tangent, an integral quadratic tangent, and the four possible factorizations of a rank-two analytic equation. All source trials start from the same B .

Step 1. In this step, we determine the local double-fivefold structure and retain its original discriminant budget. Let W be the minimal center, put $A = L|_W$, and write

$$R = \mathcal{O}_{W,x}^{\text{an}}, \quad \mathfrak{n} = \mathfrak{m}_{W,x}R, \quad g = A^5.$$

Theorem 4.13 gives a normal Cohen–Macaulay analytic domain. A general linear parameter reduction has length two. Its degree-one part has dimension $\text{edim } R - 5$, so $2 \geq 1 + \text{edim } R - 5$. Embedding dimension five would make R regular, of multiplicity one. Hence $\text{edim } R = 6$. Its convergent presentation has a height-one prime kernel in a regular six-variable power-series ring, which is factorial by Theorem 15.3, [Dem12, Chapter II, Theorem 2.10]. Consequently

$$R = \mathbb{C}\{z_1, \dots, z_6\}/(f), \quad \text{ord}_0 f = 2.$$

The full ordinary tangent ring is $\mathbb{C}[z_1, \dots, z_6]/(f_2)$, where f_2 is the quadratic initial form. Indeed, orders add in the regular ambient graded domain, so the initial ideal of (f) is (f_2) . Theorems 4.5 and 4.16 give

$$(16.1) \quad \bar{e}_1 \geq e_1 = 1, \quad 0 < b \leq 4, \quad s \leq \frac{25}{29}(n - b).$$

Here $e_1 = 1$ also follows from Theorem 4.20.

For later use, retain the original direct discriminant on W . For one of its primes P , let c_P be its old coefficient, interpreted as zero when P is absent, and put $m_P = \text{mult}_x P$. With the notation of Proposition 4.18, its original budget is

$$(16.2) \quad 2b \leq K_W = 10 - 2\bar{e}_1 - R_0 - M_0 - c_P m_P - \sum_{Q \neq P} c_Q m_Q.$$

Every subtracted term is retained. If an lc addition selects P as its minimal center, with deficit y , the same proposition gives

$$2y + (1 - c_P)m_P \leq K_W.$$

Combining the two displayed expressions, rather than replacing the old prime by a new one of coefficient zero, yields

$$(16.3) \quad 2y + m_P \leq 10 - 2\bar{e}_1 - R_0 - M_0 - \sum_{Q \neq P} c_Q m_Q \leq 8.$$

Thus the first fourfold has deficit at most $(8 - m_P)/2$. A drop to dimension at most three receives no extra adjacency cap. Embedding dimension never increases, since the local ring of a subcenter is a quotient.

Step 2. In this step, we construct complete original sources and their own lifted thresholds. For a rational $\gamma > 0$, use precisely

$$F_p(\gamma) = H^0(W, \mathcal{O}_W(pA) \otimes I_x^{[\gamma p]}).$$

The ordinary ample Hilbert polynomial and the full point quotient give

$$(16.4) \quad \dim F_p(\gamma) \geq \frac{g - 2\gamma^5}{120} p^5 - O_{W,A,x,\gamma}(p^4).$$

This is a kernel lower bound for evaluation into the whole quotient $R/\mathfrak{n}^{[\gamma p]}$. In particular, it holds without surjectivity of that evaluation. The original positive integer g is not changed by the local presentation.

There is a separate ordinary lifting cutoff. Apply Theorem 6.8, [Sta26, Lemma 30.17.1, Tag 0B5U], to the fixed coherent ideal $\mathcal{I}_W$ on the projective X , with its original ample line bundle. It gives

$$H^1(X, \mathcal{I}_W \otimes \mathcal{O}_X(pL)) = 0 \quad (p \geq p_{\text{lift}}).$$

Thus the full ordinary restriction map onto $H^0(W, pA)$ is surjective. We use the full inverse image of $F_p(\gamma)$ under this map. In particular, the lifts need not have ambient point order γp .

For a nonzero source above this cutoff, take a sufficiently large general tuple from its full inverse image and normalize its sum to an effective rational $H_\gamma \sim_{\mathbb{Q}} L$. Its restriction does not

vanish identically on W , and has point order at least γ . On a common principalization of B and the full systems, dilute the tuple so that each nonfixed prime of $H_\gamma|_W$ has coefficient less than $1/10$. General moving members have no common prime; all fixed orders remain unchanged.

Let t_γ be the actual threshold of H_γ against the whole B . Old lc places through x have centers containing W , so their H_γ -order is zero. The threshold is therefore positive; it is finite since H_γ vanishes at x , and rational by the common principalization. An old lc place dominating W persists, whereas a new threshold place has positive H_γ -order and cannot dominate W . Theorem 3.11, [GL24, Lemma 2.7], applied near x on the smooth ambient variety with its effective rational Cartier boundary, gives a properly smaller minimal center $Z_\gamma \subset W$. With $y = \text{def}_x(B + t_\gamma H_\gamma)$, interpreted as zero for a point center, Theorem 3.8 gives

$$(16.5) \quad y \leq b - \gamma t_\gamma, \quad s + t_\gamma \leq s + \frac{b - y}{\gamma}.$$

For the cited restricted-order theorem the klt background is $(X, 0)$, its effective boundary is B , and its rational Cartier addition is $t_\gamma H_\gamma$, which does not contain W ; these are exactly the hypotheses of [YZ20, Lemma 2.10].

When $Z_\gamma = P$ is a fourfold prime, let η_P be its coefficient in $H_\gamma|_W$. The direct discriminant definition in Theorem 4.1, [FH23, Theorem 1.2 and Definition 1.3], and its Cartier-addition compatibility in Proposition 4.18 give the exact identity

$$(16.6) \quad c_P + t_\gamma \eta_P = 1.$$

Indeed, adding the restriction shifts each generic discriminant threshold by $t_\gamma \eta_P$. The new coefficient is at least one by detection of the center and at most one by lc. This uses the same discriminant and compatible moduli before and after addition.

Step 3. In this step, we compute the complete descendant costs and all first-successor states. We spell out the finite calculation used below. For a state $\nu = (d_\nu, e_\nu, m_\nu)$, let $T_\nu = T(e_\nu - d_\nu, m_\nu)$ and $\beta_\nu = d_\nu - 2T_\nu/m_\nu$, as in Definition 5.2. The following table contains all positive states that can be first successors of W , and all states needed below them. Here $C_\nu = \beta_\nu$ for $d_\nu \leq 3$, and $C_\nu = \min(\beta_\nu, (8 - m_\nu)/2)$ for $d_\nu = 4$. The number q_ν is a rational upper bound for m_ν^{1/d_ν} .

Replace the radii in Definition 5.2 by q_ν , and denote the resulting upper envelope by $\hat{\mathcal{F}}_\nu$. For $a \geq 1$ and $0 \leq c \leq \beta_\nu$, set

$$J_\nu(a, c) = \max_{0 \leq z \leq c} (\hat{\mathcal{F}}_\nu(z) - az).$$

For every smaller-dimensional state τ with $e_\tau \leq e_\nu$, put

$$c_{\nu\tau} = \begin{cases} \min\{\beta_\tau, (d_\nu m_\nu - 2T_\nu - m_\tau)/m_\nu\}, & d_\tau = d_\nu - 1, \\ \beta_\tau, & d_\tau \leq d_\nu - 2. \end{cases}$$

Nonpositive caps are discarded. The exact recurrence is

$$(16.7) \quad J_\nu(a, c) = (q_\nu - a)_+ c + \max \left(0, \max_\tau J_\tau(\max(a, q_\nu), \min(c, c_{\nu\tau})) \right).$$

This is Lemma 16.1, applied to the complete successor graph with edge caps $c_{\nu\tau}$. Dimension decreases in every term, so the graph is acyclic. Increasing each radius increases every nonnegative deficit-drop cost. Theorem 5.4 therefore bounds all actual tails by these envelopes.

d_ν	e_ν	m_ν	T_ν	C_ν	q_ν	$J_\nu(50/49, C_\nu)$
1	1	1	0	1	1	0
2	2	1	0	2	1	0
2	3	2	1	1	$\frac{283}{200}$	$\frac{3867}{9800}$
2	4	3	2	$\frac{2}{3}$	$\frac{1733}{1000}$	$\frac{11639}{24500}$
2	5	4	3	$\frac{1}{2}$	2	$\frac{24}{49}$
2	6	5	4	$\frac{2}{5}$	$\frac{2237}{1000}$	$\frac{59613}{122500}$
3	3	1	0	3	1	$\frac{3867}{9800}$
3	4	2	1	2	$\frac{63}{50}$	$\frac{70137}{98000}$
3	4	3	3	1	$\frac{1443}{1000}$	$\frac{20707}{49000}$
3	5	3	2	$\frac{5}{3}$	$\frac{1443}{1000}$	$\frac{8797}{9800}$
3	5	4	4	1	$\frac{397}{250}$	$\frac{118353}{196000}$
3	5	5	6	$\frac{3}{5}$	$\frac{171}{100}$	$\frac{10137}{24500}$
3	5	6	8	$\frac{1}{3}$	$\frac{909}{500}$	$\frac{19541}{73500}$
3	6	4	3	$\frac{3}{2}$	$\frac{397}{250}$	$\frac{12953}{12250}$
3	6	5	5	1	$\frac{171}{100}$	$\frac{4579}{6125}$
3	6	6	7	$\frac{2}{3}$	$\frac{909}{500}$	$\frac{19541}{36750}$
3	6	7	9	$\frac{3}{7}$	$\frac{1913}{1000}$	$\frac{131211}{343000}$
3	6	8	11	$\frac{1}{4}$	2	$\frac{12}{49}$
3	6	9	13	$\frac{1}{9}$	$\frac{2081}{1000}$	$\frac{17323}{147000}$
4	4	1	0	$\frac{7}{2}$	1	$\frac{70137}{98000}$
4	5	2	1	3	$\frac{119}{100}$	$\frac{317993}{294000}$
4	5	3	3	2	$\frac{1317}{1000}$	$\frac{91599}{98000}$
4	5	4	6	1	$\frac{283}{200}$	$\frac{13467}{19600}$
4	6	3	2	$\frac{5}{2}$	$\frac{1317}{1000}$	$\frac{384791}{294000}$
4	6	4	4	2	$\frac{283}{200}$	$\frac{57241}{49000}$
4	6	5	6	$\frac{3}{2}$	$\frac{187}{125}$	$\frac{123699}{122500}$
4	6	6	8	1	$\frac{783}{500}$	$\frac{49857}{61250}$
4	6	7	11	$\frac{1}{2}$	$\frac{1627}{1000}$	$\frac{10727}{19600}$

Here are explicit formulas for checking the entire table and the later ones. A surface of multiplicity k has $e = k + 1$, cap $2/k$, and

$$J_{(2,k+1,k)}(a, c) = (q_{(2,k+1,k)} - a)_+ \min(c, 2/k).$$

For a threefold $\nu = (3, e, m)$, set $h_k = \min(2/k, (3m - 2T_\nu - k)/m)$. Then

$$(16.8) \quad J_\nu(a, c) = (q_\nu - a)_+ c + \max\left(\{0\} \cup \left\{(q_{(2,k+1,k)} - \max(a, q_\nu))_+ \min(c, h_k) : 1 \leq k \leq e - 1, h_k > 0\right\}\right).$$

A fourfold substitutes this formula for every threefold of no larger embedding dimension, with its displayed adjacent cap, and also includes every direct surface with cap $2/k$. Curves add zero

to these support functions, since their radius is one. Thus these formulas cover every dimension skip as well as every adjacent transition.

The moment rule gives threefold multiplicities 1; 2, 3; 3, ..., 6; 4, ..., 9 at embeddings 3, 4, 5, 6, respectively. The next multiplicity in each range has nonpositive intrinsic cap; $T(c, m)/m$ is nondecreasing thereafter. For fourfolds, embedding five permits $m = 2, 3, 4$, while embedding six permits $m = 3, \dots, 7$ by the first cap; $m \geq 8$ has no positive first deficit. Embedding equal to dimension forces multiplicity one. This proves exhaustiveness of the 28 states. In particular the $m = 8, 9$ threefold rows are present: their moments give $z \leq 1/4, 1/9$. Lemma 5.10 also bounds their entire tails by $2z, 9z/4$, respectively. Its embedding-seven caps $1/2, 1/3$ belong to the ambient table; no such embedding can occur below the present W .

Every radius comparison in the table is the integer inequality $q_{\nu}^{d_{\nu}} \geq m_{\nu}$. Every last entry follows from (16.7) and (16.8), with the complete ranges just specified. For example, the largest entry is

$$\left(\frac{1317}{1000} - \frac{50}{49}\right) \frac{5}{2} + \left(\frac{397}{250} - \frac{1317}{1000}\right) \frac{4}{3} + \left(2 - \frac{397}{250}\right) \frac{1}{2} = \frac{384791}{294000} < \frac{131}{100}.$$

The other 27 entries displayed above are smaller.

Step 4. In this step, we specify the finite checks for every state, flag, price, and arithmetic record. The supplementary family `16-double-twenty-nine/states` contains the 28 state records, all 170 permitted edges through their defining rule, and all 739 directed flags, including every flag that terminates immediately. Its canonical records are in `supplement/16-double-twenty-nine/data/tables.json` and `supplement/16-double-twenty-nine/data/flags.json`. The state enumeration starts at $m = e - d + 1$, with the regular and curve cases as above, and stops at the first nonpositive cap. Monotonicity of $T(e - d, m)/m$, together with the decreasing fourfold cap, proves that this stopping rule omits no later positive state. All radii are the least multiples of $1/1000$ whose d -th powers are at least m .

The family `16-double-twenty-nine/prices`, in `supplement/16-double-twenty-nine/data/queries.json`, gives 596 specified comparisons. There are 21 sets of 28 state queries: the terminal order, the two general source orders, the square tangent, the bound $b \leq 3$, the two orders in each of the first-power, distinct-factor and singular-power cases, and the orders $23/25, 19/20$ at each smooth exponent $3, \dots, 7$. There are also the three affine-tail corrections, the low-deficit quartic threefold, the non-Cartier quartic fourfold, and an empty, a point and a segment comparison. Each record specifies the baseline, cap, exact value and an attaining flag and tuple; the empty record has no value or witness.

For each flag, the checker enumerates every nonincreasing tuple from $\{0, C_0, \dots, C_r\}$, retains precisely those satisfying all caps, and evaluates the entire linear functional. By Lemma 16.1 this checks its maximum on the whole closed polytope, including degenerate faces. It also compares an independent optimization over all cap levels and the delivered value. Every expected state, flag, query and group must occur exactly once. A missing, duplicate, conflicting or out-of-domain record is rejected. The group maxima check all 26 general, 12 square-tangent and 8 replacement entries displayed here. For a denominator D_0 , their strict upward numerator is $\lfloor D_0 v \rfloor + 1$; this rule remains strict when $D_0 v$ is an integer. The small tables in the article are derived views of these same canonical records.

The family `16-double-twenty-nine/arithmetic`, in `supplement/16-double-twenty-nine/data/arithmetic.json`, records the 45 exact identities used for the affine prices, fixed orders, full fifth-power interval losses and positive reserves below. The checker recomputes their defining rational expressions and checks the displayed strict margins. Positive parts are evaluated before taking powers, so a zero endpoint is included. All arithmetic is rational: rounding of table ceilings is the only numerical enlargement, and there is no floating-point error. The separate geometric $O(p^4)$ errors and both cutting errors are accounted for below.

These checks imply precisely the finite inequalities once the actual center, old-coefficient and full-interval reductions in this proof have been established. The all-order moment stopping

argument, the analytic factorization and local algebra, the convexity and monotonicity over unbounded residual orders, and the uniform source and lifting cutoffs remain arguments in the article. In particular, the finite smooth-exponent checks do not replace the proof for all exponents.

Step 5. In this step, we show that a terminal intrinsic section gives generation. A nonzero original $F_p(49/50)$, at any integer $p \geq 1$, already proves generation. Its powers give nonzero sections at arbitrarily large degrees jp , since

$$j[49p/50] \geq [49jp/50].$$

Choose such a degree above the separate lifting cutoff. For its own threshold, (16.5) and the full table give an upper scalar price

$$\begin{aligned} s + t + \widehat{\mathcal{F}}_Z(y) &\leq s + \frac{50}{49}b + \frac{131}{100} \\ &\leq (n-4)\frac{25}{29} + \frac{200}{49} + \frac{131}{100} \\ (16.9) \quad &\leq n + 1 - \frac{3149}{142100}. \end{aligned}$$

The gap increases by $4/29$ when n increases by one. Both cutting errors in Theorem 3.14, [GL24, Proposition 3.2 and Remark 3.3], are included as follows. We apply it with the singleton $\{x\}$, the original ample Cartier L , and the entire current boundary. Once the actual radius ρ_i of a cut is known, choose its approximation parameter so that the sum of the weighted errors $\sum_i (1 + \rho_i)\epsilon_i$ is less than a specified allowance. There are at most four further cuts. For (16.9) use allowance $1/1000$; the gap is greater than $1/100 + 1/1000$. The complete-tail theorem then gives a point boundary of full coefficient less than $n + 1 - 1/100$, and Proposition 3.12 proves generation.

If $g \geq 2$, (16.4) with $\gamma = 49/50$ supplies such a section at every sufficiently large original degree. We may henceforth suppose, for contradiction, that generation fails. Then

$$(16.10) \quad g = 1, \quad F_p(49/50) = 0 \quad \text{for every } p \geq 1.$$

Step 6. In this step, we compute the full prices at the two initial source levels. Fix

$$\lambda = \frac{1731}{2000}, \quad \mu = \frac{23}{25}, \quad C_0 = 1 - 2\lambda^5 = \frac{458771731542349}{16000000000000000}.$$

For a first state ν , (16.5) and (16.1) bound the whole scalar price by

$$(16.11) \quad \mathcal{B}_\gamma(b_*, \nu) = (n - b_*)\frac{25}{29} + \frac{b_*}{\gamma} + J_\nu(1/\gamma, \min(b_*, C_\nu)), \quad b \leq b_*, \quad 1/\gamma > 25/29.$$

The right side increases with the upper bound b_* . At $b_* = 4, n = 7$, the following strict ceilings, in units $1/10000$, follow from the same finite formulas. The cubic fourfold here has the general cap $5/2$.

First state or group	$\gamma = \lambda$	$\gamma = \mu$
$d \leq 2$	76405	73941
$d = 3, e = 3$	74675	72621
$d = 3, e = 4$	76536	75167
$d = 3, e = 5$	78805	77208
$(3, 6, 4)$	80628	78916
$(3, 6, m), 5 \leq m \leq 9$	78205	76151
$d = 4, e = 4$	76536	75167
$d = 4, e = 5$	78845	78160
$(4, 6, 3), C_\nu = 5/2$	81792	80765
$(4, 6, 4), C_\nu = 2$	81061	79692
$(4, 6, 5), C_\nu = 3/2$	80152	78440
$(4, 6, 6), C_\nu = 1$	78869	76815
$(4, 6, 7), C_\nu = 1/2$	76877	74481

The exact general cubic-fourfold values are $821169547/100398000$ and $32322053/4002000$. The quartic-fourfold values are $406913591/50199000$ and $5315403/667000$. At λ , the quintic-fourfold value is $1005876349/125497500$, and the direct quartic-threefold value is $101185003/12549750$. These numbers, the state table and the two finite formulas give the full certificate for the grouped ceilings. Increasing n adds $(n-7)25/29$ to every entry.

The complete tails of the three exceptional fourfold states satisfy

$$(16.12) \quad \widehat{\mathcal{F}}_{(4,6,m)}(y) \leq q_m y + p_m, \quad \begin{array}{c|ccc} m & 3 & 4 & 5 \\ \hline q_m & \frac{1317}{1000} & \frac{283}{200} & \frac{187}{125} \\ p_m & \frac{851}{1500} & \frac{379}{1000} & \frac{741}{2500} \end{array}$$

For $m = 3, 4$, the largest correction passes through the quartic threefold at deficit $4/3, 1$, respectively, and then the quartic surface at deficit $1/2$. For $m = 5$, the direct quintic surface at deficit $2/5$ gives the largest correction. Substitution of every other child in (16.7) gives a smaller value.

A direct quartic threefold with $y \leq 4/3$ has λ -price at most

$$(n-7)\frac{25}{29} + \frac{120336203}{15059700} < (n-7)\frac{25}{29} + \frac{7999}{1000}.$$

If $y > 4/3$, its intrinsic cap is $y \leq 3/2$, and its actual full coefficient $c = s + t_\lambda$ satisfies

$$(16.13) \quad c + y < (n-7)\frac{25}{29} + \frac{1054271}{150597} < n + \frac{1}{1000}.$$

Indeed, substitute $b \leq 4$ and $y > 4/3$ in $c + y \leq (n-b)25/29 + (b-y)/\lambda + y$. It follows that

$$c + \frac{5000}{3997}y < n + \frac{1}{1000} + \frac{3009}{7994} < n + \frac{49267}{107919}.$$

Theorem 11.1 applies to this very $B + t_\lambda H_\lambda$: its center has state $(3, 6, 4)$, $y > 1$, and every positive original degree and full tangent is allowed. It gives a point continuation retaining this whole boundary. Proposition 3.12 contradicts nongeneration. Thus this codimension-two outcome is always accounted for, without an ambient multiplicity assumption on the lift.

In all scalar payments below reserve total weighted cutting error less than $1/10000$. Each displayed paid scalar bound has room for this allowance and a positive point-evaluation margin. The affine quartic-threefold theorem includes its own cutting allowance.

Step 7. In this step, we derive actual full fixed orders from the retained old coefficients. Here is the calculation used in each remaining prime case. Suppose P has state $(4, 6, m)$, $m = 3, 4, 5$, and the applicable local inequality is

$$b + Kc_P \leq 4, \quad K > 0.$$

For $\varepsilon \in \{0, 1/500\}$, set

$$A_m = q_m - \frac{25}{29}, \quad C_m(\varepsilon) = 8 - \varepsilon - \frac{175}{29} - p_m.$$

Nongeneration implies that the scalar price $s + t_\gamma + q_m y + p_m$ cannot be less than $n + 1$. Together with (16.5), it therefore implies the weaker, useful inequality

$$0 < (q_m \gamma - 1)t_\gamma \leq A_m b - C_m(\varepsilon).$$

All uses below have $q_m \gamma > 1$ and $C_m(\varepsilon) > (4 - K)A_m > 0$. In particular $b > 4 - K$. The exact old-coefficient identity (16.6) now gives

$$(16.14) \quad \begin{aligned} \eta_P &\geq \frac{b - 4 + K}{K} \frac{q_m \gamma - 1}{A_m b - C_m(\varepsilon)} \\ &\geq \frac{q_m \gamma - 1}{4A_m - C_m(\varepsilon)}. \end{aligned}$$

The last step uses $b \leq 4$: the derivative of $(b - 4 + K)/(A_m b - C_m(\varepsilon))$ has the negative sign of $(4 - K)A_m - C_m(\varepsilon)$. All denominators are positive. This derivation charges $1 - c_P$, not an unearned full unit.

For $\varepsilon = 1/500$, $K = m/2$, the exact right sides are

$$(16.15) \quad \frac{\gamma = \lambda}{\left| \begin{array}{ccc} m=3 & m=4 & m=5 \\ \hline \frac{3476607}{10528000} & \frac{2606317}{7275600} & \frac{2137213}{6297400} \end{array} \right|} > \frac{33}{100}; \quad \gamma = \mu, m=3: \quad \frac{460317}{921200} > \frac{49}{100}.$$

For $\varepsilon = 0$, the constants used with $K_3 = 2, K_4 = 2, K_5 = 5/2$ are

$$(16.16) \quad \frac{m}{\gamma = \lambda} \quad \left| \begin{array}{ccc} 3 & 4 & 5 \\ \hline \frac{24336249}{73348000} & \frac{2606317}{7252400} & \frac{164401}{483300} \end{array} \right|$$

$$\text{strict lower bound} \quad \left| \begin{array}{ccc} 33/100 & 7/20 & 17/50. \end{array} \right|$$

The last strict comparison retains the difference $79/483300$. The fixed dilution below $1/10$ shows that every prime satisfying one of these bounds is fixed in the entire source. Thus

$$N_P = \text{ord}_P F_p(\gamma) \in \mathbb{Z}_{\geq 1}, \quad \eta_P = N_P/p$$

is its full order at this degree, not an asymptotic order.

Step 8. In this step, we compare complete sources through full point intervals and fix their intrinsic cutoffs. At a fixed degree with $\gamma p, \gamma' p$ integral, if the entire $F_p(\gamma)$ has prime order at least N_P , then local evaluation gives an injection

$$(16.17) \quad \frac{F_p(\gamma)}{F_p(\gamma')} \hookrightarrow \frac{P^{(N_P)} \cap \mathfrak{n}^{\gamma p}}{P^{(N_P)} \cap \mathfrak{n}^{\gamma' p}}, \quad \gamma < \gamma'.$$

Its kernel before taking the quotient is exactly the complete $F_p(\gamma')$: a section maps to zero precisely when its ordinary point order is at least $\gamma' p$. The specific interval estimate is supplied by the appropriate theorem of Section 15. All fixed, residual and embedded conditions remain in the source. The target is the entire symbolic interval. Finite analytic jets agree with those of the original local algebraic ring.

All source levels and positive rational reserves in the following cases are fixed first. Their Hilbert errors depend only on (W, A, x) ; the local interval errors are uniform in the selected prime and its symbolic order. In rank two they may also depend on the fixed analytic germ. Thus one intrinsic cutoff can be chosen before selecting a degree, tuple, prime or threshold place. For a finite list of losses $L_i p^5/120 + C_i(p+1)^4$ and initial error $C_{\text{in}}(p+1)^4$, a sufficient explicit requirement for a reserve $\delta > 0$ is

$$120 \left(C_{\text{in}} + \sum_i C_i \right) \frac{(p+1)^4}{p^5} < \frac{\delta}{2}.$$

Take this cutoff together with, but separately from, p_{lift} , and then choose any sufficiently large 2000-divisible original degree p . Every level below has integral endpoint at this degree. Ceilings in (16.4) give the same all-large-degree statement before this divisibility choice.

Step 9. In this step, we treat the quadratic-rank alternatives and reduce the remaining case to a rank-two equation.

Case 9.1. The quadratic tangent is a square. Suppose $f_2 = \ell_0^2$, with $\ell_0 \neq 0$. The whole tangent ring is nonreduced. The minimal-multiplicity conclusion of Theorem 4.20 implies

$$b \leq \frac{7}{2}.$$

In fact, if $b > 3$, its exceptional cycle is a doubled linear component, and that theorem gives $b \leq 3 + 1/2$; for $b \leq 3$ the assertion already holds. For a descendant of embedding six, the surjection of full tangent rings is an isomorphism in degree one. The image of ℓ_0 is nonzero and square-zero. Thus this descendant also has nonreduced full tangent. If its state is $(4, 6, 3)$ or $(3, 6, 4)$, it has minimal multiplicity; its Cohen–Macaulay tangent has no embedded or vertex

layer that could support the nilpotent without a repeated component. The same theorem gives, respectively,

$$y \leq \frac{7}{3}, \quad y \leq \frac{5}{4}.$$

For deficits at most $d - 2$ these bounds hold as well.

Put $\gamma_1 = 87/100$. Its complete original source is nonzero for every sufficiently large p , because

$$1 - 2\gamma_1^5 = \frac{15790793}{5000000000} > 0.$$

At its own threshold, use $b_* = 7/2$ in (16.11), and the two improved first caps just proved. At $n = 7$, the first two terms of that expression sum to $1225/174$. The full recurrence gives the following upper ceilings, in units $1/100000$.

First state or group	Ceiling numerator
$d \leq 2$	747526
$d = 3, e = 3$	730581
$d = 3, e = 4$	749788
$d = 3, e = 5$	772286
$(3, 6, m), 4 \leq m \leq 9$	779445
$d = 4, e = 4$	749788
$d = 4, e = 5$	773479
$(4, 6, 3), C_\nu = 7/3$	799858
$(4, 6, 4)$	795038
$(4, 6, 5)$	785650
$(4, 6, 6)$	772521
$(4, 6, 7)$	752302

The exact maximum is

$$\frac{1225}{174} + \left(\frac{1317}{1000} - \frac{100}{87} \right) \frac{7}{3} + \frac{851}{1500} = \frac{2087627}{261000}.$$

For comparison, the quartic- and quintic-fourfold maxima are $230561/29000$ and $1708787/217500$; the whole embedding-six threefold group has maximum $226039/29000$. All other entries use the surface and threefold formulas already displayed, with every listed child and every skip. The gap below $n + 1$ is at least

$$\frac{373}{261000} + \frac{4(n-7)}{29} > \frac{1}{2000} + \frac{1}{10000}.$$

Both cutting errors fit in this gap. The entire boundary reaches a point and gives generation, contradicting our assumption.

Case 9.2. The quadratic rank is at least three. Suppose the quadratic rank is at least three. Use the general λ, μ table without a factoriality assumption. For $b \leq 3$, its largest λ -price at $n = 7$ is

$$\frac{791719547}{100398000} < \frac{78859}{10000},$$

obtained from the same recurrence with $b_* = 3$; hence these deficits are paid. At any other deficit, the direct quartic threefold is paid by (16.13) and the following affine comparison. The only unpaid λ -outcomes can therefore be fourfold primes of state $(4, 6, m)$, $m = 3, 4, 5$. At μ only $(4, 6, 3)$ can remain.

For each such prime, (16.2) gives $b + (m/2)c_P \leq 4$. Formula (16.15) therefore applies. At λ , the full fixed order N_P is greater than $33p/100$; at μ , the cubic fixed order is greater than $49p/100$. The corrected Theorem 15.9 applies to the actual R : it is a normal complex analytic hypersurface domain of dimension five, multiplicity two and quadratic rank at least three. Its interval bound is uniform over every height-one prime and requires neither factoriality nor parity of the prime multiplicity.

At the first transition the symbolic shift is $N_P m_P / (2p) > 99/200$. The full interval injection (16.17) gives a leading loss at most

$$(16.18) \quad L_1 = 2 \left[\left(\mu - \frac{99}{200} \right)^5 - \left(\lambda - \frac{99}{200} \right)^5 \right] = \frac{220301263580299}{1600000000000000}.$$

The positive-part difference decreases as the shift increases, as follows by integrating $5(t - \text{shift})_+^4$ from the lower to the upper endpoint. Since $C_0 - L_1 > 1/100$, the entire $F_p(\mu)$ is nonzero at the same sufficiently large degree.

Take its own threshold from B . If it is unpaid, the new cubic prime Q has full order $N_Q > 49p/100$. It need not equal P . Its shift is greater than $147/200$, and the entire interval from μ to $49/50$ loses at most

$$(16.19) \quad L_2 = 2 \left[\left(\frac{49}{50} - \frac{147}{200} \right)^5 - \left(\mu - \frac{147}{200} \right)^5 \right] = \frac{53282823}{40000000000} < \frac{1}{500}.$$

The exact remaining leading mass is

$$C_0 - L_1 - L_2 = \frac{4343146775241}{3200000000000000} > \frac{1}{100}.$$

The uniform errors were absorbed before choosing p, P, Q, N_P, N_Q . Consequently $F_p(49/50) \neq 0$, contrary to (16.10). This treats every quadratic rank at least three, including ranks three and four.

Case 9.3. The quadratic rank is two. The only remaining quadratic rank is two. Theorem 15.1, [GP26, Theorem 2.4], applies to the actual convergent equation $f \in (z_1, \dots, z_6)^2$, with Hessian rank two. It requires no isolated singularity. A convergent coordinate change therefore gives

$$R = \mathbb{C}\{u, v, z_1, \dots, z_4\} / (uv - \psi(\mathbf{z})), \quad \psi \in (z_1, \dots, z_4)^3.$$

Here $\mathbf{z} = (z_1, \dots, z_4)$. If $\psi = 0$, the nonzero classes of u, v have product zero, contradicting the domain property. Choose its $r_{\text{fac}} \geq 1$ distinct irreducible factors in the convergent factorial base ring and write

$$\psi = \epsilon \prod_{j=1}^{r_{\text{fac}}} h_j^{\ell_j}, \quad D = \text{ord}_0 \psi = \sum_j \ell_j \text{ord}_0 h_j \geq 3,$$

where ϵ is a unit, the h_j are pairwise nonassociate irreducible nonunits, and $\ell_j \geq 1$. The analytic change $u' = u/\epsilon(\mathbf{z})$ absorbs the unit. These are isomorphisms of the actual local ring and its ordinary point quotients. They do not alter W, A or any global section space.

There are exactly four alternatives: at least two distinct factors; one irreducible first power; a power of a singular irreducible factor; or a power of a smooth parameter. We treat all four below. A smooth first power has residual order one, and a smooth square has quadratic rank three. Neither belongs to the present $D \geq 3$, rank-two case.

Step 10. In this step, we bound the entire rank-two symbolic interval uniformly over its endpoint parameters. We record the analytic-parameter comparison before using it. For fixed $0 < a_0 < a_1$ and $\eta > 0$, set, for $z \geq 0$,

$$\Phi_\eta(z) = (a_1 - \eta z)_+^5 - (a_0 - \eta z)_+^5.$$

It is nonnegative, decreasing and convex in z , and decreases with η . For an embedding-six prime of multiplicity $m \geq 3$, Theorem 15.13 supplies

$$\alpha + \beta = m, \quad 1 + \frac{1}{D} \leq \beta \leq \frac{m}{2}, \quad \zeta = \frac{(D-1)(m-\beta) - \beta}{D-2}.$$

The normalized leading bound for its full interval is

$$(16.20) \quad K_D(\eta, \beta) = \frac{D}{D-1} \Phi_\eta(\beta) + \frac{D-2}{D-1} \Phi_\eta(\zeta).$$

Its hypotheses hold for every prime we select: the prime is the actual fourfold minimal center of the whole augmented boundary, so its quotient is normal Cohen–Macaulay near x , and its

embedding dimension is six. The theorem includes every component of the pullback and the whole opposite-character module. We use the interval bound itself, including its lower endpoint.

The expression (16.20) is convex in β , so its maximum is at one of the two endpoints. At $\beta = m/2$, it is $2\Phi_\eta(m/2) \leq \Phi_\eta(1) + \Phi_\eta(m-1)$. At the other endpoint put $w_D = 1/D$, $0 < w_D \leq 1/3$. The expression becomes

$$\frac{\Phi_\eta(1+w_D)}{1-w_D} + \frac{1-2w_D}{1-w_D} \Phi_\eta\left(m-1 + \frac{(m-3)w_D}{1-2w_D}\right).$$

The second summand decreases with w_D . For the first, use

$$\frac{\Phi_\eta(1+w_D)}{1-w_D} = \int_{a_0}^{a_1} \frac{5(t - \eta(1+w_D))_+^4}{1-w_D} dt.$$

On its positive part the derivative of the integrand with respect to w_D has the sign of $t - 5\eta + 3\eta w_D$. This is nonpositive if $a_1 \leq 4\eta$; at zero the integrand cannot increase. Taking w_D down to zero therefore proves

$$(16.21) \quad K_D(\eta, \beta) \leq \Phi_\eta(1) + \Phi_\eta(m-1) \quad \text{if } a_1 \leq 4\eta.$$

All applications have $a_1 \leq 19/20$ and $\eta \geq 33/100$, so this condition holds. This argument covers every integer D , every factor exponent and every character, including the unbounded range of D .

Step 11. In this step, we treat all four factorizations of the actual rank-two residual equation.

Case 11.1. The residual equation is an irreducible first power. Suppose $r_{\text{fac}} = 1, \ell_1 = 1$. The factoriality assertion in Theorem 15.13 applies to this convergent irreducible ψ . Hence every height-one prime of R is principal. We need two restrictions on its actual first centers. If $P = (h)$ has embedding-six quotient, then $h \in \mathfrak{n}^2$. The fourfold ring $R/(h)$ is a complete intersection. A reduction by four general linear parameters is an Artinian complete intersection in two variables, defined by equations of orders at least two. Its first three layers have dimensions at least 1, 2, 1; its length cannot be three. Thus a Cartier cubic fourfold is impossible.

If this quotient has multiplicity five, its Artinian reduction is Gorenstein of length five with first two layers 1, 2. The Hilbert function (1, 2, 2) is impossible, because its square would give a socle of dimension at least two. The degree-two part is nonzero by Nakayama. Therefore the remaining two terms have total first moment at least $2 + 3$, and the entire first moment is at least seven. Theorems 4.5 and 4.16, using [Han26, Lemmas 4.2 and 4.4], give

$$14 \leq 2\bar{e}_1(P) \leq 5(4 - y), \quad y \leq \frac{6}{5}.$$

The parameters here are a regular sequence in the Cohen–Macaulay complete intersection; no regularity of that local ring is claimed.

The general λ, μ table now pays every μ -outcome. At λ , the improved quintic cap gives

$$\frac{75}{29} + \frac{14}{5} \frac{2000}{1731} + \frac{187}{125} \frac{6}{5} + \frac{741}{2500} = \frac{993053071}{125497500} < 7.914$$

at $n = 7$. The direct quartic threefold is already covered by (16.13). Only a quartic fourfold prime can remain at λ . For $n \geq 8$ even this λ -row is paid: its ceiling 81061/10000 at $n = 7$ loses 4/29 relative to the available budget for each increase of n .

For $n = 7$, let $P = (h)$ be the remaining quartic prime. Its old inequality is $b + 2c_P \leq 4$. Formula (16.14) with $\varepsilon = 1/500$ gives

$$\eta_P = \frac{N_P}{p} \geq \frac{2606317}{7275600} > \frac{7}{20}.$$

Use the corrected principal interval theorem 15.10, with $\text{ord}_{\mathfrak{n}} h \geq 2$. It bounds the full interval by the twofold fifth-power interval with shift $2N_P$, even if the initial form of h is a zerodivisor. Thus the entire λ -to- μ loss is at most the coarser normalized quantity

$$2\left(\mu - \frac{7}{10}\right)^5 = \frac{161051}{156250000} < \frac{1}{50} < C_0.$$

We only enlarged the interval downward; the full lower endpoint is still present in the theorem used. The full kernel comparison forces $F_p(\mu) \neq 0$ at the same degree. Its own threshold from B pays every changed center by the preceding table, a contradiction.

Case 11.2. The residual equation has at least two distinct irreducible factors. Suppose $r_{\text{fac}} \geq 2$, with arbitrary positive factor exponents. Theorem 15.13 gives, for a non-Cartier embedding-six prime,

$$2b + (m_P + 1)c_P \leq 8, \quad y \leq \frac{7 - m_P}{2}.$$

For a Cartier prime it gives

$$b + \lceil m_P/2 \rceil c_P \leq 4, \quad y \leq 4 - \lceil m_P/2 \rceil.$$

An embedding-six Cartier cubic is impossible by the complete-intersection length argument above. Thus the cubic fourfold cap is two; the non-Cartier quartic cap is $3/2$, whereas a Cartier quartic retains cap two. The quintic cap is one for either type, the sextic cap is at most one, and multiplicity seven or more has nonpositive cap. All lower-dimensional first states remain in the general table.

For clarity the changed rows of that table have the following strict ceilings, again in units $1/10000$:

first state and cap	λ	μ
$(4, 6, 3), 2$	80984	79615
$(4, 6, 4), 2$	81061	79692
$(4, 6, 5), 1$	78449	76395
$(4, 6, 6), 1$	78869	76815

Every other permitted row is unchanged from the full table. For a non-Cartier quartic, its exact λ -bound with cap $3/2$ is

$$\frac{800795597}{100398000} < \frac{7999}{1000};$$

its μ -bound is no larger than the quartic entry above. The exact cubic values with cap two are $101632183/12549750$ at λ and

$$\frac{75}{29} + \frac{4}{\mu} + \left(\frac{1317}{1000} - \frac{1}{\mu} \right) 2 + \frac{851}{1500}$$

at μ . This supplies the replacements using precisely the same finite recurrence and all its children. The maximal μ -price is $5315403/667000 < 7.9692$. At λ , only a cubic fourfold, a Cartier quartic fourfold, or the already treated direct quartic threefold can fail the scalar comparison. At $n \geq 8$, even the largest λ -ceiling $81061/10000$ leaves a gap greater than $3/100$, because the relative gain is $4(n-7)/29$. Thus it suffices to treat $n = 7$.

For either remaining prime type, $b + 2c_P \leq 4$. Formula (16.16) gives

$$\eta_P \geq \frac{24336249}{73348000} > \frac{33}{100} \quad (m_P = 3), \quad \eta_P \geq \frac{2606317}{7252400} > \frac{7}{20} \quad (m_P = 4).$$

These are the actual full fixed orders N_P/p . Use $(a_0, a_1) = (\lambda, \mu)$ in (16.21). For the cubic prime the exact retained mass is

$$(16.22) \quad C_0 - \Phi_{33/100}(1) - \Phi_{33/100}(2) = \frac{126108747621}{32000000000000} > \frac{3}{10000}.$$

For the Cartier quartic use Theorem 15.10 with an equation of order at least two. Its complete interval loses at most $2\Phi_{7/20}(2)$, and

$$(16.23) \quad C_0 - 2\Phi_{7/20}(2) = \frac{446253304953}{16000000000000} > \frac{1}{100}.$$

These are complete fifth-power interval identities. They include the lower endpoint in each loss. The uniform interval error at this fixed D , together with (16.4), forces $F_p(\mu) \neq 0$ at the same original degree. Its own threshold from B has every possible changed center paid by the μ column.

The direct quartic-threefold alternative at λ is paid with the actual coefficient $c = s + t_\lambda$ by (16.13). This covers every factorization with at least two distinct factors.

Case 11.3. The residual equation is a power of a singular irreducible factor. Suppose $r_{\text{fac}} = 1$, $\ell_1 \geq 2$ and $\text{ord}_0 h_1 \geq 2$. Set $\ell = \ell_1$ and $r_h = \text{ord}_0 h_1$, so $D = \ell r_h \geq 4$. The singular-power part of Theorem 15.13 applies to this actual analytic germ. Its factorial cover is

$$rt' = h_1(\mathbf{z}), \quad u = r^\ell, \quad v = (t')^\ell.$$

All components of the pullback and every character are part of that theorem. For a non-Cartier prime of character $k \in \{1, \dots, \ell - 1\}$, its old coefficient and successor deficit obey

$$b + \left(1 + \frac{r_h k(\ell - k)}{\ell}\right) c_P \leq 4, \quad y \leq 3 - \frac{r_h k(\ell - k)}{\ell} \leq 2.$$

For every embedding-six prime the endpoint inequalities also give $b + (m_P/2)c_P \leq 4$. A Cartier cubic is impossible, so the cubic first cap is two. We retain the general caps $2, 3/2, 1, 1/2$ for multiplicities $4, 5, 6, 7$; in particular we have not imported the distinct-factor quintic cap one.

The entire scalar table is therefore the general table with only its cubic-fourfold row replaced by 80984, 79615. Its μ -maximum is again 5315403/667000. For an explicit check of the other near-terminal first states, the λ -values for the quintic and septic fourfolds are

$$\frac{1005876349}{125497500}, \quad \frac{154364177}{20079600},$$

respectively. All $m = 8, 9$ threefolds and all direct surfaces remain in the recurrence. At $n \geq 8$ the λ source pays every outcome by the same $4(n - 7)/29$ gain. At $n = 7$, only fourfold primes of multiplicity $3, 4, 5$, or the already paid direct quartic threefold, need attention.

For these three primes the old inequalities permit $K_3 = 2, K_4 = 2, K_5 = 5/2$ in (16.14). The full orders in (16.16) are therefore valid, including the strict quintic lower bound $17/50$. With $(a_0, a_1) = (\lambda, \mu)$, the complete interval comparison (16.21) gives the following three exact reserves:

$$(16.24) \quad \begin{array}{c|c|c} m_P & \eta \text{ used} & C_0 - \Phi_\eta(1) - \Phi_\eta(m_P - 1) \\ \hline 3 & \frac{33}{100} & \frac{126108747621}{320000000000000} \\ 4 & \frac{7}{20} & \frac{157041436918849}{320000000000000} \\ 5 & \frac{17}{50} & \frac{99572486514949}{320000000000000} \end{array}$$

They exceed $3/10000, 4/1000, 3/1000$, respectively. The comparison proving these bounds was continuous in $1/D \in (0, 1/3]$; it applies in particular to every $D = \ell r_h \geq 4$, without a bound on ℓ, r_h or the character. Use a common leading reserve $1/100000$ in the intrinsic cutoff. The full injection then forces $F_p(\mu) \neq 0$ in all three prime cases. Its own threshold from the original B pays every successor. The actual direct quartic-threefold outcome is covered by the same full coefficient (16.13). This gives the contradiction for every singular irreducible power.

Case 11.4. The residual equation is a power of a smooth parameter. Suppose $r_{\text{fac}} = 1$, $\text{ord}_0 h_1 = 1$, and put $\ell = \ell_1$. The analytic inverse function theorem makes h_1 an actual coordinate. Since $D \geq 3$, now $\ell = D \geq 3$, and the germ is

$$R = \mathbb{C}\{u, v, z_0, t_1, t_2, t_3\}/(uv - z_0^\ell).$$

The smooth-power part of Theorem 15.13 supplies the regular cover, all full character modules, the interval formula, and the old inequalities

$$b + \alpha c_P \leq 4, \quad b + \left(1 + \frac{k(\ell - k)}{\ell}\right) c_P \leq 4 \quad (k \neq 0).$$

Its embedding-six cubic prime is non-Cartier, with successor cap

$$y \leq 2 + \frac{1}{\ell}.$$

For every embedding-six prime, $b + (m_P/2)c_P \leq 4$ follows from $\alpha \geq m_P/2$. The complete interval has $\beta \geq 1 + 1/\ell$.

Use the general λ, μ table, improving only the cubic cap to $2 + 1/\ell$. At λ , every unpaid first prime has multiplicity 3, 4, 5; the direct quartic threefold is covered by (16.13). Formula (16.15) forces all three fixed orders to exceed $33p/100$. The uniform full-interval bound (16.21), with $(a_0, a_1) = (\lambda, \mu)$, is at most $\Phi_{33/100}(1) + \Phi_{33/100}(2)$, since $m_P \geq 3$. The reserve (16.22) therefore earns the entire μ source at this same p , for every $\ell \geq 3$.

If $\ell \geq 7$, its cubic cap is at most $15/7$. Its whole μ -price at $n = 7$ is at most

$$(16.25) \quad \frac{75}{29} + \frac{4}{\mu} + \left(\frac{1317}{1000} - \frac{1}{\mu} \right) \frac{15}{7} + \frac{851}{1500} = \frac{111976393}{14007000} < 7.9944.$$

Every other row is below 7.9692 by the complete table. Thus every changed center is paid, including the entire unbounded range $\ell \geq 7$.

It remains to keep the stronger mass estimate when $3 \leq \ell \leq 6$. For an interval (a_0, a_1) , define

$$K_\ell^*(\eta; a_0, a_1) = \frac{\ell}{\ell-1} \Phi_\eta(1 + 1/\ell) + \frac{\ell-2}{\ell-1} \Phi_\eta(2),$$

where Φ_η uses those two endpoints. For any multiplicity $m_P \geq 3$, the low- β endpoint in (16.20) has second argument at least two, so it is bounded by K_ℓ^* . At the balanced endpoint the bound is $2\Phi_\eta(m_P/2) \leq 2\Phi_\eta(3/2)$. The two weights defining K_ℓ^* sum to two and their weighted mean argument is $3/2$; convexity gives the same bound. Hence the entire interval is bounded by K_ℓ^* . The derivative calculation preceding (16.21), with $m = 3$, shows that K_ℓ^* increases with ℓ . Consequently $K_\ell^* \leq K_6^*$ for all four integers in this range, with no selection of a character.

Set $\theta = 19/20$. The first interval loss has the sharper bound

$$(16.26) \quad L_1^* = K_6^*(33/100; \lambda, \mu) = \frac{360271187894299}{1600000000000000}.$$

This leaves positive mass, so the whole μ source exists at the same original degree. Take its own threshold from B . If it is unpaid, the general table leaves only a cubic fourfold prime. Its old coefficient is still measured against B , and (16.15) gives a full fixed order greater than $49p/100$. The second prime may differ from the first. The entire μ -to- θ loss is at most

$$(16.27) \quad L_2^* = K_6^*(49/100; \mu, \theta) = \frac{11331153881}{36000000000000}.$$

The exact remaining mass is

$$(16.28) \quad C_0 - L_1^* - L_2^* = \frac{8665174751849}{2880000000000000} > \frac{1}{500}.$$

Both uniform errors fit in the intrinsic cutoff fixed before either prime. Thus the complete original $F_p(19/20)$ is nonzero.

At this level the worst cubic cap for all $\ell \geq 3$ is $7/3$. Its full price at $n = 7$ is

$$(16.29) \quad \frac{75}{29} + \frac{4}{\theta} + \left(\frac{1317}{1000} - \frac{1}{\theta} \right) \frac{7}{3} + \frac{851}{1500} = \frac{13192471}{1653000} < 7.9810.$$

All other rows are no larger than their already paid μ values. To check this monotonicity without a new optimization, write their price as the maximum, over $0 \leq y \leq C_\nu \leq 4$, of

$$\frac{75}{29} + \frac{4-y}{\gamma} + \hat{\mathcal{F}}_\nu(y).$$

Every term decreases with γ . For ambient dimension $n \geq 7$, add only $(n-7)25/29$ to (16.25) and (16.29). Their strict gaps absorb both cutting errors, and the entire boundary gives a point continuation. This is the required contradiction for every smooth power.

Step 12. In this step, we combine all quadratic-rank cases at the original whole-boundary price. The square and integral quadratic cases exhaust ranks one and at least three. At rank two the actual nonzero convergent residual has one of the four factorizations just treated. Thus every possible analytic germ of the given double fivefold has been covered. Each successful scalar or intrinsic-source branch retains the entire input B . When a source trial is used only to bound an

interval, its addition is not charged to a different trial: the latter takes its own threshold from B . The direct quartic-threefold branch instead continues its actual $B + t_\lambda H_\lambda$, with the exact coefficient computed in (16.13). Every final point boundary has coefficient strictly below $n + 1$ after its cutting errors. Proposition 3.12 therefore contradicts nongeneration in every case and proves the theorem. $\square$

17. AMBIENT PROFILE ENDPOINTS

Throughout this section X is a smooth complex projective sevenfold, L is ample Cartier with $L^7 = 1$, and $x \in X$. We retain the complete original sources of Definition 2.4. The orders used here are

$$R = \frac{9}{10}, \quad u = \frac{26}{25}, \quad r = \frac{27}{25}, \quad T = \frac{29}{25}.$$

Each threshold is taken at its own original degree. Comparisons between two thresholds concern their profiles on the same original section ring. An eligible observation at u, r, T uses a nonzero complete original source in a sufficiently large degree divisible by 25. An initial observation uses degree $20i$, for sufficiently large i . Each cutoff is fixed before the corresponding tuple, threshold, center and computing place; a later source is earned before its observation. The two main endpoints treat minimal centers of embedding dimension seven and smooth divisors at r . The final statement records the resulting restriction on initial centers. Supplementary certificates are identified at their point of use; their records use the integration conventions stated here.

17.1. Original profiles and complete source bounds. For a divisorial valuation w whose center contains x , write

$$\phi_w(t) = \inf_{j \geq 1, 0 \neq f \in U_j(t)} \frac{w(f)}{jA_X(w)}.$$

Here $w(f)$ is evaluated on the entire divisor of the section, and an empty infimum is $+\infty$. In terms of the original profile α_w in the $a = 1$ specialization of Theorem 6.13, $\phi_w = \alpha_w/A_X(w)$. For a prime divisor P on X , we write $\phi_P = \phi_{\text{ord}_P}$; its discrepancy is one. On a fixed compact subinterval of the big range, Theorem 6.13 supplies a constant C and a cutoff, before any place is selected. Dividing its estimate $0 \leq w(\mathbf{b}(U_j(t))) - j\alpha_w(t) \leq CA_X(w)$ by $jA_X(w)$ gives

$$(17.1) \quad 0 \leq \frac{w(\mathbf{b}(U_j(t)))}{jA_X(w)} - \phi_w(t) \leq \frac{C}{j}.$$

We use this comparison only after fixing that compact interval. The error in this display is the complete-series error of Theorem 6.13. It is obtained with its fixed global twist, and is not an individual-level approximation assertion for an arbitrary graded sequence of ideals.

Lemma 17.1. *Suppose $L^7 = 1$ and $K_X + 8L$ is not generated at x . There is an integer J , fixed before all subsequent degrees, averages, centers and places, such that for every $i \geq J$ the complete source $U_{20i}(9/10)$ is nonzero. Every pure actual threshold average $\Gamma \sim_{\mathbb{Q}} sL$ of that source has minimal-center dimension in $\{2, 3, 4, 5\}$, deficit $b > 4/25$, and entire coefficient $s < 38/5$.*

Proof. We bound the complete descendant cost and then exclude the point, curve and divisorial alternatives. The ordinary jet count gives

$$\dim U_{20i}(9/10) \geq \frac{(20i)^7}{7!} \left(1 - \left(\frac{9}{10}\right)^7\right) + O(i^6),$$

so this source is nonzero at every sufficiently large original degree.

Step 1. In this step, we bound every radius in the complete descendant state set. For a state $(d, e, m) \in \mathcal{P}_7$ in Theorem 5.4, put $c = e - d$. Its minimum first moment satisfies

$$T(c, m) \geq 2m - (c + 2), \quad T(c, m) \geq 3m - \left(3 + 2c + \binom{c+1}{2}\right).$$

Indeed, for each sequence defining that moment, write $\sum_j j h_j = am - \sum_{j < a} (a - j) h_j + \sum_{j > a} (j - a) h_j$, for $a = 2, 3$, and use $h_0 = 1$, $h_1 \leq c$, $h_2 \leq \binom{c+1}{2}$. The condition $T(c, m) < dm/2$ consequently gives

d	1	2	3	4	5	6
m at most	1	6	11	14	19	6.

For $d = 6, c = 1$, use $T(1, m) = m(m - 1)/2$; for $c = 0$ one has $m = 1$. Each displayed row satisfies $m^{1/d} \leq \sqrt{6}$. Since every original ample degree is a positive integer, every actual radius is at most $\sqrt{6}$. Induction in the full recurrence, retaining all adjacent caps and all skips, therefore gives $\mathcal{F}_p(z) \leq \sqrt{6}z$: a successor with deficit $y \leq z$ costs at most $\sqrt{6}(z - y) + \sqrt{6}y$. The direct point alternative satisfies the same inequality.

Step 2. In this step, we obtain the strict deficit and coefficient bounds and exclude point and curve centers. Proposition 9.1 gives $9s/10 + b \leq 7$. If $b \leq 4/25$, the entire coefficient and descendant cost are bounded by

$$\frac{70}{9} + \left(\sqrt{6} - \frac{10}{9}\right)b \leq \frac{70}{9} + \left(\sqrt{6} - \frac{10}{9}\right)\frac{4}{25} < \frac{70}{9} + \left(\frac{5}{2} - \frac{10}{9}\right)\frac{4}{25} = 8.$$

Choose both cutting errors inside this strict gap before applying Theorem 5.4. It gives generation, a contradiction. Thus $b > 4/25$ and $s \leq (10/9)(7 - b) < 38/5$. A curve has radius at most one and complete tail at most b , giving $s + b \leq 70/9 - b/9 < 8$. A point has $s \leq 70/9 < 8$. The point boundaries in all these cases give generation by Proposition 3.12.

Step 3. In this step, we exclude divisorial centers using the entire fixed-degree estimate. Finally the full moving-volume estimate of Proposition 6.19, at order $9/10$, gives

$$\limsup_{i \rightarrow \infty} \frac{F_{20i}(9/10) \cdot L^6}{20i} \leq 1 - \left(1 - \left(\frac{9}{10}\right)^7\right)^{1/7} < \frac{1}{10}.$$

The last inequality is equivalent to $2(9/10)^7 < 1$. Increase J so the full fixed degree is less than $1/10$. If a threshold center were a divisor P , its coefficient in the full fixed part, divided by $20i$, would be $1/s$. Its positive integer degree would give full fixed degree at least $1/s \geq 9/70 > 1/10$, a contradiction. The remaining dimensions are exactly those asserted. No localization of the original base ideal has been assumed primary. $\square$

Lemma 17.2. *Let t_1, t_2 lie in a fixed compact subinterval of the original big range. For $i = 1, 2$, let k_i exceed the cutoff in (17.1), suppose $k_i t_i$ is integral, and let $B_i \sim_{\mathbb{Q}} s_i L$ be a pure actual threshold average of the full ideal $I_i = \mathfrak{b}(U_{k_i}(t_i))$. Let w_i compute its threshold, with center containing x . Then, for every divisorial valuation w with center containing x ,*

$$\phi_w(t_i) \leq \frac{w(I_i)}{k_i A_X(w)} \leq \frac{1}{s_i} \leq \phi_{w_i}(t_i) + \frac{C}{k_i}.$$

In particular,

$$\phi_{w_1}(t_2) \leq \phi_{w_2}(t_2) + \frac{C}{k_2}.$$

This comparison allows unrelated degrees and unrelated centers.

Proof. Write $\tau_i = \text{lct}_x(I_i)$, so that $s_i = k_i \tau_i$. The infimum defining ϕ_w includes the degree k_i . Log canonicity of the full threshold ideal gives $\tau_i w(I_i) \leq A_X(w)$. Equality holds for w_i . The last inequality is therefore (17.1) applied to w_i . Applying the first conclusion with $i = 2$ and $w = w_1$ proves the second. Every s_i is the coefficient of its entire average. $\square$

Lemma 17.3. *Let Q be a prime divisor through x , smooth there, and let $W \subset X$ be integral, contain x , and satisfy $W \not\subset Q$. Let w be a divisorial valuation centered on W . Then $w(Q) = 0$. The noncontainment hypothesis holds in particular if $\text{edim } \mathcal{O}_{W,x} = 7$. If p_Q is a local equation*

of Q , division by p_Q^j on a space of divisible germs induces an injection from its ordinary initial image in degree ℓ into degree $\ell - j$. For every such germ f ,

$$w(f/p_Q^j) = w(f).$$

The vanishing of this valuation debit follows from the actual incidence.

Proof. In the embedding-seven case, containment would give a surjection $\mathcal{O}_{Q,x} \rightarrow \mathcal{O}_{W,x}$, hence a surjection of cotangent spaces. Since $\mathcal{O}_{Q,x}$ is regular of dimension six, the embedding dimension of its quotient is at most six. Thus containment is impossible in that case. Under noncontainment, p_Q is a unit at the generic point of W , so $w(p_Q) = 0$.

The full ring $\text{gr}_{\mathfrak{m}_x} \mathcal{O}_{X,x}$ is a polynomial ring in seven variables. The initial form of p_Q is a nonzero linear form. Multiplication by its j -th power is injective and increases ordinary order by exactly j . This proves the assertion about initial images and the displayed valuation identity. $\square$

Lemma 17.4. *Suppose $L^7 = 1$, t is rational, and $\pi^*L - tE_x$ is big. For finitely many distinct prime divisors Q_i through x , put $g_i = L^6 \cdot Q_i$. Then*

$$V(t) \leq \left(1 - \sum_i g_i \phi_{Q_i}(t)\right)_+^7.$$

The original ample degrees occur as weights in this inequality.

Proof. Let P_k be the entire fixed divisor of $U_k(t)$. The definition of the profiles gives

$$\sum_i g_i \phi_{Q_i}(t) \leq \frac{P_k \cdot L^6}{k} \leq 1 - \left(\frac{M_k^7}{k^7}\right)^{1/7},$$

where the second inequality and the convergence $M_k^7/k^7 \rightarrow V(t)$ are Proposition 6.19. Taking the limit proves the claim. All other fixed components enter P_k before its intersection number is bounded from below by the indicated sum. $\square$

Lemma 17.5. *Let $v > 0$ be rational with $\pi^*L - vE_x$ big. For a nonzero complete original source $U_k(v)$, let I_k be its full base ideal, P_k its entire fixed divisor, and $C_k = I_k \mathcal{O}_X(P_k)$. On a smooth projective principalization $\rho: Y_k \rightarrow X$, write*

$$\begin{aligned} I_k \mathcal{O}_{Y_k} &= \mathcal{O}_{Y_k}(-\rho^*P_k - G_k), & C_k \mathcal{O}_{Y_k} &= \mathcal{O}_{Y_k}(-G_k), \\ A_k &= k\rho^*L, & M_k &= A_k - \rho^*P_k - G_k. \end{aligned}$$

Suppose M_k is big. Set

$$\begin{aligned} \mu_k &= \frac{M_k^7}{k^7}, & p_k &= \frac{P_k \cdot L^6}{k}, \\ \Delta_k &= p_k^2 - \frac{P_k^2 \cdot L^5}{k^2}, & D_k &= \frac{1}{k^2} \sum_Z e(C_k \mathcal{O}_{X, \eta_Z}) (L^5 \cdot Z), \end{aligned}$$

where Z ranges over all codimension-two components of $V(C_k)$, and the empty sum is zero. Then $0 \leq p_k < 1$, $\Delta_k, D_k \geq 0$, and

$$(17.2) \quad 0 < \mu_k \leq \frac{[(1 - p_k)^2 - \Delta_k - D_k]_+^6}{(1 - p_k)^5}, \quad (1 - p_k)^2 - \Delta_k - D_k > 0.$$

For every $\varepsilon > 0$, a cutoff depending only on $(X, L, x, v, \varepsilon)$ ensures that every permitted degree k beyond it has big M_k , $|\mu_k - V(v)| < \varepsilon$, and, for every $0 \leq d_k \leq D_k$,

$$(17.3) \quad V(v) < \frac{((1 - p_k)^2 - d_k)_+^6}{(1 - p_k)^5} + \varepsilon.$$

Here a permitted degree satisfies $kv \in \mathbb{Z}$.

The fixed square in this estimate includes every companion and all their cross terms. The cutoff precedes any choice of a center or place.

Proof. We compute the first two moving intersections and then compare their successive ratios.

Step 1. In this step, we compute the first two moving intersections while retaining the full residual cycle. Suppress the degree index. The original sections generate M , so A, M are nef and $A - M = \rho^*P + G$ is effective. Put $s_j = A^{7-j}M^j$. Effectivity and nefness give $s_j \geq s_{j+1} \geq s_7 > 0$. Since C has no divisorial support, $\rho_*G = 0$ as a divisor. Thus

$$\frac{s_1}{k^7} = 1 - p > 0.$$

We retain the full residual multiplicity cycle in the next intersection. At a codimension-two component Z , $\mathcal{R} = \mathcal{O}_{X, \eta_Z}$ is regular of dimension two and $J = C\mathcal{R}$ is primary to its maximal ideal. On the ordinary ideal blowup $\text{Proj } \mathcal{R}[Jt]$, let G_J be the full exceptional Cartier scheme. Its tautological bundle is $\mathcal{O}(-G_J)$. The Hilbert polynomial

$$\text{length}_{\mathcal{R}}(J^a/J^{a+1}) = e(J)a + O(1)$$

is the Hilbert polynomial of this full exceptional scheme with that bundle. Consequently $G_J \cdot (-G_J) = e(J)$, including all scheme multiplicities and residue-field degrees. Both A and ρ^*P are trivial over $\text{Spec } \mathcal{R}$, so M restricts to $-G_J$. Projection formula on the principalization gives coefficient $e(J)$ at Z in $\rho_*(G \cdot M)$. A general member of the generated M meets every component of G properly; the resulting cycle is effective. All other contributions have nonnegative L -degree. Hence

$$A^5GM \geq k^5 \sum_Z e(C\mathcal{O}_{X, \eta_Z})(L^5 \cdot Z) = k^7D.$$

This uses the full residual ideal. In particular, a component Z contained in P retains this same charge. Moreover, $A^5\rho^*PG = 0$ by $\rho_*G = 0$, so

$$A^5\rho^*PM = k^6(P \cdot L^6) - k^5(P^2 \cdot L^5).$$

Combining these two terms in $s_1 - s_2 = A^5(\rho^*P + G)M$ yields

$$(17.4) \quad 0 < \frac{s_2}{k^7} \leq 1 - 2p + \frac{P^2 \cdot L^5}{k^2} - D = (1 - p)^2 - \Delta - D.$$

Step 2. In this step, we derive the mixed-intersection bound and prove nonnegativity of the fixed-square deficit. We derive the mixed-intersection inequalities from Theorem 6.3. For two nef rational classes B_1, B_2 on a smooth projective sevenfold, perturb both by δH with H ample and $\delta > 0$ rational. For $1 \leq j \leq 6$, take a smooth complete intersection surface formed from sufficiently divisible general members with total class proportional to $(B_1 + \delta H)^{6-j}(B_2 + \delta H)^{j-1}$. On that surface the square root of the volume is concave, and volume equals the ordinary square for ample classes. For the restrictions b_1, b_2 , differentiating $\sqrt{(b_1 + tb_2)^2}$ twice at zero gives $(b_1^2)(b_2^2) - (b_1 \cdot b_2)^2 \leq 0$. Cancel the positive complete-intersection multiples and let δ tend to zero. This proves

$$(B_1^{7-j}B_2^j)^2 \geq (B_1^{8-j}B_2^{j-1})(B_1^{6-j}B_2^{j+1}).$$

In particular s_j/s_{j-1} is nonincreasing, whence $s_7 \leq s_1(s_2/s_1)^6$. Together with (17.4), this proves (17.2).

To see $\Delta \geq 0$, apply the same inequality on X , with $j = 1$, to L and $L + \epsilon P$. The latter is ample for sufficiently small positive rational ϵ , without any nefness assumption on P . After expanding and cancelling the constant and linear terms, one obtains $P^2 \cdot L^5 \leq (P \cdot L^6)^2$, because $L^7 = 1$. This is the square of the entire P ; no companion intersection has been discarded.

Step 3. In this step, we obtain the uniform finite-degree volume comparison. Finally Proposition 6.19 gives eventual bigness and $\mu_k \rightarrow V(v)$ along the permitted original degrees. Fix its cutoff for the prescribed ε before any observation. Dropping $\Delta_k \geq 0$ only in (17.2), and then

replacing D_k by $d_k \leq D_k$, proves (17.3). The finite intersection estimate itself does not require kv to be integral. $\square$

Lemma 17.6. *Suppose $L^7 = 1$ and $\pi^*L - TE_x$ is big. Let $W \subset X$ be integral through x , let w be a divisorial valuation centered on W , and let Q be a prime divisor through x , smooth there, with $W \not\subset Q$. Fix a rational $R_* > T$ exceeding every attained original point order divided by degree. Let F, G be positive affine functions on $[T, R_*]$ such that $\phi_Q(t) \geq F(t)$ and $\phi_w(t) \geq G(t)$ on every attained order in this interval. Put*

$$J(H) = \{z \in \mathcal{O}_{X,x} : w(z) \geq A_X(w)H\} \quad (H > 0).$$

Suppose there are $\nu, \kappa > 0$, $\gamma \geq 0$, and a constant C_0 such that, for every integer $D \geq 1$ and $H > 0$ with $H/D \geq \gamma$,

$$(17.5) \quad \dim_{\mathbb{C}}(\text{in}_{\mathfrak{m}_x} J(H))_D \leq \frac{\nu}{720} (D - \kappa H)_+^6 + C_0(D + 1)^5.$$

If $\gamma > 0$, assume also that some $\eta > \gamma$ satisfies $G(t) \geq \eta(t - F(t))_+$ on $[T, R_]$. Then*

$$V(T) \leq 7\nu \int_T^{R_*} (t - F(t) - \kappa G(t))_+^6 dt.$$

Only existing local images are divided in this comparison.

Proof. We bound every ordinary layer and then sum the entire outgoing filtration. Fix the profiles and the interval before taking a degree limit.

Step 1. In this step, we bound every outgoing layer after actual smooth-prime division. For an original degree p and $\lceil pT \rceil \leq \ell < \lceil pR_* \rceil$, put

$$t = \ell/p, \quad j = \lfloor pF(t) \rfloor, \quad D = \ell - j.$$

Every section of $U_p(t)$ is locally divisible by p_Q^j . Because $w(p_Q) = 0$, its divided germ belongs to $J(pG(t))$. Division induces an injection on the degree- ℓ initial image, with target the degree- D initial image of this full ideal. This follows from the polynomial-ring argument in Lemma 17.3, using only the assumed noncontainment. If $D < 0$, the original initial image is zero. If $D = 0$, it has dimension at most one.

For $D \geq 1$, the threshold condition in (17.5) holds for all sufficiently large p . Indeed, if $\gamma > 0$, the positive minimum of G on the fixed interval and $D \leq p(t - F(t)) + 1$ show that

$$pG(t) - \gamma D \geq p(G(t) - \gamma(t - F(t))_+) - \gamma \geq p \left(1 - \frac{\gamma}{\eta}\right) \min_{[T, R_*]} G - \gamma > 0.$$

For $\gamma = 0$ there is no additional condition. The capacity estimate, including the at-most-one error in j , now gives

$$\dim \frac{U_p(\ell/p)}{U_p((\ell+1)/p)} \leq \frac{\nu p^6}{720} (t - F(t) - \kappa G(t))_+^6 + O(p^5).$$

The same error covers $D = 0$.

Step 2. In this step, we telescope the entire outgoing filtration and pass to the volume limit. All constants are fixed before this degree limit. There are $O(p)$ layers, so their errors sum to $O(p^6)$. The final source $U_p(R_*)$ is zero by the choice of R_* . The sum therefore bounds $\dim U_p(T)$, without discarding any terminal space. Along degrees divisible by the denominator of T , Theorem 6.4 identifies its normalized limit with $V(T)$. Multiplication by $7! = 7 \cdot 720$ gives the claimed integral.

Each target ideal is used only for an upper bound. The divided image retains all the original conditions on its germs; no assertion of global surjectivity or of completeness of a residual series enters. $\square$

The finite endpoint R_* can be chosen from the original geometry. For example, choose a so that aL is very ample. For each effective divisor $D \in |pL|$, intersect six general members of $|aL|$ through x , chosen smooth and transverse there, so that the resulting curve is not contained in D . Its local intersection with D at x is at least $\text{mult}_x D$, whereas its total intersection is $a^6 p L^7$. Thus any rational $R_* > \max(T, a^6)$ works. This fixes the outgoing interval before any valuation is observed.

Lemma 17.7. *Suppose $L^7 = 1$ and $K_X + 8L$ is not generated at x . Let q be a positive integer divisible by 25, suppose $U_q(T) \neq 0$, and take a pure actual threshold average $B_T \sim_{\mathbb{Q}} s_T L$. Let W_T be its minimal center and let $b_T = \text{def}_x(B_T)$. Then either $\dim W_T = 6$, or W_T is a fivefold with the following possible states:*

$\text{edim } \mathcal{O}_{W_T, x}$	$\text{mult}_x W_T$	lower bound for b_T
7	3	28725/11134
7	4	3467/1328
7	5	37483/15020.

In the nondivisorial cases, $1/s_T \geq 3004/11665$.

The multiplicity-two case is paid at the later threshold's own price.

Proof. Proposition 9.1 gives, for the whole B_T ,

$$\text{mult}_x B_T \geq T s_T, \quad \text{mult}_x B_T + b_T \leq 7, \quad T s_T + b_T \leq 7.$$

Proposition 5.9 leaves a divisor or one of four fivefold states. If the multiplicity is two, Theorem 16.2 applies with $n = 7$ and $B = B_T$, contradicting nongeneration. Its statement covers all quadratic ranks and analytic alternatives.

For the other three states, substitute $u = T$ in the respective lower deficit bounds of Proposition 5.9. This gives the displayed fractions. The first two exceed $5/2$ and the third is less than $5/2$. Consequently

$$\frac{1}{s_T} \geq \frac{T}{7 - b_T} \geq \frac{T}{7 - 37483/15020} = \frac{3004}{11665}.$$

The scalar census already includes every descendant and both cutting errors through Theorem 5.4; no additional cut is taken here. $\square$

Lemma 17.8. *Suppose that $K_X + 8L$ is not generated at x . Let $\Gamma \sim_{\mathbb{Q}} sL$ be a pure actual threshold average of a nonzero complete source $U_k(v)$, where $v \geq 26/25$ and kv is integral. If its minimal center is a fourfold, then its embedding dimension and multiplicity belong to*

$$(e, m) = (6, 3), (6, 4), (7, 4), (7, 5), (7, 6).$$

Its entire ordinary tangent ring is reduced Cohen–Macaulay, all ordinary powers of its point ideal are integrally closed, and its ordinary Rees ring is normal Cohen–Macaulay. In the two cases $e = 6$, its deficit satisfies $b > 39/20$. At $v = 27/25$, only $(6, 3), (7, 4), (7, 5)$ remain, and in the last two cases $1/s \geq 15392/72425 > 17/80$.

Proof. We record the descendant bounds, and then use the resulting slack on the ordinary blowup.

Step 1. In this step, we identify the five surviving fourfold states and their slack bounds. Proposition 5.9 gives $b > 3/2$. The states in Definition 5.2 whose fourfold cap exceeds $3/2$ and

the affine bounds needed for their full descendant envelopes are

e	m	β	a	p	$\mathcal{F}_{(4,e,m)}(z) \leq az + p \quad (0 \leq z \leq \beta).$
4	1	4	1	757/1000	
5	2	3	119/100	287/500	
5	3	2	1317/1000	171/500	
6	3	8/3	1317/1000	71/125	
6	4	2	283/200	379/1000	
6	5	8/5	187/125	297/1000	
7	4	5/2	283/200	29/50	
7	5	2	187/125	17/40	
7	6	5/3	783/500	331/1000	

These are bounds for the complete recurrence, including a successor of dimension two or one. To check an affine bound, substitute the successor envelope in that recurrence and test its affine pieces at their endpoints and at the caps $(4m - 2T(e - 4, m) - m')/m$. The root radii are bounded above by the displayed a . Lower ample degree bounds are all one. The threefold states of multiplicity eight and nine are included with the distinct embedding-six and embedding-seven caps of Lemma 5.10.

Put $M = \text{mult}_x \Gamma$ and $\sigma = 7 - M - b$. Failure of the paid continuation gives

$$8 \leq s + ab + p, \quad vs + b \leq 7, \quad 0 \leq \sigma \leq 7 - b + v(ab + p - 8).$$

A strict reverse inequality in the first comparison permits both cutting errors by Theorem 5.4, and then gives generation by Proposition 3.12. Since $a > 25/26$, the maximum of $(7 - z)/(26/25) + az + p$ occurs at the stated cap. In the rows (4, 1), (5, 2), (5, 3), (6, 5), its values are respectively

$$\frac{75}{26} + \frac{4757}{1000}, \quad \frac{12984}{1625}, \quad \frac{125}{26} + \frac{372}{125}, \quad \frac{135}{26} + \frac{13453}{5000};$$

each is less than eight. Thus precisely the five asserted rows remain. In their displayed order, substitution at the caps gives

$$\sigma \leq \frac{481}{1875}, \quad \frac{217}{12500}, \quad \frac{2311}{5000}, \quad \frac{2921}{12500}, \quad \frac{2699}{37500}.$$

Every bound is less than 1/2. Indeed, the expression bounding σ increases with b and decreases with v , because $av > 1$ and $a\beta + p < 8$.

Step 2. In this step, we obtain the ordinary-ring conclusions and the stronger source-specific deficit bounds. The same boundary has $\delta = 7 - M = b + \sigma > 3/2$. The $\delta > 1$ assertion of Theorem 4.20 therefore applies to the ordinary blowup: it gives all integral-closure equalities, a normal Rees ring with rational singularities, and a Cohen–Macaulay full tangent ring. Its $\sigma < 1/2$ assertion makes the entire exceptional scheme reduced and then makes this full tangent ring reduced. In particular this argument includes its affine vertex. The Rees ring is Cohen–Macaulay because it has rational singularities.

For either embedding-six row the failed-tail inequality gives

$$b \geq \frac{(8 - p)v - 7}{av - 1}.$$

This increases with v ; at 26/25 its values are 9116/4621 and 11573/5895, both greater than 39/20. At $v = 27/25$ the five lower bounds are

$$\frac{25664}{10559}, \quad \frac{30767}{13205}, \quad \frac{5068}{2641}, \quad \frac{29525}{15392}, \quad \frac{32063}{17282}.$$

The second and fifth exceed their caps. In the two embedding-seven rows, $1/s \geq v/(7 - b)$ consequently gives respectively 2641/12425 and 15392/72425. The former is larger, and

$$\frac{15392}{72425} - \frac{17}{80} = \frac{27}{1158800} > \frac{1}{100000}. \quad \square$$

Lemma 17.9. *Under nongeneration with $L^7 = 1$, one has $V(26/25) > 3/40$.*

Proof. Apply Proposition 7.28 to every actual layer through $26/25$, and telescope. The leading loss is

$$\left(\frac{9}{10}\right)^7 + \frac{61}{100} \left[\left(\frac{99}{100}\right)^7 - \left(\frac{9}{10}\right)^7 \right] + \frac{11}{25} \left[\left(\frac{26}{25}\right)^7 - \left(\frac{99}{100}\right)^7 \right] < \frac{37}{40}.$$

This is an inequality of rational numbers; it follows by multiplying by $40 \cdot 100^8$. The sum of the uniform layer errors is $O(q^6)$, whereas ample Riemann–Roch has leading coefficient $q^7/7!$. The complete volume limit proves the claim. $\square$

Source histories and certificate data. The arguments below use three different histories. An eligible embedding-seven observation earns the reserve $V(T) > 1/64$. An initial fivefold at R first earns $V(r) > 6/25$ and then $V(T) > 1/500$. A smooth original divisor at r is treated for every initial history, using its own fixed mass and incidence constraints to obtain the reserve $V(T) > 1/1000$. The later threshold is taken only after the relevant reserve has been earned. The geometric arguments, parameter domains, capacities, slope tails and consumed margins are given in the corresponding proofs.

The finite witnesses are data objects in the accompanying supplement. Their identifiers have the form `17-ambient-endpoints/key`; the corresponding file is `data/key.txt` in that section's supplement. The local references following each calculation specify its keys. There are twenty-five objects, with separate dictionaries and source histories. In particular the early retained-parameter certificate has 1099 cells; it is distinct from the 1031-cell source calculation used in the earlier original-source theorem.

Definition 17.10. Let Ω be a set of admissible real parameters contained in a finite union of prescribed closed rational boxes. For each $\omega \in \Omega$, let

$$L_m(\omega) = 7 \int_0^{z_m} g_m(t, \omega) dt + M_m(\omega) \quad (m \in \mathcal{M}),$$

where $\mathcal{M}$ is finite, $z_m > 0$ is rational, g_m is nonnegative and integrable, and $M_m \geq 0$. A finite witness consists of a finite subdivision of the root boxes, with both closed children of every split retained, and the following records at each terminal box B . An empty record proves $B \cap \Omega = \emptyset$ by a stated strict incompatibility. Every other record chooses $m = m(B)$, gives a rational bound $\overline{M}_B \geq M_m(\omega)$ on $B \cap \Omega$, and partitions $[0, z_m]$ into adjacent intervals I . For each I it gives a rational number $U_{B,I}$ satisfying

$$7 \int_I g_m(t, \omega) dt \leq U_{B,I} \quad (\omega \in B \cap \Omega).$$

A coordinate interval of length zero remains part of its box.

Lemma 17.11. *In Definition 17.10, suppose that every terminal record is valid and that, for prescribed rational numbers c_m , every nonempty record satisfies*

$$U_B := \overline{M}_B + \sum_I U_{B,I} < c_{m(B)}.$$

Then there is a rational $\delta > 0$ such that every $\omega \in \Omega$ satisfies $L_m(\omega) \leq c_m - \delta$ for at least one $m \in \mathcal{M}$.

Proof. The closed children of a split have union equal to their parent. Induction on the finite tree therefore puts every admissible parameter in a terminal box. A strict incompatibility cannot remove that parameter, including when one or more coordinates are fixed. Summing its interval inequalities and its mass inequality gives $L_{m(B)}(\omega) \leq U_B$. The minimum of the finitely many positive rationals $c_{m(B)} - U_B$ is a suitable δ . If there are no admissible parameters the assertion is vacuous. $\square$

Exact verification of a terminal record. We verify the integral comparisons, domain coverage, and complete error allowance of each terminal record.

Step 1. In this step, we compute the exact cell contributions for each type of polynomial bound. The local formulas below give either a polynomial majorant on each time interval, several competing majorants, or a Bernstein enclosure. For an unweighted polynomial majorant $P(t) = \sum_j p_j t^j$, the cell contribution is

$$7 \int_l^v P(t) dt = 7 \sum_j \frac{p_j}{j+1} (v^{j+1} - l^{j+1}).$$

The linked-profile dictionaries instead store the polynomial $\hat{P}(t) = 7P(t) = \sum_j \hat{p}_j t^j$, which already includes the factor seven. Its cell contribution is

$$\int_l^v \hat{P}(t) dt = \sum_j \frac{\hat{p}_j}{j+1} (v^{j+1} - l^{j+1}).$$

This value is compared directly with the recorded upper bound. The later affine and indexed dictionaries store unweighted polynomials and use the first formula, multiplying their integrals by seven. For several competing majorants, the minimum of their integrals still bounds the integral of their pointwise minimum. This is the rule used by the deterministic-tree records. For a degree-six Bernstein enclosure $P \leq \sum_{j=0}^6 b_j B_j^6$ on $[l, v]$, nonnegativity and the integral of the Bernstein basis give $7 \int_l^v P \leq (v-l) \sum_j b_j$. The same argument bounds a maximum of polynomials if the same coefficients dominate each of them.

Step 2. In this step, we verify the complete box and interval coverage with every branch convention retained. Each test reconstructs the box and its effective endpoints from the prescribed root, and verifies its dictionaries, every split, every terminal record and every cell. All changes of formula are split at their exact affine roots, capacity thresholds and anchors; adjacent pieces may merge only when their polynomials agree. A zero affine coefficient is evaluated by the constant formula, including the zero coefficient in a retained slice integral. At a jump, the applicable one-sided bounds are retained; a single time endpoint has zero integral. Parameter faces remain covered by the closed boxes. Normalized coordinates representing a collapsed physical interval are retained with its constant image, and the interior kink in a linked slope cap is included when enclosing that image. The family proofs specify the geometric empty tests and all denominators that must stay positive.

Step 3. In this step, we account for every rounding and geometric error before deducing the full-domain comparison. An integer ceiling supplies a weak rational enclosure; a strict cell bound uses the stated strict inequality. The final comparison always uses the complete sum, its family-specific fixed mass, and the strict margin stated in that proof. Thus no rounding error is unaccounted for. The separate errors in complete original profiles, residual orders and finite-degree volume comparisons are fixed before the original observations and are paid from the geometric margins there. Applying Lemma 17.11 to the compact domain, and then the stated analytic slope tails, proves the asserted loss or volume comparison for the entire domain. The complete-filtration arguments turn precisely that comparison into the claimed source or contradiction.

Definition 17.12. For $c \geq 0$, set $E_j(c) = (1 - jc)_+^6$, $N_q(c) = E_{7,q}(c)$, and

$$\begin{aligned} Q(c) &= \max\{E_1(c), (1 - 2c)_+^5 (1 + 10c)\}, \\ D_0(c) &= 6E_2(c) - 8E_3(c) + 3E_4(c), \\ M_0(c) &= E_1(c) + 3E_2(c) - 5E_3(c) + 2E_4(c), \\ M_1(c) &= \begin{cases} 1, & c < 2/11, \\ E_1(c) + E_2(c) - E_3(c), & c \geq 2/11, \end{cases} \\ D(c) &= \begin{cases} D_0(c), & c < 1/6, \\ \min\{D_0(c), 138259/900000\}, & c \geq 1/6, \end{cases} \\ M(c) &= \min\{M_0(c), N_3(c), 1, 256/729 + 18(1/6 - c)_+, M_1(c)\}. \end{aligned}$$

Here $E_{7,q}$ is the envelope of Definition 7.7. In certificate records, the capacity letters G, Q, D, M, T mean, respectively, N_5, Q, D, M, K_5 , where K_5 is defined in Theorem 7.18. The letter T in a capacity column is a table code; all time endpoints are written separately.

The capacity G applies to every initial center in Lemma 17.1, by appending zero normal weights. The capacity Q requires only a height-two prime. For D and M , the full reduced Cohen–Macaulay tangent hypotheses follow from Lemma 17.8. Their bounds are Theorem 7.18, Proposition 7.19, and Proposition 7.21. In each case the leading bound applies to every full primary comparison ideal, with an error $O((\ell + 1)^5)$ independent of its exponent and computing place.

Lemma 17.13. Fix increasing big anchors $r_0, \dots, r_v$, positive rational base heights m_i , and $\varepsilon > 0$. Suppose that pure actual thresholds of complete original sources supply computing profiles ϕ_i with $\phi_i(r_i) > m_i$, and that each normalized complete-source error is less than ε . Let a_i be supporting slopes at their respective anchors. Define

$$h_0 = m_0, \quad h_i = \max(\{m_i\} \cup \{h_j + a_j(r_i - r_j) - \varepsilon : j < i\}), \quad H_i(t) = (h_i + a_i(t - r_i))_+.$$

If K_i is any applicable nonnegative nonincreasing capacity for the full primary ideal at the center of ϕ_i , then at every fixed finite endpoint z , the complete original filtration satisfies

$$\liminf_{p \rightarrow \infty} \frac{7!}{p^7} \dim U_p(z) \geq 1 - 7 \int_0^z t^6 \min\{B(t), K_i(H_i(t)/t) : 0 \leq i \leq v\} dt.$$

Here $L^7 = 1$, B is the actual layer baseline of Proposition 7.28, and the displayed integral is used only within its stated baseline interval. On a closed rectangle $l_i \leq a_i \leq u_i$, a pointwise upper bound for this integral is obtained by computing the linked heights with l_j , and using u_i before r_i and l_i after r_i in its affine profile.

Proof. We obtain the linked profile bounds, sum their common original images, and then compare rectangle endpoints.

Step 1. In this step, we prove that the linked heights give lower supporting profiles. Lemma 17.2 and induction give $h_i < \phi_i(r_i)$. A supporting line lowered to height h_i remains a lower bound on every attained original order, including those outside the known big interval. This last assertion follows directly by taking powers and products in the original section ring, as in Theorem 6.13.

Step 2. In this step, we telescope the common full-image bounds to obtain the source estimate. For the original layer of degree p and order ℓ , put $t = \ell/p$. Every actual germ belongs, for each positive $H_i(t)$, to

$$J_{i,p,\ell} = \{f : w_i(f) \geq pA_X(w_i)H_i(t)\}.$$

This ideal is primary to the actual center prime: elements outside that prime have valuation zero, and a sufficiently high power of the prime has the required positive valuation. Since

$w_i(J_{i,p,\ell}) \geq pA_X(w_i)H_i(t)$, testing its generic threshold with w_i gives the upper bound

$$\frac{A_X(w_i)}{w_i(J_{i,p,\ell})} \leq \frac{1}{pH_i(t)}.$$

The actual ordinary layer injects into the full initial image of each such ideal. Thus the minimum of the capacity bounds and the baseline bounds this same image. At height zero use the full ordinary layer bound. Every original fixed, residual, embedded and remote condition remains in the image being bounded.

Telescope all layers through $[pz]$, starting with the whole complete linear system. The degree-zero layer is at most one and the remaining errors sum to $O(p^6)$. Riemann–Roch has leading coefficient $p^7/7!$, so the Riemann sum gives the claimed lower bound. In particular a positive lower bound produces an actual global section; it does not lift a local comparison image. The ratios $H_i(t)/t$ are nondecreasing, since an actual supporting slope is at least h_i/r_i . Their capacities have bounded variation, with only the stated finite jumps, uniformly in the actual slopes. This also makes the Riemann-sum error uniform.

Step 3. In this step, we prove the pointwise rectangle comparison. The linked heights increase with every earlier slope and height. Replacing these by lower rectangle endpoints therefore decreases them. Before an anchor the coefficient of its slope is negative; afterwards it is positive. The asserted choice of endpoints decreases each affine line on its whole time interval. Monotonicity of the capacities reverses these inequalities and proves the rectangle comparison, including all rectangle boundaries. $\square$

How the rational certificates are read. The five **reserve-*** objects and three **point-*** objects in the **17-ambient-endpoints** supplement use the following format. Each object has its own rational and polynomial dictionaries. Within a dictionary entry, a tree, or a cell trace, line wraps have no mathematical meaning: concatenate wrapped portions before decoding them. Its symbols q_j encode exact rational numbers and are unrelated to an original degree. A symbol p_j encodes one polynomial name from the following list. For a selected rectangle profile of height h , anchor r_i and endpoint slope a , put

$$d = ar_i - h, \quad Y(t) = at - d, \quad P_j(t) = 7((1 - ja)t + jd)_+^6.$$

The prefix i : selects this profile; the position of the cell relative to r_i selects its endpoint slope. The names mean

$E1$	P_1
$E2$	P_2
C	$(7/6)^6 P_1$
$Nqlo$	$7t^6 - 7(7q/(q+1))^q t^{6-q} Y(t)^q$
$Nqhi$	$7t^6 - \frac{7t^{6-q}}{(6(q+1))^q} ((6q-1+7a)t - 7d)^q$
$Qbal$	$7((1-2a)t + 2d)^5 ((1+10a)t - 10d)$
Dlo	$6P_2 - 8P_3 + 3P_4$
Dhi	$7(138259/900000)t^6$
$Mspan$	$P_1 + 3P_2 - 5P_3 + 2P_4$
$Mref$	$P_1 + P_2 - P_3$
$H3$	$7(256/729)t^6$
$H3lip$	$7(256/729 + 3 - 18a)t^6 + 126dt^5$
$Tlow$	$3P_2 - 2P_3$
$Ttwo$	$2P_2$.

A baseline name Bq means $7qt^6$ on its indicated constant baseline interval. The names $Bhigh$ and $Bzero$ mean $7((125 - 52t)/73)^6$ before $125/52$, and zero afterwards, on the last interval of the baseline.

Every polynomial is selected only on its applicable branch. The names $Nqlo$, $Nqhi$ and C require respectively $0 < c \leq 1/7$ and $1/7 < c < 1/2$; the generic-profile name $E1$ requires $c \geq 1/2$. The names Dlo and $Mspan$ are available at every positive height. The names Dhi , $H3$ require $c \geq 1/6$, $H3lip$ requires $c < 1/6$, and $Mref$ requires $c \geq 2/11$. For a Q profile, $Qbal$ is used only at $c \leq 16020581569/90194313216$, and $E1$ only at $c \geq 8010290789/45097156608$. These rationals isolate the unique positive crossing of $(1 - c)^6$ and $(1 - 2c)^5(1 + 10c)$. Indeed the logarithmic derivative of their second-to-first ratio is

$$\frac{6 - 72c}{(1 - 2c)(1 + 10c)(1 - c)}.$$

That ratio starts at one, increases until $1/12$, and then decreases to zero. Direct substitution at the two displayed rationals gives opposite signs relative to one. The supplementary cell traces select other available bounds between the isolators. For a later K_5 profile, $Tlow$, $Ttwo$ are used below $4/25$, and $E2$ at or above $4/25$. At height zero the baseline is available. All required roots of the positive parts and all relevant anchors are cell endpoints.

The initial grid specifies closed Cartesian rectangles, numbered from one on each axis. Their indices, joined by periods, are their initial tags. A split tag i/q_j replaces its parent by the two rectangles obtained by cutting coordinate i at q_j ; their tags append 0 and 1, respectively. Coordinates are numbered from zero. Their closed union is the parent, so the subdivision tree recorded in each data object certifies full coverage of its prescribed root grid.

Every final record in these objects lists all lower and upper slopes, then all linked heights, then a strict upper numerator N with denominator 10^{10} , and its complete ordered cell trace. A trace entry q_j/p_k means that the polynomial p_k is used until the right endpoint q_j . Its left endpoint is the preceding right endpoint, initially zero. The last endpoint is the family's stated integration endpoint. Every polynomial bounds the minimum on the entire cell, whether or not it is the smallest available polynomial there. If it expands as $\sum_j c_j t^j$, the cell contribution is exactly

$$(17.6) \quad \sum_j \frac{c_j}{j+1} (v^{j+1} - l^{j+1}).$$

For each such record, the sum over its entire trace is strictly less than $N/10^{10}$. Thus the dictionaries, subdivision tree and traces specify finite rational inequalities, including every polynomial and every integration interval.

Lemma 17.14. *Let $(A, \mathfrak{m})$ be a smooth complex local ring of dimension n , let p be a regular parameter, and let P be a prime of height $2 \leq c \leq n - 1$ containing p . Let w be a divisorial valuation centered on P , put $\alpha = w(p)/A_A(w)$, and set $J_q = \{f : w(f) \geq qA_A(w)\}$, for $q > 0$. Then $0 < \alpha < 1$. For $e \geq 0$ and $\ell \geq 1$,*

$$\dim_{\mathbb{C}}(\text{in}_{\mathfrak{m}}((p^e) \cap J_q))_{\ell} \leq \frac{\ell^{n-1}}{(n-1)!} \sup \Pr \left\{ Z_0 \geq \frac{e}{\ell}, \beta Z_0 + \sum_{i=1}^{c-1} b_i Z_i \geq \frac{q}{\ell} \right\} + C_n(\ell + 1)^{n-2}.$$

The supremum ranges over $\beta \geq \alpha$, $b_i \geq 0$, $\beta + \sum b_i = 1$, with the other $n - c$ weights zero; Z is uniform on the n -coordinate unit simplex. The constant precedes all choices of parameter, prime, valuation, threshold and exponent. A genuine division by p^e has normalized valuation debit $e\alpha$.

Proof. We preserve the parameter through two full-ideal specializations, and then separate a segment from the resulting Newton polyhedron.

Step 1. In this step, we preserve the retained parameter and relative threshold bounds through both specializations. The ideal J_q is P -primary: elements outside P have valuation zero, and a high enough power of P lies in J_q . At the generic point of P , complete p to regular parameters $p, z_2, \dots, z_c$. The simple normal crossing discrepancy inequality gives

$$A_A(w) \geq w(p) + \sum_{i=2}^c w(z_i) > w(p) > 0.$$

For each $0 \leq b < 1$, the same valuation gives

$$\text{lct}_{A_P}((A, b(p=0)); J_q A_P) \leq \frac{1 - b\alpha}{q}.$$

Consider the extended point Rees algebra $\mathcal{D} = \sum_{j \in \mathbb{Z}} \mathfrak{m}^j t^{-j}$, where $\mathfrak{m}^j = A$ for $j \leq 0$, and contract the full ideals J_q, P from its generic fiber. Their quotients are t -torsion-free. The special fibers are their entire point initial ideals J, Q , with $Q^N \subset J \subset Q$ for some N . At a height- c minimal prime T_0 of Q , the total local ring at (t, T_0) is regular: its quotient by the nonzerodivisor t is the regular local ring of the smooth ambient tangent space at T_0 . Lifting its parameters together with t gives a regular system of parameters in the total ring.

The retained element is p/t ; its special fiber p_0 is a nonzero linear coordinate. Since $p_0 \in T_0$, it remains a regular parameter after localization. Thus the retained divisor and the special fiber meet as smooth parameters. Theorem 6.1, namely [Laz09, Theorem 3.2 and the ideal version after Remark 3.7], also applies to mixed nonnegative rational exponents: clear a common denominator and apply it to the product of the indicated ideal powers. A unit restricted multiplier ideal makes the ambient one a unit by Nakayama's lemma. Localization at the horizontal generic prime would then contradict the displayed threshold inequality. It follows that the same relative threshold bound holds at every such component of the full point initial support. A log resolution verifies its preservation under the coefficient-field extensions in these localizations.

Next take a common homogeneous monomial order for the full ideals J, Q , retaining p_0 as a coordinate. One integer weight realizes both finite Gröbner bases. Its weight family preserves their full graded Hilbert functions and the sandwich of their powers. At each height- c coordinate support prime, a horizontal minimal support prime in the total local family has height c , by dimension and strict inclusion in the height- $c + 1$ special-fiber prime. The same regular-local restriction argument therefore applies. If K is the full resulting monomial ideal, then, at this coordinate prime,

$$\text{lct}((\mathbb{A}^n, b(p_0 = 0)); K) \leq \frac{1 - b\alpha}{q} \quad (0 \leq b < 1).$$

First prove this for rational b ; the finite set of discrepancy expressions on a log resolution gives continuity and then the limit as b tends to one.

Step 2. In this step, we separate the excluded Newton segment and obtain the retained normal coefficient. The mixed form of the Newton formula in Theorem 7.1 [How01, Main Theorem] subtracts b times the first normal weight from each discrepancy. Consequently the interior of the localized Newton polyhedron avoids

$$\left[q(1, \dots, 1), \left(0, \frac{q}{1 - \alpha}, \dots, \frac{q}{1 - \alpha} \right) \right].$$

Indeed its points are $q(1 - b, 1, \dots, 1)/(1 - b\alpha)$. Separate this compact segment from the open convex interior. Upward closure gives a nonzero normal $v_i \geq 0$ whose supporting value h_v satisfies

$$h_v \geq q \sum_i v_i, \quad h_v \geq \frac{q}{1 - \alpha} \sum_{i > 0} v_i.$$

Primariness of the localization gives $h_v > 0$ and $v_i > 0$: a pure power of a zero-weight coordinate would contradict the supporting inequality. Set $d_i = qv_i/h_v$, keep $b_i = d_i$ for $i > 0$, and increase the first coefficient to $\beta = 1 - \sum_{i > 0} d_i$. Then $\beta \geq \alpha$, and every monomial of the entire K satisfies $\beta u_0 + \sum b_i u_i \geq q$.

Step 3. In this step, we count the full intersection image and retain the actual division debit. The full initial image of $(p^e) \cap J_q$ is contained in $(p_0^e) \cap J$. Its monomial initial space has the same dimension and lies in $(p_0^e) \cap K$. This uses inclusion only; it does not identify initial ideals of intersections with intersections of initial ideals. Every monomial counted has $u_0 \geq e$ and the displayed supporting inequality. Eliminate a coordinate of zero weight, which exists since $c \leq n - 1$. The resulting simplex region has only the simplex faces and two additional halfspaces.

Project each boundary hyperplane along a largest normal coefficient. Its other slopes have absolute value at most one, so at most $C_n(\ell + 1)^{n-2}$ unit cubes meet its boundary. This proves the uniform volume-to-lattice comparison and the claim. Finally $w(f/p^e) = w(f) - ew(p)$ proves the asserted actual debit. The auxiliary β has not replaced it. $\square$

Lemma 17.15. *Suppose $L^7 = 1$ and $K_X + 8L$ is not generated at x . For every original prime divisor P through x smooth there, one has $\phi_P(26/25) < 1/5$. There is one cutoff before all degrees and primes for which its coefficient in the entire fixed part of every $U_k(t)$, $0 \leq t \leq 26/25$, divided by k , is also less than $1/5$.*

Proof. We compare actual partial division when the old place has small incidence with P , and the retained-parameter capacity when its incidence is large. Put

$$R = 9/10, \quad U = 26/25, \quad h = 263/2000, \quad h_P = 1/5.$$

Lemma 17.9 gives $V(U) > 3/40$. Fix a compact big interval extending beyond U and all complete-profile comparisons first. Suppose $F(U) = \phi_P(U) \geq 1/5$. The full fixed-mass inequality of Lemma 17.4 and the positive integer degree $L^6 P$ give

$$(17.7) \quad V(U) \leq (4/5)^7.$$

Choose an initial actual threshold of degree $20i$ and a computing place w centered on its minimal center. Lemma 17.1 gives codimension between two and five and full coefficient less than $38/5$. Choose the source error below $1/40000$. Since $5/38 - h = 3/38000 > 1/40000$, its original normalized profile G satisfies $G(R) > h$. Take supporting slopes $a \geq h/R$ and $d \geq h_P/U$, and set

$$H_a(t) = (h + a(t - R))_+, \quad F_d(t) = (h_P + d(t - U))_+.$$

These lower bounds apply to every actual original section order. Put $\theta = w(P)/A_X(w) \in [0, 1]$.

Case 1. The actual incidence satisfies $\theta \leq 3/5$.

Step 1. In this step, we bound partial division with the actual incidence debit. Suppose $\theta \leq 3/5$. For any legal portion $0 \leq q(t) \leq F_d(t)$, divide the actual layer by $p^{\lceil kq(t) \rceil}$. The residual point degree is at most $k(t - q(t))$, and its normalized valuation is at least $k(H_a(t) - (3/5)q(t)) - 1$. Padding zero normal weights permits $\mathbb{M}_{7,5}$ in every initial dimension. Its homogeneous exact capacity $C(x, y) = x^6 \mathbb{M}_{7,5}(y/x)$ is nondecreasing in x and satisfies

$$C(x, (y - \eta)_+) \leq C(x, y) + 30x^5\eta.$$

To see the constant, choose a largest normal weight, at least $1/5$, and an unused coordinate. Conditional on the other five coordinates, their sum S has density $30S(1 - S)^4$, so $\mathbb{E}(1/S) = 6$. A threshold strip has probability at most $5\eta/S$. The degree monotonicity follows by eliminating the unused coordinate and enlarging its simplex. This controls rounding before inserting any piecewise envelope.

For $X = t - q$, $Y = H_a - (3/5)q$, use the following applicable polynomial majorants of $C(X, Y)$:

$$\begin{array}{l|l} Y \leq 0 & X^6 \\ Y \geq X \text{ or } X \leq 0 & 0 \\ 0 \leq 7Y \leq X & X^6 - (35/6)^5 XY^5 \\ X \leq 7Y, Y \leq X & X^6 - X((29X + 7Y)/36)^5 \\ X \leq 7Y, Y \leq X & (7/6)^6 (X - Y)^6 \\ X \leq 2Y \leq 2X & (X - Y)^6. \end{array}$$

The first line is used only for $X > 0$; a nonpositive residual degree has its at-most-one contribution in the error. Theorem 7.9 proves each bound. The actual layer also has the baseline $S(t) = 1, 61/100, 11/25, 17/50, 1$ on the consecutive intervals cut by $R, 99/100, U, 29/25$; the last value is a permissible weakening of the ordinary layer bound.

Step 2. In this step, we exclude all supporting slopes in the small-incidence case. If $a \geq 3$, put $D = aR - h$ and $t_a = D/(a - 1/7) < R$. The full profile loss through all orders is at most

$$t_a^7 \left(1 + \frac{6}{7(a-1)} \right) < R^7 \frac{10}{7} < \frac{5}{7}.$$

This integrates the full jet bound before t_a and the convex-order bound afterwards. If $d \geq 5$, division by the same prime bounds the entire post- U loss by $(U - h_P)^7/(d - 1) < 3/40$. Either case contradicts the complete telescope at a fixed endpoint above every attained original order. Such an endpoint was constructed after Lemma 17.6.

The remaining ranges are certified by the next two finite tables. Set $d_* = 241/100$. On an old-slope rectangle use its upper endpoint before R and lower endpoint afterwards; do the same for the prime slope at U . Denote the resulting lower profiles by H, F . The three legal portions are

$$q = 0, \quad F, \quad \min\{F, \max\{(7H - t)/(16/5), 0\}\}.$$

Split the integration interval at every anchor and baseline change, the zeros of H, F , and where the last portion equals zero or F . For each portion split again at the zeros of $X, Y, Y - X, 7Y - X, 2Y - X$. Bisect each resulting interval. On each half integrate every applicable polynomial and the baseline exactly, then choose the smallest integral. Each candidate bounds that entire half, so no intersection of competing degree-six polynomials has to be located. For a polynomial $\sum c_j t^j$, its integral on $[l, v]$ is $\sum c_j (v^{j+1} - l^{j+1})/(j + 1)$. The following table gives strict upper bounds $N/10^6$ for the entire incoming loss with prime slope d_* .

a_-	a_+	N
263/1800	623/1920	787771
623/1920	42517/115200	781712
42517/115200	23827/57600	784110
23827/57600	7241/14400	790166
7241/14400	3789/6400	785541
3789/6400	19619/28800	777696
19619/28800	2063/2400	776559
2063/2400	5663/3600	787788
5663/3600	3	754890

If $d \leq d_*$, this is an upper bound for the actual incoming loss. All entries are less than $1 - (4/5)^7 = 61741/78125$, contradicting (17.7). For $d_* \leq d \leq 5$, the same rule, integrated

through $(d_-U - h_P)/(d_- - 1)$, gives

a_-	a_+	d_-	d_+	N
263/1800	623/1920	241/100	823/320	998684
263/1800	623/1920	823/320	2187/800	978712
263/1800	7241/14400	2187/800	1223/400	999409
263/1800	7241/14400	1223/400	741/200	978277
263/1800	5663/3600	741/200	5	994678
623/1920	42517/115200	241/100	7971/3200	992893
623/1920	23827/57600	7971/3200	823/320	992851
623/1920	23827/57600	823/320	2187/800	986367
42517/115200	23827/57600	241/100	7971/3200	995259
23827/57600	17597/38400	241/100	7971/3200	994889
23827/57600	7241/14400	7971/3200	823/320	992398
23827/57600	7241/14400	823/320	2187/800	986921
17597/38400	7241/14400	241/100	7971/3200	993485
7241/14400	3789/6400	241/100	823/320	998556
7241/14400	19619/28800	823/320	2187/800	989973
7241/14400	2063/2400	2187/800	1223/400	992431
7241/14400	2063/2400	1223/400	741/200	978213
3789/6400	19619/28800	241/100	823/320	988370
19619/28800	2063/2400	241/100	2187/800	987879
2063/2400	5663/3600	241/100	741/200	992675
5663/3600	3	241/100	5	833099

The rows cover the entire remaining rectangle. Beyond their indicated endpoint the prime lower profile is at least the ordinary order, so all subsequent leading layers vanish. The maximum $999409/10^6 < 1$ contradicts the full original telescope. Thus $\theta > 3/5$.

Case 2. The actual incidence satisfies $\theta > 3/5$.

Step 3. In this step, we obtain the joint capacity while retaining the smooth parameter. Now $w(P) > 0$ gives containment of the initial center in P . Its codimension is between two and five. Apply Lemma 17.14, and append zero weights to the remaining normal coordinates until there are four of them. This gives, on the same actual image, the homogeneous capacity

$$J(t, f, H) = t^6 \sup \Pr\{tZ_0 \geq f, t[\beta Z_0 + (1 - \beta) \sum_{i=1}^4 \lambda_i Z_i] \geq H\},$$

where $\beta \in [3/5, 1]$, $\lambda_i \geq 0$ and $\sum \lambda_i = 1$. Here $f \leq F_d(t)$ and $H = H_a(t)$. This padding also covers initial dimensions four and five; no separate initial-dimension exclusion is needed.

Cover the β -interval by the sixteen closed intervals $[3/5 + j/40, 3/5 + (j+1)/40]$. On the j -th interval put $B = 3/5 + (j+1)/40$ and $C = 2/5 - j/40$. Then $\beta e + (1 - \beta)y \leq Be + Cy$ for nonnegative coordinates. The coefficient sum is $41/40$ and remains so. Conditional on the retained coordinate e , put $X = t - e$, $Y = (H - Be)/C$. The remaining slice has the upper capacity $X^5 E_{6,4}(Y/X)$. Explicit choices are

$$\begin{array}{l|l} Y \leq 0 & X^5 \\ Y \geq X \text{ or } X \leq 0 & 0 \\ 0 \leq 6Y \leq X & X^5 - (24/5)^4 XY^4 \\ X \leq 6Y, 2Y \leq X & X^5 - X((19X + 6Y)/25)^4 \\ X \leq 6Y, 2Y \leq X & (6/5)^5 (X - Y)^5 \\ X \leq 2Y \leq 2X & (X - Y)^5. \end{array}$$

The whole enlarged j -th region is bounded by

$$(t - \max\{f, H/B\})_+^6 + 6 \int_f^{\min(t, H/B)} X^5 E_{6,4}(Y/X) de,$$

where reversed intervals contribute zero. Its maximum over the sixteen intervals bounds J . This is an undivided intersection estimate; B is not an actual valuation debit.

Step 4. In this step, we verify the incoming bounds and exclude all slopes in the large-incidence case. For completeness, the accompanying certificate gives every cell of the following two incoming inequalities, uniformly for $263/1800 \leq a \leq 3$:

$$7 \int_0^U \min\{t^6 S(t), J(t, F_3(t), H_a(t))\} dt < 789/1000,$$

$$7 \int_0^U \min\{t^6 S(t), J(t, 0, H_a(t))\} dt < 169/200.$$

Its exact verification rule is as follows. For an old-slope interval use its upper endpoint before R and lower endpoint afterwards. Split at the anchors and zeros of these lower profiles and, for each B, C , at every root involving

$$e_z(t) = \frac{H(t) - Czt}{B - Cz}, \quad z = 1, 1/2, 1/6, 0,$$

and the floor f . Include roots of $H - Bt$, $f - t$ and $f - e_z$, then refine to pieces of length at most $1/40$. On the middle e -range between $e_{1/2}$ and $e_{1/6}$, bisect and choose on each half either available polynomial. The other ranges have the displayed fixed branches. Integrate in e ; each result $P_j(t)$ is a polynomial of degree at most six on that entire cell.

If $P_j(t) = \sum_m p_{jm} t^m$, its degree-six Bernstein coefficients on $[l, v]$ are

$$c_{ji} = \sum_{k=0}^i \frac{\binom{i}{k}}{\binom{6}{k}} (v-l)^k \sum_{m \geq k} p_{jm} \binom{m}{k} l^{m-k}.$$

The certificate's seven integers M_i assert $c_{ji} \leq M_i/10^6$ for every one of the sixteen choices j . Nonnegativity of the Bernstein basis gives

$$7 \int_l^v \max_j P_j(t) dt \leq \frac{v-l}{10^6} \sum_{i=0}^6 M_i \leq \frac{D}{10^8}.$$

An S -row instead records $s(v^7 - l^7) \leq D/10^8$. Every endpoint, middle choice, integer coefficient and cell integral is retained in the data object. For example, the cell $[5137/9850, 1401/2650]$ has

$$(M_0, \dots, M_6) = (20120, 20396, 20676, 20959, 21246, 21537, 21832).$$

Their sum is 146766, and its integral enclosure is $274158888/261025000000 \leq 105032/10^8$, as is checked by $2741588880000000 \leq 27415977800000000$. The fourteen full row sums give maxima $78813928/10^8 < 789/1000$ and $84492074/10^8 < 169/200$, respectively.

For $d \leq 3$, the first incoming inequality yields $V(U) > 211/1000 > (4/5)^7$, with gap $803/625000$, contradicting (17.7). For $d \geq 3$, use the second incoming inequality and the entire outgoing loss of this same prime, at most $(21/25)^7/2$. Their sum is less than one, with gap $364005211/48828125000$. Telescoping at a fixed endpoint above all attained orders is again impossible. All slopes and incidences have now been covered.

Step 5. In this step, we convert the strict profile ceiling into one uniform finite-degree ceiling. Finally the fixed finite support from Theorem 6.13 contains every prime in every sufficiently late source on the fixed interval. The finitely many smooth primes through x have positive gaps $1/5 - \phi_P(U)$. Choose the common error below their minimum. Monotonicity of the profiles then proves the simultaneous finite-degree assertion. Each actual comparison used full original images and its error summed to $O(k^6)$.

Complete retained-parameter certificate. *Certificate data.*

17-ambient-endpoints/retained-parameter. They use the domain, functions and margin specified above, with the verification rule of Lemma 17.11. $\square$

Lemma 17.16. *Suppose $L^7 = 1$ and $K_X + 8L$ is not generated at x . Beyond one cutoff fixed before all original degrees and places, every pure actual threshold of $U_k(26/25)$, with $25 \mid k$, has a nondivisorial minimal center.*

Proof. We first exclude a singular divisor by its full equation, and then use the smooth-prime ceiling. Write $U = 26/25$.

Step 1. In this step, we exclude a prime profile of multiplicity-weighted height at least $37/100$. A prime whose original profile has multiplicity-weighted height at least $h = 37/100$ gives a convex weighted profile with a supporting slope $a \geq h/U$. Dividing every actual ordinary layer by the required power of its whole equation gives the leading bound $\min\{t^6 B(t), (t - (h + a(t - U))_+)^6\}$. We show that its full loss through $13/5$ is less than $199/200$.

Put $p = U - h = 67/100$, $\kappa = 461/500$, $\gamma = p/U$. For $h/U \leq a \leq 11/10$, the incoming loss is at most $11(387/550)^7 - 10p^7 < 17/50$. The outgoing loss, using the final baseline after its crossing with $(\gamma t)^6$, is at most

$$\frac{125}{52}(\gamma t_*)^6 - Up^6, \quad t_* = \frac{125}{52 + 73\gamma} = \frac{13000}{10299}.$$

Here $29/25 < t_* < 125/52$, $(125/52)(\gamma t_*)^6 < 7/10$, and $Up^6 > 9/100$. The total is less than $19/20$. For $11/10 \leq a \leq 11/5$, set $z = a - 1$. The full profile-only integral is

$$U_1(z) = \frac{(p + Uz)^7}{z(z + 1)^6}.$$

Its derivative has the sign of $(6U - 7p)z - p$; hence its maximum on $[1/10, 6/5]$ occurs at an endpoint. Those two values are less than $19/20$ and $3/4$. For $a \geq 11/5$, $z \geq 6/5$, the baseline $61/100 < \kappa^6$ on $[9/10, U]$ gives total loss at most

$$(1 - \kappa^6)(9/10)^7 + U_\kappa(z), \quad U_\kappa(z) = \frac{\kappa^6(p + Uz)^7}{z(z + \kappa)^6}.$$

This follows by splitting at $(p + Uz)/(z + \kappa) \in (9/10, U)$ and integrating the two linear seventh powers. Its derivative has the sign of $(6U\kappa - 7p)z - p\kappa$, positive for $z \geq 6/5$. Thus it is bounded by its limit, and the total is less than

$$(9/10)^7 + \kappa^6(U^7 - (9/10)^7) < 199/200.$$

The last exact comparison, after clearing denominators, is $2423979354162758872184776736600 < 242919921875000000000000000000$. The earlier endpoint comparisons are seventh-power rational inequalities of the displayed constants. Every range leaves positive full section mass at $13/5$, and Corollary 5.12 gives generation.

Step 2. In this step, we exclude both singular and smooth actual divisorial thresholds. Now let a divisorial threshold at U have full coefficient s , deficit b , center multiplicity m , and full fixed coefficient $\nu = 1/\tau$. The scalar census gives $b > 7/5$, $m \leq 6$, and $Us + b \leq 7$. If $m \geq 2$, after choosing the normalized source error less than $1/10000$, its weighted height is greater than

$$\frac{2U}{7 - b} - \frac{6}{10000} > \frac{13}{35} - \frac{6}{10000} > \frac{37}{100}.$$

The preceding argument excludes it. If $m = 1$, the divisor is smooth. The complete smooth-divisor tail bound $(5/4)b + 187/250$, together with its own U -price, gives $s \leq 749/150$. Its prime profile is therefore greater than $1/5$ after a uniform error smaller than $150/749 - 1/5$, contradicting Lemma 17.15. All coefficients used here are those of the entire actual threshold boundary. The full equation division creates only a local image comparison. $\square$

17.2. An embedding-seven observation and its later threshold.

Theorem 17.17. *Suppose $L^7 = 1$ and $K_X + 8L$ is not generated at x . There is a cutoff J , fixed before the following degrees, thresholds, centers and places. Suppose a pure actual threshold of a nonzero $U_k(26/25)$, for $k \geq J$ and $25 \mid k$, has a fourfold minimal center of embedding dimension seven, or such a threshold of $U_k(27/25)$ has a minimal center of embedding dimension seven. Then*

$$V(t) > 1/64 \quad (0 \leq t \leq 29/25).$$

There is a further uniform cutoff K such that at every original integer $q \geq K$, simultaneously on this interval,

$$\dim U_q(t) \geq \frac{q^7}{128 \cdot 7!}, \quad \frac{F_q(t) \cdot L^6}{q} < 9/20,$$

where $F_q(t)$ is the entire divisorial fixed part. For every $q \geq K$ divisible by 25, let $\Gamma \sim_{\mathbb{Q}} sL$ be its pure actual $29/25$ threshold boundary and $b = \text{def}_x(\Gamma)$. Every divisorial minimal center P has $L^6 P = 1$. If P is smooth at x , then $s \leq 749/225$, $b \geq 17654/5625$, and its original prime profile satisfies $\alpha_{\text{ord}_P}(29/25) > 3/10$. If P is singular, then $2 \leq \text{mult}_x P \leq 4$, $s \leq 874/225$, $b \geq 14029/5625$, and $\alpha_{\text{ord}_P}(29/25) > 257/1000$. Every nondivisorial minimal center has dimension five, embedding dimension seven and multiplicity 3, 4, or 5; its entire tangent ring is reduced Cohen–Macaulay, all ordinary point-ideal powers are integrally closed and its original Rees ring is normal Cohen–Macaulay. Moreover $1/s \geq 3004/11665$; every computing place centered on it has normalized original profile at $29/25$ greater than $257/1000$. All observations retain their own entire coefficients and cutoffs.

Proof. We first obtain the volume reserve from complete original layers. We then pass to all original degrees and use the entire fixed part to constrain each later threshold at its own price.

Step 1. In this step, we obtain the actual observation heights and enumerate the five profile families. Write $R = 9/10$, retain u, r, T as above, and put $\varepsilon = 1/100000$. The initial source is big by the jet count; Lemma 17.9 gives $V(u) > 3/40$, and Theorem 9.14 gives $V(r) > 1/32$. Choose the uniform comparisons at these three big anchors before choosing any source or place. Increase the cutoff until their normalized errors are less than ε and less than each of the finitely many strict height gaps below. Bigness at T is a conclusion of this argument; no threshold there is selected yet.

Choose an initial source of degree $20i$, with i above the cutoff, and one of its actual computing places. Its full coefficient is less than $38/5$, by Lemma 17.1, so its normalized original profile has height greater than $131/1000$ at R . Its center may have any of the four initial dimensions.

At u , Lemma 17.16 excludes divisorial centers. The remaining alternatives are fivefolds, embedding-six fourfolds, and embedding-seven fourfolds. Their respective deficit bounds are $b > 7/5$, $b > 39/20$, and $b > 3/2$, by Proposition 5.9 and Lemma 17.8. Since the whole coefficient satisfies $us + b \leq 7$, their inverse heights exceed, respectively,

$$\frac{13}{70} > \frac{37}{200}, \quad \frac{104}{505} > \frac{41}{200}, \quad \frac{52}{275} > \frac{189}{1000}.$$

The capacities for these three alternatives are Q, M, D in Definition 17.12.

When the specified observation is at r , its original degree is sufficiently late and divisible by 25, as required in Proposition 14.2. An embedding-seven center is either a fourfold, whose inverse height is at least $15392/72425 > 17/80$, or the cubic fivefold in Proposition 14.2, whose inverse height is at least $4321/13425 > 1609/5000$. The latter retains its nonreduced tangent and uses the unrestricted height-two capacity Q . These alternatives follow from Theorem 10.1 and Lemma 17.8.

If the specified observation occurs at u , use that embedding-seven fourfold directly. If it occurs at r , choose an independent eligible degree divisible by 25, and then an actual threshold at u , with its tuple and place. An embedding-seven fourfold there gives the preceding case, and

otherwise its capacity is Q or M . Thus the following five lists exhaust the hypotheses. Every list starts with $(G, 131/1000, R)$.

family	K_1	m_1	r_1	K_2	m_2	r_2
$D26$	D	189/1000	u		absent	
$Q26_D27$	Q	37/200	u	D	17/80	r
$M26_D27$	M	41/200	u	D	17/80	r
$Q26_C27$	Q	37/200	u	Q	1609/5000	r
$M26_C27$	M	41/200	u	Q	1609/5000	r

The linked heights of Lemma 17.13 compare these independent full thresholds. No inclusion between their centers or equality between their degrees enters.

Step 2. In this step, we bound the complete loss over all bounded and unbounded slope ranges. Apply that lemma through T . For the supporting slopes a_i , it remains to bound

$$I = 7 \int_0^T t^6 \min\{B(t), K_i(H_i(t)/t) : 0 \leq i \leq v\} dt < \frac{63}{64}.$$

The reserve data objects cover the closed boxes

$$\frac{131}{900} \leq a_0 \leq 16, \quad \frac{m_1}{u} \leq a_1 \leq 8, \quad \frac{m_2}{r} \leq a_2 \leq 12$$

with the last coordinate omitted for $D26$. Their contents, read using (17.6), are

family	rectangles	cells	max N
$D26$	133	1310	9842943211
$Q26_D27$	787	8590	9843582362
$M26_D27$	237	2694	9843027297
$Q26_C27$	277	2489	9843540583
$M26_C27$	371	3821	9843229660.

Every numerator has denominator 10^{10} , and each maximum is less than $9843750000 = 10^{10} \cdot 63/64$. The supplement contains the complete dictionaries, trees and cell traces. The displayed counts identify the domains checked by Lemma 17.11.

The complement of the boxes is controlled by the reached volumes. For a positive affine profile of height h at v ,

$$7 \int_v^\infty (t - j(h + a(t - v)))_+^6 dt = \frac{(v - jh)^7}{ja - 1}$$

whenever $ja > 1$ and $v - jh > 0$, by linear substitution. If $a_0 \geq 16$, the full initial jet bound before R and the convex-order capacity afterwards give

$$I \leq R^7 + \left(\frac{7}{6}\right)^6 \frac{(R - 131/1000)^7}{15} = \frac{353441252142374119485321361}{6998400000000000000000000000000} < \frac{63}{64}.$$

If $a_1 \geq 8$, subtract the whole post- u loss from $V(u) > 3/40$. For capacities Q, M, D , respectively, the available tails are

$$\frac{(u - m_1)^7}{7}, \quad \frac{(u - m_1)^7}{7} + \frac{(u - 2m_1)^7}{15} - \frac{(u - 3m_1)^7}{23}, \quad \frac{6(u - 2m_1)^7}{15}.$$

The applicable endpoint branches hold throughout this outgoing interval: m_1/u is beyond the Q crossing or beyond $2/11$ in the M case. Subtraction from $3/40$ leaves, respectively,

$$\frac{2444639182386909}{8960000000000000000}, \quad \frac{165134863173366609}{51520000000000000000}, \quad \frac{1029536443789265789}{195312500000000000000}.$$

Each is greater than $1/64$. If $a_2 \geq 12$, subtract the D tail $6(r - 2m_2)^7/23$, or the Q tail $(r - m_2)^7/11$, from $V(r) > 1/32$. The two differences are

$$\frac{2613812134297567}{14720000000000000000}, \quad \frac{15602257000426184167212369}{85937500000000000000000000000000}.$$

again greater than $1/64$. Increasing a slope after its anchor decreases each of these capacity tails. Integrating past T only increases the loss bound. This covers every unbounded direction and proves $V(T) > 1/64$.

Step 3. In this step, we obtain the source and full fixed-divisor bounds in every original degree. There are finitely many strict bounds in the tables and in the unbounded comparisons. Their margins have a positive minimum. The uniform layer errors and the uniform bounded-variation argument in Lemma 17.13 therefore give, at every sufficiently late original integer q ,

$$\dim U_q(T) \geq \frac{q^7}{128 \cdot 7!}.$$

This is a sum of actual layers at that integer degree; it needs no replacement by a Veronese source. The ceiling at qT contributes $O(q^6)$. Monotonicity gives the same bound simultaneously for all real $0 \leq t \leq T$.

Proposition 6.19 applies to the whole fixed divisor and gives convergence of the generated moving volume in these same original degrees. Since

$$\left(\frac{11}{20}\right)^7 = \frac{19487171}{1280000000} < \frac{1}{64} < V(T),$$

its inequality $M_q^7/q^7 \leq (1 - F_q(T) \cdot L^6/q)^7$ gives $F_q(T) \cdot L^6/q < 9/20$ after a uniform cutoff. The componentwise inequality $F_q(t) \leq F_q(T)$ extends this to the entire interval, including remote fixed primes.

Step 4. In this step, we derive the center and profile bounds at each later threshold's own price. Only now fix uniform comparisons on a compact big interval containing $[0, T]$, and enlarge K before selecting any later threshold. Its full coefficient satisfies $Ts + b \leq 7$. The smooth divisorial state has complete tail at most $(5/4)b + 187/250$. Nongeneration forces

$$8 \leq s + \frac{5}{4}b + \frac{187}{250}, \quad s \leq \frac{749}{225}, \quad b \geq \frac{17654}{5625}.$$

For a singular divisor the states of multiplicities two through six have respective deficit caps 5, 4, 3, 2, 1, and respective tail lines

$$\frac{5}{4}b + \frac{499}{500}, \quad \frac{5}{4}b + \frac{993}{1000}, \quad \frac{63}{50}b + \frac{121}{125}, \quad \frac{327}{250}b + \frac{801}{1000}, \quad \frac{5}{3}b + \frac{42}{125}.$$

These are evaluations of the entire recurrence of Definition 5.2, with both multiplicity-eight and multiplicity-nine threefold cases retained. On the stated caps all five lines are bounded by $(5/4)b + 499/500$. Combining this with its own T -price gives

$$s \leq \frac{874}{225}, \quad b \geq \frac{14029}{5625}.$$

The latter bound excludes multiplicities five and six. For either kind of divisor, $g = L^6P$ is a positive integer and the same full fixed coefficient gives

$$\frac{g}{s} \leq \frac{F_q(T) \cdot L^6}{q} < \frac{9}{20}.$$

Since $2(225/874) > 9/20$, it follows that $g = 1$. The uniform comparison and the strict inequalities $225/749 > 3/10$, $225/874 > 257/1000$ give the two asserted prime heights.

For a nondivisorial center, use Lemma 17.7 and Proposition 7.26. The double case is paid by Theorem 16.2, using the same boundary and $Ts + b \leq 7$. The remaining states have all the asserted full-ring properties and $1/s \geq 3004/11665$. The fixed positive gap above $257/1000$ supplies the uniform height for every computing place centered on that actual center. All cuts used in these necessary failure inequalities are charged through Theorem 5.4; both errors fit inside each strict payment margin.

The five complete reserve certificates. *Certificate data.*

17-ambient-endpoints/reserve-d26. They use the domain, functions and margin specified above, with the verification rule of Lemma 17.11.

Certificate data.

17-ambient-endpoints/reserve-q26-d27. They use the domain, functions and margin specified above, with the verification rule of Lemma 17.11.

Certificate data.

17-ambient-endpoints/reserve-m26-d27. They use the domain, functions and margin specified above, with the verification rule of Lemma 17.11.

Certificate data.

17-ambient-endpoints/reserve-q26-c27. They use the domain, functions and margin specified above, with the verification rule of Lemma 17.11.

Certificate data.

17-ambient-endpoints/reserve-m26-c27. They use the domain, functions and margin specified above, with the verification rule of Lemma 17.11.

□

Theorem 17.18. *Suppose $L^7 = 1$ and $V(29/25) > 1/64$. There are cutoffs J, K , fixed before the observations below. Suppose at some $k \geq J$ divisible by 25 a pure actual threshold of $U_k(26/25)$ has a fourfold minimal center of embedding dimension seven, or a pure actual threshold of $U_k(27/25)$ has a minimal center of embedding dimension seven. If at some $q \geq K$ divisible by 25 a pure actual threshold of $U_q(29/25)$ has a nondivisorial minimal center or a divisor singular at x , then $K_X + 8L$ is generated at x . Each observation uses its nonzero complete source and own entire threshold boundary. No incidence, equality of degrees or initial-fivefold history is required.*

Proof. Suppose that generation fails. We use the specified earlier observation and the specified later threshold to bound every original layer through $13/5$, and then apply Corollary 5.12 to the resulting actual section.

Step 1. In this step, we obtain three profile constraints on the original section ring. Set $R = 9/10$, $z = 13/5$, and $\varepsilon = 1/100000$. The assumed volume at T makes all observation anchors interior to the big range. Fix their complete-source comparisons before every degree, center and place. Choose an initial threshold in degree $20i$ above its cutoff. Lemma 17.1 gives a profile of height greater than $131/1000$ at R . The specified earlier observing degree is sufficiently late and divisible by 25. At that same observation the applicable height and full-tangent bounds give exactly one of

family	r_1	m_1	K_1
$D26$	$26/25$	$189/1000$	D
$D27$	$27/25$	$17/80$	D
$C27$	$27/25$	$1609/5000$	Q .

The $D26$ and $D27$ rows follow from Lemma 17.8. The $C27$ row follows from Proposition 14.2 at its $27/25$ anchor and the same sufficiently late original degree divisible by 25. It uses Q on its actual nonreduced tangent.

For the later threshold, Theorem 17.17 gives a normalized computing height greater than $257/1000$ at its own anchor T . It also shows that every remaining nondivisorial center is a fivefold with a full reduced Cohen–Macaulay tangent ring. A double fivefold would instead be paid by Theorem 16.2, with its own price $Ts + b \leq 7$. For each remaining fivefold, the capacity K_5 in Theorem 7.18 applies to every full primary comparison ideal. For a singular divisor of multiplicity $m \geq 2$, division by its required full local equation power injects the ordinary initial image into degree $\ell - mj$. Its capacity is $E_m \leq E_2 \leq K_5$, so the same larger bound K_5 applies to that whole case. The ambient associated graded ring is a polynomial ring; this division retains the entire initial equation, including its scheme multiplicity.

Take supporting slopes at the three anchors and the linked heights of Lemma 17.13, with base heights $131/1000, m_1, 257/1000$. It gives a positive complete source at z once

$$I = 7 \int_0^z t^6 \min\{B(t), N_5(H_0(t)/t), K_1(H_1(t)/t), K_5(H_2(t)/t)\} dt < 1.$$

The baseline in this display is the full actual layer bound, including its final piece, from Proposition 7.28. A current zero source has a zero image and needs no additional threshold.

Step 2. In this step, we prove the complete loss bound over every supporting-slope range. On the closed boxes

$$\frac{131}{900} \leq a_0 \leq 2, \quad \frac{m_1}{r_1} \leq a_1 \leq 8, \quad \frac{257}{1160} \leq a_2 \leq \frac{9}{4},$$

the three point-source data objects give

family	rectangles	cells	max N
$D26$	76	826	9948845674
$D27$	279	3538	9989986795
$C27$	60	606	8870441956.

Every maximum is less than 9990000000, so the complete trace inequalities imply $I < 999/1000$. The linked rectangle heights, the selection of the upper slope before an anchor, and the selection of the lower slope afterwards are precisely those of Lemma 17.13. The rational dictionaries specify every cell, including the $4/25$ change in K_5 and the end of the final baseline.

If $a_0 \geq 2$, the initial profile alone bounds the full loss by

$$R^7 + \left(\frac{7}{6}\right)^6 (R - 131/1000)^7 < 1.$$

For an explicit comparison, its two factors satisfy $R^7 < 479/1000$, $(R - 131/1000)^7 < 16/100$, and $(7/6)^6 < 253/100$, giving an upper bound $8838/10000$. If $a_1 \geq 8$, begin the telescope at its reached anchor. For a D row the entire subsequent loss is bounded by $6(r_1 - 2m_1)^7/15$. This is less than $3/40$ for $D26$, and less than $1/32$ for $D27$. For $C27$, the whole tail is at most $(27/25 - 1609/5000)^7/7 < 1/32$. The reached volume bounds are Lemma 17.9 and Theorem 9.14. Thus these three unbounded cases also leave a positive source at z .

Finally, if $a_2 \geq 9/4$, its normalized outgoing height is at least $257/1160 > 4/25$, so the E_2 branch is valid everywhere after T . Its full tail is bounded by

$$\frac{(T - 2(257/1000))^7}{2a_2 - 1} \leq \frac{366790143213462347}{27343750000000000000} < \frac{1}{64} < V(T).$$

The positive-part linear integral used in the preceding proof establishes each of these tail formulas. All unbounded directions are therefore covered using the volume already earned at the corresponding anchor.

Step 3. In this step, we obtain an actual high-order section and deduce generation. The layer telescope includes all original sections and has total error $O(p^6)$. Its strict positive lower bound therefore supplies a nonzero member of $U_p(13/5)$ at a sufficiently large original degree p divisible by 25. Its entire divisor satisfies $\text{mult}_x D \geq (13/5)p$. Corollary 5.12 gives generation, a contradiction. This endpoint uses a new complete source. The earlier and later threshold boundaries have supplied constraints on that source with their own full coefficients; neither coefficient has been substituted for the other's additive price.

The three complete point-source certificates. *Certificate data.*

17-ambient-endpoints/point-d26. They use the domain, functions and margin specified above, with the verification rule of Lemma 17.11.

Certificate data.

17-ambient-endpoints/point-d27. They use the domain, functions and margin specified above, with the verification rule of Lemma 17.11.

Certificate data.

17-ambient-endpoints/point-c27. They use the domain, functions and margin specified above, with the verification rule of Lemma 17.11. $\square$

17.3. The initial-fivefold source history.

Lemma 17.19. *Suppose that $L^7 = 1$ and generation fails at x . There are cutoffs, fixed before the source degrees and computing places, with the following property. If a pure actual threshold of a sufficiently late original source $U_{20i}(9/10)$ has a fivefold minimal center, then*

$$V(t) > \frac{6}{25} \quad (0 \leq t \leq 27/25).$$

For every sufficiently large original integer q , all these sources are nonzero and their entire fixed divisors $F_q(t)$ satisfy

$$\frac{F_q(t) \cdot L^6}{q} < \frac{13}{70} \quad (0 \leq t \leq 27/25).$$

Every sufficiently late actual threshold at a rational order $v \in [26/25, 27/25]$, in an original degree q with qv integral, has a fourfold or fivefold minimal center.

Proof. We first bound the loss imposed by the initial fivefold and then compare the resulting volume with the entire fixed divisor. Let $R = 9/10$, $r = 27/25$, and $h = 131/1000$. Lemma 17.1 and the uniform original-profile comparison give a computing place w_0 with a supporting lower profile

$$H_a(t) = (h + a(t - R))_+, \quad a \geq h/R.$$

Indeed $1/s_0 > 5/38$ and $5/38 - h = 11/19000$; choose the comparison error smaller than this fixed gap before choosing i and w_0 . The full height-two capacity of Theorem 7.9 is the function Q in Definition 17.12. Thus the complete original layer filtration gives

$$V(r) \geq 1 - I(a), \quad I(a) = 7 \int_0^r t^6 Q(H_a(t)/t) dt.$$

This uses only the full primary comparison ideal of the valuation; it imposes no primaryness on the original source.

Step 1. In this step, we bound the full initial-slope loss by $19/25$. Put $c_0 = 71/400$ and $\varepsilon = 1/5000$. The balanced branch of Q is exact for $c \leq c_0$, whereas

$$Q(c) \leq (1 - c)_+^6 + \varepsilon \quad (c \geq c_0).$$

For completeness, the difference between the balanced and endpoint branches at c_0 is

$$\frac{1110 \cdot 258^5 - 329^6}{400^6},$$

equivalently $(1 - 2c_0)^5(1 + 10c_0) - (1 - c_0)^6$; it lies between zero and $1/5000$ by substitution. On $[c_0, 4/21]$ the derivative of this difference is at most

$$-120c_0(13/21)^4 + 6(329/400)^5 < 0.$$

Above $4/21$ the endpoint branch is the larger branch by Theorem 7.9. These facts prove the stated upper bound, including the whole transition interval.

Here is the exact integration rule and its full certificate. Set $d = aR - h$, $z = R - h/a$, and define

$$B(a; l, v) = 7 \sum_{j=0}^5 \binom{5}{j} (1 - 2a)^j (2d)^{5-j} \left[\frac{(1 + 10a)(v^{j+2} - l^{j+2})}{j + 2} - \frac{10d(v^{j+1} - l^{j+1})}{j + 1} \right].$$

The incoming integral is exactly $A_*(a) = z^7 + B(a; z, R)$. If $a \leq c_0$ or $a - d/r \leq c_0$, set $B_*(a) = B(a; R, r)$. Otherwise put

$$v_0 = \frac{d}{a - c_0}, \quad v_1 = \begin{cases} r, & a \leq 1, \\ \min\{r, d/(a - 1)\}, & a > 1, \end{cases}$$

and set

$$B_*(a) = B(a; R, v_0) + D(a; v_0, v_1) + \varepsilon(r^7 - v_0^7),$$

where

$$D(a; l, v) = \begin{cases} [((1 - a)v + d)^7 - ((1 - a)l + d)^7]/(1 - a), & a \neq 1, \\ 7d^6(v - l), & a = 1. \end{cases}$$

The last term is retained even beyond the zero of the endpoint branch. The incoming integral is nondecreasing in a , and the outgoing integral is nonincreasing in a , because Q is nonincreasing. Consequently, on a slope interval $[l, v]$, $I(a) \leq A_*(v) + B_*(l)$. The following exact values are all strictly below $19/25$:

l	v	$A_*(v) + B_*(l)$
131/900	3/20	111968702950656413570657/1483154296875000000000000
3/20	4/25	7741590547219641629013/1024000000000000000000
4/25	9/50	218467457144087468347845433/2883251953125000000000000000
9/50	1/5	117817192963582900476741/156250000000000000000000
1/5	9/40	277654452441861069547/369056250000000000000000
9/40	1/4	186986397352977537897/250000000000000000000000
1/4	3/10	57102938759193373361/759375000000000000000000
3/10	2/5	48375359353392930813/640000000000000000000000
2/5	1/2	811900674634129710695788187564807/1105783372388225000000000000000000
1/2	7/10	20387087123277066859901781637412534273/2775336653601774517500000000000000000000
7/10	1	3098393891184757057783005311103072599/4354757857490092250000000000000000000000
1	3/2	8558053590267328092812691003647042504417/1267323745654960992337500000000000000000000
3/2	2	16714098849950348209644113443779/2723860357600000000000000000000000000000

Each entry follows from the displayed polynomial antiderivatives; the comparison is integer cross multiplication. For $a \geq 2$, use the incoming bound R^7 and the outgoing bound at slope two:

$$I(a) \leq R^7 + B_*(2) = \frac{560278809698942102614125624632938373597}{9118249094292696600750000000000000000000} < \frac{19}{25}.$$

Thus all finite and unbounded slopes are covered, and $V(r) > 6/25$. Monotonicity gives the full preceding interval.

Step 2. In this step, we use the entire fixed degree to exclude divisorial thresholds. On a fixed big interval extending beyond r , let $p(t) = \sum_P (L^6 P) \phi_P(t)$. Lemma 17.4 and the preceding bound give

$$p(t) < 1 - (6/25)^{1/7} < \frac{13}{70} - \frac{1}{10000}.$$

The last inequality follows from $57007/70000 < 163/200$ and $25 \cdot 163^7 < 6 \cdot 200^7$. The finite prime support and the coefficientwise uniform error in Theorem 6.13 permit one degree cutoff for which the entire finite fixed degree differs from $p(t)$ by less than $1/10000$, simultaneously for every real $t \leq r$. Complete volume convergence also supplies every original integer degree beyond a fixed cutoff.

At an actual threshold of order $v \in [26/25, 27/25]$, Proposition 5.9 leaves dimensions four, five and six. A divisorial center would have $b > 7/5$. Writing $s = q\tau$, its actual fixed multiplicity ν satisfies $\tau\nu = 1$, while $vs + b \leq 7$. Hence

$$\nu/q = 1/s > 5v/28 \geq 13/70.$$

Its positive original degree makes this a lower bound for the entire fixed degree of that same source, contradicting the finite inequality already proved. No component has been removed from either comparison. $\square$

Lemma 17.20. *Under the hypotheses of Lemma 17.19,*

$$V(t) > \frac{1}{500} \quad (0 \leq t \leq 29/25).$$

For every sufficiently large original integer q , uniformly on this interval,

$$\frac{7! \dim U_q(t)}{q^7} > \frac{1}{1000}.$$

The cutoff precedes every later threshold. At any sufficiently late own threshold at $29/25$, in a degree divisible by 25, a divisorial minimal center has multiplicity at most four and original degree at most two. A smooth such center has degree one, full coefficient at most $749/225$, and original prime profile greater than $3/10$ at $29/25$. A singular such center has profile greater than $1/4$. Every nondivisorial center has one of the four full tangent types of Proposition 7.26.

Proof. We couple the initial fivefold profile with a separately chosen actual original source at $r = 27/25$, prove the stronger reserve by a finite rational certificate, and then price its own later threshold. Keep $R = 9/10$, $h = 131/1000$, $T = 29/25$.

Step 1. In this step, we obtain the two actual profile constraints and their complete capacities. The preceding lemma supplies all original sources at r before the next degree is chosen. Choose a sufficiently late degree divisible by 25 and its own pure actual threshold. Theorem 10.1 and the preceding lemma leave a fourfold or fivefold. Its computing place has a supporting lower profile $H + b(t - r)$, with the following full capacities:

type	H	$K(c)$
fourfold of embedding dimension six	$59/250$	$M(c)$
fourfold of embedding dimension seven	$17/80$	$D(c)$
fivefold	$26/125$	$Q(c)$

Here M, D, Q are defined in Definition 17.12. The first two profiles follow from Lemma 17.8. In the last row, the full-tail bounds in Definition 5.2, applied to the fivefold rows of the ledger, give inverse coefficients at least

$$175/743, \quad 8642/36025, \quad 13637/65475, \quad 1064/5025, \quad 2452/11665$$

in the exhaustive scalar census. Each exceeds $26/125$ by more than $1/50000$. Choose the common comparison error below $1/100000$ before selecting the degree or place. The whole coefficient of each of the two observations is retained. The full tangent hypotheses for M, D follow from the same fourfold lemma; Q needs no reducedness hypothesis.

All three actual capacities are nonincreasing. At the switch $2/11$, the difference between the two branches of M is

$$2[(1 - 2c)_+^6 - 2(1 - 3c)_+^6 + (1 - 4c)_+^6] \geq 0,$$

by convexity. Its upper branch has derivative $-6(1 - c)_+^5 - 12(1 - 2c)_+^5 + 18(1 - 3c)_+^5 \leq 0$; the lower branch and D are full spanning probabilities. Consequently every original layer has leading density at most

$$t^6 \min \left\{ Q \left(\frac{(h + a(t - R))_+}{t} \right), K \left(\frac{(H + b(t - r))_+}{t} \right) \right\}, \quad a \geq h/R, \quad b \geq H/r.$$

We show that seven times its integral through T is less than $499/500$.

Step 2. In this step, we bound every slope rectangle and both unbounded ranges by exact integration. For an anchor (v, H_0) , slope a , and $d = av - H_0$, the endpoint primitives are

$$D_j(a, d; l, z) = \begin{cases} [((1 - ja)z + jd)_+^7 - ((1 - ja)l + jd)_+^7] / (1 - ja), & ja \neq 1, \\ 7(jd)_+^6(z - l), & ja = 1. \end{cases}$$

Choose all comparison constants on $[0, (T + \beta)/2]$, then the cutoff with error below $1/100000$, before the later degree or center. For an actual full coefficient $s = q\tau$, $Ts + b \leq 7$. A complete-tail pair (a, p) gives

$$b \geq \frac{(8-p)T-7}{aT-1}, \quad \frac{1}{s} \geq \frac{T}{7-b}.$$

For a divisorial center the six multiplicity rows give

m	a	p	b lower bound	$1/s$ lower bound
1	5/4	187/250	17654/5625	225/749
2	5/4	499/500	14029/5625	225/874
3	1201/1000	223/200	24665/9829	9829/38050
4	63/50	121/125	7232/2885	577/2235
5	327/250	801/1000	33771/12932	excluded
6	5/3	42/125	17721/8750	excluded.

The last two deficit bounds exceed their caps 2, 1. The other inverse coefficients exceed $1/4$ by more than the selected error, and the smooth row exceeds $3/10$. If its original degree is g , the entire fixed-mass bound gives $V(T) \leq (1 - g\phi_P(T))_+^7$. For $g \geq 3$ this is less than $(1/4)^7 < 1/500$. For a smooth center with $g \geq 2$ it is less than $(2/5)^7 < 1/500$, again a contradiction. The nondivisorial list, with its full ordinary tangent and Rees conclusions, is precisely Proposition 7.26. All scalar inequalities retain the whole threshold coefficient; their complete-tail interpretation already includes both cutting errors from Theorem 3.14. $\square$

Lemma 17.21. *Suppose that $L^7 = 1$. After cutoffs fixed before all source degrees and places, an actual initial fivefold at order $9/10$ and an own actual threshold at order $29/25$ whose center is singular divisorial or a fivefold of embedding dimension seven imply generation at x . The initial degree is $20i$; the later observing degree is divisible by 25. Each source retains its complete boundary and every fixed component.*

Proof. Assume nongeneration. The argument uses the initial fivefold, a separately chosen actual original source at $27/25$, and the own later threshold. The third observation imposes an additional constraint on the same complete original point layers. Put

$$R = 9/10, \quad r = 27/25, \quad T = 29/25, \quad Q_* = 13/5, \quad h = 131/1000, \quad H = 257/500.$$

The two preceding source lemmas give $V(r) > 6/25$, $V(T) > 1/500$, and profiles

$$P_0(t) = (h + a_0(t - R))_+, \quad P_1(t) = (H_i + a_1(t - r))_+,$$

with $a_0 \geq h/R$, $a_1 \geq H_i/r$. The second observation is made in a sufficiently late degree divisible by 25. The three alternatives (H_i, K_i) are

$$(59/250, M), \quad (17/80, D), \quad (26/125, Q),$$

as in Lemma 17.20. For every a_0 , the first profile has total loss through r less than $19/25$. These profiles use the full coefficients of the initial and auxiliary observations.

Step 1. In this step, we obtain a third profile from the entire later fixed part or the actual fivefold computing place. On the fixed big interval extending beyond T , let

$$\Phi(t) = \sum_{P \ni x} \text{mult}_x P \phi_P(t),$$

where the finite sum includes every prime in the common full fixed support. Its value at T is the limit of the entire normalized fixed-point multiplicity. If $\Phi(T) > H$, a supporting slope $A \geq H/T$ gives $\Phi(t) \geq P_2(t) := (H + A(t - T))_+$ at every attained original order. An actual section of degree q and order ℓ is divisible by the product of all the local prime equations to powers $\lceil q\phi_P(\ell/q) \rceil$. Its nonzero initial form has degree at least $qP_2(\ell/q)$. Multiplication by this form is injective in the seven-variable ordinary point-graded ring. Thus the whole original image

has leading density at most $(t - P_2(t))_+^6$. The finite sum of rounding errors contributes only $O((\ell + 1)^5)$.

A singular divisorial own threshold has multiplicity $m = 2, 3, 4$ by the preceding reserve lemma. The inverse coefficients in its three rows give unperturbed fixed-point heights

$$225/437, \quad 29487/38050, \quad 2308/2235.$$

Each exceeds H by more than $4/100000$. Fixing the common profile error below $1/100000$ before the threshold gives $\Phi(T) > H$, including all its other fixed components.

For a fivefold of embedding dimension seven, the own full tangent conclusions are Proposition 7.26. The complete-tail calculation in the preceding lemma gives

$$1/s \geq 87116/338285 > 257/1000 + 1/100000.$$

Its actual computing place consequently has a whole profile greater than $257/1000$ at T . Twice a supporting lower profile has the same form P_2 above. Theorem 7.18, applied to the full comparison ideal on the full reduced Cohen–Macaulay tangent, gives the common scaled capacity

$$K_2(c) = \begin{cases} \min\{1, 2(1 - c)_+^6\}, & 0 \leq c < 8/25, \\ (1 - c)_+^6, & c \geq 8/25. \end{cases}$$

This also bounds the fixed-divisor image just considered. No primaryness is asserted of a full or residual original source.

Step 2. In this step, we combine the three constraints and bound the unbounded slope ranges. Let

$$J = 7 \int_0^T t^6 \min\{Q(P_0(t)/t), K_i(P_1(t)/t), K_2(P_2(t)/t)\} dt.$$

Since $H/T = 257/580 > 8/25$, the outgoing third ratio remains in the factor-one range. With the full original baseline, put

$$R_2(A) = 7 \int_T^{Q_*} \min\{(t - P_2(t))_+^6, ((125 - 52t)_+/73)^6\} dt.$$

The minima bound the identical actual point quotient before it is summed. Complete volume convergence and the independent $1/500$ reserve give

$$V(Q_*) \geq 1 - \min\{J, 499/500\} - R_2(A).$$

The uniform per-layer error is $O((\ell + 1)^5)$; summing on this fixed finite order interval gives $O(q^6)$. Zero images and the degree-zero layer are included.

Here is a complete rational bound for the right-hand loss. First $Q(c) \leq (4/3)(1 - c)_+^6$: the balanced-to-endpoint ratio has its maximum $2(10/11)^5 < 4/3$ at $c = 1/12$. For $a_0 \geq 2$, the full loss through Q_* is at most

$$R^7 + (4/3)(R - h)^7 < 691/1000.$$

For $a_1 \geq 4$, use the old loss $19/25$ through r and integrate the new profile to infinity. In the order M, D, Q the bounds are

$$19/25 + (r - H_i)^7/3 + (r - 2H_i)^7/7 < 867/1000,$$

$$19/25 + 6(r - 2H_i)^7/7 < 805/1000,$$

$$19/25 + (4/3)(r - H_i)^7/3 < 931/1000.$$

The first row stays in the sharper M range since $H_i/r > 2/11$. Finally, for $A \geq 30$, the reserve and the third outgoing profile give

$$\min\{J, 499/500\} + R_2(A) \leq 499/500 + (323/500)^7/29 < 9997/10000.$$

All three unbounded ranges are therefore covered.

Step 3. In this step, we verify the bounded three-slope certificate and obtain the positive original point source. It remains to cover

$$[h/R, 2] \times [H_i/r, 4] \times [H/T, 30].$$

For a box $[l_0, v_0] \times [l_1, v_1] \times [l_2, v_2]$, partition $[0, T]$ at $0, R, r, T$, at $j/50$ for $1 \leq j \leq 57$, and at

$$\frac{av - H_0}{a - c}, \quad c \in \{0, 71/400, 2/11, 1/2, 1, 1/10, 8/25\},$$

when the denominator is nonzero and the point is interior. Here the five triples (a, v, H_0) are

$$(l_0, R, h), (v_0, R, h), (l_1, r, H_i), (v_1, r, H_i), (v_2, T, H).$$

Use the upper endpoint of an old slope before its anchor and the lower endpoint after it; use v_2 for the whole incoming third profile. These are lower profiles on each interval. Use the primitives D_j and the balanced primitive in Lemma 17.20. For K_2 , use the rational upper expression 1 through $1/10$, $2(1-c)_+^6$ on $(1/10, 8/25)$, and $(1-c)_+^6$ afterwards. Its three integrals are the seventh-power interval difference, $2D_1$, and D_1 . Monotonicity is applied to K_2 before this enlargement; no monotonicity of its enlarged expression is needed. Take the minimum of the four upper integrals, including the ordinary interval difference, on each interval and sum to obtain $C_i \geq J$.

The outgoing upper bound uses l_2 . Split at the crossing of $(1-l_2)t + l_2T - H$ and $(125-52t)/73$, if interior, and integrate the smaller positive affine root to the sixth power. This gives $R_2(l_2)$ exactly, including either root's zero and a zero slope. Thus each leaf tests the rational number

$$S_i = \min\{C_i, 499/500\} + R_2(l_2).$$

In the three nonsmooth data trees a leaf N asserts $S_i \leq N/10^6$. At each pair bisect the coordinate with largest normalized width $(v_j - l_j)/(1 + (v_j + l_j)/2)$, taking the first at a tie and the lower half on the left. This specifies every rational box. The generic, embedding-six and embedding-seven trees have 380, 195, 172 leaves and respective maxima

$$999894/10^6, \quad 999856/10^6, \quad 999856/10^6,$$

all strictly below 9999/10000. The antiderivatives and the full trees specify every integer comparison and cover the entire compact domain.

Certificate data.

17-ambient-endpoints/initial-fivefold-nonsmooth. They use the domain, functions and margin specified above, with the verification rule of Lemma 17.11.

Consequently $V(13/5) > 0$. Theorem 6.4 supplies an actual original section at this order, and Corollary 5.12 gives generation. Its complete continuation retains both published cutting errors and the full coefficient of this new source. This contradicts nongeneration and pays both stated own later outcomes. $\square$

Lemma 17.22. *Let $(A, \mathfrak{m})$ be the local ring of a smooth complex sevenfold at a closed point, and let P be a prime such that the full ring $\text{gr}_{\mathfrak{m}}(A/P)$ is reduced and Cohen–Macaulay, of embedding dimension six. Suppose its dimension is five, four or two. Let $p \in P$ be a regular parameter, let w be a divisorial valuation centered on P , and put $\alpha = w(p)/A(w)$. Then $0 < \alpha < 1$. For physical degree $D \geq 0$, real height H , and retained parameter floor $F \geq 0$, define*

$$J_s(D, F, H; \alpha) = (D - \max\{F, H\})_+^6 + 6 \int_F^{\min(D, H)} V_s\left(D - z, \frac{H - \alpha z}{1 - \alpha}\right) dz,$$

where an integral with reversed endpoints is zero. For $H \leq 0$ use $J_s(D, F, H; \alpha) = (D - F)_+^6$, and for $D \leq 0$ use zero. For positive height the remaining polynomials are

$$\begin{aligned} V_2(B, h) &= (B - 2h)_+^5, \\ V_3(B, h) &= 3(B - 2h)_+^5 - 2(B - 3h)_+^5, \\ V_5(B, h) &= 10(B - 2h)_+^5 - 20(B - 3h)_+^5 + 15(B - 4h)_+^5 - 4(B - 5h)_+^5. \end{aligned}$$

Use $s = 2, 3, 5$ for dimension five, four, two, respectively. For $s = 3$ one may also replace V_3 by $2(B - 2h)_+^5$, or take the minimum of these bounds. Every full actual image contained in $(p^e) \cap \{f : w(f) \geq qA(w)\}$ has degree- ℓ dimension at most

$$\frac{\ell^6}{6!} J_s(1, e/\ell, q/\ell; \alpha) + C(\ell + 1)^5.$$

The constant is uniform in all ideals, weights and thresholds. Before taking the displayed explicit upper bounds, the underlying joint capacity is nonincreasing in F, H, α , and its homogeneous form is nondecreasing in D . No equality between the actual incidence and any auxiliary weight is asserted.

Proof. We retain the parameter through the full degeneration, obtain simultaneous normal inequalities, and slice their common region.

Step 1. In this step, we retain the parameter through the full spanning degeneration. The mixed restriction argument in Lemma 17.14 applies to the full valuation ideal and to every generic component of each specialization. In particular its necessary relative thresholds are $(1 - b\alpha)/q$ for all $0 \leq b < 1$.

The linear initial form p_0 is in the full tangent ideal. Its quotient is reduced and Cohen–Macaulay of codimension one, two or four in the remaining six variables, with no further linear equations. Successive general regular linear sections reduce its projective scheme to a reduced spanning set in $\mathbb{P}^{s-1}$. Each quotient is Cohen–Macaulay and generically reduced, hence reduced; regularity of the sections preserves the degree-one dimension. The last graded ring has depth one and is saturated, so spanning and reducedness include the full scheme at this stage.

Implement each section as a homogeneous weight degeneration which frees the section variable. If that variable is z , saturation by z permits removal of a common z -power; the highest remaining weight is specialization at $z = 0$. Use the identical weight on the entire valuation ideal and keep p_0 as a coordinate. Both full Hilbert functions and the power sandwich of their ideals are preserved. Take s independent points to the coordinate vertices and use a diagonal family with distinct increasing weights to send the other points to those vertices. The fixed vertex branches persist; collisions and nilpotents of the full support are retained. A final common monomial term order preserves p_0 , all these branches, and the full Hilbert function of the comparison ideal.

Step 2. In this step, we obtain simultaneous retained normal inequalities at every generic branch. At every generic vertex branch the total degeneration space and its special fiber are regular, and p_0 remains a smooth parameter. A horizontal support component specializes to the branch, since flatness excludes a vertical minimal component. Thus the mixed restriction argument from the cited lemma continues to apply throughout these successive families. Separating its excluded Newton segment gives, at each vertex, one normal inequality

$$\beta_i z_0 + \sum_{j \neq i} b_{ij} z_j \geq H, \quad \beta_i \geq \alpha, \quad \beta_i + \sum_{j \neq i} b_{ij} = 1.$$

Every monomial of the same full ideal obeys all s inequalities and $z_0 \geq F$. The separate generic localizations supply necessary conditions only; the full ideal has not been replaced by their union or sum.

Step 3. In this step, we integrate the common slice bounds and retain the uniform lattice error. Fix $z_0 = z < H$. The remaining normalized inequality has threshold at least

$$\frac{H - \beta_i z}{1 - \beta_i} \geq \frac{H - \alpha z}{1 - \alpha},$$

because its derivative in β_i is $(H - z)/(1 - \beta_i)^2 > 0$. At least two of the s distinguished remaining coordinates must exceed this threshold: otherwise the inequality which omits the only such coordinate cannot hold. Inclusion–exclusion for this event gives V_2, V_3, V_5 . For three coordinates there is also the factor-two bound. Conditional on their sum B , when $2h \leq B < 3h$ the convex

feasible polygon lies in one of the three disjoint translated triangles; when $B \geq 3h$, its central triangle and three outer strips have total area at most $(B - 2h)^2$. This is twice the area of one translated simplex. Integration in the other free coordinates gives $2(B - 2h)_+^5$. Empty slices contribute zero.

For $z \geq H$, retain the whole slice. Integrating its density $6(D - z)^5$ with $z \geq F$ gives the free tail and the displayed integral. The exact feasible region shrinks when the floor, height, or lower incidence increases. Its homogeneous version increases with D , by this same slice description. These monotonicities precede the signed polynomial upper expressions; they do not assume monotonicity of each summand.

Before slicing, the region has only the simplex faces, the parameter floor and the s linear normal inequalities. Eliminate a free coordinate. Graph each boundary in a coordinate of largest normal coefficient; all resulting slopes have absolute value at most one. The number of boundary unit cubes is then bounded by $C(\ell + 1)^5$, uniformly in all coefficients. The whole leading image embeds in this region, so this proves the stated uniform finite-layer estimate, including its embedded and vertex layers. $\square$

Lemma 17.23. *Suppose that $L^7 = 1$ and generation fails at x . Fix an actual initial fivefold from a sufficiently late original source of degree $20i$ and order $9/10$. After uniform cutoffs fixed before all further degrees and computing places, every smooth divisorial minimal center P of an own actual $29/25$ threshold in a degree divisible by 25 has $L^6 P = 1$ and contains the initial center and every sufficiently late actual $27/25$ center observed in a degree divisible by 25 . The latter centers have states $(4, 6, 3)$, $(5, 6, 2)$, or $(5, 6, 3)$.*

Proof. We show that either failure of containment produces a positive original source at $13/5$. Put

$$R = 9/10, \quad r = 27/25, \quad T = 29/25, \quad Q_* = 13/5, \quad h = 131/1000, \quad h_P = 3003/10000.$$

The earlier initial-fivefold lemmas supply the two lower profiles

$$H_0(t) = (h + a_0(t - R))_+, \quad H_1(t) = (h_i + a_1(t - r))_+,$$

with $(h_i, K_i) = (59/250, M), (17/80, D), (26/125, Q)$, and $V(r) > 6/25$, $V(T) > 1/500$. They also supply $L^6 P = 1$ and $1/s_P \geq 225/749$. The gap

$$225/749 - h_P = 753/7490000 > 1/10000$$

permits a uniform comparison cutoff with $\phi_P(T) > h_P$, fixed before P and its degree. A supporting lower profile is

$$\psi(t) = (h_P + a_P(t - T))_+.$$

The entire fixed degree at r is less than $13/70$, so $\phi_P(r) < 13/70$ and $a_P > 8021/5600$.

Step 1. In this step, we obtain the full divided-image bounds under either noncontainment. If the initial center is not contained in P , then $w_0(P) = 0$. Divide every actual point-layer germ by the common power $p^{\lceil q\psi(t) \rceil}$. By Lemma 17.3, its initial image embeds in the residual ordinary degree with the same w_0 -threshold. It therefore has the joint leading bound

$$(t - \psi(t))_+^6 Q \left(\frac{H_0(t)}{(t - \psi(t))_+} \right).$$

If the auxiliary $27/25$ center is not contained in P , the identical argument gives

$$(t - \psi(t))_+^6 K_i \left(\frac{H_1(t)}{(t - \psi(t))_+} \right).$$

Both expressions are zero when their degree factor vanishes; at zero height use the full polynomial layer. No residual local germ is required to lift to a new global source. The exact homogeneous capacity is nondecreasing in the degree and nonincreasing in the height. Thus replacing the integer residual degree by its larger real bound preserves the upper inequality, with uniform error $O((\ell + 1)^5)$.

The same original image obeys the separate bounds

$$t^6, \quad t^6 Q(H_0/t), \quad t^6 K_i(H_1/t), \quad (t - \psi)_+^6.$$

Include the appropriate joint bound just obtained. Let J be seven times the integral of their minimum through T , and O the corresponding integral from T to Q_* , also including the full baseline $((125 - 52t)_+/73)^6$. Complete original-layer telescoping gives

$$V(Q_*) \geq 1 - \min\{J, 499/500\} - O.$$

All original fixed and remote conditions remain in the image to which the minimum is applied.

Step 2. In this step, we exclude every noncontainment using the complete rational slope certificate. For $a_0 \geq 2$ the loss from the initial profile alone is at most $R^7 + (4/3)(R - h)^7 < 691/1000$. For $a_1 \geq 4$, the three old-plus-new bounds are respectively $867/1000, 805/1000, 931/1000$, by the integrations in Lemma 17.21. For $a_P \geq 60$, use the $1/500$ reserve and extend the joint outgoing bound to infinity. In the old noncontainment case the loss is at most

$$499/500 + \frac{4(T - h_P - hT/R)^7}{3 \cdot 59} < 9999/10000.$$

In the new noncontainment case the corresponding bounds are

$$499/500 + \frac{(T - h_P - h_i T/r)^7 + (T - h_P - 2h_i T/r)^7}{59} < 9999/10000 \quad (K_i = M),$$

$$499/500 + \frac{6(T - h_P - 2h_i T/r)^7}{59} < 9999/10000 \quad (K_i = D),$$

$$499/500 + \frac{4(T - h_P - h_i T/r)^7}{3 \cdot 59} < 9999/10000 \quad (K_i = Q).$$

The sharper M branch applies throughout this outgoing range because its ratio already exceeds $2/11$ at T . All roots at T are positive; integration stops at their zeros. The displayed comparisons are rational power inequalities.

The remaining boxes are

$$[h/R, 2] \times [h_i/r, 4] \times [8021/5600, 60].$$

Here are exact rules for every leaf of the six incidence data trees. Partition $[0, Q_*]$ at $0, R, r, T, Q_*$ and $j/50$, $1 \leq j \leq 129$. On each interval use the upper endpoint of each slope before its own anchor and the lower endpoint afterwards. Split further at each profile zero, at each residual degree zero, and at the ratios $71/400$ and $2/11$ for the relevant kernel. Integrate every valid capacity by the primitives in Lemma 17.20, with the following general affine version of its balanced primitive:

$$\int 7(Ct + D)^5(At + B) dt = \begin{cases} \frac{A(Ct + D)^7}{C^2} + \frac{7}{6C} \left(B - \frac{AD}{C} \right) (Ct + D)^6, & C \neq 0, \\ 7D^5(At^2/2 + Bt), & C = 0. \end{cases}$$

On negative-root intervals the positive-part expression is zero. For the joint term substitute the affine degree $t - \psi$ for t , keeping the retained height unchanged. For Q retain the additive $1/5000$ term above $71/400$. Take the minimum of the five upper integrals on each interval, also including the baseline beyond T . Let their sums be $I(B)$ and $O(B)$. Then

$$\min\{J, 499/500\} + O \leq S(B) := \min\{I(B), 499/500\} + O(B).$$

The inequality uses the minimum of upper integrals, so it does not require the selected expression to minimize the capacity at every point of the interval.

Each pair in a tree bisects the side of largest normalized width $(v_j - l_j)/(1 + (v_j + l_j)/2)$, choosing the first coordinate at a tie. A leaf N asserts $S(B) \leq N/10^6$. In the recorded order, the six complete trees have 556, 292, 195, 77, 40, 15 leaves, with maxima

$$999897, 999897, 999836, 999873, 999673, 999640 \quad \text{over } 10^6.$$

Every one is below 9999/10000. The roots, the split rule, and all the leaves specify the full finite check, including every incoming layer and every compact slope boundary.

Certificate data.

17-ambient-endpoints/initial-fivefold-incidence. They use the domain, functions and margin specified above, with the verification rule of Lemma 17.11.

Thus either noncontainment would imply $V(13/5) > 0$. Complete volume convergence and Corollary 5.12 contradict nongeneration. Both centers must therefore lie in P . Their embedding dimensions are at most six, since their local rings are quotients of the smooth six-dimensional local ring of P . Theorem 10.1 and Lemma 17.19 then leave exactly the three states in the statement. $\square$

Lemma 17.24. *Let A be a regular local ring of dimension two in characteristic zero, let p be a regular parameter, and let w be a divisorial valuation centered at its closed point. Put $\theta = w(p)/A(w)$. For every primary ideal I and integer $e \geq 2$,*

$$e(I) \geq \frac{4[w(I)/A(w)]^2}{e[1 - (1 - 1/e)\theta]^2}.$$

Proof. Adjoin an e -th root of p . The resulting ring is regular and finite flat of degree e , with basis $1, p^{1/e}, \dots, p^{(e-1)/e}$. Hence the extended ideal has multiplicity $e e(I)$. For an extension $\tilde{w}$, the ramification formula gives

$$\frac{A(\tilde{w})}{\tilde{w}(I)} = \frac{A(w) - (1 - 1/e)w(p)}{w(I)}.$$

Apply Theorem 7.3 to the full extended ideal and bound its threshold by this valuation. Dividing the resulting inequality by e proves the claim. The cover changes only this local comparison. $\square$

Lemma 17.25. *Suppose that $L^7 = 1$ and generation fails at x . After the uniform cutoffs for an actual initial fivefold at 9/10, the existence of an own smooth divisorial threshold center at 29/25 forces every sufficiently late actual 27/25 center to have state $(4, 6, 3)$. The two later observing degrees are divisible by 25 and may be unrelated. The same smooth prime has original degree one and contains the initial center and every such 27/25 center.*

Proof. The preceding incidence lemma supplies the degree and containment assertions and leaves only states $(4, 6, 3)$, $(5, 6, 2)$, and $(5, 6, 3)$ at 27/25. We exclude both fivefolds by coupling the entire later fixed divisor with the retained fivefold valuation.

Step 1. In this step, we fix the two computing profiles and the actual later prime data. Put $R = 9/10$, $r = 27/25$, $T = 29/25$, and $Q_* = 13/5$. The initial height is $h_0 = 131/1000$. If the 27/25 center is a fivefold, its two complete-tail pairs are $(5/4, 92/125)$, $(623/500, 719/1000)$. Eliminating b in $8 \leq s + ab + p$, $rs + b \leq 7$ gives inverse coefficients at least $175/743$, $8642/36025$. Thus we may use $h_1 = 471/2000$, since $175/743 - h_1 = 47/1486000 > 1/100000$, fixing the common comparison error first. Let the two whole lower profiles be

$$H_i(t) = (h_i + a_i(t - r_i))_+, \quad r_0 = R, \quad r_1 = r.$$

Both have the full capacity Q . We retain the already earned facts

$$V(r) > 6/25, \quad V(T) > 1/500, \quad \sum_E (L^6 E) \phi_E(r) < 13/70.$$

Let P be the own later smooth prime, let I_2 be its actual full base ideal in degree k_2 , and write $\nu = \text{ord}_P(I_2)$, $\lambda = \nu/k_2$. The actual threshold satisfies $\tau_2 \nu = 1$. Uniform comparison gives

$$\phi_P(T) > 3003/10000, \quad \phi_P(T) \leq \lambda \leq \phi_P(T) + 1/100000.$$

For any chosen $h_P < \phi_P(T)$, let $\psi(t) = (h_P + a_P(t - T))_+$ be a lower profile. Then

$$a_P > \frac{h_P - 13/70}{T - r}, \quad V(T) \leq (1 - h_P)^7.$$

17-ambient-endpoints/initial-fivefold-fivefold. They use the domain, functions and margin specified above, with the verification rule of Lemma 17.11.

Step 3. In this step, we bound the actual incidence using the own later threshold and full residual volume. If $\phi_P(T) > 31/100$, the first certificate applies with $\Theta = 1$, contradicting either the full volume ceiling or Corollary 5.12. Hence

$$\phi_P(T) \leq 31/100, \quad \lambda \leq 3101/10000 =: \bar{\lambda}.$$

Write the entire fixed part of I_2 as F_2 , and its full residual ideal as C_2 . Lc at that very threshold gives

$$\frac{w_1(F_2 - \nu P) + w_1(C_2)}{A(w_1)} \leq \nu(1 - \theta).$$

Every term is nonnegative. If $N(t) = \sum_E \phi_E(t)E$, the lower finite-to-profile comparison for each companion and convexity give

$$\frac{w_1(N(t) - \phi_P(t)P)}{A(w_1)} \leq \frac{t}{T} \bar{\lambda}(1 - \theta) \quad (0 \leq t \leq T).$$

This correlation uses the same actual full ideal whose prime is being divided; it makes no recurrence assumption on a prime from an earlier source.

Suppose $\theta \geq 43/50$. Set $z = \phi_P(r) < 13/70$. In the original degree- m source at r , write its full fixed divisor and residual ideal as P_m, C_m , and put

$$p_m = \frac{P_m \cdot L^6}{m}, \quad \rho_m = \frac{w_1(P_m)}{mA(w_1)}.$$

Let m tend to infinity through degrees divisible by 25. The common finite support and coefficient comparison in Theorem 6.13 give

$$p_m \longrightarrow p = N(r) \cdot L^6 \geq z, \quad \rho_m \longrightarrow \rho = \frac{w_1(N(r))}{A(w_1)} \leq \theta z + (27/29)(1 - \theta)\bar{\lambda}.$$

The already selected w_1 is fixed in this limit. Every entire degree- m ideal has normalized w_1 -value at least $\phi_{w_1}(r) > h_1$; its residual value is therefore at least $(h_1 - \rho_m)_+$. At the generic point of W , the full residual ideal is either a unit or primary in the regular two-dimensional ambient local ring: any proper nonprimary support would give an omitted divisorial fixed component. Write D_m for the full residual debit defined in Lemma 17.5. Lemma 17.24 with degree six gives

$$D_m \geq \frac{4(L^5 \cdot W)}{6(1 - 5\theta/6)^2} (h_1 - \rho_m)_+^2.$$

For a unit residual ideal the positive-part term is zero. Apply Lemma 17.5 to this full fixed divisor and residual cycle, and only then use $L^5 \cdot W \geq 1$. Its moving volume tends to $V(r) > 0$, and its bound $\mu_m \leq (1 - p_m)^7$ implies $1 - p > 0$. We may therefore pass to the limit in the relaxed finite intersection bound. The lower debit has a limit, so continuity and the bound on ρ give

$$V(r) \leq \frac{[(1 - p)^2 - \frac{4}{6(1 - 5\theta/6)^2} (h_1 - \theta z - (27/29)(1 - \theta)\bar{\lambda})_+^2]^6}{(1 - p)^5}.$$

The finite brackets are positive by the same lemma, and the positive limiting volume makes the displayed bracket positive. This limit estimates the original source at r ; it does not select a new threshold or change the actual later coefficient $k_2\tau_2$. For fixed debit $D \geq 0$,

$$\frac{d}{da} \{a^{-5}(a^2 - D)^6\} = a^{-6}(a^2 - D)^5(7a^2 + 5D) > 0.$$

Thus replacing p by z enlarges the bound. Since $(27/29)\bar{\lambda} > 13/70$, both the residual height and its multiplicity coefficient increase with θ . Replacing θ by $\Theta_0 = 43/50$ gives

$$V(r) \leq \frac{[(1 - z)^2 - C_0(b_c - \Theta_0 z)^2]^6}{(1 - z)^5}, \quad C_0 = 2400/289, \quad b_c = h_1 - (27/29)(7/50)\bar{\lambda}.$$

The residual expression is positive on $[0, 13/70]$. The quadratic bracket is concave and its endpoint values exceed $3/5$, so it is positive throughout.

Step 4. In this step, we verify the residual-volume contradiction and exclude both contained fivefold states. The following two-leaf exact certificate bounds this last expression strictly below $6/25$. Define

$$D(y) = \frac{6}{25}(1 - 13y/70)^5 - [(1 - 13y/70)^2 - C_0(b_c - \Theta_0(13/70)y)^2]^6 = \sum_{k=0}^{12} a_k y^k.$$

Its Bernstein coefficients are $B_j = \sum_{k=0}^j a_k \binom{j}{k} / \binom{12}{k}$. Repeated adjacent averaging of the row $B_0, \dots, B_{12}$ gives the coefficients on the two halves of $[0, 1]$: take the first entries for the left half and the last entries in reverse order for the right half. Their respective minima are at least

$$16429905/10^9, \quad 893426/10^9.$$

These are direct rational comparisons from the displayed degree-twelve polynomial. All Bernstein basis functions are nonnegative and sum to one; hence $D > 0$. This contradicts $V(r) > 6/25$, proving $\theta < 43/50$.

Now choose $h_P = 3003/10000$ and apply the second full certificate with $\Theta = 43/50$. Its first branch contradicts the same-source volume ceiling; its second gives a positive original $13/5$ source and contradicts Corollary 5.12. Both fivefold states are excluded, and the remaining state is $(4, 6, 3)$. $\square$

Lemma 17.26. *Suppose that $L^7 = 1$. An actual initial fivefold from a sufficiently late original source of degree $20i$ at $9/10$, followed by an own smooth divisorial threshold center at $29/25$ in a sufficiently late degree divisible by 25 , implies generation at x . The cutoffs precede the degrees, averages, centers and places. Every initial and later source is complete and retains its own full coefficient.*

Proof. We use the separate $1/500$ reserve of the initial-fivefold history, force its contained $27/25$ geometry, and bound the joint images with both actual incidences retained. Assume nongeneration. Put

$$Q_* = 13/5, \quad h_0 = 131/1000, \quad h_1 = 59/250, \quad h_P = 3003/10000.$$

Step 1. In this step, we obtain the contained geometry and fix both actual computing profiles. The earlier initial-fivefold lemmas give

$$V(r) > 6/25, \quad V(T) > 1/500, \quad N(r) \cdot L^6 < 13/70.$$

Choose an auxiliary actual $27/25$ threshold in its own sufficiently late original degree divisible by 25 . Lemma 17.25 forces its center Z_1 to have state $(4, 6, 3)$. The same smooth prime P , of original degree one, contains Z_1 and the initial fivefold W_0 . The full tangent ring of Z_1 is reduced and Cohen–Macaulay by Lemma 17.8.

Let w_0, w_1 be their computing places. The inverse coefficients exceed $5/38$ and $10559/44675$, respectively. The gaps above h_0, h_1 are $11/19000, 157/446750$. Fix all comparison errors below $1/100000$ before selecting either source. We obtain whole supporting lower profiles

$$H_i(t) = (h_i + a_i(t - r_i))_+, \quad r_0 = R, \quad r_1 = r, \quad a_i \geq h_i/r_i.$$

Where $H_i > 0$, the actual profile exceeds it by at least $\varepsilon_0 = 1/200000$. This retained margin will pay integer division before the sharper discontinuous branches are used. The old unmarked capacity is $K_0 = Q$ and the new unmarked capacity is $K_1 = M$. For the own later prime, the inverse coefficient $225/749$ gives $\phi_P(T) > h_P$. Choose its lower profile

$$\psi(t) = (h_P + a_P(t - T))_+, \quad a_P > 8021/5600.$$

The last inequality follows from the full fixed degree at r , and $V(T) \leq (1 - h_P)^7$.

Step 2. In this step, we retain the prime floor and both actual incidence debits in the ten image bounds. Put $\alpha_i = w_i(P)/A(w_i) \in (0, 1)$. For $j = 1, 2, 3$, define

$$B_j(D, H, U; \alpha) = 6 \int_U^{\min(H, D)} \left(D - z - \frac{j(H - \alpha z)}{1 - \alpha} \right)_+^5 dz,$$

with reversed intervals contributing zero. The retained-parameter lemma for W_0 , and Lemma 17.22 for Z_1 , give the two joint bounds

$$\begin{aligned} S_0(D, H, U; \alpha) &= (D - \max\{U, H\})_+^6 + B_1, \\ S_1(D, H, U; \alpha) &= (D - \max\{U, H\})_+^6 + \min\{2B_2, 3B_2 - 2B_3\}. \end{aligned}$$

They bound the full original image at $(D, H, U) = (t, H_i(t), \psi(t))$, divided by $6!$ and with the uniform finite-layer error. The exact joint capacities are nonincreasing in the floor, height and lower incidence, before these polynomial upper expressions are substituted.

Use the eight closed incidence slabs formed by

$$[b_0, B_0] = [0, 3/4] \text{ or } [3/4, 1].$$

and

$$[b_1, B_1] = [0, 1/2], [1/2, 5/8], [5/8, 3/4], [3/4, 1].$$

Their interiors at the final endpoint cover every actual $\alpha_i < 1$. A joint bound uses b_i , whereas actual division uses its own α_i , bounded above by B_i only afterwards. For each i , the two legal portions

$$q_i(t) = \psi(t), \quad q_i(t) = \min\{\psi(t), H_i(t)/B_i\}$$

give the partial bounds

$$(t - q_i)_+^6 K_i \left(\frac{(H_i - B_i q_i)_+}{(t - q_i)_+} \right).$$

Indeed division by $p^{\lceil mq_i \rceil}$ debits exactly $\alpha_i \lceil mq_i \rceil$. The retained positive margin $m\varepsilon_0 - 1$ makes this an upper comparison at the displayed height for large m , even at the M -branch switch 2/11. When the displayed height is zero, use the whole residual polynomial ring. Negative residual degree gives zero image, and degree zero costs at most one. Neither actual valuation is replaced by a normal from a degeneration.

Together with the four separate bounds $t^6, t^6 Q(H_0/t), t^6 M(H_1/t), (t - \psi)_+^6$, these are ten bounds on the same image: four separate, four partial, and two joint. Let I, O be seven times the integral of their minimum on $[0, T]$, $[T, Q_*]$, with the full baseline included on the second interval. Complete original telescoping, including all incoming layers, gives

$$V(T) \geq 1 - I, \quad V(Q_*) \geq 1 - \min\{I, 499/500\} - O.$$

Thus a contradiction follows from

$$(17.9) \quad \min\{I + (1 - h_P)^7, \min\{I, 499/500\} + O\} \leq 9999/10000.$$

Step 3. In this step, we certify the full slope and incidence domain and exclude the smooth later outcome. The three unbounded ranges require no incidence comparison. For $a_0 \geq 2$, the old entire loss is below 691/1000. For $a_1 \geq 4$, the old incoming loss and the sharper M -tail give

$$19/25 + (r - h_1)^7/3 + (r - 2h_1)^7/7 < 867/1000.$$

For $a_P \geq 200$, the reserve and the prime-only outgoing bound give

$$499/500 + (T - h_P)^7/199 = \frac{1989490791196134952369386003613}{1990000000000000000000000000} < 9999/10000.$$

The remaining common root box is

$$[h_0/R, 2] \times [h_1/r, 4] \times [8021/5600, 200].$$

We specify every leaf comparison before printing the full trees. For affine $Ct + D$, write

$$\mathcal{R}(C, D; l, v) = \begin{cases} ((Cv + D)_+^7 - (Cl + D)_+^7)/C, & C \neq 0, \\ 7D_+^6(v - l), & C = 0. \end{cases}$$

Use the affine fifth-power primitive in the preceding incidence lemma to integrate $7(Ct + D)_+^5(At + B)$, splitting at the root of $Ct + D$. Denote this integral by $\mathcal{F}_5$. The unmarked integral $\mathcal{C}_i(D, H; l, v)$ is determined by splitting at the zeros of $D, H, H - cD$, where $c = 71/400$ for $i = 0$ and $c = 2/11$ for $i = 1$. Use zero for $D \leq 0$, the full D^6 integral for $H \leq 0$, and otherwise the following branches:

$$\begin{array}{l|l} \begin{array}{l} i = 0, H < (71/400)D \\ i = 0, H \geq (71/400)D \\ i = 1, H < (2/11)D \\ i = 1, H \geq (2/11)D \end{array} & \begin{array}{l} (D - 2H)_+^5(D + 10H) \\ (D - H)_+^6 + D^6/5000 \\ (D - H)_+^6 + 3(D - 2H)_+^6 - 5(D - 3H)_+^6 + 2(D - 4H)_+^6 \\ (D - H)_+^6 + (D - 2H)_+^6 - (D - 3H)_+^6. \end{array} \end{array}$$

All are integrated by $\mathcal{R}, \mathcal{F}_5$, including zero slopes and positive-part zeros.

Partition time at 0, R, r, T, Q_* and $j/50$, $1 \leq j \leq 129$. Choose the upper slope endpoint before each anchor and the lower endpoint afterwards. These profiles are lower bounds throughout the entire time interval for every actual slope triple in the box. Integrate the four independent bounds as just specified. For the four partial bounds, split at zeros of $H_i, \psi, H_i - B_i\psi$, take their nonnegative affine branches and the indicated legal portion, and use $\mathcal{C}_i(t - q_i, H_i - B_iq_i)$.

For each joint integral set $\alpha = b_i$, and split at zeros of $H_i, \psi, H_i - t, \psi - t, \psi - H_i$. The tail is the $\mathcal{R}$ -integral of $(t - \max\{H_i, \psi\})_+^6$. Put $Z = \min(H_i, t)$, using the nonnegative affine branches. If $\psi \geq Z$, there is no further contribution. Otherwise, for each needed j , put

$$c_j = jb_i/(1 - b_i) - 1, \quad G_j = t - jH_i/(1 - b_i).$$

The contribution of B_j , integrated also in time, is

$$L_j = \begin{cases} [\mathcal{R}(G_j + c_j Z) - \mathcal{R}(G_j + c_j \psi)]/c_j, & c_j \neq 0, \\ 6\mathcal{F}_5(G_j, Z - \psi), & c_j = 0, \end{cases}$$

with the current interval endpoints in each integral. This is the exact positive fifth-power integral in the retained coordinate, followed by integration in time. For the old joint bound add L_1 to the tail; for the new one add $\min\{2L_2, 3L_2 - 2L_3\}$. Each alternative bounds the whole interval, so the minimum of these integrals is valid. The lower endpoints $b_i < 1$ ensure every displayed denominator is defined, even in slabs approaching incidence one.

Take the minimum of the ten upper integrals on every interval. Beyond T , include $\mathcal{R}(-52/73, 125/73)$. Let the sums be $I(B), O(B)$. The box-specific portions are legal, so the original-image argument applies directly and the leaf quantity is

$$S(B) = \min\{I(B) + (1 - h_P)^7, \min\{I(B), 499/500\} + O(B)\}.$$

Each pair in the retained-incidence data trees bisects the largest normalized side width, with ordered ties and the lower half on the left. A leaf N asserts $S(B) \leq N/10^7$. In the order of the eight incidence slabs, the counts and largest numerators are

row	1	2	3	4	5	6	7	8
leaves	592	507	1891	1756	480	457	2449	2624
$N_{\max}$	9997553	9997807	9996498	9996477	9997951	9998671	9998644	9997516.

The complete 10756 leaves cover all eight compact domains. Every numerator is less than 9999000. The displayed primitives, slab endpoints, tree rule and leaves are the entire rational certificate for (17.9).

Certificate data.

17-ambient-endpoints/initial-fivefold-smooth. They use the domain, functions and margin specified above, with the verification rule of Lemma 17.11.

The first branch contradicts the full volume ceiling of the same own later prime. The second gives $V(13/5) \geq 1/10000$, hence an original section and generation by Corollary 5.12. All finite-layer errors sum to $O(m^6)$ on the fixed order interval and disappear after normalization. The complete coefficients $20i\tau_0, k_1\tau_1, k_2\tau_2$ were retained in every threshold comparison. The resulting point source has its own fully paid continuation, including both cutting errors. This finishes the smooth alternative. $\square$

17.4. A smooth divisor at the original 27/25 threshold.

Lemma 17.27. *Under nongeneration, an actual initial surface has multiplicity $2 \leq m \leq 6$, embedding dimension $m + 1$, and deficit $0 < b \leq 2/m$. Its full tangent ring is reduced Cohen–Macaulay, all ordinary point-ideal powers are integrally closed, and its Rees ring is normal Cohen–Macaulay. An actual 26/25 fivefold contained in a prime smooth at x is either regular or has embedding dimension six and multiplicity 2, 3, 4. In the latter cases it has these same full tangent and Rees properties.*

Proof. Case 1. We consider the actual initial surface. For an initial surface the state inequality $T(c, m) \geq 2m - (c + 2)$, together with $T(c, m) < m$, forces $m = c + 1$. A regular surface is paid by the complete tail. The remaining state caps give $2 \leq m \leq 6$ and $b \leq 2/m$. Put $M = \text{mult}_x \Gamma$ and $\sigma = 7 - M - b$. Every positive-dimensional successor is a regular curve of positive integral ample degree. Thus, if $\rho \geq \sqrt{m}$, its entire tail is at most ρb , including a direct point cut. Nongeneration and $Rs + b \leq 7$ imply

$$8 \leq s + \rho b, \quad \sigma \leq -\frac{1}{5} + (R\rho - 1)b.$$

For $m = 2, 3, 4, 5, 6$, take respectively $\rho = 283/200, 1733/1000, 2, 2237/1000, 49/20$. Substitution at $b = 2/m$ bounds σ by

$$\frac{147}{2000}, \quad \frac{2597}{15000}, \quad \frac{1}{5}, \quad \frac{5133}{25000}, \quad \frac{121}{600}.$$

These are all less than $21/100$. The minimal-multiplicity assertion of Theorem 4.20, whose deficit condition here is $b > 0$, supplies all ordinary-power equalities and the normal Cohen–Macaulay Rees ring. The full tangent ring has Hilbert series $(1 + (m - 1)z)/(1 - z)^2$; its $\sigma < 1/2$ clause makes the whole tangent ring reduced, including the vertex.

Case 2. We consider the actual later fivefold contained in a smooth prime. For the later fivefold, containment bounds the embedding dimension by six. The scalar census leaves the regular state and $(e, m) = (6, 2), (6, 3), (6, 4)$. To include the numerical exclusion explicitly, the putative $(6, 5)$ row has $b \leq 1$ and tail at most $(69/50)b + 207/500$. Its whole price is at most $6/u + 69/50 + 207/500 = 49161/6500 < 8$. For multiplicities 2, 3, 4, the full affine-tail pairs are

$$(a, p) = \left(\frac{5}{4}, \frac{92}{125}\right), \quad \left(\frac{623}{500}, \frac{719}{1000}\right), \quad \left(\frac{33}{25}, \frac{76}{125}\right).$$

They give, respectively,

$$\sigma \leq \frac{3}{10}b - \frac{1733}{3125}, \quad \sigma \leq \frac{1849}{6250}b - \frac{7153}{12500}, \quad \sigma \leq \frac{233}{625}b - \frac{2149}{3125}.$$

Here $b > 7/5$ and the caps are 4, 3, 2. The last two bounds are $3941/12500$ and $181/3125$, both less than $1/2$. The first is less than one. Since $7 - M = b + \sigma > 1$, the ordinary-blowup theorem gives all power equalities and a normal Cohen–Macaulay Rees ring. For $m = 2, b \leq 7/2$, the first bound improves to $\sigma \leq 6193/12500 < 1/2$. If $b > 7/2$, the nonreduced exceptional-component alternative in the minimal-multiplicity clause of that theorem forces $b \leq 7/2$, a contradiction. Thus the full tangent is reduced in every case. Its completed codimension-one presentation is a hypersurface of order m , so its full Hilbert series is $(1 + z + \cdots + z^{m-1})/(1 - z)^5$. Every use of a failed tail above retains the whole boundary and chooses both cutting errors inside its strict budget gap. $\square$

Lemma 17.28. *Under nongeneration, an actual initial fourfold implies $V(u) > 11/50$, $V(r) > 21/250$, and, at every sufficiently late original degree, entire normalized fixed degree less than $3/10$ through r .*

Proof. We bound the full initial-profile loss and then propagate the reserve to the later order and its fixed divisors.

Step 1. In this step, we obtain the full initial-profile capacity and split its incoming and outgoing losses. An initial computing place has $\phi(R) > h = 131/1000$. Keep one finite supporting slope $a \geq 131/900$ and put $H_a(t) = (h + a(t - R))_+$. Its generic codimension is three. The full primary and height-three bounds in Section 7 give the decreasing upper capacity

$$\kappa(c) = \begin{cases} 1 - (21c/4)^3, & 0 \leq c \leq 1/7, \\ 1 - ((17 + 7c)/24)^3, & 1/7 < c \leq 31/200, \\ 256/729 + 3 - 18c, & 31/200 < c \leq 1/6, \\ 256/729, & c > 1/6. \end{cases}$$

At $31/200$ its jump is downward, since $256/729 + 21/100 < 1 - (3617/4800)^3$; the other joins agree. Let

$$A(a) = 7 \int_0^R t^6 \kappa(H_a(t)/t) dt, \\ B(a) = 7 \int_R^{99/100} t^6 \kappa(H_a(t)/t) dt + 7 \int_{99/100}^u t^6 \min\{11/25, \kappa(H_a(t)/t)\} dt.$$

The full point filtration and the baseline give $V(u) \geq 1 - A(a) - B(a)$. The former integral increases with a ; the latter decreases.

Step 2. In this step, we bound every initial supporting slope by exact rational integration. Here is a complete rational check of every slope. Put $d = aR - h$, $C = 256/729$, and

$$\mathcal{T}(A, B; l, v) = \sum_{j=0}^3 \binom{3}{j} A^j B^{3-j} \frac{v^{4+j} - l^{4+j}}{4+j}.$$

The four branch integrals, with the factor seven included, are

$$G_1 = v^7 - l^7 - 7(21/4)^3 \mathcal{T}(a, -d; l, v), \\ G_2 = v^7 - l^7 - \frac{7}{24^3} \mathcal{T}(17 + 7a, -7d; l, v), \\ G_E = (C + 3 - 18a)(v^7 - l^7) + 21d(v^6 - l^6), \\ G_C = C(v^7 - l^7).$$

Before the positive profile use $v^7 - l^7$. Split at every interior $d/(a - c)$, for $c = 0, 1/7, 31/200, 1/6$. In the last integral of B , split instead at $c = 26528/164025, 1/6$, and use $(11/25)(v^7 - l^7)$ before the first crossing. Equal slopes cause no crossing and are evaluated on their constant branch. For successive endpoints in the list

$$\frac{131}{900}, \frac{3}{20}, \frac{1}{5}, \frac{1}{4}, \frac{3}{10}, \frac{2}{5}, \frac{1}{2}, \frac{3}{4}, 1, \frac{3}{2}, 2, 3, 5, 10, 20,$$

these primitives bound $A(a_h) + B(a_l)$, in order, by the following strict upper numerators over 10^4 :

$$\begin{array}{cccccc} 7069 & 7317 & 7543 & 7720 & 7759 & 7385 & 7461 \\ 7433 & 7525 & 7552 & 7612 & 7658 & 7694 & 7709. \end{array}$$

For example the first exact value is $977156882025138674868997/13824000000000000000000$. The same primitives cover the unbounded interval: For $a \geq 20$, one has $A(a) + B(a) \leq R^7 + B(20)$,

Indeed $8 \leq s + ab + p$ and $us + b \leq 7$ give $1/s \geq (au - 1)/(7a + p - 8)$. Every row is a full descendant bound, including the multiplicity-eight and multiplicity-nine threefolds. A strict failed inequality pays both cutting errors before the point continuation. The smallest gaps to the stated profile heights are $7/90000$, $73/2310000$, and $57/713750$, respectively, all greater than 10^{-6} . $\square$

We use one arithmetic convention for the remaining certificates. This also specifies the marked polynomials which occur below. For a physical degree Z and nonnegative threshold Y , the generic labels with codimension c mean

$$\begin{aligned} Nlo &= Z^6 - \left(\frac{7c}{c+1}\right)^c Z^{6-c} Y^c, \\ Nhi &= Z^6 - \frac{Z^{6-c}((6c-1)Z + 7Y)^c}{(6(c+1))^c}, \\ C &= (7/6)^6(Z - Y)^6, \quad A = (Z - Y)^6, \\ Q &= (Z - 2Y)^5(Z + 10Y), \\ span = R_1 &= (Z - Y)_+^6 + (Z - 2Y)_+^6 - (Z - 3Y)_+^6, \\ S &= (Z - Y)_+^6 + 3(Z - 2Y)_+^6 - 5(Z - 3Y)_+^6 + 2(Z - 4Y)_+^6, \\ H &= (256/729)Z^6, \quad Hlip = (256/729 + 3)Z^6 - 18Z^5Y. \end{aligned}$$

The polynomial label **R** means R_1 , whereas R in an anchor is $9/10$. The label **Bc** means ct^6 , **P** means the free divided bound $(t - F)_+^6$, **B1** means the full degree Z_+^6 , and **0** means zero. At $Z \leq 0$ every leading term is zero; at $Y \leq 0$ every full comparison uses Z_+^6 . The branches are $Y/Z \leq 1/7$ for **Nlo**, $1/7 \leq Y/Z \leq 1$ for **Nhi**, **C**, and $Y/Z \geq 1/2$ for the generic **A** bound. The **Q** and its endpoint **A** use the exact crossing bracket in the preceding certificate conventions; another available upper bound is used in its gap. For **M**, **S** is always available, **R** is available at $Y/Z \geq 2/11$, and **H**, **Hlip** are used respectively above and below $1/6$.

Write $J_Q = J_2$ and $J_M = J_3$ for the retained bounds of Lemma 17.22. The names **J**, **JD** stand for $J_K(t, F, Y_1; \theta_l)$ and $J_K(t - F, 0, Y_1 - \theta_h F; \theta_l)$. Their complete polynomial rule is as follows. For $J_s(D, F, H; \beta)$, start with $(D - \max(F, H))_+^6$ and put $U = \min(D, H)$. If $U > F$, for every pair

$(j, c_j) = (2, 1)$ for $s = 2$; $(2, 3), (3, -2)$ for $s = 3$; $(2, 10), (3, -20), (4, 15), (5, -4)$ for $s = 5$, put $A_j = D - jH/(1 - \beta)$, $B_j = j\beta/(1 - \beta) - 1$, and add

$$(17.10) \quad \begin{cases} c_j[(A_j + B_j U)_+^6 - (A_j + B_j F)_+^6]/B_j, & B_j \neq 0, \\ 6c_j(A_j)_+^5(U - F), & B_j = 0. \end{cases}$$

When $H \leq 0$ the result is $(D - F)_+^6$. All comparisons, affine zeros, and ratio crossings are split at their rational roots. Thus these expressions are single rational polynomials on the recorded open time intervals. Adjacent cells may merge when their coefficients agree.

Each cell with endpoints l, v , polynomial P , and integer N asserts $7 \int_l^v P < N/10^9$. Its exact evaluation is (17.6). A selected polynomial need only bound the full minimum throughout its cell. The factor seven is outside every named polynomial. The cells are adjacent from zero to their stated horizon. All trees split a parent into its two closed children on the named coordinate, retaining every other endpoint. The data retain the root, every split, every empty leaf, and every integral leaf. Coverage and exact summation are governed by Lemma 17.11.

Lemma 17.30. *In the setting of Lemma 17.29, each of the following implies $V(T) > 1/1000$: an initial center not contained in P ; a u -center not contained in P ; a regular u -center; or an initial fourfold. The regular case gives $V(T) > 1/20$, and a noncontained u -center of dimension five or embedding dimension six gives $V(T) > 1/100$.*

Proof. Fix the comparison tolerance $\varepsilon = 10^{-6}$ and its cutoffs before all observations. Let ϕ_0, ϕ_1 be initial and u -computing profiles. The common lower prime envelope is

$$(17.11) \quad F_\lambda(t) = \begin{cases} (\lambda + 12(t - r))_+, & t < r, \\ \lambda + 25(\lambda - 1/5)(t - r), & t \geq r. \end{cases}$$

It follows from the two slope bounds just proved. Its coefficients in λ are positive on both intervals.

Case 1. The u -center is not contained in P . First suppose the u -center is not contained in P . If it has embedding dimension seven, the actual 25-divisible observation meets Theorem 17.17, which gives the stronger reserve $1/64$. Otherwise use $h_0 = 131/1000$, and $h_1 = 37/200$ for a fivefold or $41/200$ for an embedding-six fourfold. These heights follow from the scalar census and the early-fourfold lemma. The cap at r gives the lower functions, with $r_0 = R, r_1 = u$,

$$(17.12) \quad H_i(t) = \begin{cases} (h_i + \frac{\lambda + \varepsilon - h_i}{r - r_i}(t - r_i))_+, & t < r_i, \\ h_i t / r_i, & t \geq r_i. \end{cases}$$

Since $w_1(P) = 0$, actual division has exactly zero debit. With $D = (t - F_\lambda)_+$, the full loss density is bounded by

$$\min\{t^6 B(t), t^6 N_5(H_0/t), t^6 K(H_1/t), D^6 K(H_1/D)\},$$

where $K = Q, M$ and B is the full baseline already defined. The four Q scalar leaves and three M scalar leaves in the corresponding data object cover $[7/30, 2/5]$. On each leaf use its upper λ endpoint in the incoming parts of (17.12), and its lower endpoint in (17.11). These lower every threshold and enlarge the available degree. Prefixes 0,1,D mean $(Z, Y) = (t, H_0), (t, H_1), (D, H_1)$, with old generic codimension five and new generic codimension three in M . All 78 cells are retained in the data object. Their maximum summed numerators over 10^9 are 988458493 and 985080068, both less than $99/100$.

Certificate data.

17-ambient-endpoints/source-noncontained26. They use the domain, functions and margin specified above, with the verification rule of Lemma 17.11.

The full filtration therefore gives $V(T) > 1/100$.

Case 2. The u -center is regular. For a regular u -center use instead $h_1 = 2777/10000$. The cap $h_1 r / u < \lambda + \varepsilon$ implies $\lambda > 36/125$, the exact slack being $4937/13000000$. Use (17.12) with this height and retain only the free divided term D^6 , together with the baseline, old N_5 , and unmarked new Q . This requires no incidence. The two leaves $[36/125, 43/125], [43/125, 2/5]$ have 20 cells. Their summed numerators are 896490743 and 933062965, both less than 950000000 .

Certificate data.

17-ambient-endpoints/source-regular26. They use the domain, functions and margin specified above, with the verification rule of Lemma 17.11.

Thus $V(T) > 1/20$.

Case 3. The initial center is not contained in P . Next suppose the initial center is not contained in P . The preceding cases allow us to assume the u -center lies in P , with type Q or M , including the regular case in the generic Q bound. Set $h_0 = 263/2000$, $h_Q = 2017/10000$, $h_M = 129/625$, and retain one supporting slope a of ϕ_0 at R . Put

$$h(a) = \max\{h_K, h_0 + a(u - R) - \varepsilon\}, \quad H_0 = (h_0 + a(t - R))_+.$$

Use (17.12) for H_1 , with its height replaced by $h(a)$. Compatibility of the two actual sources proves this height. The original r -cap bounds $h_0/R \leq a \leq (\lambda + \varepsilon - h_0)/(r - R)$. Since now $w_0(P) = 0$, the fourth term of the full minimum is $D^6 N_5(H_0/D)$. The other three are the baseline and the two unmarked bounds. The closed root is

$$[263/1800, 268501/180000] \times [7/30, 2/5] \quad \text{in } (a, \lambda).$$

On a rectangle use a_h before R , a_l after R , and $h_* = h(a_l)$. Use λ_h before u in H_1 , the radial line afterward, and F_{λ_l} . The height coefficient of the incoming H_1 is $(r - t)/(r - u) > 0$, whereas

its λ coefficient is negative. Hence these substitutions lower all thresholds. The homogeneous generic capacity increases with available degree and decreases with height, so they also bound the actual divided image. Prefix D now uses (D, H_0) with codimension five. The 13 Q rectangles and 27 M rectangles in the data have 148 and 324 cells; their maxima are $996589132/10^9$ and $998187818/10^9$, both less than $999/1000$.

Certificate data.

17-ambient-endpoints/source-noncontained-initial. They use the domain, functions and margin specified above, with the verification rule of Lemma 17.11.

Case 4. The initial center is a fourfold. Finally suppose the initial center is a fourfold. Its earlier volume lemma gives $V(r) > 21/250$, so $7/30 < \lambda < 3/10$. The outgoing slope argument strengthens to $d < 5$, since $250 \cdot 127^7 < 84 \cdot 150^7$. Thus use $F = (\lambda + 5(t - r))_+$ for $t \leq r$. If any u -center is not contained, the first case applies. Otherwise keep the same $h_0 = 263/2000$, compatible $h(a)$, and radial H_1 , but the initial capacity is now $\min\{N_3, 1, 256/729 + 18(1/6 - c)_+\}$. Use its unmarked bound, the unmarked Q or M , the baseline, and the independent free divided bound $(t - F)_+^6$. The root is $[263/1800, 56167/60000] \times [7/30, 3/10]$. The same endpoint substitutions bound the incoming loss through r . The recorded 13 Q rectangles and 36 M rectangles have 132 and 354 cells. Adding $(1 - \lambda_l)^7$ to each incoming numerator gives maxima

$$\frac{218182669169}{218700000000}, \quad \frac{545997298229}{546750000000},$$

both less than $999/1000$.

Certificate data.

17-ambient-endpoints/source-initial-fourfold. They use the domain, functions and margin specified above, with the verification rule of Lemma 17.11.

They contradict $V(r) \leq (1 - \lambda)^7$. Therefore every such u -center is noncontained, and the first case gives the required source gain.

In each case only the complete space $H^0(X, qL)$ receives its Riemann–Roch lower bound. Its actual point layers satisfy each displayed upper comparison simultaneously. Summation gives $V(v) \geq 1 - 7 \int_0^v C(t) dt$; the uniform fifth-order layer errors sum to $O(q^6)$. Combining the baseline through a fixed rational partition and refining after the degree limit gives its pointwise minimum with the local bounds. Every fixed, residual, embedded, and remote condition stays on the full image. No later threshold was used to earn any of these source bounds. $\square$

Lemma 17.31. *Under nongeneration, an eligible smooth actual 27/25 divisorial center implies $V(T) > 1/1000$ if an initial center is a threefold. The same conclusion holds for an initial surface unless it is a quintuple surface contained in the smooth prime and every sufficiently late actual 26/25 center is a contained fivefold of type $(5, 6, 2)$, $(5, 6, 3)$, $(5, 6, 4)$.*

Proof. The easy-history lemma handles noncontained centers and regular u -centers. We therefore assume both the initial center W_0 and the u -center W_1 lie in the same smooth prime P , and that W_1 is in family Q or M of the preceding comparison lemma. Its height is $h_Q = 2017/10000$ or $h_M = 129/625$, with its own whole coefficient. Use $\varepsilon = 10^{-6}$.

Step 1. In this step, we obtain the initial-center heights from their full contained descendant costs. Containment in P bounds the embedding dimension of every initial descendant by six. For an initial threefold, the filling inequality in the finite-tail definition gives surface multiplicity at most five and threefold multiplicity at most nine. Curves are regular. Put $\rho = 2237/1000$. The exact inequalities $\rho^2 = 5004169/10^6 > 5$ and $\rho^3 = 11194326053/10^9 > 9$ bound all radii on this whole subtree. Induction in the tail recurrence gives a total cost at most ρb_0 : a radius $a \leq \rho$ followed by a deficit $y \leq b_0$ costs at most $a(b_0 - y) + \rho y \leq \rho b_0$. Thus $8 \leq s_0 + \rho b_0$ and $Rs_0 + b_0 \leq 7$ give $1/s_0 \geq 10133/76590$. We may use $h_0 = 1323/10000$, because its exact gap is $143/76590000 > \varepsilon$. The initial generic codimension is four.

For an initial surface, the initial-surface lemma gives $2 \leq m_0 \leq 5$ and embedding dimension $m_0 + 1$. If $m_0 \leq 4$, use $\rho = 2$ and $h_0 = 13333/100000$; the inverse price is at least $2/15$, with gap

$1/300000 > \varepsilon$. For all $m_0 \leq 5$, the preceding $\rho = 2237/1000$, $h_0 = 1323/10000$ work. Its initial generic codimension is five. These height arguments use only positive integral ample degrees and the full tail; both published cutting errors are paid before any cut.

Step 2. In this step, we combine the capped profiles and retained-incidence bounds on every full image. Let a, b be supporting slopes at R, u , and put

$$h(a) = \max\{h_K, h_0 + a(u - R) - \varepsilon\}, \quad H_0(t) = (h_0 + a(t - R))_+, \quad H_1(t) = (h(a) + b(t - u))_+.$$

The same actual r -winner caps these slopes by

$$(17.13) \quad \frac{h_0}{R} < a < \frac{\lambda + \varepsilon - h_0}{r - R}, \quad \frac{h(a)}{u} < b < \frac{\lambda + \varepsilon - h(a)}{r - u}.$$

Compatibility supplies the height $h(a)$; in particular the old and new slopes have not been chosen independently of the actual prime cap. Let $F = F_\lambda$ be the envelope in (17.11), and put $D = (t - F)_+$. For $\theta = w_1(P)/A(w_1)$ one has $0 < \theta < 1$. The full tangent assertions proved earlier permit

$$K_Q = \min\{Q, E_1 + E_2 - E_3\}, \quad K_M = M,$$

and the retained bounds J_Q, J_M . The initial capacity K_0 is N_4 for a threefold, N_5 for a surface. Every full actual layer is bounded by the minimum of

$$t^6 B(t), \quad t^6 K_0(H_0/t), \quad t^6 K_K(H_1/t), \quad D^6, \\ D^6 K_K((H_1 - \theta F)_+/D), \quad J_K(t, F, H_1; \theta), \quad J_K(D, 0, H_1 - \theta F; \theta).$$

Division is by the actual forced power $p^{\lfloor qF(t) \rfloor}$; its exact normalized debit is $\theta \lfloor qF(t) \rfloor$. Every residual and remote condition stays on this image. The retained tangent comparison counts its full image, including the affine vertex.

Step 3. In this step, we specify the full four-parameter certificate and its endpoint comparisons. Here are the complete rules for its four-parameter certificates. On a box $[a_l, a_h] \times [b_l, b_h] \times [\lambda_l, \lambda_h] \times [\theta_l, \theta_h]$, put

$$h_* = \max\{h_K, h_0 + a_l(u - R) - \varepsilon\}, \\ a'_h = \min\{a_h, (\lambda_h + \varepsilon - h_0)/(r - R)\}, \\ b'_h = \min\{b_h, (\lambda_h + \varepsilon - h_*)/(r - u)\}, \quad b'_l = \max\{b_l, h_*/u\}.$$

The box is empty if $a_l > a'_h$, $b'_l > b'_h$, or $h_* r/u > \lambda_h + \varepsilon$. Otherwise use

$$Y_0(t) = (h_0 + a'_h(t - R))_+ \quad (t < R), \quad Y_0(t) = (h_0 + a_l(t - R))_+ \quad (t \geq R), \\ Y_1(t) = (h_* + b'_h(t - u))_+ \quad (t < u), \quad Y_1(t) = (h_* + b'_l(t - u))_+ \quad (t \geq u), \\ F_* = F_{\lambda_l}, \quad D_* = (t - F_*)_+, \quad R_* = Y_1 - \theta_h F_*.$$

These lower bounds follow from the signs before and after each anchor. Choose precisely the division amount F_* ; its remaining threshold is at least R_* . Use θ_l in both retained capacity parameters and θ_h only in this debit. Indeed, on a slice $z < H$,

$$\frac{\partial}{\partial \theta} \frac{H - \theta z}{1 - \theta} = \frac{H - z}{(1 - \theta)^2} > 0.$$

Both remaining slice capacities decrease in that threshold; for the signed M expression the derivative is $-30(B - 2h)_+^4 + 30(B - 3h)_+^4 \leq 0$. The free tail is independent of θ . Thus the substitutions bound every actual incidence in the box. At $\theta_l = 0$ the denominator is exactly one; at an upper endpoint one, only the debit uses that endpoint. Every denominator uses $\theta_l < 1$, and the uniform boundary-cube estimate in the retained lemma controls the limit toward one.

Let $I(v) = 7 \int_0^v C_{\text{box}}(t) dt$, with the substituted full minimum. The certificate alternatives are

$$(17.14) \quad I(T) < 999/1000, \quad I(r) + (1 - \lambda_l)^7 < 1.$$

The first gives the source floor, and the second contradicts the full moving-mass bound at r . The arithmetic names are those fixed above: prefix 0 uses (t, Y_0) and the indicated initial codimension; 1 uses (t, Y_1) ; D uses $(D_*, (R_*)_+)$, with codimension three for generic M . The marked names use (17.10). All rational roots, cells, empty boxes, and covering splits are retained in the corresponding data objects.

Step 4. In this step, we verify the threefold and surface certificate families and identify the sole remaining history. For the initial-threefold case, the Q root is

$$[147/1000, 267701/180000] \times [2017/10400, 198301/40000] \times [7/30, 2/5] \times [0, 1],$$

and the M root replaces the second interval by $[129/650, 193601/40000]$. These contain every parameter allowed by (17.13). There are 281 nonempty and 43 empty Q leaves, with 3065 cells, and 213 nonempty and 36 empty M leaves, with 2467 cells. The maximum full criteria are

$$\frac{8581290282848236797}{8589934592000000000}, \quad \frac{499482591}{5000000000},$$

both less than $999/1000$, so in this case even the second criterion has that stronger margin.

Certificate data.

17-ambient-endpoints/source-initial-threefold. They use the domain, functions and margin specified above, with the verification rule of Lemma 17.11.

For surfaces the Q root uses $a \in [13333/90000, 266671/180000]$ and the same Q b -interval. It covers every $m_0 \leq 4$. The M root and h_0 are the threefold values and cover every $m_0 \leq 5$. The Q family has 1075 nonempty and 278 empty leaves, with 12408 cells. Its largest r -mass criterion is $8589931765637723389/8589934592000000000 < 1$; its T -integrals are all less than $999/1000$. The M family has 611 nonempty and 112 empty leaves, with 6722 cells, and maximum full criterion $29323928302434726949/29353417113600000000 < 999/1000$.

This larger table uses literal dictionaries. A rational record **i:value** defines R_i ; a polynomial record **i:label** defines P_i . A **ROOT** record gives the eight endpoints in the order (a, b, λ, θ) . A record **S tag axis i** splits the named box at R_i , with coordinate numbers 0, 1, 2, 3. A record **E tag** asserts the explicit empty test above. A record **L tag method hIndex N upperIndex** specifies the leaf, its method in (17.14), its h_* , its cell numerator sum, and its full rational criterion. Recompute the effective endpoints from that box. A record **C rightIndex polyIndex n** gives the next right endpoint, polynomial, and upper numerator; its left endpoint is zero initially and the preceding right endpoint thereafter. Thus the dictionary notation retains every endpoint and every cell inequality.

Certificate data.

17-ambient-endpoints/source-initial-surface. They use the domain, functions and margin specified above, with the verification rule of Lemma 17.11.

The actual parameters belong to a nonempty final leaf. The original filtration gives $V(v) \geq 1 - I(v)$, so every mass leaf is impossible and every remaining leaf gives the stated source floor. This proves the threefold assertion. In the surface case, failure of the floor forces $m_0 = 5$ and excludes all M -centers at u ; the earlier cases excluded all noncontained and regular centers. Precisely the three asserted contained singular fivefold types remain. $\square$

Lemma 17.32. *Suppose $\pi^*L - uE_x$ is big. Let an actual u -threshold have a contained fivefold center $W \subset P$, and let the same original degree k have smooth actual r -divisor P . Put $\theta = w(P)/A(w)$ for an actual u -computing place, $h_k = 1/(k\tau_u)$, and let z_k, λ_k be the normalized full coefficients of P at u, r . For every $\varepsilon > 0$, there is a cutoff J , depending only on $(X, L, x, u, \varepsilon)$ and chosen before either observation, such that whenever $k \geq J$, simultaneously for every integer $e \geq 2$,*

$$V(u) < \frac{[(1 - z_k)^2 - 4[h_k - \theta z_k - (1 - \theta)\lambda_k]_+^2 / \{e(1 - \theta + \theta/e)^2\}]_+^6}{(1 - z_k)^5} + \varepsilon.$$

Proof. We compare the two full ideals at their common degree, bound the residual multiplicity, and apply the finite moving-volume estimate.

Step 1. In this step, we retain the actual companion contribution in the common-degree residual value. Write the two full ideals at this same degree as $I_- = \mathcal{O}(-F_-)C_-$, $I_+ = \mathcal{O}(-F_+)C_+$. At η_W , the entire residual C_- is either the unit ideal or primary in the regular local ring of dimension two: removing the entire divisorial fixed part removes every height-one associated support component. No component away from this localization has been removed from the global source.

The full r -ideal is lc at its own threshold $\tau_r = 1/(k\lambda_k)$. The inclusion $I_+ \subset I_-$ and the actual factor $p^{k\lambda_k}$ give

$$w(C_-)/kA(w) \geq h_k - \theta z_k - (1 - \theta)\lambda_k.$$

To see the companion term, log canonicity at r says $w(I_+) \leq k\lambda_k A(w)$; subtract its P -part, use $F_+ \geq F_-$ coefficientwise, and retain all the other components. Thus the entire companion divisor $F_- - kz_k P$ has value at most $k\lambda_k(1 - \theta)A(w)$. Subtracting the whole F_- from $w(I_-) = kh_k A(w)$ proves the displayed lower bound. This is a comparison of these two finite ideals, so it uses their equal degree.

Step 2. In this step, we obtain the residual multiplicity debit for every cyclic-root degree. For a positive right side, apply the cyclic-root bound of Lemma 17.24 with root degree e . It bounds the full generic residual multiplicity below by

$$\frac{4k^2[h_k - \theta z_k - (1 - \theta)\lambda_k]_+^2}{e(1 - \theta + \theta/e)^2}.$$

If the residual is the unit ideal the positive-part term is zero.

Step 3. In this step, we derive the uniform finite moving-volume bound from the full residual debit. Let D_k denote the full residual multiplicity debit and $p_k = F_- \cdot L^6/k$ the entire fixed degree of this original u -source. Include the displayed multiplicity with its weight $L^5 \cdot W$ in D_k , and then use $L^5 \cdot W \geq 1$. Apply Lemma 17.5, which retains the full fixed square before discarding its nonnegative deficit. Its geometric cutoff gives $V(u) < \mu_k + \varepsilon$ before any choice of e, W, w , and its finite inequality bounds μ_k using the displayed lower debit and this same p_k . Now $p_k \geq z_k$, since $L^6 \cdot P \geq 1$. The function $(A^2 - D)_+^6/A^5$ increases with A , since on its positive branch its derivative is $(A^2 - D)^5(7A^2 + 5D)/A^6 \geq 0$. Replacing $1 - p_k$ by $1 - z_k$ therefore gives the displayed upper bound for $V(u)$, with its reserved error. The quantities h_k, z_k, λ_k throughout belong to these two actual full ideals at their common degree; none has been replaced by a limiting profile value. $\square$

Lemma 17.33. *Under nongeneration, suppose an initial quintuple surface lies in a smooth actual r -divisor P , and at the same original degree as that r -observation an actual u -center is a contained fivefold of type $(5, 6, 2), (5, 6, 3), (5, 6, 4)$. Then $V(T) > 1/1000$.*

Proof. Fix $\varepsilon = 10^{-6}$, $h_0 = 1323/10000$, and $h_Q = 2017/10000$.

Step 1. In this step, we obtain both computing profiles and the common prime envelope. The initial surface and the u -fivefold have full reduced Cohen–Macaulay tangent rings by the initial-surface lemma. Their own whole-tail heights are precisely those proved in the preceding two lemmas. Let ϕ_0, ϕ_1 be their computing profiles. Put

$$\alpha = w_0(P)/A(w_0), \quad \theta = w_1(P)/A(w_1), \quad z = \phi_P(u), \quad \lambda = \phi_P(r).$$

Both incidences lie in $(0, 1)$; no equality between them is assumed. One has $0 \leq z < 1/5$, $7/30 < \lambda < 2/5$. The two supporting slopes satisfy (17.13), with $h(a) = \max\{h_Q, h_0 + a(u - R) - \varepsilon\}$. Retain $H_0 = (h_0 + a(t - R))_+$ and $H_1 = (h(a) + b(t - u))_+$ on both sides of their anchors.

Let $d_{\max}(\lambda) = 1 + 32(r - \lambda)^7$ and $c = 25(\lambda - z)$. The prime-slope lemma gives $c < d_{\max}(\lambda)$. The same prime has lower envelope

$$(17.15) \quad \begin{aligned} F(t) &= \max\{0, F_u(t), F_r(t)\}, \\ F_u(t) &= \begin{cases} z + c(t - u), & t < u, \\ zt/u, & t \geq u, \end{cases} \\ F_r(t) &= \begin{cases} \lambda + d_{\max}(\lambda)(t - r), & t < r, \\ \lambda + c(t - r), & t \geq r. \end{cases} \end{aligned}$$

Before u , the secant to r bounds the supporting slope above; after u , use the chord from the origin. Before r , use the earned upper supporting slope; after r , its secant lower bound. Thus every original section obeys (17.15).

Step 2. In this step, we retain the common-degree residual-volume bound with its two errors. The uniform finite comparisons, fixed before the common degree k , give $z \leq z_k \leq z + \varepsilon$ and $\lambda \leq \lambda_k \leq \lambda + \varepsilon$. The finite own threshold height is at least $\phi_1(u) > h(a)$. The original u -class is big by Theorem 9.14. Consequently Lemma 17.32 yields, for any integer $e \geq 2$,

$$\begin{aligned} V(u) &< A^{-5}[A^2 - 4\mu^2/K_e(\theta)^2]_+^6 + \varepsilon, \\ A &= 1 - z, \quad \mu = [h(a) - \theta z - (1 - \theta)\lambda - \varepsilon]_+, \\ K_e(\theta)^2 &= e(1 - \theta + \theta/e)^2. \end{aligned}$$

Here the weighted sum of the two finite coefficient errors is at most ε , and the moving-volume error is the separate final ε . The monotonicity in A proved there justifies the enlargement. Also $V(r) \leq (1 - \lambda)^7$.

Step 3. In this step, we bound each full original image by the ten retained and divided comparisons. The initial full spanning bound is

$$S_5 = E_1 + 10E_2 - 30E_3 + 35E_4 - 19E_5 + 4E_6.$$

Use $K_0 = \min\{N_5, S_5\}$ and $K_1 = \min\{Q, E_1 + E_2 - E_3\}$. The full tangent hypotheses permit J_5 for the initial surface and J_2 for the new fivefold. At each actual point layer divide by $p^{\lfloor qF(t) \rfloor}$, put $D = (t - F)_+$, and retain both debits $R_0 = H_0 - \alpha F$, $R_1 = H_1 - \theta F$. The full minimum has ten bounds:

$$\begin{aligned} &t^6 B(t), \quad t^6 K_0(H_0/t), \quad t^6 K_1(H_1/t), \quad D^6 K_0((R_0)_+/D), \quad D^6 K_1((R_1)_+/D), \quad D^6, \\ &J_5(t, F, H_0; \alpha), \quad J_5(D, 0, R_0; \alpha), \quad J_2(t, F, H_1; \theta), \quad J_2(D, 0, R_1; \theta). \end{aligned}$$

The full tangent rings have depth at least two and degree-zero part $\mathbb{C}$, hence connected projectivizations; their reduced, pure Cohen–Macaulay structures satisfy every hypothesis of the retained comparison. Its uniform remainder includes all embedded and vertex layers. Telescoping the original point filtration gives $V(v) \geq 1 - 7 \int_0^v C(t) dt$ for $v = u, r, T$, without assigning a complete-space lower bound to any divided image.

Step 4. In this step, we verify the full six-parameter certificate and obtain the source reserve. We give the full six-parameter certificate. Its root, in the order $(a, b, \lambda, \theta, z, \alpha)$, is

$$\begin{aligned} &[147/1000, 267701/180000] \times [2017/10400, 198301/40000] \\ &\times [7/30, 2/5] \times [0, 1] \times [0, 1/5] \times [0, 1]. \end{aligned}$$

On a box, use the effective a_h, b_h, b_l and the lower functions Y_0, Y_1 of the four-parameter proof. In addition trim

$$z_l \longleftarrow \max\{z_l, \lambda_l - (r - u)d_{\max}(\lambda_l)\}.$$

The box is empty under any previous empty test or if $z_l > z_h$. Use the following prime floor:

$$\begin{aligned} F_{u,*} &= z_l + 25(\lambda_h - z_l)(t - u) & (t < u), \\ F_{u,*} &= z_l t/u & (t \geq u), \end{aligned}$$

$$\begin{aligned}
F_{r,*} &= \lambda_l + d_{\max}(\lambda_l)(t - r) & (t < r), \\
F_{r,*} &= \lambda_l + 25(\lambda_l - z_h)(t - r) & (t \geq r), \\
F_* &= \max\{0, F_{u,*}, F_{r,*}\}.
\end{aligned}$$

The derivative in λ of the incoming F_r is positive because $t - r < 0$; the other endpoint choices follow from their displayed signs. Divide by this fixed F_* . The remaining lower thresholds are $R_{0,*} = Y_0 - \alpha_h F_*$, $R_{1,*} = Y_1 - \theta_h F_*$. Use α_l, θ_l in both pairs of retained parameters, and only the upper endpoints in the actual debits. The monotonicity proof and the endpoint-zero and endpoint-one conventions are exactly those proved for the four-parameter certificate, applied separately to the two valuations.

For the same-degree moving mass, choose

$$e = \max\{2, \lceil \theta_l / (1 - \theta_l) \rceil\}, \quad A_* = 1 - z_l, \quad \mu_* = [h_* - \theta_l z_h - (1 - \theta_l)\lambda_h - \varepsilon]_+,$$

and set

$$M_u = A_*^{-5} \left[A_*^2 - \frac{4\mu_*^2}{e(1 - \theta_l + \theta_l/e)^2} \right]_+^6 + \varepsilon.$$

Since $z_h < \lambda_h$, the residual lower bound increases with θ ; for fixed $e \geq 2$, $K_e(\theta)$ decreases. Thus M_u bounds the actual full moving mass throughout the box. A box with $M_u < 3/40$ is empty by Lemma 17.9. Every other leaf uses one of the three criteria

$$I(T) < 999/1000, \quad I(r) + (1 - \lambda_l)^7 < 1, \quad I(u) + M_u < 1.$$

The latter two contradict the corresponding full upper mass.

For the polynomial dictionary, prefixes 0,D0 use (t, Y_0) , $(D_*, (R_{0,*})_+)$, with generic codimension five and **span** equal to the homogenization of S_5 . Prefixes 1,D1 use (t, Y_1) , $(D_*, (R_{1,*})_+)$, with **span** equal to $E_1 + E_2 - E_3$. The four joint labels J0,JD0,J1,JD1 are the four displayed retained terms and use (17.10), with coefficients $(2, 10)$, $(3, -20)$, $(4, 15)$, $(5, -4)$ for the old surface and $(2, 1)$ for the new fivefold. The same positive-part and free-threshold conventions apply. Each recorded cell has its full interval, polynomial, and upper numerator over 10^9 ; the mass and root degree are recorded for every leaf which uses them.

The complete tree has 812 nonempty leaves and 105 empty boxes, with 10922 cells and 30409 branch checks. There are 573 T -integral leaves, 236 r -mass leaves, and three u -mass leaves. Their respective largest full criteria are

$$\frac{998988173}{10^9} < \frac{999}{1000}, \quad \frac{1092366576727}{1093500000000} < 1, \quad \frac{514896055565273}{524288000000000} < 1.$$

Certificate data.

17-ambient-endpoints/source-quintuple-surface. They use the domain, functions and margin specified above, with the verification rule of Lemma 17.11.

Every actual tuple belongs to a nonempty leaf. Both mass alternatives are impossible, and every remaining leaf gives $V(T) > 1/1000$, as required. The initial degree $20i$ has remained independent; only the u, r observations shared their degree in the full residual comparison. $\square$

Lemma 17.34. *Under nongeneration, every eligible smooth actual 27/25 divisorial center gives $V(t) > 1/1000$ for every $0 \leq t \leq T$. At every sufficiently large original integer q , simultaneously on this interval,*

$$\dim U_q(t) \geq \frac{q^7}{2000 \cdot 7!}.$$

The cutoffs precede every subsequent T -observation.

Proof. We combine all initial histories and then extend the resulting source floor to every original degree.

Step 1. In this step, we obtain the source floor for every possible initial history. Take the finite maximum of the cutoffs in the preceding lemmas before choosing the given 25-divisible r -degree

k . Choose an independent sufficiently late initial degree $20i$, with $5 \mid i$, and an actual initial threshold. The initial-coefficient lemma leaves dimensions two through five. An initial fivefold contradicts the given divisorial observation by Lemma 17.19. An initial fourfold or threefold gives the source floor by the preceding history lemmas. For a surface, failure of that floor forces a contained quintuple surface and only contained singular fivefold centers at u . Choose the actual u -threshold at precisely the already given degree k ; its source is nonzero and $25 \mid k$. The quintuple lemma now applies with its required common degree and gives the floor, a contradiction.

Step 2. In this step, we obtain the uniform source floor in every original integer degree. Thus $\pi^*L - TE_x$ is big before selecting a T -threshold. The complete volume limit gives the strict floor along multiples of 25. For each residue modulo 25 fix a nonzero source section at T in one sufficiently large degree in that residue, using the complete-source nonvanishing theorem. Multiplication injects each large divisible source into the corresponding residue degree: point orders add and the divisible contribution has integral order. The ratio of these two degrees tends to one. Taking the maximum of the 25 fixed cutoffs gives the displayed bound for every integer q . Source inclusions give every real $t \leq T$ simultaneously. This changes no observation degree or coefficient used to earn the floor. $\square$

Lemma 17.35. *Suppose an eligible actual $27/25$ threshold has smooth divisorial minimal center P , and an eligible actual $29/25$ threshold has divisorial minimal center $Q \neq P$. Then $K_X + 8L$ is generated at x . The later prime may be singular, and the two original degrees may differ.*

Proof. Assume nongeneration. We compare the two prime profiles, classify the earlier centers, and bound their full incoming layers.

Step 1. In this step, we derive the two-prime profile caps and volume ceilings. The all-history source lemma supplies $V(T) > 1/1000$ before the later threshold is selected. Choose the full comparison constants on a fixed big interval containing $[0, T]$, with $\varepsilon = 10^{-6}$. Put $\lambda = \phi_P(r)$, $\Lambda = \phi_Q(T)$, and $p_T = \phi_P(T)$. The earlier smooth-prime ceiling and the actual later lc comparison give

$$\frac{7}{30} < \lambda, \quad p_T > 3\lambda - \frac{2}{5}, \quad p_T \leq \Lambda + \varepsilon.$$

This comparison uses $\tau_Q \text{ord}_Q(I_T) = 1$, valid also for a singular Q on the smooth ambient variety. Both distinct primes have positive integral original degree, so the entire fixed degree at T is at least $p_T + \Lambda$. The full moving-volume inequality and $1000 \cdot 93^7 < 250^7$ force $p_T + \Lambda < 157/250$. Hence

$$\lambda < \frac{119}{500} + \frac{\varepsilon}{6}, \quad \Lambda < \frac{41}{125}, \quad d \leq \frac{\Lambda + \varepsilon - \lambda}{T - r} < \frac{71}{60} + \frac{25\varepsilon}{2} < \frac{148}{125}$$

for a supporting slope of ϕ_P at r . The fixed prime floor

$$F(t) = \left(\frac{7}{30} + \frac{148}{125}(t - r) \right)_+ \quad (0 \leq t \leq r)$$

therefore holds on every original section. It gives

$$(17.16) \quad V(u) \leq (15263/18750)^7, \quad V(r) \leq (23/30)^7.$$

Step 2. In this step, we enumerate the complete earlier-center alternatives and their heights. Choose an independent actual initial source of degree $20i$, with $5 \mid i$, and an actual u -source of sufficiently late degree divisible by 25. Their computing heights may be taken as $h_0 = 263/2000$ and the following h_1 :

family	actual u -center	h_1
NQ	fivefold not contained in P	$1857/10000$
NE	fourfold not contained in P	$189/1000$
IQ	singular fivefold contained in P	$2017/10000$
IM	fourfold contained in P	$129/625$

The noncontained heights follow from $b > 7/5$, respectively $b > 3/2$; the incident heights and full tangent properties were proved in the smooth-source comparison lemma. The actual r -cap now gives $\phi_i(r) < c_* = 23801/100000$. A regular u -fivefold is impossible, since its height $2777/10000$ has radial continuation $(2777/10000)r/u > c_*$. Thus the four rows exhaust the possible centers, including both fourfold embedding dimensions in NE .

Step 3. In this step, we obtain the full incoming-image bounds for all four families. For $r_0 = R, r_1 = u$, define

$$H_i(t) = \begin{cases} (h_i + \frac{c_* - h_i}{r - r_i}(t - r_i))_+, & t < r_i, \\ h_i t / r_i, & t \geq r_i. \end{cases}$$

The actual cap bounds every earlier incoming slope; radial convexity supplies the outgoing part. These are fixed lower cones, so no free slope or incidence is being sampled. Every family has the baseline, old $t^6 N_5(H_0/t)$, and the free divided bound $(t - F)_+^6$. In NQ , add the new Q bounds at (t, H_1) and $(t - F, H_1)$; in NE , add the corresponding N_3 bounds. Their actual division debit is zero because $w_1(P) = 0$. In IQ , add the unmarked Q bound, and in IM the unmarked height-three bound. Both incident cases also have

$$J_K(t, F, H_1; \theta) \leq J_K(t, F, H_1; 0), \quad 0 < \theta = w_1(P)/A(w_1) < 1.$$

This is the undivided retained-parameter comparison. Its monotonicity was proved above; the auxiliary endpoint zero is not used as an actual division debit.

Step 4. In this step, we verify the finite incoming-loss comparisons and contradict the prime volume ceilings. The distinct-divisor data object contains 31 cells. Its prefixes 0,1,D use $(t, H_0), (t, H_1), (t - F, H_1)$; old codimension is five, new codimension is three in NE . Its joint label is $J_K(t, F, H_1; 0)$. In this endpoint-zero case (17.10) reduces, with $U = \min(D, H)$, to the free tail plus

$$\sum_j c_j [(D - F - jH)_+^6 - (D - U - jH)_+^6],$$

when $U > F$. Thus every recorded integral is rational after the stated affine branch splits. The NQ, NE horizons are r , with total numerators 688077537, 771938221; the IQ, IM horizons are u , with totals 757580738, 752640365, all over 10^9 . Adding the corresponding upper masses from (17.16) gives respectively the strict upper bounds

$$\begin{array}{r} \frac{1845308118119}{2187000000000}, \quad \frac{2028711434027}{2187000000000}, \\ \frac{1620364553368168746227014515409}{1629441976547241210937500000000}, \quad \frac{3224629004444336245048023171443}{3258883953094482421875000000000}, \end{array}$$

each less than 999/1000.

Certificate data.

17-ambient-endpoints/distinct-divisors. They use the domain, functions and margin specified above, with the verification rule of Lemma 17.11.

The complete original filtration gives $V(v) \geq 1 - I(v)$ at the indicated horizon, contradicting (17.16) in every row. All original conditions remain imposed on each divided image; the only lower dimension estimate is for the complete ambient space. The uniform fifth-order layer errors sum to $O(q^6)$, and the baseline is combined after fixed rational partitions. This proof used no smoothness or degree-one assertion for Q . It therefore includes every singular later divisorial center. $\square$

Lemma 17.36. *Suppose eligible pure actual thresholds at 27/25 and 29/25 have the same prime divisor P , smooth at x , as minimal lc center. Then $K_X + 8L$ is generated at x , with no relation between the two original observing degrees.*

Proof. Assume nongeneration. We retain both prime anchors, classify the earlier actual centers, and compare their complete incoming loss with the full prime mass at three possible horizons.

Step 1. In this step, we derive the persistent prime's two-anchor height and slope bounds. The all-history source lemma gives $V(T) > 1/1000$. Fix the comparison errors $\varepsilon = 10^{-6}$ on a big interval containing $[0, T]$, and fix a rational upper bound for every normalized original point order before the observations. Put $\lambda = \phi_P(r)$, $\Lambda = \phi_P(T)$. The own later boundary has coefficient $s_T = k_T \tau_T$ and full smooth-sixfold tail at most $(5/4)b_T + 187/250$. Hence

$$8 \leq s_T + \frac{5}{4}b_T + \frac{187}{250}, \quad Ts_T + b_T \leq 7, \quad \frac{1}{s_T} \geq \frac{225}{749}.$$

The actual identity $\tau_T \text{ord}_P(I_T) = 1$ and the uniform comparison imply $\Lambda > 3003/10000$, since $225/749 - 3003/10000 = 753/7490000 > \varepsilon$. The earlier prime ceiling gives

$$\lambda > 7/30, \quad \Lambda > 3\lambda - 2/5.$$

The entire moving mass at T bounds $\Lambda < 157/250$. For any supporting slope $d_T > 1$, summing actual prime division on all outgoing layers gives $V(T) \leq (T - \Lambda)^7/(d_T - 1)$. Thus, also when $d_T \leq 1$,

$$(17.17) \quad d_T < 1 + 1000(T - \Lambda)^7.$$

The sum is taken on the previously fixed finite interval of possible original orders. It requires no further positive source.

Step 2. In this step, we classify every earlier history and retain its applicable incidence constraints. Choose an independent initial observation of degree $20i$ and a sufficiently late 25-divisible u -observation. For their computing profiles ϕ_0, ϕ_1 , actual lc comparison at both later anchors gives

$$(17.18) \quad \phi_i(r) \leq \lambda + \varepsilon, \quad \phi_i(T) \leq \Lambda + \varepsilon, \quad \phi_1(u) \geq \phi_0(u) - \varepsilon.$$

All initial dimensions are two through five. The generic initial height $263/2000$, the surface heights $13333/100000$ and $1323/10000$, and the following u -heights were proved using their own full tails above. A regular u -fivefold is impossible: its height gives $\lambda > 7/25$, hence $\Lambda > 11/25$, so $V(T) < (14/25)^7 < 1/20$, contrary to the regular-source case of the easy-history lemma.

The remaining configurations are recorded in this table. The initial geometry column specifies the capacity to use; a generic codimension bound is sufficient and imposes no tangent condition.

family	initial capacity	h_0	u -center relative to P	h_1^*
NQ	N_5	$263/2000$	5, not contained	$1857/10000$
NE	N_5	$263/2000$	4, not contained	$189/1000$
M	N_5	$263/2000$	4, contained	$129/625$
QT	N_4	$263/2000$	5, contained	$2017/10000$
$Q4$	N_5	$13333/100000$	5, contained	$2017/10000$
$Q5$	$\min(N_5, S_5)$	$1323/10000$	5, contained	$2017/10000$
QN	N_5	$263/2000$	5, contained	$2017/10000$

Here QT covers initial dimension at least three; $Q4$ covers a contained initial surface of multiplicity at most four; $Q5$ covers a contained quintuple surface; and QN covers a noncontained initial center. A contained surface has embedding dimension $m_0 + 1 \leq 6$, so these cases exhaust it. Contained u -fivefolds are the three singular types already proved, and contained fourfolds have embedding dimension six and multiplicity three or four. Their full tangents are reduced Cohen–Macaulay. Thus the seven rows exhaust all earlier histories.

Let $\theta = w_1(P)/A(w_1)$. It is zero in NQ, NE , and belongs to $(0, 1)$ in the other rows. In $Q5$ retain also $\alpha = w_0(P)/A(w_0) \in (0, 1)$; in QN its actual old value is zero. The other rows use no old incidence condition. The initial S_5 and both retained capacities are those proved above. In all rows use the baseline, the free prime bound, and the initial unmarked capacity. Use the new unmarked capacity at both (t, H_1) and $(t - F, H_1 - \theta F)$. It is Q in NQ , N_3 in NE , the full M capacity in M , and $\min\{Q, E_1 + E_2 - E_3\}$ in the four incident fivefold rows. Add both undivided and divided retained bounds for each incident new center. Add old division only in

QN and $Q5$, and both old retained bounds only in $Q5$. Each division is by the actual forced prime power; its debit uses the actual incidence at that place. This lists all the available upper comparisons without making an original residual image primary.

Step 3. In this step, we define the compact physical domains and the full prime-mass criteria. We next specify the compact domain and every interval substitution in its certificates. Put

$$L_* = (157/250 + 2/5 + \varepsilon)/3.$$

In physical coordinates $(a, b, \lambda, \Lambda, \theta, \alpha)$, where a, b support the two earlier profiles, the root is

$$(17.19) \quad [h_0/R, (L_* + \varepsilon - h_0)/(r - R)] \\ \times [h_1^*/u, 25(L_* + \varepsilon - h_1^*)] \times [7/30, L_*] \times [3003/10000, 157/250] \times \Theta \times A_0.$$

Here $\Theta = [0, 0]$ in NQ, NE and $[0, 1]$ otherwise; $A_0 = [0, 1]$ in $Q5$ and $[0, 0]$ otherwise. The latter interval imposes no old incidence when no old division is used. All slope upper bounds follow from (17.18), so the compact root omits no unbounded range.

Write the physical intervals as $[a_-, a_+]$, $[b_-, b_+]$, $[l_-, l_+]$, $[m_-, m_+]$, $[q_-, q_+]$, $[o_-, o_+]$. In precisely this order replace

$$\begin{aligned} m_- &\leftarrow \max\{m_-, 3l_- - 2/5 - \varepsilon\}, \\ l_+ &\leftarrow \min\{l_+, (m_+ + 2/5 + \varepsilon)/3\}, \\ h_1 &= \max\{h_1^*, h_0 + a_-(u - R) - \varepsilon\}, \\ a_+ &\leftarrow \min\{a_+, (l_+ + \varepsilon - h_0)/(r - R), (m_+ + \varepsilon - h_0)/(T - R)\}, \\ b_+ &\leftarrow \min\{b_+, (l_+ + \varepsilon - h_1)/(r - u), (m_+ + \varepsilon - h_1)/(T - u)\}, \\ b_- &\leftarrow \max\{b_-, h_1/u\}. \end{aligned}$$

Discard a box only if an interval reverses or $h_1 r/u > l_+ + \varepsilon$. These are all consequences of (17.18) and the prime secant inequality. For a surviving box put

$$d_+ = (m_+ + \varepsilon - l_-)/(T - r), \quad d_- = 25(l_- - 1/5), \quad e_+ = 1 + 1000(T - m_-)^7,$$

and take

$$\begin{aligned} F(t) &= \max\{0, l_- + d(t - r), m_- + e_+(t - T)\}, & d &= d_+ \ (t < r), \quad d = d_- \ (t \geq r), \\ H_0(t) &= (h_0 + a(t - R))_+, & a &= a_+ \ (t < R), \quad a = a_- \ (t \geq R), \\ H_1(t) &= (h_1 + b(t - u))_+, & b &= b_+ \ (t < u), \quad b = b_- \ (t \geq u). \end{aligned}$$

Each is a lower bound for the same actual profile. In a divided new bound use threshold $H_1 - q_+ F$; in its retained parameter use q_- . The old $Q5$ bounds use $H_0 - o_+ F$ and retained parameter o_- . The proof of these endpoint choices, including endpoints zero and one, was given in the incident-source lemma.

Let $I_B(v)$ be seven times an upper integral for the full minimum in this physical box. The full original filtration gives $V(v) \geq 1 - I_B(v)$. On the other hand its prime mass is at most $(1 - F(u))^7$, $(1 - l_-)^7$, or $(1 - m_-)^7$, at u, r, T , respectively. Thus the box is impossible as soon as

$$(17.20) \quad \mathcal{S}(B) = \min\{I_B(u) + (1 - F(u))^7, I_B(r) + (1 - l_-)^7, I_B(T) + (1 - m_-)^7\} \leq 9999/10000.$$

Step 4. In this step, we specify the exact polynomial integration rule for each physical box. The integral rule contains only finitely many rational polynomial comparisons. Start with knots $0, R, 99/100, u, r, T$. On each interval split at all zeros of the two earlier lines, the two prime lines, and their difference. These determine the affine branches of H_0, H_1, F . Put $D = t - F$, $R_1 = H_1 - q_+ F$, and $R_0 = H_0 - o_+ F$. For every included degree-height pair (Z, Y) , split at $Z = 0, Y = 0$ and $Y = cZ$ for

$$c \in \{1/7, 1/6, 1/5, 1/4, 1/3, 1/2, 1, 2/11, c_-, c_+\}.$$

The last two numbers are the exact Q -crossing isolators. For each retained bound also split at all the comparisons among $D, F, H, 0$ and all zeros of $A_j + B_j E$, for $E = F, D, H$, in (17.10).

These splits include every positive-part switch of S_5 . On each resulting interval bisect once at its rational midpoint and choose, on each half, the smallest exact integral of the legally available individual polynomials. Summing these minima through a horizon defines I_B . The integral of the pointwise minimum is at most each such minimum of integrals, so the rule is an upper bound on the entire interval, without sampled interpolation.

Step 5. In this step, we verify the normalized-coordinate cover and its physical enclosures. The trees for NQ, NE bisect the physical domain (17.19). The other trees use four normalized coordinates $(A, B, C, D) \in [0, 1]^4$ and the same incidence intervals. For completeness their conversion is

$$\begin{aligned}\lambda &= 7/30 + C(L_* - 7/30), \\ m_0(\lambda) &= \max\{3003/10000, 3\lambda - 2/5 - \varepsilon\}, \\ \Lambda &= m_0(\lambda) + D(157/250 - m_0(\lambda)), \\ a_0 &= h_0/R, \quad a = a_0 + A(a^* - a_0), \\ a^* &= \min\left\{\frac{\lambda + \varepsilon - h_0}{r - R}, \frac{\Lambda + \varepsilon - h_0}{T - R}, \frac{u(\lambda + \varepsilon)/r - h_0 + \varepsilon}{u - R}, \frac{u(\Lambda + \varepsilon)/T - h_0 + \varepsilon}{u - R}\right\}, \\ h(a) &= \max\{h_1^*, h_0 + a(u - R) - \varepsilon\}, \\ b^* &= \min\{(\lambda + \varepsilon - h(a))/(r - u), (\Lambda + \varepsilon - h(a))/(T - u)\}, \\ b &= h(a)/u + B(b^* - h(a)/u).\end{aligned}$$

Every actual tuple is represented: the two extra a^* inequalities are just feasibility of $b \geq h(a)/u$. Zero-width intervals are represented by coordinate zero. To enclose a normalized box physically, evaluate λ at the two C -endpoints, Λ at the corresponding (λ, D) -endpoints, and a at the corresponding (A, λ, Λ) -endpoints. All these maps increase in each argument. Put $h_- = h(a_-)$, $h_+ = h(a_+)$ and define

$$b(h, l, m, s) = (1 - s)h/u + s \min\{(l + \varepsilon - h)/(r - u), (m + \varepsilon - h)/(T - u)\}.$$

For the lower b , take the greater of h_-/u and the minimum of this expression at l_-, m_- , $h = h_-, h_+$, $s = B_-, B_+$. For the upper b , use l_+, m_+ and the same endpoints, also testing $h = (3l_+ - m_+)/2 + \varepsilon$ when it lies in $[h_-, h_+]$. The function increases in l, m , is affine in s , and is concave piecewise affine in h , with exactly that possible kink. These tests therefore enclose every actual b . Copy the incidence intervals and apply the physical reductions above.

Step 6. In this step, we verify all terminal records and exclude persistence of the smooth prime. A tree node `[i, left, right]` bisects coordinate i at its midpoint, using coordinates 0, 1, 2, 3, 4, 5 in the displayed order. An **E** leaf asserts the explicit empty test; an integer leaf N asserts $\mathcal{S}(B) \leq N/10^7$. The supplement gives the complete trees and evaluated cell witnesses, with the physical and normalized versions distinguished. Their totals are

family	nonempty	empty	cells	max N
NE	344	9	3907	9998993
M	898	0	9642	9998871
$Q4$	1331	0	12422	9998983
NQ	5	0	57	9993048
QT	372	0	3771	9998920
$Q5$	479	0	5683	9998945
QN	33	0	361	9997727

Every numerator is at most 9999000.

The rational dictionary $z_0, z_1, \dots$ in each family is local to that family. A leaf lists its twelve reduced physical endpoints, height h_1 , horizon 26, 27, 29, and the corresponding prime mass $F(u)$, l_-, m_- . Each record `end:polynomial:numerator` gives the next cell, starting at zero or the preceding right endpoint, with bound over 10^9 . The sum includes every cell, and the final test adds

the stated seventh-power mass. Prefixes 0,1,D0, D1 mean $(t, H_0), (t, H_1), (t - F, R_0), (t - F, R_1)$. The generic codimension is five for the old rows except QT , where it is four; it is three on NE, M . The **span** polynomial is S_5 on the old $Q5$ row and $E_1 + E_2 - E_3$ on a singular new fivefold. The joint labels use the four retained terms already specified, with their appropriate lower incidence. Every remaining polynomial is defined in the common arithmetic convention.

Certificate data.

17-ambient-endpoints/persistent-fourfold. They use the domain, functions and margin specified above, with the verification rule of Lemma 17.11.

Certificate data.

17-ambient-endpoints/persistent-small-surface. They use the domain, functions and margin specified above, with the verification rule of Lemma 17.11.

Certificate data.

17-ambient-endpoints/persistent-fivefold. They use the domain, functions and margin specified above, with the verification rule of Lemma 17.11.

The trees exhaust the roots, including incidence limits at one. Every actual tuple lies in a nonempty leaf and contradicts its full moving-volume ceiling by (17.20). The full point filtration used all incoming layers from zero, with uniform total error $O(q^6)$. The full source alone received Riemann–Roch; every fixed companion and every actual division debit was retained. Thus persistence of the same smooth prime is impossible under nongeneration. $\square$

Theorem 17.37. *Let $L^7 = 1$. There is a cutoff J , fixed before the degree, threshold and center, such that for every $k \geq J$ with $25 \mid k$ and $U_k(27/25) \neq 0$, a pure actual threshold with divisorial minimal center smooth at x implies generation of $K_X + 8L$ at x . Every initial history is allowed, with no hypothesis on any later center or on the original degree of the divisor.*

Proof. We first earn the original source at T from the given smooth prime, and then pay each of its own threshold outcomes.

Step 1. In this step, we obtain the original T -source before selecting its own threshold. Assume nongeneration and let P be the given actual smooth r -divisor, observed at its sufficiently late original degree divisible by 25. By Lemma 17.34, the complete original sources have volume greater than $1/1000$ through T , and dimension at least $q^7/(2000 \cdot 7!)$ at every sufficiently large integer q . Fix its new uniform cutoff before choosing an actual pure T -threshold at a degree divisible by 25. This is its own nonzero complete source and its own entire boundary.

Step 2. In this step, we exclude divisorial outcomes and the double-fivefold outcome. If its minimal center is a divisor, Lemma 17.35 applies unless that divisor is P . In the latter case, Lemma 17.36 applies. Both contradict nongeneration. The classification in Lemma 17.7 pays the double-fivefold case by Theorem 16.2 at this later boundary's own coefficient. The only other possibility is a fivefold W of embedding dimension seven and multiplicity three, four, or five, with its full reduced Cohen–Macaulay tangent ring.

Step 3. In this step, we exclude the remaining embedding-seven fivefold by its full outgoing comparison. For completeness we give its outgoing comparison. Use $\epsilon = 1/10000$, fixed before the observations, and put $\lambda = \phi_P(r) > 7/30$. If w computes this own T -threshold, put $g = \phi_w$, $\eta = g(T)$. The two actual comparisons and the smooth-prime ceiling give

$$g(r) \leq \lambda + \epsilon, \quad \eta \geq \phi_P(T) - \epsilon > 3\lambda - 2/5 - \epsilon.$$

Choose finite supporting slopes d, c for ϕ_P, g at r, T . They satisfy

$$d > 25(\lambda - 1/5), \quad c \geq (25/2)(\eta - \lambda - \epsilon).$$

For $t \geq T$, put $F(t) = \lambda + d(t - r)$, $H(t) = \eta + c(t - T)$, and

$$A_0 = T - \lambda - d(T - r) - 2\eta < 13/50 + 2\epsilon < 27/100, \quad B_0 = d + 2c - 1 > 3/2 - 50\epsilon = 299/200.$$

The first bound is $T + 6/5 - 9\lambda + 2\epsilon < 13/50 + 2\epsilon$; the second follows from $\eta > 3/10 - \epsilon$. Since W has embedding dimension seven, Lemma 17.3 gives $w(P) = 0$. Divide every actual outgoing

point layer by its forced P -power. The full spanning capacity of this height-two, embedding-seven center is at most $2(D - 2qH(t))_+^6/6! + O(q^5)$, by Theorem 7.18. The exact debit is zero. Sum the entire outgoing filtration to a fixed rational order beyond every possible original point order. Its final space is zero, and the accumulated error is $O(q^6)$. Passing to the complete volume limit gives

$$V(T) \leq 14 \int_T^\infty [A_0 - B_0(t - T)]_+^6 dt = \frac{2(A_0)_+^7}{B_0} < 2(27/100)^7 < 1/1000.$$

The last comparison is $2 \cdot 27^7 = 20920706406 < 10^{11}$. This contradicts the earned source floor. The comparison used every original condition on the divided image and introduced no later threshold or transferred boundary price. All possible own T -centers have now been paid, proving generation. $\square$

17.5. Outgoing profiles and the remaining center exclusions.

Proposition 17.38. *Suppose $L^7 = 1$ and $\pi^*L - TE_x$ is big. Fix the constant C in (17.1) on a compact interval $[0, b_0]$ in the big range, with $b_0 > T$, and put $\epsilon = 1/10000$. Let W be an integral fivefold through x of embedding dimension seven, and let w be a divisorial valuation centered on W , with $\phi_w(r) > 321/1000$. Let q exceed the fixed comparison cutoff, with $25 \mid q$ and $C/q \leq \epsilon$, and suppose a pure actual threshold of $U_q(T)$ has divisorial minimal center Q , smooth at x . If $\phi_Q(u) < 1/5$, then*

$$V(T) < \frac{11}{1000}.$$

The estimate uses the unrestricted two-normal capacity and so imposes no reducedness hypothesis on the fivefold tangent.

Proof. We compare the two actual profiles, divide the original layers, and bound the resulting tail for every supporting slope. Put

$$h = \frac{321}{1000}, \quad a_5 = \frac{h}{r} = \frac{107}{360}.$$

Choose a finite supporting slope a of ϕ_w at r . Convexity and $\phi_w(0) = 0$ give $a \geq a_5$ and

$$\phi_w(t) \geq G(t) := h + a(t - r) \quad (t \geq T).$$

These inequalities apply on every attained original order. Indeed, powers and products of original sections give convexity on the whole finite domain of the infimum profile. A supporting line at an interior big anchor is consequently a lower bound throughout that domain.

Step 1. In this step, we bound the later prime profile using its own actual threshold. Set

$$H_T(a) = h + a(T - r), \quad \lambda_0(a) = H_T(a) - \epsilon, \quad d_0(a) = \frac{\lambda_0(a) - 1/5}{T - u}.$$

Lemma 17.2, applied to the actual Q -threshold and to w , gives $\phi_Q(T) \geq \phi_w(T) - \epsilon \geq \lambda_0(a)$. Every supporting slope of ϕ_Q at T is at least its secant slope from u . Hence

$$\phi_Q(t) \geq F(t) := \lambda_0(a) + d_0(a)(t - T) \quad (t \geq T).$$

Both $\lambda_0(a)$ and $d_0(a)$ are positive already at $a = a_5$, and increase with a .

Step 2. In this step, we derive the full fivefold outgoing comparison with zero division debit. By Lemma 17.3, $w(Q) = 0$. For $H > 0$, the full ideal $J(H)$ in Lemma 17.6 is primary to I_W : a factor outside I_W has value zero, and a high enough power of I_W has value at least $A_X(w)H$. Its threshold at the generic point of W is at most $1/H$, as witnessed by w with its original discrepancy. Theorem 7.9 therefore gives (17.5) with $\nu = \kappa = 1$ and $\gamma = 4/21$. Where $t - F(t) > 0$,

$$\frac{G(t)}{t - F(t)} \geq \frac{G(t)}{t} \geq \frac{h}{r} = \frac{107}{360} > \frac{4}{21}.$$

The strict gap also handles the integer floor in the division. Lemma 17.6 yields

$$V(T) \leq 7 \int_T^{R_*} (t - F(t) - G(t))_+^6 dt.$$

Step 3. In this step, we bound the affine outgoing tail uniformly over every supporting slope.

$$A_5(a) = T - 2H_T(a) + \epsilon, \quad B_5(a) = d_0(a) + a - 1.$$

Then $t - F(t) - G(t) = A_5(a) - B_5(a)(t - T)$. At $a = a_5$,

$$A_5(a_5) = \frac{42349}{90000}, \quad B_5(a_5) = \frac{5431}{10800} > \frac{1}{2} > 0.$$

The numerator $A_5(a)$ decreases and the denominator $B_5(a)$ increases with a . Extending this nonnegative affine integral to infinity gives

[illegible]

The fraction is obtained by substituting the displayed rational values and taking the seventh power. Its strict upper bound is the positive-integer comparison after multiplication by the denominator. This treats every finite supporting slope $a \geq a_5$. $\square$

For the fourfold comparison, define

$$D_4(c) = 6(1 - 2c)_+^6 - 8(1 - 3c)_+^6 + 3(1 - 4c)_+^6 \quad (c \geq 0).$$

This is the probability that at least two of four prescribed coordinates of the uniform seven-coordinate simplex are at least c . Inclusion-exclusion gives the three coefficients 6, -8 , 3. In particular D_4 is nonnegative and nonincreasing, and $D_4(c) \leq 6(1 - 2c)_+^6$.

Proposition 17.39. *Suppose $L^7 = 1$, $V(T) > 1/64$, and $W \subset X$ is an integral fourfold through x , of embedding dimension seven and with full reduced Cohen–Macaulay ordinary tangent ring. Then there cannot exist a divisorial valuation w centered on W and a prime divisor Q through x , smooth there, such that*

$$\phi_w(r) > \frac{17}{80}, \quad \phi_Q(T) > \frac{3}{10}, \quad \phi_Q(u) < \frac{1}{5}.$$

Proof. We first estimate the outgoing tail and then use the incoming layers to handle the smaller supporting slopes. The full-image estimate is Theorem 7.18, applied to the entire reduced Cohen–Macaulay tangent of W . Put

$$h = \frac{17}{80}, \quad a_4 = \frac{h}{r} = \frac{85}{432}, \quad C_4 = \frac{138259}{900000}.$$

Choose a finite supporting slope $a \geq a_4$ of ϕ_w at r . The lower profile is $G(t) = (h + a(t - r))_+$. The secant of ϕ_Q from u to T has slope greater than $5/6$, so on the outgoing interval we can use

$$F(t) = \frac{3}{10} + \frac{5}{6}(t - T).$$

The noncontainment in Lemma 17.3 gives $w(Q) = 0$. Theorem 7.18 and $D_4(c) \leq 6(1 - 2c)_+^6$ give (17.5) with $\nu = 6, \kappa = 2, \gamma = 0$. Thus

$$V(T) \leq 42 \int_T^{R_*} (t - F(t) - 2G(t))_+^6 dt.$$

Step 1. In this step, we bound the outgoing tail and exclude supporting slopes at least one half.

$$A_4(a) = T - \frac{3}{10} - 2(h + a(T - r)), \quad B_4(a) = \frac{5}{6} + 2a - 1.$$

Here $B_4(a_4) = 49/216 > 0$. Integration of the positive affine part gives

$$V(T) \leq U_4(a) := \frac{6(A_4(a))_+^7}{B_4(a)}.$$

This function is nonincreasing on $[a_4, \infty)$, since its numerator is nonincreasing and its positive denominator increases. At the two values needed below,

$$\begin{aligned} U_4(a_4) &= \frac{233238699780054051534859}{5062294389600000000000000} < \frac{1}{20}, \\ U_4(1/2) &= \frac{81856081425519}{1600000000000000} < \frac{1}{100} < \frac{1}{64}. \end{aligned}$$

The second inequality contradicts the volume hypothesis whenever $a \geq 1/2$.

Step 2. In this step, we exclude smaller supporting slopes by combining incoming and outgoing volume bounds. For $a_4 \leq a \leq 1/2$ and $0 \leq t \leq r$,

$$G(t) \geq \frac{1}{2}(t - o)_+, \quad o = r - 2h = \frac{131}{200}.$$

Put $t_0 = 3o/2 = 393/400$. On $[t_0, r]$, this lower profile divided by t is at least $1/6$. On $[r, T]$, $G(t)/t \geq h/r = 85/432 > 1/6$. The height-three clause of Proposition 7.19 therefore bounds every full original layer on $[t_0, T]$ by capacity C_4 , up to its uniform error.

On $[0, t_0]$, use Theorem 7.18 with the lower profile $(t - o)_+/2$. Its normalized density is $7t^6$ on $[0, o]$, and on $[o, t_0]$ it is

$$7[6o^6 - 8(3o/2 - t/2)^6 + 3(2o - t)^6].$$

The three bases are nonnegative throughout the latter interval. Direct integration gives

$$\begin{aligned} &7 \int_0^{t_0} t^6 D_4\left(\frac{(t - o)_+}{2t}\right) dt \\ &= o^7 + 42(t_0 - o)o^6 - 16(o^7 - (3o/4)^7) + 3(o^7 - (o/2)^7) = \frac{11379}{1024}o^7. \end{aligned}$$

At $t = 0$ the density is interpreted as zero. Summing the entire incoming filtration and using the complete original leading term $h^0(X, pL) = p^7/7! + O(p^6)$ now yields

$$V(T) \geq 1 - L_4, \quad L_4 = \frac{11379}{1024} \left(\frac{131}{200}\right)^7 + C_4 \left[\left(\frac{29}{25}\right)^7 - \left(\frac{393}{400}\right)^7 \right].$$

Each local error has order $O(p^5)$; all $O(p)$ layers give $O(p^6)$. The ceiling at the final order has the same lower-order effect. The capacity switch is at the fixed rational point t_0 , so no limiting interval is omitted.

The exact rational values are

$$\begin{aligned} L_4 &= \frac{1287547018757097830604473}{1474560000000000000000000} < \frac{9}{10}, \\ L_4 + U_4(a_4) &= \frac{953037014798324835455360629389539}{10367578909900800000000000000000} < \frac{19}{20}. \end{aligned}$$

They follow by substitution and the binomial antiderivative used above; each comparison is between integers after multiplying by the positive denominator. The incoming estimate and the outgoing estimate would imply $1 \leq L_4 + U_4(a) \leq L_4 + U_4(a_4) < 19/20$. This excludes the whole remaining closed range $[a_4, 1/2]$. Together the two ranges exhaust every supporting slope. $\square$

Theorem 17.40. *Let $L^7 = 1$. There is a cutoff J , fixed before the original degree, tuple, threshold, center and place, such that for every $k \geq J$ with $25 \mid k$ and $U_k(27/25) \neq 0$, its pure actual threshold gives generation at x whenever its minimal center has dimension four or five and embedding dimension seven. Every earlier history and full tangent scheme is included.*

Proof. Suppose that generation fails. We obtain an own later smooth prime and then compare it with each original embedding-seven center.

Step 1. In this step, we obtain a later smooth prime after fixing the required source cutoffs. Enlarge the degree cutoff to meet the cutoffs of Theorems 17.17 and 17.18, and of Lemmas 17.8 and 17.15. These choices precede the original threshold and its computing place. Theorem 17.17 gives $V(T) > 1/64$ and supplies all sufficiently late complete original sources at T , including every degree divisible by 25. Choose one such degree, independently of the original degree, and take its own pure actual full-ideal threshold. Theorem 17.18 pays every outcome except a smooth prime Q . For this remaining outcome, Theorem 17.17 gives

$$Q \cdot L^6 = 1, \quad \phi_Q(T) > \frac{3}{10},$$

whereas Lemma 17.15 gives $\phi_Q(u) < 1/5$. The threshold comparison is applied separately to each original source, so its coefficient is never transported from one degree to the other.

Step 2. In this step, we exclude both original embedding-seven dimensions using their outgoing comparisons. For a fivefold observed in the original degree k , which is sufficiently late and divisible by 25, Proposition 14.2 gives $1/s \geq 4321/13425$; comparison with its original computing profile gives $\phi_w(r) > 321/1000$ after a fixed enlargement of the cutoff. The gap is

$$\frac{4321}{13425} - \frac{321}{1000} = \frac{463}{537000} > \frac{1}{100000}.$$

Proposition 17.38 would therefore give $V(T) < 11/1000 < 1/64$, a contradiction. For a fourfold, Lemma 17.8 gives the full reduced Cohen–Macaulay tangent ring and $\phi_w(r) > 17/80$, with a uniform strict margin after the same cutoff enlargement. Proposition 17.39 then contradicts the displayed reserve and the two smooth-prime profile bounds. Both alternatives retain the complete old boundary and the complete new source throughout. They exhaust the stated embedding-dimension-seven histories. $\square$

Proposition 17.41. *Suppose $L^7 = 1$ and $K_X + 8L$ is not generated at x . There is an integer J , fixed before all degrees, averages, centers and computing places, such that for every integer $i \geq J$ the complete original source*

$$U_{20i}(9/10) = H^0(X, 20iL \otimes \mathfrak{m}_x^{18i})$$

is nonzero. Every pure actual full-ideal threshold average of this source has minimal-center dimension 2, 3, or 4 and actual deficit greater than $4/25$. The complete coefficient and every fixed component of that same average are retained. The dimension-five exclusion is a conclusion with its own later-source proof, without any initial-fourfold history, generic-primary or selected-center assumption.

Proof. Lemma 17.1 supplies every sufficiently late original initial source and, for each of its actual pure thresholds, the strict deficit $b > 4/25$ and possible dimensions 2, 3, 4, 5. Its cutoff precedes the initial degree, tuple, center and computing place. Suppose one such center were a fivefold. Lemma 17.20 earns $V(29/25) > 1/500$ and all sufficiently late original sources there, before any later threshold is selected. Choose an own later source in a degree divisible by 25, beyond all the fixed cutoffs of the initial-fivefold lemmas.

Lemma 17.21 excludes its singular divisorial and embedding-dimension-seven outcomes, and Lemma 17.26 excludes its smooth divisorial outcome. The exhaustive own later ledger therefore leaves a double fivefold. If its own full boundary is $\Gamma_T \sim_{\mathbb{Q}} s_T L$ with deficit b_T , its actual point order gives

$$(29/25)s_T + b_T \leq 7.$$

Theorem 16.2 applies to this entire boundary at its own price, with no degree or tangent hypothesis, and gives generation. This contradiction excludes dimension five. No earlier boundary was added to or refunded by the later trial, and the surviving initial dimensions and deficit bound are exactly those asserted. $\square$

18. INTRINSIC DOUBLE-FIVEFOLD SOURCES

We return to the 27/25 price and first obtain additive continuations from complete intrinsic sections on a double fivefold, retaining the whole incoming boundary. We then compare two actual original-source profiles to prove generation when one of the observations has an initial fourfold center. The complete rational certificate is supplied with the paper; its checking rule and geometric application are given below.

Lemma 18.1. *Let X be a smooth complex projective variety, let L be ample Cartier, and let B be an effective rational boundary lc near x , with positive-dimensional minimal center W . Put $A = L|_W$ and $b = \text{def}_x(B)$. Fix a positive rational γ . Above a cutoff depending on the specified ideal I_W , every nonzero complete space $F_p(\gamma)$ has a full ambient inverse image whose sufficiently dilute general average $H \sim_{\mathbb{Q}} L$ has a positive rational threshold t from B . The pair $(X, B + tH)$ has a properly smaller minimal center and deficit y satisfying*

$$y \leq b - \gamma t.$$

Every prescribed upper bound for the coefficients of the nonfixed prime components of $H|_W$ can be imposed before this threshold. A nonzero section at any positive degree can be powered to reach the cutoff and any prescribed divisibility.

This uses ordinary restriction to the original projective center. The cutoff is separate from any estimate supplying intrinsic sections.

Proof. We obtain the full ordinary lift and then take its own threshold against the entire old boundary.

Step 1. In this step, we construct the full ambient inverse image with the prescribed dilution. Apply [Theorem 6.8](#), in the form of [[Sta26](#), Lemma 30.17.1, Tag 0B5U], to the fixed coherent ideal I_W on X . For all p above its cutoff the sequence

$$0 \longrightarrow I_W \otimes \mathcal{O}_X(pL) \longrightarrow \mathcal{O}_X(pL) \longrightarrow \mathcal{O}_W(pA) \longrightarrow 0$$

is surjective on global sections at the right. Take the full inverse image of $F_p(\gamma)$. On a common principalization with B , a sufficiently large general tuple realizes its base ideal at the threshold and avoids the strata not forced by that ideal. Increasing the size of the tuple makes every nonfixed coefficient in its normalized restriction as small as prescribed. Its fixed divisor is retained in full.

Step 2. In this step, we obtain the smaller minimal center and the restricted deficit drop. The average does not contain W . Every old lc center through x contains W , by [Theorem 3.11](#); hence every old lc place has zero order on H . On the fixed principalization this gives positivity of the threshold. The restriction vanishes at x , so every lifted member vanishes there; the ordinary point blowup gives finiteness. Rationality follows on the same principalization. An old place dominating W still has zero discrepancy, whereas a new threshold place of positive H -order cannot dominate W . Intersecting its center with W gives a properly smaller minimal center through x .

Apply [Theorem 3.8](#), namely [[YZ20](#), Definition 2.5 and Lemma 2.10], on the smooth klt background $(X, 0)$, with the two actual additions B, tH . The restricted order is at least γt ; [Lemma 3.13](#) identifies the deficit convention. This proves the inequality without changing B . Finally, a nonzero section of degree p has a nonzero power of degree jp , since W is integral, and $j[p\gamma] \geq [jp\gamma]$. $\square$

For the scalar calculations below put

$$r_0 = \frac{25}{27}, \quad B_* = \frac{18}{5}, \quad \lambda = \frac{1731}{2000}, \quad \theta = \frac{21}{20}, \quad \eta = \frac{999}{1000}.$$

A double-fivefold minimal center is normal Cohen–Macaulay by [Theorem 4.13](#). A linear parameter reduction has length two, so its embedding dimension is six. Thus its analytic ring is an order-two

hypersurface in six variables. Its first Artinian moment is one, and [Theorem 4.16](#) gives $\bar{e}_1 \geq 1$ and $0 < b \leq 4$.

Write $\omega = (d_\omega, e_\omega, m_\omega)$ for a descendant state, $T_\omega = T(e_\omega - d_\omega, m_\omega)$, and $\beta_\omega = d_\omega - 2T_\omega/m_\omega$. Use degree floor one and replace each radius by the rational upper radius q_ω in the first table below. Let $\hat{\mathcal{F}}_\omega$ denote the resulting envelope of [Definition 5.2](#), and define

$$J_\omega(h, C) = \max_{0 \leq y \leq C} (\hat{\mathcal{F}}_\omega(y) - hy), \quad 0 \leq C \leq \beta_\omega.$$

For every smaller-dimensional state ω' of no larger embedding dimension, set

$$c_{\omega\omega'} = \begin{cases} \min\{\beta_{\omega'}, (d_\omega m_\omega - 2T_\omega - m_{\omega'})/m_\omega\}, & d_{\omega'} = d_\omega - 1, \\ \beta_{\omega'}, & d_{\omega'} \leq d_\omega - 2. \end{cases}$$

Only nonpositive caps are discarded. The first-child cap is

$$C_\omega = \begin{cases} \min\{\beta_\omega, (8 - m_\omega)/2\}, & d_\omega = 4, \\ \beta_\omega, & d_\omega \leq 3. \end{cases}$$

These caps follow from [Proposition 4.18](#); all old discriminant and compatible moduli contributions have the sign required to retain them before weakening the upper bound. The point alternative has zero tail and needs no state in the tables.

Lemma 18.2. *For the states, radii and caps just specified, and every $h > 0$,*

$$(18.1) \quad J_\omega(h, C) = (q_\omega - h)_+ C + \max \left(0, \max_{\omega'} J_{\omega'}(\max(h, q_\omega), \min(C, c_{\omega\omega'})) \right).$$

The two tables below give all first-child states and their indicated exact values. Every state other than $(4, 6, 3)$ satisfies

$$J_\omega(1/\eta, C_\omega) \leq \frac{1209}{1000}.$$

The full envelopes of the states $(4, 6, 4)$ and $(4, 6, 5)$ satisfy, respectively for $0 \leq y \leq 2$ and $0 \leq y \leq 3/2$,

$$\hat{\mathcal{F}}_{(4,6,4)}(y) \leq \frac{283}{200}y + \frac{379}{1000}, \quad \hat{\mathcal{F}}_{(4,6,5)}(y) \leq \frac{187}{125}y + \frac{741}{2500}.$$

The first table includes the entire descendant calculation, with dimension skips. In the second table the columns are

$$\begin{aligned} P_\lambda(\omega) &= (7 - B_*)r_0 + B_*/\lambda + J_\omega(1/\lambda, \min(B_*, C_\omega)), \\ P_\theta(\omega) &= 3r_0 + 4/\theta + J_\omega(1/\theta, C_\omega). \end{aligned}$$

For $(3, 6, 4)$, only the first column uses the additional cap $y \leq 1$. For $(4, 6, 3)$, only the second column is replaced by the affine price $3r_0 + 4/\theta + (1317/1000 - 1/\theta)(5/2)$. The tables are rendered from the canonical dataset `18-intrinsic-double/scalar`. Its file `states.tsv` records, in order, $d_\omega, e_\omega, m_\omega, T_\omega, C_\omega, q_\omega, J_\omega(50/49, C_\omega), P_\lambda(\omega)$, and $P_\theta(\omega)$. Each of the 28 states occurs once.

ω	T_ω	C_ω	q_ω	$J_\omega(50/49, C_\omega)$
(1, 1, 1)	0	1	1	0
(2, 2, 1)	0	2	1	0
(2, 3, 2)	1	1	283/200	3867/9800
(2, 4, 3)	2	2/3	1733/1000	11639/24500
(2, 5, 4)	3	1/2	2	24/49
(2, 6, 5)	4	2/5	2237/1000	59613/122500
(3, 3, 1)	0	3	1	3867/9800
(3, 4, 2)	1	2	63/50	70137/98000
(3, 4, 3)	3	1	1443/1000	20707/49000
(3, 5, 3)	2	5/3	1443/1000	8797/9800
(3, 5, 4)	4	1	397/250	118353/196000
(3, 5, 5)	6	3/5	171/100	10137/24500
(3, 5, 6)	8	1/3	909/500	19541/73500
(3, 6, 4)	3	3/2	397/250	12953/12250
(3, 6, 5)	5	1	171/100	4579/6125
(3, 6, 6)	7	2/3	909/500	19541/36750
(3, 6, 7)	9	3/7	1913/1000	131211/343000
(3, 6, 8)	11	1/4	2	12/49
(3, 6, 9)	13	1/9	2081/1000	17323/147000
(4, 4, 1)	0	7/2	1	70137/98000
(4, 5, 2)	1	3	119/100	317993/294000
(4, 5, 3)	3	2	1317/1000	91599/98000
(4, 5, 4)	6	1	283/200	13467/19600
(4, 6, 3)	2	5/2	1317/1000	384791/294000
(4, 6, 4)	4	2	283/200	57241/49000
(4, 6, 5)	6	3/2	187/125	123699/122500
(4, 6, 6)	8	1	783/500	49857/61250
(4, 6, 7)	11	1/2	1627/1000	10727/19600

ω	$P_\lambda(\omega)$	$P_\theta(\omega)$
(1, 1, 1)	113845/15579	418/63
(2, 2, 1)	113845/15579	421/63
(2, 3, 2)	23577857/3115800	88829/12600
(2, 4, 3)	59921969/7789500	24877/3500
(2, 5, 4)	120424/15579	64/9
(2, 6, 5)	301462723/38947500	1118431/157500
(3, 3, 1)	23577857/3115800	90029/12600
(3, 4, 2)	241577027/31158000	937319/126000
(3, 4, 3)	118325497/15579000	445909/63000
(3, 5, 3)	24864887/3115800	4559/600
(3, 5, 4)	484596763/62316000	1829311/252000
(3, 5, 5)	59514527/7789500	221819/31500
(3, 5, 6)	58642937/7789500	216589/31500
(3, 6, 4)	61896863/7789500	122011/15750
(3, 6, 5)	15423584/1947375	58298/7875
(3, 6, 6)	30181687/3894750	37613/5250
(3, 6, 7)	832322881/109053000	3086557/441000
(3, 6, 8)	234269/31158	863/126
(3, 6, 9)	115447211/15579000	181243/27000
(4, 4, 1)	241577027/31158000	946319/126000
(4, 5, 2)	248772401/31158000	991997/126000
(4, 5, 3)	248400629/31158000	964913/126000
(4, 5, 4)	48978457/6231600	185029/25200
(4, 6, 3)	257954687/31158000	188971/25200
(4, 6, 4)	127838011/15579000	497167/63000
(4, 6, 5)	316054729/38947500	1212613/157500
(4, 6, 6)	155528897/19473750	588209/78750
(4, 6, 7)	48527917/6231600	25807/3600

Proof. We derive the support recurrence, evaluate its complete price tables, and verify the state coverage.

Step 1. In this step, we derive the support recurrence with the curve correction retained. Expand the first deficit drop in [Definition 5.2](#). If $q_\omega \geq h$, the outer maximum occurs at C . If $q_\omega < h$, fixing the child deficit z makes the outer maximum occur at z ; the no-child contribution is zero. This proves (18.1), also when $h < 1$. The latter range retains the curve correction in the θ -column.

Step 2. In this step, we evaluate every table entry and the two full affine-tail bounds. Here are explicit rules for checking every table entry. A curve has $J(h, C) = (1 - h)_+ C$. A surface of multiplicity k has intrinsic cap $2/k$ and only curves below it, so

$$J(h, C) = (q_{2,k} - h)_+ C + (1 - \max(h, q_{2,k}))_+ \min(C, 1).$$

For $h \geq 1$, a threefold row specializes to

$$(q_\omega - h)_+ C + \max\left(0, \max_{1 \leq k \leq e_\omega - 1} (q_{2,k} - \max(h, q_\omega))_+ \min\left(C, \frac{2}{k}, \frac{3m_\omega - 2T_\omega - k}{m_\omega}\right)\right),$$

where nonpositive adjacent caps are omitted. A fourfold tests every threefold of no larger embedding dimension with its displayed adjacent cap, every direct surface with cap $2/k$, and every direct curve. For $h < 1$, use the same recurrence with the curve term retained. These are finite rational maxima; the table specifies all their arguments. Raising each q_ω to d_ω verifies $q_\omega^{d_\omega} \geq m_\omega$.

The first table at $h_0 = 50/49$ gives the following grouped maxima and upper caps:

states	$\max J_\omega(h_0, C_\omega)$	$\max C_\omega$
$d_\omega \leq 3$	12953/12250	3
$d_\omega = 4, e_\omega \leq 5$	317993/294000	7/2
$d_\omega = 4, e_\omega = 6, m_\omega \geq 4$	57241/49000	2

For $h_1 = 1000/999 > 1$,

$$\hat{\mathcal{F}}_\omega(y) - h_1 y \leq J_\omega(h_0, C_\omega) + (h_0 - 1)C_\omega.$$

The three resulting bounds are 13703/12250, 338993/294000, 1209/1000, respectively. The first two are smaller than the third. Substitution at $h = q_{(4,6,4)}$ and $h = q_{(4,6,5)}$ gives the two last assertions. The same explicit recurrence gives both columns of the second table.

Step 3. In this step, we verify the exhaustive state ranges and every degenerate cap convention. For completeness, at embeddings 3, 4, 5, 6 the positive threefold multiplicity ranges are, respectively,

$$1; \quad 2, 3; \quad 3, \dots, 6; \quad 4, \dots, 9.$$

At the next multiplicity $\beta \leq 0$, and T/m is nondecreasing thereafter. The fourfold ranges are 1 at embedding four, 2, 3, 4 at embedding five, and 3, ..., 7 at embedding six; multiplicity at least eight has nonpositive root cap. Surfaces have $m = e - 1$. For the embedding-six threefold rows of multiplicities eight and nine, the moments are 11, 13, giving caps $1/4, 1/9$. In the first row only a multiplicity-one surface has a positive adjacent cap, and all its radii are at most two; hence the full cost is at most $2y$. In the second row every surface cap is nonpositive, so only direct curves remain and the full cost is at most $(2081/1000)y \leq 9y/4$. Embedding-seven rows have the different caps $1/2, 1/3$ of Lemma 5.10; those rows, and multiplicities ten and eleven, remain in the ambient census. They cannot be descendants of this embedding-six parent.

These ranges can also be enumerated without a trial multiplicity cutoff. For $c = e_\omega - d_\omega > 0$, choose an integer $\ell_0 > d_\omega/2$. There are $h_{\text{mon}} = \binom{c+\ell_0-1}{\ell_0-1}$ monomials of degree less than ℓ_0 . Consequently $T(c, m) \geq \ell_0(m - h_{\text{mon}})$, and positive intrinsic cap implies

$$m < \frac{\ell_0 h_{\text{mon}}}{\ell_0 - d_\omega/2}.$$

Enumerate these integers, apply the intrinsic and first-child caps, and compare the resulting state set with the dataset. The recurrence then checks every listed correction and both price columns exactly. It includes the singleton domain $C = 0$, where every term is zero; a negative cap has no feasible child. Thus enumeration and equality of all recorded values establish precisely the scalar inequalities of the lemma. $\square$

Theorem 18.3. *For $n \geq 7$, let $B \sim_{\mathbb{Q}} sL$, $s \geq 0$, be effective rational, lc near x , with a double-fivefold minimal center W . Put $A = L|_W$, $g = A^5$, $b = \text{def}_x(B)$, and assume $(27/25)s + b \leq n$. If $H^0(W, pA \otimes I_x^{[21p/20]}) \neq 0$ for some integer $p \geq 1$, the entire B has an additive point continuation of coefficient less than $n + 1 - 1/400000$. This holds in every quadratic rank, and in particular when $g \geq 3$. If the actual quadratic rank is at least three, the same conclusion holds whenever $b \leq 18/5$ or $g \geq 2$. For the low-deficit branch, at every sufficiently large degree divisible by 2000, above separate intrinsic and ordinary restriction cutoffs, the complete original intrinsic sources are considered along*

$$1731/2000, 22/25, 9/10, 23/25, 47/50, 24/25, 49/50, 1, 21/20.$$

Each reached source either gives the asserted continuation at its own threshold or earns the next complete source at that same degree. Every reached source is nonzero before its own threshold is selected.

The first assertion consumes one actual intrinsic section. The last assertion supplies that section through successive alternatives at one degree, stopping as soon as an alternative pays.

Proof. We first pay a θ -witness. We then derive the fixed orders forced by failure at a lower level and use the full symbolic interval to earn the next source. The larger-degree branch uses a single interval with its own shift.

Step 1. In this step, we pay a nonzero θ -source in every quadratic rank. Power the section above the separate restriction cutoff and apply [Lemma 18.1](#). If its threshold already gives a point, its coefficient is at most $(n-4)r_0 + 4/\theta < n+1 - 1/400000$. Otherwise let ω be the actual first state. If $\omega \neq (4, 6, 3)$, the bounds $s \leq (n-b)r_0$, $b \leq 4$ and $y \leq C_\omega$, with $1/\theta > r_0$, give

$$s + t + \widehat{F}_\omega(y) \leq (n-b)r_0 + b/\theta + \widehat{F}_\omega(y) - y/\theta \leq (n-7)r_0 + P_\theta(\omega).$$

All these entries are less than $799/100$.

If $\omega = (4, 6, 3)$, then $y \leq 5/2$. Since $1317/1000 > 1/\theta > r_0$, the new whole affine price satisfies

$$\begin{aligned} s + t + \frac{1317}{1000}y &\leq (n-b)r_0 + \frac{b}{\theta} + \left(\frac{1317}{1000} - \frac{1}{\theta}\right)y \\ &\leq (n-7)r_0 + P_\theta(\omega) \\ &= (n-7)r_0 + \frac{188971}{25200} < n + \frac{51}{100}. \end{aligned}$$

Thus [Theorem 12.3](#) applies to the entire child boundary $B + tH$, whose coefficient is $s + t$. In the scalar branches reserve total weighted cutting error less than $1/1000$. At each of the at most four remaining cuts, after its actual radius r_i is known, choose its error $\epsilon_i < 1/(4000(1+r_i))$. Both the coefficient and deficit errors of [Theorem 3.14](#) then contribute less than the reserved amount, by [Theorem 5.4](#). Every branch has the claimed margin. This construction contains all of B .

On the original W , ample Hilbert asymptotics and the full point Hilbert polynomial give

$$(18.2) \quad \dim F_p(\gamma) \geq \frac{g-2\gamma^5}{120}p^5 - O_{W,A,x}(p^4).$$

The ceiling changes only the error term. This is the kernel bound for the map to the full point quotient, so it requires no jet surjectivity or global normality of W . At $g \geq 3$, $g - 2\theta^5 \geq 715899/1600000 > 0$. This proves the degree-three assertion in every rank.

Step 2. In this step, we force a full fixed prime and a symbolic shift at every unpaid lower source. Suppose the quadratic rank is at least three and $b \leq B_*$. At every sufficiently large 2000-divisible p , (18.2) supplies $F_p(\lambda)$ with normalized leading mass

$$M_0 = 1 - 2\lambda^5 = \frac{458771731542349}{1600000000000000} > 0.$$

All intrinsic and restriction cutoffs will be fixed before the degree or any threshold prime is selected. At a reached level $\gamma \in [\lambda, \theta]$, take its own threshold from the original B , giving $B + tH$ and deficit y . The first table of prices pays every first state except $(4, 6, m)$, $m = 3, 4, 5$, and the direct quartic threefold when $y > 1$. The bound improves with γ , since the first added cost is at most $(b-y)/\gamma$. For that quartic threefold $y \leq 3/2$. With $a = 5000/3997$, its entire affine price is at most

$$(n - B_*)r_0 + B_*/\lambda + (a - 1/\lambda)\frac{3}{2} = (n-7)r_0 + \frac{463961965}{62269263} < n + \frac{49267}{107919}.$$

The strict comparison at $n = 7$ has numerator $577 \cdot 804700 - 463961965 = 349935 > 0$, since $62269263 = 577 \cdot 107919$. Apply [Theorem 11.1](#) to this actual child with $y > 1$.

It remains to consider an unpaid fourfold prime P of multiplicity $m = 3, 4, 5$. Let c_P be its old direct discriminant coefficient, zero if absent, and let ξ be its coefficient in the entire $H|_W$. Retained adjunction and the Cartier-addition law in [Proposition 4.18](#) give

$$(18.3) \quad 2b + mc_P \leq 8, \quad c_P + t\xi = 1.$$

The second equality is at the actual prime that becomes an lc center. It also follows directly by shifting every generic threshold in [Theorem 4.1](#) by the coefficient of the restricted Cartier addition, with the same moduli data.

Set			$A_m = q_m - r_0, \quad C_m = \begin{cases} 2777/2700, & m = 3, \\ 8 - 1/500 - 7r_0 - p_m, & m = 4, 5. \end{cases}$
m	q_m	p_m	
3	1317/1000	affine endpoint	
4	283/200	379/1000	
5	187/125	741/2500	

Failure of the cubic affine endpoint, or of the complete scalar tail with total error less than $1/1000$, implies

$$(18.4) \quad 0 < (q_m \gamma - 1)t \leq A_m b - C_m.$$

Using $n = 7$ in C_m only weakens the inequality for larger n . All denominators are positive, and $C_m > (4 - m/2)A_m > 0$. Thus $b > 4 - m/2$. Combining (18.3) and (18.4) gives

$$\frac{m}{2} \xi \geq \frac{(b - 4 + m/2)(q_m \gamma - 1)}{A_m b - C_m} \geq \alpha_m (q_m \gamma - 1),$$

where

$$\alpha_3 = \frac{37125}{12803}, \quad \alpha_4 = \frac{1728}{673}, \quad \alpha_5 = \frac{14175}{5617}.$$

The derivative of the affine ratio has sign $(4 - m/2)A_m - C_m < 0$, so its minimum on the allowed range occurs at B_* . For $m = 4, 5$, comparison at both endpoints of $[\lambda, \theta]$ gives

$$(18.5) \quad \alpha_m (q_m \gamma - 1) \geq Z(\gamma), \quad Z(\gamma) = \frac{37125}{12803} \left(\frac{1317}{1000} \gamma - 1 \right).$$

These are affine inequalities with rational coefficients and positive denominators. Already at λ the bounds for ξ exceed $1/4$. Choose the averages before the threshold with nonfixed coefficients less than $1/10$. Then P belongs to the full fixed divisor and $\xi = N/p$, where N is its actual integer fixed order.

Step 3. In this step, we propagate the complete intrinsic sources at one original degree. Trivialize pA near x . The germs of all sections of $F_p(\gamma)$ lie in the actual symbolic power $P^{(N)}$. For a later level γ' , the point-quotient map injects

$$F_p(\gamma)/F_p(\gamma') \longrightarrow \frac{P^{(N)} \cap \mathfrak{m}_{W,x}^{p\gamma}}{P^{(N)} \cap \mathfrak{m}_{W,x}^{p\gamma'}}.$$

Its kernel is exactly the next complete original space. Apply Theorem 15.9 to the actual prime, without a parity or factoriality hypothesis. Equations (18.5) and the monotonicity in the shift of $\int_{\gamma'}^{\gamma'} 5(t - z)_+^4 dt$ give

$$(18.6) \quad \dim F_p(\gamma) - \dim F_p(\gamma') \leq \frac{2p^5}{120} \left((\gamma' - Z(\gamma))_+^5 - (\gamma - Z(\gamma))_+^5 \right) + C_1 p^4.$$

The constant is uniform in the selected prime and its full order. Here are the eight separate transitions and their lower shifts:

γ	γ'	$Z(\gamma)$
1731/2000	22/25	11868417/29264000
22/25	9/10	590139/1280300
9/10	23/25	550341/1024240
23/25	47/50	1571427/2560600
47/50	24/25	3534003/5121200
24/25	49/50	594/775
49/50	1	4316301/5121200
1	21/20	94149/102424

With these eight rows the exact remaining normalized mass is

$$M_0 - 2 \sum_{\text{rows}} \left((\gamma' - Z(\gamma))_+^5 - (\gamma - Z(\gamma))_+^5 \right) = \frac{541084214040859614275582771919703}{70434738071778049392640000000000000} > \frac{1}{1000}.$$

Each summand is nonnegative, so every partial subtraction retains at least this mass. Before choosing p , enlarge its intrinsic cutoff so that the initial Hilbert error and eight uniform interval errors together consume less than half this reserve; equivalently, if their constants are C_0, C_1 , require $120(C_0 + 8C_1)/p < 1/2000$. Also impose the separate ordinary restriction cutoff and $2000 \mid p$. At each reached source its own threshold either pays, or the corresponding row proves the next source nonzero at this same degree. If all eight trials are unpaid, the nonzero θ -source pays by the first part of the proof. No later source is asserted after an earlier successful trial.

Step 4. In this step, we obtain the paid witness when the original degree is at least two. Now allow $b \leq 4$ and assume $g \geq 2$. The η -source is nonzero at every sufficiently large allowed degree by (18.2), since $2 - 2\eta^5 > 0$. Its noncubic successors have total ideal price at most

$$(n-7)r_0 + 3r_0 + \frac{4000}{999} + \frac{1209}{1000} = (n-7)r_0 + \frac{7982791}{999000} = n + 1 - \left(\frac{9209}{999000} + \frac{2(n-7)}{27} \right).$$

After total weighted cutting error less than $1/1000$, the gap is greater than $821/99900 + 2(n-7)/27 > 1/400000$. An unpaid trial must therefore select the actual cubic fourfold. Repeating (18.3)–(18.4) with upper deficit four gives the full symbolic shift

$$Z_2 = \frac{6750}{2411} \left(\frac{1317}{1000} \eta - 1 \right) = \frac{8523441}{9644000}.$$

Dilution again makes this an actual fixed prime. The complete interval from η to θ leaves

$$2 - 2\eta^5 - 2 \left((\theta - Z_2)^5 - (\eta - Z_2)^5 \right) = \frac{1689744238069707817908429553}{17300505882747392000000000000000} > \frac{9}{1000}.$$

Both power arguments are positive. Fixing the uniform errors before the prime earns the complete θ -source at that same degree, which pays the original B . All successful branches are additive continuations of B ; separate trials have never been summed. Point evaluation, when needed, is Proposition 3.12. $\square$

Theorem 18.4. *In the 27/25 whole-price setting of Theorem 18.3, suppose the actual tangent is reduced and the analytic local ring is factorial. If $b \leq 18/5$, the entire boundary has an additive point continuation of coefficient less than $n + 1 - 1/400000$. Every positive original degree is allowed. A nonzero original $999/1000$ intrinsic source at any positive degree suffices without the low-deficit assumption.*

Factoriality excludes the cubic fourfold successor. In rank two, the remaining unpaid successors are controlled by the full principal interval, including both components of the quadratic tangent.

Proof. We first pay a nonzero η -source in every factorial rank. We then earn it from λ in the low-deficit rank-two case.

Step 1. In this step, we exclude a principal cubic fourfold and pay every η -source. An actual fourfold prime in the factorial analytic local ring is (f) . If its quotient has embedding dimension six, then $f \in \mathfrak{m}^2$. A four-parameter reduction of this Cohen–Macaulay complete intersection leaves a two-variable regular local ring modulo two equations of order at least two. Its first graded dimensions are at least 1, 2, 1: in degree two, two equations cannot span all three quadratic monomials. Its length is at least four. Thus the state $(4, 6, 3)$ is impossible; no assertion about the parity of other primes is needed.

Given a nonzero $F_p(\eta)$, use Lemma 18.1 and Lemma 18.2. Every successor is noncubic, so the entire ideal price is at most

$$(n-7)r_0 + \frac{7982791}{999000} = n + 1 - \left(\frac{9209}{999000} + \frac{2(n-7)}{27} \right).$$

The error reservation in the preceding proof gives a final gap greater than $821/99900 + 2(n-7)/27$. This proves the one-section assertion in every rank, with no low-deficit premise.

Step 2. In this step, we obtain the witness from a lower source in the low-deficit rank-two case. If the quadratic rank is at least three, the low-deficit assertion is already [Theorem 18.3](#). In rank two, [Theorem 15.1](#), specifically [\[GP26, Theorem 2.4\]](#), gives the convergent presentation

$$R = \mathbb{C}\{u, v, z_1, \dots, z_4\}/(uv - \psi), \quad D = \text{ord } \psi \geq 3.$$

Normality and the domain property exclude $\psi = 0$. The original projective W is unchanged.

Assume $b \leq B_*$. At a sufficiently large degree divisible by 2000, [\(18.2\)](#) supplies the complete λ -source with leading mass M_0 . Its own threshold either pays, including the quartic threefold affine alternative treated above, or selects a Cartier fourfold prime of embedding six and multiplicity $m = 4, 5$. The cubic case is excluded by the first paragraph. Equations [\(18.3\)–\(18.4\)](#) apply unchanged. After dilution the actual full fixed order N obeys

$$\frac{2N}{p} \geq \min \left\{ \frac{1728}{673} \left(\frac{283}{200} \lambda - 1 \right), \frac{4}{5} \frac{14175}{5617} \left(\frac{187}{125} \lambda - 1 \right) \right\} = \frac{2426571}{4206250} =: Z_*.$$

The other entry in this minimum is 41786199/70212500. The fixed coefficient exceeds 1/4, whereas every nonfixed coefficient was chosen below 1/10.

For the actual principal prime, normality gives $P^{(N)} = (f^N)$, and embedding six gives $\text{ord}_{\mathfrak{m}} f \geq 2$. Apply [Theorem 15.10](#) to the full principal interval. Both valuation shifts are at least $2N$, and their positive weights sum to two. Therefore the exact point-quotient map, with kernel $F_p(\eta)$, gives

$$\dim F_p(\lambda) - \dim F_p(\eta) \leq \frac{2p^5}{120} \left((\eta - Z_*)^5 - (\lambda - Z_*)^5 \right) + C_D p^4.$$

The theorem permits a zero-divisor initial form of f ; neither ordinary branch nor any opposite-character contribution has been deleted. The power arguments are positive, and the exact remaining mass is

$$M_0 - 2 \left((\eta - Z_*)^5 - (\lambda - Z_*)^5 \right) = \frac{188420987150292742295489880001}{32053856100156250000000000000000} > \frac{1}{250}.$$

The fixed analytic D controls the uniform interval error, independently of f, N or its endpoint valuations. Fix an intrinsic cutoff consuming less than 1/1000 of normalized mass, and the separate ordinary restriction cutoff, before choosing p or its prime. The complete original η -source is then nonzero at this same degree. Its own threshold from B pays by the first part of the proof. This successful alternative contains B ; the earlier trial is not added to its cost. $\square$

The datasets `18-intrinsic-double/transitions` and `18-intrinsic-double/identities` record the eight intrinsic intervals and the exact endpoint values used above. An interval record gives its index, left and right orders, shift, entire loss, and remaining mass. Checking its loss formula and subtracting the records in their stated order checks every reached source, including the final one. The factorial comparison uses only the lower bounds on its two full shifts and the fact that their positive weights sum to two. Its geometric proof therefore applies to every fixed analytic order $D \geq 3$, with its own uniform error constant.

The supplementary `18-intrinsic-double/rank-two` dataset also preserves the companion calculation with deficit cap three and terminal order 19/20. Its seven smooth-power rows represent the exceptional exponents $3 \leq \mu \leq 9$. The principal-prime calculation above excludes a Cartier embedding-six cubic even when the parent is not factorial. Thus every cubic considered here is non-Cartier. The local bounds in [Theorem 15.13](#) give $y \leq 2$ in the distinct-factor and singular-power cubic cases, and $y \leq 2 + 1/\mu$ for a smooth power. For every $\mu \geq 10$, this last bound is at most 21/10; the corresponding increasing affine price is bounded by

$$4r_0 + \frac{3}{\lambda} + \left(\frac{1317}{1000} - \frac{1}{\lambda} \right) \frac{21}{10} = \frac{1169868403}{155790000} < 7 + \frac{51}{100}.$$

For the remaining interval, the lower shift comparison follows from

$$\frac{\mu^2 - 1}{\mu(2\mu - 1)} \geq \frac{80}{153} \quad (3 \leq \mu \leq 9),$$

$$-7\mu^2 + 80\mu - 153 = (9 - \mu)(7\mu - 17) \geq 0.$$

This reduction covers every smooth-power cubic exponent. The companion data and the factorial calculation retain their different deficit caps and terminal orders.

Definition 18.5. Put

$$\begin{aligned} u_0 &= 9/10, & u_1 &= 27/25, & u_2 &= 29/25, \\ \varepsilon &= 10^{-6}, & h_0 &= 263/2000, & h_1 &= 27/85 - \varepsilon = 5399983/17000000. \end{aligned}$$

For $c \geq 0$, set $E_j(c) = (1 - jc)_+^6$ and

$$N_3(c) = \begin{cases} 1 - (21c/4)^3, & 0 \leq c \leq 1/7, \\ \min\{1 - ((17 + 7c)/24)^3, (7/6)^6 E_1(c)\}, & 1/7 < c < 1/2, \\ E_1(c), & 1/2 \leq c \leq 1, \\ 0, & c \geq 1, \end{cases}$$

$$\mathcal{N}(c) = \min\{N_3(c), 256/729 + 18(1/6 - c)_+\},$$

$$\mathcal{D}(c) = \begin{cases} E_1(c) + E_2(c) - E_3(c), & 0 \leq c < 4/21, \\ E_1(c), & c \geq 4/21. \end{cases}$$

Let

$$\mathcal{B}(t) = \begin{cases} 1, & 0 \leq t < u_0, \\ 61/100, & u_0 \leq t < 99/100, \\ 11/25, & 99/100 \leq t < 26/25, \\ 17/50, & 26/25 \leq t \leq u_2. \end{cases}$$

For $a_0 \geq h_0/u_0$, put $\eta(a_0) = \max\{h_1, h_0 + (u_1 - u_0)a_0 - \varepsilon\}$. For $a_1 \geq \eta(a_0)/u_1$, define

$$\begin{aligned} f_0(t) &= (h_0 + a_0(t - u_0))_+, & f_1(t) &= (\eta(a_0) + a_1(t - u_1))_+, \\ I(a_0, a_1) &= 7 \int_0^{u_2} t^6 \min\{\mathcal{B}(t), \mathcal{N}(f_0(t)/t), \mathcal{D}(f_1(t)/t)\} dt. \end{aligned}$$

The integrand at zero is defined to be zero.

Lemma 18.6. For every pair of slopes in Definition 18.5,

$$I(a_0, a_1) < \frac{899892923}{10^9} < \frac{9}{10}.$$

The complete certificate is the dataset **18-intrinsic-double/ambient**, consisting of 237 bounded rectangles and 19 old-profile intervals, together with the two unbounded ranges in the proof.

The dataset is in **supplement/18-intrinsic-double/data/**. The file **ambient-boxes.tsv** has fields $(i, a_{0,-}, a_{0,+}, a_{1,-}, a_{1,+}, N_i)$, for exactly $1 \leq i \leq 237$; **ambient-prefix.tsv** has fields $(i, a_{0,-}, a_{0,+}, N_i)$, for exactly $1 \leq i \leq 19$. Fields are separated by tabs; rational numbers are reduced fractions with positive denominator, or integers. Each index occurs once. The following proof specifies the exact tests on these records and why they imply the bound.

Proof. We bound each rectangle by lower affine profiles, integrate the available polynomial bounds exactly, and then cover the two unbounded slope ranges.

Step 1. In this step, we derive the exact rational upper integral for each slope rectangle. For a closed rectangle $[a_{0,-}, a_{0,+}] \times [a_{1,-}, a_{1,+}]$, put

$$\eta_- = \max\{h_1, h_0 + (u_1 - u_0)a_{0,-} - \varepsilon\}, \quad a'_{1,-} = \max\{a_{1,-}, \eta_-/u_1\}.$$

If $a_{1,+} < a'_{1,-}$, the rectangle has no feasible pair. Otherwise use the old line of slope $a_{0,+}$ before u_0 and slope $a_{0,-}$ after u_0 , both of height h_0 there. Use the new line of height η_- , with slope $a_{1,+}$ before u_1 and $a'_{1,-}$ after u_1 . The signs of $t - u_i$ show that these lines are lower bounds for the actual lines. Taking positive parts preserves this order. Both capacities are nonincreasing, so their use bounds the desired integral from above. The same comparison includes $a_{1,+} = a'_{1,-}$. No feasible segment or singleton is discarded because its area is zero: the lines still bound each of its points. Every supplied rectangle has positive side lengths and a nonempty enlarged feasible set; a truly empty feasible intersection requires no estimate.

Here is the full polynomial rule. For a line $f(t) = at + \zeta$ on a subinterval where $f > 0$, write $P_j(t) = 7(t - jf(t))_+^6$. Split every positive-part root. For the old line the available polynomials and their domains are

$$\begin{array}{l|l} 7t^6 - 7(21/4)^3 t^3 f^3 & 0 < f/t \leq 1/7 \\ 7t^6 - (7/24^3)t^3(17t + 7f)^3 & 1/7 < f/t < 1 \\ (7/6)^6 P_1 & 1/7 < f/t < 1 \\ P_1 & f/t \geq 1/2 \\ 7(256/729 + 3)t^6 - 126t^5 f & 0 < f/t \leq 1/6 \\ 7(256/729)t^6 & f/t \geq 1/6. \end{array}$$

For $f/t \geq 1$ also use zero. For the new line use $P_1 + P_2 - P_3$, and additionally P_1 when $f/t \geq 4/21$. If a line is nonpositive, its candidate is $7t^6$. The baseline $7\mathcal{B}(t)t^6$ is always available. The extra old candidates after $1/2$ remain upper bounds, and the P_1 candidate supplies the smaller envelope there.

Partition $[0, u_0], [u_0, u_1], [u_1, u_2]$ at every interior multiple of $1/100$, at $99/100, 26/25$, and at every interior solution, for either line, of

$$at + \zeta = ct, \quad c \in \{0, 1/7, 1/6, 4/21, 1/3, 1/2, 1\}.$$

If $a = c$, there is no isolated crossing: either the ratio is constant or the equation has no solution. An interior test point now determines the valid candidates on a whole cell. Repeated crossing points are retained once. A zero-length integration interval contributes zero; on a constant-ratio interval the weak endpoint branch is used throughout. Isolated endpoint values do not change the integral. In particular these rules include zero slopes, coincident roots, and $f = 0$. For $p(t) = \sum_{j=0}^6 p_j t^j$, use exactly

$$\text{Int}(p; l, u) = \sum_{j=0}^6 \frac{p_j}{j+1} (u^{j+1} - l^{j+1}).$$

Take the minimum of the candidate integrals on each cell, then sum. This rational number, denoted U , is an upper bound, because the integral of a pointwise minimum is no larger than the minimum of the integrals. The row numerator is exactly $\lfloor 10^9 U \rfloor + 1$; it gives a strict upper bound even when $10^9 U$ is integral.

Step 2. In this step, we verify coverage and the integral bound on the complete bounded domain. The 237 supplied rectangles have disjoint interiors and cover

$$[263/1800, 3] \times [5399983/18360000, 20].$$

Their rational endpoints give total area $1858566687329/33048000000$, exactly the area of this rectangle. Together with containment and disjoint interiors, this verifies coverage: a missing point from the finite closed union would leave a relatively open set of positive area. All nonempty rows have numerator at most 899892923. Every feasible slope pair in the bounded domain is therefore covered, including the edges. Equivalently, split the first coordinate at every recorded endpoint. On each open strip, and separately at every endpoint fiber, sort the closed second-coordinate intervals and check that their union is the full interval $[5399983/18360000, 20]$. The active rectangles are constant on an open strip, so one interior test fiber suffices there. This is a direct finite coverage check of the interior, boundary segments and vertices.

Step 3. In this step, we bound both unbounded slope ranges. Let $A_0(a)$ be the old-profile loss from zero to u_0 , and $B_0(a; z)$ its loss from u_0 to z , both with the baseline minimum retained. Then A_0 is nondecreasing, B_0 is nonincreasing, and $A_0(a) \leq u_0^7$. The 19 supplied rows partition $[263/1800, 3]$; use the upper slope before u_0 , the lower slope after it, no new profile, and terminal point u_1 . The identical polynomial rule gives loss strictly below $891/1000$ on every such row. For $a_0 \geq 3$, discard the new profile and use all polynomial layers before u_0 and old slope three after it. Exact integration by the same rule gives

$$I(a_0, a_1) \leq u_0^7 + B_0(3; u_2) < \frac{627860905}{10^9}.$$

For $a_0 \leq 3$ and $a_1 \geq 20$, use the complete incoming bound $891/1000$. After u_1 , $f_1(t)/t \geq h_1/u_1 > 4/21$, so

$$\begin{aligned} I(a_0, a_1) &< \frac{891}{1000} + 7 \int_{u_1}^{\infty} (t - h_1 - a_1(t - u_1))_+^6 dt \\ &= \frac{891}{1000} + \frac{(u_1 - h_1)^7}{a_1 - 1} \leq \frac{891}{1000} + \frac{(u_1 - h_1)^7}{19} < \frac{898876676}{10^9}. \end{aligned}$$

Here $u_1 > h_1$ and $a_1 > 1$, which justify the positive root and the substitution. These two ranges and the bounded rectangle exhaust all slopes in the lemma.

The checker reconstructs every candidate polynomial from the displayed formulas, splits its full validity domain, and verifies each recorded numerator as $\lfloor 10^9 U \rfloor + 1$. It also checks the interval and rectangle covers and both tail formulas. Hence acceptance of all records bounds every feasible pair of slopes. The margin below $9/10$ is $107077/10^9$; it is separate from the uniform finite-degree errors used when the certificate is applied to original sections. $\square$

Proposition 18.7. *Suppose $L^7 = 1$. There is a common cutoff J , fixed before the following original observations. Suppose a pure actual threshold of $U_k(27/25)$, with $k \geq J$ and $25 \mid k$, has a double-fivefold center of quadratic rank at least two. Suppose also that a pure actual threshold of $U_{20i}(9/10)$, with $i \geq J$, has a four-dimensional center. If the $27/25$ rank is at least three, or its own deficit exceeds $18/5$, then $K_X + 8L$ is generated at x . No incidence or equality of the two observed degrees is assumed.*

Proof. Assume nongeneration. We compare the two observed profiles on the same original point-order layers, obtain a nonzero $29/25$ source, and treat every outcome of its own threshold.

Write $\Gamma_j \sim_{\mathbb{Q}} s_j L$ for the entire observed boundaries at u_j , and $b_j = \text{def}_x(\Gamma_j)$, for $j = 0, 1$. If the $27/25$ rank is at least three and its deficit is at most $18/5$, [Theorem 18.3](#) and [Proposition 3.12](#) give a contradiction already. Thus in the remaining branch the entire $27/25$ boundary has $b_1 > 18/5$, and [Proposition 9.1](#) gives $1/s_1 > 27/85$. The full tangent of this actual double fivefold is the entire hypersurface tangent ring with quadratic rank at least two, hence is reduced Cohen–Macaulay.

Step 1. In this step, we obtain the larger original source from the two actual profile constraints. The $9/10$ source is big by the ordinary point kernel. Under nongeneration [Theorem 9.14](#) also makes the $27/25$ source big. Use [Theorem 6.13](#) at fixed big rational anchors slightly above these orders. Choose the common cutoff J so that all normalized finite-degree errors at u_0, u_1 are below ε , before either degree or its place is selected. Enlarge it by the cutoffs in [Proposition 17.41](#) and [Proposition 7.28](#), using upper point-order bound $13/5$. These choices depend only on the original data.

Let w_0, w_1 be actual computing places centered on the two observed centers, and put $\phi_j = \alpha_{w_j}/A_X(w_j)$ for $j = 0, 1$. By [Proposition 17.41](#), the actual $9/10$ deficit is greater than $4/25$. Its own whole price from [Proposition 9.1](#) therefore gives

$$\frac{1}{s_0} > \frac{5}{38}, \quad \phi_0(u_0) > h_0, \quad 5/38 - h_0 = 3/38000 > \varepsilon.$$

Also $\phi_1(u_1) > h_1$. Testing the full ideal at u_1 with the old place gives $\phi_1(u_1) \geq \phi_0(u_1) - \varepsilon$. Indeed the reciprocal finite threshold is the maximum of the normalized finite orders. The finite

order at w_0 is at least $\phi_0(u_1)$, and the order at the computing place w_1 is at most $\phi_1(u_1) + \varepsilon$. Thus this comparison uses only one error. Choose finite supporting slopes a_0, a_1 at the two anchors. Convexity and $\phi_j(0) = 0$ give

$$a_0 \geq h_0/u_0, \quad \phi_1(u_1) \geq \eta(a_0), \quad a_1 \geq \eta(a_0)/u_1.$$

The supporting lines with the lower heights $h_0, \eta(a_0)$ are still lower bounds. By [Theorem 7.23](#) they control every ordinary layer of every nonzero original source, including the layers beyond the observed orders.

For $\nu \in \{0, 1\}$ and $f_\nu(\ell/q) > 0$, compare the entire layer with the ideal

$$I_{q,\ell}^{(\nu)} = \{h \in \mathcal{O}_{X,x} : w_\nu(h) \geq qA_X(w_\nu)f_\nu(\ell/q)\}.$$

An element outside the center prime has value zero, whereas a high power of that prime has value at least the displayed threshold. Thus this is a proper primary ideal of the actual center. Its generic threshold $\lambda_{q,\ell}^{(\nu)}$ satisfies $\lambda_{q,\ell}^{(\nu)} \leq 1/(qf_\nu(\ell/q))$, by the same place. The original layer injects into its full initial image. Consequently its capacity argument is at least $f_\nu(\ell/q)/(\ell/q)$. The capacities are nonincreasing, so this latter argument gives an upper bound for the image. When the profile is zero use the entire polynomial layer.

For w_0 the center has height three. [Theorem 7.9](#) gives N_3 , and [Proposition 7.21](#) gives the additional bound $256/729 + 18(1/6 - c)_+$. Their minimum is $\mathcal{N}$, with no tangent assumption on this fourfold. For w_1 , the reduced Cohen–Macaulay double-fivefold tangent has $(n, d, e) = (7, 5, 6)$. [Theorem 7.18](#), with capacity indices $(7, 1, 2)$, gives

$$\mathcal{S}_{7,1,2}(c) = E_1(c) + E_2(c) - E_3(c).$$

The exact two-normal endpoint in [Theorem 7.9](#) gives $E_1(c)$ for $c \geq 4/21$. Together these give $\mathcal{D}$. All these estimates apply to the full comparison ideals, with errors uniform in their primes and exponents. Finally, [Proposition 7.28](#) supplies $\mathcal{B}$ on the same complete original layers. Summing their pointwise minimum and applying [Lemma 18.6](#) yields

$$(18.7) \quad V(u_2) \geq 1 - I(a_0, a_1) > \frac{100107077}{10^9} > \frac{1}{10}.$$

More explicitly, the estimates bound each full image of $U_q(\ell/q)$ in $\mathfrak{m}_x^\ell/\mathfrak{m}_x^{\ell+1}$. Take their minimum at that layer and sum from $\ell = 0$ through $\lceil qu_2 \rceil - 1$; the zero layer and endpoint rounding contribute $O(q^6)$, as does the sum of all $O((\ell+1)^5)$ errors. Divide by $q^7/7!$ and let q tend to infinity through the original integers. The bounded densities have only finitely many breaks on this interval, so their Riemann sums converge to the displayed integral. This proves (18.7) in the present nongeneration branch with $b_1 > 18/5$.

Step 2. In this step, we prove generation for every outcome of the new source's own threshold. Positive volume supplies all sufficiently late original 25-divisible u_2 -sources. Fix their comparison cutoff before taking such a degree or its threshold. Let $\Gamma_2 \sim_{\mathbb{Q}} s_2L$ be its entire pure actual threshold boundary, put $b_2 = \text{def}_x(\Gamma_2)$, and choose a computing place w_2 centered on its minimal center. Write $\phi_2 = \alpha_{w_2}/A_X(w_2)$. Testing with the old place and using convexity gives

$$\phi_2(u_2) \geq \phi_1(u_2) - \varepsilon \geq \frac{29}{27}h_1 - \varepsilon = \frac{29}{85} - \frac{56}{27}\varepsilon > \frac{341}{1000} =: h_2.$$

If its center is a divisor P , use the actual prime order as computing place. The entire fixed divisor at every late degree has normalized L^6 -degree at least $(L^6 \cdot P)\phi_P(u_2) > h_2$, since the original degree of P is a positive integer. Then [Proposition 6.19](#) implies

$$V(u_2) \leq (1 - h_2)^7 = \frac{53975652627753217019}{10^{21}} < \frac{3}{50} < \frac{1}{10},$$

a contradiction. All companion and remote fixed primes are included before this inequality is weakened.

For a double-fivefold successor, its own price $(29/25)s_2 + b_2 \leq 7$ follows from its own multiplicity; [Theorem 16.2](#) gives generation. [Proposition 7.26](#), applied to this same nonzero finite source and

its own pure threshold, leaves only embedding-seven fivefolds of multiplicities 3, 4, 5. Their entire ordinary tangent rings are reduced Cohen–Macaulay; the ordinary point-ideal powers and Rees geometry are those of this same threshold. Thus [Theorem 7.18](#) applies with $(n, d, e) = (7, 5, 7)$ and gives

$$\mathcal{S}_{7,0,3}(c) = 3E_2(c) - 2E_3(c),$$

with the sharper E_2 branch for $c \geq 4/25$. The same valuation-ideal comparison therefore applies to every original layer, retaining every component and the vertex of the ordinary tangent ring. Let $a_2 > h_2/u_2$ be a supporting slope and put $u_3 = 13/5$. If $a_2 \geq 1$, all outgoing ratios exceed $4/25$, and their entire loss is at most

$$7 \int_{u_2}^{\infty} (t - 2h_2 - 2a_2(t - u_2))_+^6 dt = \frac{(u_2 - 2h_2)^7}{2a_2 - 1} \leq \left(\frac{239}{500}\right)^7 = \frac{44543599279432079}{781250000000000000} < \frac{1}{100}.$$

Subtracting this outgoing loss from the actual mass in [\(18.7\)](#) gives $V(u_3) > 9/100$.

If $h_2/u_2 < a_2 \leq 1$, put $\kappa = u_2 - h_2 = 819/1000$ and $z_0 = 25\kappa/21 = 39/40$. Before u_2 the lower profile $(t - \kappa)_+$ is valid. Use all polynomial layers up to κ , density $7[3(2\kappa - t)^6 - 2(3\kappa - 2t)^6]$ to z_0 , and density $7(2\kappa - t)^6$ to u_2 . All indicated positive parts are positive on these intervals. The complete incoming bound is

$$\begin{aligned} I_{\text{in}} &= 3\kappa^7 - 2(2\kappa - z_0)^7 + (3\kappa - 2z_0)^7 - (2\kappa - u_2)^7 \\ &= \frac{315891331947925570507}{5 \cdot 10^{20}} < \frac{2}{3}. \end{aligned}$$

After u_2 , convexity gives the radial lower profile $h_2 t/u_2$. Put $q_0 = 1 - 2h_2/u_2 = 239/580$ and $z = 125/(52 + 73q_0)$. [Proposition 7.28](#), at each original layer's own order in $[u_2, u_3]$, bounds the outgoing density by $7 \min\{q_0^6 t^6, ((125 - 52t)_+/73)^6\}$. The branches cross at $u_2 < z < 125/52 < u_3$, and the second vanishes at $125/52$. Their entire integral is

$$\begin{aligned} I_{\text{out}} &= q_0^6(z^7 - u_2^7) + \frac{(125 - 52z)^7}{52 \cdot 73^6} = q_0^6 \left(\frac{125}{52} z^6 - u_2^7 \right) \\ &= \frac{7860756587864860626248390838442045303558341047}{59119162918550643946713187123828125000000000000} < \frac{7}{50}. \end{aligned}$$

Thus the entire loss is less than $121/150$, and $V(u_3) > 29/150$. In both slope ranges choose a nonzero section in an actual original $U_q(u_3)$ at a sufficiently large integer degree. Its divisor has class qL and multiplicity at least $\lceil 13q/5 \rceil$, so [Corollary 5.12](#) gives generation. Every outcome of the new threshold therefore contradicts nongeneration. This proves generation at x , with each original trial evaluated at its own complete coefficient. $\square$

19. ALL-HISTORY LARGE-DEFICIT FEEDBACK

We retain the original spaces of [Definition 2.4](#). The comparison in this section keeps the volume at a newly reached order together with the lower height of every threshold profile at that order. Both quantities are needed to exclude a divisorial center.

Theorem 19.1. *Suppose $L^7 = 1$. There is a cutoff depending only on (X, L, x) and fixed original-source comparisons, before all adaptive degrees, centers and places, such that the following holds for every nonzero complete $U_k(27/25)$, $25 \mid k$, above the cutoff. If a pure actual full-ideal threshold has a five-dimensional minimal center with $b > 18/5$, then $K_X + 8L$ is generated at x . No multiplicity, embedding-dimension, tangent, rank or initial-history hypothesis is imposed. The same conclusion holds for every original double-fivefold center of quadratic rank at least three, at every deficit.*

The first assertion concerns an arbitrary fivefold center at the stated deficit. The second also includes the low-deficit double-fivefold case. The intermediate volume comparison has the following form.

Proposition 19.2. *Under hypothetical nongeneration in the large-deficit case of Theorem 19.1, set*

$$r = 9/10, \quad v = 27/25, \quad T = 29/25, \quad \epsilon = 10^{-6}, \quad h = 263/2000, \quad H = 27/85 - \epsilon.$$

An actual initial r -place and the actual indicated v -place give supporting slopes a, δ of their original normalized profiles, with

$$a \geq h/r, \quad \eta = \max\{H, h + (v - r)a - \epsilon\}, \quad \delta \geq \eta/v, \quad \mathcal{M} = \eta + \delta(T - v) - \epsilon > 341/1000.$$

The same $\mathcal{M}$ satisfies

$$V(T) > (1 - \mathcal{M})_+^7 + 1/10000,$$

and every sufficiently late own T -threshold has its own computing-profile height at least this $\mathcal{M}$. The T -source is earned before its threshold is selected.

The reserve is attached to the full linked height $\mathcal{M}$. We compare the two actual profiles, bound their full original images, and prove the required integral inequality with its complete rational certificate. The resulting reserve excludes every divisorial center at T . The other centers yield either a paid continuation at their own threshold or an original section of order $13/5$.

19.1. Comparison at the actual thresholds. For a divisorial place w , write $A_X(w)$ for its log discrepancy and set

$$\phi_w(t) = \inf_{q \geq 1, 0 \neq f \in U_q(t)} \frac{w(f)}{qA_X(w)}.$$

Here $w(f)$ means the order of the germ in a local trivialization near the center of w ; changing that trivialization multiplies the germ by a unit. An empty infimum is $+\infty$. We use the constants r, v, T, ϵ, h, H displayed in Proposition 19.2.

Lemma 19.3. *Let X be a smooth integral complex projective sevenfold, L ample Cartier with $L^7 = 1$, and $x \in X$ a closed point at which $K_X + 8L$ is not generated. Let $\Gamma_1 \sim_{\mathbb{Q}} s_1 L$ be a pure actual threshold average of a sufficiently late complete $U_{k_1}(v)$, with $25 \mid k_1$, whose minimal center is a fivefold and whose deficit is $b_1 > 18/5$. For every sufficiently late integer i divisible by five, the complete source $U_{20i}(r)$ is nonzero. Choose any pure actual threshold average $\Gamma_0 \sim_{\mathbb{Q}} s_0 L$ of this source and put $k_0 = 20i$. Its minimal center has dimension two, three, or four, and its deficit is $b_0 > 4/25$. Let w_0, w_1 be any places centered on the respective minimal centers and computing their full-ideal thresholds, and put $\phi_j = \phi_{w_j}$. With uniform degree cutoffs fixed before these choices, there are finite supporting slopes a, δ such that*

$$\begin{aligned} a &> h/r, & \eta &= \max\{H, h + (v - r)a - \epsilon\}, & \delta &\geq \eta/v, \\ \phi_0(t) &\geq (h + a(t - r))_+, & \phi_1(t) &\geq (\eta + \delta(t - v))_+, \\ \mathcal{M} &= \eta + \delta(T - v) - \epsilon > 341/1000. \end{aligned}$$

The profile bounds hold on every original section order. If $V(T) > 0$, then every sufficiently late pure actual full-ideal T -threshold, at a cutoff fixed before its degree and place, has every computing profile ϕ_2 satisfying $\phi_2(T) \geq \mathcal{M}$.

The initial center and its deficit are conclusions of the earlier source theorem. The two observed degrees may be unrelated. The last assertion compares the new threshold with the already selected place w_1 .

Proof. We obtain the initial heights from the whole threshold coefficients, compare the profiles on one complete source, and then repeat that comparison at the new order.

Step 1. In this step, we obtain the initial observations and their whole-coefficient profile heights. Proposition 17.41 gives the stated nonzero initial sources, all three possible center dimensions, and $b_0 > 4/25$. Its cutoff is fixed before the degree, average, center, and place. In particular the codimension of this actual initial center belongs to $\{3, 4, 5\}$. Theorem 9.14 gives $V(v) > 1/32$. Continuity of volume, in Theorem 6.3, permits a fixed big anchor larger than v . Choose rational

$b'_0 < \beta$ with $v < b'_0$ and $\pi^*L - \beta E_x$ big. Apply Theorem 6.13 with original degree multiplier one on $[0, b'_0]$. Enlarge the two degree cutoffs so that its normalized errors at r and v are less than ϵ , uniformly before either computing place is chosen. If I_j is the full base ideal at the j -th observation and τ_j its threshold, then

$$\frac{1}{s_j} = \frac{w_j(I_j)}{k_j A_X(w_j)}, \quad 0 \leq \frac{1}{s_j} - \phi_j(u_j) < \epsilon, \quad u_0 = r, \quad u_1 = v.$$

Proposition 9.1 retains separately

$$\text{mult}_x \Gamma_j \geq u_j s_j, \quad \text{mult}_x \Gamma_j + b_j \leq 7.$$

It follows that

$$\frac{1}{s_0} > \frac{5}{38}, \quad \frac{5}{38} - h = \frac{3}{38000} > \epsilon, \quad \phi_0(r) > h.$$

For the other boundary,

$$s_1 < \frac{7 - 18/5}{v} = \frac{85}{27}, \quad \phi_1(v) > \frac{27}{85} - \epsilon = H.$$

The profiles are convex, nonnegative, and zero at zero. They are finite near the two anchors, so finite supporting slopes exist. A supporting slope at r is at least $\phi_0(r)/r > h/r$. Lowering its height to h gives the bound for ϕ_0 . The value at zero is zero because sufficiently large multiples of L are generated. Products of actual sections give the same convex profile on the entire interval of nonempty original orders. Thus the supporting inequalities hold for every original section, including orders beyond the fixed interval used for the finite comparison. The value $+\infty$ is allowed after the nonempty range.

Step 2. In this step, we link the two profiles by testing the full original v -source. Testing its actual threshold with w_0 gives

$$\frac{1}{s_1} \geq \frac{w_0(I_1)}{k_1 A_X(w_0)} \geq \phi_0(v).$$

Consequently

$$\phi_1(v) \geq \phi_0(v) - \epsilon \geq h + a(v - r) - \epsilon.$$

Together with its own height bound, this proves $\phi_1(v) \geq \eta$. Choose a supporting slope $\delta \geq \phi_1(v)/v \geq \eta/v$ and lower its height to η . The resulting lower line gives

$$\mathcal{M} \geq \frac{T}{v} H - \epsilon = \frac{29}{85} - \frac{56}{27}\epsilon > \frac{341}{1000}, \quad \frac{29}{85} - \frac{341}{1000} - \frac{56}{27000000} = \frac{3}{17000} - \frac{56}{27000000} > 0.$$

Step 3. In this step, we transfer the linked height to every computing profile at the earned order. Suppose $V(T) > 0$. Choose rational $T < b''_0 < \beta'$ with $\pi^*L - \beta' E_x$ big. Fix the uniform comparison cutoff on $[0, b''_0]$, with error less than ϵ . For a subsequent original degree q , let I_2 be the full base ideal, τ_2 its own threshold, and w_2 any computing place. Testing this threshold with w_1 yields

$$\phi_2(T) \geq \frac{1}{q\tau_2} - \epsilon \geq \frac{w_1(I_2)}{qA_X(w_1)} - \epsilon \geq \phi_1(T) - \epsilon \geq \mathcal{M}.$$

Every comparison uses the same full ideal at its indicated degree. $\square$

19.2. The numerical integral and its full certificate.

Definition 19.4. Use the constants r, v, T, ϵ, h, H above. For $1/6 \leq c < 1/2$, put

$$C_5(c) = \max \left\{ (1-c)^6, 1 - \left(\frac{29-28c}{36-42c} \right)^5 \right\},$$

$$C_{\text{call}}(c) = \max \left\{ (1-c)^6, \left(\frac{7}{6} \right)^6 \left(\frac{1-2c}{1-c} \right)^5 \right\}.$$

Define the *capacity envelopes*

$$N_5(c) = \begin{cases} E_{7,5}(c), & 0 \leq c < 1/6, \\ \min\{E_{7,5}(c), C_5(c), C_{\text{call}}(c)\}, & 1/6 \leq c < 1/3, \\ (1-c)_+^6, & c \geq 1/3, \end{cases}$$

$$D(c) = \max\{(1-c)_+^6, (1-2c)_+^5(1+10c)\}.$$

Here $E_{7,5}$ is as in Definition 7.7. Set

$$\mathcal{B}(t) = \begin{cases} 1, & 0 \leq t < r, \\ 61/100, & r \leq t < 99/100, \\ 11/25, & 99/100 \leq t < 26/25, \\ 17/50, & 26/25 \leq t \leq T. \end{cases}$$

For real $a \geq h/r$, set $\eta(a) = \max\{H, h + (v-r)a - \epsilon\}$. For real $\delta \geq \eta(a)/v$, put

$$F_0(t) = (h + a(t-r))_+, \quad F_1(t) = (\eta(a) + \delta(t-v))_+,$$

$$\mathcal{M}(a, \delta) = \eta(a) + \delta(T-v) - \epsilon,$$

$$\mathcal{K}_{a,\delta}(t) = 7t^6 \min\{\mathcal{B}(t), N_5(F_0(t)/t), D(F_1(t)/t)\}, \quad t > 0,$$

$$I_{\text{in}}(a, \delta) = \int_0^v \mathcal{K}_{a,\delta}(t) dt,$$

$$I_{\text{out}}(a, \delta) = \int_v^T \mathcal{K}_{a,\delta}(t) dt.$$

Set $\mathcal{K}_{a,\delta}(0) = 0$.

The function N_5 combines the five-normal bounds, and D is the exact two-normal capacity. The baseline $\mathcal{B}$ records the independent original-layer estimates before T . We use a single certificate format for the numerical comparison of their minimum.

Definition 19.5. Let

$$\mathcal{D} = \{(a, \delta) : a \geq h/r, \delta \geq \eta(a)/v\}, \quad Q_* = [h/r, 3] \times [H/v, 20].$$

A *rational integral certificate* on Q_* consists of a finite binary subdivision of this closed rectangle into numbered closed boxes, together with the following records for every box $Q = [a_-, a_+] \times [\delta_0, \delta_+]$. Put

$$\eta_- = \max\{H, h + (v-r)a_- - \epsilon\}, \quad \delta_- = \max\{\delta_0, \eta_-/v\}, \quad \mathcal{M}_- = \eta_- + \delta_-(T-v) - \epsilon.$$

An incoming record either uses the bound $31/32$ or partitions $[0, v]$ into rational intervals; an outgoing record partitions $[v, T]$. Each interval $[l, u]$ has a polynomial p of degree at most six and an integer $N \geq 0$, satisfying

$$\mathcal{K}_{a,\delta}(t) \leq p(t) \quad ((a, \delta) \in Q \cap \mathcal{D}, l < t < u), \quad 0 \leq \int_l^u p(t) dt < N/10^{12}.$$

The polynomial inequality may exclude the finitely many prescribed breakpoints. Endpoint limits determine its integral. Write $A_Q = 31/32$ for an incoming cap, and otherwise $A_Q = \min\{\sum_{\text{in}} N/10^{12}, 31/32\}$. There are also nonnegative integers R_Q, U_Q such that

$$(1 - \mathcal{M}_-)_+^7 \leq R_Q/10^{12}, \quad A_Q + \sum_{\text{out}} N/10^{12} + R_Q/10^{12} = U_Q/10^{12}.$$

Every bisection is at a rational midpoint and retains its common boundary in both children. No numbered leaf or interval is omitted.

The certificate records upper integrals and a reserve for the same parameter box. Its coverage concerns every feasible parameter, including parameters on edges and at vertices.

Lemma 19.6. *Let $\Lambda \in \mathbb{R}$. Suppose a rational integral certificate in Definition 19.5 has $U_Q/10^{12} \leq \Lambda$ for every box. Suppose also that*

$$\min\{I_{\text{in}}, 31/32\} + I_{\text{out}} + (1 - \mathcal{M})_+^7 \leq \Lambda$$

for feasible pairs with $a \geq 3$ or $\delta \geq 20$. Then this inequality holds throughout $\mathcal{D}$.

Proof. The two children of each subdivision node cover its closed parent. Induction on the finite tree therefore gives a leaf containing each point of Q_* . For a feasible point in that leaf, $\eta(a) \geq \eta_-$, $\delta \geq \delta_-$, and $\mathcal{M}(a, \delta) \geq \mathcal{M}_-$. Integration of its complete row partitions gives

$$\min\{I_{\text{in}}, 31/32\} + I_{\text{out}} + (1 - \mathcal{M})_+^7 \leq A_Q + \sum_{\text{out}} N/10^{12} + R_Q/10^{12} \leq \Lambda.$$

An incoming cap already bounds the first term, so it needs no incoming rows. A finite set of integration endpoints contributes zero to the integral. A feasible segment or singleton in parameter space still belongs to a closed leaf and satisfies the same pointwise inequalities; no positive-area assumption has been used. An empty feasible intersection has nothing to prove. Finally $\delta \geq \eta(a)/v \geq H/v$, so every point of $\mathcal{D}$ outside Q_* belongs to one of the two assumed unbounded ranges. $\square$

Lemma 19.7. *For every real $a \geq 263/1800$ and $\delta \geq \eta(a)/v$, the quantities in Definition 19.4 satisfy*

$$\min\{I_{\text{in}}(a, \delta), 31/32\} + I_{\text{out}}(a, \delta) + (1 - \mathcal{M}(a, \delta))_+^7 \leq \frac{999894402477}{10^{12}} < \frac{9999}{10000}.$$

The estimate includes both unbounded slope ranges. The seventh-power term uses the same a, δ, η as both integrals. Its complete exact data have the stable identifier `19-large-deficit/paired-profile-v1`. The canonical record stream is `supplement/19-large-deficit/data/paired-profile.txt`; the package's `check.py` reconstructs and checks the certificate conditions in Lemma 19.6.

Proof. We prove admissibility of the polynomial densities, specify the exact integration and coverage checks on the supplementary records, and prove the two unbounded estimates.

Step 1. In this step, we prove admissibility of every polynomial density and its validity range. For a line $f(t) = At + B$, write $P_j(f; t) = 7(t - jf(t))_+^6$. The old-profile labels are

label	density	range for $c = f(t)/t$
OI	$7t^6 - 7(35/6)^5 t f(t)^5$	$0 < c \leq 1/7$
Oh	$7t^6 - (7/36^5) t (29t + 7f(t))^5$	$1/7 < c < 1$
Oc	$(7/6)^6 P_1(f; t)$	$1/7 < c < 1$
Oe	$P_1(f; t)$	$c \geq 1/3$.

There are also six densities

$$\text{Oq}_i(f; t) = 7t^6 - 7t(A_i t + B_i f(t))^5 \quad (1/6 \leq c \leq 1/4)$$

and nine densities

$$\text{Ok}_i(f; t) = 7(7/6)^6 t (C_i t + D_i f(t))^5 \quad (1/6 \leq c \leq 8/25).$$

Their entire coefficient lists are

i	point of tangency	A_i	B_i
0	1/6	2012/2523	210/841
1	9/50	44696/56169	43750/168507
2	1/5	1258/1587	875/3174
3	11/50	39236/49729	43750/149187
4	6/25	9151/11664	21875/69984
5	1/4	226/289	280/867

and

i	point of tangency	C_i	D_i
0	1/6	26/25	-36/25
1	9/50	1762/1681	-2500/1681
2	1/5	17/16	-25/16
3	11/50	1642/1521	-2500/1521
4	6/25	397/361	-625/361
5	13/50	1538/1369	-2500/1369
6	7/25	373/324	-625/324
7	3/10	58/49	-100/49
8	8/25	353/289	-625/289

For the first list, the function $q(c) = (29 - 28c)/(36 - 42c)$ has

$$q'(c) = \frac{210}{(36 - 42c)^2}, \quad q''(c) = \frac{17640}{(36 - 42c)^3} > 0.$$

Thus its tangent lines lie below it. They are positive on the indicated interval. Moreover

$$(1 - c)^6 \leq (5/6)^6 < 1 - (44/51)^5 \leq 1 - q(c)^5 \quad (1/6 \leq c \leq 1/4).$$

This proves the Oq_i upper bounds for N_5 . For the second list, $u(c) = (1 - 2c)/(1 - c)$ satisfies

$$u'(c) = -\frac{1}{(1 - c)^2}, \quad u''(c) = -\frac{2}{(1 - c)^3} < 0.$$

Its positive tangent lines lie above it. The ratio of the second branch of C_{call} to its first branch is

$$\left(\frac{7}{6}\right)^6 \frac{(1 - 2c)^5}{(1 - c)^{11}}.$$

Its logarithmic derivative is $(1 - 12c)/((1 - c)(1 - 2c))$, and at $c = 8/25$ the ratio exceeds one, since

$$(7/6)^6 (9/17)^5 > (17/25)^6.$$

This proves the Ok_i bounds on the stated interval.

For the new profile put

$$c_- = \frac{1455087}{8192000}, \quad c_+ = \frac{116407}{655360}.$$

The new-profile density labels are

$$\begin{array}{l|l} \text{Ns} & P_1(f; t) + P_2(f; t) - P_3(f; t) \\ \text{Nb} & 7(t - 2f(t))^5(t + 10f(t)) \\ \text{Ne} & P_1(f; t) \end{array} \quad \begin{array}{l} c > 0 \\ 0 < c \leq c_- \\ c \geq c_+. \end{array}$$

The ratio $(1 - 2c)^5(1 + 10c)/(1 - c)^6$ has logarithmic derivative

$$\frac{6(1 - 12c)}{(1 - 2c)(1 + 10c)(1 - c)}.$$

It first increases and then decreases to zero on $(0, 1/2)$. The balanced branch minus the endpoint branch at c_-, c_+ has the respective exact values

$$\frac{184357177943425293881926649000022111}{30223145490365729367654400000000000000000000} > 0$$

and

$$-\frac{2065795968585910646530153649}{79228162514264337593543950336000000} < 0.$$

This justifies Nb and Ne. For $0 \leq c \leq 1/2$, put $u = 1 - 2c$. Convexity gives

$$(u + c)^6 - (u - c)_+^6 = 6 \int_{-c}^c (u + z)_+^5 dz \geq 12cu^5.$$

Hence N s bounds both branches of D , including the intervening interval. For $c \geq 1/2$, the balanced branch vanishes and the same assertion holds. The label $B\beta$ means $7\beta t^6$ from the baseline; in particular $B1$ means $7t^6$. The zero density is admissible once the old ratio is at least one.

Step 2. In this step, we verify the complete integral and reserve bound on each closed box. Use the box data $\eta_-, \delta_-, \mathcal{M}_-$ of Definition 19.5. A box is infeasible if $\delta_+ < \delta_-$. Otherwise use the following lines, taking their positive parts when needed:

	before the anchor	after the anchor
old line, anchor r	$h + a_+(t - r)$	$h + a_-(t - r)$
new line, anchor v	$\eta_- + \delta_+(t - v)$	$\eta_- + \delta_-(t - v)$

Each line is a lower bound for its actual profile throughout the box. Since the numerical capacities are decreasing, the admissible densities bound the entire integral from above. The reserve is at most $(1 - \mathcal{M}_-)_+^7$.

Each row in the supplementary record stream gives its full interval $[l, u]$, a density label, and an integer N . Substitute the appropriate line from the box and expand the density as $p(t) = \sum_{j=0}^6 p_j t^j$. The row inequality is

$$(19.1) \quad \sum_{j=0}^6 \frac{p_j}{j+1} (u^{j+1} - l^{j+1}) < \frac{N}{10^{12}}.$$

All coefficients are specified by the density formulas and

$$(At + B)^k = \sum_{j=0}^k \binom{k}{j} A^j B^{k-j} t^j.$$

For positive parts, split at their zeros. Density validity is checked at the rational endpoints obtained from

$$f(t)/t \in \{0, 1/7, 1/6, c_-, c_+, 1/4, 8/25, 1/3, 1/2, 1\},$$

the baseline endpoints, r, v, T , and multiples of $1/100$. The records merge intervals only when the density is the same polynomial throughout the union. No finite-box row crosses either profile anchor. The unbounded- a rows use only the old profile, so they may cross v . Each row comparison becomes an integer inequality after multiplication by its positive common denominator.

The incoming and outgoing numerators, upper reserve numerator, and total are supplied for every box. When the incoming bound $31/32$ is used, this is stated instead of an incoming table. For example, Box 0 gives

$$\frac{31}{32} + \frac{14664236786 + 1696961152 + 1}{10^{12}} = \frac{985111197939}{10^{12}} < \frac{9999}{10000}.$$

There are exactly 301 boxes, numbered $0, \dots, 300$, and 4258 finite integration rows. Seven boxes use the incoming cap. For each row the checker reconstructs p , checks its full validity interval and positive-part signs, and evaluates (19.1) over $\mathbb{Q}$. It then sums the supplied integers, checks every reserve against the same $\mathcal{M}_-$, and checks every complete incoming and outgoing partition. The largest total, at Box 215, is 999894402477. Its gap below 999900000000 is 5597523.

Step 3. In this step, we verify coverage of the full finite parameter domain. The tree at the end of the same record stream starts from

$$[263/1800, 3] \times [5399983/18360000, 20].$$

An internal record $[0, \text{left}, \text{right}]$ bisects the a -interval and $[1, \text{left}, \text{right}]$ bisects the δ -interval, in each case at its rational midpoint. A quoted integer denotes the box with that number. The checker derives each child's endpoints from its parent and compares the entire leaf rectangle with the numbered box. It requires exactly 300 internal nodes and each of the 301 leaf indices exactly once. Thus the closed rectangle is covered, with overlaps only on boundaries. As an additional

Moreover,

$$V(T) > (1 - \mathcal{M})_+^7 + 1/10000.$$

Every sufficiently late complete T -source is nonzero before its threshold is selected. Each of its computing profiles has height at least this same $\mathcal{M}$, and no sufficiently late pure actual T -threshold has a divisorial minimal center.

The difference estimate applies before the source at T is known to be nonzero. Its kernel is the next full original point-order space.

Proof. We bound each full point-order image, sum these bounds, and combine the resulting loss with the independent volume floor at v . The same reserve then excludes every fixed prime that could be a later minimal center.

Step 1. In this step, we bound both full primary comparison images on every original layer. Put $S = \mathcal{O}_{X,x}$, with maximal ideal $\mathfrak{m}_x$, and let $\mathfrak{p}_j \subset S$ be the prime of the center of w_j . At degree q and ordinary point order ℓ , write

$$A_{q,\ell} = \text{im} \left(U_q(\ell/q) \longrightarrow \mathfrak{m}_x^\ell / \mathfrak{m}_x^{\ell+1} \right).$$

Its kernel is exactly $U_q((\ell+1)/q)$. If $F_j(\ell/q) > 0$, define the entire valuation ideal

$$J_{q,\ell}^{(j)} = \{f \in S : w_j(f) \geq qA_X(w_j)F_j(\ell/q)\}.$$

Every element outside $\mathfrak{p}_j$ has valuation zero, so this ideal is proper and $\mathfrak{p}_j$ -primary. Every original section in $U_q(\ell/q)$ has its germ in this ideal by Lemma 19.3. Consequently

$$A_{q,\ell} \subseteq (\text{in}_{\mathfrak{m}_x} J_{q,\ell}^{(j)})_\ell.$$

If λ_j is its threshold at the generic point of $\mathfrak{p}_j$, testing with w_j gives

$$\lambda_j \leq \frac{1}{qF_j(\ell/q)}, \quad \frac{1}{\ell\lambda_j} \geq \frac{F_j(\ell/q)}{\ell/q}.$$

For $F_j(\ell/q) = 0$, all polynomial jets give the bound instead. This comparison retains the full primary ideal and the entire initial image. It imposes no primaryness on the original base ideal and requires no extension of local sections to global sections.

The first center has codimension $c_0 \in \{3, 4, 5\}$. In the simplex comparison of Theorem 7.23, append $5 - c_0$ zero weights to its normal weights. The random linear form is unchanged, and its weight domain embeds in the five-normal domain of Definition 7.7. Hence $\mathbb{M}_{7,c_0} \leq \mathbb{M}_{7,5}$ on these same images. The explicit full-primary envelopes also respect this comparison. For $0 < c \leq 1/7$, the subtracted power in $B_{7,\nu}(c)$ decreases with the real parameter $\nu \geq 1$, since

$$\frac{d}{d\nu} \log \left(\frac{7c\nu}{\nu+1} \right)^\nu = \log(7c) + \log \frac{\nu}{\nu+1} + \frac{1}{\nu+1} < 0.$$

For $1/7 < c < 1/6$, put $\theta = 7(1-c)/6 \leq 1$ and $z = \theta/(\nu+1)$. The subtracted power is $(1-z)^\nu$, and its logarithmic derivative is

$$\log(1-z) + \frac{\nu}{\nu+1} \frac{z}{1-z} \leq \log(1-z) + z < 0.$$

The other branch of $E_{7,\nu}$ is independent of ν . Thus $E_{7,c_0}(c) \leq E_{7,5}(c)$ for $c < 1/6$, including $c = 0$ by continuity, and Theorem 7.9 gives that part of N_5 . For $1/6 \leq c < 1/3$, Proposition 7.22 gives the minimum of its three branches directly for every $c_0 \leq 5$; for $c \geq 1/3$ it gives the exact endpoint. The second center is a fivefold, so its codimension is two. The exact Q_7 formula in Theorem 7.9 gives D . All remainders are bounded by $C(\ell+1)^5$, with constants independent of the centers, ideals, slopes, and computing places.

Step 2. In this step, we telescope the full original layers with a uniform remainder. Proposition 7.28 bounds this same $A_{q,\ell}$ by $\ell^6 \mathcal{B}(\ell/q)/6!$, with a uniform remainder, for $0 < \ell/q < T$. Taking the minimum of the three bounds on the same image gives

$$\dim A_{q,\ell} \leq \frac{\ell^6}{6!} \min\{\mathcal{B}(\ell/q), N_5(F_0(\ell/q)/(\ell/q)), D(F_1(\ell/q)/(\ell/q))\} + C(\ell + 1)^5.$$

The order-zero image has dimension at most one. The baseline applies at the layer's own rational order ℓ/q ; if the current source is zero, the image is zero as well. Summing the exact kernels gives

$$\dim U_q(\alpha) - \dim U_q(\beta) = \sum_{\ell=q\alpha}^{q\beta-1} \dim A_{q,\ell}.$$

The error sum is $O(q^6)$ on every bounded order interval. The ratios $F_0(t)/t$ and $F_1(t)/t$ are nondecreasing, since $a \geq h/r$ and $\delta \geq \eta/v$. The capacity envelopes are bounded and decreasing, and the baseline has finitely many jumps. Their minimum has uniformly bounded variation on each baseline interval; multiplication by $7t^6$ preserves a uniform Riemann-sum error on $[0, T]$. After multiplication by $7!/q^7$, the sum of the leading upper bounds tends to the integral in (19.2), and the remainder tends to zero. This proves the difference estimate. One may equivalently start with a fixed rational mesh of size at most $1/1000$, containing $r, 99/100, 26/25, v, T$, and refine it after taking the degree limit. The estimate concerns each full layer before summation. Integer floors and ceilings change only boundedly many endpoint layers and contribute $O(q^6)$.

Step 3. In this step, we combine the starting mass and outgoing loss to obtain the strict volume reserve. The full point-order kernel, ample Riemann–Roch, and the complete-source volume comparison in Theorem 7.23 give $V(v) \geq 1 - I_{\text{in}}$, while Theorem 9.14 gives $V(v) > 1/32$. Use (19.2) to subtract the outgoing loss from the better of these two lower bounds on the same original space at v . This gives

$$V(T) \geq 1 - \min\{I_{\text{in}}, 31/32\} - I_{\text{out}}.$$

Lemma 19.7 implies

$$V(T) \geq (1 - \mathcal{M})_+^7 + \frac{105597523}{10^{12}} > (1 - \mathcal{M})_+^7 + \frac{1}{10000}.$$

In particular $\pi^*L - TE_x$ is big. Apply Theorem 6.13 at a fixed big anchor beyond T , with a cutoff fixed before the new degree, average, center and place. This gives nonzero complete original spaces at every sufficiently late allowed degree. The last part of Lemma 19.3 gives their profile height bound.

Step 4. In this step, we exclude every later divisorial center using its full fixed-prime mass. Suppose such a threshold has prime minimal center P . Then ord_P computes the full-ideal threshold, with $A_X(\text{ord}_P) = 1$. Its normalized profile satisfies $\phi_P(T) \geq \mathcal{M}$. For every original degree q , every section of $U_q(T)$ has P -order at least $q\phi_P(T)$. If P_q is its entire fixed divisor, then

$$\frac{P_q \cdot L^6}{q} \geq (L^6 \cdot P)\phi_P(T) \geq \mathcal{M},$$

because $L^6 \cdot P$ is a positive integer. Let z_q be the normalized moving volume of this same complete source, as in Proposition 6.19. It gives

$$\frac{P_q \cdot L^6}{q} \leq 1 - z_q^{1/7}, \quad z_q \longrightarrow V(T) > 0.$$

If $\mathcal{M} \geq 1$ this is impossible. Otherwise it implies $V(T) \leq (1 - \mathcal{M})^7$, contradicting the reserve. The whole fixed divisor is used, including primes not passing through x . No smoothness or degree-one assumption on P occurs. $\square$

Proof of Proposition 19.2. Assume nongeneration and fix the indicated sufficiently late fivefold observation at v , with deficit greater than $18/5$. Lemma 19.3 supplies an actual initial observation at r , together with its supporting slope a , the slope δ at v , and the full linked height $\mathcal{M} > 341/1000$. Lemma 19.8 gives

$$V(T) > (1 - \mathcal{M})_+^7 + 1/10000$$

and proves that every sufficiently late own T -threshold has every computing-profile height at least the same $\mathcal{M}$. Its source is nonzero before any such threshold is selected. The initial and v -comparison cutoffs depend only on the fixed original data. The lower volume bound $1/10000$ is uniform in a, δ , so the further comparison interval and cutoff at T are fixed from that data before any later degree or place. This proves every assertion of the proposition. $\square$

19.4. The outcomes of the earned threshold.

Lemma 19.9. *Let X be a smooth integral complex projective sevenfold, L ample Cartier with $L^7 = 1$, and $x \in X$ a closed point. Put $T = 29/25$, $R = 13/5$. Let $\mathcal{M} > 341/1000$ satisfy $V(T) > (1 - \mathcal{M})_+^7 + 1/10000$. Let $\Gamma_2 \sim_{\mathbb{Q}} s_2 L$ be a sufficiently late pure actual full-ideal threshold average of a nonzero complete T -source, with a nondivisorial minimal center W_2 . Suppose a place centered on W_2 and computing this threshold has original normalized profile ϕ_2 with $\phi_2(T) \geq \mathcal{M}$. Then $K_X + 8L$ is generated at x .*

The later source uses its own entire threshold boundary. Under nongeneration its tangent geometry and its complete original-layer baseline are supplied by the earlier propositions.

Proof. Suppose instead that generation fails at x . We pay a double-fivefold at its own price. For the other states, we treat the two supporting-slope ranges and earn a high-order original section.

Step 1. In this step, we classify the actual later center and pay the double-fivefold alternative. Let k_2 be the original degree of the observation and put $u_2 = \lceil k_2 T \rceil / k_2$. Then $u_2 \geq T$, $k_2 u_2$ is integral, and $U_{k_2}(T) = U_{k_2}(u_2)$. Proposition 7.26, applied at u_2 to this same nonzero complete source and its own threshold, makes W_2 a double-fivefold of embedding dimension six or a fivefold of embedding dimension seven and multiplicity 3, 4, 5. In every row it proves that the entire ordinary tangent ring is reduced and Cohen–Macaulay. Put $b_2 = \text{def}_x(\Gamma_2)$. Proposition 9.1 gives $\text{mult}_x \Gamma_2 \geq u_2 s_2$ and $\text{mult}_x \Gamma_2 + b_2 \leq 7$. Thus its actual whole price satisfies $T s_2 + b_2 \leq u_2 s_2 + b_2 \leq 7$.

Case 1.1. The later center is a double-fivefold. In the double case, Theorem 16.2 gives generation at x , a contradiction. This applies in every quadratic rank and uses no coefficient from a separate earlier trial.

Step 2. In this step, we bound the complete loss for an embedding-seven center in both supporting-slope ranges. The fivefold has full embedding dimension seven and the full reduced Cohen–Macaulay tangent just obtained. Theorem 7.18, with $n = 7, t = 0, s = 3$, gives the capacities

$$C_E(c) = 3(1 - 2c)_+^6 - 2(1 - 3c)_+^6, \quad (1 - 2c)_+^6 \quad (c \geq 4/25).$$

They retain all tangent components and the affine vertex, and apply to every original layer on the same computing profile. Indeed, if w_2 is its computing place and F is a nonnegative lower profile, then for $F(\ell/q) > 0$ the valuation ideal

$$J_{q,\ell} = \{f : w_2(f) \geq qA_X(w_2)F(\ell/q)\}$$

is primary to the actual center prime. Its generic threshold is at most $1/(qF(\ell/q))$, and the full exact-order image of $U_q(\ell/q)/U_q((\ell+1)/q)$ injects into its degree- ℓ initial image. For $F = 0$, use all polynomial jets. The shared uniform error sums to $O(q^6)$ over the bounded order interval. Thus the full-image argument of Theorem 7.23 gives the displayed capacities on both sides of T , without assuming that a later source is nonzero.

Since $V(T) > 0$, Theorem 6.13 makes ϕ_2 finite on a big interval containing T in its interior. Choose a finite supporting slope α at T . It satisfies $\alpha \geq \phi_2(T)/T \geq \mathcal{M}/T$. **Case 2.1.** The

supporting slope is at least one. If $\alpha \geq 1$, the lower profile $(\mathcal{M} + \alpha(t - T))_+$ has ratio at least $\mathcal{M}/T > 4/25$ for $t \geq T$. The sharper full-image bound controls the entire outgoing loss:

$$7 \int_T^\infty (t - 2\mathcal{M} - 2\alpha(t - T))_+^6 dt = \frac{(T - 2\mathcal{M})_+^7}{2\alpha - 1} \leq (T - 2\mathcal{M})_+^7 \leq (1 - \mathcal{M})_+^7.$$

The last inequality follows from $\mathcal{M} > T - 1$ and remains valid when either positive part vanishes. Subtraction from the actual volume at T , with this same $\mathcal{M}$, yields $V(R) > 1/10000$.

Case 2.2. The supporting slope is at most one. Suppose $\alpha \leq 1$, and put

$$h_2 = 341/1000, \quad \kappa = T - h_2 = 819/1000, \quad z_0 = 25\kappa/21 = 39/40.$$

In this slope range we bound the complete loss starting at zero. The number h_2 is a lower bound for the height; the reserve already proved at T remains attached to the full $\mathcal{M}$. On $[0, T]$, the supporting inequality gives the lower profile $(t - \kappa)_+$; on $[T, R]$, convexity at zero gives the lower profile $h_2 t/T$. The complete incoming density is at most $7t^6$ up to κ , then at most

$$7\{3(2\kappa - t)^6 - 2(3\kappa - 2t)^6\} \quad (\kappa \leq t \leq z_0),$$

and at most $7(2\kappa - t)^6$ on $[z_0, T]$. The latter interval starts exactly when the profile ratio reaches $4/25$. All displayed powers have nonnegative bases on their respective intervals. Their exact total integral is

$$\begin{aligned} I_0 &= 3\kappa^7 - 2(2\kappa - z_0)^7 + (3\kappa - 2z_0)^7 - (2\kappa - T)^7 \\ &= \frac{315891331947925570507}{5 \cdot 10^{20}} < \frac{2}{3}. \end{aligned}$$

On $[T, R]$, Proposition 7.28 supplies the complete density

$$7 \left(\frac{(125 - 52t)_+}{73} \right)^6.$$

It is a bound on each full original image with uniform remainder, including at a zero current source. The radial profile has ratio $h_2/T > 4/25$, so its endpoint capacity is available throughout this interval. Put $q_0 = 1 - 2h_2/T = 239/580$ and $z = 125/(52 + 73q_0)$. The two outgoing densities cross at z , and $T < z < 125/52 < R$. Taking their minimum and retaining the zero part after $125/52$ gives

$$\begin{aligned} I_1 &= q_0^6(z^7 - T^7) + \frac{(125 - 52z)^7}{52 \cdot 73^6} \\ &= q_0^6 \left(\frac{125}{52} z^6 - T^7 \right) \\ &= \frac{7860756587864860626248390838442045303558341047}{59119162918550643946713187123828125000000000000} < \frac{7}{50}. \end{aligned}$$

The simplification uses $125 - 52z = 73q_0z$. The whole loss from zero to R is therefore less than $2/3 + 7/50 = 121/150$, so $V(R) > 29/150$.

Step 3. In this step, we obtain an actual high-order section and deduce generation in every remaining case. Both slope ranges give a nonzero complete original section of a sufficiently divisible degree qL and order at least Rq . Corollary 5.12 applied to its effective divisor gives a point boundary of coefficient less than $8 - 1/100$, hence generation at x . The endpoint inputs include both cutting errors in their stated strict margins. This is a contradiction in every possible nondivisorial state. $\square$

Proof of Theorem 19.1. We first treat every fivefold of deficit greater than $18/5$, then use the intrinsic endpoint for the remaining deficits of a double-fivefold.

Step 1. In this step, we prove generation for every fivefold observation of deficit greater than $18/5$. Suppose generation fails and take the indicated actual threshold of $U_k(v)$, above the fixed cutoffs of Lemma 19.3. Proposition 19.2 earns the complete T -source with

$$V(T) > (1 - \mathcal{M})_+^7 + 1/10000.$$

Choose any sufficiently late original degree divisible by 25 and a pure average at its actual full-ideal T -threshold. Every computing profile has height at least this same $\mathcal{M} > 341/1000$. Lemma 19.8 excludes every divisorial minimal center, including singular ones. Lemma 19.9 gives generation for every nondivisorial minimal center. This is a contradiction. The argument used only the dimension and deficit of the given fivefold. The initial observation was supplied by Proposition 17.41 through every history.

Step 2. In this step, we prove generation for all deficits of a double-fivefold of quadratic rank at least three. The entire given threshold boundary has $(27/25)s + b \leq 7$ by Proposition 9.1. If $b > 18/5$, the first assertion just proved applies. If $b \leq 18/5$, Theorem 18.3 applies at this exact whole price and actual quadratic rank. It gives an additive point continuation of the entire boundary with coefficient less than $8 - 1/400000$. Proposition 3.12 gives generation at x . This endpoint includes every positive original degree of the center and its separate intrinsic and ordinary restriction cutoffs. The two cases exhaust all deficits.

All initial degree cutoffs and comparison errors were fixed before the corresponding actual observations. The later source was earned with a uniform strict margin before its own degree and threshold were chosen. Each endpoint retains its two cutting errors within its stated strict margin, and each trial retains its own whole boundary. Thus the common cutoff has exactly the dependence asserted in the theorem. $\square$

20. RANK-TWO FINITE SOURCE FEEDBACK

We retain the boundary and original-source conventions of section 2. We treat the double fivefolds of quadratic rank two and deficit at most $18/5$ left by the preceding reductions. A finite sequence of complete intrinsic sources gives their additive continuation. The old discriminant coefficient and the deficit of the actual successor together control the loss between consecutive sources.

Theorem 20.1. *Let $n \geq 7$, $B \sim_{\mathbb{Q}} sL$, $s \geq 0$, be effective rational and lc near x , with actual minimal center W of state $(5, 6, 2)$ and quadratic rank two. If*

$$(27/25)s + b \leq n, \quad b \leq 18/5,$$

the entire B has an additive point continuation of coefficient less than $n + 1 - 1/400000$. Every positive original degree and every analytic factorization of the actual convergent equation $\mathbb{C}\{u, v, z_1, \dots, z_4\}/(uv - \psi(z))$, $0 \neq \psi$, $\text{ord } \psi \geq 3$, is included.

The coefficient includes every component of the given boundary. We prove the theorem after the following source statement. Each trial in that statement begins with this same boundary.

Proposition 20.2. *In the nonfactorial case of Theorem 20.1, set*

$$\gamma_j = j/2000 \quad (1731 \leq j \leq 1966), \quad \lambda = 1731/2000, \quad \theta = 983/1000.$$

Fix an intrinsic cutoff and a separate ordinary ambient restriction cutoff before a sufficiently large 2000-divisible degree p . At each earned original complete $F_p(\gamma_j)$, take a sufficiently dilute general average from its full ambient inverse image and its own threshold from the same entire B . Either this threshold pays the asserted continuation, or its actual prime and full symbolic or principal interval earn $F_p(\gamma_{j+1}) \neq 0$ at that same p , before its threshold. If no trial pays earlier, the normalized mass $120 \dim F_p(\theta)/p^5$ is at least

$$\frac{8402435542349}{16000000000000000} + O(1/p) > 1/2000 + O(1/p),$$

with a uniform error fixed before all primes. Every own threshold of the terminal source is paid. The 235 transitions cover all factor orders, characters and endpoint valuations continuously; primes need not recur.

Thus the sequence is constructed at one degree. Its positive remaining mass permits the next threshold to be taken, regardless of which prime occurred at the preceding trial.

We use the following finite-certificate principle to pass from exact rational interval bounds to positive source dimensions.

Lemma 20.3. *Let $j_0 < j_1$ be integers, let p, Q_0 be positive integers, and let M_0, a_0, a_1 be nonnegative real numbers. For each $j_0 \leq j < j_1$, let I_j be a finite set, let $\ell_{j,\nu} \geq 0$ be rational for $\nu \in I_j$, and set*

$$L_j = \max(\{0\} \cup \{\ell_{j,\nu} : \nu \in I_j\}).$$

Suppose nonnegative integers E_j satisfy $Q_0 \ell_{j,\nu} \leq E_j$ for every $\nu \in I_j$. Let $V_{j_0} \supseteq \cdots \supseteq V_{j_1}$ be finite-dimensional vector spaces such that

$$\dim V_{j_0} \geq M_0 p^5 / 120 - a_0 p^4, \quad \dim V_j - \dim V_{j+1} \leq L_j p^5 / 120 + a_1 p^4.$$

Then, for every $j_0 \leq k \leq j_1$,

$$\frac{120 \dim V_k}{p^5} \geq M_0 - \frac{1}{Q_0} \sum_{j=j_0}^{k-1} E_j - \frac{120(a_0 + (k - j_0)a_1)}{p}.$$

In particular all these spaces are nonzero if the right-hand side at $k = j_1$ is positive.

Proof. The candidate inequalities and $E_j \geq 0$ give $L_j \leq E_j / Q_0$, including when I_j is empty. Subtract the successive dimension losses from the initial dimension bound. This gives the displayed inequality, with the empty sum interpreted as zero. Its right-hand side decreases with k , since the E_j and a_1 are nonnegative. $\square$

Proof of Proposition 20.2. We first fix the local interval bounds and both degree cutoffs. We then pay the possible changed centers, use the old coefficients to bound every unpaid interval, and sum the resulting finite certificate on the original section spaces.

Step 1. In this step, we fix the full symbolic and principal interval estimates for the actual rank-two germ. Put $A = L|_W$, $g = A^5$, and

$$R = \mathcal{O}_{W,x}^{\text{an}} = \mathbb{C}\{u, v, z_1, \dots, z_4\} / (uv - \psi), \quad \mathfrak{n} = \mathfrak{m}_{W,x} R, \quad D = \text{ord } \psi \geq 3.$$

Theorem 4.13 gives normality and Cohen–Macaulayness near x . We factor the actual convergent series ψ , absorbing its unit into a coordinate. An irreducible first power gives a factorial ring by Theorem 15.13. In the present nonfactorial case the exhaustive remaining alternatives are a product of at least two distinct irreducible factors with arbitrary positive exponents, a singular irreducible power a^{M_ψ} with $r_a = \text{ord } a \geq 2$ and $M_\psi \geq 2$, and a smooth power, analytically $z_1^{M_\psi}$ with $M_\psi = D \geq 3$.

For real $0 \leq h \leq \ell$, define

$$\Phi_{h,\ell}(w) = (\ell - w)_+^5 - (h - w)_+^5.$$

Let P be an actual fourfold prime successor of embedding dimension six, and put $\mu = \text{mult}(R/P)$. Its quotient is normal Cohen–Macaulay. Theorem 15.13, with $d = 5$, gives

$$(20.1) \quad \text{length} \frac{P^{(N)} \cap \mathfrak{n}^h}{P^{(N)} \cap \mathfrak{n}^\ell} \leq \frac{1}{120} \left\{ \frac{D}{D-1} \Phi_{h,\ell}(N\beta) + \frac{D-2}{D-1} \Phi_{h,\ell}(N\zeta) \right\} + C_D(\ell+1)^4$$

for integer $N \geq 1$ and integer interval endpoints, where

$$\alpha + \beta = \mu, \quad \alpha \geq \beta \geq 1 + 1/D, \quad \zeta = \frac{(D-1)\alpha - \beta}{D-2} \geq \beta.$$

This is an estimate for the entire interval of the full symbolic module. The constant is uniform in the prime, symbolic exponent, endpoint values, character and higher coefficients at this fixed

germ. In both power families the full opposite-character module and every pullback component are included in this estimate.

If $P = (f)$ is Cartier, embedding dimension six implies $f \in \mathfrak{n}^2$. The full principal estimate of [Theorem 15.10](#) therefore gives

$$(20.2) \quad \text{length} \frac{(f^N) \cap \mathfrak{n}^h}{(f^N) \cap \mathfrak{n}^\ell} \leq \frac{2}{120} \Phi_{h,\ell}(2N) + C'_D(\ell + 1)^4.$$

No division by an initial form is used. A Cartier cubic cannot occur: reduce the four-dimensional Cohen–Macaulay complete intersection $R/(f)$ by four general linear parameters. The result has two variables and two equations of order at least two. Its graded lengths in degrees zero, one and two are at least 1, 2, 1, so its length, and hence μ , is at least four.

Step 2. In this step, we fix the intrinsic and ordinary lifting cutoffs before choosing the original degree. The ample Hilbert polynomial on the original integral projective W , together with the full local Hilbert–Samuel polynomial, gives a constant C_0 and a fixed starting range such that

$$(20.3) \quad \dim F_p(\lambda) \geq \frac{g - 2\lambda^5}{120} p^5 - C_0 p^4 \geq \frac{m_{\text{start}}}{120} p^5 - C_0 p^4, \quad m_{\text{start}} = \frac{458771731542349}{16000000000000000}.$$

Here $g \geq 1$ is the original integral degree, and $m_{\text{start}} = 1 - 2\lambda^5 > 0$. This lower bound is the dimension of a kernel; it assumes no jet-surjectivity. Local normality at x suffices for the local term.

Let C_1 bound both errors in (20.1) and (20.2), and put $C'_1 = C_1(\theta + 1)^4$. This bounds their errors by $C'_1 p^4$ whenever $\ell \leq p\theta$ and $p \geq 1$. Choose an intrinsic cutoff p_{sec} , within the starting range of (20.3), so that

$$(20.4) \quad \frac{120(C_0 + 235C'_1)}{p} < \frac{1}{4000} \quad (p \geq p_{\text{sec}}).$$

The fixed germ, all interval constants and the finite mesh precede this choice. In particular the cutoff is independent of every prime subsequently selected.

Separately apply [Theorem 6.8](#), namely [[Sta26](#), Lemma 30.17.1, Tag 0B5U], to the fixed coherent ideal I_W on X . It gives p_{lift} with

$$H^1(X, I_W \otimes \mathcal{O}_X(pL)) = 0 \quad (p \geq p_{\text{lift}}).$$

The restriction sequence then gives

$$H^0(X, pL) \longrightarrow H^0(W, pA) \quad \text{surjective for every } p \geq p_{\text{lift}}.$$

Fix any sufficiently large p divisible by 2000 above both cutoffs. Every exponent $p\gamma_j$ is then an integer, and $F_p(\gamma_j)$ is exactly the complete source of [Definition 2.4](#). All restriction and remote conditions on W remain in it.

Step 3. In this step, we construct each actual source threshold and bound its whole continuation cost. Whenever $F_p(\gamma) \neq 0$, let

$$\mathcal{V}_p(\gamma) = \{\varsigma \in H^0(X, pL) : \varsigma|_W \in F_p(\gamma)\}$$

be its full ambient inverse image. Take a sufficiently general tuple of its actual divisors $D_1, \dots, D_Q$, and normalize

$$H = \frac{1}{pQ} \sum_{i=1}^Q D_i \sim_{\mathbb{Q}} L.$$

Choose the tuple dilute enough that every prime of $H|_W$ which is not fixed in the complete restricted source has coefficient less than $1/10$. A general tuple avoids shared moving prime components, so increasing Q achieves this. The tuple is chosen before its threshold. Principalizing the full source ideal together with B , and taking general members there, realizes its full-ideal threshold by this average.

We have $W \not\subset \text{Supp } H$ and $\text{ord}_x(H|_W) \geq \gamma$. Its own threshold t against B is positive, finite and rational. Indeed, old lc places through x have centers containing W , and hence have zero H -order. The remaining conditions on a log resolution give positivity. Every member vanishes at x , so the point blowup gives finiteness. An old lc place dominating W persists, whereas a newly computing place of positive H -order cannot dominate W . The center-intersection assertion of [Theorem 3.11](#) [[GL24](#), Lemma 2.7] therefore gives an actual minimal center $U \subsetneq W$ for $B + tH$.

Write $y = \text{def}_x(B + tH)$, setting $y = 0$ for a point center. Apply [Theorem 3.8](#), or [[YZ20](#), Definition 2.5 and Lemma 2.10], on the smooth klt background $(X, 0)$ with the additions B and tH . Lemma [3.13](#) identifies the deficit conventions. The unchanged-boundary inequality is

$$(20.5) \quad y \leq b - \gamma t, \quad t \leq \frac{b - y}{\gamma}.$$

Put

$$B_* = 18/5, \quad r_0 = 25/27, \quad \kappa(\gamma) = (7 - B_*)r_0 + B_*/\gamma.$$

Use the state set and successor caps of [Definition 5.2](#). Replace the radius of each state $v = (d_v, e_v, m_v)$ by the rational upper radius q_v in the table below, and denote the resulting majorant by $\widehat{\mathcal{F}}_v$. For $0 \leq C \leq \beta_v$, set

$$J_v(h, C) = \max_{0 \leq z \leq C} (\widehat{\mathcal{F}}_v(z) - hz).$$

If $c_{vw} = \min(\beta_w, a_{vw})$, the defining recurrence becomes

$$(20.6) \quad J_v(h, C) = (q_v - h)_+ C + \max \left(0, \max_{v \rightarrow w} J_w(\max(h, q_v), \min(C, c_{vw})) \right).$$

For a dimension skip, $a_{vw} = +\infty$. This is the support form of the already established complete-tail recurrence: if $q_v \geq h$, its outer maximum occurs at C ; otherwise, for a fixed child deficit, it occurs at that deficit. It includes every smaller dimension and every embedding dimension at most e_v , discarding only nonpositive caps.

For a first fourfold successor, [Theorem 4.16](#) and the double parent give the additional cap $(8 - m_v)/2$. Indeed, the length-two Artinian reduction has first moment one, so [Theorem 4.5](#) [[Han26](#), Lemmas 4.2 and 4.4] gives $\bar{e}_1 \geq 1$. All its hypotheses hold for the excellent normal Cohen–Macaulay local ring at x , with residue field $\mathbb{C}$. For a first successor of dimension at most three there is no adjacent root cap. Write C_v for the intersection of these bounds with β_v . By [Theorem 5.4](#), the price before cutting errors is

$$(20.7) \quad s + t + \widehat{\mathcal{F}}_U(y) \leq (n - 7)r_0 + \kappa(\gamma) + J_U(1/\gamma, \min(b, C_U)).$$

For a point, take the last term to be zero. Before maximization the γ -dependent expression is

$$(7 - B_*)r_0 + \frac{B_* - y}{\gamma} + \widehat{\mathcal{F}}_U(y).$$

It decreases with γ , since $0 \leq y \leq B_*$.

Step 4. In this step, we evaluate every changed-center state and identify the only unpaid prime types. The following is the complete first-state calculation at λ . The moment column is $T_v = T(e_v - d_v, m_v)$. The last column is $\kappa(\lambda) + J_v(1/\lambda, \min(B_*, C_v))$, except that the row $(3, 6, 4)$ uses first deficit at most one. That restriction applies only to this entry, not to its use as a descendant in [\(20.6\)](#).

(d_v, e_v, m_v)	T_v	C_v	q_v	first price
(1, 1, 1)	0	1	1	113845/15579
(2, 2, 1)	0	2	1	113845/15579
(2, 3, 2)	1	1	283/200	23577857/3115800
(2, 4, 3)	2	2/3	1733/1000	59921969/7789500
(2, 5, 4)	3	1/2	2	120424/15579
(2, 6, 5)	4	2/5	2237/1000	301462723/38947500
(3, 3, 1)	0	3	1	23577857/3115800
(3, 4, 2)	1	2	63/50	241577027/31158000
(3, 4, 3)	3	1	1443/1000	118325497/15579000
(3, 5, 3)	2	5/3	1443/1000	24864887/3115800
(3, 5, 4)	4	1	397/250	484596763/62316000
(3, 5, 5)	6	3/5	171/100	59514527/7789500
(3, 5, 6)	8	1/3	909/500	58642937/7789500
(3, 6, 4)	3	3/2	397/250	61896863/7789500 for $y \leq 1$
(3, 6, 5)	5	1	171/100	
(3, 6, 6)	7	2/3	909/500	30181687/3894750
(3, 6, 7)	9	3/7	1913/1000	832322881/109053000
(3, 6, 8)	11	1/4	2	234269/31158
(3, 6, 9)	13	1/9	2081/1000	115447211/15579000
(4, 4, 1)	0	7/2	1	241577027/31158000
(4, 5, 2)	1	3	119/100	248772401/31158000
(4, 5, 3)	3	2	1317/1000	248400629/31158000
(4, 5, 4)	6	1	283/200	48978457/6231600
(4, 6, 3)	2	5/2	1317/1000	257954687/31158000
(4, 6, 4)	4	2	283/200	127838011/15579000
(4, 6, 5)	6	3/2	187/125	316054729/38947500
(4, 6, 6)	8	1	783/500	155528897/19473750
(4, 6, 7)	11	1/2	1627/1000	48527917/6231600

Each $q_v^{d_v} \geq m_v$, so every original positive integral degree is allowed by these radius bounds. Surfaces have $m_v = e_v - 1$ and $\beta_v = 2/m_v$. For threefold embeddings 3, 4, 5, 6, positivity of β_v gives respectively $m_v = 1, 2, 3, 3, \dots, 6$, and $4, \dots, 9$. The next multiplicity has nonpositive cap and T_v/m_v does not decrease thereafter. For fourfold embedding five the range is 2, 3, 4; for embedding six the root cap excludes $m_v \geq 8$. This proves exhaustiveness of the table.

All its entries can be checked directly from (20.6). For a surface the correction is $(q_v - h)_+ \min(C, 2/m_v)$, since $h \geq 1$ in this calculation. For a threefold, test every surface multiplicity $1 \leq k \leq e_v - 1$, with cap

$$\min\left(\frac{2}{k}, \frac{3m_v - 2T_v - k}{m_v}\right),$$

discarding nonpositive values, and also the direct curve. For a fourfold, insert every threefold row with its adjacent cap and every direct surface with cap $2/k$. Direct curves are also included and give zero correction. Thus the table requires only the displayed finite rational maxima.

The canonical family is **20-rank-two-feedback/states**. Its records are in

`supplement/20-rank-two-feedback/data/state_rows.json`.

They contain these 28 states, their intrinsic and first-successor caps, both endpoint price columns, and all 170 successor records. The printed table is its first-price view. For the terminal column, (4, 6, 3) carries its affine price at deficit 7/3; every other row carries its complete scalar price. To enumerate independently, fill the degree- i slots of the Artinian moment with capacity $\binom{c+i-1}{i}$, where $c = e_v - d_v > 0$. Put $\ell_v = \lfloor d_v/2 \rfloor + 1$. Since

$$T(c, m_v) \geq \ell_v \left(m_v - \binom{c + \ell_v - 1}{\ell_v - 1} \right),$$

positivity of the intrinsic cap implies

$$m_v \leq \left\lfloor \frac{2\ell_v \binom{c+\ell_v-1}{\ell_v-1} - 1}{2\ell_v - d_v} \right\rfloor.$$

For $c = 0$, only $m_v = 1$ is allowed. This finite range, the first fourfold cap, and every smaller dimension with nonincreasing embedding dimension reproduce exactly the stated records. Evaluate (20.6) in increasing dimension and compare every supplied moment, radius, cap, successor and price. At $C = 0$ the support maximum is zero; empty successor sets also contribute zero. The recurrence retains both endpoints when a deficit interval is a segment and its sole value when the interval is a point.

In particular, the threefold rows of multiplicities eight and nine retain their complete tails. Lemma 5.10 gives respectively $y \leq 1/4$, $\mathcal{F}(y) \leq 2y$, and $y \leq 1/9$, $\mathcal{F}(y) \leq 9y/4$, with all skips. The embedding-seven caps $1/2$ and $1/3$ belong to the ambient state set in that lemma. Here every successor embeds in the actual embedding-six W , so none has embedding seven.

Except for $(4, 6, 3)$, $(4, 6, 4)$, $(4, 6, 5)$, the largest displayed first price is

$$(20.8) \quad \frac{155528897}{19473750} < \frac{799}{100}.$$

For the remaining direct quartic threefold with $1 < y \leq 3/2$, the actual whole affine price satisfies

$$\begin{aligned} s + t + \frac{5000}{3997}y &\leq (n - 7)r_0 + \kappa(\lambda) + \left(\frac{5000}{3997} - \frac{1}{\lambda}\right) \frac{3}{2} \\ &= (n - 7)r_0 + \frac{463961965}{62269263} < n + \frac{49267}{107919}. \end{aligned}$$

Theorem 11.1 applies to this entire $B + tH$, at its actual degree and with its full tangent scheme. It supplies the additive margin $1/200000$. Monotonicity of the unmaximized price extends these payments to every $\gamma \in [\lambda, \theta]$. Only actual fourfold primes of embedding six and multiplicities $\mu = 3, 4, 5$ can remain.

Step 5. In this step, we derive the unpaid deficit and full fixed-order inequalities with the old coefficient retained. The same recurrence gives, on the respective full first-deficit ranges,

$$\widehat{\mathcal{F}}_{(4,6,4)}(y) \leq \frac{283}{200}y + \frac{379}{1000}, \quad \widehat{\mathcal{F}}_{(4,6,5)}(y) \leq \frac{187}{125}y + \frac{741}{2500}.$$

For cubics use Theorem 12.3. Set

μ	3	4	5
q_μ	1317/1000	283/200	187/125
p_μ	0	379/1000	741/2500
T_μ	$7 + 51/100$	$8 - 1/500 - p_4$	$8 - 1/500 - p_5$

Here p_μ are tail intercepts, distinct from the fixed original degree p . For $\mu = 3, 4, 5$, define

$$(20.9) \quad \begin{aligned} A_\mu &= q_\mu - r_0, & C_\mu &= T_\mu - 7r_0, & D_\mu &= A_\mu B_* - C_\mu, \\ R_\mu(\gamma) &= \frac{q_\mu \gamma - 1}{D_\mu}, & Y_\mu(\gamma) &= \frac{T_\mu - \kappa(\gamma)}{q_\mu - 1/\gamma}. \end{aligned}$$

Their exact coefficients are

μ	A_μ	C_μ	D_μ
3	10559/27000	2777/2700	12803/33750
4	2641/5400	30713/27000	673/1080
5	1924/3375	41179/33750	5617/6750

All denominators and $q_\mu \gamma - 1$ are positive throughout the mesh. Reserve total weighted cutting error less than $1/1000$. If a noncubic full scalar price is at most $n + 1 - 1/500$, it is paid with

margin greater than $1/1000$. A cubic is paid whenever its whole affine price is at most $n + 51/100$. Consequently, failure of payment at this trial implies

$$(20.10) \quad y > Y_\mu(\gamma), \quad 0 < (q_\mu\gamma - 1)t < A_\mu b - C_\mu.$$

For the second inequality use $s \leq (n - b)r_0$ and (20.5) in the failed affine price. The additional term $(n - 7)(1 - r_0)$ on its lower side is nonnegative; discarding it gives the displayed weaker inequality. The first inequality follows in the same way from (20.7) before maximization.

Let c_P be the coefficient of P in the old direct discriminant, allowing $c_P = 0$, and let η_P be its coefficient in the entire $H|_W$. We spell out the same-parent addition used with Proposition 4.18. In the direct discriminant construction of Theorem 4.1 [FH23, Theorem 1.2 and Definition 1.3], let $a_{P,T}(0)$ be the generic threshold over P on an ambient lc place T dominating W . The restriction of H there is pulled back from $H|_W$. Hence $a_{P,T}(t) = a_{P,T}(0) - t\eta_P$ for every such T , and taking one minus their infimum gives the new coefficient $c_P + t\eta_P$. Choose the representatives of the adjoint class compatibly by adding the same Cartier pullback. Its contribution then cancels from the definition of the moduli part, so the moduli b-divisor remains the same. Since the pair is lc and P is its actual codimension-one lc subcenter, coefficient-one detection gives

$$(20.11) \quad c_P + t\eta_P = 1.$$

Suppose an individual retained inequality is $b + Kc_P \leq 4$, where $K > 0$ and $C_\mu > (4 - K)A_\mu$. Combining (20.10) and (20.11) gives $b > 4 - K$ and

$$(20.12) \quad \eta_P \geq \frac{(b - 4 + K)(q_\mu\gamma - 1)}{K(A_\mu b - C_\mu)} \geq \frac{B_* - 4 + K}{K} R_\mu(\gamma).$$

Indeed the derivative of $(b - 4 + K)/(A_\mu b - C_\mu)$ has numerator $(4 - K)A_\mu - C_\mu < 0$, so its minimum occurs at $b = B_*$. The individual inequality is a consequence of the retained discriminant budget; all other old primes and all compatible moduli terms remain in that budget.

The identities

$$(20.13) \quad Y_\mu = B_* - \frac{D_\mu\gamma}{q_\mu\gamma - 1}, \quad R_\mu(B_* - Y_\mu) = \gamma$$

show that every Y_μ increases strictly. In particular,

$$Y_4(\lambda) > 1, \quad Y_5(\lambda) > 1, \quad Y_3(\theta) > 7/3, \quad Y_4(\theta) > 2, \quad Y_5(\theta) > 3/2.$$

Step 6. In this step, we determine the applicable old-coefficient and successor bounds in every analytic factor family. For a Cartier prime the individual budget gives $b + (\mu/2)c_P \leq 4$: indeed $2b \leq K_W \leq 8 - \mu c_P$, retaining the other nonnegative subtracted terms in Proposition 4.18. Its full principal shift is at least $2\eta_P$. The adjacent new caps are two for $\mu = 4$ and $3/2$ for $\mu = 5$.

For non-Cartier primes in the distinct-factor family, Theorem 15.13 gives

$$b + \frac{\mu + 1}{2}c_P \leq 4, \quad y \leq \frac{7 - \mu}{2}.$$

Thus cubics have $K = 2$ and $y \leq 2$; quartics have $K = 5/2$ and $y \leq 3/2$. Quintics have $y \leq 1$, so $Y_5(\lambda) > 1$ already pays them.

For a singular irreducible power a^{M_ψ} , a non-Cartier character satisfies $1 \leq k < M_\psi$. The same theorem gives

$$b + \left(1 + \frac{r_a k(M_\psi - k)}{M_\psi}\right)c_P \leq 4, \quad y \leq 3 - \frac{r_a k(M_\psi - k)}{M_\psi} \leq 2.$$

Since $r_a \geq 2$, the old coefficient factor is at least two. For cubics we may use $K = 2$, obtaining $\eta_P \geq (4/5)R_3$. These cubics are paid once $Y_3 \geq 2$.

For a smooth power, put $\xi = 1/M_\psi \in (0, 1/3]$. Every cubic is non-Cartier, and Theorem 15.13 gives

$$\beta \geq 1 + \xi, \quad y \leq 2 + \xi, \quad b + (2 - \xi)c_P \leq 4.$$

Hence

$$(20.14) \quad \eta_P \geq \chi(\xi)R_3, \quad \chi(\xi) = \frac{8-5\xi}{10-5\xi}, \quad \xi > Y_3 - 2 \quad \text{in an unpaid trial.}$$

In particular every possible cubic in every analytic family has $y \leq 7/3$.

For quartics and quintics in either single-power family retain both endpoint comparisons of [Theorem 15.13](#):

$$(20.15) \quad b + \alpha c_P \leq 4, \quad y \leq 4 - \alpha, \quad \alpha + \beta = \mu, \quad \alpha \geq \beta.$$

These inequalities use the same original discriminant and compatible moduli data before and after addition. In particular the old comparison is with the whole B ; the new one is with the actual coefficient-one prime of $B + tH$.

For a quartic, $\alpha = 4 - \beta$, $\beta \leq 2$, and (20.15) forces $\beta > Y_4 > 1$ under failure. Apply (20.12) with $K = 4 - \beta$. The function

$$\frac{\beta(B_* - \beta)}{4 - \beta}$$

increases on $[1, 2]$: its derivative numerator is $4B_* - 8\beta + \beta^2 \geq 4B_* - 12 > 0$. Together with (20.13), this gives

$$(20.16) \quad \beta\eta_P \geq z_4(\gamma), \quad z_4(\gamma) = \frac{\gamma Y_4}{4 - Y_4}.$$

Such a failed quartic is impossible when $Y_4 \geq 2$. For a quintic, $\alpha = 5 - \beta$, $\beta \leq 5/2$, and failure forces $\beta > 1 + Y_5$. Now

$$\frac{\beta(B_* + 1 - \beta)}{5 - \beta}$$

increases on $[0, 5/2]$, since its derivative numerator is $5(B_* + 1) - 10\beta + \beta^2 \geq 17/4 > 0$. Thus

$$(20.17) \quad \beta\eta_P \geq z_5(\gamma), \quad z_5(\gamma) = \frac{\gamma(1 + Y_5)}{4 - Y_5}.$$

Such a failed quintic is impossible when $Y_5 \geq 3/2$.

All uses of (20.12) satisfy its sign condition, as the displayed fractions give

$$C_3 > \frac{7}{3}A_3, \quad C_4 > 2A_4, \quad C_5 > \frac{3}{2}A_5.$$

The smallest allowed K 's are respectively $5/3, 2, 5/2$. The resulting universal lower bounds at the first mesh point are

$$\frac{19}{25}R_3(\lambda) > \frac{1}{4}, \quad \frac{4}{5}R_4(\lambda) > \frac{1}{4}, \quad \frac{21}{25}R_5(\lambda) > \frac{1}{4}.$$

They increase thereafter. The dilution chosen before each threshold makes every nonfixed coefficient less than $1/10$. Thus an unpaid prime must be in the full fixed divisor of the original complete source. If its actual integer fixed order is N , then $\eta_P = N/p$ exactly. Every section in the source has this symbolic order, so the full interval estimates apply to its entire image.

Step 7. In this step, we bound the full symbolic intervals over all continuous endpoint parameters. Fix one consecutive interval $\gamma = j/2000 < \gamma' = (j+1)/2000$, and write

$$\Phi(w) = \Phi_{\gamma, \gamma'}(w) = \int_{\gamma}^{\gamma'} 5(t' - w)_+^4 dt'.$$

This function is nonnegative, decreasing and convex, since $\Phi''(w) = 20((\gamma' - w)_+^3 - (\gamma - w)_+^3) \geq 0$. Homogeneity and $\eta_P = N/p$ give

$$\Phi_{p\gamma, p\gamma'}(Nw) = p^5 \Phi(\eta_P w).$$

For an unpaid single-power quartic or quintic, (20.16) and (20.17) give $\eta_P\beta \geq z_\mu(\gamma)$, with $\mu = 4, 5$, respectively. The two positive coefficients in (20.1) sum to two. Since $\zeta \geq \beta$, its normalized leading contribution satisfies

$$\frac{D}{D-1}\Phi(\eta_P\beta) + \frac{D-2}{D-1}\Phi(\eta_P\zeta) \leq 2\Phi(\eta_P\beta) \leq 2\Phi(z_\mu(\gamma)).$$

Thus the actual full interval satisfies

$$\text{length } \frac{P^{(N)} \cap \mathfrak{n}^{p\gamma}}{P^{(N)} \cap \mathfrak{n}^{p\gamma'}} \leq \frac{2p^5}{120}\Phi(z_\mu(\gamma)) + C'_1 p^4,$$

with the same constant C'_1 fixed before the prime.

At $\eta = N/p$, the normalized leading expression in (20.1) is convex in $\beta \in [1 + 1/D, \mu/2]$, because both shifts are affine in β . Its maximum is at an endpoint. The balanced endpoint is $2\Phi(\eta\mu/2)$, at most $\Phi(\eta) + \Phi((\mu-1)\eta)$ by convexity.

At $\beta = 1 + 1/D$, put $z = 1/D \in (0, 1/3]$. The full expression is

$$(20.18) \quad \frac{\Phi(\eta(1+z))}{1-z} + \frac{1-2z}{1-z}\Phi\left(\eta\left[\mu-1 + \frac{(\mu-3)z}{1-2z}\right]\right).$$

If $\gamma' \leq 4\eta$, the first integrand decreases with z : where it is positive, its derivative has the sign of $t' - 5\eta + 3\eta z \leq t' - 4\eta \leq 0$. The second coefficient decreases and its shift increases for $\mu \geq 3$. The limiting value at $z = 0$ therefore bounds the entire expression in (20.18) by

$$(20.19) \quad \Phi(\eta) + \Phi((\mu-1)\eta).$$

This includes both branches. For the uses $\eta = (4/5)R_3(\gamma)$ and $\eta = (21/25)R_4(\gamma)$, substitution at λ gives $4\eta - \lambda > 1/2000$. The difference $4\eta - \gamma$ increases linearly thereafter, so $\gamma' \leq 4\eta$ on every mesh interval. Lowering an actual η_P to these bounds only increases its interval.

For a smooth-power cubic, convexity reduces $\beta \in [1 + \xi, 3/2]$ to its two endpoints. At $\beta = 1 + \xi$ the other shift is $2\eta_P$; at $\beta = 3/2$ both shifts equal $3\eta_P/2$. Put

$$\xi_* = (Y_3(\gamma) - 2)_+, \quad \chi_1(\xi) = (1 + \xi)\chi(\xi).$$

An unpaid smooth-power cubic requires $\xi > Y_3 - 2$ and $0 < \xi \leq 1/3$. By (20.14), its extreme endpoint is bounded by

$$(20.20) \quad \frac{\Phi(R_3\chi_1(\xi))}{1-\xi} + \frac{1-2\xi}{1-\xi}\Phi(2R_3\chi(\xi)).$$

The first term of (20.20) decreases throughout $[0, 1/3]$. Indeed its integrand derivative has the sign of $t' - R_3[\chi_1 + 4(1-\xi)\chi'_1]$, and

$$\chi_1 + 4(1-\xi)\chi'_1 - \frac{5}{2} = \frac{110 - 440\xi + (745/2)\xi^2 - 75\xi^3}{(10 - 5\xi)^2} \geq 0.$$

The numerator decreases on $[0, 1/3]$, since its derivative is $-440 + 745\xi - 225\xi^2 < 0$, and at $1/3$ it equals $35/18 > 0$. Also $(5/2)R_3(\lambda) - \lambda > 1/2000$, and the difference $(5/2)R_3(\gamma) - \gamma$ increases linearly, because $(5/2)q_3/D_3 > 1$. Hence $\gamma' \leq (5/2)R_3(\gamma)$ on every interval, proving the asserted monotonicity.

The function χ decreases from $4/5$ to $19/25$, and $(1 - 2\xi)/(1 - \xi)$ decreases as well. Thus, when $Y_3 < 7/3$, the two endpoints are bounded by

$$(20.21) \quad \begin{aligned} C_E &= \frac{\Phi(R_3\chi_1(\xi_*))}{1-\xi_*} + \frac{1-2\xi_*}{1-\xi_*}\Phi\left(\frac{38}{25}R_3\right), \\ C_B &= 2\Phi\left(\frac{57}{50}R_3\right). \end{aligned}$$

When $Y_3 \geq 7/3$, no smooth-power cubic remains unpaid. These comparisons range over the full real intervals for $1/D$, $1/M_\psi$ and β . They therefore cover every factor order, every character and every endpoint valuation, including unbounded orders and all higher analytic terms. An empty allowed interval for β contains no prime; if its endpoints coincide, the convexity bound evaluates

that single value. The limiting value $\xi = 0$ is included by continuity, and $\xi = 1/3$ is retained. If $\xi_* \geq 1/3$, the strict unpaid condition leaves no smooth-power cubic. Thus these reductions also cover the degenerate parameter domains and their endpoints.

Step 8. In this step, we verify every candidate interval bound and the complete positive mass reserve. At a left mesh point, evaluate R_μ, Y_μ at γ . The following table lists every candidate for the normalized leading interval loss of an unpaid prime. Include a row exactly when its middle condition holds.

family	condition	candidate bounds
smooth-power cubic	$Y_3 < 7/3$	C_E, C_B
other non-Cartier cubic	$Y_3 < 2$	$\Phi(4R_3/5) + \Phi(8R_3/5)$
distinct-factor quartic	$Y_4 < 3/2$	$\Phi(21R_4/25) + \Phi(63R_4/25)$
Cartier quartic	$Y_4 < 2$	$2\Phi(8R_4/5)$
single-power quartic	$Y_4 < 2$	$2\Phi(\gamma Y_4/(4 - Y_4))$
Cartier quintic	$Y_5 < 3/2$	$2\Phi(42R_5/25)$
single-power quintic	$Y_5 < 3/2$	$2\Phi(\gamma(1 + Y_5)/(4 - Y_5))$

The other-cubic and distinct-quartic entries use (20.12) with $K = 2, 5/2$, respectively, and (20.19). The Cartier entries use $K = \mu/2$ and the full principal shift $2\eta_P$. The single-power entries use (20.16) and (20.17). Distinct-factor quintics are already paid, and Cartier cubics do not exist. At an equality in an inclusion condition the corresponding deficit cap pays the trial, so the omitted endpoint has not been lost.

For $1731 \leq j \leq 1965$, let L_j be the maximum of zero and every included candidate in this table. There are seven family rows and eight competing expressions, since both smooth cubic expressions are retained. The maximum concerns one actual prime at one trial. The complete family `20-rank-two-feedback/mesh` is supplied in `supplement/20-rank-two-feedback/data/mesh_rows.json`. Its 235 records have exactly the indices 1731, ..., 1965, representing the consecutive intervals $[j/2000, (j+1)/2000]$ and all 236 endpoints. Each record contains its left endpoint, the three values Y_μ , every included candidate with its exact rational value, their maximum, all maximizing names, the upper integer E_j , and the remaining mass after that interval.

Here is the full checking rule. At each prescribed index, substitute $\gamma = j/2000$, $\gamma' = (j+1)/2000$ in (20.9). Apply the strict inclusion conditions in the table, and in the smooth row use $\xi_* = (Y_3 - 2)_+$ in (20.21). Require the supplied candidate names to be exactly this set and recompute every value; retaining just a recorded maximizer would not check coverage. Take the maximum together with zero, preserving all ties. Empty candidate sets give zero. Positive parts in Φ are computed by exact sign comparisons, including equality at $w = \gamma$ and $w = \gamma'$. Every denominator is positive on the corresponding inclusion range, so all comparisons reduce to integer multiplication by positive denominators.

The integer convention is ordinary upward rounding, $E_j = \lceil 10^9 L_j \rceil$. For these supplied records the stronger brackets

$$E_j - 1 < 10^9 L_j < E_j \quad (1731 \leq j \leq 1965)$$

hold. Thus ordinary ceiling and strict floor-plus-one agree on this dataset; no such agreement is assumed for an integral scaled maximum. All 1266 included candidate values satisfy the corresponding bound. In particular,

$$(20.22) \quad 10^9 L_j \leq E_j \quad (1731 \leq j \leq 1965), \quad \sum_{j=1731}^{1965} E_j = 28148081.$$

Finally recompute every prefix sum and its remainder from m_{start} . Missing, duplicate, conflicting or out-of-range records, extra fields and noncanonical rational values are rejected. The independent entry point `supplement/20-rank-two-feedback/check.py` implements these finite rules with exact rational arithmetic and also checks the state family above; its documented invocation is `check.py -data data -report report.json` with Python 3. The integer rounding is the only

numerical overestimate. The passage from these inequalities to source dimensions is [Lemma 20.3](#), applied below after the geometric interval injection has been established.

Using (20.22), the remaining normalized leading mass is

$$(20.23) \quad \begin{aligned} m_{\text{rem}} &:= m_{\text{start}} - \frac{28148081}{10^9} = \frac{8402435542349}{16000000000000000}, \\ m_{\text{rem}} - \frac{1}{2000} &= \frac{402435542349}{16000000000000000} > 0. \end{aligned}$$

All losses are nonnegative, so every partial remainder is at least this final remainder.

Step 9. In this step, we propagate the complete original sources at the fixed degree. The starting source is nonzero by (20.3), (20.4), and (20.23). Suppose that $F_p(\gamma_j)$ has been earned. Take its own full inverse-image threshold from B . If that trial is unpaid, the preceding argument gives its actual full fixed prime of integer order N . After trivializing pA at x , evaluation of germs gives an injection

$$\frac{F_p(\gamma_j)}{F_p(\gamma_{j+1})} \hookrightarrow \frac{P^{(N)} \cap \mathfrak{n}^{p\gamma_j}}{P^{(N)} \cap \mathfrak{n}^{p\gamma_{j+1}}}.$$

The kernel before taking the quotient is exactly $F_p(\gamma_{j+1})$: every section already has the fixed prime order, and the additional condition is precisely the next ordinary point-ideal exponent. Analytification preserves these finite-length quotients. This uses the full original global source; conditions away from x are unchanged.

Equations (20.1) and (20.2), with the complete candidate maximum, imply

$$(20.24) \quad \dim F_p(\gamma_j) - \dim F_p(\gamma_{j+1}) \leq \frac{L_j p^5}{120} + C'_1 p^4.$$

The full symbolic or principal module retains both branches, all opposite characters and every pullback component. The source quotient retains every fixed, residual, embedded and conductor condition. None is replaced by reduced support or by a section space on another model.

Apply [Lemma 20.3](#) to any earned partial sequence, with $Q_0 = 10^9$, $M_0 = m_{\text{start}}$, $a_0 = C_0$, and $a_1 = C'_1$. The geometric injection supplies its dimension-loss hypothesis at every unpaid step. By the uniform cutoff this gives

$$\frac{120}{p^5} \dim F_p(\gamma_{j+1}) \geq m_{\text{rem}} - \frac{120(C_0 + 235C'_1)}{p} > \frac{1}{4000}.$$

Thus every next complete source is nonzero before its threshold is selected. If no earlier trial pays, the same calculation at θ gives the asserted lower bound $m_{\text{rem}} + O(1/p)$, with a uniform signed error bounded below as displayed. No recurrence of the prime is needed.

Step 10. In this step, we pay the terminal threshold and reserve both cutting errors in the final additive boundary. Every threshold of the earned terminal source is paid. The earlier scalar and quartic-threefold payments hold throughout the mesh. For the three remaining multiplicities, the strict comparisons

$$Y_3(\theta) > 7/3, \quad Y_4(\theta) > 2, \quad Y_5(\theta) > 3/2$$

contradict the full cubic cap and the two adjacent caps in an unpaid trial. The terminal whole-price bounds can also be written explicitly: the maximum noncubic scalar price, after subtracting $(n-7)r_0$, is

$$\frac{211925069}{26541000} < \frac{7998}{1000},$$

and the cubic affine price at deficit $7/3$, after the same subtraction, is

$$\frac{199315493}{26541000} < \frac{751}{100}.$$

For any successful scalar branch apply [Theorem 5.4](#) with the published two-error cutter of [Theorem 3.14](#) [[GL24](#), Proposition 3.2 and Remark 3.3]. There are at most four further cuts. Once its actual radius ρ_i is known, choose the approximation parameter

$$0 < \epsilon_i < \frac{1}{4000(1 + \rho_i)}$$

before selecting that cutter. Both inequalities are retained:

$$t_i \leq b_i \rho_i + \epsilon_i, \quad b_{i+1} \leq b_i - t_i / \rho_i + \epsilon_i.$$

In particular $t_i \leq (b_i - b_{i+1})\rho_i + \rho_i \epsilon_i$, and the total allowance for both errors is bounded by $\sum_i (1 + \rho_i) \epsilon_i < 1/1000$. The base scalar price at most $n + 1 - 1/500$ therefore gives total coefficient

$$< n + 1 - 1/1000 < n + 1 - 1/400000.$$

The other scalar rows have the larger margin in (20.8). Cubics use [Theorem 12.3](#), and direct quartic threefolds use [Theorem 11.1](#); both endpoints already include their cutting errors and the whole incoming price.

Only the successful trial and its actual continuation enter the final additive boundary. Each trial started from the entire original B ; no old coefficient is subtracted, and the different trial divisors are never summed into one chain. This proves the proposition. $\square$

Proof of [Theorem 20.1](#). The actual center is normal Cohen–Macaulay near x by [Theorem 4.13](#). Its embedding codimension one and multiplicity two give an order-two hypersurface equation. Apply the convergent splitting assertion of [Theorem 15.1](#) [[GP26](#), Theorem 2.4] to its rank-two quadratic part. Over $\mathbb{C}$ the two quadratic coordinates can be written as uv , leaving $uv - \psi(z)$ with $\text{ord } \psi \geq 3$. Normality and integrality exclude $\psi = 0$.

If the analytic ring is factorial, its quadratic tangent $uv = 0$ is reduced, and [Theorem 18.4](#) applies with the given whole price and $b \leq 18/5$. Otherwise [Proposition 20.2](#) gives a paid trial at one sufficiently large original degree and an additive point continuation of the entire B with the stated strict margin. The factorization alternatives in that proof are exhaustive, and its only degree estimate is the original $g \geq 1$. $\square$

Finally, [Proposition 3.12](#) applies to the resulting point boundary: its whole coefficient is less than $n + 1$, so the remaining multiple of the ample L is positive. Thus the same endpoint also gives generation of $K_X + (n + 1)L$ at x .

21. FREENESS IN DIMENSION SEVEN

We retain the original section spaces and the whole-boundary conventions of Section 2. We now apply the preceding endpoints to the complete list of centers at one original source.

Theorem 21.1. *For every smooth complex projective sevenfold X and every ample Cartier divisor L , the line bundle $\mathcal{O}_X(K_X + 8L)$ is globally generated.*

In particular, the assertion includes the case $L^7 = 1$.

Proof. We first reduce to an integral sevenfold of volume one. At a hypothetical point of nongeneration, we choose one complete source after fixing all required cutoffs. We then exclude each possible minimal center and pass from generation at closed points to global generation.

Step 1. In this step, we reduce to an integral sevenfold of volume one. The finitely many connected components of a smooth projective sevenfold are open and closed. Each is integral: smoothness implies reducedness, and a regular local ring is a domain, so distinct irreducible components cannot meet. Connectedness therefore leaves one irreducible component. Each component has dimension seven, the restriction of L is ample Cartier, and the canonical divisor restricts to its canonical divisor. It suffices to prove the assertion for integral X ; we return to the components at the end.

For integral X , ampleness and the Cartier hypothesis make L^7 a positive integer. Theorem 9.5 proves the assertion if $L^7 \geq 2$. Suppose henceforth that $L^7 = 1$, and assume that $K_X + 8L$ is not generated at a closed point x .

Step 2. In this step, we choose one complete original source after fixing all cutoffs and retain its whole threshold price. Theorem 9.14 gives $V(t) > 1/32$ for $0 \leq t \leq 27/25$, together with a cutoff after which $U_k(27/25) \neq 0$ for every 25-divisible original degree k . We use this finite-degree nonvanishing assertion.

Fix the source comparisons in that theorem and in Theorems 10.1, 13.1, 17.37, 17.40, 19.1, and Proposition 14.2. Take a common integer cutoff J at least as large as each of their original-source cutoffs. These choices depend only on (X, L, x) and the fixed comparisons, and precede the source degree, its general tuple, its center and its computing places. Choose one $k \geq J$ with $25 \mid k$, and set $a = 27k/25$. Thus a is an integer and the space

$$U_k(27/25) = H^0(X, \mathcal{O}_X(kL) \otimes \mathfrak{m}_x^a)$$

is nonzero and complete.

Let $\mathfrak{b}$ be its full base ideal and $\tau = \text{lct}_x(\mathfrak{b})$ its actual local threshold. The ideal is nonzero and is contained in the proper ideal $\mathfrak{m}_x^a$; consequently τ is positive and rational. Choose a sufficiently large sufficiently general tuple of actual divisors $D_1, \dots, D_N$ from this space and put

$$(21.1) \quad \Gamma = \frac{\tau}{N} \sum_{i=1}^N D_i, \quad s = k\tau.$$

The tuple satisfies the finitely many generality requirements above simultaneously. This pure threshold average includes the entire D_i , with every fixed component, including those away from x , weighted by the displayed factor τ/N .

Proposition 9.1 makes $\Gamma \sim_{\mathbb{Q}} sL$ effective rational and lc near x , but not klt at x . On this lc neighborhood the smooth ambient is klt and Γ is $\mathbb{Q}$ -Cartier. Thus Theorem 3.11 ([Fuj11, Theorem 9.1(4)]) gives a unique minimal lc center W through x . The generic point of X is klt, so W is proper. For this same boundary set

$$M = \text{mult}_x \Gamma, \quad b = \text{def}_x(\Gamma).$$

Proposition 9.1 gives the two inequalities

$$(21.2) \quad M \geq a\tau = \frac{27}{25}s, \quad M + b \leq 7.$$

In particular,

$$(21.3) \quad \frac{27}{25}s + b \leq 7, \quad 0 < s \leq \frac{175}{27} < 8.$$

The first inequality in (21.2) is the actual multiplicity hypothesis of the cubic-fourfold endpoint; (21.3) is the whole-boundary price of the rank-two endpoint.

Step 3. In this step, we identify the exhaustive list of actual minimal centers. If $W = \{x\}$, (21.3) and Proposition 3.12, with integer coefficient 8, already contradict nongeneration. For every positive-dimensional W , Proposition 5.9 applies to the complete source above: $k(27/25)$ is integral and $27/25 \geq 26/25$. It excludes every center of dimensions one, two and three. Its conclusion uses the complete descendant costs, including all dimension skips. Hence $\dim W \in \{4, 5, 6\}$.

Put

$$d = \dim W, \quad e = \text{edim } \mathcal{O}_{W,x}, \quad m = \text{mult}_x W.$$

Theorem 10.1 applies at our preselected degree and to the same pure average. Its exhaustive list is

$$(21.4) \quad (d, e, m) \in \{(6, 6, 1), (4, 6, 3), (4, 7, 4), (4, 7, 5), (5, 6, 2), (5, 6, 3), (5, 7, 3)\}.$$

In dimension six this excludes every singular divisor; the remaining divisor is smooth at x and has original degree one. All three fourfold states have full reduced Cohen–Macaulay tangent

rings. For the fivefold states $(6, 2)$, $(6, 3)$, $(7, 3)$, the respective Hilbert numerators are $1 + z$, $1 + z + z^2$, and $1 + 2z$. The triple tangent of embedding dimension six is reduced, whereas the triple tangent of embedding dimension seven is nonreduced. Reducedness in the double case will follow below. These are conclusions about the actual full tangent rings of W .

Step 4. In this step, we exclude the divisorial, fourfold, and triple-fivefold states at their unchanged whole coefficient. We apply the endpoints to Γ in (21.1), with its unchanged coefficient s .

If $(d, e, m) = (6, 6, 1)$, the actual center is a prime divisor smooth at x . The degree k satisfies $25 \mid k$ and is above the cutoff of Theorem 17.37, and the source is nonzero, complete and at its own full-ideal threshold. That theorem gives generation at x for this divisorial case, for every earlier source history.

If $(d, e, m) = (4, 6, 3)$, apply Theorem 12.5 with $n = 7$ and $B = \Gamma$. The ambient variety is smooth projective, L is ample Cartier, and the effective rational boundary is lc near x . Its actual center has the required state, and (21.2) supplies $\text{mult}_x \Gamma \geq (27/25)s$. The theorem gives an additive continuation of the entire Γ to a point center, of coefficient less than $8 - 1/2000$. Proposition 3.12 gives generation at x . Both integral and reduced reducible cubic tangents are covered by this theorem.

If (d, e, m) is $(4, 7, 4)$, $(4, 7, 5)$, or $(5, 7, 3)$, the center has dimension four or five and embedding dimension seven. The complete source, its degree $k \geq J$ with $25 \mid k$, and its pure actual threshold satisfy Theorem 17.40. This gives generation in each of the three cases, including the full nonreduced cubic fivefold.

If $(d, e, m) = (5, 6, 3)$, the same complete source and its pure threshold meet Theorem 13.1. Its cutoff was included in J , and the actual center has exactly the prescribed dimension, embedding dimension and multiplicity. It gives generation at x for every original center degree and every full cubic tangent allowed here.

Each case contradicts nongeneration. The only state of (21.4) still to consider is

$$(d, e, m) = (5, 6, 2).$$

Step 5. In this step, we identify the actual quadratic rank of the remaining double-fivefold. Write $R = \mathcal{O}_{W,x}$, $\mathfrak{n} = \mathfrak{m}_{W,x}$, and $G = \text{gr}_{\mathfrak{n}} R$, the entire ordinary tangent ring. The original observing degree satisfies $25 \mid k$, and $k \geq J$ is above the cutoff fixed for Proposition 14.2 before choosing k and Γ . Thus that proposition applies to the pure actual threshold Γ of the complete original $U_k(27/25)$ under the assumed volume-one nongeneration. It proves that G is reduced. More explicitly, its nonreduced double case has an additive point continuation of coefficient below $79/10 < 8$, and Proposition 3.12 excludes that case. Thus the nonreduced possibility has been paid before we use reducedness.

Theorem 4.13 makes the actual minimal center normal and Cohen–Macaulay near x . Its analytic local ring R^{an} is therefore a domain of dimension five and embedding dimension six. A minimal convergent presentation has the form

$$R^{\text{an}} \simeq S/\mathfrak{p}, \quad S = \mathbb{C}\{t_1, \dots, t_6\}, \quad \text{ht } \mathfrak{p} = 1.$$

The convergent power-series clause of Theorem 15.3 ([Dem12, Chapter II, Theorem 2.10]) makes S factorial. Thus $\mathfrak{p} = (f)$ for a nonzero convergent equation f . Analytification preserves the ordinary finite jets. If q is the lowest homogeneous part of f , multiplication of lowest homogeneous parts in the polynomial domain shows that the entire initial ideal of (f) is (q) . Consequently

$$G \simeq \mathbb{C}[t_1, \dots, t_6]/(q).$$

If q has degree ν , this graded ring has Hilbert series $(1 - z^\nu)/(1 - z)^6$ and multiplicity ν . Since $m = 2$, $\nu = 2$. Thus q is the quadratic initial form of the actual hypersurface equation.

Let r be the rank of q . It is positive. A rank-one quadratic over $\mathbb{C}$ is a nonzero scalar times the square of a linear form; the quotient by it contains that nonzero nilpotent linear class.

Reducedness of G therefore gives

$$2 \leq r \leq 6.$$

This argument identifies the quadratic rank while retaining the higher terms of the convergent equation f .

Step 6. In this step, we exclude every remaining quadratic rank and deficit. If $b > 18/5$, the first assertion of Theorem 19.1 applies to the same nonzero complete original source and its actual pure threshold. We have $L^7 = 1$, $25 \mid k$, $k \geq J$, and a five-dimensional minimal center with this own deficit. It gives generation at x . If $b \leq 18/5$ and $r \geq 3$, the double-fivefold conclusion of that theorem gives generation at every deficit and every original center degree. Its source hypotheses are the same ones already checked.

It remains to consider $r = 2$ and $b \leq 18/5$. The convergent splitting clause of Theorem 15.1 ([GP26, Theorem 2.4]) applies to f : it is convergent of order two and its Hessian has rank two. After an analytic coordinate change and a linear change in the two quadratic variables, the actual ring is

$$R^{\text{an}} \simeq \mathbb{C}\{u, v, z_1, \dots, z_4\}/(uv - \psi(z)), \quad \text{ord } \psi \geq 3.$$

Here $\psi \neq 0$, since otherwise the quotient by uv would not be a domain. All factors and higher terms of ψ are retained.

We now apply Theorem 20.1 with $n = 7$ and $B = \Gamma$. The variety and polarization satisfy its ambient hypotheses. The whole $\Gamma \sim_{\mathbb{Q}} sL$ is effective rational, $s > 0$, and it is lc near x . Its actual minimal center has state $(5, 6, 2)$, quadratic rank two, and deficit $b \leq 18/5$. The required price $(27/25)s + b \leq 7$ is exactly (21.3). Finally $A = L|_W$ is ample Cartier, so the original degree $A^5 = L^5 \cdot W$ is a positive integer. The theorem allows every such degree and every factorization of the actual convergent ψ . It yields an additive point continuation of the entire Γ with coefficient less than $8 - 1/400000$. Proposition 3.12 gives generation at x , contradicting nongeneration in the last case.

Each additive endpoint used above includes both cutting errors in its stated coefficient bound. Its final boundary is lc near x and has point minimal center, and its coefficient is strictly below 8, as required for point evaluation. The source endpoints already conclude generation at their own thresholds. Each application uses the complete source specified in its hypotheses; the alternative source trials are never added together as one boundary.

Step 7. In this step, we pass from generation at closed points to global generation on every component. We have contradicted nongeneration at an arbitrary closed point when $L^7 = 1$, and Theorem 9.5 covers $L^7 \geq 2$. Hence the global evaluation morphism

$$H^0(X, \mathcal{O}_X(K_X + 8L)) \otimes_{\mathbb{C}} \mathcal{O}_X \longrightarrow \mathcal{O}_X(K_X + 8L)$$

is surjective on every closed fiber of an integral X . Its cokernel is coherent. If it were nonzero, its support would contain a closed point; Nakayama's lemma there would contradict the fiber surjectivity. Thus the evaluation morphism is surjective.

For a general smooth projective sevenfold, apply this result on each of its finitely many open and closed integral components. Global sections on their disjoint union are the direct sum of the component spaces of sections, and the evaluation morphism restricts to the corresponding surjective morphism on each component. It is therefore surjective on X , which proves the theorem. $\square$

APPENDIX A. DETAILS OF THE USE OF GENERATIVE AI

The results of this paper were obtained using Danus [Liu⁺26], an automated mathematical reasoning system built on Rethlas [Ju⁺26]. For transparency we record the experiment in detail: how the system was run (§A.1), the problem exactly as it was posed (§A.2), the guidance provided during the run (§A.3), and the finite certificates and their checking procedures (§A.4).

A.1. How the system was run. The recorded project began on 7 September 2026 at 09:58:13 UTC as a resumption of an existing six-dimensional campaign. Its six-dimensional target was accepted by the research verifier on 9 September 2026 at 08:56:09 UTC, and a separate local sixfold manuscript was completed that day. The present campaign uses that proof and its corrected manuscript as internal research material.

The arbitrary-dimensional campaign began with a new persistent objective on 12 September 2026 at 05:07:24 UTC in the same main session. Its main agent used **gpt-6-astra** at reasoning effort **ultra**. The seven original research workers used the same base model: four at **max** and three at **xhigh**. The research verifier used **gpt-6-astra** at **max**. The sevenfold theorem was accepted on 13 September 2026 at 09:15:13.650148 UTC, an elapsed time of 28 hours, 7 minutes and 49.650148 seconds from creation of the new objective. This is elapsed campaign time using inherited work, rather than an attribution of that entire interval exclusively to the sevenfold proof.

A separate native writing workspace was then used for this manuscript. It contained nineteen section writer-verifier jobs and a whole-paper verifier, while the main session supplied the outline, introduction and final editorial pass. The research workers continued with the arbitrary-dimensional objective. The recorded usage totals combine the earlier sixfold work and the continuing research; they do not give an isolated token or monetary cost for the sevenfold theorem. No comparison run is recorded.

The research and manuscript verification records describe reviews of written arguments and their uses of previously accepted statements. They do not constitute formal verification. In the reorganized paper, the article proves the applicability, coverage and error rules for the finite certificates. The complete numerical data and independently executable exact-arithmetic checkers are supplied in the supplement. The native section reviews inspect these programs and their recorded executions as well as the written proofs; the reviews themselves do not independently rerun every mathematical computation.

A.2. The problem as posed. The AI-authored problem formulation given to the research session is reproduced below in full, with the quoted direct user utterance replaced by a byte-count notice. Its arbitrary-dimensional conclusion remains open in this run. Theorem 1.1 establishes the authorized seven-dimensional intermediate objective.

```
# Current objective: Fujita freeness in arbitrary dimension
```

```
The user has authorized restarting this existing project with the following new
objective:
```

```
> For every integer  $n \geq 1$ , every smooth complex projective  $n$ -fold  $X$ , and every ample
Cartier divisor  $L$  on  $X$ , prove that  $\mathcal{O}_X(K_X + (n+1)L)$  is globally generated.
```

```
Equivalently, for every such  $n$ ,  $X$ ,  $L$  and every closed point  $x$  in  $X$ , prove surjectivity
of
```

$$H^0(X, \mathcal{O}_X(K_X + (n+1)L)) \rightarrow \mathcal{O}_X(K_X + (n+1)L) \otimes k(x).$$

```
No global-generation assumption on  $L$ , special construction class for  $X$ , hypothesis on a
selected lc center, or dimension bound is part of the final target. The target is
freeness, rather than very ampleness.
```

```
## User instruction and priorities
```

```
User instruction, preserved verbatim:
```

```
> [138 bytes of discussion have been omitted because they contain private information.]
```

Prove the arbitrary-dimensional statement. Actively extract ideas and reusable mechanisms from the completed six-dimensional campaign, its proof, and its corrected manuscript. When a direct arbitrary-dimensional argument remains out of reach, proving the seven-dimensional statement K_X+8L is an authorized intermediate objective and test case. A seven-dimensional proof is not completion of this Goal: continue toward arbitrary dimension.

The sixfold source and its full dependency proofs are a starting point for serious extension, not a reason to repeat the completed sixfold-only campaign. Adapt the strategy when new mathematical evidence warrants it, while preserving the final target.

Completion standard

Success requires an unconditional proof valid for every n , X , L and x above. A finite list of dimensions, a weaker linear coefficient, a conditional source-selection theorem, or an unproved induction bridge does not discharge this statement. Every dimension-dependent estimate, uniformity requirement, selected-source hypothesis and terminal evaluation step used in the proof must be established in its actual range.

The completed sixfold Goal remains historical completion of its own target. Start a new native persistent Goal for this newly authorized arbitrary-dimensional target in the existing project and original main session. Do not rewrite its completed sixfold Goal as if it had always been arbitrary-dimensional. Keep the new Goal active until its own target is proved, the user pauses it, or an actual blocker requires user input.

Current authorization and records

This instruction supersedes the earlier sixfold-only restriction and the later Fujita shutdown for this project's research. The existing project, seven Astra workers (four max and three xhigh), Astra ultra main, and verifier port 8407 are retained. No other project is controlled by this project's main.

Preserve the sixfold fact graph, all historical task and control records, revocations, quarantine, source namespaces, cold reviews, and completed paper. This scope change creates no mathematical fact or verification event. The existing sixfold manuscript remains intact as reference; new manuscript work uses the configured native writing workflow after the relevant proof has been established.

Read expert_guidance/GUIDANCE.md, tasks/START_HERE.md, and expert_guidance/HIGHER_DIMENSION_ANALYSIS_20260909.md. The last is advisory research analysis, not a proved fact.

A.3. Guidance given during the run. The instruction record contains the following dated directions and their outcomes. Times are UTC.

- (1) On 7 September 2026, the operator asked that the existing project resume on the new server with six-dimensional Fujita freeness as its target, setting arbitrary dimensions aside. At 10:32 that day, the main session was instructed to read its project contract and begin. This campaign produced the accepted sixfold theorem and the separate sixfold manuscript described above.
- (2) A source-fulfillment entry at 11:12 on 7 September records that the previously named paper of Jingjun Han, arXiv:2609.01574v1, had been obtained and read. It is a record of fulfilling the source direction, rather than a new mathematical instruction. Its normal Hilbert coefficient estimate and Hilbert-moment inequalities enter the present paper through [section 4](#).
- (3) At 21:34 on 8 September, the operator requested resumption of the same sixfold session and objective, retaining the seven workers, their four-max/three-xhigh allocation, and the existing source boundaries. The subsequent sixfold result completed that objective; its completion was preserved as history.

- (4) At 05:06:54.444066 on 12 September, the operator changed the research objective to Fujita freeness in arbitrary dimension, instructed the system to draw on the sixfold proof, and authorized dimension seven as an intermediate step if necessary. The accompanying execution instruction required a new persistent objective in the existing main session, preservation of the completed old objective and manuscript, and prompt continuation of all seven workers. The new objective was created at 05:07:24. The sevenfold result was accepted the next day; the arbitrary-dimensional objective remains unproved and its research continues.
- (5) At 10:49 on 13 September, the operator authorized a complete sevenfold manuscript through the native writing workflow while the higher-dimensional research continued. This superseded the earlier restriction on new manuscript drafting for this paper. The initial nineteen technical sections and the pre-restructuring manuscript passed their recorded native reviews. Those reviews apply to the earlier organization. The subsequent restructuring requires fresh review of the changed article, data and checker interfaces.
- (6) The follow-up to that writing instruction specified an isolated source view with the original project name, a regular project configuration, linked original fact and computation records, and no additional research main or workers. At 10:51, a further instruction required the alternate source root to be applied only to individual paper commands and all main-authored paper files to use the new workspace's absolute path. These directions separated this manuscript from the completed sixfold manuscript and preserved the continuing research session.
- (7) On 13 September, a further user-approved instruction required substantive reorganization of the sevenfold manuscript. The geometric proof and the correctness, coverage and error arguments for its certificates remain in the article; complete machine-readable data and independently executable checkers belong to a documented supplement. The instruction specifically required removal of the duplicate printing of 17,822 identical encoded and expanded records, while preserving both original representations and their provenance in an archive. Existing remote checkers were to be inspected and reused, with fresh checking of the article–data–checker interfaces. No arbitrary page target was imposed, and the sixfold manuscript and continuing arbitrary-dimensional research were to be preserved.

The sixfold corrections inherited by the present paper include the explicit multiplicity-eight and multiplicity-nine threefold cases. The present continuation arguments retain those cases with their derivations. The historical and manuscript source reviews were used with their stated scopes, without treating a bounded review as a certification of every historical proof.

A.4. Computations. The root `supplement/README.md` gives the command for a complete run, the package inventory and the locations of the resulting reports. Dataset identifiers below use the section directory followed by the family name.

The finite certificates are supplied in thirteen section packages. Their common entry points use Python 3.10 or later and its standard library. Integer and rational arithmetic determine the comparisons; floating-point output is not used for acceptance. Each package specifies its complete index set, input format, source mapping and checking command. A root checker runs all thirteen packages and requires fresh, complete reports with the declared families, counts and inequalities. Selecting a subset is recorded as a partial run.

The article proves why these finite predicates apply to the complete geometric sources. In particular it retains the uniform remainder estimates, both cutting errors and their strict margins. The executable checks address the finite arithmetic, coverage and record consistency; they do not prove the geometric hypotheses. The principal computations are the following.

- (1) Section 5 checks the complete scalar successor recurrences and affine prices. The ambient system has 71 states and 253 affine lines; the restricted double-fivefold system has 28 states, 170 transitions and 739 full chains. The comparisons include every permitted successor and the distinct multiplicity-eight and multiplicity-nine cases.

- (2) Section 7 checks 91 records in ten families: spanning vectors, pairs, constants, polynomial coefficients, source tables, fourfold majorants, baselines, primitives, crossings and envelopes. Exact differentiation, integration and coefficient comparisons retain the knots, all competing branches and the unbounded final interval.
- (3) Section 9 checks the scalar table, seven volume-two identities, the final 1,031-cell two-profile certificate, the incoming and outgoing incidence domains, and the separate 1,099-cell early retained-divisor certificate. The latter includes all 69,664 coefficient inequalities. Closed-domain coverage, both tails and strict rounding are checked independently.
- (4) Section 10 checks the scalar ledger and its two profile families, their old and new tails, and all three source-margin comparisons. The complete run evaluates 20,965 predicates; the mathematical allowances of $1/10^6$ remain distinct from computational failures.
- (5) Section 11 checks four families comprising 5,228 integration pieces. It reconstructs every prescribed domain, including empty sets, points and segments, and tests all applicable witnesses and strict upper bounds. The 17,822 records of the remaining family have one canonical representation. The public package retains this canonical representation and its independent checker. The original encoded and expanded versions, their dictionary and the decoder are preserved separately in a private provenance archive; checking the exact identity of the original representations is separate from checking the mathematical inequalities.
- (6) Section 12 checks the complete state exclusions, both cubic-price tables, affine corrections and integral prices. It reconstructs 332 path domains with 755 vertices, including 64 empty domains, 119 points and 40 segments, and retains all 114 surplus requests.
- (7) Section 13 checks the tail recurrences, first-source polytopes, constrained integral prices and slope intervals. The 11,472 predicates include all 39 polytopes, 466 witnesses, 17 transitions and 2,886 candidate comparisons.
- (8) Section 14 checks the scalar table and the three-profile certificate. All 510 terminal boxes, 4,714 bounded cells and 15 unbounded cells are reconstructed, with 8,050 branch comparisons. The profile reserve and the subsequent source gain retain their separate constants.
- (9) Section 16 checks the restricted states and prices, 596 surplus queries, 1,830 grids, 23,365 feasible tuples and 45 identities. These finite checks accompany the analytic arguments for all permitted degrees and exponents; the finite character checks do not replace those arguments.
- (10) Section 17 checks all 25 certificate objects for the linked profiles, the initial-fivefold histories, smooth original sources, distinct divisors and persistent centers. Exact polynomial integration, Bernstein enclosures and reconstruction of every literal subdivision verify the stated mass and point tests. The eight-slab smooth history includes all 10,756 leaves; the quintuple-surface history tests all ten competitors. The embedding-seven reserve $1/64$, initial-fivefold reserve $1/500$, later gains and the distinct original-degree conditions are not interchanged.
- (11) Section 18 checks the scalar, transition, rank-two companion, ambient and identity families. The ambient reconstruction uses all 19 prefix intervals and 237 rectangles, yielding 31,627 integration cells and 29 coverage fibers. The companion's separate parameter range and analytic large-exponent reduction remain in the proof.
- (12) Section 19 checks the paired-profile tree with 301 boxes and all 4,269 records, including 11 unbounded records. The checker reconstructs the densities and chord bounds, recomputes the exact integrals and box totals, and checks the incoming caps, linked reserve and strict integral upper bounds. The complete bound is below $9999/10000$.
- (13) Section 20 checks all 235 source intervals and 236 endpoints against all eight formulas, including 1,266 enabled candidate records. It verifies the exact ceiling brackets, prefix sums and terminal reserve. The argument for every permitted analytic exponent and the common original degree remains in the article.

The public package supplies the canonical data and executable checking programs; fresh run reports are identified in its README. The root manifest binds the complete delivered data and

executable files. Hashes record identity; the independent checking algorithms and the article's correctness arguments provide the mathematical interpretation. Historical successful reports, deliberate rejection tests for missing or duplicate coverage, altered coefficients or bounds, and malformed inputs, discovery programs and duplicate representations are preserved separately in private archival material.

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